Microsoft

Microsoft Official Course



MS-203T00

Microsoft 365 Messaging

MS-203T00
Microsoft 365 Messaging

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Module 0 Welcome

Course Introduction

Course Introduction

This course covers the following central elements of Microsoft 365 messaging administration:

- Message transport and mail flow
- Messaging Security, Hygiene, and Compliance
- Messaging infrastructure
- Hybrid Messaging

The course begins with an examination of Microsoft's messaging transport pipeline. You will learn how to plan, configure, and manage transport services to provide efficient communication between your Exchange Servers, Exchange Online, and other mail servers on the Internet. You will also learn how to manage and troubleshoot mail flow, as well as how to troubleshoot transport services when issues occur in different scenarios so that you can provide your organization with a healthy and reliable messaging infrastructure.

You will then learn about the different messaging-related security and compliance service configurations and settings with Exchange Server and Exchange Online. You will begin by examining the fundamentals of message hygiene with Exchange Online Protection and Exchange Advanced Threat Protection, as well as connection and spam filtering with Exchange. You will then examine messaging compliance solutions, such as retention and data loss prevention policies, as well as audit log investigations and eDiscovery search mechanisms from the messaging compliance perspective.

You will then transition to a study of the Microsoft messaging infrastructure. This includes managing organizational settings, mobile devices, role-based permissions, recipient objects and resources, and address lists. This section then examines public folders, including how to plan, implement, and troubleshoot your public folder hierarchy.

Finally, the course concludes with an introduction to the world of hybrid messaging. You will learn how to deploy a hybrid environment for your Exchange organization and how to migrate mailboxes to Exchange Online. You will begin by examining available mailbox migration options, and then focus on planning for a hybrid deployment. Finally, you will learn how to implement and troubleshoot a hybrid deployment.

Post-Course Access to Labs

Once you have completed the course, you can continue to access its labs in **GitHub**[1] for personal self-study.

Audience Profile

This course is designed for persons who are aspiring to be Messaging Administrators in a Microsoft 365 deployment. The Messaging Administrator deploys, configures, manages, and troubleshoots recipients, permissions, mail protection, mail flow, and public folders in both on-premises and cloud enterprise environments. Responsibilities include managing message hygiene, messaging infrastructure, and hybrid configuration and migration.

To implement a secure hybrid topology that meets the business needs of a modern organization, the Messaging Administrator must collaborate with the Security Administrator and Microsoft 365 Enterprise Administrator. The Messaging Administrator should have a working knowledge of authentication types, licensing, and integration with Microsoft 365 applications.

1 https://github.com/MicrosoftLearning/MS-203T00-Microsoft-365-Messaging/tree/master/Instructions/Labs

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Module 1 Managing the Transport Pipeline

Overview of Transport Services

Lesson Introduction

Mail flow and message transport describe the actual way in which mail is delivered from external or internal senders to a designated recipient in your organization, and from your message senders outgoing to any recipients.

In this lesson you will learn about the different transport components of Exchange, how the message routing works, and how to configure the message flow for your organization.

After this lesson, you will be able to:

- Describe the transport components of Exchange
- Plan an effective message routing for your organization
- Modify message flow for your organization
- Describe which transport agents exist and what they do

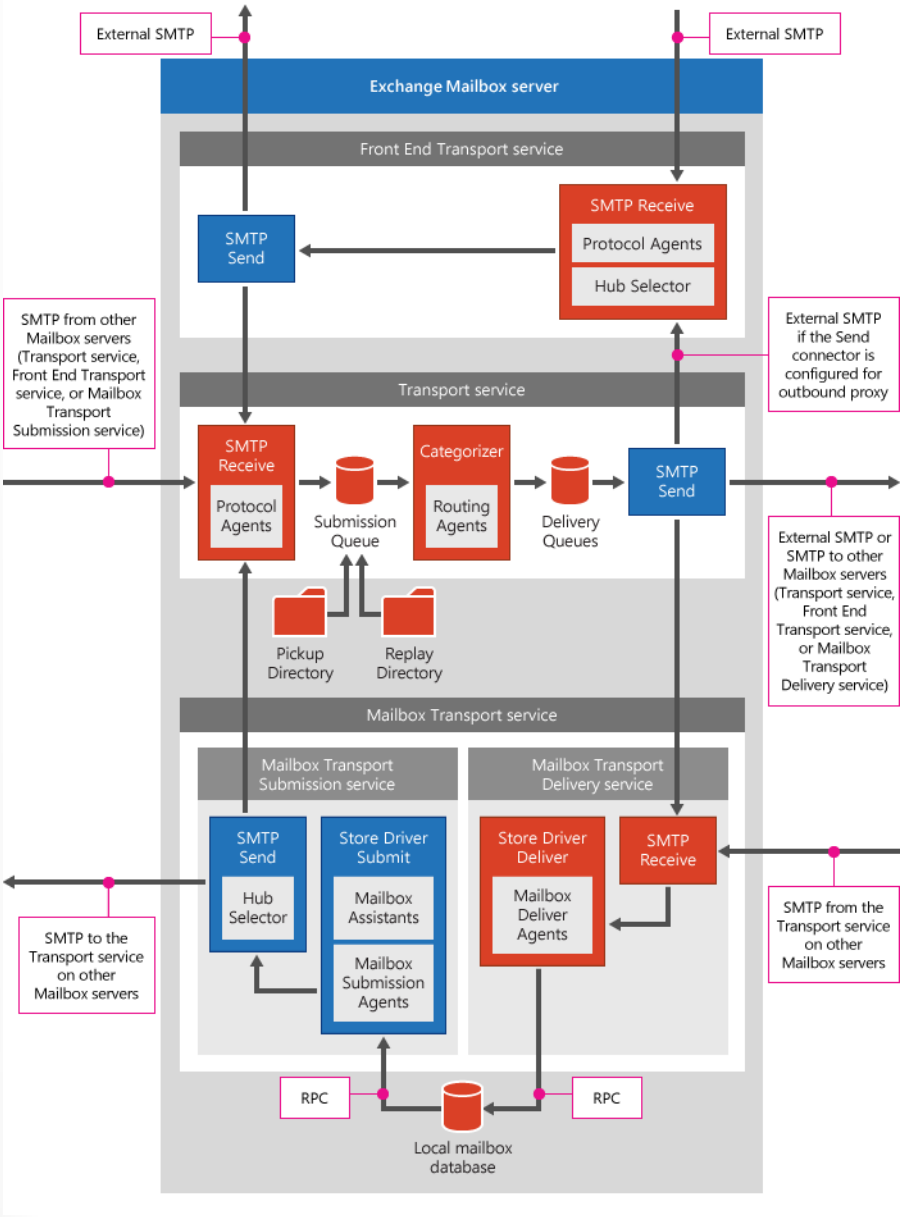
Message transport components

The process and components that Exchange Server uses to provide message routing is referred to as the transport pipeline. The transport pipeline has multiple services, queues, and connectors that perform specific roles within the message transport process. All the components work together to provide message routing.

Message transport services

The message transport process is almost the same in Exchange Server 2019 as it is in Exchange Server 2013. The notable change is that in Exchange 2019 all transport services reside on Mailbox servers, whereas in Exchange 2013, the transport services could be split between the Client Access server and Mailbox server roles.

The following diagram shows the Exchange 2019 configuration with the three transport services participating on message transport and delivery.



The transport services include:

- **Front-End Transport service**. This service proxies all inbound SMTP traffic from outside the Exchange organization to the Transport service. The Front-End Transport service does not perform any message processing or queuing. You also have the option to use the Front-End Transport service to proxy outbound mail messages as well.
- **Transport service**. This service is responsible for message categorization and content inspection. The Transport service delivers SMTP messages to other Mailbox servers, the Mailbox Transport service, or externally.

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- **Mailbox Transport service**. This logical service is responsible for message delivery to and from mailbox databases. Remote procedure call (RPC) is used for communication with the mailbox database. The Mailbox Transport service is comprised of two Windows services:
  - **Mailbox Transport Delivery service**. This service is responsible for accepting messages from the Transport service and delivering them to the mailbox database.
  - **Mailbox Transport Submission service**. This service is responsible for retrieving messages from the Outbox folder in mailboxes and delivering the messages to the Transport service.

SMTP send and receive components

The transport services use SMTP to send and receive messages. The SMTP send component in one transport service delivers messages to the SMTP receive component in another transport service. Communication between transport services by using SMTP can occur within the same server or between Mailbox servers.

Note: Communication between the Mailbox Transport service and mailbox databases uses remote procedure call (RPC) and is performed only within the same server.

Pickup and replay directories

Most messages enter the message transport pipeline through the SMTP Receive component, or by submission through the store driver. However, messages can also enter the message transport pipeline by being placed in the Pickup directory or Replay directory on a Mailbox server.

You can use the Pickup directory to submit a properly formatted text file as a message for delivery. This can be useful for validating mail flow in an organization, replaying specific messages, or returning recovered email to the message transport pipeline. In addition, some legacy applications may place messages directly into the Pickup directory for delivery, rather than communicate directly with Exchange Server SMTP Receive connectors.

The Replay directory is used to resubmit exported Exchange messages and to receive messages from foreign gateway servers. These messages are already formatted for the Replay directory. There is little or no need for administrators or applications to compose and submit new message files by using the Replay directory. You can use the Pickup directory to create and submit new message files.

Submission queue

When messages are received by the Transport service, they are placed in the submission queue. Messages wait in the submission queue until they are processed by the categorizer for delivery.

Categorizer

The categorizer is the component that is responsible for making routing decisions. Each message is retrieved from the submission queue and processed by the categorizer to determine how a message should be delivered. When the message delivery process has been determined the categorizer places the message in a delivery queue.

As part of message processing, the categorizer performs the following tasks:

- **Identifies and verifies recipients**. All messages must have a valid SMTP address to be identified.
- **Bifurcates messages that have multiple recipients**. The expansion of distribution lists enables identification of individual recipients who belong to the distribution list. In addition, the categorizer

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processes the return path for distribution-list delivery status notifications (DSNs), and it determines whether Out-of-Office messages or automatically generated replies are sent to the original message's sender.

- **Determines routing paths**. When determining the routing path, the categorizer identifies the destination, which must be a user's mailbox, a public folder, or an expansion server for distribution groups. If the categorizer cannot determine a valid destination, a non-delivery report (NDR) is generated.
- **Converts content format**. Recipients can require messages in different formats. The categorizer converts the message to an appropriate format for the recipient. Inside the Exchange organization, the recipient format is stored in Active Directory. Messages routed to the Internet are sent in the Multipurpose Internet Mail Extensions (MIME) or Secure/Multipurpose Internet Mail Extensions (S/MIME) format.
- **Applies organizational message policies**. You can use organizational policies to control messaging aspects such as size, permission to send messages to specific users, the number of message recipients, and other characteristics.

Delivery queues

Delivery queues contain messages that the Exchange Server has not delivered. Messages that are in the Delivery Queue are processed by the SMTP Send component. Depending on their intended delivery route, these messages can be forwarded to another Mailbox server or to the SMTP Receive component of another message transport service on the same Mailbox server.

Store driver

The store driver is a software component that is present in both the Mailbox Transport Submission service and the Mailbox Transport Delivery service. It's the software that understands how to communicate with the mailbox database. All communication with the mailbox database is done by using RPC.

When a user sends a message, it is placed in the Outbox folder of the user's mailbox. The Store Driver Submit component retrieves the message from the Outbox and delivers it to the Hub Selector for delivery to a Transport service, where it's placed in the submission queue for processing. After the message has been placed in the submission queue, the message is moved from the Outbox folder to the Sent Items folder.

Messages in the Outbox are stored in the Messaging Application Programming Interface (MAPI) format. The store driver must convert them to Summary Transport Neutral Encapsulation Format (STNEF) before placing them in the submission queue. The store driver performs this conversion to ensure successful delivery of the messages, regardless of the format that created the messages. A Transport Neutral Encapsulation Format (TNEF) encoded message contains a plain text version of the message, and a binary attachment that contains various other parts of the original message.

Some Microsoft Outlook features require that TNEF encoding be understood correctly by an Internet email recipient who also uses Outlook. For example, when you send a message with voting buttons to a recipient over the Internet, if TNEF is not enabled for that recipient, the voting buttons will not be received. If the store driver cannot convert the content, it generates a non-delivery report (NDR).

Message routing

The primary task of the Transport service that exists on Mailbox servers in your Exchange organization is to route messages received from users and external sources to their final destinations. Routing decisions

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are made during message categorization, which is a component of the Transport service that processes all incoming messages and determines what to do with the messages based on information about their destinations.

When a message is received by the Transport service on a Mailbox server, the message must be categorized. The first phase of message categorization is recipient resolution. After the recipient has been resolved, the ultimate destination can be determined. The next phase, routing, determines how to best reach that destination. Routing in Exchange is generalized for increased flexibility and decreased complexity by using the concepts of **routing destinations** and **delivery groups**.

Routing destinations

Each outgoing message has a source and a destination. The destination for each message in an Exchange Server organization is called a routing destination. There are several types of routing destinations, including:

- **Mailbox Database**. When a message is sent to a user with a mailbox on the Mailbox server in an Exchange organization, the routing destination for the message is the Mailbox Database. This also applies to public folders in Exchange Server.
- **Connector**. A connector is used as a routing destination when it is configured as a send connector for SMTP messages. A delivery-agent connector or a foreign connector is used as a routing destination for non-SMTP messages.
- **Distribution group expansion server**. If a distribution group has a dedicated expansion server, then that server is a routing destination for messages that are sent to the distribution group.

Delivery groups

Delivery groups represent the collection of transport servers that are responsible for delivering messages to a specific routing destination. Each routing destination has its own delivery group. Transport servers in a delivery group can be Exchange Server 2019, 2016 or 2013 Mailbox servers.

- If a message is sent to a destination within the same delivery group, no routing is required. Within the same delivery group, a message is delivered directly to the destination.
- If a message is sent to a destination in a different delivery group, routing is based on the lowest cost. By default, routing cost is based on the cost of Active Directory site connectors. Message routing is performed by the Transport service in each site as required to deliver the message to the destination delivery group. In most cases, messages are delivered directly from the source delivery group to the destination delivery group unless the default message flow has been modified.

There are several types of delivery groups in Exchange Server, including:

- **Routable database availability group (DAG).** This represents a set of Exchange Server 2019, 2016, or 2013 Mailbox servers that are members of the same DAG. All mailbox databases in the DAG are local routing destinations for this delivery group. When the message arrives, the Transport service on the Mailbox server accepts it and routes it to the Mailbox Transport Delivery service on the Mailbox server that currently holds the active copy of the destination mailbox database. The Mailbox Transport Delivery service delivers the message to the mailbox database. In this case, the DAG is the delivery group boundary.
- **Mailbox delivery group (Exchange Server 2013, 2016, and 2019).** When Mailbox servers are not DAG members, the delivery group boundary is defined by the Active Directory site. All Exchange Server 2013, 2016, and 2019 Mailbox servers in a single Active Directory site are part of the same delivery group. For example, the Transport service on an Exchange Server 2013 Mailbox server can

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deliver messages to the Mailbox Transport Delivery service on an Exchange Server 2016 Mailbox server where a mailbox is hosted.

- **Connector source servers**. The connector source servers represent a mixed set of Exchange Server 2010 Hub Transport servers, Exchange Server 2013 servers, and Exchange Server 2016 Mailbox servers that are designated as source servers for the send connector, the delivery agent connector, or a foreign connector in the same or a different AD DS site. The connector is the routing destination. When a connector is scoped, only that server in that particular AD DS site is aware of it and can route messages to it.
- **Server list**. When the recipient for a message is a distribution group, the distribution group must be expanded for delivery to the group members. If expansion servers are defined for the distribution group, the list of expansion servers is a delivery group.
- **Active Directory site.** An AD site is used as a delivery group when a message must travel through an AD site on the path for delivery. This occurs when an AD site is designated as a Hub site, or when the Exchange Edge server is subscribed to the specific site, and other sites cannot access it directly.

A server can be a member of multiple delivery groups. For example, an Exchange Mailbox server can be a DAG member, as well as the source transport server for a send connector. The server is a member of the DAG delivery group and the connector source server's delivery group.

Configuring message transport

There are different ways to configure message transport in your organization. When a message is sent to a remote delivery group, Exchange Server calculates the least-cost routing path. Exchange uses the following logic to select the routing path for a message. This logic is basically unchanged from Exchange 2010:

1. **Least-cost routing path.** Calculate the least-cost routing path by adding the cost of the IP site links that must be traversed to reach the destination. If the destination is a connector, the cost assigned to the address space is added to the cost to reach the selected connector. If multiple routing paths are possible, the routing path with the lowest aggregate cost is used.
2. **Least number of hops.** If more than one routing path has the same aggregate cost, the number of hops in each path is evaluated and the routing path with the least number of hops is used.
3. **Name of the AD site closest to the destination in alphanumeric order.** If more than one routing path is still available, the names assigned to the Active Directory sites that precede the name of the destination are considered. The routing path where the Active Directory site nearest the destination is lowest in alphanumeric order is used. If the site nearest the destination is the same for all routing paths being evaluated, an earlier site name is considered.

Note: Size limits on connectors are also a factor for transportation. Connectors that are configured with message size limits smaller than the size of the message are eliminated from consideration.

Delivery groups can span multiple Active Directory sites. For example, a DAG can have members in multiple sites. When this occurs, all Active Directory sites containing DAG members are evaluated and the Active Directory site with the lowest routing cost is selected as the primary site for message delivery.

After the least-cost route for message delivery has been selected, the Transport service in the source delivery group attempts to deliver the message directly to the destination delivery group. If the least-cost route includes multiple Active Directory sites, message delivery does not hop through each site. Other Active Directory sites on the least-cost path are used only if there is a delivery error or you have configured a modification.

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Fallback

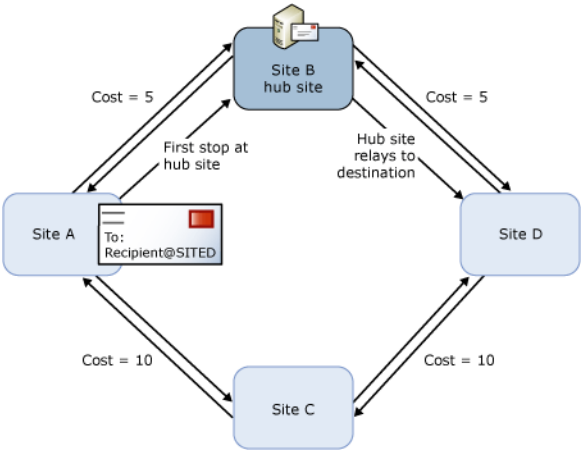
If the destination on the least-cost path cannot be contacted, then the Transport service falls back and attempts to deliver the message to the next closest site on the least-cost path. The message is queued at the next closest site until the destination is available.

Note: If the next closest site on the least-cost path is not available, then the Transport service tries one hop back again. This process continues until the message can be delivered.

Hub sites

You can configure one or more Active Directory sites in your organization as hub sites. When a hub site exists along the least-cost routing path between two Mailbox servers, the messages are routed to a Mailbox server in the hub site for processing before they are relayed to the destination server.

The Transport service routes a message through a hub site only if it exists along the least-cost routing path. The originating Mailbox server always calculates the lowest-cost route first, and then checks if any of the sites on the route are hub sites. If the lowest-cost route does not include a hub site, the Hub Transport service attempts a direct connection.



You should run the following PowerShell command to configure a site as hub site:

```
Set-ADSite –Identity <sitename> –HubSiteEnabled $true
```

You should run the following command to check whether you configured a hub site:

```
Get-AdSite | Format-List Name,HubSiteEnabled
```

Exchange-specific routing costs

You can assign an Exchange-specific cost to an Active Directory IP site link. If you assign an Exchange-specific cost to the site link, the Transport service determines the least-cost routing path by using this attribute rather than the Active Directory-assigned cost, unless the mailbox server is a member of a DAG, in which case Active Directory site link costs are always used.

You should run the following command to assign an Exchange-specific routing cost to an Active Directory IP site link:

```
Set-AdSiteLink –Identity <ADsitelinkname> –ExchangeCost <value>
```

You should run the following command to assign a maximum message size limit for messages sent between AD DS sites:

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```
Set-AdSiteLink –Identity <ADsitelinkname> –MaxMessageSize <value>
```

You should run the following command to check if you properly configured an Exchange cost:

```
Get-AdSite | Format-List Name,HubSiteEnabled
```

Expansion servers for distribution groups

By default, when a message is sent to a distribution group, the first Transport service that receives the message expands the distribution list and calculates how to route the messages to each recipient in the list. If you configure an expansion server for the distribution list, all messages sent to the distribution list are sent to the specified Mailbox server, which then expands the list and distributes the messages. For example, you can use expansion servers for location-based distribution groups to ensure that the local Transport service resolves them.

Transport agents

Transport agents process email messages that pass through the transport pipeline on Transport service components. Because all messages are processed by the Transport service for delivery, you're guaranteed that all messages are evaluated by transport agents installed on the server.

Microsoft Exchange provides a library of classes that support the extension of the Exchange transport behavior. These classes enable the reading, writing, and converting of content types. You can use these classes to create Exchange transport applications that read, write, and process messages in the transport pipeline.

Transport agents use SMTP events, such as **OnHeloCommand**, **OnDataCommand,** or **OnEndOfData**. These events are triggered as messages move through the transport pipeline. SMTP events give transport agents access to messages at specific points during the SMTP conversation and during routing of messages through the organization.

While multiple transport agents are included with Exchange Server, most of them are not manageable; therefore, they're not visible to the management cmdlets for transport agents. For example, the Journal Agent is not visible when you use the **Get-TransportAgent** cmdlet.

If anti-spam functionality is enabled on an Exchange Server Mailbox server, the anti-spam transport agents are installed.

Priority of transport agents

There are two factors that determine the order that transport agents act on messages in the transport pipeline:

- The SMTP event where the transport agent is registered, and when that SMTP event encounters messages.
- The priority value that's assigned to the transport agent if there are multiple agents registered to the same SMTP event. The highest priority is 1. A higher integer value indicates a lower agent priority.

For example, suppose you configured the following transport agents:

- Transport Agent A with a priority of 1 and Transport Agent C with a priority of 2 are registered to the OnEndOfHeaders SMTP event.
- Transport Agent B with a priority of 4 is registered to the **OnMailCommand** SMTP event.

Transport Agent B is applied to messages first because the **OnMailCommand** event encounters messages before the **OnEndOfHeaders** event. When messages reach the **OnEndOfHeaders** event, Transport

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Agent A is applied before Transport Agent C because Transport Agent A has a higher priority (lower integer value) than Transport Agent C.

Built-in transport agents

Most of the built-in transport agents on Exchange Mailbox servers are invisible and unmanageable by the transport agent management cmdlets. Virtually all the built-in transport agents that are visible and manageable are in the Transport service on Mailbox servers and on Edge Transport servers.

Two of the interesting transport agents on Exchange Mailbox servers are:

- **Transport Rule Agent**. This agent is manageable and applies transport rules to each message during processing by the Transport service. It acts on the SMTP event **OnResolvedMessage** and has a priority of 1.
- **Journal Agent**. This agent is not manageable and carries out journaling rules. It works on the SMTP event **OnRoutedMessage** and has no priority.

On Edge Transport servers, most of the built-in transport agents are visible and manageable by the transport agent management cmdlets or by other feature-specific cmdlets. Some of the interesting transport agents on Exchange Edge Servers include:

- **Connection Filtering Agent**. This agent applies the configured connection filtering settings for malware filtering. It works on SMTP events **OnConnectEvent**, **OnMailCommand**, **OnRcptComand**, **OnEndOfHeaders** and has a priority of 1.
- **Sender Filter Agent**. This agent enforces the configured sender filtering. It works on the SMTP events **OnMailCommand** and **OnEndOfHeaders** with a priority of 6.
- **Recipient Filter Agent**. This agent enforces the configured sender filtering. It works on the SMTP event **OnRcptCommand** and has a priority of 7.
- **Attachment Filtering Agent**. This agent performs the attachment filtering. It acts on the SMTP event **OnEndOfData** and has a priority of 9.

There are more agents on Mailbox and Edge servers, as well as manageable or unmanageable, and visible or invisible.

Custom transport agents

Custom transport agents provide additional functionality to Exchange Server. Custom transport agents are typically installed as part of add-on software for Exchange Server. For example, if you install anti-spam or antivirus software on Exchange Server, that software includes a transport agent to scan messages that pass through the Transport service.

You can also develop and implement your own transport agents. If you create your own custom transport agent, you must install it on each server by using the **Install-TransportAgent** cmdlet.

Note: Transport agents have unrestricted access to all messages passing through the transport pipeline on a server. You should implement only trusted transport agents. Also, be aware that a poorly implemented transport agent can affect system stability.

Viewing enabled transport agents

The transport pipeline has specific events such as **OnSubmittedMessage**. Transport agents are triggered by these events. You can view the transport agents triggered by specific events in the transport pipeline by running the following PowerShell command:

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```
Get-TransportPipeline | FL Event,TransportAgents
```

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Configuring Message Transport

Lesson Introduction

Configuring and modifying the organization settings for message transport is a recurring task for messaging administrators. This task includes the configuration of different transport options, managing connectors and domains, and overall message moderation.

In this lesson, you will examine the tasks that messaging administrators must complete to configure message transport. You will review the message transport options and learn how to configure domains and connectors and how to implement an approval workflow for messaging.

After this lesson, you will be able to:

- Configure the different transport options
- Plan and set up domains for your organizations
- Understand how receive and send connectors work
- Describe how message moderation for different recipients work

Configuring message transport options

Some transport options that are valid for entire Exchange organizations are set at the organization level. These transport settings apply only to those servers in the organization running Exchange Server. Some of these settings can be modified in the EAC, but to obtain full access to all the transport settings, you must use the **Get-TransportConfig** and **Set-TransportConfig** cmdlets in the Exchange Management Shell (EMS).

Some organization-level transport settings include:

- **Maximum message sizes**. You can configure MaxRecieveSize and MaxSendSize to control the maximum size of messages allowed in the organization. The default value for both settings is 10 megabytes (MB).
- **Maximum number of recipients**. You can configure MaxRecipientEnvelopeLimit to control the maximum number of recipients allowed for a single message. The default value is 500.
- **Postmaster addresss**. You can configure ExternalPostmasterAddress to define the email address that's used as the sender of DSN messages. If you don't define this option, postmaster@defaultAcceptedDomain is used by default
- **Shadow redundancy settings**. You can configure a variety of shadow redundancy settings, such as ShadowHeartBeatTimeoutInterval.
- **SafetyNet settings**. You can configure a variety of SafetyNet settings, such as SafetyNetHoldTime.

Even if you configure these settings at the organization level, you may still need to configure individual services or Exchange objects. For example, if you change the maximum message size for your organization, you must also change the maximum message sizes on your receive and send connectors, or those will still block your users messages if they're too large.

To configure the transport options for individual servers, you can also use the following cmdlets for the Exchange Management Shell:

- Get-/Set-TransportService
- Get-/Set-MailboxTransportService

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- Get-/Set-FrontendTransportService

Transport settings for Exchange Online

If your organization uses Exchange Online, you're limited to a subset of organization settings, because some settings are managed by Microsoft and cannot be modified, such as message size and maximum recipients per message limits. While you can use the **Get-TransportConfig** and **Set-TransportConfig** cmdlets to modify some settings for your organization, you should be aware of the Exchange Online service limits to avoid throttling of your users or confusing NDRs.

Users regularly encounter some of the following limitations in their daily work:

- **Message size limit**. Message size limits are necessary to prevent large messages from blocking delivery of other messages and affecting service performance for all users. These limits include attachments and apply organization-wide to all messages (inbound, outbound, and internal). Messages larger than this limit will not be delivered, and the sender will receive a non-delivery report (NDR). While message size limits can be configured up, down, or on a per-user basis, administrators can also create transport rules to limit the maximum size of any individual attachment.
- **Subject length limit.** The maximum number of text characters allowed in the subject line of an email message.
- **File attachments limit**. The maximum number of file attachments allowed in an email message. Even if the total size of all the file attachments doesn't violate the message size limit, there is still a limit on how many attachments are allowed in the message. This limit is controlled by the multipart message limit.
- **File attachment size limit**. The maximum file size of a single attachment.
- **Multipart message limit.** The maximum number of message body parts that are allowed in a MIME multipart message. This limit also controls the maximum number of file attachments that are allowed in a message.
- **Embedded message depth limit**. The maximum number of forwarded email messages that are allowed in an email message.
- **Receiving limits.** Apply to the number of messages that a user, group, or public folder can receive per hour. Currently 3,600 messages per hour.
- **Recipient rate limit**. To discourage the delivery of unsolicited bulk messages, Exchange Online has recipient limits that prevent users and applications from sending large volumes of email. These limits are applied per-user to all outbound and internal messages. Currently 10,000 recipients per day.
- **Recipient limit.** This is the maximum number of recipients allowed in the To:, Cc:, and Bcc: fields for a single email message. Currently 500 recipients.
- **Message rate limit.** Message rate limits determine how many messages a user can send from their Exchange Online account within a specified period of time. This limit helps prevent overconsumption of system resources by a single sender. If a user submits messages through SMTP client submission at a rate that exceeds the limit, the messages will be rejected and the client will need to retry.

Additional reading. For more information, see the following article on **Exchange Online Limits**[1].

1 https://aka.ms/AA3srac

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Configuring SMTP domains

Exchange supports two types of SMTP domains: accepted domains and remote domains. Accepted domains are further dilineated into three sub-types. This topic examines each of these SMTP domain types.

Accepted domains

An accepted domain is a domain from which the Exchange organization receives and processes messages. Depending on the configuration of the accepted domain, the message may be delivered to an internal recipient or forwarded to another messaging organization.

Accepted domains are another part of message routing and receival, because Exchange Servers only accept mails that are sent to one or more of their accepted domains. Outgoing messages are also checked against the configured accepted domains to determine if a message needs to be delivered inside your organization, to a partner organization, or if the recipients are external.

There are three types of accepted domains:

- **Authoritative domain**. An authoritative domain is one for which the Exchange organization is completely responsible. All recipients for that domain are hosted in the Exchange organization and if a recipient is not found in the organization, an NDR is sent to the message sender.
- **Internal relay domain**. An internal relay domain is used when some mailboxes are in the Exchange organization and some mailboxes are in an external organization. Messages received for an internal relay domain are first evaluated to identify whether there is a matching recipient in the Exchange organization. If there is a matching recipient, the message is delivered to that recipient. If there is no matching recipient, the message is forwarded through a send connector defined for the internal relay domain. The send connector for the internal relay domain defines how to deliver the messages to another organization.
- **External relay domain**. An external relay domain is configured when the Exchange organization needs to process messages for a domain, but no recipients for that domain are in the Exchange organization. For example, the Exchange organization may be performing anti-spam scanning for a subsidiary within a separate Exchange organization. A send connector defines where the messages for the external relay domain should be delivered.

Note: For Exchange Server deployments, you can add any accepted domain to your local Exchange organization. However, for Exchange Online deployments, you must first prove ownership of the domain before you can add it. For this reason, you need to add and verify your domain in Office 365 before it will be automatically added to your organization's accepted domains.

Authoritative domains

You configure an accepted domain as an authoritative domain when all recipients in that domain exist in your Exchange organization. By default, when you install the first Exchange Mailbox server, the fully qualified domain name (FQDN) of your forest root domain in Active Directory is configured as an authoritative domain. If you don't want to use this domain for email, you must add another authoritative domain.

An organization can be configured with multiple authoritative domains. The set of email domains for an organization are the authoritative domains. You can use authoritative domains in email address policies, and Exchange is responsible for generating NDRs for non-existent recipients in authoritative domains.

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Note: You should only configure domains that you own and that have recipients only in your Exchange organization. If there are other participating mail servers or services hosting mailboxes with this domain suffix, you should configure a relay domain.

Relay domains

You configure an accepted domain as a relay domain (also known as non-authoritative domain) when some or none of the recipients in that domain exist in your Exchange organization (for example, partners or subsidiaries). Exchange isn't responsible for generating NDRs for non-existent recipients in a relay domain. Instead, you configure a Send connector with the address space of the relay domain, and you configure this Send connector to use smart host routing to relay messages to their destination (directly or to the next hop).

You can configure a relay domain as an internal relay domain or as an external relay domain. The differences are described in the following list:

- **Internal relay domains**
  - Some of the recipients in the internal relay domain don't exist in the Exchange organization. For example:
    - You share the domain between the Exchange organization and a third-party messaging system.
    - You share the domain between Exchange organizations in different Active Directory forests.
  - Recipients in the internal relay domain can be represented as mail contacts or mail users in the Exchange organization (manually created or automatically created by using directory synchronization).
  - The Send connector that you configure for the internal relay domain is sourced on an internal Mailbox server.
  - You can use internal relay domains in email address policies.
- **External relay domains**
  - None of the recipients in the external relay domain exist in the Exchange organization (including mail contacts or mail users). For example, your Exchange organization is the central location for accepting Internet email for a group of separate organizations.
  - The Send connector that you configure for the external relay domain is sourced on an Edge Transport server or Internet-facing Mailbox server.
  - You can't use external relay domains in email address policies.

Remote domains

Remote domains are different from accepted domains because they don’t manage which mails are accepted or rejected in your organization; instead, they define settings for message delivery to SMTP domains that are external to your Exchange Server organization. You can create remote domain entries to define the settings for message transfer between the Exchange Server organization and domains outside your Active Directory forest. When you create a remote domain entry, you control the types of messages that are sent to that domain. You can also apply message-format policies and acceptable character sets for messages that are sent from your organization’s users to the remote domain.

There is one remote domain named Default that exists after Exchange Server is installed or your Tenant is created. This remote domain is defined for the domain name *, which applies to all messages to external domains. You can create remote domains for additional domains as required. Remote domains are often

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created for partner domains where you want to allow messages that are not normally allowed. For example, a remote domain for a partner organization may allow users to automatically forward messages that the Default remote domain blocks.

Some of the settings you can configure for a remote domain include:

- **AllowedOOFType**. Defines whether external or internal out of office messages are delivered to the remote domain. The default is External.
- **AutoReplyEnabled**. Defines whether automatic replies are sent to the remote domain. The default is $false.
- **AutoForwardedEnabled**. Defines whether messages can be forwarded automatically to the remote domain by using a rule. The default is $false.
- **DeliveryReportEnabled**. Defines whether delivery reports requested by clients are sent to the remote domain. The default is $true.
- **NDREnabled**. Defines whether non-delivery reports (NDRs) are sent to the remote domain. The default it $true.
- **ContentType**. Defines the format for messages sent to the remote domain. The default is MimeHtml-Text, which formats all messages as HTML unless they are text formatted.

Configuring connectors

Connectors are a collection of instructions that customize the way your email flows to and from your Exchange Server or Exchange Online organization. There are different connectors available in Exchange Server and Exchange Online that serve a similar purpose.

Connector in Exchange Server	Connector in Exchange Online	Description
Receive connector	Inbound Connector	Control incoming SMTP mail flow. They listen for incoming connections that match the configuration of the connector.
Send connector	Outbound Connector	Control outgoing SMTP mail flow. The connector is chosen based on the message recipients (e.g., target SMTP domain) and the configuration of the connector.
Delivery agents and Delivery Agent Connectors	Not available	Delivery agents and Delivery Agent connectors control outgoing mail flow to non-SMTP systems. Outgoing messages are put into message queues for delivery to the non-SMTP system. Delivery agents and Delivery agent connectors are preferred over Foreign connectors due to their improved performance and management.

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Connector in Exchange Server	Connector in Exchange Online	Description
Foreign connectors	Not available	Foreign connectors control outgoing mail flow to non-SMTP systems. Outgoing messages are written to files in a location called the Drop directory to be picked up by the non-SMTP system.

Connectors in Exchange Server

Exchange Server uses connectors to enable incoming and outgoing mail flow on Exchange servers, and between services in the transport pipeline on the local Exchange server. The most common connects in Exchange are Receive and Send Connectors.

You can create and manage connectors for Exchange Server in the Exchange Admin Center (EAC) and Exchange Management Shell (EMS) using the **Set-ReceiveConnector** and **Set-SendConnector** cmdlets.

Receive connectors

In Exchange Server, receive connectors are required to allow a Mailbox server to receive SMTP messages. The receive connectors are used to receive messages from other Exchange servers in the organization, Edge Transport servers, application servers, Post Office Protocol (POP) or Internet Message Access Protocol (IMAP) clients, and external mail servers.

Each receive connector has the following settings:

- **Remote network settings**. These define the IP addresses that are allowed to communicate by using this receive connector.
- **Network adapter bindings**. These define the local IP address and port number that the connector listens on for connections.
- **Authentication settings**. These define the authentication types that can be used with the connector, such as Transport Layer Security (TLS), basic authentication, or integrated Windows authentication.
- **Permission groups**. These define who is allowed to communicate by using this receive connector. For example, you can allow Exchange users or Anonymous users.
- **Maximum message size**. This defines the maximum message size that can be transported over this receive connector. The default is 35 megabytes (MB).
- **Protocol logging**. This setting allows you to enable or disable protocol logging for this receive connector. The default is none.

Each receive connector must have a unique combination of remote network settings and network adapter bindings. For example, multiple receive connectors can listen on the same IP address and port 25, but each receive connector must have unique remote network settings.

If there are overlapping remote network settings, the connector selected for communication is the one with the most unique address range. For example, a connector with a specific IP address listed for the remote network settings is selected over a connector with a network range listed for the remote network settings.

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When you create a new receive connector, you need to select a role for it. The two options are:

- **Hub Transport**. The Microsoft Exchange Transport service implements and controls receive connectors with this role. This is the backend transport role that is used for communication between servers.
- **Frontend Transport**. Receive connectors with this role are implemented and controlled by the Microsoft Exchange Front End Transport service. This role is typically used for communication with external applications, users, and mail servers outside of the Exchange organization.

Default receive connectors

A server running Exchange Server has five receive connectors configured automatically during installation. The default connectors allow communication between Exchange servers within the organization and allow messages to be received from external mail systems and clients.

The default receive connectors include:

- **Default Frontend ServerName**. The connector accepts connections from SMTP senders over port 25. This is the common messaging entry point into the Exchange organization. This connector accepts non-authenticated (anonymous) connections and has a Front-End Transport role.
- **Default ServerName**. Accepts authenticated connections from Mailbox servers running the Transport service and from Edge servers. This connector has the Hub Transport role, and it accepts connections on port 2525.
- **Client Frontend ServerName**. This connector accepts authenticated connections from clients such as Windows Mail for sending emails. It listens on port 587. This connector has a Front-End Transport role.
- **Client ServerName**. This connector accepts connections from front-end servers. It has the Hub Transport role, accepts connections on port 465 (Secure SMTP), and requires authentication.
- **Outbound Proxy Frontend ServerName**. The connector accepts messages from a Send Connector on a back-end server, with front-end proxy enabled. It accepts connections on port 717.

Note: Receive connectors with the Front-End Transport role and the Hub Transport role must use unique port numbers. The Frontend Transport role is implemented by the Microsoft Exchange Front End Transport service. The Microsoft Exchange Transport service implements the Hub Transport role. If the same port is used for both roles, a port conflict occurs and message reception becomes unreliable.

Send connectors

Send connectors are required for an Exchange organization to deliver messages to other email systems. Message delivery within the Exchange organization is performed automatically without manually creating send connectors. No send connectors are created by default.

Each send connector has the following settings:

- **Address space**. This setting defines the remote domains for which the send connector is used.
- **Source server**. This setting defines the specific servers that are allowed to use the send connector.
- **How to send mail**. This setting defines whether mail is routed based on MX records or through a smart host.
- **Smart host authentication**. If mail is routed through a smart host, you can define the authentication credentials that are used to authenticate to the smart host.

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- **Proxy through client access server**. When this option is selected, the outbound communication is proxied through the Front-End Transport service. If this is not selected, the Transport service delivers the message.
- **Maximum message size**. This defines the maximum message size that can be transported over this receive connector. The default is 35 megabytes (MB).
- **Protocol logging**. This setting allows you to enable or disable protocol logging for this receive connector. The default is none.

Note: Receive connectors are configured per server. Send connectors are configured for the Exchange organization.

Protocol logging

Protocol logging is a very important feature for troubleshooting, especially if any mail flow issues occur. Protocol logging records the SMTP conversations that occur between messaging servers and between Exchange services in the transport pipeline as part of message delivery. The SMTP conversations that can be recorded by protocol logging occur in the following locations:

- Send connectors and Receive connectors in the Transport service on Mailbox servers.
- Send connectors and Receive connectors in the Transport service on Edge Transport servers.
- Receive connectors in the Front-End Transport service on Mailbox servers.
- The implicit and invisible intra-organization Send connector in the Transport service on Mailbox servers.
- The implicit and invisible intra-organization Send connector in the Front-End Transport service on Mailbox servers.
- The implicit and invisible intra-organization Send connector in the Mailbox Transport Submission service on Mailbox servers.
- The implicit and invisible Mailbox Delivery Receive connector in the Mailbox Transport Delivery service on Mailbox servers.

By default, protocol logging is enabled on the following connectors:

- The default Receive connector named Default Frontend <ServerName> in the Front-End Transport service on Mailbox servers.
- The implicit and invisible Send connector in the Front-End Transport service on Mailbox servers.

By default, protocol logging is disabled on all other connectors. You need to enable or disable protocol logging on each individual connector. You configure other protocol logging options for all Receive connectors or all Send connectors that exist in each individual transport service on the Exchange server. All Receive connectors in a transport service share the same protocol log files and protocol log options. These files and options are separate from the Send connector protocol log files and protocol log options in the same transport service on the Exchange server.

Additional reading. For more information, see the following articles on **Receive connectors**[2] and **Send connectors**[3].

2 https://aka.ms/AA3t6uy
3 https://aka.ms/AA3tel8

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Connectors in Exchange Online

Connectors are a collection of instructions that customize the way your email flows to and from your Exchange Online organization. Actually, most Exchange Online organizations don't need connectors for regular mail flow. The default connectors to and from the Internet are automatically created, and additional connectors are only required for special connections such as encrypted communication to partner organizations.

You can configure inbound or outbound connectors in the EAC or using the **Set-InboundConnector** or **Set-OutboundConnector** cmdlets in Windows PowerShell.

Inbound Connectors

The following table identifies the configuration options that are available for inbound connectors in Exchange Online.

Inbound From	Description
Your organization's email server	This is an email server that you manage. It's the most highly trusted email server for EOP. You should use this connector to connect, for example, a local Exchange Server or an SMTP smart host to Exchange Online.
Partner organization	A partner can be an organization you do business with, such as a bank. It can also be a cloud email service provider that provides services such as archiving, anti-spam, and so on.
Internet	You do not need to create an inbound connector for this scenario, as it is already available by default. However, to set up this mail flow scenario, you need to update your MX record to point to Exchange Online.

Outbound Connectors

The following table identifies the configuration options that are available for outbound connectors in Exchange Online.

Outbound to	Description
Your organization's email server	This is an email server that you manage. If your domain's MX record points to Exchange Online, you must set up an alternative server (called a smart host) so that Exchange Online can send email to your organization's email server (also called on-premises server).

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Outbound to	Description
Partner organization	Create a connector only if you want to enhance security for the email messages sent between Exchange Online and your partner organization or service provider. You can create multiple connectors for this scenario, each applying to different partner organizations or service providers. Organizations typically create this outbound connector to enforce transport encryption, and in doing so, they use the TLS protocol.
Internet	You do not need to create an outbound connector to the Internet as it is available by default.

Additional reading. For more information, see the following article on **Configure mail flow using connectors in Office 365**[4].

Message moderation

Message moderation is a feature you can activate on user mailboxes and distribution groups to implement an approval workflow for messaging. You can implement message moderation to:

- Protect large distribution groups
- Require approval by a manager
- Require multiple layers of approvals
- Require approval based on message content

Message moderation for distribution groups

There are several reasons to implement an approval process for sending messages to distribution groups, such as compliance requirements to limit communication between departments or just to control the allowed senders for very large distribution groups. For example, users might send messages about selling fundraising items to the entire organization. You can also configure several permanently allowed senders for the distribution group, but maybe everyone in your organization should in fact send messages to a distribution groups and you just want to review their mails before releasing them to the whole group.

By using message moderation for a distribution groups, you can define a list of users that are allowed to send messages to the group without moderation. Anyone else that sends message to the group requires approval by a group moderator. If moderators are not defined, the owner of the group is sent the notification for approval.

You can configure one of the following notification options for a distribution group:

- Notify all senders when their messages aren't approved.
- Notify only the senders in your organization when their messages aren't approved.
- Do not notify anyone when a message isn't approved.

Note: You can set these same moderation options for individual mailboxes too. However, EAC does not expose these options; instead, you need to use EMS. To view the moderation settings for a mailbox, you should run the following command:

4 https://aka.ms/AA3t6vx

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```
Get-Mailbox <MailboxName> | FL *moderat*
```

Message moderation with transport rules

You can also use transport rules to implement more complex scenarios for message moderation. In addition to requiring moderation for messages sent to a specific recipient, you can require moderation based on anything that can be evaluated with a transport rule. Each transport rule can specify different moderators.

Some characteristics that can be evaluated by using transport rules include:

- Sender
- External or internal delivery
- Text patterns
- DLP policies

To require approval by multiple moderators, you can create multiple rules with the same conditions, but with approval requests forwarded to different moderators. The rule with the highest priority sends a request for moderator approval first. When the first moderator approves the message, the rule the second highest priority sends a request for moderator approval. The moderated message is delivered only if both moderators approve the message.

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Lab1-Ex1-Create Connectors

MS-203 Lab Environment

The labs in this course will use three virtual machines:

- Domain Controller VM (LON-DC1), Adatum's on-premises domain controller
- Exchange Server VM (LON-EX1), Adatum's on-premises Exchange Server
- Client 1 VM (LON-CL1), a client PC attached to the Adatum network

The three machines are networked together in the **adatum.com** domain.

Important: A system administrator account has been created by default in the on-premises adatum.com domain. The login for this account is **adatum\Administrator** and the password is **Pa55w.rd** (**Note**: the password for ALL on-premises user accounts in the MS-203 labs is **Pa55w.rd**.

Your Microsoft 365 tenant came with a default global administrator account already created in Microsoft 365. The name of this user account is **MOD Administrator**. The username for this MOD Administrator account is **admin@M365xZZZZZZ.onmicrosoft.com**, which is the username assigned to your tenant (where ZZZZZZ is your unique tenant suffix ID provided by your lab hosting provider). When you sign into Office 365 with this **admin@M365xZZZZZZ.onmicrosoft.com** username, your user initials of **MA** (for **M**OD **A**dministrator) will display in a circle as the user icon in the upper right corner of the Office 365 screens.

When you log into Microsoft 365, you will log in with this **MOD Administrator** account of **admin@M365xZZZZZZ.onmicrosoft.com**. The password for this account will be provided by your lab hosting provider. Your instructor will show you how to retrieve your tenant email account and password. In the labs, you will also create a Microsoft 365 user account for Holly Dickson, who is Adatum's Enterprise Administrator. You will assign Holly the Microsoft 365 Global Admin role, and you will use the Holly Dickson persona to perform the majority of administrative tasks throughout the labs in this course.

IMPORTANT: If your lab environment provides some form of an "End Lab" option, you should never select it; **doing so will end your lab instance for the current trial tenant that you were assigned, and you will lose all the data that you've entered up to that point.** The only time you would need to select this button is if your trial tenant somehow became corrupted and you needed to start your labs for this course over from scratch.

WARNING – Be prepared for UI changes

Given the dynamic nature of Microsoft cloud tools, you may experience user interface (UI) changes that were made following the development of this training content. This will manifest itself in UI changes that do not match up with the detailed instructions presented in this lab manual.

The Microsoft World-Wide Learning team will update this training course as soon as any such changes are brought to our attention. However, given the dynamic nature of cloud updates, you may run into UI changes before this training content is updated. **If this occurs, you will have to adapt to the changes and work through them in the lab exercises as needed.**

Introduction

This lab references three domains:

- **adatum.com** – This is the default on-premises domain that was provided by the lab hosting environment.

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- **M365xZZZZZZ.onmicrosoft.com** – This is the domain associated with the Microsoft 365 tenant that was provided by the lab hosting environment. The first part of this domain name (M365xZZZZZZ) is the unique tenant ID provided by the lab hosting provider. The ZZZZZZ portion of the tenant ID, which is the tenant suffix ID, will be unique for each student.

  IMPORTANT: This is critical because throughout this lab, you will be asked to enter the **M365x-ZZZZZZ.onmicrosoft.com** domain name when signing into apps with a given username (for example, holly@M365xZZZZZZ.onmicrosoft.com). When doing so, you must enter the unique tenant suffix ID that is assigned to your tenant ID in place of the ZZZZZZ.

  For example, if your Tenant Email is **admin@M365x609668.onmicrosoft.com**, the unique tenant suffix ID (ZZZZZZ) is **609668**. When logging in as Holly when entering this domain, you would replace ZZZZZZ with 609668 (for example, holly@M365a609668.onmicrosoft.com). When you come to an instruction that requires your domain suffix for the **onmicrosoft.com** domain, remember to replace the ZZZZZZ in the instruction with the tenant suffix ID provided by your lab hosting provider.

- **XXYYZZa.xxxCustomDomainxxx.xxx** – This is a custom accepted domain that you will create in the lab. Companies typically do not use the **onmicrosoft.com** domain to run their work services; rather, they prefer to have their name header display their company name in their production environment. Also, when setting up a hybrid environment, as you will in this course, both the on-premises environment and the cloud tenant must share responsibility over at least one domain. For the purposes of your hosted lab environment, instead of creating a custom domain titled @adatum.com, you will use the hosted lab domain titled XXYYZZa.xxxCustomDomainxxx.xxx, which will be shared between Adatum's on-premises and cloud environments. When you create this domain, you will replace the XXYYZZa and xxxCustomDomainxxx.xxx portions with the following values:

  - The first part of the domain name, XXYYZZa, is the UPN name that will also be unique for each student. When performing the lab steps that require you to enter this domain name, you must replace **XXYYZZa** with the UPN name that was assigned to your tenant by your lab hosting provider.

  - The second part of the domain name, xxxCustomDomainxxx.xxx, must be replaced with your lab hosting provider's custom domain, or vanity domain name. For example, let's assume your lab hosting provider is Fabrikam Inc. If the UPN number it assigns to your tenant is AMPVU3a and its domain name is fabrikam.us, then the domain name for your new accepted domain would be AMPVU3a.fabrikam.us.

RECOMMENDATION: It is highly recommended that you write down the values of the ZZZZZZ tenant ID, the XXYYZZ UPN Name, and the xxxCustomDomainxxx.xxx domain name, all of which are provided by your lab hosting provider.

You should also write down the IP Address of your parent domain (for example, 64.64.206.13). Your instructor will show you how to find this address, which will be needed when you add a new accepted domain.

Writing down this information now will save you from having to constantly retrieve this information from your VM lab environment.

IMPORTANT: In the labs for this course, you will take on the role of Holly Dickson, the Messaging Administrator for Adatum Corporation. In the fictitious organizational structure that has been created for this lab, Contoso Ltd. is the parent company of its subsidiary, Adatum Corporation. As such, the Microsoft

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365 tenant has been set up to reflect the Contoso name, which appears wherever the company name from the tenant is displayed in the Microsoft 365 environment.

Lab Introduction - Exchange admin centers

Adatum Corporation is an example of an organization with a hybrid Exchange deployment. They have Exchange Server 2019 installed on their on-premises Exchange Server (LON-EX1), and they have on-premises user mailboxes stored in Exchange Server. In addition, now that they are implementing Microsoft 365, they can create user accounts and mailboxes in Exchange Online. This hybrid deployment will require that they synchronize their on-premises and cloud deployments.

For the labs in this course, you will take on the role of Holly Dickson, Adatum's Messaging Administrator. Before Adatum rolls out their Microsoft 365 deployment to all users, they have decided to run a pilot project on key aspects of the system to identify potential issues. Holly will lead this effort on all messaging-related tasks.

In your lab environment, the trial tenant has already been selected and a default tenant admin account has already been created by your lab hosting provider. You will log into the Exchange Server VM using the Administrator account, and when you access Microsoft 365 for the first time, you will initially log in using the default tenant admin account.

This lab exercise requires you to log on to the Microsoft Labs Online environment to do the lab steps in BOTH an Exchange Server 2019 on-premises environment and in the Microsoft 365 cloud environment.

Important: When dealing with Microsoft Exchange, it's important to note that there are two very similar Exchange admin centers. These centers, which control both Exchange Online and Exchange Server (2013-2019), are Web-based admin centers that are unified through one interface called the Exchange admin center, or EAC.

- For on-premises Exchange deployments, the EAC is the web-based management console in Exchange Server.
- In Microsoft 365 deployments, the EAC is the web-based management console in Exchange Online.

In hybrid deployments (such as Adatum), either EAC can be used. The two Exchange admin centers are essentially the same, but there are some minor differences. Throughout the labs in this course, you will perform various Exchange functions within each admin center. The reason we have you use both is to give you experience using each version, and to call out those exceptions that must be performed in one EAC or the other.

To help avoid confusion, when using the on-premises EAC, you will access it on the Exchange Server (LON-EX1) VM. When accessing the EAC for Exchange Online, you will access it on either the Domain Controller (LON-DC1) or Client (LON-CL1) VM. The one exception to these rules is when you run the Hybrid Configuration Wizard to build Adatum's hybrid deployment. The wizard must be run from the EAC for Exchange Online, but it must be run on the Exchange Server.

Lab1-Ex1-Create Connectors

In this exercise, you will take on the persona of Holly Dickson, Adatum's Messaging Administrator. As part of her pilot project for Adatum's Exchange deployment, Holly wants to begin by creating custom send and receive connectors in her on-premises Exchange Server using the on-premises Exchange admin center.

Exchange uses connectors to enable incoming and outgoing mail flow on Exchange servers and between services in the transport pipeline on the local Exchange server. In this lab you will log into the Exchange

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Server (LON-EX1) VM and create custom Receive and Send connectors using the on-premises Exchange admin center (EAC).

As you complete each task, you should leave the EAC open in your browser for the next lab. This will save you from having to open the EAC at the start of each lab.

See your VM lab environment for the lab instructions.

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Managing Transport Rules

Lesson Introduction

Transport rules are a very powerful configuration to control the message flow in your organization. For example, it's possible to use transport rules for blocking messages that match certain criteria, create CC copies of messages and send them to additional recipients, or encrypt messages through Rights Management Services when they leave your organization or if they are sent to certain recipients.

After this lesson, you will be able to:

- Understand what transport rules are
- Describe how transport rules are working
- Configure custom transport rules
- Describe how transport rules can be used for data loss prevention

Overview of transport rules

You can use transport rules in Exchange to look for specific conditions on messages that pass through your organization and take action on them. Transport rules apply policies on messages while they're in transit through the transport pipeline. The purpose of transport rules is to help protect corporate network resources and data by applying an action to messages meeting specified conditions. There are two types of transport rules:

- Transport rules on mailbox servers
- Edge Transport rules on Edge Transport servers for messages sent to or received from the Internet

Transport rules

Transport rules can be created to identify and take action on messages that flow through the transport pipeline in an Exchange organization. Transport rules are configured at the mailbox server level. Transport rules configured on one Mailbox server automatically apply to all other Mailbox servers in the organization. Exchange Server stores the transport rules in the Configuration container in Active Directory and replicates them throughout the Active Directory forest so that they're accessible to all other Mailbox servers. This means that Exchange Server applies the same transport rules to all email messages that users send or receive in the organization.

On Mailbox servers, transport rules can:

- Prevent specified users from sending or receiving email from other specified users.
- Prevent inappropriate content from entering or leaving the organization.
- Apply restrictions based on message classifications to restrict the flow of confidential organization information.
- Track or journal messages that specific individuals send or receive.
- Redirect incoming and outgoing messages for inspection before delivery.
- Apply disclaimers to messages as they pass through the organization.
- Apply Active Directory Rights Management Services (AD RMS) templates to the messages based on message criteria.

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Edge Transport rules

Edge Transport rules are used to control the flow of messages sent to or received from the Internet. Edge Transport rules are configured on each Edge Transport server. Edge Transport rules that you create and configure on Edge Transport servers are stored in the local instance of AD LDS on the server. No automated replication of transport rules occurs on Edge Transport servers. Rules on the Edge Transport server apply only to messages that flow through the local server. If you need to apply the same set of mail flow rules on multiple Edge Transport servers, you can clone the Edge Transport server configuration, or export and import the mail flow rules.

On Edge Transport servers, edge transport rules can:

- Quarantine a message
- Drop or reject a message
- Append additional recipients
- Log an event

Differences between Transport rules and Inbox rules in Outlook

Transport rules are similar to the Inbox rules that are available in Outlook and Outlook on the web. The main difference between the two centers around where the rules are set up, where they’re applied, and the feature set each rule type provides.

- Outlook rules are set up on the client, which then takes action on the messages after they’re delivered to each mailbox.
- Transport rules are set up at the server end, and they’re applied to messages while they're in transit.
- Transport rules contain a richer set of conditions, exceptions, and actions, which provides organizations with the flexibility to implement more types of messaging policies as compared to Outlook rules.

How transport rules work

After you create a transport rule, it gets stored in Active Directory so that all Exchange servers can access it, and any server can apply the configured rule on messages in transport.

Keep in mind that because transport rules for Exchange mailbox servers are stored in Active Directory, Edge Transport servers that are regularly not domain joined cannot access them. Therefore, to accommodate transport rules for inbound and outbound Internet mail flow, you can create Edge Transport rules that reside on Edge Transport servers.

Transport rules consist of the following components:

- **Conditions.** Conditions identify the messages that you want to apply the actions to. Some conditions examine message header fields (for example, the To, From, or Cc fields). Other conditions examine message properties (for example, the message subject, body, attachments, message size, or message classification). Most conditions require you to specify a comparison operator (for example, equals, doesn't equal, or contains) and a value to match. If there are no conditions or exceptions, the rule is applied to all messages.
- **Exceptions.** Exceptions are an optional component, since they identify the messages that the actions shouldn't apply to. The same message identifiers that are available in conditions are also available in exceptions. Exceptions override conditions and prevent the rule actions from being applied to a message, even if the message matches all the configured conditions.

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- **Actions.** Actions specify what to do to messages that match the conditions in the rule and don't match any of the exceptions. There are many actions available, such as rejecting, deleting, or redirecting messages, adding additional recipients, adding prefixes in the message subject, and inserting disclaimers in the message body.
- **Properties.** Properties specify other rules settings that aren't conditions, exceptions, or actions. For example, when the rule should be applied, whether to enforce or test the rule, and the time period when the rule is active.

The following table identifies how multiple conditions, condition values, exceptions, and actions are handled in a rule.

Component	Logic	Comments
Multiple conditions	AND	A message must match all the conditions in the rule. If you need to match one condition or another, use separate rules for each condition. For example, if you want to add the same disclaimer to messages with attachments and messages that contain specific text, create one rule for each condition. You can easily copy a rule in the EAC.
One condition with multiple values	OR	Some conditions allow you to specify more than one value. The message must match any one (not all) of the specified values. For example, let's assume you have a rule in which the email Subject has a word match condition for "Contoso" or "stock". If an email message has the subject "Stock price information", the condition is satisfied because the subject contains at least one of the specified values.
Multiple exceptions	OR	If a message matches any one of the exceptions, the actions are not applied to the message. The message doesn't have to match all the exceptions.

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Component	Logic	Comments
Multiple actions	AND	Messages that match a rule's conditions get all the actions that are specified in the rule. For example, if the actions "Prepend the subject of the message with" and "Add recipients to the Bcc box" are selected, both actions are applied to the message. Keep in mind that some actions, such as the "Delete the message without notifying anyone" action prevent subsequent rules from being applied to a message. Other actions such as "Forward the message" do not allow additional actions. You can also set an action on a rule so that when that rule is applied, subsequent rules are not applied to the message.

Transport rules are applied by a transport agent on Mailbox servers and Edge Transport servers.

- On Mailbox servers, rules are applied by the Transport Rule agent.
- On Edge Transport servers, rules are applied by Edge Rule agent.

Although similar in functionality, the agents have some differences. The important differences are summarized in the following table.

Transport agent	SMTP or categorizer event where rules are invoked	Where rules are stored
Transport Rule agent on Mailbox servers	The **OnResolvedMessage** categorizer event. The Transport Rule agent was invoked on the **OnRoutedMessage** categorizer event. The change to **OnResolvedMessage** allowed new rule actions that can change how a message is routed (for example, require TLS).	In Active Directory. Rules are available to all Mailbox servers in the Active Directory forest.

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Transport agent	SMTP or categorizer event where rules are invoked	Where rules are stored
Edge Rule agent on Edge Transport servers	The **OnEndOfData** SMTP event.	In the local instance of Active Directory Lightweight Directory Services (AD LDS) on the server. Rules are only applied to messages that flow through the local server.

There are differences in processing different types of messages that flow through an organization. The following table identifies which messages types can be processed by transport rules.

Type of message	Can a rule be applied?
Regular messages Messages that contain a single rich text format (RTF), HTML, or plain text message body or a multipart or alternative set of message bodies.	Yes
S/MIME encrypted messages.	Rules can only access envelope headers and process messages based on conditions that inspect those headers. Rules with conditions that require inspection of the message's content, or actions that modify the message's content can't be processed.
RMS Protected messages: Messages that are protected by applying an Active Directory Rights Management Services (AD RMS) rights policy template.	Rules can always access envelope headers and process messages based on conditions that inspect those headers. For a rule to inspect or modify a protected message's content, your need to have transport decryption set to Mandatory or Optional (By default, Transport decryption is set to Optional), or you need to have the encryption key.
Clear-signed messages: Messages that have been signed but not encrypted.	Yes
UM messages: Messages that are created or processed by the Unified Messaging service in Exchange 2016, such as voice mail, fax, missed call notifications, and messages created or forwarded by using Microsoft Outlook Voice Access. (Note: Unified Messaging is not available in Exchange 2019.)	Yes
Anonymous messages: Messages that were sent by anonymous senders.	Yes
Read reports: Reports that are generated in response to read receipt requests by senders. Read reports have a message class of IPM.Note*. MdnRead or IPM.Note*.MdnNotRead.	Yes

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Configuring transport rules

Transport rules provide you with an almost limitless ability to control messaging in your Exchange Server organization. Messaging administrators must carefully plan their transport rules to ensure they behave as intended; otherwise, you could accidentally delete messages, or deliver messages to unintended recipients.

You should consider the following recommendations when you plan transport rules:

- **Plan conditions and exceptions carefully**. Transport rule conditions and exceptions define which messages are affected by the transport rule. If you implement the rules incorrectly, you may unintentionally modify or delete messages.
- **Plan for Transport rule priority and order**. In many cases, you must apply several transport rules in your organization. If these transport rules have conditions that can overlap in some cases, it's very important that you order them properly.
- **Use regular expressions to check message contents**. Use regular expressions to simplify the list of terms when you are including a text string in a condition. You can use one regular expression, rather than a list of variations on the same word. For example, when searching for a phone-number pattern, you can use the expression "\d\d\d(-|.)\d\d\d\d", which denotes a pattern of three digits, then a dot or dash, and then four digits.
- **Test application of transport rules**. Test new transport rules to ensure they behave as intended. This is important because a new transport rule could conflict with existing transport rules.
- **Plan for transport rule limitations on encrypted and digitally signed messages**. RMS integration with Exchange Server enables you to implement transport rules and messaging policies when you're using RMS Information Rights Management encryption to protect messages. Encryption through other mechanisms may prevent you from applying transport rules or records management. For example, Exchange Server may not be able to scan encrypted messages for the text string specified in a transport rule. In addition, antivirus scanners cannot scan messages with encrypted attachments.
- **Consider transport rule recovery**. Deleted transport rules are not easily recoverable. Transport rules are stored in Active Directory and restoring rules from Active Directory is a complex process. Alternatively, documented transport rules are easy to re-create, and you can export transport rules to backup files by using the **Export-TransportRuleCollection** cmdlet. However, when you import transport rules onto a Hub Transport server, the server replaces all the existing transport rules for the organization.

Transport rules can be configured through the Exchange Admin Center (EAC) and the Exchange Management Shell (EMS) by using the **New-TransportRule** and **Set-TransportRule** cmdlets.

Additional reading. For more information, see the following article on **Mail flow rule conditions and exceptions (predicates) in Exchange Server**[5].

5 https://aka.ms/AA3tevg

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Module Review

Module 1 Review

Checkbox

Item 1. Which of following services are components of the Transport Pipeline in Exchange Server? (Select 3)

- ☐ Mailbox Server role
- ☐ Front End Transport Service
- ☐ Unified Messaging service
- ☐ Transport Service
- ☐ Mailbox Transport Service
- ☐ Virtual Directories

Checkbox

Item 2. Which of following services are components of the Mailbox Transport Service in Exchange Server? (Select 2)

- ☐ Mailbox Server role
- ☐ Mailbox Transport Delivery service
- ☐ Front End Transport Service
- ☐ Mailbox Transport Submission service
- ☐ Virtual Directories

Checkbox

Item 3. Which of following Exchange Server components are routing destinations in the Exchange Server transport services? (Select 3)

- ☐ Mailbox Server role
- ☐ Mailbox Database
- ☐ Connector
- ☐ Unified Messaging service
- ☐ Distribution group expansion server
- ☐ Virtual Directories

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Multiple choice

Item 4. You are a messaging administrator for Contoso. You have just deployed an Exchange Server in your organization that uses contoso.local as the AD domain. You want to create an accepted domain contoso.com, where all recipients for that domain will be hosted in the Exchange organization. What type of accepted domain will you create?

- ☐ Remote Domain
- ☐ Internal relay domain
- ☐ Authoritative domain
- ☐ External relay domain

Multiple choice

Item 5. You are a messaging administrator for Contoso, which uses a third-party messaging solution. You have just deployed a new Exchange Server in your organization. You want to create a connector that will enable all recipients to send email to the Internet. What type of connector will you create?

- ☐ Receive Connector
- ☐ Delivery Agent Connector
- ☐ Foreign Connector
- ☐ Send Connector

Multiple choice

Item 6. You are a messaging administrator for Adatum Corporation, which uses Exchange Server. You have been informed that Adatum must meet new compliance requirements by limiting email communication between its Research and Sales departments. You need to configure Exchange Server so that if anyone from the Sales department sends a message to a user in the Research department, the mail must be approved by a moderator. What should you do to satisfy this requirement?

- ☐ Use Queue Viewer.
- ☐ Enable Protocol Logging on the custom connector to monitor communication between the two departments.
- ☐ Create a transport rule to moderate the communication between departments.
- ☐ Deny users from Sales the ability to send email to users in the Research department.

Checkbox

Item 7. What settings are part of the message transport rules? (Select 3)

- ☐ Conditions
- ☐ Protocol Logs
- ☐ Exceptions
- ☐ Client Access services
- ☐ Actions
- ☐ Databases

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Multiple choice

Item 8. You are a messaging administrator for Adatum Corporation, which uses Exchange Server. You need to create a transport rule that will search for an employee's corporate ID data in an email. In the Research department all email that is sent between employees is encrypted with S/MIME. You notice that the transport rule cannot process for member of the Research department. Why are transport rules not being processed for Research department users?

- ☐ Because protocol logging is not enabled.
- ☐ Because S/MIME encrypted messages cannot be processed by transport rules that search for content.
- ☐ Because transport rules apply only when email is sent to the Internet.
- ☐ Because transport rules apply only on email sent by anonymous senders.

Multiple choice

Item 9. You are a messaging administrator for Adatum Corporation, which uses Exchange Server. You received a business requirement that every email sent from the Finance department to the Development department must be approved by the Development department manager. What should you do to satisfy this request?

- ☐ Assign the Development department manager as a member of the Organization Management role.
- ☐ Assign the Development department manager as a member of the Discovery Management role.
- ☐ Create a transport rule that will apply moderation when email is sent from users in the Finance department to users in the Development department.
- ☐ Deny users from the Sales department from being able to send email to users in the Research department.

Multiple choice

Item 10. You are a messaging administrator for Adatum Corporation, which recently migrated from Exchange 2013 to Exchange 2019. As part of the migration, you created a custom connector for a third-party application that needs to send email through Exchange 2019. However, some users have since complained that they no longer receive notifications from this third-party application. They were receiving emails before the migration. What should you do to analyze the communication between third-party applications and Exchange Server 2019?

- ☐ Enable Protocol Logging for the custom connector on Exchange Server 2019.
- ☐ User Microsoft Remote Connectivity Analyzer.
- ☐ Enable Protocol Logging on the send connector on Exchange Server 2019.
- ☐ Use Queue Viewer.

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Multiple choice

Item 11. You are a messaging administrator for Contoso. You have just deployed an Exchange Server, and you are in the process of migrating from a third-party messaging solution. Some of the employee's mailboxes are already migrated to Exchange Server, but some of them are still located on the third-party solution. Your organization uses contoso.local as the AD domain.
You want to create an accepted domain named contoso.com, where, at the moment of migration, part of the employees' mailboxes are located on Exchange Server, and part of the mailboxes are located on the third-party solution. In the future, all recipients for that domain will be hosted in the Exchange organization. What type of accepted domain will you create?

- ☐ Remote Domain
- ☐ Internal relay domain
- ☐ Authoritative domain
- ☐ External relay domain

Multiple choice

Item 12. Which of the following performs mail routing both within an organization and between an organization and the Internet?

- ☐ Client access services
- ☐ Front-end Transport services
- ☐ Message Transport services
- ☐ Mailbox Transport service

Multiple choice

Item 13. When a Mailbox server receives a message, what passes the message to the Mailbox Transport service?

- ☐ Client access services
- ☐ Front-end Transport services
- ☐ Message Transport services
- ☐ Active Sync

Multiple choice

Item 14. What do all messaging clients connect to when accessing an Exchange Server mailbox?

- ☐ Client access services
- ☐ Front-end Transport services
- ☐ Message Transport services
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Checkbox

Item 15. Before a received message is written to the mailbox database, where does the Mailbox server write the message? (Select 2)

- ☐ To the current transaction log
- ☐ To the lagged database copy
- ☐ To the Extensible Storage Engine
- ☐ To a disk designated as the hot spare disk
- ☐ To the MetaCache database in a DAG environment
- ☐ To the memory cache

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Answers

Checkbox

Item 1. Which of following services are components of the Transport Pipeline in Exchange Server? (Select 3)

- ☐ Mailbox Server role
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- ■ Transport Service
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Item 7. What settings are part of the message transport rules? (Select 3)

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- ☐ To the MetaCache database in a DAG environment
- ■ To the memory cache

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Module 2 Managing and Troubleshooting Mail Flow

Managing Mail Flow

Lesson Introduction

Mail flow is the essential component of every messaging system such as Exchange. For this reason, understanding the details of the mail flow and how to manage your mail flow is a crucial task for every Exchange administrator.

This lesson shows you the differences between managing mail flow in Exchange Online, Exchange Server, and Exchange Hybrid deployments.

After this lesson, you will be able to:

- Manage mail flow in organizations
- Understand mail flow for Exchange Servers
- Manage mail flow for Exchange Online
- Describe and manage mail flow in hybrid environments

Introduction to Managing mail flow

An organization's mail flow basically describes the manner in which inbound and outbound emails are processed to be successfully sent or received. These processes will vary depending on whether you have an Exchange Server, Exchange Online, or hybrid Exchange deployment.

- **Exchange Server deployments.** In Exchange Server, mail flow occurs through the transport pipeline. The transport pipeline is a collection of services, connections, components, and queues that work together to route all messages to the categorizer in the Transport service and on an Exchange Mailbox server inside the organization.
- **Exchange Online deployments.** For organizations using Microsoft 365, Microsoft hosts their mail-boxes and takes care of mail flow. It's the simplest configuration and means that Microsoft 365 manages all mailboxes and filtering. However, some organizations need more complex mail flow

setups to make sure that they comply with specific regulatory or business needs, so the Exchange Online mail flow process must be modified to accommodate those requirements.

- **Hybrid Exchange deployments.** A hybrid deployment gives organizations the ability to extend the feature-rich experience and administrative control they have with their existing on-premises Microsoft Exchange organization to the cloud. With hybrid transport, messages sent between recipients in either organization are authenticated, encrypted, and transferred using Transport Layer Security (TLS), and they appear as "internal" to Exchange components such as transport rules, journaling, and anti-spam policies. The Hybrid Configuration Wizard in Exchange Server is used to configure hybrid transport.

In general, the transport services on a server are part of the entire mail flow process. However, transport is often meant for describing the message delivery processes inside a server, while mail flow is more focused on the delivery between servers and message routing.

Messaging administrators must consider the following issues to successfully manage their mail flow:

- **How are your MX records configured?** External mail servers determine where to transport messages for your organization by the MX record of your public DNS domain. The servers configured in your MX record are your perimeter servers that are the first to accept external messages and often scan them for spam and malware.
- **How are messages routed from your perimeter to your user mailboxes?** After a message was received on the perimeter network, it needs to be routed to the correct Mailbox server hosting the user mailbox or recipient.
- **How many protecting barriers are between your Mailbox Servers and the outside world?** You will use a minimum of one email gateway to scan for spam and malware, but you may have multiple gateways in your perimeter network to scan your incoming and outgoing mail flow; these can include external providers such as Exchange Online Protection and Edge servers
- **How many different rules are applied to your mail flow?** A transport or mail flow rule can modify, copy or redirect messages flowing to your mail servers and anti-spam gateways. Transport rules can also apply different rules on messages to add anti-spam header fields and possibly redirect messages. The more organizations apply rules to their mail flow, the greater the complexity when troubleshooting their mail flow.
- **Do you use a mandatory TLS encryption with a partner organization?** Sometimes you must establish TLS encrypted connections to partner organizations or, for example, with Exchange Online.

Managing mail flow in Exchange Server deployments

There are a variety of tools and approaches that can be used to manage mail flow within on-premises Exchange Server deployments. For example, you
can modify the default internal message flow of how Exchange servers communicate with each other, you can use
connectors to configure inbound and outbound routing to your Exchange server, and you can use tools to manage and
verify mail flow.

Modifying the default mail flow of Exchange Server

By default, the mail flow in Exchange is fully automated and employs the following rules:

- Least-cost route is calculated

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- Messages are sent directly to their destination

To send a message to a remote delivery group, Exchange calculates the least-cost routing path. The cost of routing paths is calculated by adding the cost of IP site links that need to be crossed to reach the destination. In cases where there are multiple paths with the same cost, the path with the lowest number of hops is preferred. Finally, if there are multiple sites with the same cost and the same number of hops, then the name of the Active Directory site is used to determine the path. The lowest alphanumeric name is preferred.

Delivery groups can span multiple Active Directory sites. For example, a DAG can have members in multiple Active Directory sites. In such a case, all Active Directory sites containing DAG members are evaluated and the Active Directory site with the lowest routing cost is selected as the primary site for message delivery.

After the least-cost route for message delivery has been determined, the Transport service in the source delivery group attempts to deliver the message directly to the destination delivery group. If the least-cost route includes multiple Active Directory sites, message delivery does not hop through each site. Other Active Directory sites on the least-cost path are used only if there is a delivery error or you have configured a modification.

Calculating the least-cost routing path takes into account the following mail flow modifications:

- **Fallback.** If the final destination on the least-cost path cannot be contacted, then the Transport service falls back and attempts to deliver the message to the next closest site on the least-cost path. The message is queued at the next closest site until the final destination is available.

  Note: If the next closest site on the least-cost path is not available, then the Transport service tries one hop back again. This process continues until the message can be delivered.

- **Hub sites.** You can configure one or more Active Directory sites in your organization as hub sites. When a hub site exists along the least-cost routing path between two Mailbox servers, the messages are routed to a Mailbox server in the hub site for processing before they are relayed to the destination server.

  The Transport service routes a message through a hub site only if it exists along the least-cost routing path. The originating Mailbox server always calculates the lowest-cost route first, and then checks if any of the sites on the route are hub sites. If the lowest-cost route does not include a hub site, the Hub Transport service attempts a direct connection.

  - You should run the following command to configure a site as hub site:

    ```
    Set-ADSite –Identity sitename –HubSiteEnabled $true
    ```

  - You should run the following command to check whether you have configured a hub site:

    ```
    Get-AdSite | Format-List Name,HubSiteEnabled
    ```

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- **Exchange-specific routing costs.** You can assign an Exchange-specific cost to an Active Directory IP site link. If you assign an Exchange-specific cost to the site link, the Transport service determines the least-cost routing path by using this attribute rather than the Active Directory-assigned cost, unless the mailbox server is a member of DAG. If the mailbox server is a member of a DAG, Active Directory site link costs are always used.
  - You should run the following command to assign an Exchange-specific routing cost to an Active Directory IP site link:

    ```
    Set-AdSiteLink –Identity ADsitelinkname –ExchangeCost value
    ```

  - You also can assign a maximum message size limit for messages sent between Active Directory sites by running the following command:

    ```
    Set-AdSiteLink –Identity ADsitelinkname – MaxMessageSize value
    ```

  - To check if you properly configured an Exchange cost, you should run following command:

    ```
    Get-AdSite | Format-List Name,HubSiteEnabled
    ```

- **Expansion servers for distribution groups.** By default, when a message is sent to a distribution group, the first Transport service that receives the message expands the distribution list and calculates how to route the messages to each recipient in the list. If you configure an expansion server for the distribution list, all messages sent to the distribution list are sent to the specified Mailbox server, which then expands the list and distributes the messages. For example, you can use expansion servers for location-based distribution groups to ensure that the local Transport service resolves them.

Manage mail flow outside the Exchange organization

You manage the mail flow to and from your Exchange organization using Send Connectors and Receive Connectors.

Connector Type	Description
Send connectors	When you install Exchange there are no visible Send connectors. Therefore, before you can send mail to someone outside of your organization, you must create one or more Send connectors on the Mailbox server. You can create an outbound Send connector by using the EAC or by using the Exchange Management Shell.

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Connector Type	Description
Receive connector	Exchange Server requires an SMTP Receive connector to accept any SMTP email. An SMTP Receive connector enables an Exchange Transport service to receive mail from any other SMTP sources, including SMTP mail programs such as Windows Mail, POP3 and IMAP4 clients, SMTP servers on the Internet, Exchange Edge Transport servers, and other Exchange Server servers.

Note: In Exchange Online the connectors are called Inbound and Outbound Connectors, but they serve the same purpose: To accept and deliver mails.

Tools to manage Mail Flow in Exchange

The following table identifies the variety of tools available to manage mail flow in Exchange.

Tool	Description	When to use
Queue Viewer	The Queue Viewer tool is included in the Exchange Toolbox that is installed on each server running Exchange. Queue Viewer is a graphical tool that you can use to see message queues and their contents on a specific server.	You can use the queue viewer to manage mail queues in the way to suspend or resume mail queues (or a specific connector). You can also use the queue viewer to remove specific messages from a message queue.

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Tool	Description	When to use
Message tracking	Message tracking logs provide much more detailed information about message delivery than delivery reports. Each server maintains its own set of message tracking logs that are retained by default for 30 days, with a maximum size for all message tracking logs of 1 gigabyte (GB). These logs contain detailed information about how messages were processed on the local server. There is no graphical tool for viewing the message tracking logs. Instead you use the **Get-MessageTrackingLog** cmdlet with filtering to identify the log events you wish to review.	You can use message tracking to make sure the mail flow is using the correct Exchange servers and connectors.
Delivery reports	You can use Delivery reports in Exchange Admin Center (EAC) to follow the message delivery path. A delivery report provides basic information about the path a message has taken through the Exchange organization and whether it was delivered. All the delivery hops in the Exchange organization are identified. Users can also view delivery reports for messages that they have sent or received.	Delivery reports can be used to verify or proof mail deliveries from their source to target Exchange servers or to the Internet. Delivery reports are based on Message Tracking, but do not provide as much detail.

Managing mail flow in Exchange Online deployments

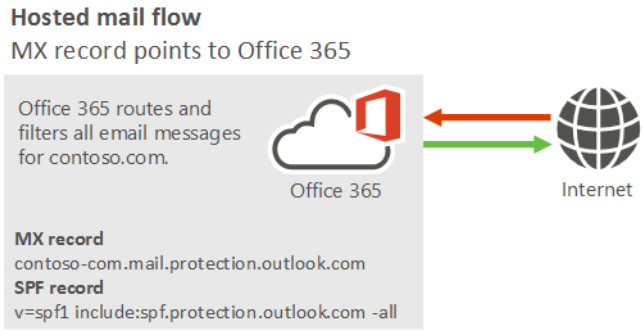
Managing mail flow for Exchange Online is the easiest way of configuring mail flow, because it's all configured by default and managed by Microsoft. Therefore, mail flow in Exchange Online is considered a "black box" because mail flow is handled internally within Exchange Online with no assistance from

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Exchange Administrators. That being said, administrators can create additional connectors to improve mail flow throughout the organization, but outside of that, they have no impact on the mail flow process.

Mail flow in Exchange Online is kept as easy as possible, so the only configuration requirement you need to configure in Exchange Online deployments is to make sure of the following:

- Your DNS MX record must point inbound to Exchange Online.
- Internet SMTP email is sent out directly from Exchange Online Protection to the recipient's mail server.



Managing Mail Flow in Exchange Online

Managing the mail flow for Exchange Online includes the creation of connectors, accepted domains, and optional mail flow rules. You can do this in the Exchange Admin Center and with Exchange Online PowerShell.

There are also several features to configure in the Security & Compliance Center to control the data loss prevention filtering for email flowing through your organization and anti-spam, anti-spoofing, and anti-malware settings. All these settings are covered in a later lesson.

Additional reading. For more information, see the following article on **Mail flow best practices for Exchange Online and Office 365 (overview)**[1].

Managing Connectors in Exchange Online

Exchange Online was developed in a way that mail flow is configured automatically for the most common scenarios. However, there may be scenarios where you need to create a connector. The following table provides an overview of these scenarios.

Scenario	Description	Connector required?	Connector settings
Send mail from your Exchange Online tenant to the Internet.	Exchange Online internal routing can route any messages from your Exchange Organization to an Internet domain to the Internet.	No	Default settings are sufficient

1 https://aka.ms/AA3tf45

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Scenario	Description	Connector required?	Connector settings
Some of your mailboxes are on your on-premises email servers, and some are in Exchange Online.	Before you manually configure connectors, verify whether an Exchange hybrid deployment better meets your business needs. For details, see the "**What if I have my own email servers?** *(https://aka.ms/AA3tf3u)*" article and the **Exchange Server Hybrid Deployments** *(https://aka.ms/AA3tf3v)* article.	Yes	**Connector for incoming email:** • **From**: Your on-premises email server • **To**: Microsoft 365 **Connector for outgoing email:** • **From**: Microsoft 365 • **To**: Your on-premises email server
All of your mailboxes are in Exchange Online, but you must send email from sources in your on-premises organization.	You don't have your own email servers, but you must send email from non-mailboxes, such as printers, fax machines, apps, or other devices. For details, see **Option 3: Configure a connector to send mail using Office 365 SMTP relay** *(https://aka.ms/AA3t7cv)*	Optional	**Only one connector for incoming email:** • **From**: Your organization's email server • **To**: Microsoft 365

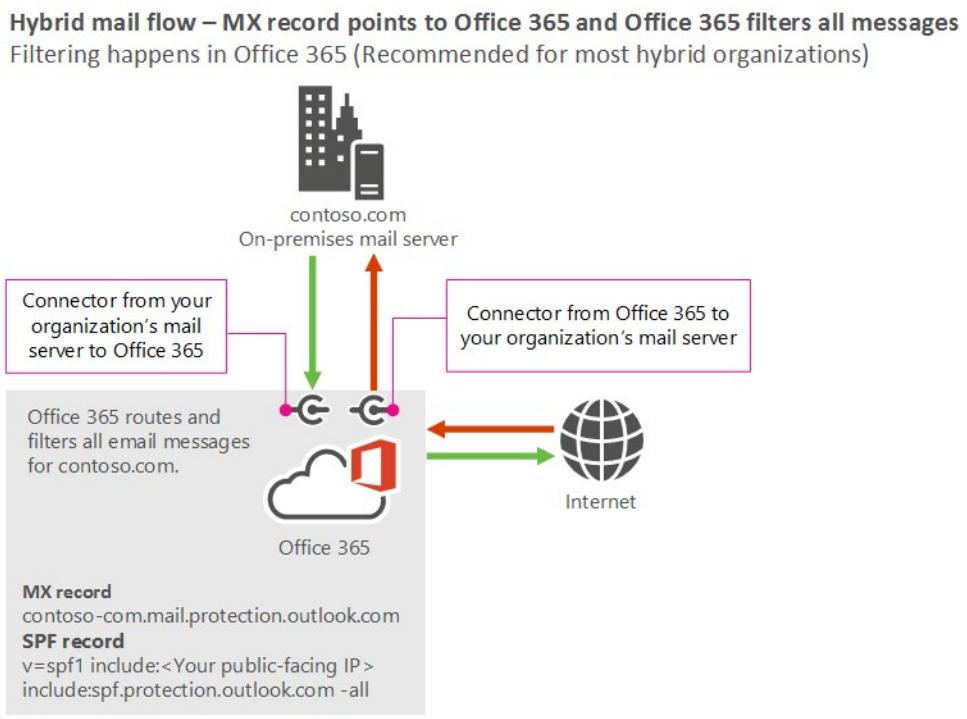
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Scenario	Description	Connector required?	Connector settings
You frequently exchange sensitive information with business partners, and you want to apply security restrictions.	You want to use Transport Layer Security (TLS) to encrypt sensitive information, or you want to limit the source (IP addresses) for email from the partner domain. For details, see **Set up connectors for secure mail flow with a partner organization** *(https://aka.ms/AA3t7cw)*.	Optional	**Connector for incoming email:** • **From**: Partner organization • **To**: Microsoft 365 **Connector for outgoing email:** • **From**: Microsoft 365 • **To**: Partner organization
You have a standalone EOP subscription.	You have your own on-premises email servers, and you subscribe to EOP only for email protection services for your on-premises mailboxes. Note: you do not have mailboxes in Exchange Online). For more information, see the topic **Exchange Online Protection overview** *(https://aka.ms/AA3tf42)* and the **How do connectors work with my on-premises email servers?** *(https://aka.ms/AA3t7d0)* section later in this topic.	Yes	**Connector for incoming email:** • **From**: Your on-premises email server • **To**: Microsoft 365 **Connector for outgoing email:** • **From**: Microsoft 365 • **To**: Your on-premises email server

Managing mail flow in hybrid Exchange deployments

A hybrid deployment is typically configured to share a common SMTP email namespace. This requires routing messages in between your on-premises Exchange servers and Exchange Online. For this reason, you must consider how to organize the email flow to and from the Internet, and between Exchange Online and the Exchange Server on-premises.

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Inbound email flow from the Internet

Inbound email can be delivered to Exchange Server on-premises, or you can target the mail exchanger (MX) resource record to Exchange Online. You should choose a method depending on your organization's requirements. The following are some considerations for each scenario:

- **Inbound to Exchange Server on-premises**. Use this option if you want to keep full control of your email domains, such as message tracking and journaling of messages in your company. Configure the MX resource record to point to your organization's SMTP smart host so thatthe hybrid deployment automatically forwards all messages for mailboxes that are located on Exchange Online. If you currently use a third-party mail filter in front of your on-premises environment, then this scenario also applies.
- **Inbound to Exchange Online**. Use this option if you want Microsoft to handle your email domains, and you want to automatically take advantage of the anti-virus and anti-spam scanning engines from Exchange Online Protection. To configure this, point the MX resource record to Exchange Online Protection. Exchange Online automatically delivers messages that are located on Exchange Server on-premises by using the Exchange Server that you define when you run the Hybrid Deployment Wizard.

Outbound email flow to the Internet

Like inbound email flow, you can configure outbound email flow from your domain to the Internet, by using one of the following options:

- **Deliver Internet-bound messages directly**. Use this option to send any outbound message that is targeted to the Internet directly from either Exchange Online or Exchange Server on-premises. If the mailbox is located on Exchange Online, the Internet messages are delivered directly to the target SMTP domain without passing through the Exchange Server on-premises environment. Messages sent

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from on-premises mailboxes are routed directly to Internet recipients without passing through Exchange Online. The benefit of this option is that the message traffic is optimized, but the drawback is that it's harder to track messages to the Internet because not every message flows through the on-premises Exchange servers.

- **Route all Internet-bound messages through your on-premises Exchange servers**. This option forces Exchange Online to send any message that is targeted to the Internet through the Exchange Server on-premises environment first. The on-premises Exchange servers then route the message to the Internet and deliver the message. The benefit of this option is that all messages pass through the Exchange servers, so you can use message tracking, journaling, and other compliance features.
- **Route all Internet-bound messages through ExchangeOnline**. This option forces the on-premises Exchange servers to send any message that is targeted to the Internet through the Exchange Online first.

Email flow between Exchange Online and the Exchange Server on-premises organization

Email flow between Exchange Online and an Exchange Server on-premises organization uses SMTP send and receive connectors that the HCW configures automatically. The connectors enforce the requirement that messages be encrypted by using the Transport Layer Security (TLS) protocol.

Additional reading. For more information, see the following article on **Manage mail flow with mailboxes in multiple locations (Exchange Online and on-premises)**[2].

2 https://aka.ms/AA3t7df

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Troubleshooting Mail Flow

Lesson Introduction

If emails are not routed correctly in or outside your organization, or secure connections cannot be established successfully, you must start with troubleshooting your mail flow. Microsoft provides several tools to help you find the root cause of your issues and fix your mail flow.

After this lesson, you will be able to:

- Understand how to troubleshoot SMTP mail flow issues
- Describe how to troubleshoot issues with a shared namespace
- Describe how to troubleshoot encryption issues with TLS

Troubleshooting SMTP mail flow

This topic examines the tools and cmdlets provided by Microsoft Exchange that enable you to troubleshoot SMTP mail flow issues.

Queue Viewer

The Queue Viewer tool is included in the Exchange Toolbox that's installed on each server running Exchange Server. Queue Viewer is a graphical tool that you can use to view the following message queues and their contents on a specific server:

- **Submission queue**. This queue is always visible in Queue Viewer. There should be a limited number of messages in the submission queue, and these messages should be removed quickly. If messages are not moving out of the submission queue, the categorizer is having problems processing the messages. A common cause of categorization problems is Active Directory connectivity issues.
- **Remote delivery queue**. If Exchange Server delivers messages directly to the Internet, you will see a remote delivery queue for each domain that has recently received messages. If there's an error delivering to a specific domain, you can view the error message for that queue. Many delivery errors to destinations on the Internet are caused by anti-spam software. If you're using an Edge Transport server or a non-Microsoft SMTP relay for Internet mail delivery, you will not see domain specific queues on the internal servers.
- **Poison message queue**. If Exchange Server determines that a message may cause the server to crash, the message is placed in the poison message queue. This queue is rarely seen.
- **Mailbox delivery queue**. There is a queue for each mailbox database for which messages are destined. On a busy server, you might see a few messages in each queue. If there are many messages in a queue, the most common cause is a dismounted mailbox database.
- **Unreachable queue**. Queue Viewer displays an unreachable queue if there are messages that cannot be routed to their destination. If configuration changes are made, the messages in the unreachable queue are reevaluated to find a routing path again. A destination might not be routable due to a disabled connector, a connector being unavailable, or an invalid expansion server on a distribution group.

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You can also manage queues and their messages from the Exchange Management Shell (EMS). The following are some of the cmdlets you can use:

- Get-Queue
- Suspend-Queue
- Resume-Queue
- Retry-Queue
- Get-Message
- Suspend-Message
- Resume-Message
- Remove-Message

Delivery reports

You can use Delivery reports in the Exchange Admin Center (EAC) to follow the message delivery path. A delivery report provides basic information about the path a message has taken through the Exchange organization and whether it was delivered. All the hops for delivery in the Exchange organization are identified. Users can also view delivery reports for messages they have sent or received.

Message tracking

Message tracking logs and message traces provide much more detailed information about message delivery than delivery reports. Message tracking is covered in detail in another lesson.

Protocol logging

You can enable protocol logging on SMTP Send and SMTP Receive connectors to log the activity of each connector. You can read more about protocol logging in the first lesson of this module.

Telnet

You can use a telnet client to connect directly to a server running Exchange Server. This enables you to verify whether the server is allowing connections on port 25 for SMTP delivery. After you're connected, you can type SMTP commands to perform delivery of a simple message. In some cases, you get more detailed error information for troubleshooting by using this method than by using other tools that interpret the SMTP errors and display them for you.

You can use the following syntax from a terminal command prompt:

```
telnet <servername/IP> 25
```

Microsoft Remote Connectivity Analyzer website

Microsoft provides the **Remote Connectivity Analyzer**[3] website for troubleshooting Exchange connectivity issues. This website enables you to test inbound and outbound Internet mail flow with your Exchange organization, and it provides a detailed report on the process.

3 https://testconnectivity.microsoft.com/

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By using the Microsoft Remote Connectivity Analyzer (RCA) tool, you can diagnose single sign-on (SSO) logon issues in a Microsoft cloud service such as Office 365, Microsoft Azure, or Microsoft Intune. Depending on the specific test that you select, you might be asked for credentials of a user account in your Exchange organization. To avoid the risk of having your working credentials exploited and possibly compromise the security of your Exchange server environment, it's strongly recommended that you create a test account for the purpose of using this tool, and then delete this account immediately after you complete the connectivity tests.

Microsoft Connectivity Analyzer Tool

The Microsoft Connectivity Analyzer Tool is a downloadable client program that's located on the Remote Connectivity Analyzer website. You can use this tool to identify connectivity issues between email clients and your Exchange Server, and to troubleshoot Exchange Server deployments. Email users can also use the Microsoft Connectivity Analyzer Tool to identify common problems.

While the Remote Connectivity Analyzer (RCA) enables you to identify connectivity issues by simulating connectivity from the Internet, the Microsoft Connectivity Analyzer Tool allows both you and your end-users to run similar tests from a client computer within the corporate network.

Back pressure

Exchange Server monitors the resources used for message transport and throttles resource usage to ensure the system remains functional. When back pressure starts limiting resource usage, you'll notice symptoms such as slow message delivery or incoming messages being blocked. The best way to avoid back pressure is to ensure the Exchange server has the necessary amount of system resources, including memory and disk space. You can verify that back pressure is throttling resource usage by looking for events in the Application log that have the event source MSExchangeTransport and the event category Resource Manager.

You can use the **Get-ExchangeDiagnostics** cmdlet that's available in the Exchange Management Shell to view the current back pressure settings on an Exchange server. You can view this information by running the following command:

```
[xml]$bp=Get-ExchangeDiagnosticInfo -Server LON-EX1 -Process EdgeTransport
-Component ResourceThrottling; $bp.Diagnostics.Components.ResourceThrot-
tling.ResourceTracker.ResourceMeter
```

Additional reading. For more information, see the following article on **Understanding back pressure**[4].

Analyze DSN/NDR messages

A DSN/NDR notification message displays a status code and possible cause for non-delivery, along with the internal or external host that generated the notification. If an external host is generating notifications, you may have to work with the receiving organization's messaging administrator to resolve the issue, or clarify the reason email is not being delivered. If a server in your Exchange organization is generating the notifications, you may have to reconfigure a transport component or troubleshoot to isolate the issue.

4 https://aka.ms/AA3t7fz

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Analyze message headers

When email messages reach their destination but were noticeably delayed, you can analyze the original message headers to determine the amount of time that lapsed during each hop of the email routing process. Analyzing message headers is a quick and effective troubleshooting technique that helps you isolate the source of the delay and enables you to determine whether the source is internal or external to your Exchange organization. By default, the original message headers are not displayed in most mail clients, but you usually can retrieve them.

If you want to retrieve them in Outlook, open the affected message, click **File**, and then click **Properties**. In the **Properties** dialog box, you will see the message headers in the **Internet headers** field. However, this information might be hard for you to decipher. Therefore, you can copy the message header text and use the Message Analyzer function of the Microsoft Remote Connectivity Analyzer tool to display the information in an easy-to-read format.

Troubleshooting shared namespace environments

Shared Namespace is a scenario where organizations share the same SMTP domain between Exchange Server and a third-party email solution, or between two different Exchange Server organizations (with two different Active Directory forests). These scenarios are common where organizations migrate from a third-party email solution to Exchange Server, or in organizations that merge. Since the domain suffix is the same for all users, Exchange Server should be configured properly to allow correct email routing.

The steps that needs to be performed in shared namespace scenarios include:

1. **Create an internal relay accepted domain.** Using either the EAC or EMS, you should create a new accepted domain that matches the shared SMTP domain namespace. The new accepted domain type should be Internal Relay. For example, to create an internal relay accepted domain Adatum.com in EMS, you would run the following command:
   ```
   New-AcceptedDomain -Name "Adatum" -DomainName Adatum.com -DomainType Internal-
   nalRelay
   ```
2. **Create a send connector.** Using either the EAC or EMS, you should create a new send connector that will route the messages to the shared SMTP domain. For example, to create a send connector in EMS for Adatum.com that will route messages through a smart host named smarthost.Adatum.com, you would run the following command:
   ```
   New-SendConnector -Name "Adatum.com Send Connector" -Custom -AddressSpace
   Adatum.com -DNSRoutingEnabled $false -SmartHosts smarthost.Adatum.com
   -SmartHostAuthMechanism ExternalAuthoritative
   ```

Troubleshooting shared name space environments includes checking whether previous steps that created and configured connectors were performed properly. It's also important to analyze whether any third-party SMTP gateway solution was involved in the message transport of the organization. For example, if an SMTP gateway performs LDAP lookups in Active Directory and determines that a user is not present in Active Directory, it may reject the email, even though the user has a mailbox located on a non-Exchange mail system.

Troubleshooting Exchange TLS

The Exchange Server Domain Security feature uses Transport Layer Security (TLS) with mutual authentication, also known as mutual TLS, to provide session-based authentication and encryption. You can configure Exchange Server to use TLS to provide security for SMTP email. By requiring TLS for all SMTP email

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sent between your organization and other specified organizations, you can enable a high security level for SMTP email.

Securing a connector to a partner organization

To secure a connector to a partner organization, you should configure mutual TLS, where each server verifies the identity of the other server by validating the certificate the other server provides. It's an easy way for administrators to manage secured message paths between domains over the Internet. This means that all connections between the partner organizations are authenticated, and that all messages are encrypted while in transit on the Internet.

With mutual TLS, each server verifies the connection with the other server by validating a certificate the other server provides. Securing a connector to a partner organization works in a similar manner to establishing a TLS connection to an SMTP Receive connector. However, because you use mutual TLS, both the sender and the recipient authenticate each other before they send data. The message takes the following route from one organization to the other:

1. The transport component on the sender Mailbox server initiates a mutual TLS session with the transport component on the target Mailbox server by exchanging and verifying the target Mailbox server's certificates. This is only established when both the sending and receiving SMTP connector can identify the sending domain. You must set the domain information on the sending side by running the **Set-TransportConfig -TLSSendDomainSecureList <domain name>** command. On the receiving side, run the **Set-TransportConfig -TLSReceiveDomainSecureList <domain name>** command to set the domain information.
2. The SMTP communication is encrypted and transferred to the target Mailbox server.
3. The message is marked as secure, which displays in Outlook 2010 or newer versions, and in Outlook on the web.

To secure a connector to a partner organization, you must perform the following steps:

1. On the Mailbox server, generate a certificate request for a TLS certificate. You can request the certificate from an internal, private Certification Authority (CA) or a commercial CA. The SMTP server in the partner organization must trust the certificate. When you request the certificate, ensure that the certificate request includes the domain name for all internal SMTP domains in your organization.
2. Import and enable the certificate on the Mailbox server. After you request the certificate, you must import the certificate on the Mailbox server, and then enable the certificate for use by the SMTP connectors that are used to send and receive domain-secured email.
3. Configure outbound connector security. To configure outbound connector security, use Exchange Management Shell commands to specify the domains to which you will send domain-secured email, and then configure the SMTP Send connector to use domain-secured email.
4. Configure inbound connector security. To configure inbound connector security, use Exchange Management Shell commands to specify the domains from which you will receive domain-secured email, and then configure the SMTP Receive connector to use domain-secured email.
5. Notify your partner to configure connector security. Connector security must be configured on both sides–sending and receiving. This means that you must also contact your partner's administrator to configure your domain for connector security.
6. Test message flow. Finally, send a message to the partner, and vice versa, to verify that domain security is working correctly.

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Troubleshooting steps for Domain Security configuration with TLS include:

- Verify whether the name of the servers is correctly configured in Domain Security settings to match the common name in the certificate used for TLS.
- Verify whether the certificate issued to the Mailbox servers in the Domain Security configuration is trusted. It's recommended to use a commercial CA for Domain Security, where the commercial CA is selected from the list of Microsoft Unified Communications certificate partners.
- Verify whether the firewall at your partner organization doesn't block connections from your organization mailbox servers.
- Verify whether your partner company is using supported versions of Exchange and TLS.
- Review send and receive protocol logs for STARTTLS events that are marked successful.

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Lab2-Ex1-Create Mail Flow Rules

Lab2-Ex1-Create Mail Flow Rules

In this lab, you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. You will use your client computer to access the Exchange admin center for Exchange Online. You will then create a series of mail flow rules that are designed to protect Adatum's messaging environment.

You will first create a mail flow rule that quarantines messages that are sent from inside the organization that have the words "Secret", "Classified", or "Sensitive" in the body or subject of the message. You will create a second rule will also quarantine messages that have attachments that can't be scanned or where the scanning can't finish. Both rules will generate a reply message that lets the sender know they have a message that is undeliverable. Finally, you will create a mail flow rule that restricts the size of emails.

See your VM lab environment for the lab instructions.

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Troubleshooting Transport Issues

Lesson Introduction

Besides issues with the mail flow, there are also some scenarios where you must troubleshoot the transport components, such as the transport agents, connectors, and general architecture issues.

After this lesson, you will be able to:

- Perform troubleshooting for network-based issues
- Describe troubleshooting procedures for connector and agent issues
- Plan troubleshooting for architectural issues
- Understand how to perform troubleshooting in coexistence

Troubleshooting network-based issues

There are multiple scenarios where communication issues can occur over an organization's network. This is due primarily to the transport pipeline which contains several different components that have multiple dependencies. These components must work together so that email can be routed efficiently through the organization.

Network connectivity

To facilitate mail flow, you must ensure the network ports that Exchange transport services use are not restricted. If firewalls or other network devices, such as a network load balancer, are present, you must ensure that necessary exceptions are in place to allow incoming and outgoing traffic on required ports.

The following table identifies the key ports used by Exchange:

Port	Service	Description
25/TCP	SMTP	The default port for any SMTP mail transmission.
443/TCP	HTTPS	Most client access is tunneled over HTTPS. This is the most important port for client access.
80/TCP	HTTP	Unencrypted HTTP is required as the Autodiscover fallback method, for Internet calendar publishing, and to access Certificate Revocation Lists.
143/TCP	IMAP	Mailbox access using the IMAP protocol.
993/TCP	secure IMAP	Encrypted mailbox access using the IMAP protocol.
110/TCP	POP3	Mailbox access using the POP3 protocol.
995/TCP	secure POP3	Encrypted mailbox access using the POP3 protocol.

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Port	Service	Description
587/TCP	authenticated SMTP	Authenticated SMTP communication to a server.
53/UDP 53/TCP	DNS	Name resolution for internal and external resources.
50636/TCP	secure LDAP	Secure LDAP connections are utilized for Edge synchronization.

DNS

The ability to resolve hostnames is critical for Exchange transport components to route mail between each other and to/from external mail relay hosts. When dealing with external mail relay hosts, DNS is required so that remote hosts can understand where to route messages for the domain names that your organization uses to send and receive email. You should have a mail exchanger (MX) record published in your organization's external DNS that points to the internet facing SMTP server that receives mail. The internet-facing server that your organization uses to relay messages to other organizations should also have a host (A) record and reverse lookup (PTR) record published which correspond to the hostname that is used during SMTP conversations with other hosts. These external DNS records help validate your server's identity to other SMTP hosts.

Another important DNS record is the Sender Policy Framework (SPF) record. It identifies the specific mail servers that are allowed to send email messages by using your domain, which is an important defense against spammers who send messages with spoofed From addresses that belong to your organization. Although SPF records are not required, there is a growing number of organizations that quarantine or reject email messages which do not pass a SPF check. Therefore, it's important for you to ensure that your organization has an accurate SPF record published that accounts for all outbound email sources from your domains. Sender reputation and other anti-spam topics will be covered in detail in a later lesson.

Troubleshooting transport connectors and agents

Exchange Server uses different types of connectors to enable inbound and outbound mail flow between Exchange servers and the transport pipeline's various components. Receive and send connectors that are configured improperly can inhibit mail flow and cause messages to become stuck in the transport pipeline. Send and receive connectors can also be responsible for messages that are rejected for exceeding the maximum message size.

Both connector types provide detailed logs for troubleshooting. These logs are located on different places in the file system, depending on the server and connector type. The folder SmtpReceive contains logs of receive connectors, and SmtpSend contain logs of send connectors.

The following transport service logs can be very helpful when troubleshooting mail flow issues:

- Front End Transport service on Mailbox servers:
  - %ExchangeInstallPath%TransportRoles\Logs\FrontEnd\ProtocolLog\SmtpReceive
  - %ExchangeInstallPath%TransportRoles\Logs\FrontEnd\ProtocolLog\SmtpSend
- Transport service on Mailbox servers:
  - %ExchangeInstallPath%TransportRoles\Logs\Hub\ProtocolLog\SmtpReceive

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    - %ExchangeInstallPath%TransportRoles\Logs\Hub\ProtocolLog\SmtpSend
- Mailbox Transport Delivery service on Mailbox servers:
    - %ExchangeInstallPath%TransportRoles\Logs\Mailbox\ProtocolLog\SmtpReceive\Delivery
- Mailbox Transport Submission service on Mailbox servers:
    - %ExchangeInstallPath%TransportRoles\Logs\Mailbox\ProtocolLog\SmtpSend\Submission
- Transport service on Edge Transport servers:
    - %ExchangeInstallPath%TransportRoles\Logs\Edge\ProtocolLog\SmtpReceive
    - %ExchangeInstallPath%TransportRoles\Logs\Edge\ProtocolLog\SmtpSend

Transport agents

Transport agents process messages when they pass through the transport pipeline. These software components are located, by default, on your Exchange server, for which they provide built-in functionality. However, you can also create your own transport agents or install agents created by third-parties. Issues with one or more transport agents can inhibit mail flow in the transport pipeline. Some of the common transport agents include:

- **Transport rule agent**. The transport rule agent is responsible for evaluating transport rules for every message that traverses the transport pipeline. It's important to note that too many transport rules can slow down mail delivery, and certain rules may block delivery altogether.
- **Malware agent**. Exchange Server 2013 first introduced the Malware agent, which is enabled by default on Exchange Mailbox servers. This agent scans messages for viruses and other malware as the messages pass through the transport pipeline. If you have other software that performs malware scanning on email, you should consider disabling the built-in malware agent to improve performance.
- **DLP policy agent**. The data loss prevention (DLP) policy agent evaluates an organization's DLP policies, and it uses transport rules to take necessary action when it detects sensitive information. Large numbers of DLP policies can slow down mail delivery, so you should only deploy the policies that your organization requires, and you should verify that they are configured properly.

Troubleshooting transport in coexistence

Message transport coexistence is automatically achieved if you have multiple Exchange servers in your deployment because they will automatically fail-over if one server fails.

You should consider the following issues when troubleshooting mail routing during coexistence:

- Send connectors are organization-wide, which means they are used by all Exchange Servers in the organization. If email is not sent through the Send Connectors, verify the Send Connectors' source servers are properly configured. In a coexistence scenario, Send Connectors may include both old and new versions of Exchange as source servers.
- Receive connectors are server-wide, which means that each server has its own set of receive connectors that must be configured separately. In a coexistence scenario, you must ensure that all receive connectors are configured properly on each Exchange Server before you switch the SMTP traffic from the old to the new versions of Exchange.
- Verify that A records and PTR records are properly configured during in a coexistence scenario. Determine which public IP address will be used for Exchange servers after previous versions of the Exchange servers are decommissioned. You must then determine whether to use the same public IP

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address and reroute the SMTP traffic internally, or assign a new public IP address where email will be routed to Exchange 2019.

- Verify that Exchange Online Protection services or any third-party SMTP gateways are reconfigured to send mail to Exchange servers and to receive mail from Exchange.
- Verify that firewalls allow SMTP traffic to and from Exchange servers.
- Be aware that all third-party applications and devices that were using the old Exchange Server version to send email used receive connectors created and configured specifically for those applications and devices. You must create, configure, and test the new receive connectors for those applications and devices on Exchange before you uninstall the older Exchange versions.

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Troubleshooting with Logs

Lesson Introduction

When all troubleshooting for the actual service availability and message transport has finished and an issue persists, or if you must find historical data about issues in the past, you should check the event, protocol, and tracking logs.

After this lesson, you will be able to:

- Create searches for the message tracking log
- Describe how to troubleshoot using the protocol logs
- Understand how to work with the event logging for Exchange

Troubleshooting with Message Tracking logs

Message tracking logs record a detailed account of all activity that takes place as email messages move through the transport pipeline. Message tracking logs and message traces can be useful to perform forensics on specific messages, as well troubleshoot the specific send and receive connectors that processed a message. Each Exchange Server maintains its own set of message tracking logs that are retained by default for 30 days, with a maximum size for all message tracking logs of 1 gigabyte (GB). These logs contain detailed information about how messages were processed on the local server.

By default, message tracking is enabled, so you can use the EAC to enable or disable message tracking and to specify the location of the message-tracking log files. However, for advanced message tracking configuration tasks, you will must use the Exchange Management Shell's **Set-TransportService** cmdlet. Additional message tracking configuration tasks include:

- Specifying the location for message tracking log files. The default location is %ExchangeInstall-Path%TransportRoles\Logs\MessageTracking.
- Specifying the maximum message tracking log file size. The default size is 10 MB.
- Specifying the message tracking logs folder size. The default size is 1000 MB.
- Specifying the message tracking log file age. The default age is 30 days.
- Enabling or disabling logging of the message subject. The default is enabled.

The following command syntax should be used to search message tracking log events:

```
Set-TransportService <Server> `
-MessageTrackingLogEnabled <$true | $false> `
-MessageTrackingLogMaxAge <dd.hh:mm:ss> `
-MessageTrackingLogMaxDirectorySize <Size> `
-MessageTrackingLogMaxFileSize <Size> `
-MessageTrackingLogPath <LocalFilePath> `
-MessageTrackingLogSubjectLoggingEnabled <$true | $false>
```

The naming convention for message tracking log files is **MSGTRKServiceyyyymmdd-nnnn.log**. The letters in the log name have following meaning:

- **MSGTRK.** Log files for the Transport service on Mailbox and Edge Transport Servers.
- **MSGTRKMA.** Log files for the approvals and rejections in moderated transport on Mailbox Servers.

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- **MSGTRKMD.** Log files for messages delivered to mailboxes by the Mailbox Transport Delivery service on Mailbox Servers.
- **MSGTRKMS.** Log files for messages sent from mailboxes by the Mailbox Transport Submission service on the Mailbox Servers.
- **Yyyymmdd.** date when the log file was created.
- **Nnnn.** The instance number, whose value starts over at 1 every day. Once the maximum log file size is reached, a new log file is created with an incremented instance number. If a log file reaches its maximum age or if log folder reaches its maximum size, then the oldest log file is deleted.

The data in the message tracking log files is written in comma-separated value (CSV) format. The first line of the log file contains the header, and every other line contains the data described in the header. The log file header has following format:

- **#Software.** The value is Microsoft Exchange Server.
- **#Version.** Version number of the Exchange Server. The format of the version number is 15.02.nnnn.nnn.
- **#Log-Type.** The value is Message Tracking Log.
- **#Date.** The UTC date-time when the log file was created. The format of the date and time is: yyyy-mm-ddThh:mm:ss.fffZ, where yyyy = year, mm = month, dd = day, T indicates the beginning of the time component, hh = hour, mm = minute, ss = second, fff = fractions of a second, and Z signifies UTC.
- **#Fields.** Field names that are used in the log files. Field names include information that can be used to analyze message transport, such as client-ip, client-hostname, server-ip, server-hostname, total-bytes, and recipient-count.

Note: For a detailed list of fields used in the message tracking logs, see the following article on **Message tracking**[5].

You can use the Exchange Management Shell's **Get-MessageTrackingLog** cmdlet to search for entries in the message-tracking log by using specific search criteria. For example, you can:

- Determine what happened to a message that a user sent to a specific recipient.
- Determine whether a transport rule acted on a message.
- Determine whether a message sent from an Internet sender made it into your Exchange Server organization.
- Find all messages that a specific user sent during a specified time period.

You can export cmdlet results to a .csv file, and then view the events in Excel for easier sorting. By default, the **Get-MessageTrackingLog** cmdlet displays the first 1000 entries on the server.

The following command syntax should be used to search message tracking log events:

```
Get-MessageTrackingLog -Server <Server> '
-ResultSize <Integer> | Unlimited] '
-Start <DateTime> '
-End <DateTime> '
-EventId <EventId> '
-InternalMessageId <InternalMessageId> '
-MessageId <MessageId> '
```

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```
-MessageSubject <Subject> `
-Recipients <RecipientAddress1,RecipientAddress2...> `
-Sender <SenderAddress>]
```

For example, to search message tracking log events on server LON-EX1, in specific time interval, where email has been sent from don@adatum.com, you would run the following command:

```
Get-MessageTrackingLog –Server LON-EX1 `
–Start "01/08/2019 10:00:00" `
–End "01/09/2019 11:00:00" `
–Sender "don@adatum.com"
```

You can also use the EAC to view information about messages sent or received by a specific mailbox for up to 14 days after the email has been sent or received. However, for most detailed analysis and troubleshooting, you should use EMS's **Get-MessageTrackingLog** cmdlet.

Message tracking logs in Exchange Online

The **Get-MessageTrackingLogs** cmdlet is not available when connected to Exchange Online. Instead of **Get-MessageTrackingLog**, you must use **Get-MessageTrace** and **Get-MessageTraceDetails** to view results for message tracking.

The message trace functionality in Exchange Online enables you to view a message's progress through the Exchange Online servers and identify whether the message was delivered. If the message was not delivered, you can investigate based on the error messages in the message trace.

Message trace in Exchange admin center

The Exchange Admin Center provides a simple user interface that can perform a message trace. You can access the message trace interface in the EAC by clicking **mail flow**, and then clicking **message trace** on the toolbar. When you perform a message trace, you can specify the following search criteria:

- Date range
- Delivery status
- Message ID
- Sender
- Recipient

Note: When you add a sender or recipient, it may appear that you are unable to add email addresses that are not part of your organization. However, you can add any email address by typing it in the box next to the **Check names** button.

Message trace in Windows PowerShell

You can use the Windows PowerShell **Get-MessageTrace** cmdlet to search for messages that have been sent or received. You then can use the **Get-MessageTraceDetail** cmdlet to view the same details that are available in the EAC.

Some of the parameters that you can use with the **Get-MessageTrace** cmdlet include:

- StartDate

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- EndDate
- MessageID
- SenderAddress
- RecipientAddress
- FromIP

For example, to perform the same search for messages sent by don@adatum.com in a specific time interval, you would run the following cmdlet:

```
Get-MessageTrace -SenderAddress "don@adatum.com" `
-StartDate "01/08/2019 10:00:00" `
-EndDate "01/09/2019 11:00:00"
```

Note: There is often a delay of 5 to 30 minutes before message trace information is available after a message is sent. This applies to both the EAC and Windows PowerShell.

Troubleshooting with Protocol logs

Protocol logging is a source of information where you can view the SMTP conversations that occur as part of message delivery between messaging servers, and between Exchange Server services in the transport pipeline. Protocol logging can be enabled on SMTP Send and SMTP Receive connectors to log the activity of each connector. These logs show the SMTP commands sent between the source and destination, and the IP addresses of the source and destination. Protocol logs are useful for identifying why connectivity is failing and for verifying that a host is communicating with the correct connector.

Protocol logging provides detailed information that helps diagnose mail-flow problems by logging the SMTP conversations between the following components:

- Send connectors and Receive connectors in the Transport service on Mailbox servers.
- Send connectors and Receive connectors in the Transport service on Edge Transport servers.
- Receive connectors in the Front-End Transport service on Mailbox servers.
- The implicit and invisible intra-organization Send connector in the Transport service on Mailbox servers.
- The implicit and invisible intra-organization Send connector in the Front-End Transport service on Mailbox servers.
- The implicit and invisible intra-organization Send connector in the Mailbox Transport Submission service on Mailbox servers.
- The implicit and invisible Mailbox Delivery Receive connector in the Mailbox Transport Delivery service on Mailbox servers.

By default, protocol logging is enabled on the following connectors only and disabled for all others:

- The default Receive connector in the Front-End Transport service on Mailbox servers. Its name is in the Default Frontend ServerName naming format.
- The implicit and invisible Send connector in the Front-End Transport service on Mailbox servers.

You have already learned that connector logs are stored in the following location in the file system, depending on the server type:

- %ExchangeInstallPath%TransportRoles\Logs*\ProtocolLog\SmtpReceive

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- %ExchangeInstallPath%TransportRoles\Logs*\ProtocolLog\SmtpSend

Consider also that the default maximum size of protocol log files is 10 MB and the default maximum age is 30 days.

The naming convention for protocol log files is **SENDyyyymmdd-nnnn.log** for Send connectors and **RECVyyyymmdd-nnnn.log** for Receive connectors. The placeholders represent the following information:

- **yyyymmdd** is the date when the log file was created.
- **nnnn** is an instance number that is reset to a value of 1 every day.

For example, let's assume a server produces 50 MB of protocol logs every day and you must search for events in the receive connector logs that happened on the late evening of 01/08/2019. You would open the log file with the name **RECV20190108-0005.log**.

Once the maximum log file size is reached, a new log file is created with an incremented instance number. If a log file reaches its maximum age, or if a log folder reaches its maximum size, then the oldest log file is deleted.

Activate or deactivate protocol logging

You should run the following commands in the Exchange Management Shell to enable or disable protocol logging:

```
Set-SendConnector <connector> -ProtocolLoggingLevel <Verbose | None>
Set-ReceiveConnector> <connector> -ProtocolLoggingLevel <Verbose | None>
```

The following command syntax should be used to configure the protocol log settings on a Mailbox server:

```
Set-MailboxTransportService <server> '
-ReceiveProtocolLogPath <LocalFilePath> '
-ReceiveProtocolLogMaxFileSize <Size> '
-ReceiveProtocolLogMaxDirectorySize <Size> '
-ReceiveProtocolLogMaxAge <dd.hh:mm:ss> '
-SendProtocolLogPath <LocalFilePath> '
-SendProtocolLogMaxFileSize <Size> '
-SendProtocolLogMaxDirectorySize <Size> '
-SendProtocolLogMaxAge <dd.hh:mm:ss>
```

Similar to message tracking logs, the data in the message protocol log files is written in comma-separated value (CSV) format. Protocol log files use the same log file header format that was described for message tracking logs in the previous topic.

Protocol logs are used to analyze and troubleshoot issues that are related to a specific connector, while message tracking logs are used for analyzing the detailed records of all message activity as mail flowed through the transport pipeline on an Exchange server. For example, if there are issues only when sending email to users on the Internet, we should analyze the protocol logs generated by the Front-End Transport service Internet send connector. However, if you suspect that a user cannot send email to different users where email is sent over multiple connectors, then you should analyze message tracking logs to troubleshoot the issue.

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Additional reading. For more information, see the following article on **Fields in the protocol log**[6].

Troubleshooting with Event logs

Microsoft Exchange Server generates different event information related to transport services in the Event Viewer. You can use the Event Viewer to view those events and to troubleshoot any potential transport issues. Examining the event logs on a regular basis can help Exchange administrators fix problems before they cause an issue or persist too long. In an enterprise environment, it's important that logs are sized properly, so that more data is preserved in the logs. Furthermore, event logs should be backed up if historical analysis is needed for troubleshooting some issues that persisted earlier.

You can view the event logs in the Event Viewer or by running the **Get-EventLog** cmdlet. The **Get-EventLog** cmdlet is not specific to Exchange Server, but you can use it to gather information about Exchange Server from the event logs. For example, to return the 10 most recent event log entries that have a source of MSExchange Common, you would run the following command:

```
Get-EventLog –LogName Application –Source "MSExchange Common" –EntryType
Error –Newest 10
```

You can modify the command to return events that contain specific error numbers or words. You can also return data from multiple computers. The following command returns the 10 most recent events from multiple servers:

```
Get-EventLog –LogName Application –Source "MSExchange Common" –EntryType
Error –Newest 10 –ComputerName LON-EX1,LON-EX2
```

You can retrieve all warning and error events quickly from a specific server's Application log by using the following command:

```
Get-EventLog –ComputerName LON-EX1 –LogName Application –Source –EntryType
Error,Warning
```

Furthermore, by running the **Set-Eventloglevel** cmdlet, you can configure a different event log level for a specified event category, such as: Lowest, Low, Medium, High, and Expert. The syntax for the **Set-Eventloglevel** cmdlet is:

```
Set-Eventloglevel -Identity <ECIdParameter> -Level <Lowest | Low | Medium |
High | Expert>
```

In addition to the Application Logs in the Event Viewer, Exchange Server records specific events to the crimson channel logs under **Applications and Services Logs > Microsoft > Exchange**.

Event log categories for Exchange included in the crimson channel include: ActiveMonitoring, Compliance, DxStoreHA, ESE, HighAvailability, MailboxAssisants, MailboxDatabaseFailureItems, ManagedAvailability, PushNotifications, and Troubleshooters.

Event logs in the Application log that are related to transport issues include the following sources:

- **Source MSExchangeADAccess**. Includes events about Active Directory access.
- **Source MSExchangeSubmission, MSExchangeDelivery**. Includes events about mail submission and mail delivery services, such as information about mail submission or mail delivery service status (started or stopped).
- **Source MSExchangeAntimalware**. Includes events about logging the status of the anti-malware agent.

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- **SourceMSExchangeTransport**. Includes events about logging the status of the configuration update, configuration of the connectors, authentication failures, TLS certificate validation errors, DNS failure during the send process, and routing issues.
- **MS Exchange Messaging Policies**. Includes information about messaging policies features, such as encryption, federation, and certificate validation.
- **MSExchange Store Driver Delivery and MSExchange Store Driver Submission**. Includes events about failures in mail delivery and submission processing.
- **MSExchangeTransportLogSearch**. Includes information about Active Directory access, insufficient permissions, message tracking log corrupted, and the Transport Log Search service not being able to start.

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Module Review

Module 2 Review

Checkbox

Item 1. You are experiencing an issue where a third-party application within your internal network needs to send a notification email to your application development team. The third-party application is configured to connect to your organization's Exchange Server receive connector in order to send the notification email. The development team reports that even when they manually trigger the email sending process from the application, no message is delivered to their Inbox.
What should you do to start troubleshooting the problem? (Select 2)

- ☐ Check whether your firewall is connected to the Internet.
- ☐ Enable protocol logging on the receive connector where the application connects to.
- ☐ Restart your Exchange Server and inform the application development team to try sending the message again.
- ☐ In the appropriate connector file location, open the log file with the file format of RECVyyyym-mdd-nnnn.log and analyze the recorded data.

Multiple choice

Item 2. You suspect that during the night, all of your Exchange Servers lost connectivity to the domain controllers in your network. You are searching the event logs for an event that indicates a potential connectivity issue between the Exchange Servers and the domain controllers.
What event source should you search for in order to filter the event log data?

- ☐ MSExchangeMessagingPolicies
- ☐ MSExchangeAntimalware
- ☐ MSExchangeDomainControllers
- ☐ MSExchangeADAccess

Multiple choice

Item 3. Your colleagues complain that email messages are arriving much later when compared to the time they were sent. You need to troubleshoot this problem, since timely communication is business critical for your organization.
Which tool should you use to troubleshoot this delay in email messages?

- ☐ Message Tracking
- ☐ DNS Management console
- ☐ Telnet
- ☐ Active Directory Users and Computers console

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Multiple choice

Item 4. You are an Exchange Server administrator for your organization. Your colleagues are complaining that email messages from external recipients didn't arrive. Email sent between users in your organization arrives normally. You tried to send email from your private account to your company email address and you noticed that the email is not deliverd as well.
Which tool will you use in order to troubleshoot this problem?

- ☐ Protocol Logging
- ☐ Jet Stress
- ☐ Microsoft Remote Connectivity Analyzer
- ☐ Queue Viewer

Checkbox

Item 5. You are responsible for configuring a shared SMTP domain namespace due to a recent merger and acquisitions between your organization and another company. Your organization is running Exchange Server, while the other company runs a third-party email system. You asked your Internet DNS provider to add the MX records with the new shared SMTP domain. However, users in both organizations started to complain that they are not receiving email when it is sent with the new SMTP domain suffix in their email address.
Which two steps should you perform to resolve the issue? (Select 2)

- ☐ Create an Internal Relay Accepted Domain with the new shared SMTP domain namespace.
- ☐ Create new Active Directory domain for the company that merged with your organization.
- ☐ Create a send connector that will route the messages to the shared SMTP domain.
- ☐ Install a new Exchange Server for the company that merged with your organization.

Multiple choice

Item 6. You manage an Exchange Server organization. You need to add the DNS records so that email from the Internet is properly routed to your organization. Which record type should you add to your DNS to support email to your custom domain?

- ☐ MX
- ☐ CNAME
- ☐ TXT
- ☐ SPF

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Multiple choice

Item 7. You manage a Microsoft 365 tenant with a custom domain. You need to configure a DNS record so that email sent from other companies is routed to your organization. Which DNS record type should you create?

- ☐ A
- ☐ CNAME
- ☐ TXT
- ☐ MX
- ☐ SRV

Multiple choice

Item 8. You are an administrator of an Exchange Server organization. An employee has complained that a message sent to a colleague never arrived. You are responsible for troubleshooting this transport issue. You need to determine the following requirements:

- What happened to the message that a user sent to a specific recipient.
- Whether a transport rule acted on the message.
- Find all messages that this specific user sent during a specified time.

Which type of logging solution will you use?

- ☐ Message Tracking Log
- ☐ Protocol Log
- ☐ Telnet
- ☐ Application Log

Checkbox

Item 9. You are migrating your messaging solution from Exchange 2013 to Exchange 2019. Before the migration, network devices were configured to send notification emails through Exchange 2013. After the migration was completed, you noticed that mailboxes that used to receive email from those applications and devices no longer receive email.
What will you do to troubleshoot the problem? (Select 3)

- ☐ Enable protocol logging on the receive connector on Exchange 2019 where the application connects.
- ☐ Restart the applications and devices and try to force sending an email to Exchange Server 2019.
- ☐ In the file location where the receive connector stores protocol logs, open the log file with the file name format RECVyyyymmdd-nnnn.log and analyze the data recorded.
- ☐ Configure authentication options on the Exchange 2019 receive connectors to accept email from third-party applications and devices.
- ☐ Restart network switch where Exchange 2019 is connected.
- ☐ Restart Exchange Server.

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Checkbox

Item 10. Your colleagues complain that they cannot send emails to a domain of a partner company named Contoso. You need to troubleshoot this problem, since Contoso is a strategic partner and communication with them is very important for your organization. The email communication between other companies works normally.
Which tools should you use to troubleshoot this email delivery issue? (Select 2)

- ☐ Queue Viewer
- ☐ Partner Send and Receive Connectors
- ☐ Telnet
- ☐ Active Directory Users and Computers console

Checkbox

Item 11. You are an Exchange administrator for your organization. After the upgrade of Exchange 2013 to Exchange 2019, you have decommissioned all Exchange 2013 servers. After the uninstall process has been completed, organization employees have complained that emails from the Internet are no loner being received. You run the Exchange Remote Connectivity Analyzer and you notice that there are DNS records for your organization. You notice that the IP addresses listed in the DNS were used by Exchange 2013 but not by Exchange 2019. You need to provide your DNS provider with information as to what records should be updated for your Exchange 2019 organization.
Which records should your DNS provider update so that email connectivity can be reestablished to your organization? (Select 2)

- ☐ MX record
- ☐ A Record
- ☐ SRV record
- ☐ PTR record
- ☐ CNAME record

Multiple choice

Item 12. Your company's networking team is deploying a new firewall that also acts as an SMTP gateway that protects your organization from malicious email threats. However, after the firewall is deployed, mail flow stops from the Internet to your organization. The networking team has checked that all necessary ports are open. Internal mail flow and emails sent to the Internet works correctly. Protocol logging is enabled on all receive connectors.
Where on Exchange Server will you locate the logs that need to be analyzed for troubleshooting this issue?

- ☐ %ExchangeInstallPath%TransportRoles\Logs\FrontEnd\ProtocolLog\SmtpReceive
- ☐ %ExchangeInstallPath%TransportRoles\Logs\FrontEnd\ProtocolLog\SmtpSend
- ☐ %ExchangeInstallPath%TransportRoles\Logs\Hub\ProtocolLog\SmtpReceive
- ☐ %ExchangeInstallPath%TransportRoles\Logs\Mailbox\ProtocolLog\SmtpReceive\Delivery

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Checkbox

Item 13. You are an administrator for an Exchange organization. You need to send and receive secure email with a partner organization. Therefore, you have configured mutual TLS, where each server verifies the identity of the other server by validating the certificate that the other server provides. However, users in your organization report that they cannot send email to the partner organization. They also report that they can send email on any other domain to the Internet.
What troubleshooting steps will you perform? (Select 2)

- ☐ Check if the name of the servers is correctly configured in Domain Security settings to match the common name in the certificate used for TLS.
- ☐ Check if the certificate issued to the Mailbox servers in Domain Security configuration is trusted.
- ☐ Check whether the firewall in your organization is blocking mail sent to the Internet.
- ☐ Run the eseutil utility.

Multiple choice

Item 14. Each Exchange Server maintains its own set of message tracking logs that are retained by default for how many days?

- ☐ 30 days
- ☐ 60 days
- ☐ 90 days
- ☐ 120 days

Multiple choice

Item 15. What tool can you use to enable or disable message tracking and to specify the location of the message-tracking log files?

- ☐ Front-End Transport services
- ☐ Extensible Storage Engine
- ☐ Remote Connectivity Analyzer
- ☐ Message Transport services
- ☐ Exchange Admin Center

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Answers

Checkbox

Item 1. You are experiencing an issue where a third-party application within your internal network needs to send a notification email to your application development team. The third-party application is configured to connect to your organization's Exchange Server receive connector in order to send the notification email. The development team reports that even when they manually trigger the email sending process from the application, no message is delivered to their Inbox.
What should you do to start troubleshooting the problem? (Select 2)

- ☐ Check whether your firewall is connected to the Internet.
- ■ Enable protocol logging on the receive connector where the application connects to.
- ☐ Restart your Exchange Server and inform the application development team to try sending the message again.
- ■ In the appropriate connector file location, open the log file with the file format of RECVyyyym-mdd-nnnn.log and analyze the recorded data.

Multiple choice

Item 2. You suspect that during the night, all of your Exchange Servers lost connectivity to the domain controllers in your network. You are searching the event logs for an event that indicates a potential connectivity issue between the Exchange Servers and the domain controllers.
What event source should you search for in order to filter the event log data?

- ☐ MSExchangeMessagingPolicies
- ☐ MSExchangeAntimalware
- ☐ MSExchangeDomainControllers
- ■ MSExchangeADAccess

Multiple choice

Item 3. Your colleagues complain that email messages are arriving much later when compared to the time they were sent. You need to troubleshoot this problem, since timely communication is business critical for your organization.
Which tool should you use to troubleshoot this delay in email messages?

- ■ Message Tracking
- ☐ DNS Management console
- ☐ Telnet
- ☐ Active Directory Users and Computers console

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Multiple choice

Item 4. You are an Exchange Server administrator for your organization. Your colleagues are complaining that email messages from external recipients didn't arrive. Email sent between users in your organization arrives normally. You tried to send email from your private account to your company email address and you noticed that the email is not deliverd as well.
Which tool will you use in order to troubleshoot this problem?

- ☐ Protocol Logging
- ☐ Jet Stress
- ■ Microsoft Remote Connectivity Analyzer
- ☐ Queue Viewer

Checkbox

Item 5. You are responsible for configuring a shared SMTP domain namespace due to a recent merger and acquisitions between your organization and another company. Your organization is running Exchange Server, while the other company runs a third-party email system. You asked your Internet DNS provider to add the MX records with the new shared SMTP domain. However, users in both organizations started to complain that they are not receiving email when it is sent with the new SMTP domain suffix in their email address.
Which two steps should you perform to resolve the issue? (Select 2)

- ■ Create an Internal Relay Accepted Domain with the new shared SMTP domain namespace.
- ☐ Create new Active Directory domain for the company that merged with your organization.
- ■ Create a send connector that will route the messages to the shared SMTP domain.
- ☐ Install a new Exchange Server for the company that merged with your organization.

Multiple choice

Item 6. You manage an Exchange Server organization. You need to add the DNS records so that email from the Internet is properly routed to your organization. Which record type should you add to your DNS to support email to your custom domain?

- ■ MX
- ☐ CNAME
- ☐ TXT
- ☐ SPF

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Multiple choice

Item 7. You manage a Microsoft 365 tenant with a custom domain. You need to configure a DNS record so that email sent from other companies is routed to your organization. Which DNS record type should you create?

- ☐ A
- ☐ CNAME
- ☐ TXT
- ■ MX
- ☐ SRV

Multiple choice

Item 8. You are an administrator of an Exchange Server organization. An employee has complained that a message sent to a colleague never arrived. You are responsible for troubleshooting this transport issue. You need to determine the following requirements:

Which type of logging solution will you use?

- ■ Message Tracking Log
- ☐ Protocol Log
- ☐ Telnet
- ☐ Application Log

Checkbox

Item 9. You are migrating your messaging solution from Exchange 2013 to Exchange 2019. Before the migration, network devices were configured to send notification emails through Exchange 2013. After the migration was completed, you noticed that mailboxes that used to receive email from those applications and devices no longer receive email.
What will you do to troubleshoot the problem? (Select 3)

- ■ Enable protocol logging on the receive connector on Exchange 2019 where the application connects.
- ☐ Restart the applications and devices and try to force sending an email to Exchange Server 2019.
- ■ In the file location where the receive connector stores protocol logs, open the log file with the file name format RECVyyyymmdd-nnnn.log and analyze the data recorded.
- ■ Configure authentication options on the Exchange 2019 receive connectors to accept email from third-party applications and devices.
- ☐ Restart network switch where Exchange 2019 is connected.
- ☐ Restart Exchange Server.

AUTHORIZED TRAINER USE ONLY. STUDENT USE PROHIBITED

Checkbox

Item 10. Your colleagues complain that they cannot send emails to a domain of a partner company named Contoso. You need to troubleshoot this problem, since Contoso is a strategic partner and communication with them is very important for your organization. The email communication between other companies works normally.
Which tools should you use to troubleshoot this email delivery issue? (Select 2)

- ■ Queue Viewer
- ■ Partner Send and Receive Connectors
- ☐ Telnet
- ☐ Active Directory Users and Computers console

Checkbox

Item 11. You are an Exchange administrator for your organization. After the upgrade of Exchange 2013 to Exchange 2019, you have decommissioned all Exchange 2013 servers. After the uninstall process has been completed, organization employees have complained that emails from the Internet are no loner being received. You run the Exchange Remote Connectivity Analyzer and you notice that there are DNS records for your organization. You notice that the IP addresses listed in the DNS were used by Exchange 2013 but not by Exchange 2019. You need to provide your DNS provider with information as to what records should be updated for your Exchange 2019 organization.
Which records should your DNS provider update so that email connectivity can be reestablished to your organization? (Select 2)

- ☐ MX record
- ■ A Record
- ☐ SRV record
- ■ PTR record
- ☐ CNAME record

Multiple choice

Item 12. Your company's networking team is deploying a new firewall that also acts as an SMTP gateway that protects your organization from malicious email threats. However, after the firewall is deployed, mail flow stops from the Internet to your organization. The networking team has checked that all necessary ports are open. Internal mail flow and emails sent to the Internet works correctly. Protocol logging is enabled on all receive connectors.
Where on Exchange Server will you locate the logs that need to be analyzed for troubleshooting this issue?

- ■ %ExchangeInstallPath%TransportRoles\Logs\FrontEnd\ProtocolLog\SmtpReceive
- ☐ %ExchangeInstallPath%TransportRoles\Logs\FrontEnd\ProtocolLog\SmtpSend
- ☐ %ExchangeInstallPath%TransportRoles\Logs\Hub\ProtocolLog\SmtpReceive
- ☐ %ExchangeInstallPath%TransportRoles\Logs\Mailbox\ProtocolLog\SmtpReceive\Delivery

AUTHORIZED TRAINER USE ONLY. STUDENT USE PROHIBITED

Checkbox

Item 13. You are an administrator for an Exchange organization. You need to send and receive secure email with a partner organization. Therefore, you have configured mutual TLS, where each server verifies the identity of the other server by validating the certificate that the other server provides. However, users in your organization report that they cannot send email to the partner organization. They also report that they can send email on any other domain to the Internet.
What troubleshooting steps will you perform? (Select 2)

- ■ Check if the name of the servers is correctly configured in Domain Security settings to match the common name in the certificate used for TLS.
- ■ Check if the certificate issued to the Mailbox servers in Domain Security configuration is trusted.
- ☐ Check whether the firewall in your organization is blocking mail sent to the Internet.
- ☐ Run the eseutil utility.

Multiple choice

Item 14. Each Exchange Server maintains its own set of message tracking logs that are retained by default for how many days?

- ■ 30 days
- ☐ 60 days
- ☐ 90 days
- ☐ 120 days

Multiple choice

Item 15. What tool can you use to enable or disable message tracking and to specify the location of the message-tracking log files?

- ☐ Front-End Transport services
- ☐ Extensible Storage Engine
- ☐ Remote Connectivity Analyzer
- ☐ Message Transport services
- ■ Exchange Admin Center

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Module 3 Managing Message Hygiene

Planning for Message Hygiene

Lesson Introduction

Microsoft Exchange Online Protection (EOP) is a cloud-based email filtering service that provides anti-malware and anti-spam policies to protect your organization against spam and malware. EOP includes features to safeguard your organization from messaging-policy violations.

In this lesson, you will learn about EOP features and functionality and how to plan messaging routing for this service.

After this lesson, you will be able to:

- Explain the use and features of Exchange Online Protection
- Plan message routing for Exchange Online Protection
- Investigate the available EOP reports and logs

Additional reading. For more information, see the following article on **Microsoft Exchange Online Protection**[1]

Exchange Online Protection Overview

Current messaging environments require a robust anti-virus and anti-spam solution to minimize the impact of malicious email. Exchange Online Protection (EOP) is an anti-virus, anti-spam service that is included with Exchange Online that you can purchase for your on-premises Exchange Server environment. EOP is a hosted service that requires no hardware or software installation.

The primary ways in which you can use EOP for messaging protection include:

- **In a standalone scenario.** EOP provides cloud-based email protection for your on-premises Exchange Server environment, legacy Exchange Server versions, or any other on-premises SMTP email solution.
- **As a part of Microsoft Exchange Online.** By default, EOP protects Exchange Online cloud-hosted mailboxes.

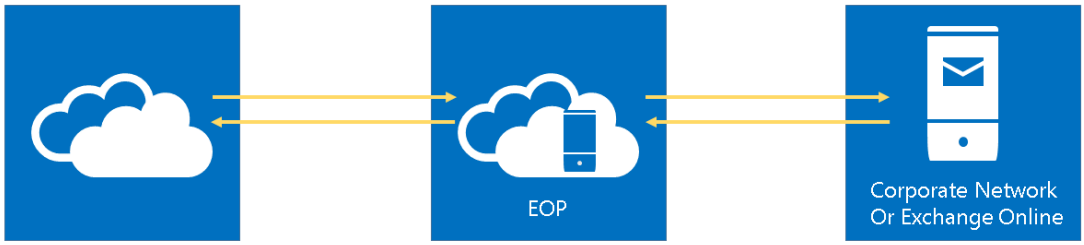
1 https://products.office.com/en-us/exchange/exchange-email-security-spam-protection

- **In a hybrid deployment.** EOP can be configured to protect your messaging environment and control mail routing when you have a mix of on-premises and cloud mailboxes.

Exchange Online Protection includes the following functionality:

- Scans incoming, outgoing, and internal email messages. This scanning helps protect your organization from malicious content that originates inside your organization.
- Multiple anti-virus engines help catch email-borne viruses and other malicious code.
- Proprietary anti-spam technology is used to achieve high-accuracy rates.
- All functionality is built into the service. No configuration is necessary to start or to maintain the filtering technology. EOP requires only an EOP Send connector in the on-premises Exchange Server environment so that messages are sent to the EOP domain for scanning. If you use only Exchange Online and your DNS MX record points to Microsoft 365, then no additional configuration is needed.
- Customizable filters help you comply with corporate policies and with government regulations.
- Directory Based Edge Blocking (DBEB) is used to reject messages sent to invalid recipients.
- Anti-spoofing protection by supporting Sender Policy Framework (SPF), DomainKeys Identified Mail (DKIM), and Domain-based Message Authentication, Reporting, and Conformance (DMARC).
- In combination with Azure AD Connect and Exchange on-premises, safe lists and block lists are automatically synchronized and used in EOP. For more information, see the following article on **Manage mail users in EOP**[2].

If you register for Exchange Online or Microsoft 365, you automatically use EOP for any message that is received or sent from your online tenant; no extra configuration is needed. If you configure a hybrid deployment using the Hybrid Configuration Wizard, any mail that is sent between on-premises Exchange and Exchange Online is subject to inspection by EOP.



Inbound and Outbound email filtered through EOP

2 https://docs.microsoft.com/en-us/office365/SecurityCompliance/eop/manage-mail-users-in-eop

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Additional reading. For more information, see the following article on **Exchange Online Protection Service Description**[3].

EOP functionality can optionally be extended by using another service called Microsoft 365 Advanced Threat Protection (ATP). ATP is an email filtering service that provides additional protection against specific types of advanced threats, such as:

- **Protection against unknown malware and viruses.** EOP employs a robust and layered anti-virus protection solution powered with three different engines against known malware and viruses. ATP extends this protection through a feature called Safe Attachments, which protects against unknown malware and viruses and provides better zero-day protection to safeguard your messaging system. All messages and attachments that do not have a known virus/malware signature are routed to a special hypervisor environment where behavior analysis is performed using a variety of machine learning and analysis techniques to detect malicious intent. If no suspicious activity is detected, the message is released for delivery to the mailbox.
- **Real time, time-of-click protection against malicious URLs.** EOP scans each message in transit in Office 365 and provides time-of-delivery protection, thus blocking any malicious hyperlinks in a message. Sometimes attackers try to hide malicious URLs with seemingly safe links that are redirected to unsafe sites by a forwarding service after the message has been received. The Safe Links feature of ATP proactively protects your users if they click such a link. That protection remains every time they click the link; malicious links are dynamically blocked while good links can be accessed.
- **Rich reporting and URL trace capabilities.** ATP also offers rich reporting and tracking capabilities so that you can gain critical insights into who is getting targeted in your organization and the category of attacks you are facing. Reporting and message tracing allow you to investigate messages that have been blocked due to an unknown virus or malware, while the URL trace capability allows you to track individual malicious links in the messages that have been opened.

A brief description of Microsoft 365 Advanced Threat Protection and its features will be covered in a later lesson.

Additional reading. For more information, see the following article on **Exchange Online Protection overview**[4].

Planning message routing for EOP

Exchange Online Protection (EOP) is enabled by default for every message in Exchange Online. If your company only has cloud-only users and Exchange Online mailboxes, all messages sent to your organization pass through EOP before your users see them, and you don't need to configure any special mail routing.

EOP is not limited to just Exchange Online; in fact, there are three different ways of deploying Exchange Online Protection:

- **EOP with Exchange Online mailboxes**. This scenario is appropriate if you want EOP protection and all your mailboxes hosted in Exchange Online. It can help you reduce complexity because you don't have to maintain on-premises messaging servers.
- **EOP with on-premises mailboxes (also known as EOP standalone)**. This scenario is appropriate if you have an existing mail-hosting infrastructure you want to use, or if you have business requirements that necessitate keeping mailboxes on-premises.

3 https://docs.microsoft.com/en-us/office365/servicedescriptions/exchange-online-protection-service-description/exchange-online-protection-service-description

4 https://docs.microsoft.com/en-us/office365/securitycompliance/eop/exchange-online-protection-overview

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- **EOP with Exchange Hybrid Deployment**. Some organizations want cloud mailboxes in Exchange Online for the majority of their users, but they also want to keep some mailboxes on-premises only for certain users.

Considerations for configuring your mail flow using EOP

Before you start the actual configuration of your mail routing, you should first review your mail flow requirements and answer the following questions:

- **Inbound mail flow.** Do you still want to use your on-premises anti-spam solution for inbound mail flow, or do you want to replace it with EOP completely?
- **Outbound mail flow.** Do you want to route outbound mail flow over your on-premises solution, or should all mail be routed over EOP equally?
- **Secure messaging with a trusted partner.** Do you use connectors with mandatory TLS encryption with a trusted partner that you need to consider?
- **Conditional mail routing.** Do you want to route emails for a defined recipient domain, or from a defined security group of senders over EOP?
- **Anti-spoofing.** Are you already using anti-spoofing solutions such as SPF, DKIM, and DMARC that you need to reconfigure when adding another trusted system like EOP to your mail flow?

Once you answer these questions, you'll have a better idea as to your mail routing needs, and you can start configuring your solution.

Configuring mail flow using EOP with Exchange Online mailboxes

When all your users have mailboxes in Exchange Online and all your domain's MX-records point to Microsoft 365, EOP is already set up to scan your inbound and outbound messages. No additional configuration is necessary and no on-premises systems are present that you need to consider.

Configuring mail flow using EOP Standalone or a Hybrid Deployment

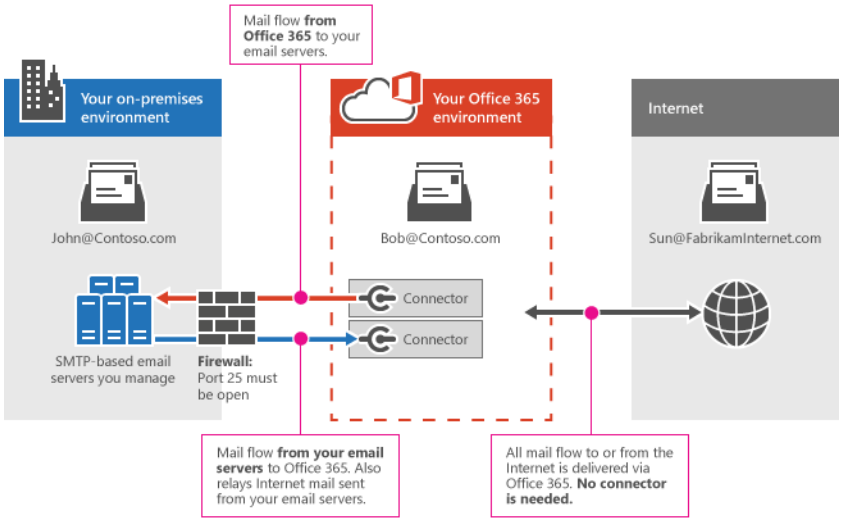
When analyzing whether to use EOP standalone or hybrid, you can decide for yourself if you want to replace your on-premises solution, or if you want to use both simultaneously to add a second layer to your mail protection.

Note: You can use two layers of anti-spam and email protection, but keep in mind that each solution uses its individual scanning and quarantine settings. As a result, implementing two solutions together can cause unpredictable mail flow issues if the scanning and quarantine settings of each solution conflict with one another. Therefore, it's recommended that organizations use EOP as their exclusive in- and outbound solution for message protection is recommended.

If you want to use EOP and you are still running your own email servers, whether they are on-premises or in a private cloud outside Microsoft 365, you need to set up connectors to enable mail flow between EOP and your environment. For a hybrid deployment, those connectors are created by the Hybrid Configuration Wizard (HCW); however, if you are using EOP standalone, you must create them yourself.

The following diagram shows how connectors work to switch your mail routing to EOP:

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For connecting your on-premises environment to EOP, you must consider the role that the following connectors play in routing messages:

- **EOP outbound connector.** Delivers emails to your on-premises mail servers.
- **EOP inbound connector.** Accepts emails from your on-premises mail servers.
- **On-premises connector.** Send connector to send all outgoing mail to EOP.

While the connectors are responsible for routing messages between your controlled environments, you must also consider the outside world in the Internet, where message routing is not performed with connectors, but with MX records. Mail servers deliver email for domains to the servers that are resolved by checking the public domains' MX records. Therefore, you must configure your MX records for delivering messages to EOP or on-premises.

When EOP is implemented in a hybrid environment, it can be administered by using either the Exchange Admin Center or Windows PowerShell. However, if you are using EOP standalone, you can only use PowerShell cmdlets to configure your service.

Additional reading. For more information, see the following article on **Set up connectors to route mail between Office 365 and your own email servers**[5].

Monitoring and Reporting

Microsoft Exchange Online Protection (EOP) offers many different reports that can help you determine the overall status and health of your organization. Protection reports are available in the Microsoft Security & Compliance Center.

Security & Compliance Center reports

The Reports Dashboard in the Security & Compliance Center provides different information about Microsoft 365 features, including message traffic, spam and malware detections, and messages affected by Exchange Transport Rules or Data Loss Prevention (DLP) policies. The enhanced reports for protection,

5 https://docs.microsoft.com/en-us/exchange/mail-flow-best-practices/use-connectors-to-configure-mail-flow/set-up-connectors-to-route-mail

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rules, and DLP offer an interactive reporting experience for EOP admins. These reports provide summary data and the ability to drill down into details about individual messages.

Note: At one time the Exchange Admin Center included an "action center," which was used to report blocked senders and to unblock them. This blocked sender list has now been moved from the EAC to the Security & Compliance Center, where it's available under **Threat management** > **Review**.

The following table describes the types of reports that are available, how to find them, and where to go to learn more about each report.

Type of information	How to get there in the Security & Compliance Center	Where to go to learn more
Threat management dashboard (this is also referred to as the **Security dashboard** and the **Threat Intelligence dashboard**). Threat detections, malware trends, top targeted users, details about sent and received email messages, and more.	Go to **Threat management** > **Dashboard**.	**Security dashboard overview** *(https://aka.ms/AA3qwxj)*
Advanced Threat Protection and email security reports Email security and threat protection reports (including malware, spam, phishing, and spoofing reports).	Go to **Reports** > **Dashboard**.	**View reports for Office 365 Advanced Threat Protection** *(https://support.office.com/article/e47e838c-d99e-4c0b-b9aa-e66c4fae902f)* **View email security reports in the Security & Compliance Center** *(https://aka.ms/AA3qwxk)*
Mail flow Information about sent and received email messages, recent alerts, top senders and recipients, email forwarding reports, and more.	Go to **Mail flow** > **Dashboard**.	**Mail flow insights in the Office 365 Security & Compliance Center** *(https://docs.microsoft.com/en-us/microsoft-365/security/office-365-security/mail-flow-insights-v2?view=o365-worldwide)*

Scheduled mail protection reports

Messaging Admins are responsible for monitoring their organization's mail flow, as well as determining how much spam and malware is being detected, and how often their rules and policies are being matched. This information, and more, is available in the mail protection reports produced by the Security & Compliance Center. These reports can provide detailed and summarized views of the messages that Microsoft 365 has delivered or rejected based on spam or malware characteristics, rules, or data loss prevention (DLP) policies.

- You can choose to either schedule mail protection reports to be sent to your Inbox automatically, or you can view them at your own discretion in the Security & Compliance Center. You can perform the following tasks to manage reports in the Security & Compliance Center: Set up and download a custom report

- Download existing reports
- Manage schedules for multiple reports

Reporting using web services

In an Exchange Online or hybrid deployment, you can use the REST/OData Tenant Reporting web service to programmatically collect summary and detailed reports about messaging data, and you can display the data on a web page in a custom management Web portal. Reporting using web services is not available to EOP standalone customers.

The Microsoft 365 Reporting web service enables developers to integrate information on email and spam, anti-virus activity, compliance status, and Skype for Business Online conferences and sessions into their custom service reporting applications and web portals. All the reports available in the admin portal, within the downloadable Microsoft Excel spreadsheets, and those accessed through Windows PowerShell cmdlets are accessible using the Reporting web service. Users accessing the Reporting web service must have administrative rights in the organization.

Additional reading. For more information, see the following site **to retrieve a complete Reporting web service description document for your tenant**[6].

Additional reading. For more information, see the following article on **Office 365 Reporting web service**[7].

Message trace

The message trace feature in the EAC and Security & Compliance Center enables you as a messaging administrator to follow email messages as they pass through Exchange Online Protection. This can help you determine whether a targeted email message was received, rejected, deferred, or delivered by the service. It also shows what actions have occurred to the message before reaching its final status. Obtaining detailed information about a specific message lets you efficiently answer your user's questions, troubleshoot mail flow issues, validate policy changes, and alleviates the need to contact technical support for assistance.

6 https://aka.ms/AA3r4jk
7 https://aka.ms/AA3qwxp

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Managing Anti-Malware and Anti-Spam Policies

Lesson Introduction

Anti-malware and anti-spam defenses are a critical part of any modern messaging system. Exchange Server and Exchange Online provide highly effective tools for minimizing the number of unwanted messages that reach user mailboxes while providing strong defenses against malicious software.

In this lesson, you will review the anti-malware and anti-spam protection that Exchange Server and Online Protection provide, and you will learn how to configure SPAM and malware filters, policies, and settings to provide protection for your users.

After this lesson, you will be able to:

- Understand the different message header fields relevant for spam and spoofing protection
- Configure anti-spam and anti-malware filters in Exchange Server
- Using additional features for outbound spam filtering and quarantine
- Implementing protection features against phishing and spoofing
- Create transport rules for custom requirements

Message headers and spam confidence levels

Exchange uses several anti-spam technologies to minimize incoming spam messages. It scans incoming messages and stores the results in the anti-spam message headers that are part of every SMTP message. Additionally, it saves the spam confidence level (SCL) as part of this header that includes the likeliness that the message is spam.

Anti-Spam Message Headers

When an inbound email message is scanned, the **X-Forefront-Antispam-Report** header is inserted into the message. The fields in this header can help provide administrators with information about the message and about how it was processed, and it contains entries such as the SCL for the message.

The fields in the **X-Microsoft-Antispam** header provide additional information about bulk mail and phishing. In addition to these two headers, Exchange Online Protection also inserts email authentication results for each message it processes in the **Authentication-results** header.

You can view the headers in a message by using any text editor, such as Notepad or the Message Header Analyzer in the **Microsoft Remote Connectivity Analyzer**[8]. Please note, however, that text editors such as Notepad only show you the raw header, the Message Header Analyzer analyzes the raw data and displays it in a more readable way.

The **X-ForeFront-Antispam-Report** message header can contain the following entries:

8 http://aka.ms/RCA

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Message field	Description
CTRY	The region from which the message connected to the service. This is determined by the connecting IP address, which may not be the same as the originating sending IP address.
LANG	The language in which the message was written as specified by the country code (for example, "ru-RU" for Russian).
SCL	The Spam Confidence Level (SCL) value of the message.
SRV:BULK	The message was identified as a bulk email message. If the **Block all bulk email messages** advanced spam filtering option is enabled, it will be marked as spam. If it is not enabled, it will only be marked as spam if the rest of the filtering rules determine that the message is spam.
SFE	Filtering was skipped, and the message was let through because it originated from a safe sender. For more information about safe senders, see the **Safe Sender and Blocked Sender Lists FAQ** *(https://aka.ms/AA3q791)*.
BLK	Filtering was skipped, and the message was blocked because it originated from a blocked sender.
SPM	The message was marked as spam by the spam filter.
SKS	The message was marked as spam prior to being processed by the content filter. This includes messages where the message matched a Transport rule to automatically mark it as spam and bypass all additional filtering.
NSPM	The message was marked as non-spam and was sent to the intended recipients.

The **X-Microsoft-Antispam** message header contains the following information regarding the Bulk Complaint Level (BCL) and the Phishing Confidence Level (PCL):

- **Bulk Complaint Level (BCL).** The BCL indicates the likeliness of a message being a bulk email. A BCL of 0 indicates the message isn't from a bulk sender, while a BCL of 9 indicates the message is from a bulk sender who has generated numerous complaints.
- **Phishing Confidence Level (PCL).** The PCL indicates the likeliness of a message being a phishing message.
  - 0-3: The message's content isn't likely to be phishing.
  - 4-8: The message's content is likely to be phishing.
  - -9990: (Exchange Online Protection only) The message's content is likely to be phishing.

Note: The remaining fields in the **X-Microsoft-Antispam** message header are used exclusively by the Microsoft anti-spam team for diagnostic purposes.

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Spam Confidence Levels (SCL)

As incoming messages go through spam filtering, they are assigned a spam score in the **X-Fore-Front-Antispam-Report**. This score maps to an SCL, as recorded in the X-header.

SCL Rating	Interpretation	Default Action
-1	Message comes from safe sender, safe recipient, or safe IP address	Deliver to Inbox
0, 1	Message scanned and found to be clean	Deliver to Inbox
5, 6	Spam	Send to Junk Email
9	High confidence spam	Send to Junk Email

Note: Exchange Online Protection does not use SCL levels 2, 3, 4, 7, and 8.

You can use spam filtering policies to specify what happens with high confidence spam; for example, deleting the message and not sending it to junk mail. You can also set SCL conditions in transport rules.

Additional reading. For more information, see the following articles on **Anti-spam message headers**[9] and **Use mail flow rules to set the spam confidence level (SCL) in messages**[10].

Anti-spam and Anti-malware Overview

The anti-spam and anti-malware protection that is included in Exchange Server and Exchange Online provide a set of default rules for protection from malware and spam.

Exchange Server and Exchange Online slightly differ in the message hygiene options that are available. While Exchange Online uses EOP, which is continuously updated with new features and fueled by AI and machine learning, the built-in protection features of Exchange Server are just used for basic tasks. If you need the features of a modern anti-spam and anti-malware protection, you can also use EOP standalone or hybrid with Exchange Server by routing your mail flow through EOP, or you can utilize a third-party solution on your on-premises perimeter network.

Because Exchange Online Protection and ATP are covered in other lessons, this lesson will describe the built-in Exchange Server features for anti-spam and anti-malware protection.

Anti-spam and anti-malware filtering in Exchange Server

The basic protection in Exchange Server is provided by the following two features:

9 https://aka.ms/AA3r4jl
10 https://aka.ms/AA3qwxr

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Feature name	Description	Used at…
Anti-spam protection	Exchange uses transport agents to provide anti-spam protection, and the built-in agents that are available in Exchange Server are relatively unchanged from Exchange Server 2010, but in modern Exchange deployments, configuration and management of these agents is available only in the Exchange Management Shell.	These agents are enabled by default on Edge Transport servers, and you can enable them on Exchange Mailbox servers.
Anti-malware protection	Anti-malware protection in Exchange Server helps combat viruses and spyware in your email messaging environment. Viruses infect other programs and data, and they spread throughout your computer looking for programs to infect. Spyware gathers personal information (for example, sign-in information and personal data) and sends it back to its author. The anti-malware protection in Exchange Server was introduced in Exchange 2013 and is provided by the Transport agent titled Malware Agent. The agent scans messages as they travel through the Transport service on a Mailbox server. Anti-malware protection is configured and managed through anti-malware policies, server settings, and scripts.	The Malware agent is available and enabled by default on Exchange Mailbox servers.

Managing Anti-spam protection

Anti-spam agents are typically enabled on a Mailbox server if your organization doesn't have an Edge Transport server, or if it doesn't do other anti-spam filtering on incoming messages. In these instances, you should consider implementing the following agents on the Mailbox servers and Edge Transport servers:

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Anti-spam agents on Mailbox servers	Anti-spam agents on Edge Transport servers
Sender Filter agent. Sender filtering compares the sending server to a list of senders or sender domains that are prohibited from sending messages to your organization. For more information, see **Sender filtering** *(https://aka.ms/AA3px3y)*.	**Connection Filtering agent.** Connection filtering uses an IP block list, IP allow list, IP block list providers, and IP allow list providers to determine whether a connection should be blocked or allowed. For more information, see **Connection filtering on Edge Transport servers** *(https://aka.ms/AA3px42)*.
Sender ID agent. Sender ID relies on the IP address of the sending server and the Purported Responsible Address (PRA) of the sender to determine whether the sending email address is spoofed. For more information, see **Sender ID** *(https://aka.ms/AA3px43)*.	**Recipient Filter agent.** Recipient filtering uses a recipient block list to identify messages that aren't allowed to enter the organization. The recipient filter also uses the local recipient directory to reject messages sent to invalid recipients. For more information, see **Recipient filtering procedures on Edge Transport servers** *(https://aka.ms/AA3px48)*.
Content Filter agent. The Content filtering agent assigns a spam confidence level (SCL) to each message based on data from both legitimate messages and spam messages. For more information, see **Content filtering** *(https://aka.ms/AA3ppwf)*. Spam quarantine, which is a component of the Content Filter agent, reduces the risk of losing legitimate messages that are incorrectly classified as spam. Spam quarantine provides a temporary storage location for suspicious messages, so an administrator can review the messages. For more information, see **Spam quarantine in Exchange Server** *(https://aka.ms/AA3px4c)*. Content filtering also uses the safelist aggregation feature. Safelist aggregation collects safe list data that users configure in Microsoft, Outlook, and Outlook on the web and makes this information available to the Content Filter agent. For more information, see **Safelist aggregation** *(https://aka.ms/AA3ppw9)*.	**Attachment Filtering agent.** Attachment filtering blocks messages or attachments based on the attachment file name, extension, or MIME content type. For more information, see **Attachment filtering on Edge Transport servers** *(https://aka.ms/AA3ppwb)*.
Protocol Analysis agent (sender reputation). The Protocol Analysis agent is the agent that provides sender reputation. Sender reputation uses several tests to calculate a sender reputation level (SRL) on incoming messages that determines the action to take on those messages. For more information, see **Sender reputation and the Protocol Analysis agent** *(https://aka.ms/AA3px46)*.	

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Strategy for implementing an anti-spam solution

Anti-spam is a balancing act between blocking unwanted messages and allowing legitimate messages. If you configure the anti-spam features too aggressively, you'll likely block too many legitimate messages (false positives). If you configure the anti-spam features too loosely, you likely allow too much spam into your organization.

These are some best practices to consider when configuring the built-in anti-spam features in Exchange:

- Reject messages that are identified by the Connection Filtering agent, Recipient Filter agent, and Sender Filter agent rather than quarantining the messages or applying anti-spam stamps. This approach is recommended for the following reasons:
  - Messages that are identified by the default settings of the connection filtering, recipient filtering, or sender filtering typically don't require further tests to determine if they're unwanted. For example, if you configured sender filtering to block specific senders, there's no reason to continue to process messages from those senders. If you didn't want the messages rejected, you wouldn't have put the senders on the blocked senders list in the first place.
  - Configuring a more aggressive level for the anti-spam agents that encounter messages early in the transport pipeline saves processing, bandwidth, and disk resources. The further a message travels along the transport pipeline, the greater number of variables that the remaining anti-spam features must evaluate to successfully identify the message as spam. Therefore, the basic rule of thumb is to reject obvious messages early so that you can process ambiguous messages later.
- You need to monitor the effectiveness of the anti-spam features at their current configuration levels. Monitoring allows you to react to trends and increase or decrease the aggressiveness of the settings. You should start with the default settings to minimize the number of false positives. As you monitor the amount of spam and false positives, you can increase the aggressiveness of the settings based on the type of spam and spam attacks that your organization experiences.

Managing Anti-malware protection

Anti-malware policies are configured in the EAC, and the Malware agent scans messages as they travel through the Transport service on a Mailbox server. Malware filtering is implemented by configuring the following components:

- **Anti-malware policies.** These policies specify inbound and outbound scanning and notification options for malware filtering. There's a default policy that applies to all recipients in the Exchange organization, and you can create additional policies that are applied in a specific order.
- **Anti-malware server settings.** These settings specify the error and retry actions, and the engine and definition update settings for malware filtering. The Malware agent uses Internet access on TCP port 80 (HTTP) to check for engine and definition updates every hour.
- **Anti-malware scripts.** These scripts enable or disable malware filtering on the server, and they manually download engine and definition updates.

Each of these anti-malware components are described in greater detail below.

Anti-malware policies

Anti-malware policies control the actions and notification options for malware detections. The important settings in anti-malware policies are outlined in the following table.

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Setting	Description	Options
Action	Specifies what action to perform when a message is found that contains malware.	The options are: • Delete the message (this is the default value) • Replace all attachments with a text file that contains this default text • Replace all attachments with a text file that contains the custom text you specify
Notifications	When an anti-malware policy is configured to delete messages, you can choose whether to send a notification message to the sender.	You can send notification messages based on whether the sender is internal or external. The default notification message has the following properties: • **From** : Postmaster postmaster@[defaultdomain].com • **Subject** : Undeliverable message • **Message text** : This message was created automatically by mail delivery software. Your email message was not delivered to the intended recipients because malware was detected. You can customize the message properties for internal and external notifications. You can also specify additional recipients (administrators) to receive notifications for undeliverable messages from internal or external senders.

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Setting	Description	Options
Recipient filters	For custom anti-malware policies, you can specify recipient conditions and exceptions that determine who the policy applies to.	You can use the following properties for conditions and exceptions: • By recipient • By accepted domain • By group member-ship **Note**: You can only use a condition or exception once, but the condition or exception can contain multiple values. Multiple values of the same condition or exception use OR logic (for example, [recipient1] or [recipi-ent2]). Different conditions or exceptions use AND logic (for example, [recipient1] and [member of group]).
Priority	If you create multiple custom anti-malware policies, you can specify the order in which they should be applied.	

The basic elements of an anti-malware policy include:

- **The malware filter policy.** Specifies the action and notification options for malware filtering.
- **The malware filter rule.** Specifies the priority and recipient filters (who the policy applies to) for a malware filter policy.

Every Mailbox server has a built-in anti-malware policy named Default that has the following properties:

- The malware filter policy named Default is applied to all recipients in the Exchange organization, even though there's no malware filter rule (recipient filters) associated with the policy.
- The policy named Default has the custom priority value of Lowest, which you can't modify (the policy is always applied last). Any custom anti-malware policies that you create always have a higher priority than the policy named Default.
- The policy named Default is the default policy (the IsDefault property has the value True), and you can't delete the default policy.

You also need to decide, if you are configuring anti-malware policies with the EAC or with the Exchange Management Shell:

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When using the EAC	When using the Exchange Management Shell
When you create an anti-malware policy in the EAC, you're actually creating a malware filter rule and the associated malware filter policy at the same time using the same name for both. When you modify an anti-malware policy in the EAC, settings related to the name, priority, enabled or disabled, and recipient filters modify the malware filter rule. Other settings (actions and notification options) modify the associated malware filter policy. When you remove an anti-malware policy from the EAC, the malware filter rule and the associated malware filter policy are removed.	In the Exchange Management Shell, you create the malware filter policy first, then you create the malware filter rule that identifies the policy that the rule applies to. In the Exchange Management Shell, you modify the settings in the malware filter policy and the malware filter rule separately. When you remove a malware filter policy from the Exchange Management Shell, the corresponding malware filter rule isn't automatically removed, and vice versa. You manage malware filter policies by using the **-MalwareFilterPolicy** cmdlets, and you manage malware filter rules by using the **-MalwareFilter-Rule** cmdlets.

Anti-malware server settings

You can use the **Get-MalwareFilteringServer** and **Set-MalwareFilteringServer** cmdlets in the Exchange Management Shell to view and configure the update, timeout, and download settings for the Malware agent on the Mailbox server.

Anti-malware scripts

Exchange includes three Exchange Management Shell scripts that you can use to manage malware filtering:

- **Disable-Anti-malwarescanning.ps1**. Disables the Malware agent, malware engine, and definition updates on the Mailbox server.
- **Enable-Anti-malwarescanning.ps1**. Enables the Malware agent, malware engine, and definition updates, and runs engine and definition updates on the Mailbox server.
- **Update-MalwareFilteringServer.ps1**. Manually runs malware engine and definition updates on the Mailbox server.

Using outbound spam settings, quarantine and best practices

In addition to the anti-spam and anti-malware filters in Exchange, you can also configure other features, such as how to handle outbound spam, and how to quarantine and release messages or unblocked users in the Security & Compliance Center.

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Outbound Spam

Outbound spam filtering is a feature found only in Exchange Online. It's always enabled if you use the Exchange Online service for sending outbound email, thereby protecting organizations using the service and their intended recipients. Like inbound filtering, outbound spam filtering is comprised of connection filtering and content filtering; however, the outbound filter settings are not configurable. If an outbound message is determined to be spam, it is routed through the higher risk delivery pool, which reduces the probability of the normal outbound-IP pool being added to a block list. If a customer continues to send outbound spam through the service, they will be blocked from sending messages. Although outbound spam filtering cannot be disabled or changed, you can configure several company-wide outbound spam settings through the default outbound spam policy.

This outbound spam filter has the following options that control what the system does when it identifies a message containing outbound spam:

Outbound Spam Preference	Description
Send a copy of all suspicious outbound email messages to the following email address or addresses	This option enables you to specify the email address or addresses of administrators who will receive copies of all suspicious outbound messages. If you specify multiple addresses, separate each address with a semicolon.
Send a notification to the following email address or addresses when a sender is blocked for sending outbound spam.	This option enables you to specify the email address or addresses of administrators who will be notified when outbound messages identified as spam are blocked. Again, use a semicolon to separate multiple addresses.

Note: Outbound spam filtering is not available in Exchange Server deployments without utilizing Exchange Online Protection (EOP).

Quarantine

If you set your content policy to direct spam messages into quarantine and then your organization receives a message that is classified as spam, that message will end up in the quarantine area. Messages from transport rule matches can also end up in quarantine.

The quarantine feature is limited in Exchange Server deployments. While in Exchange Server, you can define a single mailbox as the quarantine mailbox, which is only accessible by messaging administrators through the Microsoft Outlook client.

In contrast, messages that are quarantined in Exchange Online can be released by messaging administrators, as well as by users who received a spam message that has been quarantined. The spam quarantine can be accessed through the Quarantine page in the Security & Compliance Center.

Note: Messages that are quarantined due to a transport rule match are not included in end-user spam quarantined messages; only spam-quarantined messages are listed. A messaging administrator as well as an end user can then inspect the messages before choosing to release them on to their original addressees or mark them as false positives and release to the addressees. Alternatively, you can simply leave the messages in quarantine until they expire.

The messaging administrator or the user can then perform one of the following tasks on the message:

- Analyze the message by viewing the message header or previewing the message
- Release the message

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With either option, you can release the message to all recipients or just to selected recipients. However, if you release the message to one recipient and then subsequently release it to all recipients, the first recipient will not receive the message a second time.

To assist with managing quarantined messages, there is also an advanced search function that enables you to filter messages based on the following criteria:

- Message ID
- Sender email address
- Recipient email address
- Subject
- Received (by day)
- Expires (by day)
- Type (Spam, Transport rule, Bulk or Phish)

Note: False positive reports will not be processed if the message was quarantined because of an advanced spam filter option, or if it was quarantined due to a transport rule match.

Unblock users from the SCC

If a user continuously sends email messages from Exchange Online that is classified as spam, the user will be blocked from sending any more messages. The user will also be listed in the service as a bad outbound sender, and he or she will receive a Non-Delivery Report (NDR) that states:

- Your message couldn't be delivered because you weren't recognized as a valid sender. The most common reason for this is that your email address is suspected of sending spam, and it's no longer allowed to send messages outside of your organization. Contact your email admin for assistance. Remote Server returned '550 5.1.8 Access denied, bad outbound sender'

Messaging administrators can configure their outbound spam policy settings so that they receive a notification when an Exchange Online user is blocked from sending email. After the problem with the user's mailbox is resolved, the messaging admin can remove the block on that sender through the Security & Compliance Center.

Note: There's a limit to the number of times that an account can be unblocked by the messaging admin. If the limit for a user has been exceeded, an error message appears, and the messaging admin will then need to contact Microsoft Support to unblock the user. It may take up to 1 hour before the user is unblocked.

Best Practices for Deploying an Anti-Spam Solution

Anti-spam protection requires ongoing monitoring of the anti-spam solution reports. Messaging administrators must evaluate anti-spam settings and adjust the configuration according to current Internet spam threats and the users' feedback. For example, an organization's users might complain that they receive more than five spam messages per day, which indicates that anti-spam configuration should be enhanced with additional settings.

When configuring anti-spam settings, consider the following best practices:

- **Update anti-spam definitions**. Anti-spam software uses definitions to scan email for content that is likely to be spam. However, spam senders are continuously trying to use new techniques to hide the spam content to avoid anti-spam softer filters. Therefore, anti-spam software vendors must remain

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diligent in updating their anti-spam definitions. Consequently, organizations should regularly update their anti-spam definitions to stay abreast of the latest changes from their anti-spam vendors.

- **Monitor anti-spam reports**. Messaging administrators should regularly monitor anti-spam software reports to evaluate the total number of messages received from Internet, the number of blocked messages due to spam, and the number of quarantined messages.
- **Regularly read about latest Internet security and spam threats**. Messaging administrators and security administrators should regularly update their knowledge about the latest security, spam, and malware threats. Anti-spam settings should be reconfigured according to latest best practices and recommendations.
- **Regularly evaluate end users' feedback**. User feedback related to the number of spam messages received per day or per week and the number of spam messages quarantined per day or per week is critical when you evaluate the effectiveness of your anti-spam solution. Messaging administrators should regularly evaluate end users' feedback on their everyday experience, and then reconfigure their solution, if necessary, to provide better protection. For example, users might complain about the excessive number of spam messages received each day. Conversely, users might mention that they do not receive email from business partners; this would indicate that anti-spam software should be reconfigured with less aggressive protection settings.
- **Use multi-layered anti-spam protection**. It is recommended that spam be stopped before it enters the internal network. One way that an organization can address this is by deploying the Edge Transport role in their perimeter network, outside of the organization's internal Active Directory forest. Another way is by deploying hybrid anti-spam protection; in other words, by using both cloud-based Exchange Online Protection and Exchange on premises anti-spam features. A third option would be to deploy a SMTP gateway with anti-spam functionality that is in the perimeter network, in addition to the anti-spam features in an Exchange Server on-premise deployment only.

Managing Spoofing protection

Spoofing is the term used for the creation of email messages with a forged sender address. The idea is to convince the recipient that the email was sent by somebody else. The following technologies, all supported by Office 365, ensure that your domain cannot be spoofed by others:

- Sender Policy Framework (SPF)
- DomainKeys Identified Mail (DKIM), and
- Domain-based Message Authentication, Reporting, and Conformance (DMARC)

Configuring Sender Policy Framework

Configuration for SPF is identical for Exchange Server deployments on-premises, Exchange hybrid deployments, and Exchange Online only.

You configure SPF using a DNS TXT record for your public DNS domain, so the target mail servers can retrieve your SPF configuration information by doing just a normal public DNS lookup. Because SPF consists of simple TXT records in DNS, it is mandatory to use the following syntax:

```
v=spf1 <IP> <enforcement rule>
```

The following table explains the different parts of an SPF record:

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Part of SPF	Examples	Description
Declaration	v=spf1	This defines the TXT record as an SPF TXT record.
Trusted hosts and networks	ip4:192.168.0.1/32 or ip-v6:ip6:1080::8:800:200C:417A/96 mx a	Lists all trusted hosts, trusted network mail sources, and other DNS records with values that shall be trustworthy.
Include	include:spf.protection.outlook.com	Include additional SPF records from other domains.
Enforcement rule	-all	**-all** : Indicates hard fail. If you know all authorized IP addresses for a domain. Also recommended when using SPF only, without DMARC or DKIM. **~all** : Indicates soft fail. Used when not all possible sending hosts are known yet. Also, if you also using DMARC with p=quarantine or p=reject, then ~all is required. **?all** : Indicates neutral. This is used when testing SPF. Not recommend for live deployment.

For example, the default SPF TXT record for Exchange Online or Exchange Online Protection is:

```
v=spf1 include:spf.protection.outlook.com -all
```

Once you have configured the SPF DNS records in your environment, you are raising your domain reputation significantly, and the mail addresses of your domain are much harder to spoof for malicious and phishing email senders.

Nevertheless, you should be extremely careful when configuring SPF records in your live environment if you have complex mail flow scenarios and many different sending hosts for application emails, automatic forwarding, and so on. For example, if a server in your environment is sending mail with sender addresses from your environment and this server is not included in your SPF record, mail from this host will be flagged as a phishing attack with a spoofed sender address.

Additional reading. For more information, see the following article on **How Office 365 uses Sender Policy Framework (SPF) to prevent spoofing**[11].

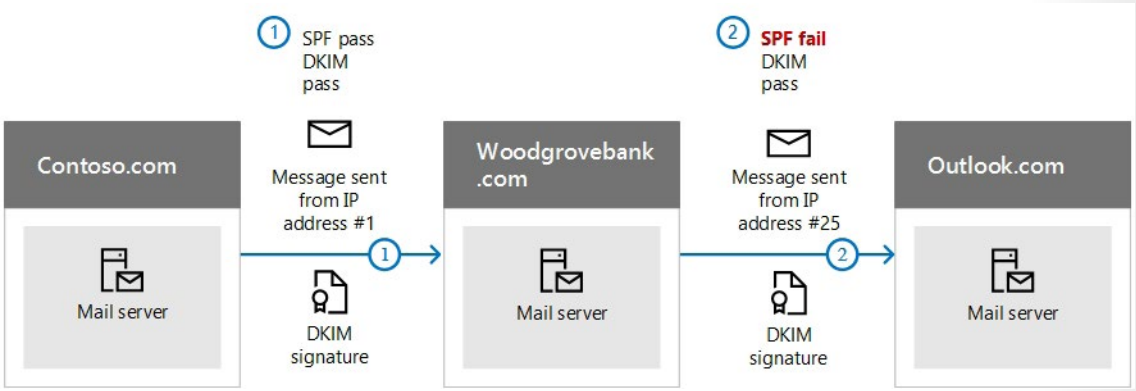
11 https://aka.ms/AA3r76n

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Configuring DomainKeys Identified Mail (DKIM)

Configuring DKIM is currently not possible with the native tools in Exchange Server; instead, it's only available when using Exchange Online or a hybrid deployment.

DKIM adds an encrypted entry to the message header called the DKIM-Signature entry. When you forward a message, portions of that message's envelope can be stripped away by the forwarding server. Since the digital signature stays with the email message because it's part of the email header, DKIM works even when a message has been forwarded as shown in the following example.



In this example, if you had only published an SPF TXT record for your domain, the recipient's mail server could have marked your email as spam and generated a false positive result. The addition of DKIM in this scenario reduces false positive spam reporting. Because DKIM relies on public key cryptography to authenticate rather than just IP addresses, DKIM is considered a much stronger form of authentication than SPF. It is recommended that organizations use both SPF and DKIM, as well as DMARC in their deployments.

When you configure DKIM, you authorize your domain to associate, or sign, its name to an email message by using cryptographic authentication. Email systems that receive email from your domain can use this digital signature to help determine whether incoming email is legitimate.

Basically, you use a private key to encrypt the header in your domain's outgoing email. You publish a public key to your domain's DNS records that receiving servers use to decode the signature. They use the public key to verify that the messages are coming from you and not from someone spoofing your domain.

DKIM is enabled by default for Office 365 domains that are named <;tenant>;.onmicrosoft.com. You only need to enable DKIM to any additional domains that you have added to your Office 365 tenant.

To enable signing DKIM for a domain in Office 365, you need to create two CNAME records.

Host name	Record type	Value	TTL
selector1._domain-key.<domain>	CNAME	selector1-<domain-GUID>._domain-key.<initialDomain>	3600
selector2._domain-key.<domain>	CNAME	selector2-<domain-GUID>._domain-key.<initialDomain>	3600

In this table, the <domainGUID> and <initialDomain> parameters are defined as follows:

- **domainGUID.** Can be taken from DNS MX record that points to Office 365 (e.g., **contoso-com**.mail.protection.outlook.com)

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- **initialDomain**. Your default Office 365 domain that is named <tenant>.onmicrosoft.com (for example, contoso.onmicrosoft.com)

For example, the following two DNS records are required to enable DKIM for the domain **contoso.com**:

- First public DNS domain record:
    - **Host name** : selector1._domainkey.contoso.com
    - **Record type** : CNAME
    - **Value** : selector1-contoso-com._domainkey.contoso.onmicrosoft.com.
    - **TTL** : 3600
- Second public DNS domain record:
    - **Host name** : selector2._domainkey.contoso.com
    - **Record type** : CNAME
    - **Value** : selector2-contoso-com._domainkey.contoso.onmicrosoft.com.
    - **TTL** : 3600

You would then perform the following steps to enable DKIM for the domain:

1. Open the **Exchange Admin Center** and go to **Protection** > **dkim**.
2. Select the domain for which you want to enable DKIM and then, for **Sign messages for this domain with DKIM signatures** , choose **Enable**.

Additional reading. For more information, see the following article on **Use DKIM to validate outbound email sent from your custom domain in Office 365**[12].

Configuring Domain-based Message Authentication, Reporting, and Conformance (DMARC)

DMARC works like SPF with public DNS records. it is intended to give recommendations to the recipient about how to utilize SPF and DKIM for the recipient's domain. It is therefore available for all Exchange implementations, but to use it, your MX record must point to Exchange Online because on-premises Exchange Servers cannot currently use DMARC records.

Like the DNS records for SPF, the record for DMARC is a DNS text (TXT) record that helps prevent spoofing and phishing. DMARC TXT records validate the origin of email messages by verifying the IP address of an email's author against the alleged owner of the sending domain. The DMARC TXT record identifies authorized outbound email servers. Destination email systems can then verify whether messages they receive originate from authorized outbound email servers.

To configure DMARC for your SMTP mail domains, you must have a DNS TXT record configured by your Internet DNS provider for each domain. Each DNS TXT record should be configured per the following format:

```
\_dmarc.domain TTL IN TXT "v=DMARC1; pct=100; p=policy; ruf=failure\_re-
port; rua=aggregate\_report; fo=failure\_option"
```

The following table describes the syntax used to format a DNS TXT record for DMARC:

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Syntax	Example	Description
domain	Contoso.com	This is the domain you want to protect. By default, the record protects mail from the domain and all subdomains. For example, if you specify _dmarc.contoso.com, then DMARC protects mail from the domain and all subdomains, such as marketing.contoso.com and itdepartment.contoso.com.
pct=100	pct=100	Indicates how many messages, as a percentage, that must be checked: 100 means every email must be checked.
policy	p=none	Specifies what policy you want the receiving server to follow if DMARC fails. You can set the policy to **None** , **Quarantine** , or **Reject** : • **None**. Monitor only the impact of implementing DMARC. • **Quarantine**. Request that external mail systems quarantine mail that fails DMARC. • **Reject**. Request that external mail systems not accept messages that fail DMARC. Remember, you won't be able to effectively quarantine or reject mail by using DMARC unless you also implement SPF and DKIM.
Failure report(optional)	ruf=mailto:fail@contoso.com	When specified, a message is sent every time an SPF, DKIM, or DMARC check fails. You should only use this option when investigating issues.
Aggregate report(recommended)	rua=mailto:agg@contoso.com	When specified, a message including an overview report is sent once a day.

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Syntax	Example	Description
Failure Option(optional)	fo=0	Specifies which failure options will send a DMARC message: • **0** - Generate report if all authentication mechanisms fail. • **1** - Generate report if any authentication mechanism fails. Be aware that you should enable this option only to for investigating issues, as it will generate a large number of messages.

For example, the following DNS record configured DMARC for the domain contoso.com and all subdomains just to monitor the impact of DMARC by using a "none" policy, and send an aggregate report to agg@contoso.com:

```
\_dmarc.contoso.com 3600 IN TXT "v=DMARC1; pct=100; p=none; rua=mailto:agg@
contoso.com; fo=0"
```

As a best practice when implementing DMARC, you should start and configure DMARC policy as none, measure the impact, and then consider moving to the more restrictive policies such as quarantine or reject.

Additional reading. For more information, see the following article on **Use DMARC to validate email in Office 365**[13].

Managing Phishing protection

Phishing attacks attempt to steal sensitive information through emails, websites, text messages, or other forms of electronic communication that often look to be official communication from legitimate companies or individuals. Phishing and email spoofing work together, where spoofing is the term used for the creation of email messages with a forged sender address. The idea is to convince the recipient that the email was sent by somebody else.

Anti-phishing policies provide additional protection against impersonation and phishing attacks, above and beyond the anti-spoofing solutions covered in the previous topic. Anti-phishing protection was originally available only for Advanced Threat Protection, but it's now available for all EOP users.

Anti-phishing policies are managed in the Security & Compliance Center. The default policy applies to all users within the organization and is a single view where you can fine-tune anti-phishing protection. Custom policies can be created and configured for specific users, groups, or domains within the organization and will take precedence over the default policy for the scoped users.

As you set up or edit your anti-phishing policies, you can choose from several options, as described in the following table.

13 https://aka.ms/AA3qzna

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This setting	Does this	Use when you want to:
Applied to	Defines the recipients whose incoming email messages will be subject to the rules of the policy. You can create conditions and exceptions for the recipients associated with the policy. For example, you can create a global policy for your organization by applying the rule to all recipients in your domain. You can also create exception rules, such as a rule that does not scan email messages for a specific group of recipients.	Each policy must be associated with a set of users; for example, users in a group or domain.

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This setting	Does this	Use when you want to:
Choose actions	Choose the action to take when Microsoft 365 detects an intra-org or external-org spoofing attempt against your users. These actions apply to any incoming email that has been identified by Microsoft 365 as a spoofing attempt for users who are under the protection of this anti-phishing policy. The actions include: • **Quarantine message.** Email will be sent to Microsoft 365 quarantine. When you choose this option, the email is not sent to the original recipient. • **Move message to the recipients' Junk email folder.** Email will be sent to the recipients' Junk email folder. When you choose this option, the email is still sent to the original recipient, but it's not placed in the recipient's inbox. • **Don't apply any action.** Email will be delivered to the original recipient's inbox. No other action will be taken on the email message.	When you want to take an action on messages that Microsoft 365 has determined to be a spoofing attempt of internal or external domains as defined in the policy.

Configuring transport rules

Transport rules are another mechanism that can be used to improve defenses against spam. They can also be used to carry out additional functions, such as:

- Prevent inappropriate content from entering or leaving the organization.
- Filter confidential organization information.

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- Track or archive copied messages that are sent to or received from specific individuals.
- Redirect inbound and outbound messages for inspection prior to delivery.
- Append disclaimers to messages as they pass through the organization.

Each transport rule consists of:

- Message characteristics to which you can apply a transport rule action.
- Further conditions that can exclude messages from the transport rule action.
- Actions which apply to messages that match the conditions and exceptions.

This topic specifically concentrates on two applications of transport rules: Bypass spam filtering and Add disclaimers.

Bypass Spam Filtering configuration

Bypass spam filtering enables you to allow single senders or whole sender domains and ensure that messages from those senders are always accepted without any spam filtering. However, you should be aware that SMTP address spoofing can lead to messages being accepted that do not originate from the specified domain.

Note: Before bypassing spam filtering in productive systems, you should evaluate the reasons for bypassing a security system. Sometimes the business requirements force you to apply a configuration like this, but you should always try to avoid bypassing a security system if possible.

To bypass spam filtering and the allow list for a specific domain, you should perform the following steps:

1. Log in to the Exchange Admin Center and select **Mail flow**.
2. In **Mail flow**, under **Rules**, select the + sign and select **Bypass spam filtering**.
3. In the **Name** field, enter a descriptive name, such as "Allow list all partner domains".
4. Under **Apply this rule if**, select the drop down arrow in the field, hover your mouse over **The sender**, then select **Domain is**.
5. In the **Specify domain** box, enter the domains that you want to add to the Allow list and select the **+** sign.
6. Select **OK** when you have finished adding domains.
7. Under **Do the following**, the action should already be set to **Set the spam confidence level (SCL) to** and the value set to **Bypass spam filtering**.
8. Under **Except if**, select **add exception** and include any of the options for excluding people within those domains.
9. Select the option for **Audit this rule with severity level** and select an auditing level if you want to have the bypass spam filtering action audited.
10. Under **Choose a mode for this rule**, ensure that **Enforce** is selected, unless you only want to test the effect, in which case select **Test without Policy Tips**.
11. Under **Match sender address in message**, ensure that **Header** is selected and then select **Save**.

All the domains that you specified are now allow listed and messages sent from those domains will go directly to the recipients without having spam filtering applied.

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Add Disclaimers configuration

Disclaimers provide a mechanism for organizations to attach standard text to the bottom of an email, such as explaining what to do if the recipient is sent the message in error or disclaiming responsibility for any actions the recipient may carry out because of the contents of the email. Typically, legal disclaimers try to address five main areas:

- **Breach of confidentiality, whether accidental or deliberate**. For example, do not forward this message to people outside the company.
- **Transmission of viruses**. For example, we cannot guarantee this message is virus-free.
- **Entering contracts**. For example, this email does not form a contract.
- **Negligent misstatement**. For example, this email does not consist of advice and should not be used as such.
- **Employer's liability**. For example, do not send offensive emails and company email systems are subject to monitoring.

The main problem with disclaimers is that courts may take a very different view of the legal position of the content of the message as compared to the information set out in the disclaimer. However, they can indicate that the sender's organization has attempted to prevent, for example, a virus outbreak at a client site by warning the recipient to have the incoming email scanned.

Disclaimers can also be added for marketing reasons, although these are more like footnotes or email signatures.

To use the Exchange Admin Center to set up a disclaimer that bypasses spam filtering, perform the following steps:

1. Login to the Exchange Admin Center and select **Mail flow**.
2. In **Mail flow**, under **Rules**, select the + sign and select **Apply disclaimers**.
3. In the **Name** field, enter a descriptive name, such as "Corporate Disclaimer".
4. Under **Apply this rule if**, select the drop down arrow in the field and then select **The recipient is located**.
5. In the **select recipient location** box, select **Outside the organization** and select **OK**.
6. Under **Do the following**, the action should already be set to **Append the disclaimer**.
7. Select **Enter text**, and in the **Specify disclaimer text** box, enter the text of the disclaimer, then select **OK**.
   Note: In the **Specify disclaimer text** box, you can insert plain text, as well as HTML and inline CSS, including user based dynamic information from Active Directory.
8. To the right of **Fall back to action**, select **Select one**, then in the **Specify fallback action** box, select **Wrap**, **Ignore**, or **Reject**, and then select **OK**.
9. Under **Except if**, select **Add exception** and include any of the options for excluding senders, recipients, or messages.
10. Select the option for **Audit this rule with severity level** and select an auditing level if you want to have the bypass spam filtering action audited.
11. Under **Choose a mode for this rule**, ensure that **Enforce** is selected, unless you only want to test the effect, in which case select **Test without Policy Tips**.
12. Select **More options** to specify additional information, such as times when this rule will be activated or deactivated, and then select **Save**.

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Additional reading. For more information, see the following article on **Organization-wide disclaimers, signatures, footers, or headers in Exchange Server**[14] and **Manage mail flow rules**[15].

14 https://aka.ms/AA3rbb8
15 https://aka.ms/AA3rbb9

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Lab3-Ex1-Create Hygiene Filters

Lab3-Ex1-Create Hygiene Filters

In this lab, you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. Adatum has experienced a recent rash of malware infections. The company's CTO has asked Holly to investigate the various options that are available in Exchange Online to fortify Adatum's messaging environment.

You will access the Exchange admin center for Exchange Online from your client computer and create a series of hygiene filters that are designed to protect Adatum's messaging environment. You will create a malware filter, a connection filter, and a spam filter.

See your VM lab environment for the lab instructions.

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Managing Advanced Threat Protection

Lesson Introduction

The previous lessons examined how Exchange Online Protection (EOP) uses a multi-layered approach to protect users from a variety of threats including phishing, spoofing, spam, bulk email, and malware. Advanced Threat Protection (ATP) extends the protection provided by EOP by filtering targeted attacks that could pass through EOP's line of defenses, including advanced threats such as zero-day attacks in email attachments and Office documents and time-of-click protection against malicious URLs. When integrated together, EOP and ATP represent the anti-malware pipeline in Microsoft 365 that provides the most efficient level of protection against commodity and advanced targeted threats.

Microsoft 365 ATP protects users from advanced threats through features such as safe attachments and safe links. ATP also generates reports which provide administrators with insight into attacks targeting their tenants through email.

After this lesson, you will be able to:

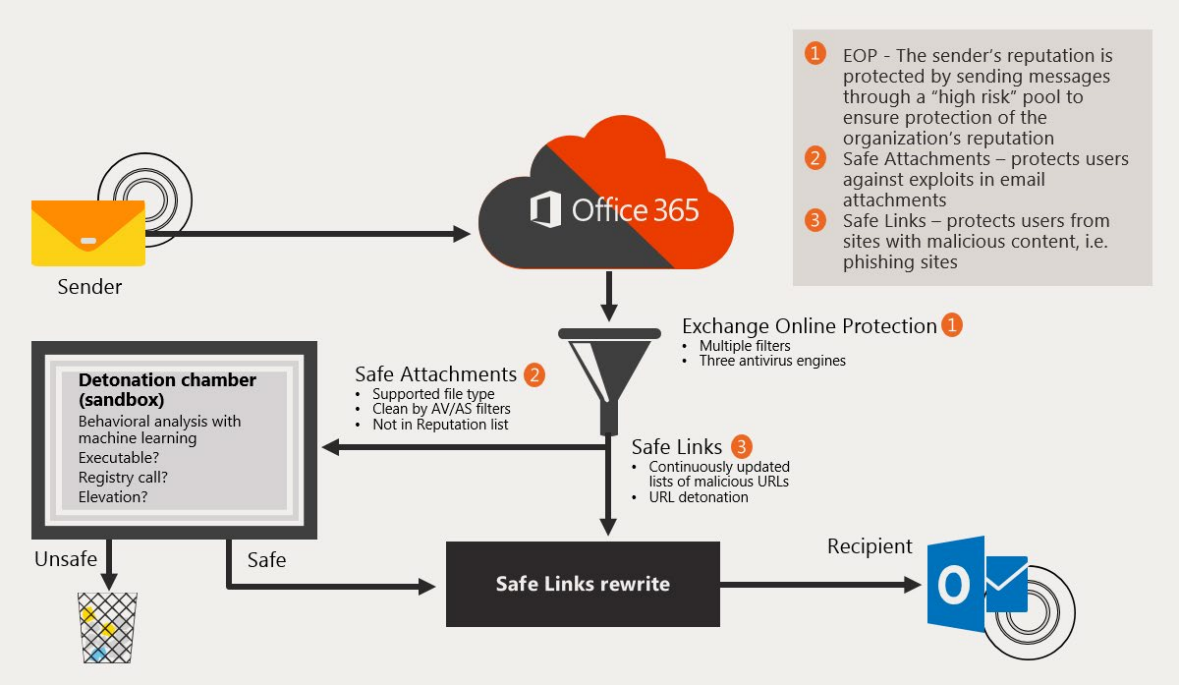
- Describe the features of Advanced Threat Protection
- Describe the protection provided by Safe Attachment and Safe Links policies
- Understand the spoof intelligence features
- Know how ATP anti-phishing policies work

Advanced Threat Protection Overview

Microsoft 365 Advanced Threat Protection (ATP) complements Exchange Online Protection (EOP) by providing protection against specific types of advanced threats, including phishing and zero-day attacks. ATP employs the following tools to provide a wide range of threat protection:

- **ATP Safe Attachments.** ATP uses this tool to scan email attachments for malware. Safe Attachments protects against unknown malware and viruses and provides zero-day protection to safeguard your messaging system. All messages and attachments that don't have a known virus/malware signature are routed to a special environment where ATP uses a variety of machine learning and analysis techniques to detect malicious intent. If no suspicious activity is detected, the message is released for delivery to the mailbox.
- **ATP Safe Links.** ATP uses this tool to scan web addresses (URLs) in email messages and Office documents. Safe Links proactively protects your users from malicious hyperlinks in a message. The protection is enforced every time they click the link. Malicious links are dynamically blocked while good links are accessible.
- **ATP for SharePoint, OneDrive, and Microsoft Teams.** ATP uses these tools to identify and block malicious files in online libraries.
- **ATP anti-phishing capabilities in Microsoft 365.** ATP uses this functionality to detect when someone attempts to impersonate your users and your organization's custom domains. The ATP anti-phishing capabilities apply a set of machine learning models and impersonation detection algorithms to incoming messages to provide protection for commodity and spear phishing attacks.
- **Spoof intelligence.** ATP uses this functionality to check email messages for spam with unauthorized spoofing.

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Microsoft 365 Advanced Threat Protection (ATP) is included in Office 365 E5 and available as a standalone offering.

Safe Attachments

People regularly send, receive, and share attachments, such as documents, presentations, spreadsheets, and more. It's not always easy to tell whether an attachment is safe or malicious just by looking at an email message. In fact, the rising number of ransomware and trojan attacks nowadays proves it's even hard for professionals to spot potentially harmful attachments.

Safe Attachments is a feature in Microsoft 365 Advanced Threat Protection that opens every attachment of a supported file type in a special hypervisor environment, checks to see if the attachment is malicious, and then takes appropriate action to protect your organization. It is designed to protect organizations by detecting malicious attachments even before anti-virus signatures are made available. That makes it an ideal solution to protect against unknown attack vectors and to react fast on new threats, such as modified harmful PDF documents or Word files to perform ransomware attacks. Safe Attachments analyzes attachments that are common targets for malicious content, such as different versions of Office files such as Word, PowerPoint, and Excel, PDFs, executable file types, and Flash files.

Safe Attachment Policies

To implement Safe Attachments, you must first define ATP Safe Attachments policies, which control how Safe Attachments protection is employed.

- **Selecting attachments to test**. Safe Attachments works on email attachments that are received from senders in your organization and external senders who are outside of the organization. The feature protects your organization according to policies that are set by your Microsoft 365 Enterprise or security administrators. When an ATP Safe Attachments policy is in place and someone covered by that policy views their email in Microsoft 365, their email attachments are checked, and appropriate actions are taken based on the organization's ATP Safe Attachments policies.

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- **Attachment testing**. Attachments are tested in virtual environments that run different versions of the Windows operating system and applications. The attachments are executed, or detonated, and undergo a behavioral analysis to determine if the file executes malicious behavior such as installing a Trojan horse or a virus that makes changes to the registry or system settings that result in the system and/or network being more vulnerable to attack.

Depending on how your policies are defined, people can continue working without ever knowing they were sent malicious files.

For example, suppose that George receives an email message that has an attachment. It's not obvious to George whether that attachment is safe or contains malware designed to steal his user credentials. In George's organization, a security administrator defined an ATP safe attachments policy a few days ago. Now, with the ATP Safe Attachments feature in place, the email attachment is opened and tested in a Microsoft 365 virtual environment before George receives it. If the attachment is safe, it will open as expected when George clicks on it. If the attachment is determined to be malicious, it will be removed and can be researched by the organization's security administrators and analysts.

Dynamic Delivery

When ATP Safe Attachments policies are set up for your organization, there are several options for how email attachments are handled. These include **Block**, **Replace**, and **Dynamic Delivery**. Depending on how ATP Safe Attachments policies are configured, email recipients can experience a minor delay in email delivery while their attachments are scanned. To avoid message delays, you can choose Dynamic Delivery.

Dynamic Delivery eliminates email delays by sending the body of an email message through to the recipient with a placeholder for each email attachment. The placeholder remains until a copy of the attachment is scanned and determined to be safe by ATP Safe Attachments. Most PDFs and Office documents can be previewed in safe mode while ATP scanning is underway. If an attachment is not compatible with the Dynamic Delivery previewer, email recipients see an attachment placeholder until ATP Safe Attachments scanning is complete.

- As each attachment is cleared, it is available to be opened or downloaded.
- If an attachment is determined to be malicious, it is sent to quarantine, where someone on your organization's security team (such as a Microsoft 365 Global Administrator or Security Administrator) can manage quarantined messages in Microsoft 365.

With Dynamic Delivery, email recipients can read and respond to their email messages right away, knowing that their attachments are being analyzed.

There are certain scenarios in which Dynamic Delivery is not supported, including:

- Email messages that are in public folders
- Email messages that are routed out of and then back into the user's mailbox using custom rules
- Messages that are moved (automatically or manually) out of the hosted mailbox and into other locations, including archive folders
- Messages that are deleted
- A user's mailbox search folder that is in an error state
- Messages encrypted with Secure/Multipurpose Internet Mail Extensions (S/MIME)

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Safe Links

Safe Links is a feature in Microsoft 365 Advanced Threat Protection (ATP) that protects users from malicious URLs that are commonly used in phishing attacks to extract sensitive information from a user, deliver malicious code to end-user machines, or trick users into actions that cause financial damages.

The ATP Safe Links feature works by replacing embedded URLs from incoming messages and testing the URLs before a user can use them. If the URLs are not harmful, users can click the URLs in messages and get redirected to their target without any interruption. If a URL is potentially dangerous or malicious, a warning message is displayed, and the user is not redirected automatically, but can optionally choose to visit the URL anyway or to leave it unvisited.

Considerations for implementing ATP Safe Links

It's important that you perform the following prerequisite tasks before deploying ATP Safe Links in your environment:

- Verify that your organization has Microsoft 365 Advanced Threat Protection licensed.
- Verify that you have the necessary permissions to define or edit ATP policies. This requires you to be a member of the Company administrators or Security admins role group.
- Review your policies to ensure they're defined in the proper sequence. Policies are enforced in the order they're listed in the Security & Compliance Center.
- Verify that Office clients are configured to use Modern Authentication (this is for ATP Safe Links protection in Office documents).
- Allow up to 30 minutes for your new or updated policies to be propagated to all Microsoft 365 datacenters.
- Verify that a default policy is in place that that applies to everyone, while additional policies can be defined for specific recipients.

When deploying ATP Safe Links, you should begin by checking the default policy and then configure additional policies according to your organization's needs. For example, your default policy can block users from overriding blocked URLs, while additional policies can block specific URLs automatically and skip rewriting for specific URLs.

Default policy options

Default policy options apply to everyone in your organization.

Default policy option	Effect
Block the following URLs	Enables your organization to have a custom list of URLs that are automatically blocked. When users click a URL in this list, they'll be taken to a warning page that explains why the URL is blocked.

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Default policy option	Effect
Microsoft 365 Apps for enterprise (formerly Office 365 ProPlus), Office for iOS and Android	When this option is selected, ATP Safe Links protection is applied to URLs in documents that are open in Microsoft 365 Apps for enterprise (Word, Excel, and PowerPoint on Windows or Mac OS), Office documents on iOS, or Android devices, Visio 2016 on Windows, and Office Online (Word Online, PowerPoint Online, Excel Online, and OneNote Online), provided the user has signed into Microsoft 365.
Don't track when users click ATP Safe Links	When this option is selected, click data for URLs in Word, Excel, PowerPoint, and Visio documents is not stored.
Don't let users click through ATP Safe Links to original URL	When this option is selected, users cannot proceed past a warning page to a URL that is determined to be malicious.

Additional recipient-based policies

Policy options that apply to specific email recipients:

Custom policy option	Effect
Off	Does not scan URLs in email messages. Enables you to define an exception rule, such as a rule that does not scan URLs in email messages for a specific group of recipients.
On	Rewrites URLs to route users through ATP Safe Links protection when the users click URLs in email messages. Checks a URL when clicked against a list of blocked or malicious URLs.
Use Safe Attachments to scan downloadable content	When this option is selected, URLs that point to downloadable content are scanned.
Apply Safe Links to messages sent within the organization	When this option is available and selected, ATP Safe Links protection is applied to email messages sent between people in your organization, provided the email accounts are hosted in Office 365.
Do not track user clicks	When this option is selected, click data for URLs in email from external senders is not stored. URL click tracking for links within email messages sent within the organization is currently not supported.
Do not allow users to click through to original URL	When this option is selected, users cannot proceed past a warning page to a URL that is determined to be malicious.

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Custom policy option	Effect
Do not rewrite the following URLs	Leaves URLs as they are. Keeps a custom list of safe URLs that don't need scanning for a specific group of email recipients in your organization. See Set up a custom "Do not rewrite" URLs list using ATP Safe Links for more details, including recent changes to support for wildcard asterisks (*).

URL detonation

URL detonation is a capability that combines elements of Safe Links and Safe Attachments into a single feature to protect users in the event a link points to a malicious file on a web server. When a user clicks a link to a file on a web server, the file is downloaded into the Safe Attachments sandbox environment and detonated. As the content is being analyzed, a web page is presented to the user explaining that a scan is in process. If the file is ultimately determined to be malicious, the user is redirected to a warning page advising the user that the site is malicious.

Note: This option can only be configured in a custom policy you create. The option is not available in the default policy.

Safe Links in Microsoft 365 Apps for enterprise

Safe Links is also capable of validating URLs in documents that are opened in Microsoft 365 Apps for enterprise (formerly Office 365 ProPlus) such as Word, Excel, PowerPoint, and Visio running on Windows.

Note: This option can only be configured in the default, organization-wide policy.

Spoofing Intelligence

When a sender spoofs an email address, they appear to be sending mail on behalf of one or more user accounts within one of your organization's domains, or an external domain sending to your organization. The spoofer's goal is to masquerade as a trusted sender to avoid being targeted as a potential threat; thereby, allowing the spoofer's malicious email to be delivered.

As previously discussed, spoofing is controlled by the built-in protection provided by Exchange Online Protection (EOP) and by implementing authentication techniques such as SPF, DKIM, and DMARC. The Spoof Intelligence feature can provide additional control and insight into senders who are spoofing your domain. You can review senders who are spoofing your domain and then choose to either allow the sender to continue or to block the sender.

Spoof intelligence is available as part of Office 365 E5 or separately as part of Advanced Threat Protection (ATP) and as of October 2018 Exchange Online Protection (EOP).

Plan for spoof intelligence

For domains you own, you can review senders who are spoofing your domain and then choose to either allow the sender to continue or block the sender. For external domains, you can allow the sender domain combined with the sending infrastructure, although not an individual sending email address.

Surprisingly, there are some legitimate business reasons for spoofing. For example, in these cases, you wouldn't block the sender from spoofing your domain:

- You have third-party senders who use your domain to send bulk mail to your own employees for company polls.

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- You have hired an external company to generate and send out advertising or product updates on your behalf.
- An assistant who must regularly send email for another person within your organization.
- An application that is configured to spoof its own organization in order to send internal notifications by email.

External domains frequently send spoofed email, and many of these reasons are legitimate. For example, here are some legitimate cases when external senders send spoofed email:

- The sender is on a discussion mailing list, and the mailing list is relaying the email from the original sender to all the participants on the mailing list.
- An external company is sending email on behalf of another company (for example, an automated report, or a software-as-a-service company).

You need a way to ensure that the mail sent by legitimate spoofers doesn't get caught up in spam filters in Microsoft 365 or external email systems. Normally, Microsoft 365 treats these email messages as spam. As a Microsoft 365 admin, you can prevent this by setting up spoof filters in the Security & Compliance Center. If you own the domain, you can configure SPF, DKIM, and DMARC to allow for these senders.

Managing spoof intelligence

The spoof intelligence policy you set up is always enforced by Microsoft 365. You cannot disable it, but you can choose how much you want to actively manage it.

You can review the senders who are spoofing your domain, or external domains, and then decide whether each sender should be allowed to do so by using the Security & Compliance Center. For each spoofed user account that a sender spoofs from your domain or an external domain, you can view the information in the following table.

Parameter	Description
Sender	Also called the true sender. This is usually the domain from which the spoof email originates. Microsoft 365 determines the domain of the pointer (PTR) DNS record of the sending IP address that is spoofing your organization. If no domain is found, the report displays the sender' IP address instead.

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Parameter	Description
Spoofed user	The user account that is being spoofed by the sender. • **Internal tab only**. This field contains a single email address, or if the sender is spoofing multiple user accounts, it contains more than one. • **External tab only**. External domains only contain a sending domain, and do not contain a full email address. **Note**: The spoofed user is the "From (5322.From)" address which is also the address displayed as the "From" address by the mail client. This is sometimes called the "header.from" address. The validity of this address is not checked by SPF.
Number of messages	The number of mail messages sent by the sender to your organization on behalf of the identified spoofed sender or senders within the last 30 days.
Number of user complaints	Complaints filed by users against this sender by your users within the last 30 days. Complaints are usually in the form of junk submissions to Microsoft.
Authentication result	This value is Passed if the sender passed Exchange Online Protection (EOP) sender authentication checks, such as SPF or DKIM, Failed if the sender failed EOP sender authentication checks, or Unknown if the result of these checks isn't known.
Decision set by	Shows whether the Microsoft 365 administrator or the spoof intelligence policy determined whether the sender is allowed to spoof the user.
Last seen	The last date on which a message was received by this sender on behalf of this spoofed user.

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Parameter	Description
Allowed to spoof?	Displays whether this sender is allowed to send email on behalf of the spoofed user. Possible values include: • **Yes:** All spoofed addresses from this spoofing sender will be allowed to spoof your organization. • **No:** Spoofed addresses from this spoofing sender won't be allowed to spoof your organization. Instead, messages from this sender will be marked as spam by Microsoft 365. If a sender is spoofing multiple users, some spoofed addresses from this sender will be allowed to spoof your organization; the rest will be marked as spam. Use the Detailed tab to see the specific addresses.
Spoof Type	This value is Internal if the domain is one of your organization' provisioned domains, otherwise the value is External.

ATP anti-phishing

Phishing is similar to spoofing in that the sender of an email masquerades as a trustworthy entity, but in the case of phishing, the purpose of the email is to try and acquire sensitive information such as user-names, passwords, credit card information, social security numbers, and so on. Phishing attacks come in a variety of forms, from commodity-based attacks to targeted spear phishing or whaling. With the growing complexity of phishing attacks, it's difficult for even a trained eye to identify some of these sophisticated attacks.

By default, Microsoft 365 includes built-in features that help protect your users from phishing attacks. Anti-phishing policies increase this protection; for example, by refining settings to better detect and prevent impersonation and spoofing attacks.

ATP is available for Office 365 E5 users and as standalone solution. The less powerful anti-phishing is based on ATP anti-phishing; it is available for all EOP users and was already covered in the previous lesson.

Managing ATP anti-phishing policies

ATP anti-phishing is managed in the Security & Compliance Center. The default policy applies to all users within the organization and is a single view where you can fine-tune anti-phishing protection. Custom policies, which can be created and configured for specific users, groups, or domains within the organization, take precedence over the default policy for the scoped users.

There are several prerequisites to managing ATP anti-phishing policies that you must be aware of, including:

- You must be a member of the Company administrators or Security admins role group.
- Most organizations typically set up multiple anti-phishing policies. Microsoft 365 enforces these policies in the order they're listed on the Anti-phishing page and ATP anti-phishing pages in the

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Security & Compliance Center. Once you've reviewed the policy options, take some time to determine how many policies you'll need and the priority for each.

- Allow up to 30 minutes for your new or updated policy to be propagated to all Microsoft 365 data-centers.

ATP anti-phishing policy options

As you set up or edit your ATP anti-phishing policies, you can choose from several options that provide the most sophisticated and comprehensive protection, as described in the following table.

This setting	Does this	Use when you want to:
Add users to protect	Defines which email addresses will be protected by the policy. **Note**: You can add up to 60 internal and external addresses that you want to protect from impersonation.	When you want to ensure that mail from outside your organization isn't an impersonation of one of the users on the list of users you are protecting. Examples of users you might want to protect are high-level executives, business owners, external board members, and so on. This list of protected users is different from the list of people to which the policy applies, or rather, for which the policy is enforced. You define the applies to list in the Applied to section of the policy options. For example, if you add Jenna Glover <jenglov@contoso.com> as a user to protect, then apply the policy to the group “All Users”. This would ensure that a mail that appeared to impersonate "Jenna Glover" sent to a user in the “All Users” group would be acted on by the policy.

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This setting	Does this	Use when you want to:
Add domains to protect	Allows you to choose which domains you want to protect from impersonation. You can specify that the policy includes all your custom domains, a comma-separated list of domains, or a combination of the two. If you choose the **Automatically include domains that I own** option, and you later add a domain to your Microsoft 365 organization, this anti-phishing policy will be in place for the new domain.	Whenever you want to ensure that mail from outside your organization isn't an impersonation of one of the domains defined in your list of verified domains or that of a partner domain.

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This setting	Does this	Use when you want to:
Choose actions	Choose the action to take when Microsoft 365 detects an impersonation attempt against the users and domains you added to the policy. You can choose different actions for users and domains in the same anti-phishing policy. These actions apply to any incoming email that has been identified by Microsoft 365 as impersonating a user account or domain that is under the protection of this anti-phishing policy. The actions you can choose from include: • **Quarantine message.** Email will be sent to Microsoft 365 quarantine. When you choose this option, the email is not sent to the original recipient. • **Redirect message to another email address.** Email will be sent to the email address you specify. You can specify multiple email addresses. When you choose this option, the email is not sent to the original recipient. • **Move message to the recipients' Junk email folder.** Email will be sent to the recipients' Junk email folder. When you choose this option, the email is	When you want to take an action on messages that Microsoft 365 has determined to be an impersonation of a user or domain as defined in the policy.

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This setting	Does this	Use when you want to:
Enable mailbox intelligence	Enables or disables mailbox intelligence for this policy. You can only enable mailbox intelligence for cloud-based accounts, that is, accounts whose mailbox is hosted entirely in Microsoft 365.	When you want to enhance impersonation results for users based on each user's individual sender map. Mailbox intelligence is built around the people you send and receive mail from. This intelligence allows Microsoft 365 to customize the impersonation policy at a user-level in order to better handle false positive results.
Add trusted senders and domains	Defines email addresses and domains that will not be considered impersonations by this policy. Messages from the sender email addresses and domains you add as trusted senders and domains won't ever be classified as an impersonation-based attack. As a result, the actions and settings in this policy won't be applied to messages from these senders and domains.	When users interact with domains or users that trigger impersonation but are considered to be safe. For example, if a partner has the same/similar display name or domain name as a user defined on the list.
Applied to	Defines the recipients whose incoming email messages will be subject to the rules of the policy. You can create conditions and exceptions for the recipients associated with the policy. For example, you can create a global policy for your organization by applying the rule to all recipients in your domain. You can also create exception rules, such as a rule that does not scan email messages for a specific group of recipients.	Each policy must be associated with a set of users, for example, users in a particular group or domain.

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This setting	Does this	Use when you want to:
Advanced phishing thresholds	Defines the level of settings for how phishing messages are handled. These include: • **Standard.** Email suspected to be phishing is handled in the standard way. • **Aggressive**. Email suspected to be phishing with a high or very high degree of confidence are handled by the system in the same way. • **More aggressive.** Email suspected to be phishing with a medium, high, or very high degree of confidence are handled by the system in the same way. • **Most aggressive.** Email suspected to be phishing with a low, medium, high, or very high degree of confidence are handled by the system in the same way.	When you want to be more aggressive in the treatment of potential phishing messages within Microsoft 365. For example, messages with a very high probability of being phish will have the most aggressive actions taken on them, while messages with a low probability have less aggressive actions taken. This setting also impacts other parts of the filtering system that combine signals together. The chance of moving good messages increases as the level of settings increases.

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Module Review

Module 3 Review

Checkbox

Item 1. Which of the following are key functionalities of Exchange Online Protection? (Select 3)

- ☐ Scan incoming, outgoing, and internal email messages.
- ☐ Encrypt email messages with certificates.
- ☐ Proprietary anti-spam technology used to achieve high-accuracy rates.
- ☐ Multiple antivirus engines to scan passing messages.
- ☐ On-premises Active Directory synchronization to Azure AD.
- ☐ Role Based Access Control (RBAC) for granular permission management in Exchange Online.

Multiple choice

Item 2. You are a messaging administrator for Contoso Ltd. What is the initial configuration step that you must complete to use Exchange Online Protection for your on-premises Exchange organization?

- ☐ Install EOP on all Exchange Mailbox Servers in your organization.
- ☐ Install EOP on the Exchange Edge servers in your perimeter network.
- ☐ Configure the organization's MX record to EOP and create connectors for message routing through EOP.
- ☐ Create a transport rule to route all email messages to mail.protection.outlook.com.

Multiple choice

Item 3. You are a message administrator for Contoso Ltd. You have been tasked with configuring mail routing for a hybrid environment with Exchange Online. As you develop your implementation plan, which best practice should you plan to perform for the initial configuration of your mail routing to Exchange Online?

- ☐ Create all on-premises connectors manually and create a support case for creating connectors in Exchange Online Protection.
- ☐ Run the Hybrid Configuration Wizard (HCW).
- ☐ Create all on-premises connectors manually and use the Exchange Management Shell to create the Exchange Online protection connectors.
- ☐ You don't need to configure any connectors when deploying a hybrid organization.

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Multiple choice

Item 4. Which of the following DNS record types do you have to configure to route messages for your authoritative domains to Exchange Online Protection?

- ☐ A records
- ☐ MX records
- ☐ CNAME records
- ☐ AAAA records
- ☐ SRV records

Multiple choice

Item 5. You work as a messaging administrator for Adatum Corp. Your company uses Exchange Online, and all of its users are in Exchange Online. Management has asked that you provide a report about all emails sent by a certain employee on a certain day during the prior week. Which tool should you use to deliver the requested report?

- ☐ You search the message tracking log using the Exchange Management Shell.
- ☐ You perform a message trace in the SCC.
- ☐ You configure a report in the Microsoft 365 Admin Center.
- ☐ You can use the REST/OData Tenant Reporting Web service.

Multiple choice

Item 6. You work as a messaging administrator for Contoso Ltd. Several users report that legitimate messages are being sent to their Junk folders. The affected users sent you the message headers of the original email and you analyzed them with the help of RCA (Remote Connectivity Analyzer). Which field in the **X-ForeFront-Antispam-Report** *message header is responsible for sending the messages to the Junk folder?*

- ☐ The SCL field
- ☐ The LANG field
- ☐ The CTRY field
- ☐ The SFE field

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Checkbox

Item 7. Which of the following Exchange Management Shell scripts manage anti-malware setting in Exchange Server? (Select 3)

- ☐ Disable-Antimalwarescanning.ps1
- ☐ Get-UCPool.ps1
- ☐ Import-RetentionTags.ps1
- ☐ Enable-Antimalwarescanning.ps1
- ☐ Update-MalwareFilteringServer.ps1
- ☐ Disable-OutsideIn.ps1
- ☐ ConfigureAdam.ps1

Checkbox

Item 8. Contoso wants to implement common techniques to prevent phishing and spoofing of their domains. As Contoso's messaging administrator, which technologies do you recommend to protect their domains against spoofing? (Select 3)

- ☐ Simple Mail Transfer Protocol (SMTP)
- ☐ DomainKeys Identified Mail (DKIM)
- ☐ Mail Exchanger (MX)
- ☐ Sender Policy Framework (SPF)
- ☐ Pointer (PTR)
- ☐ Domain-based Message Authentication, Reporting, and Conformance (DMARC)

Checkbox

Item 9. Which of the following actions can you perform by using transport rules? (Select 3)

- ☐ Prevent messages with certain content from entering or leaving your organization.
- ☐ Delete messages inside user mailboxes.
- ☐ Track or archive messages that are sent to or received from specific individuals.
- ☐ Move messages from shared mailboxes to user mailboxes.
- ☐ Open and read the body of a message that is encrypted with an end-user certificate.
- ☐ Append disclaimers to messages as they pass through the organization.

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Checkbox

Item 10. Which of the following are components of a transport rule? (Select 2)

- ☐ Inserts
- ☐ Conditions
- ☐ Views
- ☐ Actions
- ☐ Policies

Checkbox

Item 11. You are the messaging administrator for Contoso Ltd. You must explain to IT management the main advantages of using Advanced Threat Protection for Exchange in contrast to Exchange Online Protection. Which features do you describe as exclusively available when using ATP for Exchange? (Select 2)

- ☐ Anti-malware
- ☐ Anti-Spam
- ☐ Safe Attachments
- ☐ Safe Links

Checkbox

Item 12. Which licenses does your company need to own for its employees to use the Advanced Threat Protection for Exchange features? (Select 2)

- ☐ Office 365 E3
- ☐ Office 365 E5
- ☐ Office 365 Business Premium
- ☐ Advanced Threat Protection Standalone

Multiple choice

Item 13. You work as a messaging administrator for Contoso Ltd. which has just bought Advance Threat Protection for Exchange. You have asked a colleague to configure some ATP anti-phishing policies. Which of the following role groups from the Security & Compliance Center do you need to assign to his user account so that he has permission to configure ATP anti-phishing?

- ☐ Records Management
- ☐ Security Administrator
- ☐ MailFlow Administrator
- ☐ Service Assurance User

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Multiple choice

Item 14. What does "spoofing" mean for your messaging environment?

- ☐ When you spoof your organization's domains, you relay emails for your domains to an external Exchange organization.
- ☐ When a sender spoofs an email address, they appear to be sending mail on behalf of one or more user accounts within one of your organization's domains.
- ☐ When your organization is spoofed it means that your anti-malware policies are outdated and that you need to perform updates.
- ☐ When a recipient spoofs an email address, the mails in their Inbox appear to be sent from a different sender than they actually have been sent from.

Multiple choice

Item 15. Which of the following statements describe how "URL detonation" works?

- ☐ The feature scans all Exchange virtual directories for possible configuration issues and generates a report with all configured URLs.
- ☐ When using URL detonation, you can configure all virtual directories in your Exchange organization at once by running a single command in the Exchange Management Shell.
- ☐ When a user clicks a link to a file on a web server, the file is downloaded into the Safe Attachments sandbox environment, where it is then analyzed and only allowed to be downloaded by the user if it is considered safe.
- ☐ The term describes a best practice of deploying multiple data loss prevention (DLP) policies in your organization.

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Answers

Checkbox

Item 1. Which of the following are key functionalities of Exchange Online Protection? (Select 3)

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- ☐ Encrypt email messages with certificates.
- ■ Proprietary anti-spam technology used to achieve high-accuracy rates.
- ■ Multiple antivirus engines to scan passing messages.
- ☐ On-premises Active Directory synchronization to Azure AD.
- ☐ Role Based Access Control (RBAC) for granular permission management in Exchange Online.

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- ☐ Install EOP on the Exchange Edge servers in your perimeter network.
- ■ Configure the organization's MX record to EOP and create connectors for message routing through EOP.
- ☐ Create a transport rule to route all email messages to mail.protection.outlook.com.

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- ☐ Create all on-premises connectors manually and create a support case for creating connectors in Exchange Online Protection.
- ■ Run the Hybrid Configuration Wizard (HCW).
- ☐ Create all on-premises connectors manually and use the Exchange Management Shell to create the Exchange Online protection connectors.
- ☐ You don't need to configure any connectors when deploying a hybrid organization.

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Multiple choice

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- ■ You perform a message trace in the SCC.
- ☐ You configure a report in the Microsoft 365 Admin Center.
- ☐ You can use the REST/OData Tenant Reporting Web service.

Multiple choice

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- ☐ The LANG field
- ☐ The CTRY field
- ☐ The SFE field

AUTHORIZED TRAINER USE ONLY. STUDENT USE PROHIBITED

Checkbox

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- ☐ Import-RetentionTags.ps1
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- ■ Update-MalwareFilteringServer.ps1
- ☐ Disable-OutsideIn.ps1
- ☐ ConfigureAdam.ps1

Checkbox

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- ☐ Simple Mail Transfer Protocol (SMTP)
- ■ DomainKeys Identified Mail (DKIM)
- ☐ Mail Exchanger (MX)
- ■ Sender Policy Framework (SPF)
- ☐ Pointer (PTR)
- ■ Domain-based Message Authentication, Reporting, and Conformance (DMARC)

Checkbox

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- ■ Track or archive messages that are sent to or received from specific individuals.
- ☐ Move messages from shared mailboxes to user mailboxes.
- ☐ Open and read the body of a message that is encrypted with an end-user certificate.
- ■ Append disclaimers to messages as they pass through the organization.

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Checkbox

Item 10. Which of the following are components of a transport rule? (Select 2)

- ☐ Inserts
- ■ Conditions
- ☐ Views
- ■ Actions
- ☐ Policies

Checkbox

Item 11. You are the messaging administrator for Contoso Ltd. You must explain to IT management the main advantages of using Advanced Threat Protection for Exchange in contrast to Exchange Online Protection. Which features do you describe as exclusively available when using ATP for Exchange? (Select 2)

- ☐ Anti-malware
- ☐ Anti-Spam
- ■ Safe Attachments
- ■ Safe Links

Checkbox

Item 12. Which licenses does your company need to own for its employees to use the Advanced Threat Protection for Exchange features? (Select 2)

- ☐ Office 365 E3
- ■ Office 365 E5
- ☐ Office 365 Business Premium
- ■ Advanced Threat Protection Standalone

Multiple choice

Item 13. You work as a messaging administrator for Contoso Ltd. which has just bought Advance Threat Protection for Exchange. You have asked a colleague to configure some ATP anti-phishing policies. Which of the following role groups from the Security & Compliance Center do you need to assign to his user account so that he has permission to configure ATP anti-phishing?

- ☐ Records Management
- ■ Security Administrator
- ☐ MailFlow Administrator
- ☐ Service Assurance User

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Multiple choice

Item 14. What does "spoofing" mean for your messaging environment?

- ☐ When you spoof your organization's domains, you relay emails for your domains to an external Exchange organization.
- ■ When a sender spoofs an email address, they appear to be sending mail on behalf of one or more user accounts within one of your organization's domains.
- ☐ When your organization is spoofed it means that your anti-malware policies are outdated and that you need to perform updates.
- ☐ When a recipient spoofs an email address, the mails in their Inbox appear to be sent from a different sender than they actually have been sent from.

Multiple choice

Item 15. Which of the following statements describe how "URL detonation" works?

- ☐ The feature scans all Exchange virtual directories for possible configuration issues and generates a report with all configured URLs.
- ☐ When using URL detonation, you can configure all virtual directories in your Exchange organization at once by running a single command in the Exchange Management Shell.
- ■ When a user clicks a link to a file on a web server, the file is downloaded into the Safe Attachments sandbox environment, where it is then analyzed and only allowed to be downloaded by the user if it is considered safe.
- ☐ The term describes a best practice of deploying multiple data loss prevention (DLP) policies in your organization.

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Module 4 Managing Compliance

Messaging Compliance in the SCC

Lesson Introduction

Due to the high relevance of messaging in businesses of all sizes, compliance has become an essential topic for every messaging administrator. Most countries and regions have their individual requirements for data protection, such as the General Data Protection Regulation (GDPR) within the European Union. Besides legal requirements, many companies must also meet regulatory requirements in their industry sector. These requirements oftentimes limit access to messages for certain business roles or require special retention periods for messages.

This lesson describes the different compliance features in the Security & Compliance Center (SCC) that messaging administrators can use to comply with legal and regulatory requirements.

After this lesson, you will be able to:

- Describe different policy and compliance features for messaging
- Evaluate the different roles in the Security & Compliance Center
- Plan retention policies for Exchange Online mailboxes
- Configure data loss prevention (DLP) policies for data in Microsoft 365
- Create message traces to understand the mail flow in your Exchange Online organization

Message Policy and Compliance

Email has become a reliable and ubiquitous communication medium for information workers in organizations of all sizes. Messaging stores and mailboxes have become repositories of valuable corporate data. It's important for organizations to formulate messaging policies that dictate the fair use of their messaging systems, provide user guidelines for how to act on the policies, and where required, provide details about the types of communication that may not be allowed.

Organizations must also create policies to manage email lifecycle, retain messages for the length of time based on business, legal, and regulatory requirements, preserve email records for litigation and investiga-

tion purposes, and be prepared to search and provide the required email records to fulfill eDiscovery requests.

The following messaging policy and compliance features in Exchange Online can be configured in the Security & Compliance Center:

- **In-Place Archiving.** Helps you regain control of your organization's messaging data by eliminating the need for personal store (.pst) files and allowing users to store messages in an archive mailbox accessible in Outlook and Outlook on the web.
- **Retention Policies.** Enables you to retain or delete mailbox content based on policies to comply with regulatory and legal rules.
- **Data loss prevention (DLP).** Includes sensitive information types that are ready to use for prevention of accidental or intentional data loss in messages.
- **eDiscovery.** Allows you to search mailbox data across your Exchange organization and create eDiscovery cases to organize compliance relevant search operations.
- **Office 365 Message Encryption.** Uses Azure RMS to encrypt and control access to emails and attachments sent out to internal and external recipients, such as Do Not Forward protection and content encryption.

These features follow a unified approach that is not limited to Exchange messaging only. While traditional messaging administrators and pure on-premises-only administrators configure most of the features in the Exchange Admin Center, you should use the Security & Compliance Center when using Microsoft 365 services. This enables you to also to implement unified protection solutions, even if you are not managing Exchange Online only, but also when you are responsible for Exchange hybrid solutions.

Note: There are even more features for security and compliance for Microsoft 365 services that are not part of this messaging administrator course. Microsoft 365 is continuously being improved, existing features are being updated, and new features are being added. You should regularly monitor the Message center in the Microsoft 365 Admin Center to stay updated on all the changes and new features that are being planned for release.

Encryption for data at rest and data in transit

Encryption in Microsoft 365 is separated into two different areas:

- **Data at rest.** This includes all information storage objects, containers, and types that exist statically on physical media, whether magnetic or optical disk. For example, disk encryption for Exchange Online.
- **Data in transit.** When data is being transferred between components, locations, or programs, it's in transit. One of the features for encrypting data in transit is TLS encryption of the data stream.

Encryption of Microsoft 365 customer data at rest is provided by multiple service-side technologies, including BitLocker, DKM, Azure Storage Service Encryption, and service encryption in Exchange Online, Skype for Business, OneDrive for Business, and SharePoint Online. Microsoft 365 Service Encryption includes an option to use customer-managed encryption keys that are stored in Azure Key Vault. This customer-managed key option, called Customer Key, is available for Exchange Online, SharePoint Online, and OneDrive for Business.

Note: Features like Azure Information Protection, Azure Storage Service Encryption, and Customer Key are typically the responsibility of the Security Administrator, but there is some overlapping for messaging administrators.

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Feature	Description
BitLocker	Microsoft 365 servers use BitLocker to encrypt the disk drives containing customer data at rest at the volume-level. BitLocker encryption is a data protection feature that is built into Windows. BitLocker is one of the technologies used to safeguard against threats in case there are lapses in other processes or controls (for example, access control or recycling of hardware) that could lead to someone gaining physical access to disks containing customer data. In this case, BitLocker eliminates the potential for data theft or exposure because of lost, stolen, or inappropriately decommissioned computers and disks.
Distributed Key Manager (DKM)	In addition to BitLocker, Microsoft uses a technology called Distributed Key Manager. DKM is a client-side functionality that uses a set of secret keys to encrypt and decrypt information. Only members of a specific security group in Active Directory Domain Services can access those keys to decrypt the data that is encrypted by DKM. In Exchange Online, only certain service accounts under which the Exchange processes run are part of that security group. As part of standard operating procedure in the datacenter, no person is given credentials that are part of this security group and therefore no person has access to the keys that can decrypt these secrets.
Microsoft 365 Message Encryption	Microsoft 365 Message Encryption allows email users to send encrypted email messages to anyone in- and outside your organization. Office Message Encryption uses protection features from Azure Information Protection (Azure RMS).
Secure/Multipurpose Internet Mail Extensions (S/MIME)	S/MIME is a client-side solution to protect sensitive information by sending signed and encrypted email. Encryption and signing are handled through personal certificates.

Evaluating RBAC Roles in the SCC

The Microsoft 365 Security & Compliance Center is used to grant permissions to users who perform compliance tasks such as data loss prevention, eDiscovery, retention, and so on. These people can perform only the tasks that are explicitly grant to them. To configure your organization's compliance features in the Security & Compliance Center, you must be a Microsoft 365 global administrator or a member of one or more Security & Compliance Center role groups.

Permissions in the Security & Compliance Center are based on the Role Based Access Control (RBAC) permissions model. This is the same permissions model that's used by Exchange, so if you're familiar with Exchange, granting permissions in the Security & Compliance Center will be very similar. It's important to

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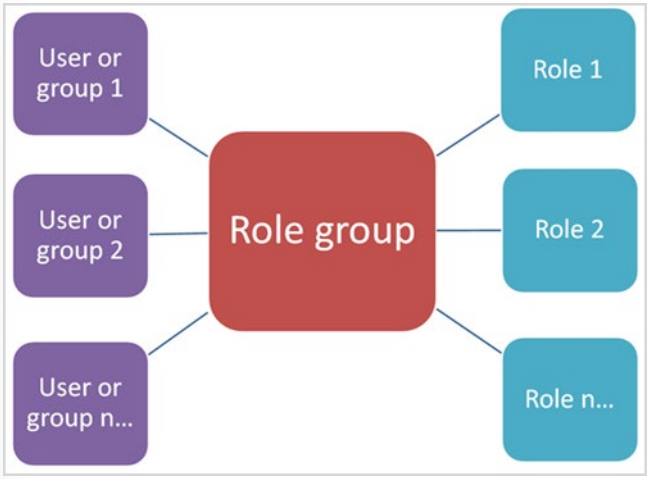
remember, however, that Exchange role groups and Security & Compliance Center role groups do NOT share membership or permissions. While both have an Organization Management role group, they are not the same. The permissions they grant, and the members of the role groups, are different. The Security & Compliance Center role groups are examined in greater detail below.

Relationship of members, roles, and role groups

The RBAC permissions model that is used in the SCC is based on roles and role groups.

- **Role.** A role grants permissions to perform a set of tasks. For example, the Case Management role lets people work with eDiscovery cases.
- **Role group.** A role group is a set of roles that lets people perform their job across the Security & Compliance Center. The roles that make up a role group are typically associated with all the tasks needed to perform a specific job function. For example, the Compliance Administrator role group includes the roles for Case Management, Content Search, and Organization Configuration (plus others) because someone who's a compliance admin will need the permissions necessary to perform all those tasks.

The Security & Compliance Center includes default role groups for the most common tasks and functions that you'll need to assign people to. As a best practice, you should simply add individual users as members to the default role groups.



Note: While you can edit or delete the existing default role groups, it is not recommended that you do so. Instead of editing a default role group, it is recommended that you copy it, modify it, and then save it with a different name.

Default role groups in SCC

The following table identifies the default role groups that are available in the Security & Compliance Center. To grant permissions to a user to perform a compliance task, you should add them to the appropriate Security & Compliance Center role group.

Role group	Description
Compliance Administrator	Members can manage settings for device management, data loss prevention, reports, and preservation.

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Role group	Description
eDiscovery Manager	Members can perform searches and place holds on mailboxes, SharePoint Online sites, and OneDrive for Business locations. Members can also create and manage eDiscovery cases, add and remove members to a case, create and edit Content Searches associated with a case, and access case data in Microsoft 365 Advanced eDiscovery. An eDiscovery Administrator is a member of the eDiscovery Manager role group who has been assigned additional permissions. In addition to the tasks that an eDiscovery Manager can perform, an eDiscovery Administrator can: • View all eDiscovery cases in the organization. • Manage any eDiscovery case after they add themselves as a member of the case. The primary difference between an eDiscovery Manager and an eDiscovery Administrator is that an eDiscovery Administrator can access all cases that are listed on the eDiscovery cases page in the Security & Compliance Center. An eDiscovery manager can only access the cases they created or the cases in which they are a member.
Organization Management	Members can control permissions for accessing features in the Security & Compliance Center, as well as manage settings for device management, data loss prevention, reports, and preservation. This role group is designed for a user who is not a global administrator to see the list of devices managed by MDM for Microsoft 365 and perform actions on these devices, such as retiring a device from MDM for Microsoft 365. The user must be an Exchange administrator. **Note**: Microsoft 365 global admins are automatically added as members of this role group.
Records Management	Members can manage and dispose record content.

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Role group	Description
Reviewer	Members can only view the list of cases on the eDiscovery cases page in the Security & Compliance Center. They can't create, open, or manage an eDiscovery case. The primary purpose of this role group is to allow members to view and access case data in Advanced eDiscovery. This role group has the most restrictive eDiscovery-related permissions.
Security Administrator	Membership in this role group is synchronized across services and managed centrally. This role group is not manageable through the administrator portals. Members of this role group may include cross-service administrators, as well as external partner groups and Microsoft Support. By default, this group may not be assigned any roles. However, it will be a member of the Security Administrators role groups and will inherit the capabilities of that role group. This role group possesses all of the read-only permissions of the Security reader role, plus a number of additional administrative permissions for the same services, such as Azure Information Protection, Identity Protection Center, Privileged Identity Management, Monitor Microsoft 365 Service Health, and Microsoft 365 Security & Compliance Center.
Security Reader	Members have read-only access to several security features of Identity Protection Center, Privileged Identity Management, Monitor Office 365 Service Health, and Microsoft 365 Security & Compliance Center. Membership in this role group is synchronized across services and managed centrally. This role group is not manageable through the administrator portals. Members of this role group may include cross-service administrators, as well as external partner groups and Microsoft Support. By default, this group may not be assigned any roles. However, it will be a member of the Security Reader role group and it will inherit the capabilities of that role group.

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Role group	Description
Service Assurance User	Members can access the Service assurance section in the Microsoft 365 Security & Compliance Center. Service assurance provides reports and documents that describe Microsoft's security practices for customer data that's stored in Microsoft 365. It also provides independent third-party audit reports on Microsoft 365.
Supervisory Review	Members can create and manage the policies that define which communications are subject to review in an organization.
MailFlow Administrator	Members have read-only permission to the mail flow dashboard.

Configuring retention policies in the SCC

For most organizations, the volume and complexity of their data increases daily. This includes their email, documents, instant messages, and more. Effectively managing or governing this information is important because you must:

- **Comply proactively with industry regulations and internal policies** that require you to retain content for a minimum period of time - for example, the Sarbanes-Oxley Act might require you to retain certain types of content for seven years.
- **Reduce your risk in the event of litigation or a security breach** by permanently deleting old content that you're no longer required to keep.
- **Help your organization to share knowledge effectively and be more agile** by ensuring that your users work only with content that's current and relevant to them.

A retention policy in Microsoft 365 can help you achieve these goals by managing content with two commonly required actions:

- **Retaining** content so that it can't be permanently deleted before the end of the retention period.
- **Deleting** content permanently at the end of the retention period.

With a retention policy, you can:

- Decide proactively whether to retain content, delete content, or both retain and then delete the content.
- Apply a single policy to the entire organization or just specific locations or users.
- Apply a policy to all content or just content meeting certain conditions, such as content containing specific keywords or specific types of sensitive information.

Note: The term Retention Policy is used in context of Message Records Management (MRM) in Exchange and in context of the Security & Compliance Center; however, they are not identical. Retention Policies and Retentions Tags are available in Exchange only, while Retention Policies in SCC are usable across multiple services, such as Exchange Online, SharePoint Online, Teams, and so on. Retention policies and tags in context of Message Records Management (MRM) will be covered in a later lesson.

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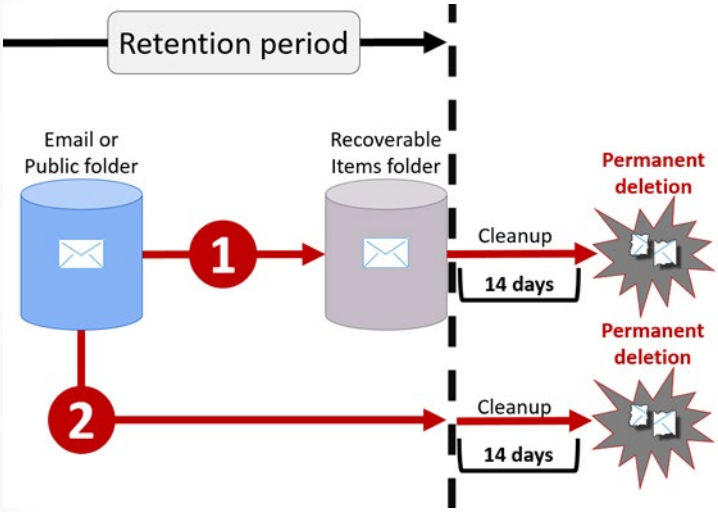
How a retention policy works with content in place

For a user's mail, calendar, and other items, a retention policy is applied at the mailbox level. For a public folder, a retention policy is applied at the folder level, not the mailbox level. Both a mailbox and a public folder use the Recoverable Items folder to retain items. Only users who have been assigned eDiscovery permissions can view items in another user's Recoverable Items folder.

By default, when a user deletes a message in a folder other than the Deleted Items folder, the message is moved to the Deleted Items folder. When a user deletes an item in the Deleted Items folder, the message is moved to the Recoverable Items folder. A process periodically evaluates items in the Recoverable Items folder. If an item doesn't match the rules of at least one retention policy, the item is permanently deleted (also known as hard deleted) from the Recoverable Items folder.

When a user attempts to change certain properties of a mailbox item — such as the subject, body, attachments, senders and recipients, or the date sent or received for a message — a copy of the original item is saved to the Recoverable Items folder before the change is committed. This happens for each subsequent change. At the end of the retention period, copies in the Recoverable Items folder are permanently deleted.

If a user leaves your organization and their mailbox is included in a retention policy, the mailbox becomes an inactive mailbox when the user's Microsoft 365 account is deleted. The contents of an inactive mailbox are still subject to any retention policy that was placed on the mailbox before it was made inactive, and the contents are available to an eDiscovery search.



After a retention policy is assigned to a mailbox or public folder, content can follow one of two paths:

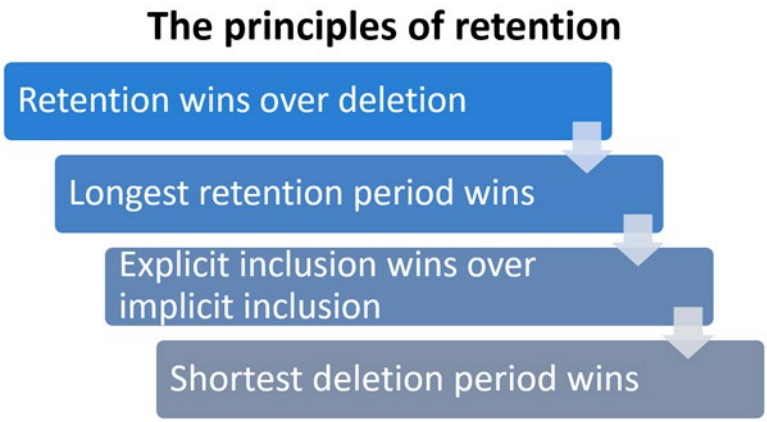
- If the item is modified or permanently deleted by the user (either SHIFT+DELETE or deleted from Deleted Items) during the retention period, the item is moved (or copied, in the case of an edit) to the Recoverable Items folder. There, a process runs periodically and identifies items whose retention period has expired, and these items are permanently deleted within 14 days of the end of the retention period. 14 days is the default setting, but it can be configured up to 30 days.
- If the item is NOT modified or deleted during the retention period, the same process runs periodically on all folders in the mailbox and identifies items whose retention period has expired. These items are then permanently deleted within 14 or more days of the end of the retention period.

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The principles of retention

It's possible or even likely that content might have several retention policies applied to it, each with a different action (retain, delete, or both) and retention period. When this occurs, the obvious question is "Which policy takes precedence?" At the highest level, rest assured that content being retained by one policy can't be permanently deleted by another policy.

The following diagram displays a simplified version of the order in which multiple retention policies are applied to content:



To understand how different retention policies are applied to content, keep these principles of retention in mind:

1. **Retention wins over deletion**. Suppose that one retention policy says to delete Exchange email after three years, but another retention policy says to retain Exchange email for five years and then delete it. Any content that reaches three years old will be deleted and hidden from the users' view, but still retained in the Recoverable Items folder until the content reaches five years old, when it will be permanently deleted.
2. **The longest retention period wins**. If content's subject to multiple policies that retain content, it will be retained until the end of the longest retention period.
3. **Explicit inclusion wins over implicit inclusion**. If a retention policy includes a specific location, such as a specific user's mailbox or OneDrive for Business account, that policy takes precedence over another retention policy that applies to all users' mailboxes or OneDrive for Business accounts but doesn't specifically include that user's mailbox.
4. **The shortest deletion period wins**. Similarly, if content's subject to multiple policies that delete content (with no retention), it will be deleted at the end of the shortest retention period.

Understand that the principles of retention work as a tie-breaking flow from top to bottom: If the rules applied by all policies or labels are the same at one level, the flow moves down to the next level to determine precedence for which rule is applied.

Retention Policies in Exchange

Retention Policies from the SCC can be used to manage the retention of Exchange Online mailboxes, public folders, Microsoft 365 Groups, Teams, and other services of Microsoft 365. But these policies are only able to retain the content of mailboxes that are in scope of Exchange Online; the policies cannot manage the retention of content from any on-premises mailboxes. To protect Exchange Server on-prem-

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ises or mailboxes still hosted on-premises in Exchange hybrid, you need to combine Retention Policies from the SCC and Retention Policies from MRM.

Note: Message Records Management (MRM) will be covered in a later lesson.

If an organization is hosting its entire Exchange messaging system in Exchange Online, you should consider configuring your retention and deletion policies in the SCC and select all required locations. This will replace the following features with retention policies that are centralized in the SCC:

- eDiscovery searches from the EAC (eDiscovery hold)
- In-Place Hold and Litigation Hold (eDiscovery hold)
- Retention tags and retention policies, also known as messaging records management (MRM) (Deletion only)

Configuring DLP Policies

Data loss prevention (DLP) is an important component of messaging systems. Since companies make extensive use of email to send and receive business-critical enterprise data, that data needs to be protected against accidental or intentional loss. DLP policies are designed to protect your enterprise data before it leaves your company and area of influence, while not hindering the productivity of your workers. You can configure DLP policies in the Security & Compliance Center.

A DLP policy contains the following components:

- **Locations.** Define where to protect the content, such as Exchange Online, SharePoint Online, and OneDrive for Business sites.
- **Rules.** Rules define the "when" and "how" to protect content, and they consist of conditions and actions.
  - **Conditions.** Defines the requisite state the content must match before the rule is enforced. For example, look only for content containing Social Security numbers that have been shared with people outside your organization.
  - **Actions.** The steps that you want the rule to perform when content matching the conditions is found. For example, block access to the document and send both the user and compliance officer an email notification.

You can use a rule to meet a specific protection requirement, and then use a DLP policy to group together common protection requirements, such as all the rules needed to comply with a specific regulation.



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Note: In contrast to DLP policies in Exchange, DLP policies in the Security & Compliance Center can cover many different locations and services across the entire tenant. DLP policies in the Exchange Admin Center will be covered in a later lesson, but the principle of rules is also used by DLP policies in the EAC.

DLP Rules

Rules are what enforce your business requirements on your organization's content. A policy contains one or more rules, and each rule consists of conditions and actions. When the conditions are met for each rule, the actions are taken automatically. Rules are executed sequentially, starting with the highest-priority rule in each policy.

A rule also provides options to notify users (with policy tips and email notifications) and admins (with email incident reports) that content has matched the rule.

Rule part	Description
Conditions	Conditions are important because they determine what types of information you're looking for, and when to take an action. For example, you might choose to ignore content containing passport numbers unless the content contains more than ten such numbers and is shared with people outside your organization. Conditions focus on the content, such as what types of sensitive information you're looking for, and also on the context, such as who the document is shared with. You can use conditions to assign different actions to different risk levels. For example, sensitive content shared internally might be a lower risk and require fewer actions than sensitive content shared with people outside the organization. The conditions now available can determine if: • Content contains a type of sensitive information • Content contains a label • Content is shared with people outside or inside your organization

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Rule part	Description
Sensitive information types	A DLP policy can help protect sensitive information, which is defined as a sensitive information type. Microsoft 365 includes definitions for many common sensitive information types across many different regions that are ready for you to use, such as a credit card number, bank account numbers, national ID numbers, and passport numbers. When a DLP policy looks for a sensitive information type such as a credit card number, it doesn't simply look for a 16-digit number. Each sensitive information type is defined and detected by using a combination of: ● Keywords ● Internal functions to validate checksums or composition ● Evaluation of regular expressions to find pattern matches ● Other content examination This helps DLP detection achieve a high degree of accuracy while reducing the number of false positives that can interrupt peoples' work.
Actions	When content matches a condition in a rule, you can apply actions to automatically protect the content. For example, you can restrict access to the content. For emails, this action blocks the message from being sent. Depending on how the DLP rule is configured, the sender will see an NDR or (if the rule uses a notification) a policy tip and/or email notification.

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Rule part	Description
User notifications and user overrides	You can use notifications and overrides to educate your users about DLP policies and help them remain compliant without blocking their work. For example, if a user tries to send an email containing sensitive information, a DLP policy can display a policy tip that explains, why content conflicts with a DLP policy. If you choose, the email notification and policy tip can allow users to override a rule by reporting a false positive or providing a business justification. This can help you educate users about your DLP policies and enforce them without preventing people from doing their work. Information about overrides and false positives is also logged for reporting (see below about the DLP reports) and included in the incident reports (next section), so that the compliance officer can regularly review this information.
Incident reports	When a rule is matched, you can send an incident report to your compliance officer (or any people you choose) with details of the event. This report includes information about the item that was matched, the actual content that matched the rule, and the name of the person who last modified the content. For email messages, the report also includes as an attachment the original message that matches a DLP policy.

Additional reading. For more information, see the following article on **What the sensitive information types look for**[1].

DLP reports

DLP reports gain insight about the number of DLP policy and rule matches and the number of false positives and overrides. For each report, you can filter those matches by location, time frame, and even narrow it down to a specific policy, rule, or action.

The following business insights are available with DLP reports:

- Focus on specific time periods and understand the reasons for spikes and trends.
- Discover business processes that violate your organization's compliance policies.
- Understand any business impact of the DLP policies.

In addition, you can use the DLP reports to fine tune your DLP policies as you run them.

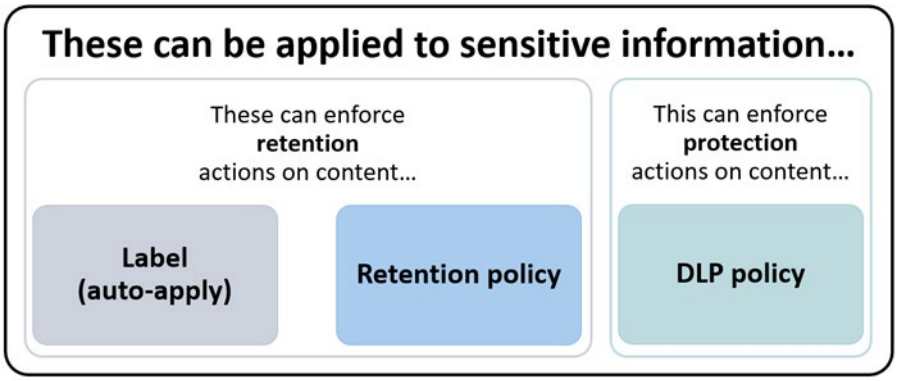
1 https://aka.ms/AA3rbdb

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How DLP relates to other features

Several features can be applied to content containing sensitive information:

- A retention label and a retention policy can both enforce retention actions on this content.
- A DLP policy can enforce protection actions on this content. And before enforcing these actions, a DLP policy can require other conditions to be met in addition to the content containing a label.



Note that a DLP policy has a richer detection capability than a label or retention policy applied to sensitive information. A DLP policy can enforce protective actions on content containing sensitive information, and if the sensitive information is removed from the content, those protective actions are undone the next time the content's scanned. But if a retention policy or label is applied to content containing sensitive information, that's a one-time action that won't be undone even if the sensitive information's removed.

By using a label as a condition in a DLP policy, you can enforce both retention and protection actions on content with that label. You can think of content containing a label exactly like content containing sensitive information - both a label and a sensitive information type are properties used to classify content so that you can enforce actions on that content.

Performing a Message Trace

Sometimes an email message gets lost in transit, or it can take a lot longer than expected for delivery. In either case, your users are left wondering what happened. As a messaging administrator, you can use the message trace feature to follow messages as they pass through your Exchange Online or Exchange Online Protection service. With message tracing, you can determine whether a targeted email message was received, rejected, deferred, or delivered by the service. It also shows what events have occurred to the message before reaching its final status. Getting detailed information about a specific message lets you efficiently answer your user's questions, troubleshoot mail flow issues, validate policy changes, and eliminate the need to contact technical support for assistance.

Message traces can be started in the Security & Compliance Center and in the EAC. Trace data is available for the past 90 days and if a message is older than 7 days, you can only view the results in a downloadable .CSV file.

Note: Message traces are used with Exchange Online, while investigations using the message tracking log are for on-premises Exchange Servers.

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Relevant information fields of message traces

The following information is available when viewing message trace results for messages that are less than seven days old.

Details	Description
Message size	The size of the message, including attachments, in kilobytes (KB), or, if the message size is greater than 999 KBs, in megabytes (MB).
Message ID	This is the Internet message ID (also known as the Client ID) found in the header of the message with the "Message-ID:" token. The form of this varies depending on the sending mail system.
To IP	The IP address or addresses to which the service attempted to deliver the message. If there are multiple recipients, these are displayed. For inbound messages sent to Exchange Online, this value is blank.
From IP	The IP address of the computer that sent the message. For outbound messages sent from Exchange Online, this value is blank.
Date	The date and time that the event occurred.

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Details	Description
Event	This field provides a summarized description of what happened with the message. For example, if the message was received by the service, if it was delivered or failed to be delivered to the intended recipient, and so on. The following are examples of events that may be listed: • **RECEIVE.** The message was received by the service. • **SEND.** The message was sent by the service. • **FAIL.** The message failed to be delivered. • **DELIVER.** The message was delivered to a mailbox. • **EXPAND.** The message was sent to a distribution group that was expanded. • **TRANSFER.** Recipients were moved to a bifurcated message because of content conversion, message recipient limits, or agents. • **DEFER.** The message delivery was postponed and may be re-attempted later. • **RESOLVED.** The message was redirected to a new recipient address based on an Active Directory lookup. When this happens, the original recipient address is listed in a separate row in the message trace along with the final delivery status for the message.
Action	This field shows the action that was performed if the message was filtered due to a malware or spam detection or a rule match. For example, it will let you know if the message was deleted or if it was sent to the quarantine.
Detail	This field provides detailed information that elaborates on what happened. For example, it may inform you which specific mail flow rule (also known as a transport rule) was matched, and what happened to the message as a result of that match. It can also inform you which specific malware was detected in which specific attachment, or why a message was detected as spam. If the message was successfully delivered, it can tell you the IP address to which it was delivered.

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Message trace results for messages that are older than 7 days are provided in .CSV files. If you didn't include routing details when running the message trace, the following information is included in the .CSV file, which you can open in an application such as Microsoft Excel.

Details	Description
origin_timestamp	The date and time at which the message was received by the service, using the configured UTC time zone.
sender_address	The email address of the sender in the form alias@domain.
Recipient_status	The status of the delivery of the message to the recipient. If the message was sent to multiple recipients, it will show all the recipients and the corresponding status against each, in the format: <email address>##<status>. Message delivery statuses include: ● **##Receive, Send.** Indicates the message was received by the service and sent to the intended destination. ● **##Receive, Fail.** Indicates the message was received by the service but failed to be delivered to the intended destination. ● **##Receive, Deliver.** Indicates the message was received by the service and delivered to the recipient's mailbox.
message_subject	The subject line text of the message. If necessary, this is truncated to the first 256 characters.
total_bytes	The size of the message, including attachments, in bytes.
message_id	This is the Internet message ID (also known as the Client ID) found in the header of the message with the "Message-ID:" token. The form of this varies depending on the sending mail system. The following is an example: <08f1e-0f6806a47b4ac103961109ae6ef@server.domain>.
network_message_id	This is a unique message ID value that persists across copies of the message that may be created due to bifurcation or distribution group expansion. An example value is 1341ac7b13fb42ab4d-4408cf7f55890f.
original_client_ip	The IP address of the sender's client.
directionality	This field denotes whether the message was sent inbound (1) to your organization, or whether it was sent outbound (2) from your organization.

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Details	Description
connector_id	The name of the source or destination Send connector or Receive connector. For example, ServerName \ ConnectorName or Connector-Name.
delivery_priority	Denotes whether the message was sent with **High** , **Low** , or **Normal** priority.

If you included routing details when running the message trace, all information from the message tracking logs is included in the .CSV file, which you can open in an application such as Microsoft Excel.

Run a message trace

You can create message traces in the Security & Compliance Center, inside the Exchange Admin Center, and with the Exchange Management Shell.

Additional reading. For more information, see the following articles on **Message tracking**[2] and **Message Trace FAQ**[3].

2 https://aka.ms/AA3rbde
3 https://aka.ms/AA3qtt0

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Messaging Compliance in Exchange

Lesson Introduction

This lesson describes the compliance features available in the Exchange Admin Center for Exchange Server and hybrid deployments.

While compliance settings in the Security & Compliance Center only apply to messages stored in Exchange Online that are routed through Exchange Online / Exchange Online Protection, they can't be applied to messages in on-premises Exchange servers. If you are running a hybrid environment and you are still routing your mail flow over your on-premises servers, or if you still maintain mailboxes outside of Exchange Online, you need to combine the compliance features from the Exchange Admin Center with the modern compliance features from the SCC.

After this lesson, you will be able to:

- Describe litigation and in-place holds in Exchange Server
- Plan retention and deletion with Message Records Management (MRM)
- Protect your mail flow with data loss prevention policies in Exchange Server
- Investigate the message tracking log in your Exchange organization

In-place and Litigation holds

When a reasonable expectation of litigation exists, organizations are required to preserve electronically stored information, including email that's relevant to the case. This expectation often exists before the specifics of the case are known, and preservation is often broad; for example, organizations may need to preserve all email related to a specific topic or all email for certain individuals. Depending on the organization's electronic discovery (eDiscovery) practices, the following measures can be adopted to preserve email:

- End users may be asked to preserve email by not deleting any messages. However, users can still delete email knowingly or inadvertently.
- Automated deletion mechanisms such as messaging records management (MRM) may be suspended. This could result in large volumes of email cluttering the user mailbox, and thus impacting user productivity. Suspending automated deletion also doesn't prevent users from manually deleting email.
- Some organizations copy or move email to an archive to make sure it isn't deleted, altered, or tampered with. This increases costs due to the manual efforts required to copy or move messages to an archive, or third-party products used to collect and store email outside Exchange.

Failure to preserve email can expose an organization to legal and financial risks such as scrutiny of the organization's records retention and discovery processes, adverse legal judgments, sanctions, or fines.

Differences between in-place and litigation hold

There are two types of holds available in Exchange Server: Litigation Hold and In-Place Hold. When Litigation Hold is enabled, all mailbox items are placed on hold. In contrast, you can use an In-Place Hold to preserve only those items that meet that the criteria of a search query that you define by using the In-Place eDiscovery tool. You can place multiple In-Place Holds on a mailbox, but Litigation Hold is either enabled or disabled for a mailbox.

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Messaging administrators should be aware of the following facts concerning In-place and Litigation holds:

- Litigation hold uses the LitigationHoldEnabled property of a mailbox and holds all items.
- In-place hold works as search filter to only hold specific mailbox items.
- You can use In-Place hold to place a user on multiple holds.
- If you remove a litigation hold from a mailbox that also has in-place holds applied, the items in scope of the in-place hold will remain on hold.
- The maximum number of keywords in all query-based (In-place) holds placed on a mailbox is 500. If there are more than 500 keywords, then all content in the mailbox is placed on hold (not just that content that matches the search criteria). All content is held until the total number of keywords is reduced to 500 or less.
- When you put a mailbox on Litigation hold or In-Place hold, the hold is placed on both the primary and archive mailboxes.
- In-place holds are most likely used in context of eDiscovery, which requires in-place holds in litigation cases to prevent the deletion of case relevant data.

Hold goals and features

You can use Litigation Hold and In-Place Hold to accomplish the following goals:

- Place user mailboxes on hold and preserve mailbox items immutably.
- Preserve items indefinitely or for a specific duration.
- Preserve mailbox items deleted by users or automatic processes such as MRM.
- Preserve messages that are forwarded to another mailbox.
- Use query-based In-Place hold to search for and retain items matching specified criteria (you can also place all items hold by including all mailbox content when you create the hold).
- Place a user on multiple holds for different cases or investigations.
- Keep holds transparent from the user by not having to suspend MRM.
- Use In-Place eDiscovery to search for items that are preserved by being placed on hold.

In short, every time you need to prevent deletion of any mailbox item, you should place the entire mailbox on litigation hold. If you only need to prevent deletion of specific mailbox items, define an appropriate search string and hold only the items you need.

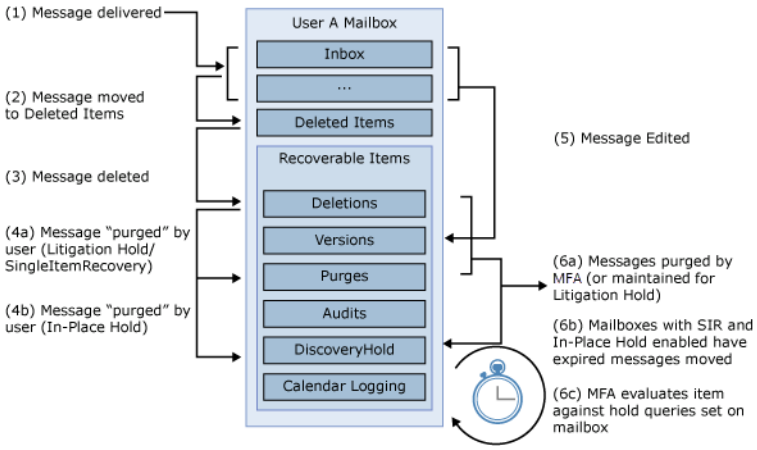
Holds and the Recoverable Items folder

Both Litigation hold and In-Place hold use the Recoverable Items folder to preserve items. The Recoverable Items folder is hidden from the default view of Outlook, Outlook on the web, and other email clients.

By default, when a user deletes a message from a folder other than the Deleted Items folder, the message is moved to the Deleted Items folder. When the mail is then removed from the deleted items folder, it is "soft deleted," which means that it's moved to the recoverable items folders.

Items in the Recoverable Items folder are retained for the deleted item retention period that's configured on the user's mailbox database. By default, the deleted item retention period is set to 14 days for mailbox databases.

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The Recoverable Items folder contains the following subfolders that are used to store deleted items in various sites and facilitate Litigation hold and In-Place hold.

Folder Name	Description
Deletions	Items removed from the Deleted Items folder or soft-deleted from other folders are moved to the Deletions subfolder and are visible to the user when using the Recover Deleted Items feature in Outlook and Outlook on the web. By default, items reside in this folder until the deleted item retention period configured for the mailbox database or the mailbox expires.
Purges	When a user deletes an item from the Recoverable Items folder (by using the Recover Deleted Items tool in Outlook and Outlook on the web), the item is moved to the Purges folder. Items that exceed the deleted item retention period configured on the mailbox database or the mailbox are also moved to the Purges folder. Items in this folder aren't visible to users if they use the Recover Deleted Items tool. When the mailbox assistant processes the mailbox, items in the Purges folder are purged from the mailbox database. When you place the mailbox user on Litigation hold, the mailbox assistant doesn't purge items in this folder.

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Folder Name	Description
DiscoveryHolds	If a user is put on an In-Place hold, deleted items are moved to this folder. When the mailbox assistant processes the mailbox, it evaluates messages in this folder. Items that match the In-Place hold query are retained until the hold period specified in the query. If no hold period is specified, items are held indefinitely or until the user is removed from the hold. If you put a user who was already on an In-Place hold on Litigation hold, the Litigation hold takes preference. Therefore, deleted items are moved to the Purges folder instead.
Versions	When a user is put on In-Place hold or Litigation hold, mailbox items must be protected from tampering or modification by the user or by a process. This is done by using a copy-on-write process. When a user or a process changes specific properties of a mailbox item, a copy of the original item is saved in the Versions folder before the change is committed. This process is repeated for subsequent changes. Items captured in the Versions folder are also indexed and returned in In-Place eDiscovery searches. After the hold is removed, copies in the Versions folder are removed by the Managed Folder Assistant.

Although the DiscoveryHolds, Purges, and Versions folders aren't visible to the user, all items in the Recoverable Items folder are discoverable by using In-Place eDiscovery. After a mailbox user is removed from In-Place hold or Litigation hold, items in the DiscoveryHolds, Purges, and Versions folders are purged by the Managed Folder Assistant.

If a mailbox isn't placed on Litigation hold or In-Place hold, items in the Purges folder are permanently deleted from the Recoverable Items folder on a first-in, first-out basis when the item has resided in the folder for longer than the deleted item retention period.

Holds and other features

Items in the Recoverable Items folder aren't included when calculating the user's mailbox quota. In Exchange, the Recoverable Items folder has its own quota. For Exchange, the default values for the RecoverableItemsWarningQuota and RecoverableItemsQuota mailbox properties are set to 20 GB and 30 GB, respectively. To modify these values for a mailbox database for Exchange Server, use the **Set-MailboxDatabase** cmdlet. To modify them for individual mailboxes, use the **Set-Mailbox** cmdlet.

When a user's Recoverable Items folder exceeds the warning quota for recoverable items (as specified by the RecoverableItemsWarningQuota parameter), an event is logged in the Application event log of the Mailbox server. When the folder exceeds the quota for recoverable items (as specified by the Recovera-

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bleItemsQuota parameter), users won't be able to empty the Deleted Items folder or permanently delete mailbox items. Also copy-on-write won't be able to create copies of modified items. Therefore, it's critical that you monitor Recoverable Items quotas for mailbox users placed on In-Place hold.

Users with mailboxes on Exchange Server can use Outlook and Outlook on the web to set up email forwarding for their mailbox. Email forwarding lets users configure their mailbox to forward received email messages to another mailbox located in or outside of their organization. Administrators can also set up transport rules to forward message to another mailbox. In both cases, email forwarding can be configured so that any message sent to the original mailbox isn't copied to that mailbox and is only sent to the forwarding address.

If email forwarding is set up for a mailbox and messages aren't copied, what happens if the mailbox is on hold? During the delivery process, the hold settings for the mailbox are checked. If the message meets the hold criteria for the mailbox, a copy of the message is saved to the Recoverable Items folder. That means you can use In-Place eDiscovery to search the original mailbox to find messages that were forwarded to another mailbox.

Deleting a mailbox on hold

If you delete a user account that has a mailbox, the Exchange Information store will eventually detect that the mailbox is no longer connected to a user account, in which case it will mark that mailbox for deletion, even if the mailbox is on hold. If you want to preserve the mailbox, you must perform the following tasks:

1. You must disable the user account rather than delete it.
2. Change the properties of the mailbox to restrict the use and access to the mailbox. For example, set send and receive quotas equal to 1, block who can send messages to the mailbox, and restrict who can access the mailbox.
3. Retain the mailbox until all data has been removed or until preserving the data is no longer required.

Migrating mailboxes on hold from Exchange Server to Microsoft 365

If you have an Exchange hybrid deployment, the following conditions are true when you move (onboard) an on-premises Exchange Server mailbox to Exchange Online in Microsoft 365:

- If the on-premises mailbox is on Litigation hold or In-Place hold, the hold settings are preserved after the mailbox is moved to Exchange Online.
- If the on-premises mailbox is on Litigation hold or In-Place hold, any content in the Recoverable Items folder is moved to the Exchange Online mailbox.
- Hold settings and content in the Recoverable Items folder are also preserved when you move (off-board) an Exchange Online mailbox to your on-premises Exchange Server organization.

Configuring Message Records Management

Messaging records management (MRM) helps organizations to manage the email lifecycle and reduce legal risks associated with email and other communications. MRM makes it easier to keep messages needed to comply with company policy, government regulations, or legal needs, and to remove content that has no legal or business value.

In Exchange Server, MRM is accomplished using retention tags and retention policies. Retention tags are used to apply retention settings to an entire mailbox and default mailbox folders such as Inbox and Deleted Items. You can also create and deploy retention tags that Outlook and Outlook on the web users

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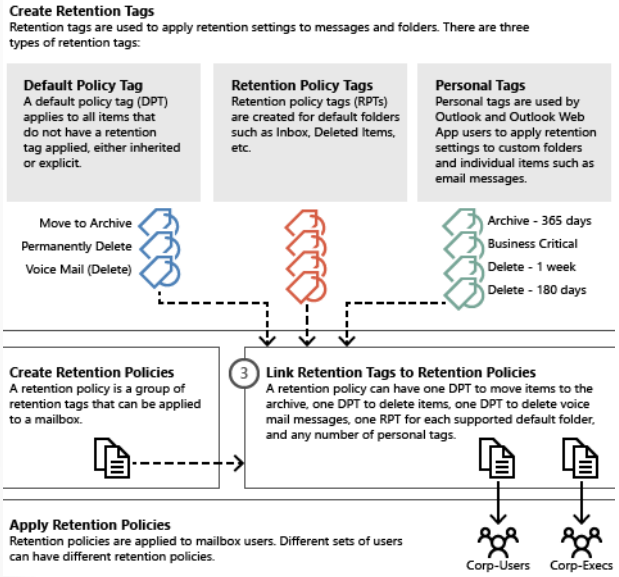
can use to apply to folders or individual messages. After they're created, you add retention tags to a retention policy and then apply the policy to users. The Managed Folder Assistant processes mailboxes and applies retention settings in the user's retention policy.

When a message reaches the retention age specified in the applicable retention tag, the Managed Folder Assistant takes the retention action specified by the tag. Messages can then be deleted permanently or deleted with the ability to recover them. If an archive has been provisioned for the user, you can also use retention tags to move items to the user's In-Place Archive.

Messaging Records Management features

Messaging records management consists of the following features:

- **Retention policies.** These policies are used to apply sets of retention tags to user mailboxes.
- **Retention policy tags (RPTs).** These are created for default folders, such as the Inbox and Deleted Items.
- **Default policy tags (DPTs).** Are used by mailboxes to manage the retention of all untagged items.
- **Personal tags.** Are used to apply retention settings to custom folders and individual items, such as email messages.



By using the features of MRM, you can implement different strategies for message retention, deletion, and general optimization tasks, as defined in the following table.

Retention strategy	Description
Remove all messages after a specified period	In this strategy, you implement a single MRM policy that removes all messages after a certain period. In this strategy, there's no classification of messages. You can implement this policy by creating a single default policy tag (DPT) for the mailbox. However, this doesn't ensure that messages are retained for the specified period. Users can still delete messages before the retention period is reached.

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Retention strategy	Description
Move messages to archive mailboxes	In this strategy, you implement MRM policies that move items to the user's archive mailbox. An archive mailbox provides additional storage for users to maintain old and infrequently accessed content. Retention tags that move items are also known as archive policies. Within the same retention policy, you can combine DPT and personal tags to move items, and DPT, RPTs, and personal tags to delete items.
Remove messages based on folder location	In this strategy, you implement MRM policies based on email location. For example, you can specify that messages in the Inbox are retained for one year and messages in the Junk Email folder are retained for 60 days. You can implement this policy by using a combination of RPTs for each default folder you want to configure and a DPT for the entire mailbox. The DPT applies to all custom folders and all default folders that don't have an RPT applied.
Allow users to classify messages	In this strategy, you implement MRM policies that include a baseline retention setting for all messages but allow users to classify messages based on business or regulatory requirements. In this case, users become an important part of your records management strategy - often they have the best understanding of a message's retention value. Users can apply different retention settings to messages that need to be retained for a longer or shorter period. You can implement this policy using a combination of the following: • A DPT for the mailbox • Personal tags that users can apply to custom folders or individual messages • Additional RPTs to expire items in specific default folders (Optional)
Retain messages for eDiscovery purposes	In this strategy, you implement MRM policies that remove messages from mailboxes after a specified period but also retain them in the Recoverable Items folder for In-Place eDiscovery purposes in Exchange Server, even if the messages were deleted by the user or another process.

You can meet this requirement by using a combination of retention policies and In-Place hold and Litigation hold in Exchange Server or Litigation hold. Retention policies remove messages from the mailbox after the specified period. A time-based In-Place hold or Litigation hold preserves messages that were deleted or modified before that period. For example, to retain messages for seven years, you can create a retention policy with a DPT that deletes messages in seven years and a Litigation hold to hold

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messages for seven years. Messages that aren't removed by users will be deleted after seven years; messages deleted by users before the seven-year period will be retained in the Recoverable Items folder for seven years.

Optionally, you can use RPTs and personal tags to allow users to clean-up their mailboxes. However, In-Place hold and Litigation hold continues to retain the deleted messages until the hold period expires.

Note: Retention policies and retentions tags of MRM are available in Exchange Server and Exchange Online equally. However, in Exchange Online you should primarily use them for clean-up purpose to move user messages to their in-place archive and use Retention Policies in the SCC for retention and compliance purposes. In Exchange Server they serve both tasks - the clean-up for performance and retention for compliance reasons.

Configuring Retention Policies and Retention Tags

To apply one or more retention tags to a mailbox, you need to add them to a retention policy and then apply the policy to mailboxes. A mailbox can't have more than one retention policy. Retention tags can be linked to or unlinked from a retention policy at any time, and the changes automatically take effect for all mailboxes that have the policy applied.

Note: Although a retention policy doesn't need to have any retention tags linked to it, this is not recommended. If mailboxes with retention policies don't have retention tags linked to them, this may cause mailbox items to never expire.

The following tables describe the available actions for retention tags and how the different actions are utilized in the different retention tag types.

Retention action	Action taken...	Except...
Move to archive	Moves the message to the user's archive mailbox. Only available for DPTs and personal tags.	If the user doesn't have an archive mailbox, no action is taken.
Delete and allow recovery	Emulates the behavior when the user empties the Deleted Items folder. Items are moved to the Recoverable Items folder in Exchange Server in the mailbox and preserved until the deleted item retention period. Provides the user a second chance to recover the item using the Recover Deleted Items dialog box in Outlook or Outlook on the web.	If you've set the deleted item retention period to zero days, items are permanently deleted.

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Retention action	Action taken...	Except...
Permanently delete	Permanently deletes messages. You can't recover messages after they're permanently deleted.	If the mailbox is placed on In-Place hold and Litigation hold in Exchange Server or Litigation hold, items are preserved in the Recoverable Items folder based on the hold parameters. In-Place eDiscovery in Exchange Server will still return these items in search results.
Mark as past retention limit	Marks a message as expired. In Outlook and Outlook on the web, expired items are displayed with the notification stating, 'This item has expired' and 'This item will expire in 0 days'.	not applicable

The following table describes the different types of retention tags that are available.

Type of retention tag	Applied...	Available actions...
Default policy tag (DPT)	Automatically to entire mailbox. A DPT applies to untagged items, which are mailbox items that don't have a retention tag applied directly or by inheritance from the folder.	Move to archive Delete and allow recovery Permanently delete
Retention policy tag (RPT)	Automatically to a default folder. Default folders are folders created automatically in all mailboxes. For example, Inbox, Deleted Items, and Sent Items.	Delete and allow recovery Permanently delete
Personal tag	Manually to items and folders. Users can automate tagging by using Inbox rules to either move a message to a folder that has a particular tag or to apply a personal tag to the message.	Move to archive Delete and allow recovery Permanently delete

Managed Folder Assistant

The Managed Folder Assistant is a mailbox assistant that runs on Mailbox servers and processes mailboxes that have a retention policy applied. The Managed Folder Assistant applies the retention policy by inspecting items in the mailbox and determining whether they're subject to retention. It then stamps items subject to retention with the appropriate retention tags and takes the specified retention action on items past their retention age.

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The Managed Folder Assistant is a throttle-based assistant. Throttle-based assistants are always running and don't need to be scheduled. The system resources they can consume are throttled. You can configure the Managed Folder Assistant to process all mailboxes on a Mailbox server within a certain period (known as a work cycle). Additionally, at a specified interval (known as the work cycle checkpoint), the assistant refreshes the list of mailboxes to be processed. During the refresh, the assistant adds newly created or moved mailboxes to the queue. It also reprioritizes existing mailboxes that haven't been processed successfully due to failures and moves them higher in the queue, so they can be processed during the same work cycle.

You can also use the **Start-ManagedFolderAssistant** cmdlet to manually trigger the assistant to process a specified mailbox.

Note: The default work cycle for the MFA in Exchange Server is once a day, while in Exchange Online it runs every 30 minutes.

Retention policies in Exchange hybrid

MRM retention policies are invaluable for managing the message lifecycle and retention in Exchange Server, but they only apply to messages that are stored in the same Exchange organization. Therefore, in Exchange hybrid you must manage retention policies in Exchange Server and Exchange Online simultaneously, and while you will use Retention Policies in the SCC for Exchange Online, you still need to manage retention in Exchange Server with MRM.

Note: When planning your retention strategy, you need to consider the limits of retention policies and how to ensure message retention and deletion across your on-premises and your Exchange Online organizations.

Retention policies and retention tags are not synchronized between your organizations, but there are two ways of transferring them manually:

- When running the Hybrid Configuration Wizard (HCW), you can utilize the Organization Configuration Transfer (OCT) to transfer all retention policies and tags to Exchange Online.
- You can use the Exchange scripts **Export-RetentionTags.ps1** and **Import-RetentionTags.ps1** from your Exchange installation directory to export and then import all MRM configuration of your organization.

Additional reading. For more information, see the following article on **Export and import retention tags**[4].

Configuring Data Loss Prevention Policies in the EAC

Data loss prevention (DLP) in the Exchange Admin Center works similar to data loss prevention in the SCC, except that it’s not able to protect as many different locations.

DLP policies are simple packages that are collections of mail flow rules (also known as transport rules) that contain specific conditions, actions, and exceptions that filter messages and attachments based on their content. DLP policies can use the full power of mail flow rules to detect and then act on messages in transit. For example, a mail flow rule can perform deep content analysis through keyword matches, dictionary matches, text pattern matches through regular expressions, and other content examination techniques to detect content that violates an organization's DLP policies. Document fingerprinting is also an available feature of DLP for Exchange to detect sensitive information in standard forms.

4 https://aka.ms/AA3r7pz

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Data loss prevention in Exchange Server includes the following considerations:

- DLP is a premium feature that requires an Exchange Enterprise Client Access License (CAL).
- In hybrid environments where some mailboxes are in on-premises Exchange and some are in Exchange Online, DLP policies are only applied in Exchange Online. Messages that are sent between on-premises users don't have DLP policies applied, because the messages don't leave the on-premises environment.
- Document fingerprinting is no longer available in Exchange Online, but it's still available for Exchange Server deployments.

Sensitive information types in DLP policies

Data loss prevention includes 80 sensitive information types that are ready for you to use in your DLP policies. A sensitive information type is defined by a pattern that can be identified by a regular expression or a function. In addition, corroborative evidence such as keywords and checksums can be used to identify a sensitive information type. Confidence level and proximity are also used in the evaluation process.

Additional reading. For more information, see the following article on **Sensitive information types in Exchange Server**[5].

Document fingerprinting

You can use document fingerprinting to identify a document format that should be monitored. If your organization has a standard form for an information type such as a customer record or financial reports, you can use document fingerprinting to allow DLP to recognize that information type. You can then prevent documents that match the fingerprint from being sent outside the organization.

Note: Document fingerprinting does not detect sensitive information if the file is password protected, contains only images, or does not contain all the text that was in the original fingerprinted document.

DLP policies in Exchange hybrid

In Exchange hybrid, DLP policies work through mail flow rules. As a result, they can only act on messages that are routed through the organization's transport servers.

Performing Message Tracking

The message tracking log is a detailed record of all activity as mail flows through the transport pipeline on Mailbox servers and Edge Transport servers. You can use message tracking for message forensics, mail flow analysis, reporting, and troubleshooting.

By default, Exchange uses circular logging to limit the message tracking log based on file size and file age; this helps control the hard disk space that's used by the log files.

Note: Investigating the message tracking logs is available in on-premises Exchange Server deployments only. To investigate message tracking in Exchange Online, you need to use Message Traces instead.

5 https://aka.ms/AA3r05a

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Structure of the message tracking log files

By default, the message tracking log files exist in the following path:

- Path: **%ExchangeInstallPath%TransportRoles\Logs\MessageTracking**
- Log file naming syntax: **MSGTRKServiceyyyymmdd-nnnn.log**

File name	Servers	Description
MSGTRK	Mailbox servers and Edge Transport servers	Log files for the Transport service.
MSGTRKMA	Mailbox servers	Log files for the approvals and rejections in moderated transport.
MSGTRKMD	Mailbox servers	Log files for messages delivered to mailboxes by the Mailbox Transport Delivery service.
MSGTRKMS	Mailbox servers	Log files for messages sent from mailboxes by the Mailbox Transport Submission service.

The message tracking log stores each message event on a single line in the log. The message event information is organized by fields that are separated by commas. The field name is generally descriptive enough to determine the type of information that it contains. However, some fields may be blank, or the type of information in the field may change based on the message event type and the service that recorded the event.

Additional reading. General descriptions of the fields that are used to classify each message tracking event are explained in the following article titled **Message tracking**[6].

Configure message tracking

You cannot modify the message trace settings because they are only available as a service by Microsoft. However, you can modify the message tracking log settings on your Exchange Servers.

You can use the **Set-TransportService** cmdlet in the Exchange Management Shell on Mailbox servers and Edge Transport servers for all message tracking configuration tasks, such as:

- Enable or disable message tracking (default: enabled)
- Specify the location of the message tracking log files
- Specify a maximum size for the individual message tracking log files (default: 10 MB)
- Specify a maximum size for the directory that contains the message tracking log files (default: 1000 MB)
- Specify maximum age for the message tracking log files (default: 30 days)
- Enable or disable message subject logging in the message tracking logs (default: enabled)

No message content is stored in the message tracking log. By default, the subject line of an email message is stored in the message tracking log. You may need to disable subject logging to comply with increased security or privacy requirements.

6 https://aka.ms/AA3r7q3

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Note: On Mailbox servers, you can also use the Exchange admin center (EAC) to enable or disable message tracking, and to specify the location of the message tracking log files.

Search message tracking logs

Message tracking records the message activity as mail flows through the transport pipeline on Mailbox servers and Edge Transport servers. You can use the **Get-MessageTrackingLog** cmdlet in the Exchange Management Shell to search for entries in the message tracking log by using specific search criteria.

You can use the message tracking log to perform the following tasks:

- Find out what happened to a message that was sent by a user to a specific recipient.
- Find out if a transport rule acted on a message.
- Find out if a message sent from an Internet sender made it into your Exchange organization.
- Find all messages sent by a specified user during a specified time period.

The following example displays the command to search the message tracking log on a single server named "Mailbox01":

```
Get-MessageTrackingLog -Server Mailbox01 -Start "03/13/2018 09:00:00" -End
"03/15/2018 17:00:00" -Sender "john@contoso.com"
```

The following example displays the command to search all messages in the tracking log related to the MessageId "ba18339e-8151-4ff3-aeea-87ccf5fc9796@mailbox01.contoso.com":

```
$Servers = Get-ExchangeServer; $Servers | where {$_.isHubTransportServer
-eq $true -or $_.isMailboxServer -eq $true} | Get-MessageTrackingLog -Mes-
sageId ba18339e-8151-4ff3-aeea-87ccf5fc9796@mailbox01.contoso.com | Se-
lect-Object Timestamp,ServerHostname,ClientHostname,Source,EventId,Recipi-
ents | Sort-Object -Property Timestamp
```

The following example displays the command to search the message tracking log on all Exchange servers:

```
Get-ExchangeServer | Get-MessageTrackingLog -Start "03/13/2018 09:00:00"
-End "03/15/2018 17:00:00" -Sender "john@contoso.com"
```

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Managing Exchange Online Archiving and Auditing

Lesson Introduction

Because of the complex retention requirements of modern messaging environments, it's essential to understand how archiving is performed with Exchange so that you can provide an efficient and compliant environment to your users. You must also understand how additional archive storage is provided to your users and how messages are automatically processed and archived.

Another key aspect of messaging compliance is the audit logging in Exchange that provides information about administrator, delegate, and user actions in user mailboxes and your Exchange organization.

After this lesson, you will be able to:

- Describe what in-place archiving is and how it works
- Understand the differences between journaling and archiving
- Know what the mailbox and administrator audit logs are used for

Planning and implementing in-place archiving

In-Place Archiving in Exchange helps organizations regain control of their messaging data by eliminating the need for personal store (.pst) files and allowing users to store messages in an archive mailbox. The archive mailbox is an additional mailbox that's enabled for a user's primary mailbox. The archive mailbox is accessible in Outlook and Outlook on the web. Each user can view his or her archive mailbox and move or copy messages between their primary mailbox and their archive mailbox.

Messaging administrators have several choices when it comes to locating archive mailboxes. You can provision a user's archive mailbox on:

- the same mailbox database as the user's primary mailbox.
- a different mailbox database on the same Mailbox server.
- on a mailbox database on a different Mailbox server in the same Active Directory site.

The mailbox database for the archive mailbox is chosen automatically in Exchange Online. In hybrid deployments, you can also provision a cloud-based archive mailbox for primary mailboxes located in your on-premises Exchange organization.

In-Place Archiving provides the following advantages:

- You can avoid client issues with very large mailboxes by decreasing the size of the cached, primary mailbox.
- Users will experience improved client performance when working with smaller primary mailboxes and when searching for recent data.
- Because archive mailboxes for on-premises users in a hybrid deployment can already exist in Exchange Online, on-premises Exchange Servers and databases are relieved, and the users don't see a difference when working with their on-premises primary mailbox and their cloud archive. In other words, the local server isn't taxed with the data load of the archive mailbox because it doesn't have to ping that specific data load every time the user accesses his or her mailbox, thereby relieving some of the network data traffic on the on-premises environment.

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- Archive mailboxes enable organizations to eliminate legacy (.pst) files that are often damaged on file servers or lost on local storage.
- Archive mailboxes also provide advantages to stay compliant with regulatory requirements, because received emails can stay in the user's mailboxes, and by utilizing other features, all versions are saved and all user correspondence is searchable through eDiscovery.

Implementing and managing in-place archiving is a process that requires considerable planning; therefore, messaging administrators must understand how in-place archives work. For example, if you want to deploy archives in your Exchange Server deployments, what will be the increased effect on your storage requirements? Or how will your organization's licensing be affected for Exchange Server and Exchange Online?

When planning for In-Place Archiving, you must consider the following:

- An in-place archive technically works as an additional mailbox for a user that already has a primary mailbox connected to your messaging system.
- In-place archives can be accessed using Outlook for Windows, Outlook for Mac, or Outlook on the web, but only when connected to Exchange because no offline access is possible.
- As soon as a user is activated for an in-place archive, the default MRM retention policies will move all the user's messages that are older than two years to the archive mailbox.
- To use archives in Exchange Server deployments, you need enterprise licenses and enterprise client access licenses (CALs).
- To use archives in Exchange Online, you need Exchange Online Plan 2 or an Exchange Online Plan 1 license and an additional Exchange Online Archiving license.
- From a technical perspective, an archive mailbox is a secondary mailbox that's connected to a user that already has a primary mailbox. This secondary mailbox can reside in a different database than the primary mailbox, and in a hybrid deployment, it can even reside in Exchange Online while the primary mailbox remains on-premises.
- Auto-expanding archives are available in Exchange Online, and they automatically grow when the first 100 GB limit is reached. When that occurs, additional mailbox space is provisioned to the user; however, there may be a delay of up to 30-days until the additional storage is available.
- Once auto-expanding archiving is activated, it cannot be turned off. In addition, access to an auto-expanding archive is only possible with Outlook 2016 and Outlook 2019.
- When an archive mailbox exceeds the specified archive quota, messages are no longer moved to the archive, a warning event is logged in the Application event log, and a quota message is sent to the mailbox user.

Additional reading. For more information, see the following article on **Overview of unlimited archiving in Office 365**[7] and **In-Place Archiving in Exchange Server**[8]

Moving messages to the archive mailbox

There are several ways to move messages from a user's primary mailbox to their archive mailbox:

- **Move or copy messages manually.** Users can manually move or copy messages from their primary mailbox or a mounted (.pst) file to their archive mailbox in their Outlook client. The archive mailbox is

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8 https://aka.ms/AA3r05o

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displayed like any other mailbox in Outlook and Outlook on the web, and it supports drag and drop functionality.

- **Move or copy messages using Inbox rules.** Users can create Inbox rules in Outlook or Outlook on the web to automatically move messages to a folder in their archive mailbox.
- **Move messages using retention policies.** The default MRM retention policy moves messages to the in-place archive as soon as their creation timestamp exceeds two years. You can also apply personal tags to move messages to the archive mailbox.
- **Import messages from (.pst) files.** In Exchange Server, administrators can also use import requests to import messages from a (.pst) file to a user's archive or primary mailboxes. It's also possible to import mailbox items from (.pst) files by using the Import function in the Security & Compliance Center.

Archive policies with Message Records Management are implemented by creating retention tags that use the **Move to Archive** retention action.

Messages are moved to a folder in the archive mailbox that has the same name as the source folder in the primary mailbox. If a folder with the same name doesn't exist in the archive mailbox, it's created when the Managed Folder Assistant moves a message. Re-creating the same folder hierarchy in the archive mailbox allows users to easily find messages.

The Default MRM Policy contains retention tags with various **Move to Archive** actions as defined in the following table.

Retention tag name	Tag type	Description
Default 2 year move to archive	Default (DPT)	Messages are automatically moved to the archive mailbox after two years. Applies to items in the entire mailbox that don't have a retention tag applied explicitly or inherited from the folder.
Personal 1 year move to archive	Personal	Messages are automatically moved to the archive mailbox after one year.
Personal 5 year move to archive	Personal	Messages are automatically moved to the archive mailbox after five years.
Personal never move to archive	Personal	Messages are never moved to the archive mailbox.
Recoverable Items 14 days move to archive	Recoverable Items Folder	Messages are moved from the Recoverable Items folder in the user's primary mailbox to the Recoverable Items folder in the archive mailbox. Users attempting to recover deleted items in their archive mailbox must use the Recover Deleted Items tool in the archive mailbox.

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Managing in-place archiving

To enable archive mailboxes for users that already have a primary mailbox, you can use the Exchange admin center, the Exchange Management Shell, or for Microsoft 365, the Security & Compliance Center.

In Exchange, creating and managing archive mailboxes is integrated with common mailbox management tasks, such as:

- **Creating an archive mailbox.** You can enable an archive mailbox for an existing mailbox, or you can create an archive mailbox when creating a new mailbox.
- **Moving an archive mailbox.** You can move a user's archive mailbox to another mailbox database on the same Mailbox server or to another server, independent of the primary mailbox. To move a user's archive mailbox, you must create a mailbox move request. For details, see **Manage on-premises mailbox moves in Exchange Server**[9].
- **Disabling an archive mailbox.** You may want to disable a user's archive mailbox for troubleshooting purposes or if you're moving the primary mailbox to a version of Exchange that doesn't support In-Place Archiving. Disabling an archive is like disabling a primary mailbox. In on-premises deployments, a disabled archive mailbox is retained in the mailbox database until the deleted mailbox retention period for that database is reached. During this period, you can reconnect the same disabled archive mailbox to a user's primary mailbox. When the deleted mailbox retention period is reached, the disconnected archive mailbox is purged from the mailbox database.
- **Retrieving mailbox statistics and folder statistics.** You can retrieve mailbox statistics and mailbox folder statistics for a user's archive mailbox by using the Archive switch with the **Get-MailboxStatistics** and **Get-MailboxFolderStatistics** cmdlets.
- **Testing archive connectivity.** In Exchange Server, you can use the **Test-ArchiveConnectivity** cmdlet to test connectivity to a specified user's on-premises or cloud-based archive mailbox.

Introduction to Journaling

Journaling in Exchange can help your organization respond to legal, regulatory, and organizational compliance requirements by recording all or targeted email messages. Journaling in Exchange Server is basically unchanged since Exchange Server 2010.

It's important to understand the difference between journaling and archiving when it comes to email messages:

- **Journaling** refers to recording email communications as part of the organization's email retention strategy.
- **Archiving** refers to removing email messages from their native location (for example, a user's primary mailbox), and storing them elsewhere.

Many organizations need to maintain records of the email communication that occurs as employees perform their daily business tasks. You can use Exchange journaling as a tool in your email retention or archival strategy.

Although a regulation may not specifically require journaling, Exchange journaling can help your organization achieve compliance with the regulation. For example, corporate officers in some financial sectors can be held liable for claims that are made by their employees to customers. Designated compliance managers can use journaling to collect and regularly review the email messages that are sent by employees to customers as part of their greater employee-to-customer communications review. The compliance

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managers can report their approval to the corporate officer, and the corporate officer can then report compliance to the regulating body.

Journaling in Exchange consists of the features identified in the following table.

Journaling feature	Description
Journaling agent	The Journaling agent is the built-in Exchange transport agent that processes messages as they flow through the Transport service on Mailbox servers. The journaling configuration settings are stored in Active Directory and are read by the Journaling agent.
Journal rule replication	Because journal rules are stored in Active Directory, they're read and applied by the Transport service on all Mailbox servers in the organization. When you create, modify, or remove a journal rule, the change is replicated between the domain controllers in your organization. This allows Exchange to provide a consistent set of journal rules across the organization.
Journal reports	A journal report is the message that the Journaling agent generates when a message matches a journal rule and is to be submitted to the journaling mailbox. The original message that matches the journal rule is included unaltered as an attachment to the journal report. The body of a journal report contains information from the original message such as the sender email address, message subject, message-ID, and recipient email addresses.
Journal rules (Journal recipient)	The journal recipient specifies who you want to journal. Messages that are sent to or received by the journal recipient are journaled (the direction doesn't matter). You can configure a journal rule to journal messages for all senders and recipients in the Exchange organization, or you can limit a journal rule to an Exchange mailbox, group, mail user, or mail contact. If you specify a distribution group, you enable journaling for the members of the distribution group (not for the group itself). **Note**: In Exchange Online you can only configure Journaling on a per-user basis and not organization wide.

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Journaling feature	Description
Journal rules (Journal rule scope)	After you define who you want to journal, you need to define the scope of the messages to journal. Possible scopes are: • **Internal messages only.** The source or destination of the message is inside your Exchange organization. • **External messages only.** The source or destination of the message is outside your Exchange organization. • **All messages.** The source or destination of the message doesn't matter.
Journal rules (Journaling mailbox)	The journaling mailbox is used to collect journal reports. How you configure the journaling mailbox depends on your organization's policies, regulatory requirements, and legal requirements. You can specify one journaling mailbox to collect messages for all the journal rules configured in the organization, or you can use different journaling mailboxes for different journal rules or sets of journal rules. **Note**: In Microsoft 365, you can't designate an Exchange Online mailbox as a journaling mailbox. You can deliver journal reports to an on-premises archiving system or a third-party archiving service. If you're running an Exchange hybrid deployment with your mailboxes split between on-premises servers and Microsoft 365, you can designate an on-premises mailbox as the journaling mailbox for your Exchange Online and on-premises mailboxes.

Considerations for implementing Journaling

Messaging administrators should take into account the following considerations before activating Journaling:

- **Journal reports and IRM-protected messages.** When implementing journaling, you must consider journaling reports and IRM-protected messages. IRM-protected messages affect the search and discovery capabilities of third-party archiving systems that don't have RMS support built-in. In Microsoft 365, you can configure Journal Report Decryption to save a clear-text copy of the message in a journal report.
- **Alternate journaling mailbox.** When the journaling mailbox is unavailable, you'll want to avoid having undeliverable journal reports collect in mail queues on Mailbox servers. Instead, you can configure an alternate journaling mailbox to store those journal reports. The alternate journaling mailbox receives the journal reports as attachments in the non-delivery reports (NDRs) generated when the journaling mailbox or the server on which it's located refuses delivery of the journal report or becomes unavailable.

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- **Storage requirements for journaling mailbox.** If the Journaling mailbox becomes unavailable, the alternate Journaling mailbox can take over. However, if the alternate mailbox also becomes unavailable, the rejected journal reports are lost and can't be retrieved.
- **Directly addressed messages.** The alternate journaling mailbox should be treated as a special dedicated mailbox. Any messages addressed directly to the alternate journaling mailbox aren't journaled.
- **Maximum journal rules.** In Microsoft 365, the maximum number of journal rules that you can create is 10.

Managing admin and mailbox auditing

Audit logging is an essential component of any messaging compliance solution. It plays a key role in troubleshooting configuration issues because it can track specific changes made by administrators, and it helps organizations meet regulatory, compliance, and litigation requirements. Exchange provides two types of audit logging:

- **Administrator audit logging.** Administrator auditing records any action, based on an Exchange PowerShell cmdlet, performed by an administrator. This can help you troubleshoot configuration issues or identify the cause of security-related or compliance-related problems. In Exchange Online, actions performed by Microsoft administrators and delegated administrators are also recorded.
- **Mailbox audit logging.** Mailbox auditing records whenever a mailbox is accessed by an administrator, a delegated user, or the person who owns the mailbox. This can help you determine who has accessed a mailbox and what they've done.

Both audit logs are of great value to track changes inside mailboxes and around the organization and individual recipient configuration in Exchange.

Administrator and mailbox audit logs provide great value in a variety of real-world scenarios, including:

- When users report that mailbox items are missing or were moved without their knowledge, you can check the mailbox audit log to see what happened and who performed the action. This can be very helpful when users forget that an inbox rule exists that moves specific items automatically, or they forget about delegates that still have access to their mailbox. With the mailbox audit log, you can help users understand who performed the unexpected actions inside their mailboxes.
- When configuration changes cause issues with the mail flow, you can check the administrator audit log to see who changed something and what they did. With that information, you can undo incorrect administrator actions and identify the administrator who performed them.
- In some cases, both audit logs work together to help solve issues. For example, consider the scenario where an administrator grants himself access to a mailbox and then removes some items from that mailbox. The administrative action of granting access is logged in the administrator audit log, and the changes that were made in the mailbox are logged in the mailbox audit log.

Differences between Exchange Server and Exchange Online

While the audit logging technology that’s used is the same for Exchange Server and Exchange Online, there are some important differences you need to consider:

- Administrator audit logging is always activated by default, no matter if you are working in Exchange Server or Exchange Online.

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- Mailbox audit logging is activated by default in Exchange Online, but it must be activated manually per mailbox in Exchange Server. You can't use the Exchange admin center (EAC) to enable or disable mailbox audit logging; instead, you must use the Exchange Management Shell to enable mailbox audit logging.
- If you don't activate the mailbox audit log in Exchange Server before you need to search it, the log file will be empty when you try to search it.
- Both audit logs can be searched in the Exchange Admin Center and through the Exchange Admin Shell.
- For Microsoft 365, the administrator and mailbox audit logs are both part of the unified auditing strategy. You need to activate unified audit log recording in the Security & Compliance Center before you can use the ***-UnifiedAuditLog** cmdlets.
- By default, the logs are available in Exchange Online for 90 days. However, with an Office 365 E5 license, you can extend the time to one year.

Additional reading. For more information, see the following article on **Search the audit log in the Office 365 Security & Compliance Center**[10].

Auditing reports in the EAC

The following reports are available on the Auditing page in the EAC. The results are displayed in the details pane of the report.

Report name	Description	Use this report to...
Run a non-owner mailbox access report	Searches mailbox audit logs for mailboxes that have been opened by someone other than the owner. You must enable mailbox audit logging for each mailbox that you want to run a non-owner mailbox access report for. If mailbox audit logging isn't enabled for a mailbox, you won't get any results for it when you run this report.	Use this report to find mailboxes that have been accessed by someone other than the person who owns the mailbox.
Run an administrator role group report	Searches the administrator audit log for changes made to role groups, which are used to assign administrative permissions to users.	Use this report to search for changes made to administrator role groups.
Run an in-place eDiscovery and hold report	Searches the administrator audit log for changes made to In-Place eDiscovery searches and In-Place holds.	Use this report to find mailboxes that have been put on, or removed from, In-Place hold.

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Report name	Description	Use this report to…
Run a per-mailbox Litigation Hold report	Searches the administrator audit log to determine if a Litigation hold was enabled or disabled for a user's mailbox.	Use this report to find mailboxes that were put on, or removed from, Litigation hold.
Run the administrator audit log report	Views entries from the administrator audit log about configuration changes made by administrators in your organization.	Use this report to view entries from the administrator audit log. Instead of exporting the administrator audit log, which can take up to 24 hours to receive in an email message, you can run this report in the EAC. This report records configuration changes made by administrators in your organization. Up to 5000 entries will be displayed on multiple pages. To narrow the search, you can specify a date range.
Run the external administrator audit log report	Views entries from the administrator audit log about configuration changes made to your Exchange Online services by Microsoft or by a delegated administrator.	This report is only available in Exchange Online and Exchange Online Protection. Actions performed by Microsoft administrators or delegated administrators are logged in the administrator audit log. Use the external administrator audit log report to search for and view the actions that administrators outside your organization performed on your Exchange Online deployment.

Run a non-owner mailbox access report

The **Non-Owner Mailbox Access** report in the Exchange admin center (EAC) lists the mailboxes that have been accessed by someone other than the person who owns the mailbox. When a non-owner accesses a mailbox, Microsoft Exchange logs information about this action. Exchange stores this mailbox audit log as an email message in a hidden folder in the audited mailbox. The **Non-Owner Mailbox Access** report displays entries from this log file for any mailboxes that were accessed by a non-owner. For each entry, the report displays the mailbox, who accessed it, when the access occurred, and whether or not the action was successful.

Exchange logs specify actions by both owners and non-owners, including administrators and users who have been assigned permissions to a mailbox they don't own (who are referred to as delegated users). You can also narrow the search to users inside or outside your organization. By default, Exchange retains entries in the mailbox audit log for 90 days.

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Managing Content Search

Lesson Introduction

If your organization adheres to legal discovery requirements (related to organizational policy, compliance, or lawsuits), eDiscovery for Microsoft Exchange can help you perform discovery searches for relevant content within mailboxes.

In-place eDiscovery is the basic technology to perform compliance searches across your organization. When using Exchange Online, eDiscovery is implemented in a much more powerful way. It can be used to not only search Exchange recipient objects, but to also perform unified searches across all Microsoft 365 services. Nevertheless, eDiscovery is still based on the in-place eDiscovery feature. The different types of searches that can be performed in the Microsoft 365 Security & Compliance Center, and which are covered in this lesson, include:

- Content searches
- eDiscovery cases
- Advanced eDiscovery

After this lesson, you will be able to:

- Understand content searches to search for messages in your organization
- Describe eDiscovery cases and in-place eDiscovery for Exchange
- Manage Advanced eDiscovery cases in the Security & Compliance Center

Designing content searches

Content searches are a basic part of eDiscovery in both on-premises and web-based implementations. However, content searches in web-based implementations are much more powerful as they can search across all Microsoft 365 services. Search performance is very fast with few limitations, which makes it ideal for quick and very large search operations.

There are several situations where a simple and fast search operation is very useful; for example, if you need to bulk delete certain emails or other potentially harmful or high-risk messages from different mailboxes.

You must be a member of the eDiscovery Manager role group in the Security & Compliance Center to perform searches and to preview and export search results in the Content search page. You don't have to assign additional search permissions in Exchange Online.

Note: Because "search and purge" is a powerful feature that allows anyone who is assigned the necessary permissions to delete email messages from all mailboxes in your organization, you must be very careful when assigning this role.

Limitations for content search

Content search is not limited in the number of mailboxes it can search, nor is there a limit in the number of concurrent content searches in your organization. There is only a limit of 10 searches a single user can run at the same time and various limitations for the preview of your search results. However, those limits only apply when previewing search results prior to the results being exported.

A Content Search that also deletes items is known as a "search and purge." This type of search, which is limited to 50,000 mailboxes, is accomplished by using the **New-ComplianceSearchAction -Purge**

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command. If a Content Search that includes a purge action is performed on more than 50,000 mailboxes, then the purge action will fail.

Note: The limit of 10 searches that a single user can start simultaneously is oftentimes hit when the user tries to start multiple searches by using the **Get-ComplianceSearch** | **Start-ComplianceSearch** command in the Security & Compliance Center PowerShell. You should keep this limitation in mind when planning content searches.

Planning cloud-based mailbox searches for on-premises users in Microsoft 365

If your organization has an Exchange hybrid deployment and has enabled Microsoft Teams, users can use the Teams chat application for instant messaging. For both cloud-based and on-premises users, the Teams chat data (also called 1xN chats) is saved to the user's primary mailbox. For cloud-based users, their primary mailbox is located in the cloud, whereas the primary mailbox for on-premises users is stored on-premises. This dual location of primary mailboxes limited searches of Teams chat data to cloud-based users only because the Content Search tool in the Microsoft 365 Security & Compliance Center only targets cloud-based mailboxes.

To address this limitation, Microsoft has released a new feature where a cloud-based storage area (called a cloud-based mailbox for on-premises users) is created to store Teams chat data for on-premises users. This lets you use the Content Search tool in the Microsoft 365 Security & Compliance Center to search and export Teams chat data for both on-premises and cloud-based users.

Messaging administrators must take into account the following considerations when setting up and searching cloud-based mailboxes for on-premises users:

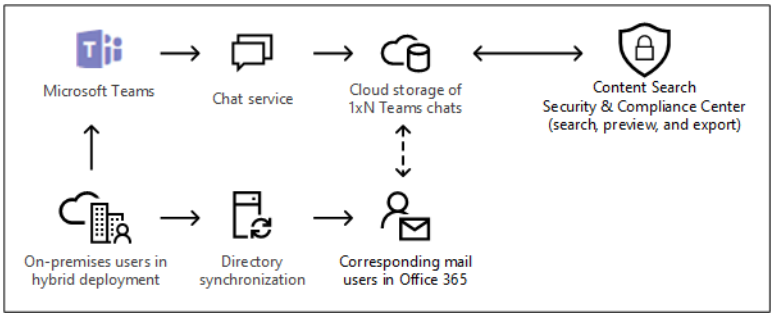
- The user accounts in your on-premises directory service (such as Active Directory) must be synchronized with Azure Active Directory, which is the directory service in Microsoft 365. This means that a mail user account is created in Microsoft 365 for each user whose primary mailbox is in the on-premises organization.
- The cloud-based mailbox for on-premises users is only used to store Teams chat data. An on-premises user cannot sign in to the cloud-based mailbox or access it in any way; for example, it can't be used to send or receive email messages.
- Placing a cloud-based mailbox for an on-premises user on a hold associated with an eDiscovery case or assigning it to a Microsoft 365 retention policy is not supported.
- The content location picker for eDiscovery holds displays on-premises users and will let you select them, even though an eDiscovery hold cannot be applied to the cloud-based mailboxes for selected on-premises users.
- You must submit a request to Microsoft Support to enable your organization to search for Teams chat data in the cloud-based mailboxes for on-premises users.

Note: Teams channel conversations are always stored in the cloud-based mailbox that's associated with the Team; therefore, you can use Content Search to search channel conversations without having to file a support request.

Teams chat data is indexed for search after it's stored in the cloud-based mailbox. This lets you use Content Search (and searches associated with eDiscovery cases) to search, preview, and export Teams chat data for on-premises users. You can also use the **ComplianceSearch** cmdlets in the Microsoft 365 Security & Compliance Center PowerShell to search for Teams chat data for on-premises users.

The following graphic shows the workflow of how Teams chat data for on-premises users is made available for search, preview, and export.

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Designing Your Content Search

You should consider the following questions when designing for a content search:

- Who should create and run the content search?
- What type of content search do you want to create (i.e., New search, Guided search, Search by ID list)?
- What keywords should be used for the search?
- What conditions should be used (for example, type of data, sender, date, subject, and so on)?
- Do you want to search all locations, or only specific locations (for example, Exchange Online, SharePoint Online, Microsoft Teams, and so on)?
- Which action, if any, should be performed on the items that are found (for example, soft delete, purge, and so on)?

Content Search is useful when you need to perform large searches, or when you simply need to search for specific data in your organization without needing to apply an eDiscovery Hold on the content or create an eDiscovery case.

After you run a Content Search, the number of content locations and an estimated number of search results are displayed in the details pane on the Content search page in the SCC. After you run a Content Search you can preview the results, export the results to a local computer, or prepare the results for analysis.

Additional reading. For more information, see the following article on **Limits for Content Search in the Office 365 Security & Compliance Center**[11].

Managing content searches

Content searches can be created within the Security & Compliance Center or through PowerShell by utilizing the **New-ComplianceSearch** cmdlet. It enables you to run organization-wide searches of all mailboxes and other destinations by using Windows PowerShell. By using the Compliance Search cmdlets, you can create scripts to integrate Content Searches into your eDiscovery workflow. For example, you could regularly run a Content Search to search all mailboxes in your organization to identify those that contain specific search results.

Cloning a Content Search

Although enhanced by scalability and performance improvements, searching a large number of mailboxes or SharePoint sites when using Content Search may still take a while to complete. Cloning a Content

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Search can be used to break up a large, single search in to multiple, smaller searches that run more quickly.

When a you clone a search, a new search is created with a different name that contains the same properties as the original search. You can subsequently edit the new search by changing various properties, such as the keyword query or the date range, before running the search.

Cloning a Content Search may also be advantageous when you need to:

- Compare the results of different keyword search queries run on the same content locations.
- Re-enter many content locations when creating a new search, thereby saving time and effort.
- Decrease the size of the search results. For example, if you have a search that returns too many results to export, you can clone the search and then add a search condition based on a date range to reduce the number of search results.

Cloning a Content Search is accomplished by copying and running the script that appears in the **Clone a Content Searchpage in the Office 365 Security & Compliance Center**[12]. This page includes a list of prerequisites to running the script, a copy of the script itself, and instruction on editing and running the script.

Once you run the script to create the clone, you can edit and run the new search in the Security & Compliance Center.

Searching for and deleting email messages

Occasionally, you may need to find and remove potentially harmful or high-risk email such as:

- Messages that contain a dangerous attachment or virus
- Phishing messages
- Messages that contain sensitive data

You can use Content Search to search for and delete email messages from all the mailboxes in your organization.

The first step is to create and run a Content Search to find the messages that you want to remove from mailboxes in your organization. You can create the search by using the Content Search feature in the Security & Compliance Center or by running the **New-ComplianceSearch** and **Start-ComplianceSearch** cmdlets. The messages that match the query for this search are deleted by running the **New-ComplianceSearchAction** cmdlet.

Important: When you run the **New-ComplianceSearchAction** cmdlet, the content locations that are searched in the Content Search must include only mailboxes and public folders. You cannot include other locations in a search that deletes emails.

After you have created and refined a Content Search to return the message that you want to remove, the final step is to connect to the Security & Compliance Center with remote PowerShell and then run the **New-ComplianceSearchAction** cmdlet to delete the message. Deleted messages are moved to a user's Recoverable Items folder.

Additional reading. For more information, see the following article on **Search for and delete email messages in your Office 365 organization - Admin Help**[13].

12 https://go.microsoft.com/fwlink/?linkid=839274
13 https://aka.ms/AA3rizm

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Managing eDiscovery cases

To comply with regulatory and legal requirements, the features that exist in In-place eDiscovery and eDiscovery cases refer to the same technique of searching through multiple mailboxes, Exchange recipient objects, and even other Microsoft 365 services.

- **In-place eDiscovery.** This feature in Exchange is available in the EAC and through the Exchange Management Shell. It's used to search for certain messaging objects, put them on In-place hold, and process them for further use by exporting them.
- **eDiscovery case.** This feature is available in the Microsoft 365 Security & Compliance Center (SCC) and serves the same purpose as In-place eDiscovery, but with a different web interface and the ability to search across all Microsoft 365 services in a single, unified search operation.

Note: When creating an eDiscovery case in the SCC, the searches you create are actually content searches that are only visible from inside the case you created. You won't be able to see those content searches on the Content search page in the SCC.

Differences for eDiscovery in Exchange deployments

When you are working with Exchange Online only or in hybrid deployment, you will use the more unified eDiscovery search approach and create eDiscovery cases in the Security & Compliance Center. This provides a better search experience by arranging all data in search cases, and search operations can search across all Microsoft 365 services.

Note: In Exchange Server deployments you can still perform eDiscovery searches with the **New-MailboxSearch** PowerShell cmdlet. This cmdlet is no longer available in Exchange Online, and you must instead use the **New-ComplianceSearch** and **New-CaseHoldPolicy** cmdlets from the SCC to create new eDiscovery cases.

In a hybrid deployment, where some mailboxes exist on your on-premises Exchange Servers and some mailboxes exist in Exchange Online, you can also perform In-Place eDiscovery searches of your cloud-based mailboxes using the EAC in your on-premises organization. If you intend to copy messages to a discovery mailbox, you must select an on-premises discovery mailbox. Messages from cloud-based mailboxes that are returned in search results are copied to the specified on-premises discovery mailbox.

To successfully perform cross-premises eDiscovery searches in an Exchange Server hybrid organization, you must configure OAuth (Open Authorization) authentication between your Exchange on-premises and Exchange Online organizations so that you can use In-Place eDiscovery to search on-premises and cloud-based mailboxes. OAuth authentication is a server-to-server authentication protocol that allows applications to authenticate to each other.

OAuth authentication supports the following eDiscovery scenarios in an Exchange hybrid deployment:

- Search on-premises mailboxes that use Exchange Online Archiving for cloud-based archive mailboxes.
- Search on-premises and cloud-based mailboxes in the same eDiscovery search.

How In-Place eDiscovery works

In-Place eDiscovery uses the content indexes created by Exchange Search. Role Based Access Control (RBAC) provides the Discovery Management role group to delegate discovery tasks to non-technical personnel, without the need to provide elevated privileges that may allow a user to make any operational changes to Exchange configuration. The Exchange admin center (EAC) for Exchange Server and the SCC for Exchange Online provide an easy-to-use search interface for non-technical personnel, such as legal and compliance officers, records managers, and human resources (HR) professionals.

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Authorized users can perform an In-Place eDiscovery search by selecting the mailboxes, and then specifying search criteria such as keywords, start and end dates, sender and recipient addresses, and message types. After the search is complete, authorized users can then select one of the following actions:

- **Estimate search results.** This option returns an estimate of the total size and number of items that will be returned by the search based on the criteria you specified.
- **Preview search results.** This option provides a preview of the results. Messages returned from each mailbox searched are displayed.
- **Copy search results.** This option lets you copy messages to a discovery mailbox.
- **Export search results.** After search results are copied to a discovery mailbox, you can export them to a PST file.

In-Place eDiscovery permissions

For authorized users to perform In-Place eDiscovery searches, you must add them to one of the following role groups, depending on whether you're searching in Exchange only or in the SCC:

- In Exchange (In-place eDiscovery):
  - **Discovery Management.** This role group consists of two management roles: The Mailbox Search Role, which allows a user to perform an In-Place eDiscovery search, and the Legal Hold Role, which allows a user to place a mailbox on In-Place hold and Litigation hold.
- In the Security & Compliance Center (eDiscovery cases):
  - **eDiscovery Manager**. Members of this role group can create and manage eDiscovery cases. They can add and remove members, place content locations on hold, create and edit Content Searches associated with a case, export the results of a Content Search, and prepare search results for analysis in Advanced eDiscovery.
  - **eDiscovery Administrator**. This group can perform all case management tasks that an eDiscovery Manager can do. Additionally, an eDiscovery Administrator can access and manage all eDiscovery cases.
  - **Reviewer**. This role group has the most restrictive eDiscovery-related permissions. Members of this group can only see and open the list of cases on the eDiscovery page in the Security & Compliance Center that they are members of. They can't create cases, add members to a case, create holds, create searches, export search results, or prepare results for Advanced eDiscovery. However, members can access cases in Advanced eDiscovery to perform analysis tasks.

By default, permissions to perform In-place eDiscovery-related tasks aren't assigned to any user or messaging administrator. Messaging administrators who are members of the Organization Management role group can add users to the Discovery Management role group and create custom role groups to narrow the scope of a discovery manager to a subset of users. The reason for this is that in most region, administrators can manage the messaging systems and mailboxes, but they cannot access any content inside mailboxes.

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In-Place eDiscovery prerequisites and limits

To work with eDiscovery in your organization, you must first understand its prerequisites and limitations, including:

- In-place eDiscovery is available for all Exchange deployments, whether you are using Exchange Standard or Enterprise, Registered Education Savings Plans, Exchange Online Plan 1 or Exchange Online Plan 2 licenses.
- If you need to use in-place eDiscovery searches not only to search for content but also to place content on-hold, you need to license the In-place hold feature. This is an Enterprise feature in Exchange Server. For Exchange Online, it requires an Exchange Online Plan 2 license.
- In Exchange Server, the resources that In-Place eDiscovery uses are controlled with throttling policies. In the following list you can see some of the relevant default settings:

Parameter	Description	Default value
DiscoveryMaxConcurrency	The maximum number of In-Place eDiscovery searches a user can perform concurrently.	2
DiscoveryMaxMailboxes	The maximum number of mailboxes that can be searched in a single In-Place eDiscovery search. Public folder mailboxes are also counted against the source mailbox limit.	10,000
DiscoveryMaxStatsSearchMailboxes	The maximum number of mailboxes that can be searched in a single In-Place eDiscovery search that still allows you to view keyword statistics.	100 **Note**: After you run an eDiscovery search estimate, you can view keyword statistics. These statistics show details about the number of items returned for each keyword used in the search query. If more than 100 source mailboxes are included in the search, an error is returned if you try to view keyword statistics.
DiscoveryMaxKeywords	The maximum number of keywords that can be specified in a single In-Place eDiscovery search.	500
DiscoveryMaxSearchResultsPageSize	The maximum number of items displayed on a single page when previewing In-Place eDiscovery search results.	200
DiscoverySearchTimeoutPeriod	The number of minutes that an In-Place eDiscovery search will run before timing out.	10 minutes

- There are also limits for creating eDiscovery cases in Exchange Online, including:

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Description of limit	Limit
Maximum number of cases for an organization	No limit
Maximum number of case holds for an organization	10,000
Maximum number of mailboxes in a single case hold	1,000
Maximum number of SharePoint and OneDrive for Business sites in a single case hold	100

Managing Advanced eDiscovery

Organizations oftentimes must respond to legal cases involving certain executives or other employees. This might involve quickly finding and retaining specific information in email, documents, instant messaging conversations, and other content locations used by people in their day-to-day work tasks.

With Advanced eDiscovery, you can better understand your Microsoft 365 data and reduce your eDiscovery costs. Advanced eDiscovery helps you analyze unstructured data within Microsoft 365, perform more efficient document reviews, and make decisions to reduce data for eDiscovery. Advanced eDiscovery helps you analyze large data sets and find content that's most relevant to a case. It can also organize your content, making the legal review process easier and more efficient.

You can work with data stored in Exchange Online, SharePoint Online, OneDrive for Business, Skype for Business, Microsoft 365 Groups, and Microsoft Teams. You can perform an eDiscovery search in the Microsoft 365 Security & Compliance Center to search for content in groups, individual mailboxes, and sites, and then analyze the search results with Advanced eDiscovery.

Note: When you prepare search results for analysis in Advanced eDiscovery, Optical Character Recognition enables the extraction of text from images. This feature allows the powerful text analytic capabilities of Advanced eDiscovery to be applied to image files.

Advanced eDiscovery streamlines and speeds up the document review process by identifying redundant information with features like near-duplicates detection and Email Thread analysis. The Relevance feature applies predictive coding technology to identify relevant documents.

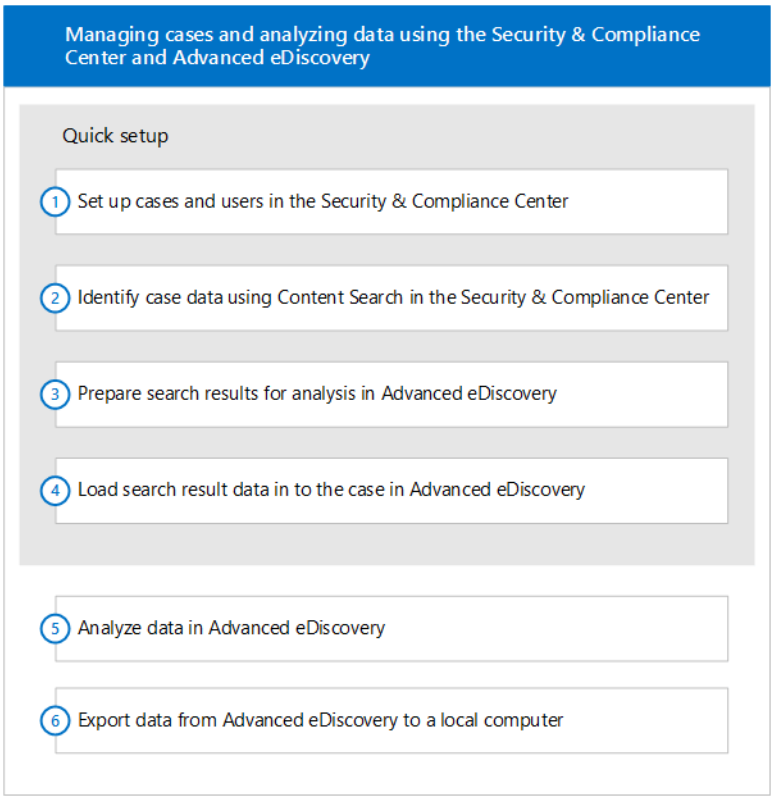
Advanced eDiscovery learns from your tagging decisions on sample documents and applies statistical and self-learning techniques to calculate the relevance of each document in the data set. This enables you to focus on key documents, make quick yet informed decisions on case strategy, cull data, and prioritize review.

Note: To analyze a user's data using Advanced eDiscovery, the user (the custodian of the data) must be assigned an Office 365 E5 license. Alternatively, users with an Office 365 E3 license can be assigned an Advanced eDiscovery standalone license. Administrators and compliance officers who are assigned to cases and use Advanced eDiscovery to analyze data don't need an E5 license.

Workflow for Advanced eDiscovery

The following diagram illustrates the common workflow for managing and using eDiscovery cases in the Security & Compliance Center and Advanced eDiscovery.

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As Advanced eDiscovery is built on the eDiscovery functionality in the Security & Compliance Center, the basic steps to create an eDiscovery case remain the same.

After a search that's associated with an eDiscovery case in the Security & Compliance Center is successfully run, you can prepare the search results for further analysis with Microsoft 365 Advanced eDiscovery. This feature lets you analyze large, unstructured data sets and reduce the amount of data that's relevant to a legal case. Advanced eDiscovery features include:

- **Optical character recognition.** When you prepare search results for Advanced eDiscovery, optical character recognition (OCR) functionality automatically extracts text from images and includes it with the search results that are loaded in to Advanced eDiscovery for analysis. OCR is supported for loose files, email attachments, and embedded images. This allows you to apply the text analytic capabilities of Advanced eDiscovery (near-duplicates, email threading, themes, and predictive coding) to the text content in image files.
- **Near-duplicate detection.** This feature lets you structure your data review more efficiently so that one person reviews a group of similar documents. This helps prevent multiple reviewers from having to view different versions of the same document.
- **Email threading.** This feature helps you identify the unique messages in an email thread so you can focus on only the new information in each message. In an email thread, the second message contains the first message. Likewise, later messages contain all the previous messages. Email threading removes the need to review every message in its entirety in an email thread.
- **Themes.** Themes help you gain valuable insight about your data beyond just keyword search statistics. Themes help investigations by grouping related documents so that you can look at the documents in context. When using themes, you can view the related themes for a set of documents, determine any overlap, and then identify cross-sections of related data.

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- **Predictive coding.** This feature lets you train the system on what you're looking for, by allowing you to make decisions (about whether something is relevant or not) on a small set of documents. Advanced eDiscovery then applies that learning (based on your guidance) when analyzing all the documents in the data set. Based on that learning, Advanced eDiscovery provides a relevance ranking so that you can decide which documents to review. This ranking is based on its determination as to which documents will most likely be relevant to the case.
- **Exporting data for review applications.** You can export data from Advanced eDiscovery and Microsoft 365 after you've completed your analysis and reduced the data set. The export package includes a CSV file that contains the properties from the exported content and analytics metadata. This export package can then be imported to an eDiscovery review application.

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Lab4-Ex1-Managing Messaging Compliance

Lab4-Ex1-Managing Messaging Compliance

In this lab you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. You have been tasked with maintaining message traffic and reducing the number of undeliverable emails in your environment.

To complete this task, you will create data loss prevention policies and conduct message tracking to familiarize yourself with the tools provided by Microsoft 365 that assist you with managing message compliance.

See your VM lab environment for the lab instructions.

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Module Review

Module 4 Review

Checkbox

Item 1. Encryption in Microsoft 365 involves which the following two areas? (Select 2)

- ☐ Data at rest
- ☐ Data in-transit
- ☐ Data in third-party solutions
- ☐ Data of external users

Multiple choice

Item 2. Which of the following enable you to retain or delete mailbox content based on policies to comply with regulatory and legal rules?

- ☐ eDiscovery
- ☐ Retention Policies
- ☐ Office 365 Message Encryption
- ☐ In-Place Archiving
- ☐ Data loss prevention (DLP)

Checkbox

Item 3. Which of the following features describe the permission model that is used by Microsoft 365? (Select 2)

- ☐ Permissions are managed centrally for all services in the Security & Compliance Center.
- ☐ Roles are assembled into role groups that reflect business job roles.
- ☐ Role groups in the Security & Compliance Center are used to manage access to features in the Security & Compliance Center Admin portal.
- ☐ Microsoft 365 is using ACLs (Access Control Lists) to manage access to various Security & Compliance features and functions.

Checkbox

Item 4. Which of the following are the two actions a retention policy is designed for? (Select 2)

- ☐ Retaining
- ☐ Deleting
- ☐ Classification
- ☐ Searching

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Multiple choice

Item 5. Which of the following is a security feature that protects sensitive data from being accidentally or intentionally shared with unauthorized users?

- ☐ Data loss prevention (DLP)
- ☐ Advanced Threat Protection (ATP)
- ☐ Messaging Record Management (MRM)
- ☐ In-Place Archiving

Multiple choice

Item 6. You work as a messaging administrator for Contoso Ltd., where all user mailboxes are hosted in Exchange Online. You need to verify why a specific email did not arrive in the user's Inbox. Which action should you perform to verify the email delivery?

- ☐ Create a massage tracking report using the Exchange Management Shell.
- ☐ Perform a message trace in the Security & Compliance Center.
- ☐ Create a mail flow report in the Security & Compliance Center.
- ☐ View the Usage report in the Microsoft 365 Admin Center.

Multiple choice

Item 7. You are the messaging administrator for Contoso Ltd. The compliance department requires you to provide support in a legal investigation of a user mailbox. All messages of the user's mailbox shall be preserved, and the user must be prohibited from permanently deleting any mailbox item. To accommodate this request, which of the following should you configure for the user mailbox?

- ☐ A content search in the Security & Compliance Center
- ☐ An In-Place Hold or Litigation Hold on the mailbox
- ☐ A Data loss prevention policy
- ☐ An In-place archive for the user

Multiple choice

Item 8. Fill in the blank. After a mailbox user is removed from In-Place Hold or Litigation Hold, items in the DiscoveryHolds, Purges, and Versions folders are automatically purged by _____.

- ☐ the user
- ☐ the Exchange Mailbox role
- ☐ a delegate
- ☐ the Managed Folder Assistant

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Multiple choice

Item 9. As a messaging administrator for Contoso Ltd. you must provide a solution for users in your Exchange Hybrid organization that will archive very large amounts of data (greater than 100 GB in total) to their archive mailbox. Which feature do you activate for them?

- ☐ Auto-expanding archives
- ☐ Retention Policies for deletion
- ☐ Journaling
- ☐ Data loss prevention policies

Multiple choice

Item 10. Fill in the blank. Journaling refers to _____ email communications as part of the organization's email retention strategy.

- ☐ deleting
- ☐ recording
- ☐ moving
- ☐ tagging

Checkbox

Item 11. You are working as a messaging administrator for Contoso Ltd. You have been tasked with providing a solution for the Compliance department to audit actions of administrators and regular users. To accomplish this, which logging features do you need to check, enable, and configure in your Exchange organization? (Select 2)

- ☐ Journaling
- ☐ Administrator audit logging
- ☐ Mailbox audit logging
- ☐ Usage reports
- ☐ Sent and received email reports

Checkbox

Item 12. Which of the following are parts of Message Records Management (MRM) for Exchange? (Select 3)

- ☐ Default policy tags (DPTs)
- ☐ Admin roles
- ☐ Journal rules
- ☐ Retention policy tags (RPTs)
- ☐ Recovery Point Objective (RPO)
- ☐ Connection filters
- ☐ Personal tags

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Multiple choice

Item 13. If you remove litigation hold from a mailbox that also has in-place holds applied, what will happen to the items in scope of the in-place hold?

- ☐ They will be deleted.
- ☐ They will be moved to a hold library.
- ☐ They will remain on hold.
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Multiple choice

Item 14. How many days are message trace reports preserved for?

- ☐ 14 days
- ☐ 90 days
- ☐ 180 days
- ☐ 365 days

Multiple choice

Item 15. You must be a member of which role group to have access to the Content search page to perform searches, as well as preview and export search results?

- ☐ Security Administrator
- ☐ eDiscovery Manager
- ☐ Organization Management
- ☐ MailFlow Administrator
- ☐ Records Management
- ☐ Service Assurance User

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Answers

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Module 5 Managing Organizational Settings

Managing Authentication for Messaging

Lesson introduction

Security is a great concern for every organization. In today's world where employees are mobile and work from multiple locations and multiple devices, it is even more important that they sign in and access data and resources in a secure manner.

As a messaging administrator, you need to ensure that user accounts are well protected and secure. However, you should also ensure that deploying multiple security features doesn't introduce unnecessary complexity in users' everyday work, resulting in lower business productivity and new security risks.

After completing this lesson, you will be able to:

- Configure password policy options
- Configure self-service password management
- Implement multi-factor authentication
- Plan password policies

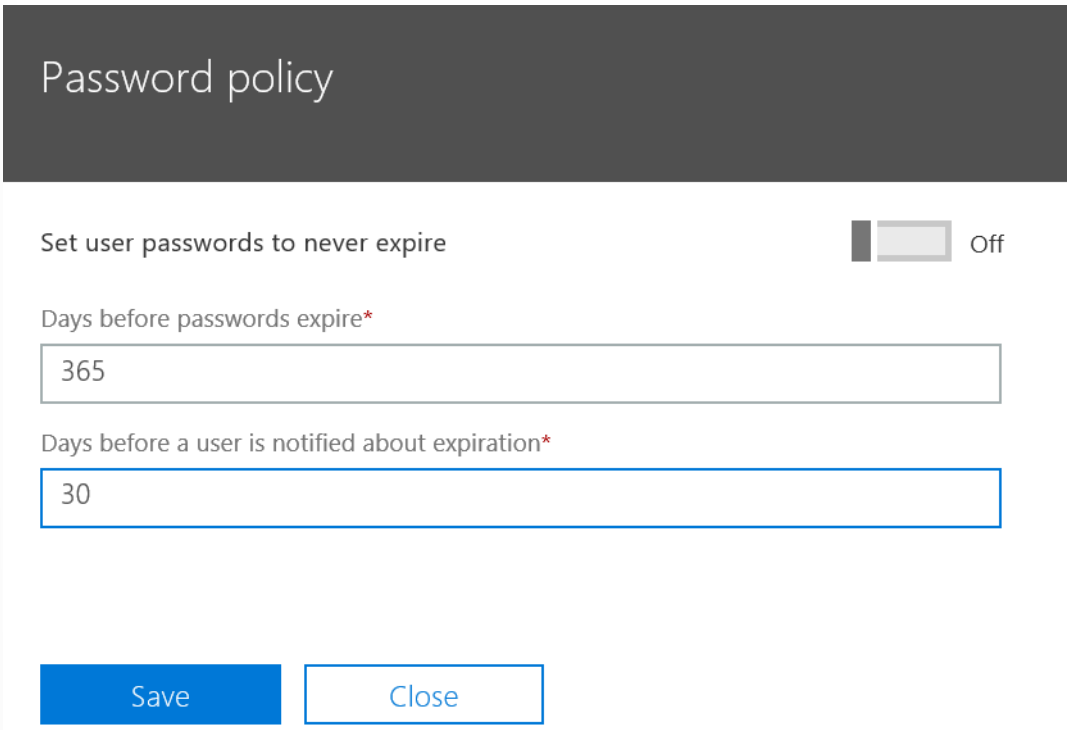
Password policy options

Setting password policy for users in your organization is one of the key responsibilities for administrators in a Microsoft 365 organization. The primary goal of a more secure password system is password diversity. You want your password policy to contain many different and hard-to-guess passwords. Here are a few recommendations for keeping your organization as secure as possible:

- Maintain an 8-character minimum length requirement for passwords (longer isn't necessarily better)
- Don't require character composition requirements within passwords, such as * & (^ %$
- Don't require mandatory periodic password resets for user accounts
- Ban common passwords to keep the most vulnerable passwords out of your system
- Educate your users to not re-use their organization passwords for non-work-related purposes

- Enforce registration for multi-factor authentication
- Enable risk-based multi-factor authentication challenges

Password policies control password properties for users in your organization, such as password complexity and expiration date. Some organizations also mandate password expiry and changes to comply with auditing requirements. In Microsoft 365, password policy configuration enables you to set user passwords expire after a certain number of days or set passwords to never expire. To configure password policy in Microsoft 365, you must be assigned a Microsoft 365 Global Administrator role.



You must perform the following steps to configure password expiration in Microsoft 365:

1. Go to the Microsoft 365 Admin center.
2. In the Microsoft 365 admin center, navigate to **Settings** > **Org settings**, and then select the **Security and privacy** tab.
3. Select **Password expiration policy**.
4. In the **Password expiration policy** window, if you don't want to force users to change passwords, leave the **Set user passwords to expire after a number of days** check box blank and close the window.
5. However, if you want user passwords to expire, then continue performing the following steps:
   - Select the **Set user passwords to expire after a number of days** check box.
   - In the **Days before passwords expire** box, enter the number of days before passwords expire. Enter a number from **14** to **730** (days).
   - In the **Days before a user is notified about expiration** box, enter the number of days before users' passwords are set to expire when users are notified that their password will expire. Enter a number from **1** to **30** (days).
6. Select **Save changes**.

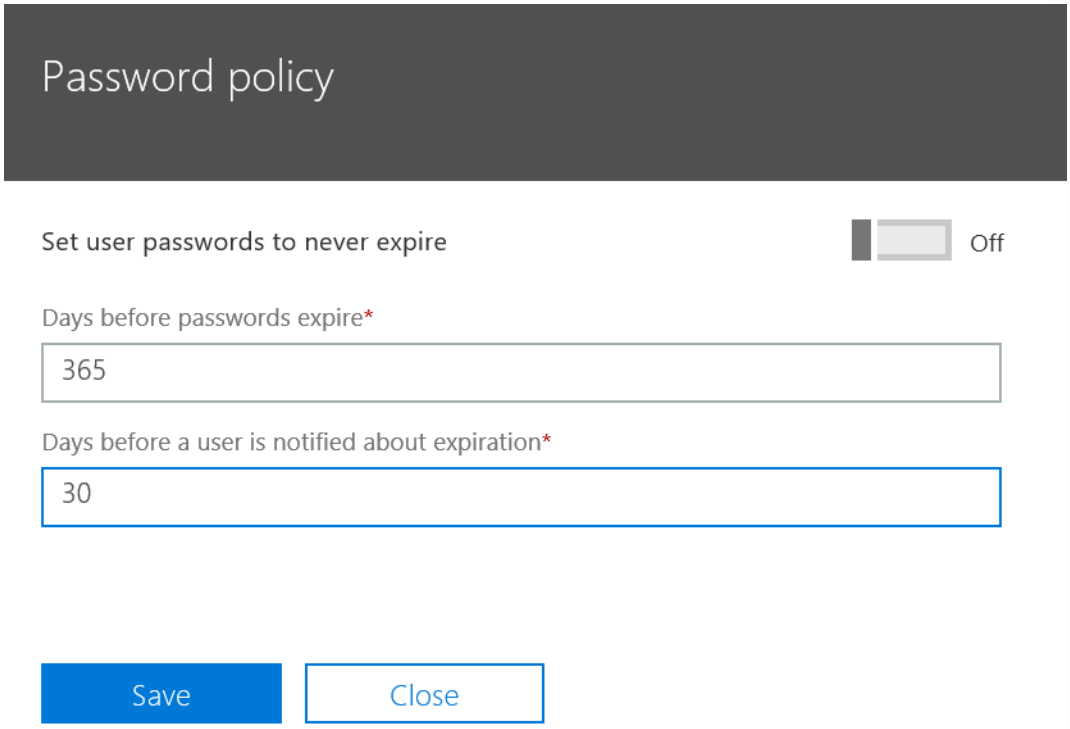
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If passwords are set to expire and the current date falls within the notification period for a user, a notification will appear in the lower right corner of the user's screen. If a user doesn't change his or her password and it eventually expires, the user will receive a similar notification.

In Exchange Online deployments, you can configure additional password policy settings, such as password length and complexity, but you must use Azure Active Directory. These policies will apply to cloud-only accounts, to administrator accounts responsible for replying to emergencies in enterprise environments, and to smaller organizations that don't use Azure AD Connect to synchronize Active Directory.

Configuring Complex Passwords for Active Directory

In on-premises environments, including those synchronizing Active Directory to Azure AD, password policies are configured in Active Directory. In these cases, the password policy applied to on-premises accounts will also apply to Azure AD accounts. This includes scenarios involving Password Hash Synchronization, Pass Through Authentication, and Active Directory Federation Services (AD FS). To configure password policies in AD, edit the **Default Domain Policy** in the Group Policy Management Console (GPMC), and in the left navigation pane, navigate to **Computer Configuration**>**Policies**>**Windows Settings**>**Security Settings**>**Account Policies**>**Password Policy**.



Additional reading. For more information, see the following article on **Password and account lockout policies on managed domains**[1].

Self-service password management

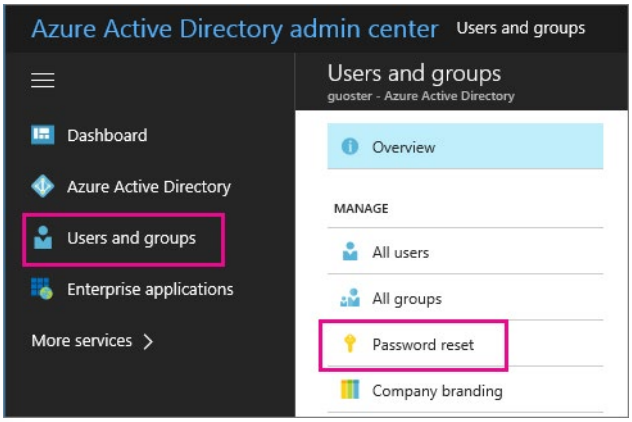
In most organizations, users often forget their passwords and then send a request to administrators for password reset. In Microsoft 365, you can configure self-service password reset (SSPR) so that users can

1 https://aka.ms/AA3s0kl

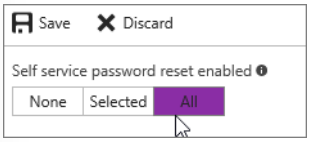
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reset their own passwords. SSPR is useful especially in medium and enterprise organizations, where the number of password reset requests can be very large. While self-service password reset can also be configured in hybrid environments, it is not available in Exchange Server on-premises environments.

SSPR is configured in the Azure Active Directory admin center. For cloud-only accounts, this is included within Microsoft 365 enterprise and business subscriptions. For synchronized accounts, a subscription to Azure AD Premium Plan 1 is required for each user.



On the **Properties** page, select **All** to enable it for everyone in your business, and then select **Save**.



When your users sign in to Microsoft 365, they will be prompted to enter additional contact information that will help them reset their password in the future. The following steps describe workflow logic of the password reset page:

1. The user selects the **Can't access your account** link or goes directly to the **password reset page**[2].
2. The user enters a user ID and passes a captcha.
3. Azure AD verifies that the user is enabled to use this feature by performing the following checks:
   - Verifies that the user has this feature enabled and has an Azure AD license assigned.
   - Verifies that the user has the right authentication methods defined on his or her account in accordance with administrator policy.
   - Verifies whether the user's password is managed on-premises (federated, pass-through authentication, or password hash synchronized) and if writeback is deployed and the user's password is managed on-premises.
4. If it's determined that the user can successfully reset their password, then the user is guided through the reset process.
5. If the checks are not validated, the user will be advised to contact their administrator to reset their password.

2 https://aka.ms/sspr

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Authentication methods

If SSPR is enabled, you must select at least one of the following options for the authentication methods, sometimes referred to as "gates." It is highly recommended that you choose two or more authentication methods, because in this scenario you will provide your users with more flexibility in case they are unable to access one when they need it.

- Mobile app notification
- Mobile app code
- Email
- Mobile phone
- Office phone
- Security questions

Users can only reset their password if they have data present in the authentication methods that the administrator has enabled.

Number of authentication methods required

This option determines the minimum number of the available authentication methods a user must go through to reset or unlock their password. It can be set to either one or two.

Registration

Enabling this option requires a user to complete the password reset registration if they sign in to any applications using Azure AD. This includes the following applications:

- Microsoft 365
- Azure portal
- Access Panel
- Federated applications
- Custom applications using Azure AD

Set the number of days before users are asked to reconfirm their authentication information

This option determines the period of time between setting and reconfirming authentication information and is available only if you enable the **Require users to register when signing in** option.

Valid values are 0 to 730 days, with "0" meaning users are never asked to reconfirm their authentication information.

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Notifications

Notify users on password resets

If this option is set to **Yes**, then users resetting their password will receive an email notifying them that their password has been changed. The email is sent through the SSPR portal to their primary and alternate email addresses that are on file in Azure AD. No one else is notified of the reset event.

Notify all admins when other admins reset their passwords

If this option is set to **Yes**, then all administrators receive an email that is sent to their primary email address on file in Azure AD. The email notifies them that another administrator has changed their password by using SSPR.

Write back passwords to your on-premises directory

This control determines whether password writeback is enabled for this directory. If writeback is on, it indicates the status of the on-premises writeback service. This is useful if you want to temporarily disable password writeback without having to reconfigure Azure AD Connect.

Allow users to unlock accounts without resetting their password

This control designates whether users who visit the password reset portal should be given the option of unlocking their on-premises Active Directory accounts without having to reset their password. By default, Azure AD unlocks accounts when it performs a password reset. You use this setting to separate those two operations.

- If set to **Yes**, then users are given the option to reset their password and unlock the account, or to unlock their account without having to reset the password.
- If set to **No**, then users are only be able to perform a combined password reset and account unlock operation.

Implementing MFA for Exchange

Multi-factor authentication (MFA) is a method of authentication that requires the use of more than one verification method and adds a second layer of security to user sign-ins and transactions. Once enabled, MFA will require any two or more of the following verification methods:

- A randomly generated pass code
- A phone call
- A smart card (virtual or physical)
- A biometric device

You can configure multi-factor authentication in Microsoft 365 to provide an additional layer of security during the sign-in process. Microsoft 365 offers the following subset of Azure multi-factor authentication capabilities as a part of the subscription:

- Enable and enforce multi-factor authentication for end users
- Mobile app (online and one-time password [OTP]) as a second authentication factor

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- Phone call as a second authentication factor
- Short Message Service (SMS) message as a second authentication factor
- Application passwords for non-browser clients
- Default Microsoft greetings during authentication phone calls

To configure multi-factor authentication in Microsoft 365, you must be a Microsoft 365 Global Administrator. You can configure multi-factor authentication in the Microsoft 365 admin center by performing the following steps:

1. Go to the Microsoft 365 admin center.
2. Navigate to **Users** > **Active users**.
3. Select **Multi-factor authentication**.
4. Select the check box next to each user for whom you want to enable MFA.
5. When you select a user, the user's detail pane appears on the right. In the detail pane, under **quick steps**, select **Enable**.
6. In the dialog box that opens, select **enable multi-factor auth**.

In larger environments, you may want to bulk-enable or disable MFA for multiple users. To do so, you must first create a CSV file that contains the user names of the existing accounts. To bulk-enable or disable MFA, you should perform following steps:

1. On the **multi-factor authentication** page, select the **bulk update** button.
2. In the **Select a CSV file** dialog box, select **Browse for file**.
3. Browse for the file that contains the updates, then select **Open**. The column headings in your file must match the column headings in the following example:

	A	B
1	Username	MFA Status
2	chris@contoso.com	Enabled
3	ben@contoso.com	Disabled
4	kyle@contoso.com	Disabled
5	kenny@contoso.com	Enabled
6	eric@contoso.com	Enabled

4. Select the **Next** arrow.
5. After the file is verified, select the **Next** arrow to update the accounts.
6. When the process is finished, select the **Done** checkmark.

Planning password policies

Detailed planning of your organization's password policy is essential to maximizing the security of your users' credentials. Poorly planned password policies can lead to successful password attacks. However, policies that are too restrictive can also lead to risk, since the users will find it hard to comply with the complexity, and they'll try different ways to store passwords, such as writing them down.

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Password management should be guided by the following best practices:

- Educating users where to enter passwords, such as trusted devices that have been approved by the organization, or web sites that are validated and trusted. For example, users should avoid signing in to organizational email through a public kiosk computer or the computer of any non-trusted person.
- Educating users to choose passwords where length, complexity, and uniqueness should be used, and to avoid using a common combination of letters that can be guessed. For example, Alex1234 is not a good choice for a password, since it contains the name of the user and a number combination can be easily seen typed on the keyboard by another user.
- Educating users about social networking and how this type of attack is difficult to detect. For example, a phone call to an administrator that claims that general manager needs password reset.
- Enforcing password diversity, which means that users create passwords that are different and hard to guess.
- Enforce at least an 8-character minimum length requirement, but the length shouldn't be larger since it will force users to repeat patterns that are easier to guess.
- Periodic password resets shouldn't occur on very short intervals. This practice will make users write down passwords or create predictable passwords that can easily be guessed.
- Educate users to not re-use their organization passwords for non-work-related purposes.
- Enforce registration for multi-factor authentication.
- Educate users not to use passwords that are easy to guess by someone who knows a lot about the user, such as family member names and birthdays.

Additional reading. For more information on configuring a custom password complexity policy for your organization, see **Azure AD B2C: Configure complexity requirements for passwords**[3].

3 https://go.microsoft.com/fwlink/?linkid=864494

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Configuring Organizational Settings

Lesson introduction

As a messaging administrator, you may need to configure settings that apply to the entire organization or to many users in the organization. For example, you should configure mailbox quotas to control disk usage on the Exchange Server, and you may also need to configure throttling policies to prioritize some workloads that affect the entire organization.

In some organizations, you must integrate Exchange Server with Skype for Business Server and SharePoint Server to enable unified eDiscovery search for compliance reasons. Organizational settings should be carefully planned and tested because they typically affect most of users in the organization.

After completing this lesson, you will be able to:

- Configure workload policies and throttling
- Configure quota configurations
- Deploy Microsoft 365 add-ins

Configuring workload policies and throttling

Exchange Server uses a concept in monitoring and management called Workload Management. An Exchange workload is an Exchange Server feature, protocol, or service that's been explicitly defined for the purposes of Exchange system resource management. Each Exchange workload consumes system resources such as CPU, mailbox database operations, or Active Directory requests to run user requests or background work.

You can manage workloads in Exchange Server in the following ways:

- **Monitoring system resources**. To monitor the Exchange workload, resources used by it are monitored, such as CPU usage, memory consumption, and network utilization, among others. If server resources are highly utilized, Exchange Server progressively slows down the lowest priority workloads. Priorities are defined by the classification assigned to each workload: Urgent, Customer Expectation, Internal Maintenance, and Discretionary, where the Urgent classification has the highest priority and Discretionary classification has the lowest priority. System resource thresholds, where utilization is measured, have three levels: Underloaded, Overloaded, and Critical.
- **Controlling how individual users consume resources**. This method of managing workloads introduces different types of workload usage by users, including:
    - **Burst allowances**. Exchange Server allows users to have greater resource consumption for short periods of time without throttling.
    - **Recharge rate**. Exchange server uses a resource budget system, where administrators set a rate where users' budgets are recharged in defined periods of time. For example, a value of 300,000 milliseconds means that users' budgets are recharged on five minutes of usage per hour.
    - **Traffic shaping**. Exchange Server delays the user whenever a user reaches the configured limit for the defined time interval. This type of workload usage prevents users from overloading the performance of the server. Usually, users' business tasks are not affected because the delays are short and almost undetectable.
    - **Maximum usage**. Exchange Server temporarily blocks users from performing their tasks when they have reached their threshold in resource usage. Users are unblocked the moment their budget is recharged.

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Exchange workload management, which was previously known as user throttling, is configured in the Exchange Management Shell by creating or changing the workload management policy settings. These setting can be configured at the organizational level and applied to all Exchange Servers in the organization, or at the server level and applied only to that specific server.

The cmdlets used to manage resource policy include:

- New-ResourcePolicy
- Remove-ResourcePolicy
- Get-ResourcePolicy
- Set-ResourcePolicy

Cmdlets used to manage workload management policy include:

- New-WorkloadManagementPolicy
- Remove-WorkloadManagementPolicy
- Get-WorkloadManagementPolicy

Cmdlets used to manage workload policies include:

- New-WorkloadPolicy
- Remove-WorkloadPolicy
- Get-WorkloadPolicy
- Set-ResourcePolicy

Throttling policies are managed and assigned by using the following cmdlets:

- New-ThrottlingPolicy
- Get-ThrottlingPolicy
- Set-ThrottlingPolicy
- Remove-ThrottlingPolicy
- Get-ThrottlingPolicyAssociation
- Set-ThrottlingPolicyAssociation

To display current workload management policies, you should use the following cmdlet:

```
Get-WorkloadManagementPolicy
```

To change the default workload management policy for your organization's Outlook Web App workload, you should use the following cmdlet:

```
New-WorkloadPolicy OrgOWAWorkloadPolicy -WorkloadType OWA -WorkloadClassifi-
cation Discretionary -WorkloadManagementPolicy GlobalOverrideWorkloadMan-
agementPolicy
```

To create a workload management policy for Outlook on the Web for a specific server, you should perform the following steps:

1. Create a custom workload management policy that will be applied later to a specific server by running the following command:

   ```
   New-WorkloadManagementPolicy LondonWorkloadManagementPolicy
   ```

2. Create a new Outlook Web App workload policy by running the following command:

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```
New-WorkloadPolicy LondonOWAWorkloadPolicy -WorkloadType OWA -WorkloadClas-
sification Discretionary -WorkloadManagementPolicy LondonWorkloadManagement-
Policy
```

3. Apply the custom workload management policy you just created to a specific server by running the following command:

```
Set-ExchangeServer -WorkloadManagementPolicy LondonWorkloadManagementPolicy
-Identity EXCH1
```

It is recommended that you never modify the default throttling policy, because changes to the default policy could be overwritten by future Exchange updates. Instead, you should create custom throttling policies that contain customized settings.

Configuring quota configurations

In Exchange Server, storage quotas are used to control the size of mailboxes and manage the growth of mailbox databases.

In Microsoft 365, Exchange Online storage quotas have a fixed maximum for the primary mailbox and depend on subscription type, such as 50 GB for Business subscriptions and 100 GB for Enterprise subscription. However, these quotas can be altered on a per-mailbox basis if your organization wants to control mailbox growth.

When a mailbox reaches or exceeds a specified storage quota, Exchange sends a descriptive notification to the mailbox owner. Storage quotas are typically configured on a per-database basis because quotas configured for a mailbox database apply to all mailboxes in that database. However, you can also configure storage quotas on a specific user mailbox if you want to create an exception for that particular mailbox. For example, all users located on the Marketing database have a 5GB quota, while the VP of Marketing requires a quota of 10GB.

You can use the Exchange admin center (EAC) or the Exchange Management Shell to customize the mailbox storage quotas for specific mailboxes.

Use the **Limits** tab on the **mailbox database properties** to specify the storage limits, the warning message interval, and the deletion settings for a mailbox database. If you want to configure storage limits per user, use the **mailbox usage** tab on a **user mailbox properties**.

- **Issue warning at (GB).** Select this check box to automatically warn mailbox users that their mailbox is approaching its storage limit. To specify the storage limit, select the check box, and then specify in gigabytes (GB) how much content can be stored in the mailbox before a warning email message is sent to the mailbox users. You can enter a value from 0 through 2,097,151 megabytes (MB) (2.0 terabytes).
- **Prohibit send at (GB).** Select this check box to prevent users from sending new email messages after the size of their mailbox reaches its maximum limit. To specify this limit, select the check box, and then type the size limit of the mailbox in GB. When a user's mailbox reaches this limit, the user will be prohibited from sending new email messages, and a warning message will be issued to the user. You can enter a value from 0 through 2,097,151 MB (2.0 terabytes).
- **Prohibit send and receive at (GB).** Select this check box to prevent users from sending and receiving email messages after their mailbox size reaches the specified limit. To specify this limit, select the check box, and then type the size limit of the mailbox in GB. When a user's mailbox reaches this limit, the user will be prohibited from sending and receiving new email messages, and a warning message will be issued to the user. You can enter a value from 0 through 2,097,151 MB (2.0 terabytes).

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Use the EAC to set storage quotas for a database or a mailbox

You should perform the following steps to set storage quotas for a database or mailbox by using the Exchange Admin Center:

1. In the EAC, navigate to **Servers** > **Databases** for database configuration or **Recipients** > **Mailboxes** for user mailbox configuration.
2. In the list of databases or user mailboxes, select the database or mailbox that you want to change the storage quotas for, and then select **Edit**.
3. Configure appropriate limits where the value range for any of the storage quota settings is from 0 through 2047 gigabytes (GB).
   - **Issue a warning at (GB).** This box displays the maximum storage limit before a warning is issued to the user. If the mailbox size reaches or exceeds the value specified, Exchange sends a warning message to the user. The message associated with the **Issue warning** quota won't be sent to the user unless the value of this setting is greater than 50% of the value specified in the **Prohibit send** quota. For example, if you set the **Prohibit send** quota to 8 MB, you must set the **Issue warning** quota to at least 4 MB. If you don't, the **Issue warning** quota message won't be sent.
   - **Prohibit send at (GB).** If the mailbox size reaches or exceeds the specified limit, Exchange prevents the user from sending new messages and displays a descriptive error message.
   - **Prohibit send and receive at (GB).** If the mailbox size reaches or exceeds the specified limit, Exchange prevents the mailbox user from sending new messages and won't deliver any new messages to the mailbox. Any messages sent to the mailbox are returned to the sender with a descriptive error message.
4. Select **Save** to save your changes.

Use the Exchange Management Shell to configure storage quotas for a mailbox database

This example sets the Issue Warning, Prohibit Send, and Prohibit Send and Receive quotas for the database DB1 mailbox to 15 GB, 20 GB, and 25 GB, respectively.

```
Set-MailboxDatabase -Identity DB1 -IssueWarningQuota 15gb -ProhibitSendQuo-
ta 20gb -ProhibitSendReceiveQuota 25gb
```

Use the Exchange Management Shell to configure storage quotas for a mailbox

This example sets the Issue Warning, Prohibit Send, and Prohibit Send and Receive quotas for Alex Darrow's mailbox to 24.5 GB, 24.75 GB, and 25 GB, respectively. To verify the custom settings for the mailbox are used rather than the mailbox database defaults, you must set the **UseDatabaseQuotaDefaults** parameter to **$false** , as shown in the following command:

```
Set-Mailbox -Identity "Alex Darrow" -IssueWarningQuota 24.5gb -Prohibit-
SendQuota 24.75gb -ProhibitSendReceiveQuota 25gb -UseDatabaseQuotaDefaults
$false
```

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Deploying Microsoft 365 add-ins

Office add-ins help you personalize your documents and streamline the way you access information on the web. As a Microsoft 365 admin, you can deploy Microsoft add-ins for the users in your organization using the Centralized Deployment feature in the Microsoft

- A Global admin can assign an add-in directly to a user, to multiple users through a group, or to everyone in the tenant.
- When the relevant Office application starts, the add-in automatically downloads for the user. If the add-in supports add-in commands, the add-in automatically appears in the Ribbon within the Office application.
- Add-ins do not appear for users if the admin turns off or deletes the add-in, or if the user is removed from Azure Active Directory or from a group that the add-in is assigned to.

Consider rolling out add-ins in a phased approach to help ensure your add-in deployment goes smoothly. The following roll-out plan is recommended:

1. Roll-out the add-in to a small set of business stakeholders and members of the IT department. Evaluate whether the deployment was successful, and if so, move on to step 2.
2. Roll-out to a larger set of individuals within the business who will be using the add-in. Again, evaluate results and, if all went well, go to the next step of a full deployment.
3. Full rollout to the full target audience of users.

To deploy Microsoft 365 add-ins, you should perform the following steps:

1. In Microsoft 365 Admin Center, in the navigation menu, navigate to **Settings** > **Add-ins**.
2. Select **+ Deploy Add-in** that appears on the menu bar above the list of add-ins.
3. On the **Deploy a new add-in** page, select **Next**.
4. Choose from one of the following options:
   - Deploy from the Store - select the **Choose from Store** button
   - Deploy a custom add-in - select the **Upload custom apps** button.
5. If you selected the option to add an add-in from the Microsoft Store, you can now make your add-in selection in the **Select add-in** page. Notice that you can view the available add-ins through categories of **Suggested for you**, **Rating**, or **Name**.

   NOTE: With the Microsoft Store option, updates and enhancements to the add-in will automatically be made available to users without your intervention.
6. On the **Select add-in** page, select **Add**.
7. On the **License terms and Privacy policy** page, select **Continue**.
8. On the **Configure add-in** page, select either **Everyone** or **Specific Users/Groups** or **Just me**. Also select the method that you want to use to deploy the add-in.
9. When finished, select **Deploy**, review the add-in settings, and then select **Close**.

   You should now see your add-in along with other apps in the **Add-ins** page.

It's recommended that you inform the users and groups where the add-in will be deployed so they know that it's available. You may consider training them on how to use the add-in before the deployment.

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Considerations when assigning an add-in to users and groups

Admins can assign an add-in to everyone or to specific users and groups. Each option has implications:

- **Everyone.** This option assigns the add-in to every user in the tenant. It's recommended to be used only for add-ins that are universal to your organization.
- **Users.** If you assign an add-in to an individual user, then to deploy the add-in to a new user, you will need to first add that user. The same goes for removing users.
- **Groups.** If you assign an add-in to a group, users who are added to the group will automatically be assigned the add-in. When a user is removed from a group, the user loses access to the add-in. In either case, no additional action is required from you as the admin.

The option that is right for your organization depends on your configuration. However, it's recommended that you create assignments through groups. As an admin, you might find it easier to manage add-ins using groups and control the membership of those groups rather than having to change the users assigned each time. On the other hand, in some situations you may want to restrict access to a very small set of users and therefore make assignments to specific users. As a result, you must manage the assigned users manually.

Add-in states

The following table describes the states that apply to an add-in.

State	How the state occurs	Impact
Active	Admin uploaded the add-in and assigned it to users or groups.	Users and groups assigned to the add-in see it in the relevant clients.
Turned off	Admin turned off the add-in.	Users and groups assigned to the add-in no longer have access to it. If the add-in state is changed to Active, the users and groups will have access to it again.
Deleted	Admin deleted the add-in.	Users and groups assigned the add-in no longer have access to it.

Consider deleting an add-in if no one is using it anymore. Turning off an add-in may also make sense if the add-in is used only during specific times of the year.

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Configuring Organizational Sharing

Lesson introduction

In an Exchange organization, users can share information such as their calendar free/busy information (known as calendar availability), and they can choose to share detailed calendar information. When two or more organizations that work closely together use Exchange server or Exchange Online as their messaging system, you can provide a similar sharing functionality. In other words, users in one organization can request the free/busy calendar information of users in the other organization, and they can configure policies to allow read-only calendar sharing. This functionality to share information between Exchange organizations is called federation or federated sharing.

In this lesson, you will begin by examining federation delegation, which allows users from one federated organization to be trusted by another federated organization. With those trust relationships established, users within each organization can begin sharing information. To support federated sharing, you will learn about federated sharing components, as well as considerations for designing and implementing federated sharing. Finally, you will learn how to implement the organization relationships and sharing policies that are needed to begin sharing information across trusted organizations.

After completing this lesson, you will be able to:

- Provide an overview of Exchange federated delegation sharing features
- Describe federated sharing components
- Explain considerations for designing and implementing federation trusts and certificates
- Implement organization relationships
- Implement sharing policies

Overview of Exchange Federated Delegation features

Users in one organization often need to collaborate with users from other organizations, such as vendors, partners, or customers. As part of this collaboration, users often need to share their availability information, calendars, and contacts with users in other organizations. By default, sharing address lists or availability data with users who are outside an Exchange Server organization is not possible. However, organizations can enable collaboration between their Exchange 2010 or higher organization and users who are in another Exchange Server 2010 or higher organization or located in Exchange Online.

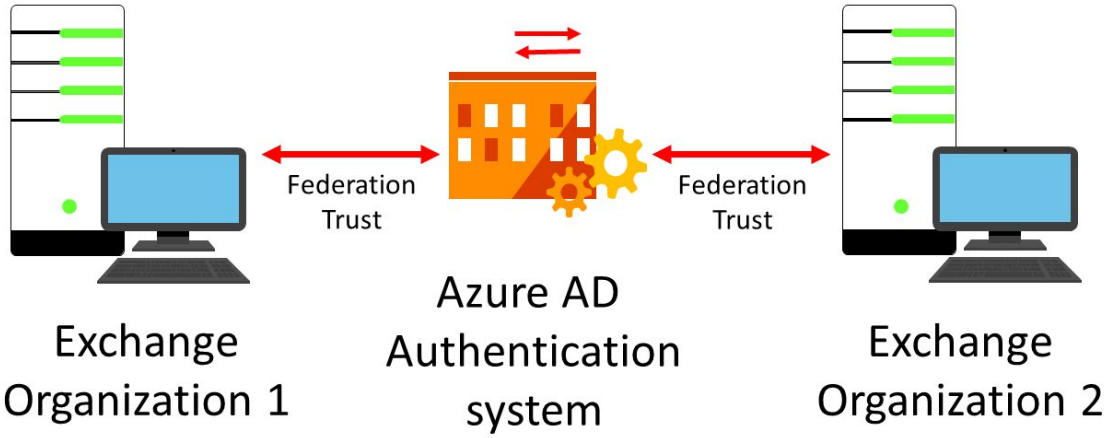
Developing Federated Trusts

Federation refers to the underlying trust infrastructure that supports federated sharing. When you configure federation, you configure a federated trust with the Microsoft Azure Active Directory (Azure AD) authentication system. When other organizations configure the same federated trust, the Azure AD authentication system provides a security token that contains claims (an assertion of information about the subject that has been authenticated) that enable users in one organization to access information in the other organization.

Each Exchange organization must develop federated trusts using the business instance of the Microsoft Azure AD authentication system. The following Exchange organizations use the business instance of the Microsoft Azure AD authentication system: Exchange Online and Exchange Server 2010 or later organizations that run the Federated Trust Wizard.

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Note: Microsoft has discontinued federation trust support for the consumer instance of the Microsoft Federation Gateway, which was used in Exchange 2010 and earlier implementations.



After you configure the federated trust, you must configure organization relationships or sharing policies. These settings define the organizations where you want to share information. When both organizations define these policies, users in the organizations can share their availability information, calendars, and contacts.

Federation Sharing Components

To configure Exchange federated sharing successfully, you must first become familiar with the components that are involved. The components used in federated sharing include:

- **Federation trust with the Azure AD authentication system**. Used as a trust broker.
- **AppID.** A unique number generated by the Azure Active Directory authentication system to identify Exchange organizations.
- **OrgID.** Defines which of the authoritative accepted domains configured in an organization are enabled for federation.
- **Self-signed certificate.** Signs and encrypts delegation tokens.
- **TXT resource record in DNS.** Validates domain ownership.

Federation trust with the Azure AD authentication system

The most important component of federated sharing is the federation trust with the Azure AD authentication system. The Azure AD authentication system is the new name for what was previously known as the Microsoft Federation Gateway.

The Azure AD authentication system is an Azure-based service, which is a free Microsoft offering that acts as a trust broker between two federated Exchange Server organizations. Instead of establishing a direct trust as with forest trusts in Active Directory Domain Services (AD DS), you establish a trust between your

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Exchange server environment and the Azure AD authentication system. Other organizations must establish the same federated trust relationship with an Azure AD authentication system before you can start sharing information with them. When you have an Azure AD authentication system acting as the trust broker for your organization, you do not need to establish multiple or individual trust relationships with other organizations.

After you configure a trust with the Azure AD authentication system, the Azure AD authentication system service issues a Security Assertion Markup Language (SAML) delegation token for each user that the on-premises Active Directory domain controllers authenticate. This token enables the authenticated user to access shared resources within federated Exchange Server organizations.

If you establish a trust with the Azure AD authentication system, your organization exchanges digital certificates with the Azure AD authentication-system certificate, including federation metadata.

Self-signed certificates

You must run the Federated Trust wizard to create the trust with the Azure AD authentication system. You can use either the Exchange Admin Center (EAC) or the Exchange Management Shell (EMS) to run the wizard; if you use the EMS, you must run the **New-FederationTrust** cmdlet. The Federated Trust wizard creates a self-signed certificate for the purpose of federation, which is replicated to all Exchange Servers. It uses this certificate to sign and encrypt delegation tokens from the Azure AD authentication system that enables external federated organizations to trust end-users.

AppID and TXT resource records

When you create a federation trust with the Azure AD authentication system, Azure AD creates an object called an application identifier (AppID). You can reach this object by running the **Get-FederationTrust** cmdlet in the Exchange Management Shell. AppID uniquely identifies your Exchange Server organization on the Azure AD authentication system's side when establishing relationships with another Exchange Server organization. AppID also provides valid proof that an organization is the owner of the domain that it uses for federation. This proof of ownership is accomplished by creating a text (TXT) resource with the AppID in the public DNS zone for each federated domain. To obtain the content of the TXT record, you should run the following command:

```
Get-FederatedDomainProof –domainname <NameOfDomain>
```

This **Get-FederatedDomainProof** cmdlet returns the value of the TXT resource record that you place in the DnsRecord field in the DNS. You can then use the DNS Manager to create a TXT resource record in your public DNS that contains the domain proof's content.

OrgID

When you configure federation, you must define which of your organization's authoritative accepted domains to use and enable for federation. This domain is then stored in a federated organization identifier (OrgID). It's important that you define this parameter because only users that have email addresses for the domain that's configured in the OrgID can use the Exchange Server federation features. OrgID is a combination of a predefined string and an accepted domain. The domain that is in OrgID is the domain that you select as the primary shared domain in the federation trust wizard when you create a federated trust.

For example, if you specify the federated domain adatum.com as the primary shared domain in your organization, this automatically creates the FYDIBOHF25SPDLT.adatum.com account namespace as the OrgID for your Exchange Server organization's federation trust. You can set OrgID by running the following command:

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```
Set-FederatedOrganizationIdentifier –Enabled $true
```

Important: To enable or disable all federation sharing features in your organization, you must enable or disable the OrgID for the federation trust.

Please remember that federated sharing uses Autodiscover to verify your domain; therefore, Autodiscover must be implemented and it must be working from the Internet. You can use the **Remote Connectivity Analyzer**[4] tool to verify this for your Exchange organization.

Considerations for designing and implementing Federation Trusts and Certificates

To perform federated sharing, you must first configure a federation trust with the Azure AD authentication system, which is the central point for federation trusts. However, before you configure federation between the two Exchange Server organizations, you must verify which Azure AD authentication system instance each Exchange organization uses; this enables you to determine whether a federation trust between the two organizations already exists.

If a federation trust to the Azure AD authentication system already exists for both Exchange organizations, you don't need to change the federation trust; rather, you just have to configure an organization relationship or sharing policy. To verify the federation trust, you must run the following command to determine the value of **<uri:federation:MicrosoftOnline>** for the **TokenIssuerURIs** parameter:

```
Get-FederationInformation –DomainName <hosted Exchange domain namespace>
```

To establish a federation trust with the Azure AD authentication system, you can use a self-signed certificate or an X.509 certificate signed by a certificate authority (CA). You must ensure that this certificate is created and installed on the Exchange Server that you used to create the trust.

We recommend that you use the self-signed certificate, that was automatically created and installed when you ran the Federation Trust Wizard. When the certificate is generated by the Federation Trust wizard, it is automatically deployed to all the Exchange Servers in your organization.

You can also generate a certificate using the **New-ExchangeCertificate** cmdlet. The following example creates a new certificate for federated sharing, and then deploys it to your Exchange Servers as the federation trust certificate:

```
New-ExchangeCertificate -FriendlyName "Exchange Federated Sharing" -Domain-
Name $env:USERDNSDOMAIN -Services Federation -KeySize 2048 -PrivateKeyEx-
portable $true -SubjectKeyIdentifier $ski

Get-ExchangeCertificate | ?{$_.FriendlyName -eq "Exchange Federated Shar-
ing"} | New-FederationTrust -Name "Azure AD Authentication"
```

Azure AD uses this certificate to sign and encrypt delegation tokens for federated sharing; it does not use it to identify your organization. Therefore, the name in the certificate is not relevant.

As an alternative to using self-signed certificates generated either through the Federated Trust wizard or by running the **New-ExchangeCertificate** cmdlet, you can also use an external x.509 certificate for federation. However, if you go this route, you must ensure that the correct subject alternative names (SAN) in the x.509 certificate are available. The downside to using an external x.509 certificate is that the process to create one is far more complex compared to creating a self-signed certificate within Exchange.

4 http://aka.ms/RCA

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Implementing Organization Relationships

You need to use organization relationships to enable federated sharing with another Exchange Server organization or Exchange Online. Each organization relationship is for a single external organization, which Azure AD identifies by its domain name and application identity. Therefore, not only does your organization need to create a federated trust relationship with Azure AD authentication, but so too does the external Exchange Server organization that you want to collaborate with. Both organizations must have bi-directional connectivity in place so that Exchange Servers can reach each organization's respective Exchange Web Services and Autodiscover URLs.

Note: Exchange Online, by default, has a trust in place with the Azure AD authentication system.

When you create an organization relationship in the Exchange Admin Center, you configure the following options:

- **Relationship name**. This is the name that you choose. You can, for example, use the domains to share.
- **Domains to share with**. Type the fully qualified domain name (FQDN) of the domain or domains with which you want to establish federation. This would be the domain name the other organization uses for its federation trust.
- **Enable calendar free/busy information sharing**. This turns on information sharing. If you enable this option, choose one of the following options:
  - Calendar free/busy information with time only
  - Calendar free/busy information with time, subject and location
  - Share calendar free/busy information for:
    - Everyone in your organization
    - A specified security group

When you configure the organization relationship, the Azure AD authentication system checks your DNS zone and searches for the appropriate TXT resource record that has the content that validates the domain. Create this record before you create the organization relationship.

Note: Even if an organization relationship specifies that all user calendars are shared, users can override this setting. Users can configure the default permissions for their own calendars to prevent sharing. However, changing the default permission also affects sharing with internal users.

To identify the external organization with which you want to create an organization relationship, you typically use the domain name of the external organization to automatically populate the necessary information into the organization relationship. If you specify the domain name, the wizard for establishing federation obtains all the necessary configuration information from the Azure AD authentication system.

If you use the Exchange Management Shell (EMS) to create the organization relationship, use the **Get-FederationInformation** cmdlet to obtain the federation information for the external organization. You can pipe this information to the **New-OrganizationRelationship** cmdlet when you create the organization relationship.

The following command provides an example ofcreating an organization relationship to share Free/Busy information with the domain adatum.com:

```
Get-FederationInformation -DomainName adatum.com | New-OrganizationRela-
tionship
```

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```
-Name "Adatum" -DomainNames adatum.com -FreeBusyAccessEnabled $true
-FreeBusyAccessLevel LimitedDetails
```

You can use Autodiscover to obtain the uniform resource locator (URL) for the external organization's Availability Web Service. If the external organization does not have Autodiscover configured for access from the Internet, you can enter the URL manually.

For example, let's assume you want to configure the organization relationship for the following settings:

- The Exchange Web Services application URL is mail.adatum.com
- The Autodiscover URL is **https://mail.adatum.com/autodiscover/autodiscover.svc/wssecurity**

To configure the relationship for this example, you should run the following command:

```
Set-OrganizationRelationship -Identity "Adatum" -TargetAutodiscoverEpr
"https://mail.adatum.com/autodiscover/autodiscover.svc/wssecurity" -Targ-
etApplicationUri "mail.adatum.com"
```

Implementing Sharing Policies

If you want to control federated sharing, sharing policies are an alternative to organization relationships. The main difference is that with sharing policies users in your organization send sharing invitations to external users with whom they want to share calendars or contacts.

You can assign a sharing policy to specific mailboxes, or you can use a sharing policy to determine what information a user can share with external users. Although the organization that contains the external user's mailbox does not need to have a federation trust, you should configure a federation trust to enable a two-way sharing relationship.

Sharing Policies available within Exchange include the Default Sharing policies, applied to all users, and specific Sharing Policies which are assigned to selected users. This allows you to ensure that sharing policies can be granular, and not automatically available for everybody in an external organization automatically as when you configure organization relationships.

Each Sharing Policy – both the default sharing policy and additional policies you create – consist of two core components:

- domains you wish to share with
- the level of sharing you wish to allow for those domains

You can configure the following settings for domains:

- Sharing with all domains
- Sharing with a specific domain

For the domains in-scope within the policy, you can then configure the following levels of sharing allowed:

- Calendar free/busy information with time only
- Calendar free/busy information with time, subject, and location
- All calendar appointment information, including time, subject, location and title

The default sharing policy automatically applies to all Exchange Server mailboxes, and unless you modify it, it enables the sharing of free/busy information with all domains. The default sharing policy enables users to share their free/busy information with external users immediately after a federation trust is created. Therefore, changing the default sharing policy will, by default, apply to every mailbox in your Exchange organization as well.

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While the default sharing policy is automatically applied to all mailboxes, any additional sharing policies must be applied to mailboxes to take effect. In addition, you can apply only a single sharing policy in each mailbox. A policy is applied to a mailbox by using the policy's properties or the recipient's properties.

You can use the Exchange admin center or the Exchange Management Shell to create sharing policies and assign them to specific mailboxes. For example, if you want to create a sharing policy for adatum.com that includes sharing free/busy details, you would run the following command:

```
New-SharingPolicy -Name "Adatum.com"
 -Domains 'adatum.com: CalendarSharingFreeBusyDetail'
```

Once you created a new sharing policy, you can apply it to one or more mailboxes. For example, you would run the following command to apply the sharing policy for adatum.com to a user named Alex in your organization:

```
Set-Mailbox -Identity Alex -SharingPolicy "Adatum.com"
```

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Module Review

Module 5 Review

Multiple choice

Item 1. You are a messaging administrator for Contoso, which recently deployed Microsoft 365. While all users are managed in Microsoft 365, you are concerned about the security of their credentials. You want all users to have to change their password every 180 days. Which of the following tasks should you perform to accomplish this requirement?

- ☐ Create a Group Policy in Active Directory that will enforce users to change passwords every 180 days.
- ☐ In the Microsoft 365 Admin Center, change the password policy to 180 days before passwords expire.
- ☐ Create a logon script that will require users to change passwords every 180 days.
- ☐ Update registry settings on all users' computers by running a script that will require users to change passwords every 180 days.

Multiple choice

Item 2. You are a messaging administrator for Adatum, which has deployed Microsoft 365. All users are managed in Microsoft 365. A situation has developed in which users are regularly forgetting their passwords, and then sending requests to administrators for password reset. With tens of thousands of employees world-wide, the rate of password reset requests has gone from being a simple annoyance to a critical issue that is consuming far too much time and resources. Management has decided that if users forget their passwords, they will need to reset passwords themselves. Which of the following tasks should you perform to enforce this new policy?

- ☐ Configure self-service password reset in the Azure Active Directory admin center.
- ☐ Configure self-service password reset in Active Directory.
- ☐ Configure self-service password reset in the Exchange Admin Center.
- ☐ Configure self-service password reset in the Exchange Management Shell.

Multiple choice

Item 3. You are a messaging administrator for Contoso, which has deployed Microsoft 365. All users are managed in Microsoft 365. Given your concern for the security of users' credentials, you have proposed implementing a policy change in which users must be authenticated not only by their username and password, but also through a PIN that they will receive on their mobile devices. Which of the following tasks should you perform to enforce this new policy?

- ☐ Create a Group Policy in Active Directory that requires users to enter a PIN after entering their username and password.
- ☐ In the Microsoft 365 Admin Center, configure multi-factor authentication for users in your organization.
- ☐ Create a logon script that requires users to enter a PIN after they enter their username and password.
- ☐ Create registry settings on computers using a script that requires users to enter a PIN after they enter their username and password.

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Checkbox

Item 4. You are a messaging administrator for Contoso, which has deployed Microsoft 365. All users are managed in Microsoft 365. You are considering ways in which you can increase the security of your users' credentials. Which of the following plans will help you achieve that goal? (Select 2)

- ☐ Educating users as to where they should enter passwords, such as on trusted organization approved devices or web sites that are validated and trusted.
- ☐ Educating users to use organizational email for public purposes through public kiosk computers or on any non-trusted person's computer.
- ☐ Educating users that they should reset passwords if they receive a phone call that a user needs password reset.
- ☐ Enforce at least an 8-character minimum length requirement using special Characters or phrases for passwords, but the length shouldn't be larger than that since longer passwords typically result in users repeating patterns that are easier to guess.
- ☐ Change passwords every two weeks.

Multiple choice

Item 5. Which of the following steps can you perform to increase the security of your users' individual passwords?

- ☐ In the Microsoft 365 Admin Center, change password policy to 180 days before passwords expire.
- ☐ Configure self-service password reset in the Azure Active Directory admin center.
- ☐ In the Microsoft 365 Admin Center, configure multi-factor authentication for users in your organization.
- ☐ Configure password policies in Group Policy Management Console (GPMC).

Multiple choice

Item 6. You are a messaging administrator for Contoso Ltd. You have been asked to configure User workload management so that you can control how Exchange system resources are consumed by users. Which of the following steps will help you achieve this goal?

- ☐ Customize the default throttling policy according to your needs.
- ☐ Customize the default email address policy according to your needs.
- ☐ Customize the default address book policy according to your needs.
- ☐ Customize the default role assignment policy according to your needs.

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Multiple choice

Item 7. You are a messaging administrator for Contoso Ltd,where you need to plan the storage requirements for your Exchange Online users. Your company is using Office 365 E3 licenses. What is the default mailbox quota limit for your Exchange Online users?

- ☐ 10 GB per user mailbox
- ☐ 200 GB per user mailbox
- ☐ 30 GB per user mailbox
- ☐ 100 GB per user mailbox

Checkbox

Item 8. You are a messaging administrator for Contoso Ltd. You are currently planning the storage and quota limits for your Exchange 2019 on-premises environment. What are the different quotas that you can configure on a user mailbox? (Select 3)

- ☐ Prohibit send quota
- ☐ Prohibit read quota
- ☐ Issue warning quota
- ☐ Prohibit send and receive quota
- ☐ Prohibit create quota
- ☐ Prohibit receive quota

Multiple choice

Item 9. You are a messaging administrator for Contoso Ltd. You have been tasked with deploying third-party add-ins based on group membership. What should you do?

- ☐ Always deploy add-ins to everyone with no filtering.
- ☐ Deploy add-ins on an individual user level.
- ☐ Deploy add-ins on a group level.
- ☐ Do not deploy any add-ins because they are a legacy feature.

Multiple choice

Item 10. You are a messaging administrator for Contoso Ltd. You must configure the federation trust for your newly acquired Microsoft 365 tenant. Which federation trust do you need to configure?

- ☐ Federation between your organization and Azure AD.
- ☐ Federation between your organization and foreign mail servers.
- ☐ Federation between Azure AD and foreign trust providers.
- ☐ Federation between your organization with Azure AD and all external organizations in trust with Azure AD.

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Multiple choice

Item 11. What does the Federated Trust wizard create for the purpose of federation, which is then replicated to all Exchange Servers?

- ☐ an AppID number
- ☐ a self-signed certificate
- ☐ an OrgID number
- ☐ a Sharing policy

Multiple choice

Item 12. Which of the following is a type of workload usage in which Exchange Server allows users to have greater resource consumption for short periods of time without throttling?

- ☐ Traffic shaping
- ☐ Sharing policies
- ☐ Burst allowances
- ☐ Maximum usage

Multiple choice

Item 13. Fill in the blank. To configure an organization relationship with an external partner company, you typically use the _____ to populate the required information automatically in the organization relationship.

- ☐ Domain name
- ☐ Display name of a user
- ☐ Microsoft 365 tenant name
- ☐ Description

Multiple choice

Item 14. Fill in the blank. The default sharing policy automatically applies to ____ Exchange Server 2019 mailboxes, and unless you modify it, it enables the sharing of free/busy information with all domains.

- ☐ legacy
- ☐ all
- ☐ some
- ☐ no

AUTHORIZED TRAINER USE ONLY. STUDENT USE PROHIBITED

Answers

Multiple choice

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- ☐ Configure self-service password reset in Active Directory.
- ☐ Configure self-service password reset in the Exchange Admin Center.
- ☐ Configure self-service password reset in the Exchange Management Shell.

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Checkbox

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- ☐ Educating users that they should reset passwords if they receive a phone call that a user needs password reset.
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- ■ Prohibit send and receive quota
- ☐ Prohibit create quota
- ☐ Prohibit receive quota

Multiple choice

Item 9. You are a messaging administrator for Contoso Ltd. You have been tasked with deploying third-party add-ins based on group membership. What should you do?

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- ■ Deploy add-ins on a group level.
- ☐ Do not deploy any add-ins because they are a legacy feature.

Multiple choice

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- ☐ Federation between Azure AD and foreign trust providers.
- ☐ Federation between your organization with Azure AD and all external organizations in trust with Azure AD.

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Multiple choice

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Multiple choice

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- ☐ legacy
- ■ all
- ☐ some
- ☐ no

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Module 6 Managing Mobile Devices

Mobile Device Mailbox Policies

Lesson introduction

In the modern organization, users have many devices from which they can perform everyday tasks. These tasks are not limited to communication, such as sending and receiving email, collaborating with chat, and audio and video conferencing, but also involve production tasks, such as updating and sharing files, updating applications, visiting websites, and installing apps.

Since the number of devices used by employees in an organization can be very large, it's imperative that administrators use a mobile device management solution that enables centralized and automated management of these devices. In this lesson, you will examine Mobile Device Management in Microsoft 365, as well as how Exchange ActiveSync and mobile device mailbox policies support this effort.

After completing this lesson, you will be able to:

- Describe how Exchange ActiveSync works
- Configure mobile device mailbox policies
- Understand Mobile Device Management in Microsoft 365

How Exchange ActiveSync works

When mobile devices, such as smartphones, connect to Exchange to get email, they often use a protocol named Exchange ActiveSync (EAS), also known simply as ActiveSync. ActiveSync is used whether the mobile device is using native apps (such as Mail on iOS) or apps from the app store (such as the Outlook app). The core functionality provides users with access to email, contacts, tasks, and calendaring. Additionally, it provides limited mobile device management (MDM) functionality.

Microsoft designed ActiveSync to function on high-latency and low-bandwidth networks. It uses HTTP and XML, which keeps the protocol lightweight. Many vendors have standardized on building in ActiveSync support on their devices, so it is very common for any modern device to have built-in ActiveSync support.

By default, Exchange ActiveSync is enabled. All users who have an Exchange mailbox can synchronize their mobile device with Microsoft Exchange Server.

You can perform the following tasks within Exchange ActiveSync:

- Enable and disable Exchange ActiveSync for users
- Set policies such as minimum password length, device locking, and maximum failed password attempts
- Initiate a remote wipe to clear all data from a lost or stolen mobile phone
- Run a variety of reports for viewing or exporting into a variety of formats
- Control which types of mobile devices can synchronize with your organization through device access rules

Some of the key features of ActiveSync include:

- **Remote device wipe**. If a mobile device that is connected to your Exchange environment is lost or stolen, the user or an administrator can remotely wipe the device to erase all its data.
- **MDM features**. You can enforce passwords on connected devices, enforce specific settings such as encryption, and set minimum password policies. While the built-in features don't rival the enterprise MDM products, they often complement them. In smaller organizations, it is common to solely use ActiveSync's built-in MDM functionality. The MDM features are implemented by using mobile device mailbox policies.
- **Direct Push**. When new content is ready to be synced to a device, Exchange Server can use a feature named Direct Push to notify the client. The client, after notification, then initiates a sync operation to obtain the latest content from the Exchange Server.

As a messaging administrator, you will be responsible for managing mobile device settings, including:

- **Policies with personal devices**. Many organizations choose to use mobile device mailbox policies to ensure a secure environment. Messaging administrators often have to deal with the fallout of such policies, since this has been known to cause consternation among employees who use a personal device for their work email. Not only do they take exception to having their personal device managed by their company, they're also worried about losing their personal data or losing control of their phone. Others aren't sure what it means to have their personal device managed by the email system.
- **Third-party apps**. Some users want to use third-party apps to connect to their mailbox. In many cases, the IT team will be familiar with the most common apps. However, because there are hundreds of apps available, you will occasionally have to work with apps that you aren't familiar with. It can be challenging for IT teams to support different apps and all the different devices types and operating system versions that are available today.
- **Connectivity**. When devices connect to Exchange Server from the internet, you have very little control over the quality of the connection. Users can connect from public Wi-Fi networks, from the middle of a national park, or from a congested network. Some users may not have enough data in their plan to effectively use email on their device. As the administrator, you should be familiar with these common scenarios so that you can help users during troubleshooting situations.

Note: It's important that you understand what happens when a remote wipe is initiated, especially if you allow users to use ActiveSync with their personal devices. Remote wipe can clear all data on the mobile phone, including installed applications, photos, and personal information.

The Active Sync connection process includes the following steps:

1. The mobile device uses HTTPS to connect to the Microsoft-Server-ActiveSync IIS virtual directory on the client access services of the Mailbox server, where the client is authenticated.

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2. After the client is successfully authenticated, the client access services proxy the client's request to the Mailbox server where the active database copy of the user's mailbox is located and retrieves the mailbox data.
3. Once the client has established the ActiveSync connection to the Exchange server, it downloads email, calendar items, contacts, and other configured items.

The ActiveSync protocol provides many functionalities. Some of the most important features of Exchange ActiveSync in Exchange include:

- Support for HTML-formatted messages
- Support for follow-up flags on messages
- Conversation grouping of email messages
- Ability to synchronize or not synchronize an entire conversation
- Synchronization of Short Message Service (SMS) messages with a user's Exchange mailbox
- Support for viewing message reply status
- Support for fast message retrieval
- Meeting attendee information
- Enhanced Exchange Search
- PIN reset
- Enhanced device security through password policies
- Autodiscover for over-the-air provisioning
- Support for setting automatic replies when users are away, on vacation, or out of the office
- Support for task synchronization
- Direct Push
- Support for free/busy information for contacts
- Global address list (GAL) photos
- A means of sending only the new portion of an email and avoiding redundant information
- A method to apply digital rights management control and encryption to email messages that are sent and received

Configuring mailbox policies for mobile devices

A mobile device mailbox policy lets you apply a common set of security and mobile device settings to a group of users. Each recipient in your organization must be assigned a mobile device mailbox policy. In fact, when you install Exchange Server or provision Exchange Online, a default mobile device mailbox policy is created, and users are automatically assigned this policy. However, you can also create multiple mobile device mailbox policies that you can apply to different types of users.

Mobile device mailbox policies can be created in the Exchange Administration Center (EAC) or the Exchange Management Shell (EMS). If you create a policy in the EAC, you can configure only a subset of the available settings; however, you can configure all available settings using EMS. For a complete list of policy settings, see **Mobile Device Mailbox Settings**[1].

1 https://aka.ms/AA3sklh

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Edit an existing policy

You can use the EAC or PowerShell to edit a mobile device mailbox policy. In the EAC, you should perform the following steps to edit a mobile device mailbox policy:

1. In the EAC, select **mobile** > **mobile device mailbox policies**.
2. Select a policy from the List view and select the **Edit** button.
3. Use the **general** and **security** tabs to edit the mobile device mailbox policy settings. For example, in the **security** tab, select **Require a password** to enable a mobile device password and enable minimum password length for 6 characters.
4. Select **Save** to update the policy.

You can edit a mobile device mailbox policy by using the **Set-MobileDeviceMailboxPolicy** cmdlet in the EMS. For example, you should run the following command to edit the default mobile device mailbox policy that will configure these settings:

- Enable passwords with alphanumerical characters
- Store the recovery password in Exchange
- Allow two weeks of email items to synchronize to the mobile device
- Allow wireless internet access
- Allow mobile devices to access information stored on a storage card

```
Set-MobileDeviceMailboxPolicy -Identity:Default '
-PasswordEnabled:$true '
-AlphanumericPasswordRequired:$true '
-PasswordRecoveryEnabled:$true '
-MaxEmailAgeFilter:TwoWeeks '
-AllowWiFi:$True '
-AllowStorageCard:$true '
-IsDefault:$true
```

Create a new policy

You can use the EAC to create a new mobile device mailbox policy by performing the following steps:

1. In the EAC, select **mobile** > **mobile device mailbox policies**, and then select the **plus (+) sign** icon.
2. Use the various check boxes and drop-down lists to configure the settings for the mobile device mailbox policy.
3. Select **Save**.

You can make the new mobile device mailbox policy the default mobile mailbox policy by selecting **This is the default policy**. After you make a mobile mailbox policy the default policy, all new users will be automatically assigned this policy when their user account is created.

You can create mobile device mailbox policy by using the **New-MobileDeviceMailboxPolicy** cmdlet. For example, you should run the following command to create a new mobile device mailbox policy that will configure these settings:

- Create a new mobile device mailbox policy named "Management"
- Allow Bluetooth, browser, and camera

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- Enable password with alphanumerical characters
- Store recovery password in Exchange
- Allow two weeks of email items to synchronize to the mobile device
- Allow wireless internet access
- Allow mobile device to access information stored on a storage card

```
New-MobileDeviceMailboxPolicy -Name:"Management" `
-AllowBluetooth:$true `
-AllowCamera:$true `
-PasswordEnabled:$true `
-AlphanumericPasswordRequired:$true `
-PasswordRecoveryEnabled:$true `
-MaxEmailAgeFilter: TwoWeeks `
-AllowWiFi:$true `
-AllowStorageCard:$true
```

Add or Remove Users from a Policy

You can use the EAC to assign a user's mobile device mailbox policy by performing the following steps:

1. In the EAC, select **recipients** > **mailboxes** and then select a mailbox.
2. In the Details pane, scroll to **Phone and Voice Features** and select **View details** to display the **Mobile Device Details** page.
3. The mobile device mailbox policy that's currently assigned is displayed. To change the mobile device mailbox policy, select **Browse**.
4. Choose the appropriate mobile device mailbox policy from the list, select **OK** and then select **Save**.

You can also use PowerShell to assign a user's mobile device mailbox policy. For example, to assign a mobile device mailbox policy named "Sales" to user Tony, run the following command.

```
Set-CASMailbox -Identity tony@contoso.com `
-ActiveSyncMailboxPolicy "Sales"
```

Mobile Device Management in Microsoft 365

Active Sync provides basic functionality to secure mobile devices that contain your company messaging data, but it is not as powerful as some full-featured MDM solutions. For example, Active Sync cannot protect your users from accessing documents or downloading and storing company data on potentially unsafe locations. Microsoft provides a solution for those scenarios called Mobile Device Management (MDM) for Microsoft 365. When configured, MDM replaces Active Sync.

Mobile Device Management for Microsoft 365 is included with many of the Microsoft 365 commercial subscriptions.

Note: Policies and access rules created in MDM for Microsoft 365 override Exchange ActiveSync mobile device mailbox policies and device access rules created in the Exchange Admin Center. After a device is enrolled in MDM for Microsoft 365, any Exchange ActiveSync mobile device mailbox policy or device access rule applied to the device will be ignored.

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When planning your Mobile Device Management strategy, you may choose between built-in Microsoft 365 mobile device management capabilities or using another product, such as the more powerful Microsoft Intune or other third-party products, depending on your organization's requirements.

Considerations for deploying MDM for Microsoft 365

- When you create a new policy, you may want to set the policy to allow access and report a policy violation when a user's device isn't compliant with the policy. This lets you see how many mobile devices would be impacted by the policy without blocking access to Microsoft365.
- Before you deploy a new policy to everyone in your organization, we recommend you test it on the devices used by a small number of users.
- Also, before you deploy policies, let your organization know the potential impacts of enrolling a device in MDM for Microsoft 365. Depending on how you set up the policies, devices that don't comply with these policies (non-compliant devices) could be blocked from accessing Microsoft 365. Non-compliant devices can also have apps installed, photos, and other personal information which, on an enrolled device, could be deleted if the device is wiped.

How to configure Mobile Device Management for Microsoft 365

To configure MDM for Microsoft 365, you should perform the following steps:

1. **Configure domains for MDM**. First, you must add two publicly accessible DNS CNAME records in your custom public DNS zone that point to Mobile Device Management for Microsoft 365. Those records, which are used by clients to access and enroll in MDM, are called **EnterpriseEnrollment** and **EnterpriseRegistration**. The following table displays the records needed for MDM in Microsoft 365.

Host name	Record type	Address	TTL
EnterpriseEnrollment	CNAME	EnterpriseEnrollment.manage.microsoft.com	3600
EnterpriseRegistration	CNAME	EnterpriseRegistration.windows.net	3600

2. **Configure an APNs Certificate for iOS devices**. To manage iOS devices like iPad and iPhones in Mobile Device Management for Microsoft 365, you must create an APNs certificate or token in the Azure portal by performing the following steps:

   - In the Azure portal, on the **Notification Hub** page, select **Apple (APNS)** on the left menu.
   - For **Authentication Mode**, select either **Certificate** or **Token**.
   - If you select **Certificate**:
     - Select the file icon, and then select the .p12 file you want to upload.
     - Enter a password.
     - Select **Sandbox** mode. Or, to send push notifications to users who purchased your app from the store, select **Production** mode.
   - If you select **Token**:
     - Enter the values for **Key ID**, **Bundle ID**, **Team ID**, and **Token**.

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- Select **Sandbox** mode. Or, to send push notifications to users who purchased your app from the store, select **Production** mode.

3. **Set up multi-factor authentication (MFA)**. While this is an optional step, it's recommended that you implement MFA because it provides an additional layer of security. MFA asks users for a second form of authentication, such as a phone call, text message, or app notification on their mobile device after correctly entering their work account and password.
4. **Manage device security policies**. You should create and deploy device security policies according to your organization security business requirements. To do this in the Microsoft 365 admin center, navigate to the **EndPoint Manager admin center**, select **devices**, and then select **Compliance policies**.

   - Select **Create Policy** and then select the platform the policy will apply to.
   - Select **Create**.
   - Enter a name and optionally a description for the policy.
   - Configure the policy settings, which are divided into functional areas such as **Device healt, Device properties, Configuration Manager Compliance, and System Security.**
   - Select the action you want performed on noncompliant devices.
   - Select who to assign the policy to (**Selected groups** or **All users**).
   - Review and create the policy.

In Microsoft 365 you can also subscribe to Intune, which is a more advanced Mobile Device Management and Mobile Application Management (MAM) solution. For more details about Microsoft Intune, see the next lesson titled "Managing Mobile Device Access".

Additional reading. For more information, see the following article on **Choose between MDM for Office 365 and Microsoft Intune**[2].

2 https://aka.ms/AA3skmn

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Managing Mobile Device Access

Lesson Introduction

Mobile phones, tablets, and other devices can store sensitive corporate data and provide access to many corporate resources. One of the key responsibilities of the messaging administrator is to configure infrastructure requirements so that mobile devices can connect to Exchange from both the internal corporate network and the Internet. However, messaging administrators should also protect and secure devices so that users can access corporate data and resources, but at the same time be protected by any malware, unauthorized access, and theft.

In this lesson, you will examine how to manage and troubleshoot mobile device access. This includes configuring both access and infrastructure for mobile devices, understanding the implications of mobile device remote wipe, and learning about alternative methods for mobile device management.

After this lesson, you will be able to:

- Configure access for mobile devices.
- Understand components of mobile device infrastructure.
- Tell how a mobile device remote wipe works.
- Describe alternatives for mobile device management.
- Troubleshoot mobile device access.

Configuring access for mobile devices

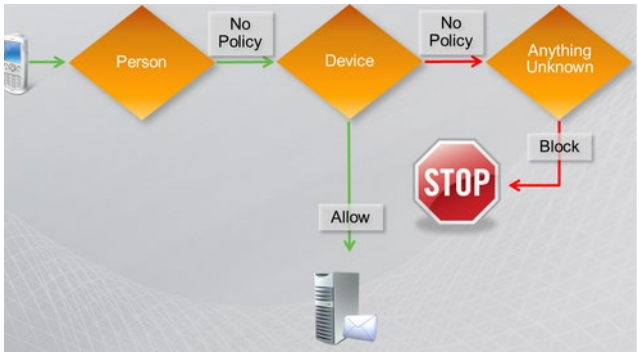
The default settings in Exchange allow any mobile device to connect if it adheres to the company's required security policy. Exchange Server lets administrators create allow lists, block lists, and quarantine lists that specify which mobile devices can access their Exchange mailboxes. This allows you to identify the devices that users can use to connect to the Exchange. For example, you can specify that only devices that are running a specific operating system or that are company-owned can connect to Exchange.

This capability is achieved by defining the device access state for each mobile device that connects to Exchange. A device access state is the status of a particular device. You can control device access states in several ways, and a mobile device will behave differently in each access state. The access state of a device can be one of the following:

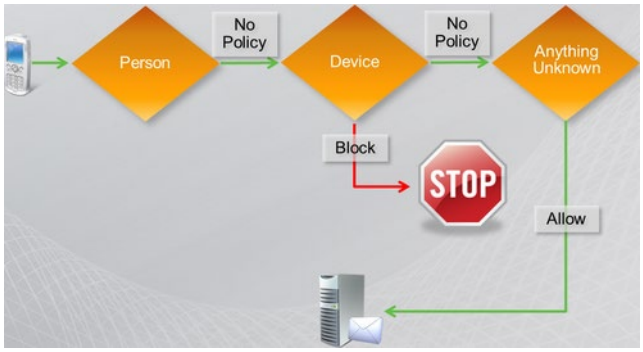
- **Allowed**. In the Allowed access state, a mobile device can synchronize through Exchange ActiveSync and connect to Exchange to retrieve email and work with calendar information, contacts, tasks, and notes.
- **Blocked**. If the device access rule specifies a device that should be blocked, that device cannot connect to Exchange Online, and it will receive an HTTP 403 forbidden error. You can block a device for a variety of reasons:
  - You can block a device based on the device family
  - You can block a specific device model
  - You can block a device because it fails to apply the mobile device mailbox policies
- **Quarantined**. When a mobile device is in a quarantined state, it can connect to Exchange, but with limited access to data. The user can add content to their calendar, contacts, tasks, and notes folders, but Exchange will not allow the device to retrieve any content from the user's mailbox.

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In a restrictive organization, the IT department will only choose to allow the devices it supports, and all other devices are blocked (as shown in the following flowchart).



Permissive organizations typically allow all, or most, devices to connect to Exchange. In these cases, the Allow, Block, Quarantine (ABQ) list can help organizations block a particular device or set of devices from connecting. This is useful if there is a security vulnerability or if the device is putting a particularly heavy load on the Exchange server. In these cases, the IT department can identify the misbehaving device and block that device until a fix or update for that device brings it into compliance. All other devices, including the unknown devices, are given access (as shown in the following flowchart).



Quarantining devices is useful when an IT department wants to monitor new devices connecting to their organization and decide before any company data is synchronized to new devices. Both permissive and restrictive organizations may choose to employ this mechanism. In a permissive organization, quarantine can be used so that IT administrators know what devices, and which users, are making new connections. In restrictive organizations, this can be used to see who is trying to work around policy and gauge demand from "Bring Your Own Device" (BYOD) users. This logic is depicted in the following flowchart. Note that you could also choose to quarantine at the device/device family level if desired (this scenario is not shown in the diagram for simplicity sake).



You can change the default access level from Allow to Quarantine by using the Exchange Admin Center or the **Set-ActiveSyncOrganizationSettings** cmdlet in Exchange Management Shell.

The following example uses the Exchange Management Shell to set the default access level across the Exchange organization to Quarantine and sets two administrative email addresses.

```
Set-ActiveSyncOrganizationSettings `
-DefaultAccessLevel Quarantine `
```

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```
-AdminMailRecipients will@contoso.com, roger@contoso.com
```

The following example uses the Exchange Management Shell to set the default access level to Access for a specific device type:

```
New-ActiveSyncDeviceAccessRule `
-QueryString "Outlook for iOS and Android" `
-Characteristic DeviceModel `
-AccessLevel Allow
```

Configuring infrastructure for mobile devices

To provide an environment where users can successfully connect with their mobile devices, access data, and securely collaborate with email, you must ensure that you have properly configured the necessary infrastructure components.

To successfully plan and configure your infrastructure, it's essential that you consider the following requirements:

- **Network infrastructure**. Mobile devices in Exchange use the HTTPS protocol to connect to client access services on the mailbox server. Therefore, you need to ensure that port 443, which is used by HTTPS protocol, must be open on the firewalls that connect mobile devices with the Exchange Mailbox Server role.
- **DNS requirements**. The Autodiscover record should be configured in both your internal and public DNS servers to provide automatic email account configuration for mobile devices. Furthermore, you should configure the CNAME records that are required when mobile devices enroll for an MDM solution.
- **High availability**. To provide high availability for users that connect with mobile devices, you should either use Network Load Balancers or configure round robin with multiple DNS records pointing to a single server name such as mail.contoso.com.
- **Devices support**. To protect corporate data and resources, organizations should only allow supported and approved devices to connect to Exchange.
- **Management solutions**. MDM and MAM solutions should be used so that security on mobile devices is enforced, and only approved applications are deployed for each mobile device that connects to the organization.

Mobile device remote wipe

When an ActiveSync connection is established between a mobile device and Exchange, the mobile device stores company data from the user's mailbox, including the user's domain credentials, which are the user name and password needed to authenticate to Exchange. If a device is lost or stolen, that data can be compromised.

Because the risk of losing a mobile device is always possible, it's essential that you secure the data on mobile devices. You can secure mobile devices by enforcing an ActiveSync policy that specifies password requirements for the device. However, this does not prevent data from being compromised when devices are lost or stolen.

For cases when a device is lost or stolen, Exchange provides an option called Remote Wipe to protect your company data. When the Remote Wipe command is issued, it deletes all data on the phone and storage cards and resets all settings to factory defaults. Restoring settings to factory defaults prevents

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any unauthorized user from accessing your account data or data cached on the device. If you're performing a remote device wipe on a mobile phone in your possession and you want to keep the data on the storage card, remove the storage card before you initiate the remote device wipe.

Since some smartphones do not have removable storage, **it's important to remember that, by default, Remote Wipe destroys all data on the device, including personal photos and data!** If you allow employee owned devices (BYOD) and you do not educate your users about the situations where a remote wipe is performed and what it does with the personal data, you can get into legal problems in several countries/regions. For example, consider the scenario where a user loses his phone, which triggers the messaging administrator to perform a remote wipe on the device per the company's policy. The next day the user finds his phone, only to discover that all his personal photos and files have been deleted.

The Remote Wipe command can be issued in either of two ways:

- By the user of a specific mobile device through the Outlook on the Web interface.
- By the administrator through the Exchange Admin Center (EAC) or Exchange Management Shell. The Remote Wipe command will only be accepted by the device if it still has a connection with the Exchange server, either by data (3G, LTE, or similar mobile data service) or by Wi-Fi. Remote Wipe will not work if the connection is lost; for example, when the Subscriber Identity Module, or SIM, card is removed, or the ActiveSync account is manually removed from the device. Therefore, if you plan to issue a Remote Wipe command, you should do so as soon as possible.

Note: After a remote device wipe, data recovery is very difficult, if not impossible, especially when the device supports a strong encryption of all device data. If the device was not encrypted, it may still be possible to retrieve data from a device using sophisticated forensic investigation tools.

Using the Exchange Admin Center to wipe a user's phone

You can use the EAC to wipe a user's phone or cancel a remote wipe that has not yet completed by performing the following steps:

1. In the EAC, navigate to **Recipients** > **Mailboxes**.
2. Select the user, and under **Mobile Devices**, choose **View details**.
3. On the **Mobile Device Details** page, select the lost mobile device, and then select **Wipe Data**.
4. Select **Save**.

Using the Exchange Management Shell to wipe a user's phone

You can use the **Clear-MobileDevice** cmdlet in the EMS to wipe a user's phone. For example, the following command wipes the device named WM_TonySmith and sends a confirmation message to admin@contoso.com.

```
Clear-MobileDevice -Identity WM_TonySmith -NotificationEmailAddresses admin@
contoso.com
```

Deleting only company data

Starting with Exchange ActiveSync version 16.1 it's possible to perform an "Account-only remote wipe" that deletes only Exchange company data and leaves personal data untouched. You can issue an account only remote wipe by adding the -AccountOnly switch when using the **Clear-MobileDevice** cmdlet.

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The previous command will now appear as follows assuming Contoso only requires company data to be removed when performing a remote wipe:

```
Clear-MobileDevice -Identity WM_TonySmith -AccountOnly -NotificationEmailAd-
dresses admin@contoso.com
```

Using Outlook on the web to wipe a user's phone

By performing the following steps, your users can wipe their own phones using Outlook on the web:

1. In Outlook on the web, select **Settings > Phone > Mobile devices**.
2. Select the mobile phone.
3. Click or tap the **Wipe Device** icon.

Verifying the remote wipe completed successfully

There are several ways to verify that the remote wipe completed:

- Run the **Clear-MobileDevice** cmdlet with the –NotificationEmailAddresses parameter configured. A message will be sent to the supplied email address when the remote wipe has completed.
- In the EAC, check the status of the mobile device. The status will change from **Wipe Pending** to **Wipe Successful**.
- In Outlook on the web, check the status of the mobile device. The status will change from **Wipe Pending** to **Wipe Successful**.

Local device wipe

Local device wipe occurs when a mobile device wipes itself without the request coming from the server. For example, your organization can implement a mobile device mailbox policy that specifies that when a maximum number of unsuccessful password attempts is exceeded, the mobile device should perform a local device wipe.

The result of a local device wipe is the same as that of a remote device wipe - the device is returned to its factory default condition. When a mobile device performs a local device wipe, no confirmation is sent to the Exchange server.

Alternatives for mobile device management

Both Mobile Device Management (MDM) for Microsoft 365 and Microsoft Intune provide the ability to manage mobile devices in an organization; in fact, both can be used in the same Microsoft 365 tenant. Setting up both Intune and MDM lets you decide which solution is best for specific users and their devices. If you have both options available, you can choose whether to manage a user's devices with MDM for Microsoft 365 or the more feature-rich Intune solution. However, there are key differences between the two solutions, each of which is described in the following table.

Feature area	MDM for Microsoft 365	Microsoft Intune
Cost	Included with many Microsoft 365 commercial subscriptions.	Requires a paid subscription for Microsoft Intune or can be purchased with Enterprise Mobility Suite.

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Feature area	MDM for Microsoft 365	Microsoft Intune
How you manage devices	Manage devices using the Security and Compliance Center in Microsoft 365.	If you use Intune by itself, you manage devices using the Intune admin console. If you integrate Intune with System Center 2012 Configuration Manager, you use the Configuration Manager console to manage devices on-premises and in the cloud.
Devices you can manage	Cloud-based management for iOS, Android, and Windows devices.	Cloud-based management for iOS, Mac OS X, Android, Windows 8.1 (Phone and PC) and later to include Windows 10

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Feature area	MDM for Microsoft 365	Microsoft Intune
Key capabilities	Help ensure that Microsoft 365 corporate email and documents can be accessed only on phones and tablets that are managed by your company and that are compliant with your IT policies. Set and manage security policies, like device level pin lock and jailbreak detection, to help prevent unauthorized users from accessing corporate email and data on a device when it is lost or stolen. Remove Microsoft 365 company data from an employee's device while leaving their personal data in place. Details are included in **capabilities of Built-in Mobile Device Management for Office 365** *(https://aka.ms/AA3s29a)*.	MDM for Microsoft 365 capabilities, plus: Help users securely access corporate resource with certificates, Wi-Fi, VPN, and email profiles. Enroll and manage collections of corporate-owned devices, simplifying policy and app deployment. Deploy your internal line-of-business apps and apps in stores to users. Enable your users to more securely access corporate information using the Office mobile and line-of business apps they know, while ensuring security of data by helping to restrict actions like copy, cut, paste, and save as, to only those apps managed by Intune. Enable more secure web browsing using the Intune Managed Browser app. Manage PCs from the cloud with no infrastructure required using Intune, or connect Intune to Configuration Manager to manage all of your devices including PCs, Macs, Linux and UNIX servers, and mobile devices from a single management console.

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Troubleshooting mobile device access

ActiveSync is most commonly used to provide email access to mobile devices. The access is through the built-in email app or through a third-party email app. ActiveSync is the protocol used to facilitate the access. There are several tools available to troubleshoot ActiveSync, including:

- **PowerShell**. You can use PowerShell to quickly run through some basic ActiveSync tasks to determine if there are issues. If things are healthy, the results for each test will be Success. Optionally, you can opt to not filter the output and you can get another column of information which lists errors. You would run the following command to initiate the ActiveSync test on the lon-ex1.adatum.com client access server:

  ```
  Test-ActiveSyncConnectivity -ClientAccessServer lon-ex1.adatum.com | FT
  CasServer,LocalSite,Scenario,Result -AutoSize
  ```

- **Smartphone**. Don't overlook the value of testing from a client device such as a smartphone. You can quickly set up a new email profile and attempt to connect with ActiveSync. By testing with a smartphone, you can determine whether the issue you are troubleshooting is limited to only one device or device type (such as all Apple iPhones). The downside to smartphone testing is that you will get limited troubleshooting information. In many scenarios, you will only receive a message that there was a failure to connect, without any supporting information.

- **Microsoft Remote Connectivity Analyzer**. The Remote Connectivity Analyzer mimics a smartphone to run through ActiveSync testing. The upside is that you get detailed troubleshooting information. Instead of just finding out that the connection failed, you receive support information that enables you to find out where in the connection it failed, and often why it failed.

Note: Most troubleshooting tools require that you specify credentials to run their specific test. It's recommended that you create a temporary test account for testing, and that you delete the account when the testing is complete.

In addition to determining whether ActiveSync itself is functional, you must also analyze the user configuration. This is especially true if just one (or a few) users are reporting the issue. At the user level, ActiveSync can be enabled or disabled. In the following screen capture, Beth Burke is currently disabled for ActiveSync. You can figure that out looking at the Mobile Devices section towards the bottom of the screenshot. This section includes the option to "Enable Exchange ActiveSync", which indicates that ActiveSync is currently disabled for Beth's mailbox.

Beyond In addition to determining whether a user is enabled for access, you can also test individual user access by using the Remote Connectivity Analyzer test. This can help you ensure that a user has functional ActiveSync access without reliance on their device. If you opt to do this, you should reset the user's password to a temporary password, perform the test, then reset the password again (at which point, the user can reset the password back to what they had or want).

Note: In several countries/regions it can be considered a violation of the user's data protection rights if you reset their password and access their messaging data, even if it's just for troubleshooting purposes. You should always be familiar with the data protection regulations and laws of the country/region in which your company resides.

Debug logging

If a specific mobile device is having problems with Exchange ActiveSync connectivity, you can enable Exchange ActiveSync debug logging. The debug log is a text file that contains detailed information about the command and responses between the mobile device and the Exchange server. Users can enable their own debug logging in Outlook on the web and retrieve the log as an email attachment.

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You can also enable debug logging, but it must be done in the Exchange Management Shell by running the following commands (in this example, sending email notification to Contoso's administrator):

```
Set-CASMailbox -Identity <user> -Exchange ActiveSyncDebugLogging $true
Get-MobileDeviceStatistics -Mailbox <user> -GetMailboxLog $true -Notifica-
tionEmailAddress administrator@contoso.com
```

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Lab5-Ex1-Implement ActiveSync

Lab5-Ex1-Implement ActiveSync

In this lab you will be guided through the process of enabling and disabling Exchange ActiveSync. ActiveSync is a client protocol that enables users to synchronize their Exchange mailbox with a mobile device. In this lab, you will log into the Exchange Server (LON-EX1) VM and set up ActiveSync for a single mailbox as well as for multiple mailboxes using the on-premises Exchange admin center.

See your VM lab environment for the lab instructions.

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Module Review

Module 6 Review

Checkbox

Item 1. Which of the following are functionalities of Active Sync? (Select 2)

- ☐ It is designed to function on high-latency and low-bandwidth networks.
- ☐ It is designed for low-latency and high-bandwidth networks only.
- ☐ It is available only on Microsoft hardware.
- ☐ Many vendors have standardized Active Sync support on their devices.

Checkbox

Item 2. Which of the following are features of ActiveSync? (Select 2)

- ☐ Enables large file transfers between sites
- ☐ Supports remote device wipe
- ☐ Provides Direct Push functionality
- ☐ Is based on the SMTP protocol

Checkbox

Item 3. You are a messaging administrator for Contoso Ltd., and you need to configure mobile device mailbox policies for your user's devices. Which of the following are features that you can control through mobile device mailbox policies? (Select 3)

- ☐ Allow Bluetooth and camera
- ☐ Send all pictures from the device to an administrator
- ☐ Block the private use of custom business line apps
- ☐ Allow wireless internet access
- ☐ Deploy wireless profiles on mobile devices
- ☐ Enable password with alphanumerical characters

Checkbox

Item 4. You are a messaging administrator for Contoso Ltd., and you need to configure MDM for Microsoft 365. What are the two DNS records that you must add to your custom public DNS zone to allow enrollment of clients to MDM? (Select 2)

- ☐ A CNAME record with the name EnterpriseEnrollment
- ☐ A MX record with the name EnterpriseEnrollment
- ☐ A SPF records with the name EnterpriseEnrollment
- ☐ A CNAME record with the name EnterpriseRegistration

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Multiple choice

Item 5. You are a messaging administrator for Contoso Ltd. You have been tasked with managing iOS devices with MDM for Microsoft 365. What do you need to do first before you can manage Apple devices?

- ☐ Install an APN certificate for iOS devices on your on-premises servers.
- ☐ Create an APN certificate in the Microsoft 365 admin center.
- ☐ There are no special requirements to manage Apple devices.
- ☐ You cannot manage Apple devices with MDM for Office 365.

Multiple choice

Item 6. What is the name of the more advanced mobile device management solution offered by Microsoft that is also able to perform mobile application management?

- ☐ Microsoft Intune
- ☐ Enterprise Mobility + Security
- ☐ Microsoft Active Directory Federation Services (ADFS)
- ☐ Microsoft 365 Apps for Enterprise

Multiple choice

Item 7. You are a messaging administrator for Contoso Ltd. You must deploy a solution that allows mobile devices to access your Exchange servers. In addition, before users can sync email through Active Sync using their personal devices, they must first receive administrator approval. What is the name of the feature that you should implement that allows administrators to decide whether a user can sync email through their personal device using Active Sync?

- ☐ Unknown
- ☐ Allowed
- ☐ Blocked
- ☐ Quarantine

Multiple choice

Item 8. You are a messaging administrator for Contoso Ltd., and you need to ensure access for mobile devices to your Exchange organization. To accomplish this, you must contact your Security department and inform them that they must ensure a specific port be open on the firewalls that connect mobile devices with the Exchange Mailbox Server role. Which port are you referring to?

- ☐ port 25 (SMTP)
- ☐ port 443 (HTTPS)
- ☐ port 80 (HTTP)
- ☐ port 53 (DNS)

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Multiple choice

Item 9. You are a messaging administrator for Contoso Ltd. A user reports that he lost his mobile phone, which is synchronized with the company's messaging environment. While you need to remove all company data from his phone, you have not yet introduced the Account-Only remote wipe functionality at Contoso. What will you tell the user?

- ☐ His personal data will remain intact when performing a remote wipe.
- ☐ Only the settings of the phone are wiped, and all applications and data will remain intact.
- ☐ Everything will be deleted from the mobile phone's internal storage.
- ☐ You cannot perform a remote wipe on personal devices.

Multiple choice

Item 10. You are a messaging administrator for Contoso Ltd., and you are currently evaluating whether you want to implement Account-Only remote wipe functionality. You are using the Remote Connectivity Analyzer (RCA) to check the Active Sync version of your Exchange environment. What is the earliest version of Exchange ActiveSync that you can implement that includes Account-Only remote wipe functionality?

- ☐ 1.0
- ☐ 16.1
- ☐ 2.0
- ☐ 25.75

Multiple choice

Item 11. Which of the following statements is correct regarding the costs of MDM for Microsoft 365?

- ☐ MDM for Microsoft 365 needs to be bought separately and installed on an on-premises server.
- ☐ MDM for Microsoft 365 is included in Enterprise subscriptions.
- ☐ MDM for Microsoft 365 is deprecated and cannot be bought anymore.
- ☐ MDM for Microsoft 365 requires Enterprise Mobility + Security licenses.

Multiple choice

Item 12. What do you need to configure to restrict Active Sync access for individual users?

- ☐ You need to use role assignment policies.
- ☐ You need to use retention policies.
- ☐ You need to use address book policies.
- ☐ You can enable and disable Active Sync on individual mailboxes.

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Multiple choice

Item 13. You are a messaging administrator for Contoso Ltd. Because you are encountering Active Sync errors, you want to investigate the debug log for Active Sync of a user. What do you need to do first?

- ☐ Log in to an Exchange Mailbox server.
- ☐ Log in to your Microsoft 365 tenant.
- ☐ Activate debug logging for the mailbox.
- ☐ Grant write access to drive C on the Mailbox server with the user's mailbox.

Multiple choice

Item 14. Where should the Autodiscover record be configured?

- ☐ In your internal DNS servers only
- ☐ In your Edge Transport servers
- ☐ In your Mailbox servers
- ☐ In both your internal and public DNS servers
- ☐ In your perimeter network
- ☐ In Azure Active Directory

Multiple choice

Item 15. Fill in the blank. You can secure mobile devices by enforcing a(n) _____ policy that specifies password requirements for the device.

- ☐ Intune
- ☐ Autodiscover
- ☐ Multi-Factor Authentication
- ☐ Group
- ☐ ActiveSync
- ☐ DNS

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Answers

Checkbox

Item 1. Which of the following are functionalities of Active Sync? (Select 2)

- ■ It is designed to function on high-latency and low-bandwidth networks.
- ☐ It is designed for low-latency and high-bandwidth networks only.
- ☐ It is available only on Microsoft hardware.
- ■ Many vendors have standardized Active Sync support on their devices.

Checkbox

Item 2. Which of the following are features of ActiveSync? (Select 2)

- ☐ Enables large file transfers between sites
- ■ Supports remote device wipe
- ■ Provides Direct Push functionality
- ☐ Is based on the SMTP protocol

Checkbox

Item 3. You are a messaging administrator for Contoso Ltd., and you need to configure mobile device mailbox policies for your user's devices. Which of the following are features that you can control through mobile device mailbox policies? (Select 3)

- ■ Allow Bluetooth and camera
- ☐ Send all pictures from the device to an administrator
- ☐ Block the private use of custom business line apps
- ■ Allow wireless internet access
- ☐ Deploy wireless profiles on mobile devices
- ■ Enable password with alphanumerical characters

Checkbox

Item 4. You are a messaging administrator for Contoso Ltd., and you need to configure MDM for Microsoft 365. What are the two DNS records that you must add to your custom public DNS zone to allow enrollment of clients to MDM? (Select 2)

- ■ A CNAME record with the name EnterpriseEnrollment
- ☐ A MX record with the name EnterpriseEnrollment
- ☐ A SPF records with the name EnterpriseEnrollment
- ■ A CNAME record with the name EnterpriseRegistration

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Multiple choice

Item 5. You are a messaging administrator for Contoso Ltd. You have been tasked with managing iOS devices with MDM for Microsoft 365. What do you need to do first before you can manage Apple devices?

- ☐ Install an APN certificate for iOS devices on your on-premises servers.
- ■ Create an APN certificate in the Microsoft 365 admin center.
- ☐ There are no special requirements to manage Apple devices.
- ☐ You cannot manage Apple devices with MDM for Office 365.

Multiple choice

Item 6. What is the name of the more advanced mobile device management solution offered by Microsoft that is also able to perform mobile application management?

- ■ Microsoft Intune
- ☐ Enterprise Mobility + Security
- ☐ Microsoft Active Directory Federation Services (ADFS)
- ☐ Microsoft 365 Apps for Enterprise

Multiple choice

Item 7. You are a messaging administrator for Contoso Ltd. You must deploy a solution that allows mobile devices to access your Exchange servers. In addition, before users can sync email through Active Sync using their personal devices, they must first receive administrator approval. What is the name of the feature that you should implement that allows administrators to decide whether a user can sync email through their personal device using Active Sync?

- ☐ Unknown
- ☐ Allowed
- ☐ Blocked
- ■ Quarantine

Multiple choice

Item 8. You are a messaging administrator for Contoso Ltd., and you need to ensure access for mobile devices to your Exchange organization. To accomplish this, you must contact your Security department and inform them that they must ensure a specific port be open on the firewalls that connect mobile devices with the Exchange Mailbox Server role. Which port are you referring to?

- ☐ port 25 (SMTP)
- ■ port 443 (HTTPS)
- ☐ port 80 (HTTP)
- ☐ port 53 (DNS)

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Multiple choice

Item 9. You are a messaging administrator for Contoso Ltd. A user reports that he lost his mobile phone, which is synchronized with the company's messaging environment. While you need to remove all company data from his phone, you have not yet introduced the Account-Only remote wipe functionality at Contoso. What will you tell the user?

- ☐ His personal data will remain intact when performing a remote wipe.
- ☐ Only the settings of the phone are wiped, and all applications and data will remain intact.
- ■ Everything will be deleted from the mobile phone's internal storage.
- ☐ You cannot perform a remote wipe on personal devices.

Multiple choice

Item 10. You are a messaging administrator for Contoso Ltd., and you are currently evaluating whether you want to implement Account-Only remote wipe functionality. You are using the Remote Connectivity Analyzer (RCA) to check the Active Sync version of your Exchange environment. What is the earliest version of Exchange ActiveSync that you can implement that includes Account-Only remote wipe functionality?

- ☐ 1.0
- ■ 16.1
- ☐ 2.0
- ☐ 25.75

Multiple choice

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- ☐ DNS

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Module 7 Managing Role-Based Permissions

Managing Admin Roles

Lesson Introduction

Permission management is an essential task for any messaging administrator. Exchange Server and Exchange Online both use the same permission model that was originally introduced in Exchange 2010 called Role Based Access Control (RBAC).

This lesson examines the basics of admin roles that are managed through RBAC.

After this lesson, you will be able to:

- Describe how RBAC is used to assign roles to users
- Understand what management role group for administrative tasks are
- Assign the built-in management roles for administration
- Create custom management roles and assign them through role assignment policies to users
- Troubleshoot RBAC management roles

Introduction to role-based access control

Exchange includes a large set of predefined permissions based on the Role Based Access Control (RBAC) permissions model. This enables you to easily grant permissions to your administrators and users; thereby, enabling you to get your organization up and running quickly.

RBAC is the permission model that's used in most of the Microsoft services, such as Exchange Server, Exchange Online, Microsoft 365 Admin Center, and the Security and Compliance Center. That being said, it's important to note that role groups for one service do not necessarily grant permissions for other services.

RBAC enables you to control what administrators and end-users can do at both broad and granular levels. RBAC also enables you to more closely align the roles you assign users and administrators to the actual roles they hold within your organization.

Management roles grant permissions to perform tasks in Exchange, but you need an easy way to assign them to administrators and users. There are two primary ways of assigning permissions to users in your organization with RBAC, depending on whether the user is an administrator, specialist user, or end-user:

- Management role groups
- Management role assignment policies

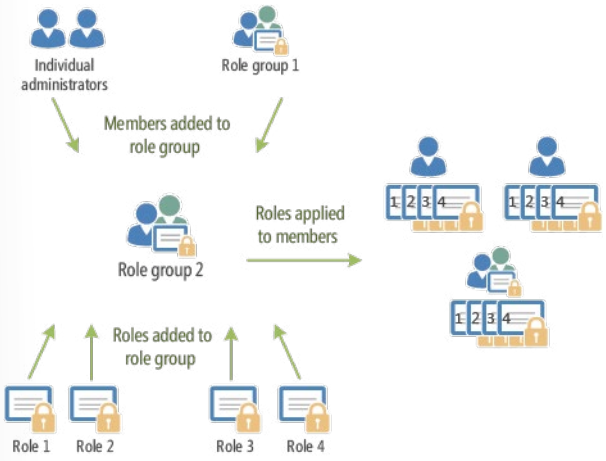
A third, more advanced method, direct user role assignment, can also be used. This method is covered in the next lesson.

Management role groups

Every administrator who manages Exchange must be assigned at least one or more roles. Messaging administrators tend to have more than one role because they typically perform job functions that span multiple areas in Exchange. For example, if a messaging administrator manages both recipients and transport hygiene features in the Exchange organization, he or she may be assigned both the Mail Recipients and Hygiene Management roles.

To make it easier to assign multiple roles to an administrator, Exchange includes role groups. When a role is assigned to a role group, the permissions granted by the role are granted to all the members of the role group. This enables you to assign many roles to many role group members at once. Role groups typically encompass broader management areas, such as recipient management, and they're only used with administrative roles, and not end-user roles. Role group members can be Exchange users as well as other role groups.

The following diagram depicts the relationship between users, role groups, and roles.



There is also a set of built-in user roles in Exchange, which are covered in a later lesson.

Management role assignment policies

Exchange provides role assignment policies so that you can control what settings your users can configure on their own mailboxes, what functionality is exposed, and what control they possess over distribution groups they own. These settings include their display name, contact information, voice mail settings, and distribution group membership.

Your Exchange organization can have multiple role assignment policies that provide different levels of permissions for the different types of users in your organizations. Some users can be allowed to change their address or create distribution groups, while others can't, depending on the role assignment policy

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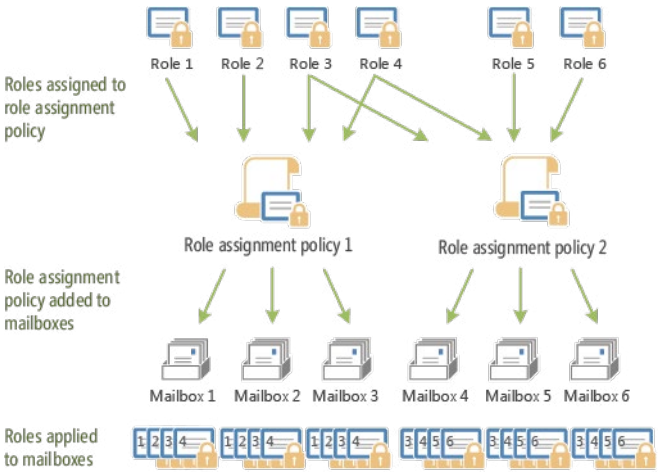
associated with their mailbox. Role assignment policies are added directly to mailboxes, and each mailbox can only be associated with one role assignment policy at a time.

Of the role assignment policies in your organization, one is marked as default. The default role assignment policy is associated with new mailboxes that aren't explicitly assigned a specific role assignment policy when they're created. The default role assignment policy should contain the permissions that should be applied to the majority of your mailboxes.

Permissions are added to role assignment policies using end-user roles. End-user roles begin with "My", and they grant permissions for users to manage only the mailbox or distribution groups they own. They can't be used to manage any other mailbox. Only end-user roles can be assigned to role assignment policies.

When an end-user role is assigned to a role assignment policy, all the mailboxes associated with that role assignment policy receive the permissions granted by the role. This enables you to add or remove permissions to sets of users without having to configure individual mailboxes. This relationship is depicted in the following diagram, which shows the hierarchical relationship between end-user roles, role assignment policies, and roles assigned to mailboxes:

- **End-user roles are assigned to role assignment policies.** Role assignment policies can share the same end-user roles.
- **Role assignment policies are associated with mailboxes.** Each mailbox can only be associated with one role assignment policy.
- **After a mailbox is associated with a role assignment policy, the end-user roles are applied to that mailbox.** The permissions granted by the roles are granted to the user of the mailbox.



Introduction to management role groups

Management role groups associate management roles to a group of administrators or specialist users that need to perform administrative tasks:

- Administrators manage a broad Exchange organization or recipient configuration.
- Specialist users can either manage the specific features of Exchange, such as compliance, or they may have limited management abilities, such as Help desk members, but aren't assigned broad administrative rights.

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Role groups typically include administrative management roles that enable administrators and specialist users to manage the configuration of their organization and recipients. For example, role groups control whether administrators can manage recipients or use mailbox discovery features.

Adding or removing users to or from role groups is how you most often assign permissions to administrators or specialist users.

Role groups consist of the following components that define what administrators and specialist users can do:

- **Management role group.** The management role group is a special universal security group (USG) that contains mailboxes, users, other USGs, and other role groups that are members of the role group. This is where you add and remove members, and it's also what management roles are assigned to. The combination of all the roles on a role group defines everything that users added to a role group can manage in the Exchange organization.
- **Management role.** A management role is a container for a grouping of management role entries. Roles are used to define the specific tasks that can be performed by the members of a role group that's assigned the role. A management role entry is a cmdlet, script, or special permission that enables each specific task in a role to be performed.
- **Management role assignment.** A management role assignment links a role and a role group. Assigning a role to a role group grants members of the role group the ability to use the cmdlets and parameters defined in the role. Role assignments can use management scopes to control where the assignment can be used.
- **Management role scope.** A management role scope is the scope of influence or impact on a role assignment. When a role is assigned with a scope to a role group, the management scope targets specifically what objects that assignment can manage. The assignment, and its scope, are then given to the members of the role group and restrict what those members can manage. A scope can consist of a list of servers or databases, organizational units (OUs), or filters on server, database or recipient objects.

When you add a user to a role group, the user is given all the roles assigned to the role group. If scopes are applied to any of the role assignments between the role group and the roles, those scopes control what server configuration or recipients the user can manage.

If you want to change what roles are assigned to role groups, you need to change the role assignments that link the role groups to roles. Unless the assignments built into Exchange don't suit your needs, you won't have to change these assignments.

Managing permissions with RBAC to fulfil your organization’s business requirements can be very complicated. Management role groups are created, assigned, and removed through the Exchange Admin Center or with the Exchange Management Shell. As a messaging administrator, it’s recommended that you use the Exchange Management Shell for working with advanced RBAC needs, because several tasks can only be performed with PowerShell, such as adding delegates and working with complex scopes. Messaging admins should become very familiar with the following PowerShell cmdlets for managing role groups:

- *-RoleGroup
- *-RoleGroupMember
- *-ManagementRole
- *-ManagementScope
- *-ManagementRoleEntry
- *-ManagementRoleAssignment

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Management role scopes

Management role scopes enable you to define the specific scope of impact or influence of a management role when a management role assignment is created. When you apply a scope, the role assignee assigned to the role can only modify the objects contained within that scope. A role assignee can be a management role group, management role, management role assignment policy, user, or universal security group (USG).

Every management role, whether it's a built-in role or a custom role, has management scopes. Management scopes can be either of the following:

- **Regular scope.** A regular scope isn't exclusive. It determines where, in Active Directory, that objects can be viewed or modified by users assigned the management role. In general, a management role indicates what you can create or modify, and a management role scope indicates where you can create or modify. Regular scopes can be either implicit or explicit scopes.
- **Exclusive scope.** An exclusive scope behaves almost the same as a regular scope. The primary difference is that an exclusive scope enables you to deny users access to objects contained within the exclusive scope if those users aren't assigned a role associated with the exclusive scope. All exclusive scopes are explicit scopes.

Scopes can be inherited from the management role, specified as a predefined relative scope on a management role assignment, or created using custom filters and added to a management role assignment. Scopes inherited from management roles are called implicit scopes while predefined and custom scopes are called explicit scopes.

The following table provides an overview of existing management scopes.

Scope type	Description	Example
Implicit scopes	Implicit scopes are the default scopes that apply to a management role type. Because implicit scopes are associated with a management role type, all the parent and child management roles with the same role type also have the same implicit scopes. Implicit scopes apply to both built-in management roles and to custom management roles.	**Organization** to create or modify recipient objects across the Exchange organization. **OrganizationConfig** to create or modify any server or database configuration object across the Exchange organization.
Explicit scopes	Explicit scopes are scopes that you set yourself to control which objects a management role can modify. Although implicit scopes are defined on a management role, explicit scopes are defined on a management role assignment. This enables the implicit scopes to be applied consistently across all management roles unless you choose to use an overriding explicit scope.	See predefined relative scopes and custom scopes.

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Scope type	Description	Example
Predefined relative scopes	Relative scopes are explicit scopes that are referring to the user the role is assigned to. They are "relative" to the user. Predefined relative scopes can only be used to scope recipient objects. Predefined relative scopes can't be used to scope configuration objects.	**Self** to modify only the properties of the current user's mailbox. **MyDistributionGroups** to create or modify distribution list objects owned by the current user.
Custom scopes	Custom scopes are explicit scopes that are needed when neither the implicit write scope nor the predefined relative scopes meet the needs of your business. Custom scopes enable you to define, at a granular level, the scope to which your management role will be applied.	**OU scope** to modify only recipient objects within an OU. **Recipient filter** scope to manage only a certain recipient type or other recipient properties such as department, manager, location, and more. **Configuration scope** to target specific servers based on server lists or filterable properties that can be defined on servers, such as an Active Directory site or a server role.

Note: There are also "unscoped top-level management roles", which are a special type of management role that enables you to grant access to custom scripts and non-Exchange cmdlets to users using RBAC. If you need to provide access to non-Exchange cmdlets that run on an Exchange server, or if you need to publish a script that can be run by your users, you need to add them to an unscoped role.

Management role hierarchy

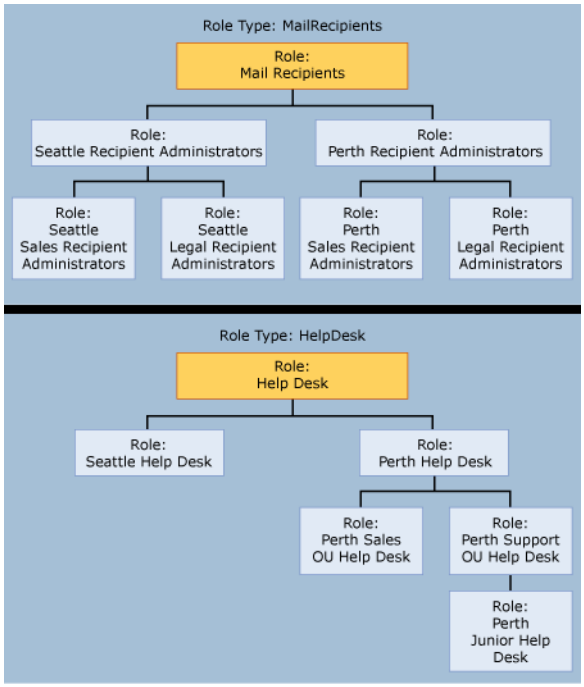
Management roles exist in a parent and child hierarchy. At the top of the hierarchy are the built-in management roles provided by default in Exchange. When you create a role, a copy of a parent role is made. The new role is a child of the role you copied it from. You can then customize the new role to suit the needs of the administrators or users that will be assigned to the role

Customized roles can be used to create roles. When you create a role from an existing customized role, the existing customized role remains a child of its parent role, but also becomes the parent for the new role. Each time a role is copied, the new child role can contain only the role entries that exist in the immediate parent role.

Each management role is given a role type that can't be changed. The role type defines the base context of use for the role. The role type is copied from the parent role when the child role is created.

The following diagram illustrates the hierarchical relationship of several management roles. The Help Desk role is a built-in role, and all the child roles derived from this role inherit the role type of the built-in role. All child roles derived either directly or indirectly from the Help Desk role inherit the HelpDesk role type.

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Built-in management role groups

Exchange includes several built-in role groups, each of which provides permissions to manage specific areas in Exchange. Some role groups may overlap with other role groups. The following table lists each role group with a description of its use.

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Role group	Description	Available in...
Organization Management	Administrators who are members of the Organization Management role group have administrative access to the entire Exchange organization and can perform almost any task against any Exchange Server object, with some exceptions, such as the Discovery Management role. **Important**: Because the Organization Management role group is a powerful role, only users or USGs that perform organizational-level administrative tasks that can potentially impact the entire Exchange organization should be members of this role group. **Note**: The administrative access does not include the Discovery Management role to access mailbox content.	Exchange Server and Exchange Online
View-Only Organization Management	Administrators who are members of the View Only Organization Management role group can view the properties of any object in the Exchange organization.	Exchange Server and Exchange Online
Recipient Management	Administrators who are members of the Recipient Management role group have administrative access to create or modify Exchange recipients within the Exchange organization.	Exchange Server and Exchange Online

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Role group	Description	Available in...
Help Desk	By default, the Help Desk role group enables members to view and modify the Outlook on the web (formerly known as Outlook Web App) options of any user in the organization. These options can include modifying the user's display name, address, and phone number. They don't include options that aren't available in Outlook on the web options, such as modifying the size of a mailbox or configuring the mailbox database on which a mailbox is located.	Exchange Server and Exchange Online
Hygiene Management	Members of this management role group can manage Exchange anti-spam features and grant permissions for antivirus products to integrate with Exchange.	Exchange Server and Exchange Online
Records Management	Users who are members of the Records Management role group can configure compliance features, such as retention policy tags, message classifications, and transport rules.	Exchange Server and Exchange Online
Discovery Management	Administrators or users who are members of the Discovery Management role group can perform searches of mailboxes in the Exchange organization for data that meets specific criteria and can also configure legal holds on mailboxes.	Exchange Server and Exchange Online
Public Folder Management	Administrators who are members of the Public Folder Management role group can manage public folders on servers running Exchange Server.	Exchange Server only
Server Management	Administrators who are members of the Server Management role group can configure server-specific configuration of transport, client access, and mailbox features such as database copies, certificates, transport queues and Send connectors, virtual directories, and client access protocols.	Exchange Server only

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Role group	Description	Available in...
Delegated Setup	Administrators who are members of the Delegated Setup role group can deploy servers running Exchange Server that have been previously provisioned by a member of the Organization Management role group.	Exchange Server only
Compliance Management	Users who are members of the Compliance Management role group can configure and manage Exchange compliance settings in accordance with their organization's policy.	Exchange Server and Exchange Online
Company Administrators (TenantAdmins_ <unique value>)	The Company Administrators role group is a special role group that ties together the Global administrators Office 365 role and the Organization Management Role Exchange Online role group. The Company Administrators role group doesn't have any roles assigned to it. However, it's a member of the Organization Management role group and inherits the permission provided by that role group. This role group can't be managed in Exchange Online. You can add members to this role group by adding users to the Global administrator Office 365 role.	Exchange Online only

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Role group	Description	Available in...
Help Desk Administrators (HelpdeskAdmins_ <unique value>)	The Help Desk Administrators role group doesn't have any roles assigned to it. However, it's a member of the View-Only Organization Management role group and inherits the permissions provided by that role group. This role group can't be managed in Exchange Online. You can add members to this role group by adding users to the Password administrator Office 365 role.	Exchange Online only
UM Management	Administrators who are members of the UM Management role group can manage features in the Exchange Online organization such as UM properties on mailboxes, UM prompts, and UM auto attendant configuration. **Note**: The UM Management role is not available in Exchange 2019.	Exchange Online only

If you work in a small organization that has only a few administrators, you may need to add those administrators to just the Organization Management role group, and you may never need to use the other role groups. If you work in a larger organization, you may have administrators who perform specific tasks administering Exchange, such as recipient or organization-wide Public Folder configuration. In those cases, you may add one administrator to the Recipient Management role group and another administrator to the Public Folder Management role group. Those administrators can then manage their specific areas of Exchange, but they won't have permissions to manage areas they're not responsible for.

If the built-in role groups in Exchange don't match the job function of your administrators, you can create custom role groups and add roles to them.

Microsoft 365 permissions in Exchange Online

The roles in the Microsoft 365 Admin Center, Security & Compliance Center, and the service specific RBAC roles in Exchange Online and SharePoint Online typically don’t interfere with each other, although there are some exceptions.

When you create a user in Microsoft 365, you can choose whether to assign various administrative roles in the Microsoft 365 Admin Center, such as Global administrator, Service administrator, Password administrator, and so on, to the user. Some, but not all, Microsoft 365 roles grant the user administrative permissions in Exchange Online, as defined in the following table.

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Microsoft 365 Admin Center role	Exchange Online role group
Global administrator	Organization Management **Note**: The Global administrator role and the Organization Management role group are linked together using a special Company Administrator role group. The Company Administrator role group is managed internally by Exchange Online and can't be modified directly.
Billing administrator	No corresponding Exchange Online role group
Password administrator	Help Desk administrator
Service administrator	No corresponding Exchange Online role group
User management administrator	No corresponding Exchange Online role group

Note: Role Based Access Control is available within other services, such as Azure Active Directory and Microsoft Teams. However, these services do not implement RBAC in the same way as Microsoft Exchange, and function in a similar way to the Microsoft 365 Admin Center. You can assign users to roles, but you do not have the ability to create fine-grained roles that allow or restrict administrative capability or user capability to the same level of detail as Exchange Online.

Working with admin roles for role assignment policies

You need to create custom management role groups if the built-in RBAC management role groups do not provide sufficient permissions for all your company job roles. However, before you start creating completely new management roles, you should always remember the following two best practices:

- **Don’t change the existing built-in role groups.** If you change the default groups, you will most likely end up in a situation you can no longer receive support involving role group issues because you have modified the default roles.
- **Don’t create new roles and role groups from the scratch.** Instead, you should find an existing role group that most closely matches your needs, copy this role group as a new custom role group, and then add or remove roles to this new role group until it meets your business requirements. You should avoid creating roles and role groups from the scratch because your custom roles and role groups should fit into the existing permission structure rather than creating a completely new hierarchy of permission roles in parallel to the built-in roles.

Plan and implement your management role groups

Before you start creating any roles or role groups, you should match your business job roles with the built-in roles. If you have any job roles remaining that could not be matched to a built-in role, you should then answer the following questions:

- **How many role groups do you need?** Always start with a single role group, and then add another role group if that one is not enough.
- **What roles will you add to each role group?** Decide what roles you want to add to the custom role group, try to use the built-in roles, and create custom roles only if the built-in roles are not suitable.
- **What scopes do you require for each role group?** Define a scope, such as a database scope, only if you need it.

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As soon as you finished matching all job roles to built-in and custom roles, you can then begin creating the custom management roles and role groups that will map to your unmatched job roles.

To manage your permissions using role groups in Exchange, you can use the Exchange Management Shell and the EAC. Because assigning roles with PowerShell is different from working with the EAC, it's recommended that you use the EAC to manage your role groups.

For your new role group, you'll need to:

1. Choose a name for your role group.
2. Select the roles you want to add to the role group.
3. Add members to the role group.
4. Save the role group.

If there's an existing role group that has some, but not all, of the permissions you need, you can copy it as a new role group and then make your necessary changes to the new role group. Copying an existing role group lets you create a customized version of the role group without affecting the original role group. When you copy the existing role group, you can assign a new name and description to the new group, add and remove roles to and from the new role group, and add new members. When you create or copy a role group, you use the same dialog box that's shown in the preceding figure.

Existing role groups can also be modified. You can add and remove roles from existing role groups and add and remove members from it at the same time using an EAC dialog box like the one in the preceding figure. By adding and removing roles to and from role groups, you turn on and off administrative features for members of that role group.

After you create a new, custom role group, you will manage it like any other role group.

Management role assignments

A management role assignment is the link between a management role and a role assignee. A role assignee is a role group, role assignment policy, user, or universal security group (USG). A role must be assigned to a role assignee for it to take effect.

You can create the following types of role assignments:

- Regular and delegating role assignments
- Exclusive role assignments

When you change role assignments, the changes you make will probably be between role groups and role assignment policies. By adding, removing, or modifying role assignments to or from these role assignees, you can control what permissions are given to your administrators and users, in effect turning on and off management of related features.

You can also assign roles directly to users or USGs. This is a more advanced task that enables you to define at a granular level what permissions your users are given. Although this provides you with flexibility, it also increases the complexity of your permissions model. For example, if the user changes jobs, you might need to manually reassign the roles assigned to that user to another user. For this reason, it is recommended that you use role groups and role assignment policies to give permissions to your users. You can assign the roles to a role group or role assignment policy, and then just add or remove members of the role group or change role assignment policies as needed.

You can add, remove, and enable role assignments, modify the management scope on an existing role assignment, and move role assignments to other role assignees. The process of assigning roles to role

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groups, role assignment policies, users, and USGs is largely the same for each role assignee, although there are a couple of exceptions:

- Role assignment policies can only be assigned end-user management roles.
- Role assignment policies cannot be assigned delegating role assignments.
- You can't specify a management scope when creating a role assignment to role assignment policies.

Note: Role assignment policies for end-user roles will be covered in a later lesson.

The two types of role assignments are available:

- **Regular and delegating role assignments.** Regular role assignments enable the role assignee to access the management role entries made available by the associated management role. If multiple management roles are assigned to a role assignee, the management role entries from each management role are aggregated and applied.
- **Exclusive role assignments.** Exclusive role assignments are created when you associate an exclusive scope with a role assignment. Exclusive scopes work like regular scopes and enable role assignees to manage recipients that match the exclusive scope. However, unlike regular scopes, all other role assignees are denied the ability to manage the recipient, even if the recipient matches scopes applied to their role assignments.

Troubleshooting RBAC

If you encounter issues with RBAC in Exchange, your first troubleshooting step is to gather data. RBAC issues are often related to administrators not being assigned the right permissions. To track that down, you need to gather a bunch of data.

To do so, you should begin by asking the following questions:

- **Which task are you trying to perform?** The goal here is to find out which task isn't working and to eventually figure out which management role entries are associated with the task.
- **Which object are you trying to create, modify, or delete?** You need to know which object and the location of the object (mailbox server, database, and OU location) that's affected. This can help you determine if the management scope is the problem. For example, if a recipient administrator is trying to delete a mailbox on a certain OU but his scope is limited to another OU, then it won't work.
- **What error message are you seeing?** This helps you rule out issues such as trying to run some actions from the Exchange Management Shell without elevation (when you might have to use an elevated Exchange Management Shell for the task).

There are many components to RBAC. Let's review some methods that are very helpful in obtaining information when troubleshooting RBAC issues.

Troubleshooting oftentimes begins by determining what roles are assigned to a specific role group. For example, running the following command provides a list of the assigned roles for the Recipient Management role group:

```
Get-RoleGroup 'Recipient Management'
```

```
[PS] C:\>Get-RoleGroup 'Recipient Management'

Name                AssignedRoles
----                -------------
Recipient Management {Distribution Groups, Mail Recipient Creation, Mail Recipients, Message Tracking, Migration, Move
                    Mailboxes, Recipient Policies, Team Mailboxes}
```

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By running the following command, you can determine which management roles have the New-Mailbox cmdlet.

```
Get-ManagementRoleEntry '*\New-Mailbox' | Format-List Name,Role,Parameters
```

This is useful if you want to figure out the management role with the least amount of permissions while still providing the cmdlet and functionality required.

```
Name       : New-Mailbox
Role       : Mail Enabled Public Folders
Parameters : {Debug, HoldForMigration, IsExcludedFromServingHierarchy, Name, PublicFolder, Verbose, WarningAction,
             WarningVariable}

Name       : New-Mailbox
Role       : Mail Recipient Creation
Parameters : {AccountDisabled, ActiveSyncMailboxPolicy, AddressBookPolicy, Alias, Arbitration, ArbitrationMailbox, Archive,
             ArchiveDatabase, ArchiveDomain, AuditLog, AuxAuditLog, Confirm, Database, Debug, Discovery, DisplayName...}

Name       : New-Mailbox
Role       : Mail Recipients
Parameters : {EnableRoomMailboxAccount}

Name       : New-Mailbox
Role       : Public Folders
Parameters : {Debug, HoldForMigration, IsExcludedFromServingHierarchy, Name, PublicFolder, Verbose, WarningAction,
             WarningVariable}

Name       : New-Mailbox
Role       : Retention Management
Parameters : {EnableRoomMailboxAccount}
```

Let's say you ran the prior command and you discovered which roles have the New-Mailbox cmdlet. After reviewing the list of roles, the Mail Enabled Public Folders role caught your attention, and you want to see which other cmdlets are assigned it. To find out that information, you should run the following command:

```
Get-ManagementRoleEntry 'Mail Enabled Public Folders*' | Select Name
```

```
[PS] C:\>Get-ManagementRoleEntry 'Mail Enabled Public Folders\*' | select Name

Name
----
Start-AuditAssistant
Set-UnifiedAuditSetting
Set-MailPublicFolder
Remove-SyncMailPublicFolder
New-SyncMailPublicFolder
New-Mailbox
Get-User
Get-UnifiedAuditSetting
Get-Mailbox
Get-MailPublicFolder
Enable-MailPublicFolder
Disable-MailPublicFolder
```

You've now run into a situation where you must determine whether the scope of the Mail Enabled Public Folders role is creating an RBAC issue that you've encountered. When roles are scoped to the organization level, they are effectively un-scoped because there aren't any scoping limitations. You can run the following command to check the scoping of the Mail Enabled Public Folders role:

```
Get-ManagementRole 'Mail Enabled Public Folders' | Format-List scope
```

```
[PS] C:\>Get-ManagementRole 'Mail Enabled Public Folders' | FL *scope*


ImplicitRecipientReadScope  : Organization
ImplicitRecipientWriteScope : Organization
ImplicitConfigReadScope     : OrganizationConfig
ImplicitConfigWriteScope    : OrganizationConfig
```

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You can get quite granular in your information gathering. For example, if you wanted to find out which parameters are enabled for New-Mailbox in the Mail Enabled Public Folders role, you would run the following command:

```
(Get-ManagementRoleEntry 'Mail Enabled Public Folders\New-Mailbox').parame-
ters
```



The commands that were identified in the previous examples can provide you with a good starting point when gathering data to troubleshoot an RBAC issue. All this RBAC data is stored in Active Directory. When you are troubleshooting, it's important to remember that if there are connectivity issues (or latency issues), or if there are performance issues with a domain controller that Exchange uses, it might cause some strange issues, even with RBAC.

You should also use auditing to track down recent changes to the configuration and/or roles. There are two key auditing reports that you can run from the EAC:

- **Administrator role group report.** This search generates a report that displays all the role groups that have changed. You can specify a date or date range filter, and you can limit the search to a specific role group or to all role groups.
- **Administrator audit log report.** This report displays all the changes made by administrators in your organization. You can narrow down the results to a specific date or date range. You can also use PowerShell (**Search-AdminAuditLog**) to filter results by cmdlet name, the administrator who ran the command, or other properties.

There are a couple of other cmdlets that you can use to track down RBAC information, including:

- **Get-RoleGroupMember**. You can use this cmdlet to find out which users are members of a role group. For example, you would run the following command to see which users are members of the Organization Management role group:

  ```
  Get-RoleGroupMember -Identity 'Organization Management'
  ```

- **Get-ManagementRoleAssignment**. You can run just this cmdlet name to see all the assignments, or you can filter the results based on any of the role properties.

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Managing User Roles

Lesson Introduction

RBAC is primarily designed for delegating and managing administrative permissions. But in Exchange it is also used to assign self-service permissions to end-users. By allowing users to manage some of their own user attributes and even perform basic diagnostics on their own mailbox, you can reduce the work of your HelpDesk.

After this lesson, you will be able to:

- Describe the built-in end-user roles
- Configure role assignment policies
- Create new custom roles and role assignment policies

Built-in role assignment policies

Just as there are built-in management role groups to administrate your Exchange environment, so too are there default role assignment policies and several built-in user roles.

The following table identifies the built-in user roles that are available in RBAC to support your Exchange environment.

User role	Description what end-users can do	Enabled by default
MyContactInformation MyAddressInformation MyMobileInformation MyPersonalInformation	Modify their contact information. Edit address, mobile, and personal information.	Yes
MyProfileInformation MyDisplayNam eMyName	Change their name and display name.	No
MyDistributionGroups	Create, modify, and view distribution groups, and manage members of groups they own.	No
MyDistributionGroupMembership	View and modify their membership in distribution groups, such as by joining or leaving a group.	Yes
MyCustomApps	View and modify their custom apps.	Yes
MyMarketplaceApps	View and modify their marketplace apps.	Yes

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User role	Description what end-users can do	Enabled by default
My ReadWriteMailbox Apps	This role will allow users to install apps with ReadWriteMailbox permissions.	Yes
MyBaseOptions	View and modify basic configuration, and access settings for Outlook on the web.	Yes
MyRetentionPolicies	View and modify retention tags and retention tag settings.	No
MyTextMessaging	Manage options for text messaging.	Yes
MyVoiceMail	Manage options for voice mail.	Yes
MyDiagnostics	Perform basic diagnostics on their mailbox.	No
MyMailboxDelegation	This role enables administrators to delegate mailbox permissions.	No
MyTeamMailboxes	Create site mailboxes and connect them to Microsoft SharePoint Services sites.	Yes

You can use the EAC to assign default roles to a role assignment policy. To view all user roles available in your organization, you can run the following command in the Exchange Management Shell:

```
Get-ManagementRole | where { $_.IsEndUserRole -eq $true }
```

Working with management role assignment policies

You have three options if the default role assignment policy is not enough to fulfill all your business requirements:

- Add or remove user roles from the default role assignment policy.
- Create a new role assignment policy from the built-in roles and assign it to individual users.
- Create new end-user roles and add them to a role assignment policy.

The default role assignment policy that you use should contain the permissions that apply to your broadest set of users. You should then create role assignment policies for each of your specialized user groups and tailor those role assignment policies to grant more or less restrictive permissions to them.

By organizing your role assignment policies this way – that is, by explicitly assigning role assignment policies to your specialized users while the majority of your users receive the more common permissions provided by the default role assignment policy – you reduce the complexity of your organization's role-based permission structure.

A mailbox can have only one role assignment policy. All users, including administrators and specialist users are assigned one role assignment policy. If you want a user to have a different set of permissions, you must assign that user's mailbox another role assignment policy with the required permissions.

To add a new role assignment policy, you must first create one and then decide whether it should be the default role assignment policy. After you create a role assignment policy, you must assign management

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roles to it, and then assign the role assignment policy to mailboxes. You can later choose to add or remove management roles or choose a different role assignment policy to be the default.

Note: You can only create explicit assignment policies using the Exchange Administration Center (EAC). If you want to create a new default assignment policy, you must use the Exchange Management Shell.

For example, you can run the following Exchange Management Shell command to create the explicit assignment policy “Limited Mailbox Configuration” and assign the “MyBaseOptions”, “MyAddressInformation”, and “MyDisplayName” roles to it:

```
New-RoleAssignmentPolicy "Limited Mailbox Configuration" -Roles MyBaseOptions, MyAddressInformation, MyDisplayName
```

You can change the management role assignment policy assigned to a mailbox. When you change a mailbox's assignment policy, the change takes effect as soon as the user refreshes the connection, such as the next time they log into their mailbox or open the mailbox options page.

For example, you can run the following command to set the assignment policy to “Limited Mailbox Configuration” on the mailbox named “Brian”:

```
Set-Mailbox Brian -RoleAssignmentPolicy "Limited Mailbox Configuration"
```

Creating new roles, role assignments, and scopes

The Exchange Admin Center only allows you to assign default user roles to role assignment policies. Therefore, while you can use the EAC to create and manage custom role assignment policies, it does not allow you to assign custom roles to them. To create new user roles and assign them to role assignment policies, you need to use the Exchange Management Shell.

If you want to customize the permissions that you assign to a group of end users, you must create a new custom management role assignment policy. The assignment policy you create can be customized to suit your end user's specific requirements.

Note: You can only create **explicit** assignment policies using the Exchange Administration Center (EAC). If you want to create a new **default** assignment policy, you must use the Exchange Management Shell.

Change a mailbox role assignment policy

You can use the EAC and Exchange PowerShell to assign an individual role assignment policy to user mailboxes. The following command assigns the role assignment policy named “Limited Mailbox Configuration” to the mailbox of Brian:

```
Set-Mailbox Brian -RoleAssignmentPolicy " Limited Mailbox Configuration"
```

If you are already working with custom role assignment policies, you can replace a certain policy with another using the following command:

```
Get-Mailbox | Where { $_.RoleAssignmentPolicy -Eq "Limited Mailbox Configuration" } | Set-Mailbox -RoleAssignmentPolicy "Mailbox Self Service"
```

Create custom role assignment policies

The following command creates a new role assignment policy named “Limited Mailbox Configuration” and assigns it the three roles "MyBaseOptions", “MyAddressInformation” and "MyDisplayName":

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```
New-RoleAssignmentPolicy "Limited Mailbox Configuration" -Roles MyBaseOp-
tions, MyAddressInformation, MyDisplayName
```

The following command creates the default assignment policy “Limited Mailbox Configuration” and assigns the "MyBaseOptions", “MyAddressInformation” and "MyDisplayName" roles to it:

```
New-RoleAssignmentPolicy "Limited Mailbox Configuration" -Roles MyBaseOp-
tions, MyAddressInformation, MyDisplayName -IsDefault
```

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Exchange Setup - RBAC and AD Split Permission

Lesson Introduction

Exchange is completely dependent on Active Directory and stores relevant settings, such as proxy addresses and SendAs permission not in Exchange but in Active Directory. This interconnection requires several considerations in enterprise environments because the job roles of a messaging administrator and an Active Directory administrator differ substantially. A messaging administrator must plan and configure permissions carefully so as not to put their environment or their entire Active Directory at risk.

After this lesson, you will be able to:

- Understand the differences between shared permissions and split permissions
- Describe multi-forest permissions
- Identify the differences between the permission models

Introduction to split permissions

Organizations that separate the management of Exchange objects and Active Directory objects use what's called a split permissions model. Split permissions enable organizations to assign specific permissions and related tasks to specific groups responsible for identity and messaging within the organization. This separation of work helps to maintain standards and workflows and helps to control change in the organization.

The highest level of split permissions is the separation of Exchange management and Active Directory management. Many organizations have two groups of administrators:

- Administrators that manage the organization's Exchange infrastructure, including servers and recipients.
- Administrators that manage the Active Directory infrastructure.

This is an important separation for many organizations because the Active Directory infrastructure often spans many locations, domains, services, applications, and even Active Directory forests. Active Directory administrators must ensure that changes made to Active Directory don't negatively impact any other services. As a result, only a small group of administrators is typically allowed to manage that infrastructure.

At the same time, the infrastructure for Exchange, including servers and recipients, can also be complex and require specialized knowledge. Exchange also stores extremely confidential information about the business of the organization. Exchange administrators can potentially access this information. By limiting the number of Exchange administrators, the organization limits who can make changes to Exchange configuration and who can access sensitive information.

Split permissions typically make a distinction between the creation of security principals in Active Directory, such as users and security groups, and the subsequent configuration of those objects. This helps to reduce the chance of unauthorized access to the network by controlling who can create objects that grant access to it. Most often only Active Directory administrators can create security principals while other administrators, such as Exchange or Messaging administrators, can manage specific attributes on existing Active Directory objects.

To support the varying needs to separate the management of Exchange and Active Directory, Exchange lets you choose whether you want a shared permissions model or a split permissions model, where the

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split permissions model is also separated in two types: RBAC Permissions and Active Directory Permissions.

By default, if you don't configure anything else, Exchange uses a shared permissions model.

Introduction to multiple-forest permissions

Many organizations deploy multiple Active Directory forests to create security boundaries within their organizations. Using multiple forests helps administrators to define these security boundaries to better match their requirements, whether that's ensuring the fewest number of people have access to resources, or segmenting divisions within an organization.

Microsoft Exchange Server supports two types of multiple forest topologies:

- **Cross-forest.** This topology type can have multiple forests, each with their own installation of Exchange.
- **Resource forest.** This topology type has an Exchange forest and one or more accounts forests.

For the purposes of this topic, the forest that contains the universal security groups (USGs) and users outside of the forest where Exchange is installed, whether it's an accounts forest or other resource forest, is called a foreign forest.

Configuration of permissions in a multiple forest topology relies on the correct configuration of forest trusts and global address list (GAL) synchronization for the creation of linked mailboxes. The Exchange forest must trust the foreign forest that contains the USGs associated with linked role groups and users associated with linked mailboxes.

Exchange uses a Role Based Access Control (RBAC) permissions model. The management role groups that administrators are members of, and the management role assignment policies that end users are assigned, determine what each administrator and end user can do. To understand multiple-forest permissions, you need to be familiar with RBAC.

Permissions in a multiple forest topology

RBAC applies permissions to all Exchange objects within a single forest, and the RBAC configuration in each forest is configured independently of all other forests. When you create a role group in one forest, that role group doesn't exist in any other forest and the permissions granted by that role group apply only to the forest in which it was created. For example, a member of a role group that grants permissions to create a mailbox can create a mailbox only in the forest that contains that role group.

If you have multiple Exchange forests and you want to configure permissions identically within each forest, you must apply the same configuration explicitly in each forest. For example, if you have two Exchange forests and you want to create a Compliance Management role group to manage permissions for your legal department, you must do the following:

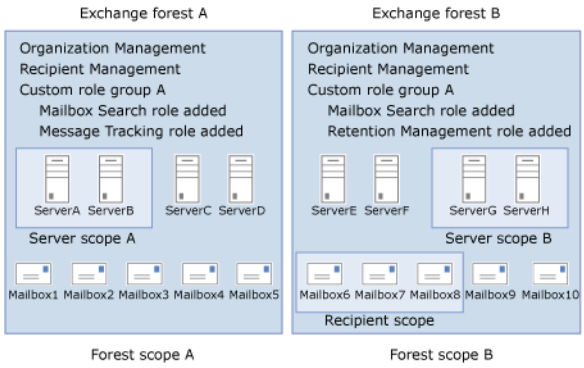
- In each forest, create a role group named Compliance Management. If your administrators are in a separate foreign forest from either Exchange forest, create both role groups as linked role groups.
- In each forest, create role assignments between the new role groups and the roles that you want to use.
- As part of the new role assignments, optionally add management scopes that encompass the server and recipient objects within each forest.
- If you created the role groups as linked role groups, add members to the associated USG in the foreign forest.

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The following diagram shows how the role groups configured within Exchange forests are bound to their respective forests. The Organization Management role group in Exchange forest A grants permissions only to manage the mailboxes and servers that are within that forest. Likewise, the role groups in Exchange forest B grant permissions only to the mailboxes and servers within that forest.

The diagram also shows that the Custom role group A is created in each forest. Even though they were created with the same name, each is its own separate entity. In fact, as the figure shows, each can be assigned different management roles in their respective forests. Custom role group A in Exchange forest A is assigned the Mailbox Search and Message Tracking roles while Custom role group A in Exchange forest B is assigned the Mailbox Search and Retention Management roles.

Finally, management scopes created in each forest are also bound by the forest. Server scopes created in each forest can only contain the servers that are members of that forest. In the diagram, Server scope A can contain only servers within Exchange forest A, while Server scope B can contain only servers that are within Exchange forest B. Similarly, the recipient scope in Exchange forest B can only contain mailboxes that are within Exchange forest B.



Additional reading. For more information, see the following article on **Cross-boundary permissions**[1].

Differences in permissions

If you decide to use a split permissions model, you can choose between RBAC split permissions and Active directory split permissions. In this topic you will learn the details of each to understand the differences and when to use each for enterprise Exchange deployments.

Note: RBAC split permissions is the recommended model to choose in most deployments, but you need to analyze and understand your company's individual business needs to decide which model will serve your company best for managing Exchange permissions.

Configure RBAC split permissions

The RBAC security model modifies the default management role assignments to separate who can create security principals in the Active Directory domain partition from those who administer the Exchange organization data in the Active Directory configuration partition. Security principals, such as users with mailboxes and distribution groups, can be created by administrators who are members of the Mail Recipient Creation and Security Group Creation and Membership roles. These permissions remain separate from the permissions required to create security principals outside of the Exchange management tools. Exchange administrators who aren't assigned the Mail Recipient Creation or Security Group Creation and Membership roles can still modify Exchange-related attributes on security principals. Active

1 https://aka.ms/AA3r0l9

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Directory administrators also have the option of using the Exchange management tools to create Active Directory security principals.

Exchange servers and the Exchange Trusted Subsystem also have permissions to create security principals in Active Directory on behalf of users and third-party programs that integrate with RBAC.

RBAC split permissions is a good choice for your organization if the following are true:

- Your organization doesn't require that security principal creation be performed using only Active Directory management tools and only by users who are assigned specific Active Directory permissions.
- Your organization allows services, such as Exchange servers, to create security principals.
- You want to simplify the process required to create mailboxes, mail-enabled users, distribution groups, and role groups by allowing their creation from within the Exchange management tools.
- You want to manage the membership of distribution groups and role groups within the Exchange management tools.
- You have third-party programs that require that Exchange servers be able to create security principals on their behalf.

If your organization requires a complete separation of Exchange and Active Directory administration where no Active Directory administration can be performed using Exchange management tools or by Exchange services, see the Configure for Active Directory Split Permissions section later in this topic.

Switching from shared permissions to RBAC split permissions is a manual process where you remove the permissions required to create security principals from the role groups that are granted them by default. The following table shows the roles that enable the creation of security principals in Exchange and the management role groups they're assigned to by default.

To configure RBAC split permissions, you must do the following:

1. Disable Active Directory split permissions if it's enabled.
2. Create a role group, which will contain the Active Directory administrators that will be able to create security principals.
3. Create regular and delegating role assignments between the Mail Recipient Creation role and the new role group.
4. Create regular and delegating role assignments between the Security Group Creation and Membership role and the new role group.
5. Remove the regular and delegating management role assignments between the Mail Recipient Creation role and the Organization Management and Recipient Management role groups.
6. Remove the regular and delegating role assignments between the Security Group Creation and Membership role and the Organization Management role group.

After doing this, only members of the new role group that you create will be able to create security principals, such as mailboxes. The new group will only be able to create the objects. It won't be able to configure the Exchange attributes on the new object. An Active Directory administrator, who is a member of the new group, will need to create the object, and then an Exchange administrator will need to configure the Exchange attributes on the object. Exchange administrators won't be able to use the following cmdlets:

- New-Mailbox
- New-MailContact

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- New-MailUser
- New-RemoteMailbox
- Remove-Mailbox
- Remove-MailContact
- Remove-MailUser
- Remove-RemoteMailbox

Configure Active Directory split permissions

With Active Directory split permissions, the creation of security principals in the Active Directory domain partition, such as mailboxes and distribution groups, must be performed using Active Directory management tools.

Several changes are made to the permissions granted to the Exchange Trusted Subsystem and Exchange servers to limit what Exchange administrators and servers can do. This is because Exchange performs the actions in Active Directory using these contexts, with RBAC filtering administrative permissions, rather than as the actual administrator accounts themselves.

The following changes in functionality occur when you enable Active Directory split permissions:

- Creation of mailboxes, mail-enabled users, distribution groups, and other security principals is removed from the Exchange management tools.
- Adding and removing distribution group members can't be done from the Exchange management tools.
- All permissions granted to the Exchange Trusted Subsystem and Exchange servers to create security principals are removed.
- Exchange servers and the Exchange management tools can only modify the Exchange attributes of existing security principals in Active Directory.

For example, to create a mailbox with Active Directory split permissions enabled, a user must first be created using Active Directory tools by a user with the required Active Directory permissions. Then, the user can be mailbox-enabled using the Exchange management tools. Only the Exchange-related attributes of the mailbox can be modified by Exchange administrators using the Exchange management tools.

Active Directory split permissions is a good choice for your organization if the following are true:

- Your organization requires that security principals be created using only the Active Directory management tools or only by users who are granted specific permissions in Active Directory.
- You want to separate the ability to create security principals from those who manage the Exchange organization.
- You want to perform all distribution group management, including creation of distribution groups and adding and removing members of those groups, using Active Directory management tools.
- You don't want Exchange servers, or third-party programs that use Exchange on their behalf, to create security principals.

After you enable Active Directory split permissions, the following cmdlets are no longer available:

- New-Mailbox
- New-MailContact
- New-MailUser

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- New-RemoteMailbox
- Remove-Mailbox
- Remove-MailContact
- Remove-MailUser
- Remove-RemoteMailbox

After you enable Active Directory split permissions, the following cmdlets are accessible, but you cannot use them to create distribution groups or modify distribution group membership:

- Add-DistributionGroupMember
- New-DistributionGroup
- Remove-DistributionGroup
- Remove-DistributionGroupMember
- Update-DistributionGroupMember

Some cmdlets, although still available, may offer only limited functionality when used with Active Directory split permissions. This is because they may allow you to configure recipient objects that are in the domain Active Directory partition and Exchange configuration objects that are in the configuration Active Directory partition. They may also allow you to configure Exchange-related attributes on objects stored in the domain partition. Attempts to use the cmdlets to create objects, or modify non-Exchange-related attributes on objects, in the domain partition will result in an error.

For example, the Add-ADPermission cmdlet will return an error if you attempt to add permissions to a mailbox. However, the Add-ADPermission cmdlet will succeed if you configure permissions on Exchange related objects, such as a Receive connector. This is because a mailbox is stored in the domain partition while Receive connectors are stored in the configuration partition.

Additionally, the associated features in the Exchange Administration Center and Outlook Web App, such as the New Mailbox wizard, will also no longer be available or will generate an error if you try to use them.

Exchange administrators will, however, be able to create and manage Exchange-specific objects, such as transport rules, and so on.

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Lab6-Ex1-Manage Roles and Permissions Policies

Lab6-Ex1-Manage Roles and Permissions Policies

In this exercise you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. Holly has been tasked with managing admin roles and permission policies for Adatum. The CTO has asked Holly to create a new management role group that allows an administrator to remotely access a mailbox without having the password.

In this exercise Holly will return to the Exchange Admin Center for Exchange Online, but this time she will access it from her client computer (LON-CL1).

See your VM lab environment for the lab instructions.

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Module Review

Module 7 Review

Multiple choice

Item 1. You are the messaging administrator for Contoso Ltd. You have been tasked with using Role-based Access Control (RBAC) to assign permissions to other administrators with different responsibilities. To accomplish this task, what do you need to configure in Microsoft 365?

- ☐ Role assignment policies
- ☐ Management role groups
- ☐ Access Control Lists (ACL)
- ☐ End-user roles

Multiple choice

Item 2. You are a messaging administrator for Contoso Ltd. You must assign your company data protection officer a role that grants rights to use the Search-Mailbox cmdlets in the Exchange Management Shell. Which of the following roles do you assign to this user that provides her with this functionality?

- ☐ Hygiene Management
- ☐ Server Management
- ☐ Discovery Management
- ☐ Recipient Management

Checkbox

Item 3. When troubleshooting RBAC issues, which of the following questions are important to ask to find the root cause of the RBAC issues? (Select 3)

- ☐ Which task does the user try to perform?
- ☐ Which workstation is the affected user currently working from?
- ☐ Does the password of the affected user meet the complexity requirements?
- ☐ Is the mailbox of the affected user still below the quota limit?
- ☐ Which object is the user trying to create, modify, or delete?
- ☐ What error message does the user see?

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Multiple choice

Item 4. You are the messaging administrator for Contoso Ltd. You plan to implement custom role assignment policies to provide some self-service capabilities to your end-users. Which role assignment policy is assigned by default to all users?

- ☐ No policy is assigned by default to user mailboxes.
- ☐ The Default role assignment policy is assigned to all mailboxes.
- ☐ The first role assignment policy you create will be assigned by default to all mailboxes.
- ☐ You need to choose between different default role assignment policies for every mailbox when creating new mailboxes.

Multiple choice

Item 5. How many role assignment policy/policies can a mailbox have?

- ☐ Only one
- ☐ Two
- ☐ An unlimited number
- ☐ Ten

Multiple choice

Item 6. Which is the default permissions model for Exchange?

- ☐ RBAC split permissions
- ☐ Active Directory split permissions
- ☐ Shared permissions
- ☐ ACLs

Multiple choice

Item 7. You are the messaging administrator for Contoso Ltd., a large enterprise organization. You have been tasked with planning the deployment of a new Exchange organization. Your IT department has a group of employees dedicated to administrating Active Directory. Which permissions model you should choose in this scenario?

- ☐ FullAccess and SendAs permissions
- ☐ Active Directory split permissions
- ☐ RBAC split permissions
- ☐ Shared permissions

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Multiple choice

Item 8. Fill in the blank. With Active Directory split permissions, the creation of security principals in the Active Directory domain partition, such as mailboxes and distribution groups, must be performed by using _____.

- ☐ Exchange Management Shell
- ☐ Active Directory management tools
- ☐ Exchange Admin Center
- ☐ Exchange Online PowerShell.

Multiple choice

Item 9. As a messaging administrator for Contoso Ltd. must deploy a solution for end-users that allows them to change different settings on their own mailboxes. Which of the following will provide a solution for this task?

- ☐ Management role groups
- ☐ Role assignment policies
- ☐ Different customizations of the Access Control Lists (ACL)
- ☐ Split permissions

Checkbox

Item 10. Which of the following RBAC management roles are only available in Exchange Server and not in Exchange Online? (Select 2)

- ☐ Compliance Management
- ☐ Delegated Setup
- ☐ Server Management
- ☐ Records Management

Multiple choice

Item 11. As the messaging administrator for Contoso Ltd., you are responsible for its Exchange Online environment. You get a new colleague that requires full administration rights in Exchange Online. Which Exchange Online management role group do you assign to his user account to grant him full administration rights in your organization?

- ☐ Recipient Management
- ☐ Records Management
- ☐ Discovery Management
- ☐ Organization Management

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Checkbox

Item 12. Every management role, whether it's a built-in role or a custom role, has management scopes. Which of the following are valid management scopes for management roles? (Select 2)

- ☐ Regular
- ☐ Exclusive
- ☐ Default
- ☐ Hierarchical

Multiple choice

Item 13. As a messaging administrator for Contoso Ltd., you have been tasked with changing the default role assignment policy for your organization's users to allow end-users to change their DisplayName by themselves. To accomplish this, which user role should you include in the default policy that is not activated by default?

- ☐ MyContactInformation
- ☐ MyCustomApps
- ☐ MyProfileInformation
- ☐ MyBaseOptions

Checkbox

Item 14. As a messaging administrator for Contoso Ltd., you discover that the default role assignment policy assigned to certain employees does not provide them with sufficient permissions to accomplish all their job functions. Which options do you have to change the role assignment policies in your organization? (Select 3)

- ☐ Add or remove user roles from the default role assignment policy.
- ☐ Create a management role to change mailbox role assignment policies.
- ☐ Assign the Organization Management role to all users in your organization.
- ☐ Create a new role assignment policy from the built-in roles and assign it to individual users.
- ☐ Create new end-user roles and add them to a role assignment policy.
- ☐ Create a newsletter that informs all users they should change their role assignment policies by themselves.

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Checkbox

Item 15. Which of the following statements should be true for your organization's requirements to deploy RBAC split permissions as the preferred split permissions model over Active Directory split permissions? (Select 3)

- ☐ Your organization allows services, such as Exchange servers, to create security principals.
- ☐ You want to manage the membership of distribution groups and role groups within the Exchange management tools.
- ☐ Creation of mailboxes, mail-enabled users, distribution groups, and other security principals need to be removed from the Exchange management tools.
- ☐ You have third-party programs that require that Exchange servers be able to create security principals on their behalf.
- ☐ All permissions granted to the Exchange Trusted Subsystem and Exchange servers to create security principals need to be removed.
- ☐ Exchange servers and the Exchange management tools can only modify the Exchange attributes of existing security principals in Active Directory.

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Answers

Multiple choice

Item 1. You are the messaging administrator for Contoso Ltd. You have been tasked with using Role-based Access Control (RBAC) to assign permissions to other administrators with different responsibilities. To accomplish this task, what do you need to configure in Microsoft 365?

- ☐ Role assignment policies
- ■ Management role groups
- ☐ Access Control Lists (ACL)
- ☐ End-user roles

Multiple choice

Item 2. You are a messaging administrator for Contoso Ltd. You must assign your company data protection officer a role that grants rights to use the Search-Mailbox cmdlets in the Exchange Management Shell. Which of the following roles do you assign to this user that provides her with this functionality?

- ☐ Hygiene Management
- ☐ Server Management
- ■ Discovery Management
- ☐ Recipient Management

Checkbox

Item 3. When troubleshooting RBAC issues, which of the following questions are important to ask to find the root cause of the RBAC issues? (Select 3)

- ■ Which task does the user try to perform?
- ☐ Which workstation is the affected user currently working from?
- ☐ Does the password of the affected user meet the complexity requirements?
- ☐ Is the mailbox of the affected user still below the quota limit?
- ■ Which object is the user trying to create, modify, or delete?
- ■ What error message does the user see?

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Multiple choice

Item 4. You are the messaging administrator for Contoso Ltd. You plan to implement custom role assignment policies to provide some self-service capabilities to your end-users. Which role assignment policy is assigned by default to all users?

- ☐ No policy is assigned by default to user mailboxes.
- ■ The Default role assignment policy is assigned to all mailboxes.
- ☐ The first role assignment policy you create will be assigned by default to all mailboxes.
- ☐ You need to choose between different default role assignment policies for every mailbox when creating new mailboxes.

Multiple choice

Item 5. How many role assignment policy/policies can a mailbox have?

- ■ Only one
- ☐ Two
- ☐ An unlimited number
- ☐ Ten

Multiple choice

Item 6. Which is the default permissions model for Exchange?

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- ☐ Different customizations of the Access Control Lists (ACL)
- ☐ Split permissions

Checkbox

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- ☐ Records Management
- ☐ Discovery Management
- ■ Organization Management

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- ☐ MyContactInformation
- ☐ MyCustomApps
- ■ MyProfileInformation
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- ■ Add or remove user roles from the default role assignment policy.
- ☐ Create a management role to change mailbox role assignment policies.
- ☐ Assign the Organization Management role to all users in your organization.
- ■ Create a new role assignment policy from the built-in roles and assign it to individual users.
- ■ Create new end-user roles and add them to a role assignment policy.
- ☐ Create a newsletter that informs all users they should change their role assignment policies by themselves.

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Checkbox

Item 15. Which of the following statements should be true for your organization's requirements to deploy RBAC split permissions as the preferred split permissions model over Active Directory split permissions? (Select 3)

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- ■ You want to manage the membership of distribution groups and role groups within the Exchange management tools.
- ☐ Creation of mailboxes, mail-enabled users, distribution groups, and other security principals need to be removed from the Exchange management tools.
- ■ You have third-party programs that require that Exchange servers be able to create security principals on their behalf.
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- ☐ Exchange servers and the Exchange management tools can only modify the Exchange attributes of existing security principals in Active Directory.

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Module 8 Managing Recipient Objects and Resources

Exchange Recipients

Lesson introduction

When you manage Exchange, it's important that you understand the available recipient types, such as mailboxes and Microsoft 365 groups. Two of the most common tasks that Exchange administrators perform are creating and configuring email recipients. As organizations hire new employees or employees change positions within the organization, the Exchange administrators must ensure the users have the messaging functionality they require.

Most users in an organization will use Exchange mailboxes, although Exchange also provides various other mailbox options that administrators can configure. For an example, you can use a shared mailbox if you want your users to monitor and use one mailbox to receive and send emails, (such as info@contoso.com). Or you can use a room mailbox if you want to allow your Outlook users to book a conference room from their calendar.

This lesson examines the different types of Exchange Server recipients, including how they differ from each other.

After this lesson, you will be able to:

- Describe the different recipient objects in Exchange
- Describe resource mailboxes
- Describe shared mailboxes
- Describe linked mailboxes and site mailboxes
- Describe groups

Recipient types overview

Exchange provides the following recipient types that can be used by organizations in different scenarios.

Recipient Types	Description	When to use
Mailboxes	This is a mailbox that you assign to an individual user in your Exchange organization. This is the most common type of recipient in Exchange.	To any user who wants to use Outlook, Outlook on the Web, or Outlook Mobile to send or receive personal messages.
Shared mailboxes	Mailboxes that can be accessed by multiple users as an additional mailbox in Outlook.	Team, department, or project mailboxes to centrally store, respond to, or manage messages. Organizations often use shared mailbox in order to provide one specific mailbox for dedicated group of people such as sales, help desk, or general information requests.
Room mailboxes	Used to book meetings or conference rooms in Outlook. Can be configured to automatically process booking requests.	Conference or meeting rooms, special team rooms, or buildings. You can include resource mailboxes as resources in meeting requests, which provides a simple and efficient way to schedule resource usage.
Equipment mailboxes	Used to book equipment. Can be configured to automatically process booking requests.	Use it for equipment that can be moved between locations, for example, a projector or a car.
Distribution groups	Distribute messages to a set of users or contacts.	The most common form of a globally available distribution list for sending out one-to-many e-mail messages or where you need to use the membership list elsewhere, such as to determine who Transport Rules should apply to. It can also be used for reaching out to a team as a contact address or used to allow send-as an additional email alias without creating a shared mailbox.
Security groups	Distribute messages and provide access to resources to a set of users.	Used when you need to apply permissions to resources such as a shared mailbox or Public Folder. But can also be used the same time as a distribution group, and be used outside Exchange Online, for example in SharePoint Online.

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Recipient Types	Description	When to use
Dynamic distribution groups	Dynamically create a set of users based on an LDAP search in Active Directory (AD) to distribute messages.	Commonly used to create a global company distribution list using attributes in the company address book, such as region, office, department, and roles.
Microsoft 365 groups	These groups are available only in Microsoft 365 and provide team collaboration such as document storage, a centralized calendar, a OneNote notebook, working collaborative on project plans, and team email distribution in one space.	Excellent for team or department collaboration where additional resources such as a OneNote or OneDrive is needed. Because it provides hooks into multiple Microsoft 365 services, it can be used as a tool to aid adoption and drive business value.
Mail contacts	Contacts that contain information about people or organizations that exist outside an Exchange organization and that have an external email address.	Show external contacts in Global Address List so that your users can add them to their distribution lists. Exchange routes all messages sent to the mail contact to this external email address.
Mail users	Users who have a user account in the organization but have an external email address. All messages sent to the mail user are routed to this external email address.	A mail user is typically created for temporary employees or external vendors who need to access resources in the organization but use their own email address which is located outside your organization.
Mail-enabled Public Folders	This is a folder in the public folder hierarchy that has an email address assigned thus can receive email messages.	If your team already works extensively with public folders.
Site Mailboxes	Site mailboxes are available in Exchange Server but have been deprecated in Office 365. These types of mailboxes include both an Exchange Server mailbox and a Microsoft SharePoint site. This is an add-on to your SharePoint team site that allows team members to receive and respond to email using Outlook or SharePoint.	If your team uses SharePoint team sites and wants to use an email address to receive or respond to messages; for example, a centralized support team. Also, SharePoint team site users who prefer to work in Outlook may find site mailboxes particularly valuable, as they provide direct access to the Document Library from within Outlook.

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Recipient Types	Description	When to use
Linked mailboxes	User mailboxes that are associated with specific users in a separate, trusted forest.	When you create a linked mailbox, a disabled user account is created in the Exchange organization, and a user account from a trusted forest is given access to the mailbox.
Remote mailboxes	Mailboxes which are located in the Exchange Online environment in a hybrid Exchange Server deployment.	You can create and manage remote mailboxes in the Exchange Online environment by using the Exchange Admin Center and the Exchange Management Shell

User mailboxes

A user mailbox is the most common type of recipient because one is created for every user in an organization. In Exchange Online, each user mailbox is associated with a user created in Microsoft 365; whereas in Exchange Server, each user mailbox is associated with an Active Directory user account. The user can use the mailbox to send and receive messages, to store messages, schedule meetings and appointments, maintain task lists, write notes, store documents, and manage contacts. User mailboxes can be accessed from different types of clients such as Outlook, Outlook on the Web, Outlook Mobile, or third-party clients.

You can associate an archive mailbox with each user mailbox. An archive mailbox provides additional storage, and you can use it for storing older messages. An archive mailbox is a specialized mailbox that appears alongside the users' primary mailbox in Outlook and Outlook on the Web. An archive mailbox can be accessed and searched in the same way as the primary mailbox, and users can move or copy messages between their primary mailbox and their archive mailbox. Furthermore, you can assign retention policies that will move messages to archive mailbox automatically; for example, moving all messages that are older than two years from the primary mailbox to the archive mailbox. However, an archive mailbox is available only if the user is connected to Exchange, which means it cannot be accessed in Outlook cached mode if the user is not connected to Exchange.

Resource mailboxes

Resource mailboxes are specific types of mailboxes that you can use to represent meeting rooms or shared equipment, and you can include them as resources in meeting requests. You can create two different types of resource mailboxes in Exchange:

- **Room mailboxes.** These are resource mailboxes that you can assign to meeting locations, such as conference rooms, auditoriums, and training rooms. On a room mailbox, you can configure settings such as booking delegates and booking options.
- **Equipment mailboxes.** These are resource mailboxes that you can assign to resources that are not location specific, such as portable computer projectors, microphones, or company cars. On an equipment mailbox, you can configure similar settings as on room mailboxes.

You can include both types of resource mailboxes as resources in meeting requests, which provides a simple and efficient way for users to book these resources. For example, to book a room or a piece of

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equipment open Outlook from your computer and schedule a new meeting where you must add the room or equipment to the meeting, similar to how you would invite other users:

1. Open **Outlook** on your computer.
2. On the **Home** tab, choose **New Items** > **Meeting**.
3. In the **To** field, type the name of the conference room or equipment you want to reserve, in addition to any attendees you'd like to invite. Optionally, you may use the **Rooms** button if you want to search for all room mailboxes or use the **Address Book** button to search for all room and equipment mailboxes.

Resource mailboxes have properties such as location and size that help you to search for meeting rooms that meet their requirements. You can also use the **Scheduling Assistant** option to see a calendar view of the room as well as the equipment's availability.

Introduction to shared mailboxes

Shared mailboxes are special types of mailboxes that multiple users can access to read and respond to messages. For example, an organization may provide an email address such as info@contoso.com on a public web site. The organization may want to have several users monitor the mailbox associated with this email address to ensure prompt replies to potential customers. Shared mailboxes can be used, for an example, for dedicated departments such as services, marketing, or logistics.

Shared mailboxes provide:

- A generic email address, such as marketing@contoso.com or sales@contoso.com to field customer inquiries.
- A way for teams or departments that provide centralized services, such as helpdesks or support, to respond to requests from employees or customers.
- Support for multiple users to monitor and reply to external or internal email addresses.

A shared mailbox doesn't have its own User name and password like a User mailbox does. You can't sign in to a shared mailbox directly using Outlook, Outlook Web App, Exchange ActiveSync, Exchange Web Services (EWS), or any other Exchange protocol. You must first be granted permission to the shared mailbox, and then you access it by using Outlook or Outlook Web App. Shared mailboxes aren't supported on mobile devices. To access shared mailbox from a mobile device, you must use Outlook on the Web. Once a user is granted permission to the shared mailbox, it will automatically be added to the user's Outlook profile and opened alongside to the user's primary mailbox.

Shared mailboxes are a good solution for groups of people that need to share a common email identity and set of messages, including a mailbox folder structure they can work in. When a user replies to a message sent to a shared mailbox, the reply appears to come from the shared mailbox address. Also, all users who have access to that shared mailbox can see the messages that have been sent to that account.

Shared mailboxes do not require a Microsoft 365 license, so you can consider shared mailboxes to be free mailboxes in your Microsoft 365 tenant. However, if you want to use some advanced Exchange Online functionality such as in-place archive or in-place hold, you need an Exchange Online Plan license.

Overview of linked and site mailboxes

Linked mailboxes provide mailboxes for users whose primary accounts are located in a separate, trusted forest. Users with linked mailboxes sign in to their local AD DS domain by using the local credentials, and those credentials are then used to access a mailbox in an Exchange organization in a different forest.

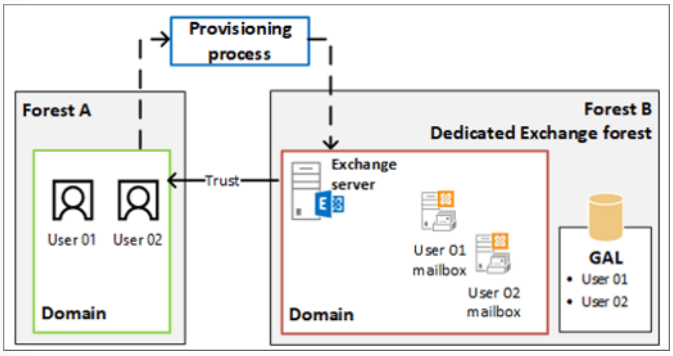
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Linked mailboxes can be useful in the following two scenarios:

- **Organizations deploy Exchange in a resource forest.** When organizations deploy Exchange Server in a resource forest, they deploy Exchange Server into one AD DS forest while allowing access to the Exchange mailboxes to user accounts that are located in one or more trusted forests (called account forests).
- **Organizations use linked mailboxes in a merger or acquisition scenario.** In this scenario, both organizations may have deployed Exchange Server before the merger or acquisition. Linked mailboxes provide the opportunity to remove the Exchange Server deployment from one of the organizations. The users from one of the organizations can be configured with linked mailboxes in the other organization. This ensures that users from both organizations are listed in a single GAL, making availability information accessible to all users.

When deploying linked mailboxes, messaging administrators must carefully plan security permissions. Several approaches to managing the messaging environment are available:

- Organizations can decide which administrator's team will manage user accounts in the account forest and which administrator's team will manage recipients in the resource forest. Both teams should work together in administering the messaging environment; however, administrators should have just enough administration permissions to perform their tasks so that unauthorized access to each forest is not allowed.
- Another approach is that one administration team manages both forests.
- You may also consider trust relationship's selective authentication. In this scenario, sign-in's from the resource forest are restricted to specific hosts in the production forest.



When configuring a linked mailbox, the user account that accesses the linked mailbox does not exist in the forest where Exchange is deployed. When you create the linked mailbox, a disabled user account is created in the domain where Exchange is deployed and associated with the linked mailbox. The user account from the account forest is granted full control of the mailbox.

To implement linked mailboxes, you must perform the following steps:

1. Configure a one-way trust in which the domain where Exchange is deployed trusts the domain where the user account exists. This can be an external or forest trust. Note that a one-way trust is required.
2. Verify that the user account exists in the account forest before you create a linked mailbox. You cannot create the user account when you create the linked mailbox.
3. In addition to configuring the one-way trust, you should consider creating a two-way trust between the domains. The two-way trust is not required, but the account that creates the linked mailbox must

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have permissions to modify the user object in the account forest. If you do not implement a two-way trust, you must provide account forest administrator credentials when you create the linked mailbox.

Site mailboxes

A site mailbox provides integration between a SharePoint site and an Exchange Server mailbox, which results in an integrated experience for users who need to collaborate. Site mailboxes are only available in on-premises environments and are deprecated in Office 365. Site mailboxes enable users to access both documents stored on SharePoint Server and email stored on an Exchange Server mailbox by using the same client interface such as Outlook and Outlook on the Web. For example, one challenge that users face when they work collaboratively is accessing information that is stored in several different locations. Users who work on the same project may need to exchange emails related to the project, and they may also need to access shared documents stored on file shares or on a SharePoint Server site.

A site mailbox is an add-on to your SharePoint team site that allows team members to receive and respond to email from the site mailbox. You can then view your email on SharePoint and view site documents in Outlook.

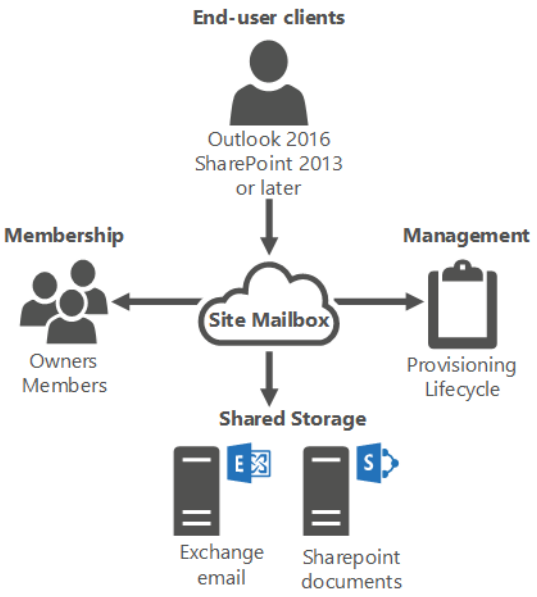
A site mailbox has the following features:

- Access, forward, and edit documents directly from Outlook, SharePoint Online, or Outlook on the web (OWA)
- Can be part of eDiscovery Search Scopes for legal holds
- End users can create and manage site mailboxes

For example, let's assume that you have a group of users working on a project that requires email communication and a document-review process. With site mailboxes, users can send and read email messages in the site mailbox, and they can post and review documents on the SharePoint site.

Site mailboxes are available in Outlook and can be used to view both the email messages in the mailbox and the documents stored in the SharePoint site. The same content can also be accessed directly from the SharePoint site. With site mailboxes, Exchange Server stores the email, providing users with the same email conversations that they use every day for their own mailboxes. SharePoint Server stores the documents and provides advanced document-management tools such as version control. Site mailboxes provide this integration on the user interface layer, while leaving the content in the optimized stores, such as Exchange Server for email and SharePoint Server for documents.

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Additional information. For more information, see the following article on **Site mailboxes**[1].

Introduction to Groups

Exchange provides different types of groups so that you can group together recipients and make them available in your global address list. This topic focusses on groups that can be used for Exchange to address a group of recipients. Additionally, you can create a security group that is not mail-enabled and cannot be used for distributing email to multiple recipients in Microsoft 365. These groups are commonly used to provide access to resources, such as control access to OneDrive and SharePoint, and can be used for Mobile Device Management.

The following table identifies the different types of groups that you can create and manage in the Exchange admin center (EAC) portal.

Group Type	Description	Membership Management
Distribution lists (Distribution Groups)	These groups can only be used to distribute messages to a set of recipients.	Configure if owner approval is required to join the group (open, closed, or owner approval)
Mail-enabled security groups	These groups can be used to distribute messages and to provide access to resources, such as control access to OneDrive and SharePoint.	Configure if owner approval is required to join.
Dynamic distribution lists (Dynamic distribution groups)	These groups do not have a predefined member list because they use recipient filters and conditions that you define to dynamically determine membership at the time that messages are sent.	Membership is dynamic, based on a query against the Azure Active Directory or Active Directory Domain Services.

1 https://aka.ms/AA3qu5c

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Group Type	Description	Membership Management
Microsoft 365 groups	These groups are available in Microsoft 365 only. They provide distribution list team collaboration functionality for small to medium teams when the requirements include functionality such as conversations, co-authoring of documents, a shared OneNote notebook, or task-based planning.	Membership is managed by the administrator or self-managed, thus users can search for a Microsoft 365 group and join it. Dynamic group membership is possible but requires an Azure AD Premium license.

Additional reading. For more information, see the following article on **What to create: Office 365 groups, shared mailboxes, security groups, and distribution lists**[2].

2 https://aka.ms/AA3r1qv

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Creating and Managing Exchange Recipients

Lesson Introduction

Managing recipients is a crucial day-to-day task for Exchange and Microsoft 365 administrators. For example, you may need a new e-mail address when a new employee joins a team, or your users might need to add a new e-mail address when your organization starts a new marketing campaign, or you need to create a new contact that is available throughout your organization.

This lesson examines the various tasks that require you to create and manage Exchange recipients in Exchange.

After completing this lesson, you will be able to create and manage the following Exchange objects:

- Mailbox settings
- Resource mailboxes
- Shared mailboxes
- Mail contacts and mail users
- Recipient permissions
- Groups

Creating and managing user mailboxes

One of the primary tasks involved in managing Exchange recipients is creating and managing user mailboxes. User mailboxes, or just mailboxes, are the most common requirement for providing your users with e-mail functionality. While Exchange Server mailboxes are created in the Exchange admin center (EAC), Exchange Online mailboxes are created in the Microsoft 365 admin center and assigned a Microsoft 365 license that includes an Exchange Online plan for that user. You can also create user mailboxes in the Exchange Management Shell and Exchange Online PowerShell. Furthermore, Exchange Management Shell enables you to create scripts to automate the mailbox creation process and perform bulk mailbox creation operations.

Creating a user mailbox

When you create a mailbox in Exchange Server, you have the option to either associate the mailbox with an existing AD user account or create a new AD account that will be associated with the mailbox. Furthermore, you can also configure different mailbox properties, such as choosing a specific mailbox database for the mailbox.

If you create or enable a user mailbox through the Exchange Management Shell,you can configure multiple attributes at the same time that you create the account.

In the following example, the **New-Mailbox** command creates a use account and mailbox in the contoso.com domain for a user named Alex Darrow. The command assigns alexd as the user account name, , alexd@contoso.com as the alias, and specifies dc1.contoso.com as the domain controller that will be used to read data from or write data to Active Directory.

```
New-Mailbox -Name AlexD -UserPrincipalName alexd@contoso.com -Password
(ConvertTo-SecureString -String 'P@ssw0rd' -AsPlainText -Force) -Alias
alexd@contoso.com -FirstName Alex -LastName Darrow-DisplayName "Alex Dar-
row" -DomainController dc1.contoso.com
```

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If you want to create a mailbox for an existing user account, you can use the EAC or the **Enable-Mailbox** cmdlet. Once the user mailbox is created, you can use the **Set-Mailbox** cmdlet to modify the attributes of the mailbox. This cmdlet allows you to update more attributes than is allowed in the EAC.

Managing user mailboxes

After you create the mailbox, you can manage the mailbox settings by using the Exchange Admin Center or the Exchange Management Shell. The following list contains some of the mailbox configuration options available in Exchange Server and Exchange Online:

Tab	Configuration settings
general	User names and custom attributes
mailbox usage	Displays the last logon information In Exchange Online - View mailbox size and limits In Exchange Server – View and configure mailbox size and limits
contact information	Configure information such as address and phone number
organization	Configure the title, department, company, and manager settings
email address	Configure the email addresses assigned to the mailbox Can include Single Mail Transfer Protocol (SMTP), Exchange Unified Messaging addresses, or addresses associated with other messaging systems
mailbox features	Configure the policies that apply to the mailbox Configure mail flow settings including delivery options, message size, and delivery restrictions Configure different types of access such as mobile devices, MAPI, Outlook on the Web Configure Archiving
member of	View the groups to which the user account belongs
MailTip	Configure MailTip for the mailbox to be displayed when users add this recipient as a message recipient
mailbox delegation	Configure Send As, Send on Behalf of, and Full Access permissions to the user mailbox

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A user's Archive mailbox can be enabled during the mailbox creation process or after the mailbox is created by using the EAC or Exchange PowerShell. For example, to create an archive mailbox for an existing user mailbox, such as Alex Darrow, you would run the following command:

```
Enable-Mailbox "Alex Darrow" -Archive
```

In Exchange Server, the mailbox size for different types of users in an organization plays an important role in planning the disk space needed. It's recommended that you use the Exchange Server Role Requirements Calculator for estimating the optimal disk size and configuration to accommodate disk space for users' mailboxes.

In Exchange Online, depending on the subscription, users may have different mailbox size limits: For example, in Office 365 Business Essentials and Business Premium subscriptions, users have a 50 GB primary mailbox limit and a 50 GB archive mailbox limit, whereas in Office 365 E3 and E5 subscriptions, users have a 100 GB primary mailbox limit and unlimited storage for the archive mailbox.

Deleting a user mailbox

A user mailbox can be disabled or deleted, depending on whether the user account is still needed in the organization. Disabling a user mailbox deletes the mailbox, but the associated user account will still be present.

It's important to note that disabling a mailbox is different than disabling a user account in Active Directory. When mailbox is disabled, all Exchange attributes are removed from the associated user account in Active Directory, and the disconnected mailbox is hidden and marked for removal. However, the user in Active Directory is not disabled and can sign in to the domain. The disconnected mailbox is permanently deleted (purged) based on the **MailboxRetention** property value for the mailbox database, which by default is 30 days. Within this 30 day timeframe, you can reconnect the disconnected mailbox to a new or existing user account that doesn't already have an associated mailbox.

You can disable a user mailbox in the EAC or Exchange PowerShell by using following command:

```
Disable-Mailbox alexd@contoso.com
```

If both a user account and its associated user mailbox are no longer needed, such as when an employee leaves the organization, you can delete the mailbox by using the EAC or Exchange PowerShell. For example, to remove the mailbox for Alex Darrow, you would run the following command:

```
Remove-Mailbox -Identity "Alex Darrow"
```

The deleted mailbox is permanently deleted (purged) based on the **MailboxRetention** property value for the mailbox database, which by default is 30 days. Within this 30 day timeframe, you can reconnect the disconnected mailbox to another user account that doesn't already have an associated mailbox.

If a mailbox has an associated archive mailbox, the **Remove-Mailbox** cmdlet also marks the associated archive mailbox for deletion. Alternatively, you can disable the archive mailbox rather than deleting it by running the following command:

```
Disable-Mailbox "Alex Darrow" -Archive
```

Additional information. For more information, see the following article on **Create user mailboxes in Exchange Server**[3].

3 https://aka.ms/AA3qqg9

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Creating and managing resource mailboxes

When you create a resource mailbox, you can also configure settings that determine how the resource mailbox will respond to meeting requests. You can configure resource mailboxes to process incoming meeting requests for all users automatically, or you can restrict who can book the meeting room. You can configure delegates who must approve all meeting requests, and you can also configure the resource mailbox to accept only certain types of meetings. For example, you can configure a conference room to automatically accept incoming meeting requests but not accept recurring meeting requests.

You can create and manage resource mailboxes in the Exchange admin center or with Exchange PowerShell using the **New-Mailbox** cmdlet with the **-Room** parameter for a room mailbox or the **-Equipment** parameter for an equipment mailbox.

To create a new room mailbox with the name ConfRoom1 and the display name "Conference Room 1", you would run the following command:

```
New-Mailbox -Name ConfRoom1 -DisplayName "Conference Room 1" -Room
```

To create a new equipment mailbox with the name "Projector1" and display name "Projector 1", you would run the following command:

```
New-Mailbox -Name Projector1 -DisplayName "Projector 1" -Equipment
```

Managing resource mailboxes

When managing a resource mailbox through the Exchange admin center, you can configure the following settings that define how the mailbox will accept meeting requests.

Settings	Description
General	Specify Name, capacity, hide from address lists, department, company, address book policy, and custom attributes.
Booking delegates	Accept booking requests automatically, select delegates, or customize an acceptance policy for this mailbox.
Booking options	Allow repeated meetings, only schedule during working hours, maximum booking lead time, maximum meeting duration, and a customized reply to the meeting organizer.
Contact information	Add street, Zip/post code, city and so on if required.
Email address	Add further email addresses if required.
MailTip	Create a MailTip to provide additional information that users can see when they select this address in an email.
Mailbox delegation	Configure Send As, Send on Behalf Of, and Full Access permission for this mailbox, as with shared mailboxes.

In addition to the settings that are available to be updated in the EAC, you can also configure several additional settings that control how the resource mailbox will respond to meeting requests. These settings are configured by using the **Set-CalendarProcessing** cmdlet. Some of the available options include:

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Configuration option	Sample command	When to use
Allow conflicting meetings.	Set-CalendarProcessing –id MeetingRoom1 –AllowConflicts $true	If an organization wants to allow room bookings even when there is a conflict in the booking calendar so that conflicts will be manually resolved by a booking delegate.
Allow certain users to request meetings that do not follow the policies regarding maximum lead time or maximum meeting limits.	Set-CalendarProcessing –id MeetingRoom1 –RequestOutOf-Policy adam	If an organization needs specific users to have permission to book a room regardless of the book-ing policy.
Prevent the meeting room from automatically accepting meeting requests.	Set-CalendarProcessing -Identity MeetingRoom1 -AutomatePro-cessing:None	If an organization needs to assign a delegate who will be responsible for room booking.

Messaging administrators should consider the following factors when designing how meeting requests will be accepted:

- **Who can schedule a resource mailbox**. You might accept the default settings for most resources in the organization but consider restricting who can book heavily used or important resources. For example, if you use a resource room mailbox to manage the schedule for a large conference room, you may want to restrict who can book meetings in the conference room.
- **When users can schedule the resource mailbox**. You may want to set restrictions on the time of day when meetings can be booked with a resource or restrict the meeting length or meeting recurrence.
- **How to configure the automatic acceptance policy for the resource mailbox**. By default, all resource mailboxes are configured to accept all new appointment requests and to block conflicting requests. You can change this so that all meeting requests are accepted as tentative, or to allow users to book the meeting resource for the same time.

Creating and managing shared mailboxes

In Exchange, creating a shared mailbox is a single-step process performed by using the Exchange admin center or Exchange PowerShell. You can create a shared mailbox and grant users Full Access and Send As mailbox permissions when you create the mailbox. The following screenshot display the **Add a mailbox** page that's used to add a mailbox in the EAC.

You can also create and manage shared mailboxes with PowerShell by using the **New-Mailbox** command with the **-Shared** parameter. For example:

```
New-Mailbox -Shared -Name "HR Department" -DisplayName "HR Department"
-Alias HR@contoso.com | Set-Mailbox -GrantSendOnBehalfTo HRSG | Add-Mail-
boxPermission -User HRSG -AccessRights FullAccess -InheritanceType All
```

Managing shared mailboxes

When you grant a user Full Access permission to the shared mailbox, the delegated user can access the mailbox and view and manage all messages in the mailbox. Granting Full Access permissions does not grant the delegated user the right to send mail as the selected mailbox. To allow a user to send mail from a delegated mailbox, you must also assign Send As permissions. When a user with Send As permissions sends a message from the delegated mailbox, any message sent from the mailbox will appear as if it were sent by the mailbox owner.

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Shared mailboxes can have the following delegate permissions:

- **Full Access.** Users with full access permission can log on and carry out actions consistent with a mailbox owner. However, to send mail, users with Full Access permission must also have Send As or Send on Behalf of permission. You can configure Full Access permission through the EAC or PowerShell.
- **Send As.** Users with Send As permission can impersonate the mailbox when sending mail. Messages received are "from" the mailbox, so they appear to come directly from, say, marketing@contoso.com. You can configure Send As permission through the EAC or PowerShell.
- **Send on Behalf of.** Send on Behalf of permission grants the right to send messages, but those messages are stamped as coming from one user on behalf of another; for example, from "Sara Davis on behalf of IT Department." Send on Behalf of permission is only configurable through PowerShell.

Creating and managing mail contacts and mail users

Contacts are Exchange recipients that consist of name fields, an alias, and an external email address. There are two types of contacts available in Exchange: mail contacts and mail users.

Mail Contacts

Mail contacts do not have a user account in Active Directory (AD) or Microsoft 365; therefore, they cannot log on to a mailbox using, for example, Outlook. However, mail contacts do appear in the global address list (GAL) throughout the organization and can be added to mail-enabled security groups, distribution groups, or dynamic distribution groups (but not security groups). As a result, you can use contacts as you might use entries in your contacts folder in Outlook, with the difference being that AD or Microsoft 365 contacts are managed centrally.

You can also use contacts within your own hierarchy and assign them a manager. This approach is useful if your organization engages outside contractors or associates. After you create a contact, you can add some optional fields, such as contact information, phone numbers, notes, title, department, company, manager and direct reports. Finally, you can configure a MailTip that appears when someone sends a message to that person.

Mail Users

A mail user combines some of the attributes of a full mailbox user along with the characteristics of a contact. Mail users enable administrators to provide logon facilities to Microsoft 365 while continuing to provide an external email address.

Organizations that use associates often use mail user accounts to provide logon facilities to these personnel while forwarding their emails to their external email address. The mail user accounts can be assigned to a manager and department for administrative purposes.

The characteristics of a mail user include:

- They can log into AD or Microsoft 365 and access resources such as OneDrive for Business or SharePoint Online.
- They have an email address that is external to Microsoft 365, registered against the **ExternalEmailAddress.**
- They can have a secondary email address for the default <tenantname>.onmicrosoft.com domain.

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Creating and Managing Contacts

You can create or manage a contact after deciding what type of contact you need, Contacts can be added and maintained through either the Exchange Admin Center (EAC) or Windows PowerShell.

To create a contact in the Microsoft 365 EAC, you should perform the following procedure:

1. In the **Microsoft 365 admin center**, select **Show all** and then under **Admin centers** select **Exchange**.
2. In the **Exchange admin center**, select **recipients** and then select **contacts**.
3. Select the **+ (new)** icon and then in the drop-down menu, select either **Mail contact** or **Mail user**.
4. In the **New mail contact** page (or **New mail user** page), enter a First name, Initials, and Last name.
5. The **Display name** is autogenerated based on those first three fields in the form of First name, middle initial, Last name, but you can change that format.
6. In **Alias**, enter a unique value.
7. In **External email address** , enter the address to which you want mail for that user to be sent.
8. Select **Save**.

Depending on the type of contact you create, there are different settings that can be configured. Once you have created the new mail contact, you can then edit the details to add or change further information in the following tabs.

Tab	Configuration settings
general	Name fields, alias, and external SMTP address, hide from address list
contact information	Add street, Zip/post code, city and so on if required
organization	Add manager and department information
MailTip	Create MailTip to provide additional information that users can see when they select this address in an email

When you have created the new mail user, you can then edit the details to add or change further information in the following tabs.

Tab	Configuration settings
general	Name fields, alias, and external SMTP address, hide from address list
contact information	Add street, Zip/post code, city and so on if required
organization	Add manager and department information
email addresses	Add further email addresses if required
Mail flow settings	Restrict who can and can't send email to this account.
Member of	Add to distribution groups
MailTip	Create MailTip to provide additional information that users can see when they select this address in an email

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You also can use Windows PowerShell to create mail users. You might want to consider this approach if you're an experienced administrator or prefer to use Windows PowerShell to perform tasks. To create users in PowerShell, use the **New-MailContact** or **New-MailUser** cmdlets. For example, the following command creates a new mail user in your Microsoft 365 tenant named JohnDoe, and then forwards all messages to JohnDoe@contoso.com:

```
New-MailUser -Name \<name\> -WindowsLiveID \<Microsoft ID\> -Password
(ConvertTo-SecureString -String '\<password\>' -AsPlainText -Force) -Exter-
nalEmailAddress johndoe@contoso.com
```

Deleting a contact is as simple as selecting the contact and clicking the Delete icon in EAC or using the **Delete-MailUser** or **Delete-MailContact** cmdlets.

Bulk Importing Contacts

Adding contacts individually can be a lengthy and boring process, so if you have several contacts to import, it is best to do this in bulk by using Windows PowerShell. Here, the **Import-CSV** cmdlet comes in handy.

To bulk import contacts, perform the following process:

1. Create a CSV file containing the necessary information.
2. Use PowerShell to create the contacts.
3. Customize the newly created contacts using PowerShell.

At a minimum, you must provide values for the following fields in the CSV file:

- FirstName
- LastName
- Name
- ExternalEmailAddress

You then connect to Exchange Online using PowerShell and run the following cmdlet to create the contacts:

```
Import-Csv .\ExternalContacts.csv| foreach{New-MailContact -Name $\_.Name
-DisplayName $\_.Name -ExternalEmailAddress $\_.ExternalEmailAddress
-FirstName $\_.FirstName -LastName $\_.LastName}
```

Of course, if you want to create mail users, you need to exchange the **New-MailContact** cmdlet with **New-MailUser**.

Additional reading. For more detailed instructions on how to bulk import contacts as well as a sample CSV file that you can use to import contacts, see the following site titled **Bulk import external contacts to Exchange**[4].

Hiding Contacts

Since contacts appear in the Global Address List (GAL), there may be situations when you want to hide contacts from your users. For example, if you want to use contacts for e-mail forwarding only, your users should not add them to any distribution list or group.

4 https://go.microsoft.com/fwlink/?linkid=833321

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To hide contacts, you can also use Windows PowerShell with the **Set-MailContact** or **Set-MailUser** cmdlet and the **-HiddenFromAddressListsEnabled $true** parameter.

For example, the following command would hide the mail contact of Bob Kelly:

```
Set-MailContact BOBK -HiddenFromAddressListsEnabled $true
```

Managing permissions for recipients

You can use the Exchange Admin Center or the Exchange Management Shell to assign permissions to users or groups that will allow them to receive and respond to messages and meeting requests and responses on your behalf. Users or groups who receive permission to act on behalf of someone else are commonly referred to as delegates. You can also grant additional permissions that allow delegates to read, create, or have more control over items in your Exchange Server mailbox. For example, a department manager may delegate permissions to the assistant to read mail, respond to email on manager's behalf, or manage the manager's calendar.

Permissions can be assigned to user mailboxes, linked mailboxes, resource mailboxes, and shared mailboxes. You can also assign permissions to distribution groups, dynamic distribution groups, and mail-enabled security groups to allow delegates to send messages on behalf of the group.

The following permissions are used to access mailboxes or send messages on behalf of mailboxes or groups:

- **Full Access.** This permission allows a delegate to open a user's mailbox and access the contents of the mailbox.
- **Send As.** This permission allows delegates to use the mailbox to send messages.
- **Send on Behalf.** After this permission is assigned to a delegate, the From address in any message sent by the delegate indicates that the message was sent by the delegate on behalf of the mailbox owner.

The following procedure shows how to assign permissions to a user mailbox.

Note: You follow a similar procedure to assign permissions to resource or shared mailboxes by navigating to the **Resources** or **Shared** page in the EAC and selecting the mailbox to assign the permissions to.

1. In the EAC, navigate to **recipients** > **mailboxes**.
2. In the list of mailboxes, select the mailbox that you want to assign permissions to, and then select the **pencil (Edit)** icon.
3. On the mailbox properties page, select **mailbox delegation**.
4. To assign permissions to delegates, select the **plus (+) sign** icon under the appropriate permission to display a page that lists all recipients in your Exchange organization that can be assigned the permission. Select the recipients you want, add them to the list, and then select **OK**. You can also search for a specific recipient by typing the recipient's name in the search box and then selecting the **Search** icon.
5. To remove a permission for a recipient, under the appropriate permission, select the recipient and then select the **dash (Remove)** icon.
6. Select **Save** to save your changes.

The following example gives the Exchange Online user, Ayla Kol, Send As permission for the mailbox Help Desk:

```
Add-RecipientPermission "Help Desk" -AccessRights SendAs -Trustee "Ayla
Kol"
```

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The following example assigns the Send As permission to the Helpdesk group on the shared mailbox Helpdesk Support Team in an on-premises Exchange organization:

```
Add-ADPermission -Identity helpdesksupport -User helpdeskgroup -ExtendedRi-
ghts "Send As"
```

The following example assigns the delegate Holly Holt the Send on Behalf permission to the mailbox of Sean Chai:

```
Set-Mailbox -Identity seanc@contoso.com -GrantSendOnBehalfTo hollyh
```

Automapping is a feature that allows Outlook clients to automatically map to any mailbox to which a user has Full Access permissions. Therefore, if a user has Full Access permissions to another user's mailbox or to a shared mailbox, Outlook automatically loads all the mailboxes to which the user has full access.

Sometimes it's necessary to disable automapping. For example, a user may have Full Access permissions to so many mailboxes that it causes performance issues when starting Outlook.

You can disable automapping by using PowerShell. For example, if you want to grant the user Ed Jones full access permission to Kim Chastain's mailbox and disable the auto-mapping feature, you would run the following command:

```
Add-MailboxPermission -Identity KimC -User 'Ed Jones' -AccessRight FullAc-
cess -InheritanceType All -Automapping $false
```

Creating and managing groups

Exchange provides different types of groups to be able to group recipients. The following screenshot shows the groups tab in the Exchange Admin Center, which allows you to decide what type of group you want to create.

Managing Microsoft 365 groups

Exchange includes security and distribution groups that are also available in Exchange Online, but Microsoft 365 also includes a special group with additional functionality that is not available in Exchange Server. These are known as Microsoft 365 groups.

Microsoft 365 groups provide team collaboration such as document storage, a centralized calendar, a OneNote notebook, and working on project plans. They can also be used to distribute emails to the team in one space.

The difference between a Microsoft 365 group and a distribution list is that Microsoft 365 groups not only include a distribution list but also a shared:

- Inbox for group email communication
- Calendar for scheduling group meetings and events
- OneDrive for storing files and folders
- OneNote notebook for taking project and meeting notes
- Planner tool for organizing and assigning tasks and getting updates on project progress

When you join a group, your email address is automatically added to the distribution list and you gain access to all group information. Although you can create many Microsoft 365 groups, it's important to define an organization-wide naming convention; for example, consider a "team" Microsoft 365 group, rather than "team files", "team plan" and "team calendar" Microsoft 365 groups. The single group can fulfill multiple needs for the same group of people.

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You can create a Microsoft 365 group in the Microsoft 365 admin center, Exchange admin center, or by using Windows PowerShell. Remember, if you create groups in the EAC, they cannot be edited using the Microsoft 365 admin center even though the groups appear in the Security Groups list of the portal's Users and Groups section.

The following table identifies the Windows PowerShell cmdlets that are used to create and manage Microsoft 365 groups.

Cmdlet	Description
New-/Get-/Set-UnifiedGroup	Create/view/configure Microsoft 365 groups and their settings.
Remove-UnifiedGroup	Delete a Microsoft 365 group.
Add-/Get-/Remove-UnifiedGroupLinks	Add/view/remove owners, members and subscriber list.

For example, to create a Microsoft 365 group named "IT Administrators" with the alias ITAdmins, you should run the following command:

```
New-UnifiedGroup -DisplayName "IT Administrators" -Alias ITAdmins
```

You can also create Microsoft 365 groups with rule-based memberships using the Azure Management Portal. This allows easy management of larger groups or the creation of groups that always reflect the organization's structure. For more information, see the following article on **Dynamic membership rules for groups in Azure Active Directory**[5].

Note: When you enable dynamic membership for Microsoft 365 groups, you require Azure AD Premium licenses in your Microsoft 365 tenant.

After creating a mail-enabled distribution group, you can manage it by using the EAC or PowerShell. Management options will vary depending of the group type. For example, in mail-enabled security groups, you can manage the following settings:

Settings	Description
General	Change the display name, alias, email address, description, and the option to hide the group from address lists.
Ownership	Modify the owners of the group.
Membership	Modify the group membership.
Membership approval	Specify the options for joining or leaving the group.
Delivery management	Specify whether external addressees or only internal users can email this group.
Message approval	Configure moderation, specifying who can moderate the group and who can send messages to the group without moderation.
Email options	Add additional email addresses for the group.
MailTip	Add a MailTip to specify what displays when users send messages to the group.

5 https://aka.ms/AA3qqj8

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Settings	Description
Group delegation	Specify Send As and Send on Behalf Of permission for users or groups.

Managing Security and Distribution Groups

In the Exchange Admin Center or Microsoft 365 admin center, you can create mail-enabled security groups and add users to those groups. If you have synchronized your Microsoft 365 account with your on-premises Active Directory, security groups created in Active Directory are also synchronized across to Microsoft 365.

You can create mail-enabled security groups in the EAC or in Windows PowerShell using the **New-DistributionGroup** cmdlet.

Group types	PowerShell cmdlet
Mail-enabled security group	New-DistributionGroup -Type Security
Mail-enabled distribution groups	New-DistributionGroup
Mail-enabled dynamic distribution group	New-DynamicDistributionGroup

For example, to create a new security group that can also be used as a distribution group for your finance department, you should run the following command:

```
New-DistributionGroup –Name "Finance Department" –Alias FDep –Type security
```

Dynamic distribution groups change their membership depending on a query against account types and additional criteria. Because dynamic distribution lists can be quite large, it is important to design them correctly.

Creating dynamic distribution lists in the EAC is similar to a distribution list, except for setting up the criteria. When selecting Members, you can select one, some or all of the following options:

- Users with Exchange mailboxes
- Mail users with external email addresses
- Resource mailboxes
- Mail contacts with external email addresses
- Mail-enabled groups

You can create a dynamic distribution group by using PowerShell with the following command:

```
New-DynamicDistributionGroup –IncludedRecipients MailboxUsers –Name "Sales
Users Dynamic Group" –Department Sales
```

Note: Due to the nature of dynamic distribution groups, a user can become a member of the group without explicit notice or knowledge. This can occur if a user's account properties are changed to match the dynamic distribution group filters or conditions. In this case, the user would start to receive messages sent to the dynamic group without the manager of the group knowing this. Therefore, it's important to understand that dynamic distribution groups memberships are managed outside of the group, unlike a normal distribution group.

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Migrating Distribution Lists to Microsoft 365 Groups

Microsoft 365 offers the ability to migrate a distribution list to a Microsoft 365 group. You can either use the EAC or Windows PowerShell to migrate distribution lists. However, you only can convert a distribution group to Microsoft 365 Groups if the following conditions are met:

- Only distribution groups can be converted, not security groups or dynamic distribution groups
- Distribution group may contain mailboxes, shared mailboxes, site mailboxes or mail users
- Distribution group must be visible in the global address list
- No nested or moderated distribution groups can be converted
- No send-on-behalf setting can be configured on the distribution group

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Lab7-Ex1-2-Create Recipient Objects and Resources

Lab7-Ex1-Create Exchange Recipients

Adatum Corporation plans to create a hybrid Exchange deployment. They have Exchange Server 2019 and their on-premises user mailboxes installed on their on-premises Exchange Server (LON-EX1). Now that they are implementing Microsoft 365, they can also create user accounts and mailboxes in Exchange Online.

In your role as Holly Dickson, Adatum's Messaging Administrator, you are interested in seeing how mailboxes are maintained in both Adatum's on-premises Exchange Server deployment as well as its new Microsoft 365 deployment. In the earlier lab on Implementing ActiveSync, you created on-premises user mailboxes for Holly, Paul Wimmer, and Jessica Hofer. Each mailbox was created in the on-premises adatum.com domain on LON-EX1. In this lab, you will create an Exchange Online user mailbox for Holly, which will be added to the M365xZZZZZZ.onmicrosoft.com domain (where ZZZZZZ is the tenant ID provided by your lab hosting provider).

See your VM lab environment for the lab instructions.

Lab7-Ex2-Create Groups

In this exercise, you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. As part of her pilot project for deploying Microsoft 365, Holly wants to examine how creating groups in her on-premises Exchange environment differs from creating groups in Microsoft 365 for Exchange Online. She will begin by creating an on-premises distribution group and then a cloud distribution group. She will finish by creating a Microsoft 365 group.

See your VM lab environment for the lab instructions.

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Managing Email Addresses, Lists, and Resources

Lesson Introduction

In Exchange, all recipients are listed in address lists that are accessible to users from Outlook and Outlook on the Web. Address lists simplify the process of finding recipients in an organization. However, in large organizations the address lists may contain thousands of recipients. Therefore, you may create custom address list that include users based on difference criteria, such as geographical location or department. Outlook users can download an offline copy of the address lists that is called an offline address book (OAB). This enables a user running Outlook in cached mode to access the address book while disconnected from Exchange.

Address book policies (ABPs) allow you to segment users and provide customized views of the organization's global address list (GAL). ABPs are used in scenarios when two separate companies work in one Exchange organization, and each of those two companies need to access only their own address lists. ABPs may be also used in schools, where students in each class need to see only the other students in their own class, including the teacher and the principal.

Because many messaging systems host multiple SMTP domains, messaging administrators must manage the email addresses assigned to the Exchange recipients. To ensure that recipients have the appropriate email addresses, you can configure multiple accepted domains in Exchange Server and create and apply email address policies that define the email address' format.

After completing this lesson, you will be able to:

- Describe address lists
- Explain how to configure address lists
- Describe address book policies
- Explain how to configure offline address books
- Describe email address policies

Address Lists Overview

Address lists are used to group recipient objects based on a LDAP query for specific user attributes. You can use address lists to sort the GAL into multiple views, which makes it easier to locate recipients. This is especially helpful for very large or highly segmented organizations.

You can configure address lists with recipient filters that determine which objects belong in each address list. Address lists are evaluated every time a mail-enabled account is modified to determine the address lists on which it should appear.

There are two types of recipient filters:

- **Pre-canned.** This is a commonly used Exchange filter that you can use to meet a variety of recipient-filtering criteria for creating address lists. For example, using pre-canned filters, you can determine the scope of recipients and add conditional filtering based on properties such as company, department, and state or region.
- **Custom.** If pre-canned filters don't meet your needs for creating address lists, you can create a custom filter by using the RecipientFilter parameter. For example, you can use the Filter parameter to filter information about users whose title contains the word "manager".

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The following examples describe when and how address lists are used:

- **Custom address lists.** Consider an organization that has two large divisions and one Exchange organization. One division, named Fourth Coffee, imports and sells coffee beans. The other division, Contoso, Ltd., underwrites insurance policies. Because of the different nature of each business, the employees rarely communicate with each other. To make it easier for employees to find recipients who exist only in their division, you can create two new custom address lists, one for Fourth Coffee and one for Contoso, Ltd. When employees search for recipients in their division, these custom address lists allow them to select only the address list that is specific to their division. However, if an employee is unsure about the division in which the recipient exists, the employee can search within the GAL that contains all recipients in both divisions.
- **Hierarchical address lists.** You can use subcategories of address lists, which are known as hierarchical address lists. For example, you can create an address list that contains all recipients in Vancouver and another address list that contains all Seattle recipients. You also can create another list called Research and Development within the Vancouver address list, which contains all employees who work in Vancouver's Research and Development department. This allows employees to easily find the information they need.

When creating address lists, you should consider the following recommendations:

- Address lists should make it easier for users to find recipients; therefore, avoid creating so many address lists that users can't decide which list to use.
- Use a naming convention and location hierarchy for your address lists so users can easily determine which recipients are included in the list.
- Remind users that they can find anyone in their organization by using the GAL.

Additional reading. For more information, see the following article on **Address lists in Exchange Online**[6].

Creating and managing address lists

To create and manage address lists, you can use Windows PowerShell for Exchange Online or Exchange Server. In addition, for Exchange Server you can also use EAC to manage address lists, but that option is not available in Exchange Online.

This topic examines the most common address list management tasks performed by Messaging Administrators, including:

- Creating address lists
- Modifying address lists
- Deleting address lists
- Hiding mailboxes from address lists
- Updating address list membership

Creating address lists

You can use the **New-AddressList** cmdlet to create an address list. For example, to create an address list that includes all users who work in Contoso Asia offices in Japan and India, you would run the following command:

6 https://aka.ms/AA3qqk5

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```
New-AddressList -Name "Asia Offices" -IncludedRecipients MailboxUsers -Condi-
tionalStateorProvince "Japan","India"
```

Once you created the address list, you can verify the creation of the address list by using the following command:

```
Get-AddressList -Identity "Asia Offices" | Format-List Name,RecipientFilter-
Type,RecipientFilter,IncludedRecipients,Conditional*
```

If you want to verify membership of the address list you created, you would run the following commands:

```
$AL = Get-AddressList -Identity "Asia Offices"
Get-Recipient -ResultSize unlimited -RecipientPreviewFilter $AL.Recipient-
Filter | select Name,PrimarySmtpAddress,HiddenFromAddressListsEnabled
```

Modifying address lists

To modify the address list, you can use @{Add=Value} ore Remove={Value} syntax. For example, to modify an address list that will includes all users who work in Singapore offices in the "Asia Offices" address list, you would run the following command:

```
Set-AddressList -Identity "Asia Offices" -ConditionalStateOrProvince @{Ad-
d="Singapore"}
```

Deleting address lists

Once the address list isn't needed anymore, you can use the **Remove-AddressList** cmdlet to remove the address list. For example, you nee to create address lists for multiple regions of Asia, and you want to remove the "Asia Offices" address list. To removethe "Asia Offices" address list, you would run the following command:

```
Remove-AddressList -Identity " Asia Offices"
```

Hiding mailboxes from address lists

In some scenarios, you may need to hide some user mailboxes from an address list.

Note: Hiding the recipient from an address list doesn't prevent the recipient from receiving email messages; it only hides the recipient from all address lists and the GAL.

To hide a mailbox user from address lists using the EAC, on the general tab of the user mailbox properties, select the "Hide from address lists" checkbox. You may also use Exchange Online PowerShell with following command:

```
Set-Mailbox -Identity <a href="mailto:sara.davis@contoso.com" title=""
target="_blank" data-generated=''>sara.davis@contoso.com</a> -HiddenFromAd-
dressListsEnabled $true
```

Updating address list membership

In addition to the most common address lists management tasks that are available in Exchange Online, there are additional tasks that you can perform in Exchange Server by using the EAC and the Exchange Management Shell. The most common of these tasks involves updating address list membership.

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After creating or modifying the GAL, you must start the GAL update process. For example, after creating a GAL named "Contoso GAL", you must update it by using following command:

```
Update-GlobalAddressList -Identity "Contoso GAL"
```

Note: While updating the address lists can also be done in the Exchange admin center, it is recommended that you use the Exchange Management Shell if the address list contains more than 3000 recipients. For example, to update the address list named "Marketing", you would run the following command:

```
Update-AddressList -Identity "Marketing"
```

Address book policies

Address-book policies (ABPs) can limit the information that users see in their Global Address List. Some organizations require that certain users be prohibited from seeing all the other users in the GAL. For example, a large investment organization may have several divisions that are competitors in selected markets; therefore, allowing communication between investors in each division might violate trading laws. Other organizations that have extremely large GALs may want to limit the size of the offline address book for users. Limiting what users can see in the GAL is called GAL segmentation.

In Exchange, you can use address-book policies to configure GAL segmentation. When configuring an ABP, you assign a GAL, an offline address book, a room list, and one or more address lists to the policy. You can then assign the address-book policy to mailbox users, which means that the users can only see the objects in the GAL that are part of their policy.

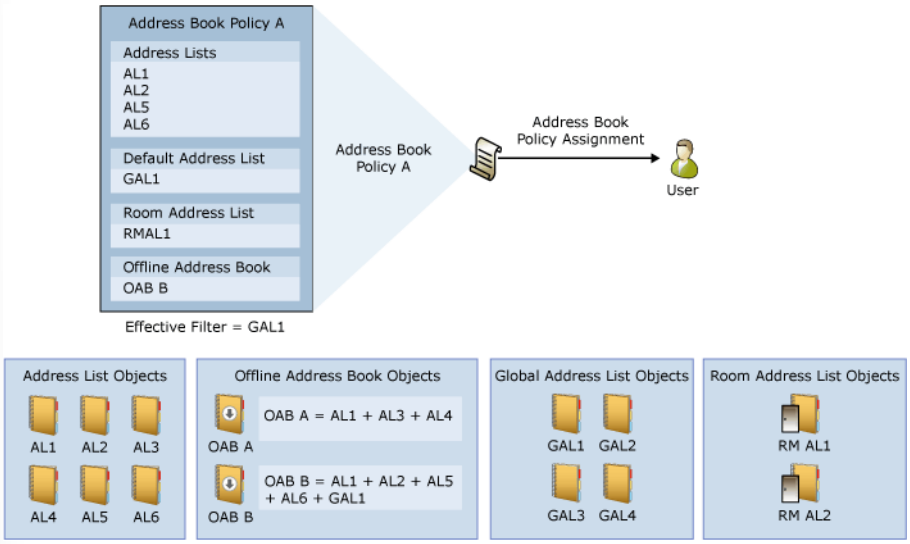
Note: ABPs provide a virtual GAL segmentation, and not a legal separation. This means that users might sometimes be aware of other recipients in the organization that are not part of their address-book policy. For example, a distribution group that is included in the ABP might include recipients from another ABP. If one of those recipients has an out-of-office message configured, the out-of-office message will be sent to anyone who sends an email to the distribution group.

An ABP contains the following components:

- One GAL
- One OAB
- One room list
- One or more address lists

The following diagram shows how ABPs work. The user is assigned "Address Book Policy A", which has the following components: Global address list "GAL1", OAB "OAB B", Room address list "RMAL1" and Address lists "AL1", "AL2", "AL5" and "AL6". When the ABP is created and assigned to the user, the ABP becomes the scope of the address lists that the user can view. APBs take effect when a user connects to his or her Exchange Mailbox. If you modify an ABP, the updated APB will be applied to a user when the user restarts or reconnects his or her email client app.

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ABP configuration tasks include following steps that can be performed through Exchange Online Power-Shell in either Exchange Online or in the Exchanger Management Shell in Exchange Server.

1. Turn on ABP routing using the following command:

   ```
   Set-TransportConfig -AddressBookPolicyRoutingEnabled $true
   ```

2. Create an Address Book Policy. For example, to create an ABP for our example in the illustration, you would run the following command:

   ```
   New-AddressBookPolicy -Name "Address Book Policy A" -AddressLists "\AL1","\
   AL2 ","\ AL5 ","\AL6" -OffineAddressBook "\OAB B" -GlobalAddressList "\GAL1"
   -RoomList "\RMAL1"
   ```

3. Assign an ABP to a mailbox user. For example, to assign an ABP "Address Book Policy A" to laura@contoso.com, you would run the following command

   ```
   Set-Mailbox -Identity laura@contoso.com -AddressBookPolicy " Address Book
   Policy A"
   ```

You can also use a list of specific mailboxes or you can filter mailboxes by different attributes to assign an ABP to multiple user mailboxes.

1. Change an ABP settings. For example, if you want to replace the "OAB B" address book with "OAB A" in ABP "Address Book Policy A", you would run the following command:

   ```
   Set-AddressBookPolicy -Identity "Address Book Policy A" -OffineAddressBook
   "\OAB A"
   ```

2. Remove an ABP. You can remove an ABP only if it is not assigned to any user mailbox. For example, once you confirm that the ABP "Address Book Policy A" is not assigned to any mailbox, you may remove it with following command:

   ```
   Remove-AddressBookPolicy -Identity "ABP Address Book Policy A"
   ```

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Additional reading. For more information, see the following article on **Address book policies in Exchange Online**[7].

Configuring offline address books

Offline address books (OABs) are downloadable copies of the address lists that are used by Outlook clients when they are configured in cached mode and they are not connected to Exchange. OABs are also queried first when Outlook clients are connected to Exchange; this is done to reduce the workload on Exchange Server and prevents the client from causing additional network traffic if the local OST includes the search result. The default offline address book contains the entire GAL, which includes all recipients in the Exchange organization. You can create additional address lists and add them to a custom offline address book.

By default, the offline address book is generated on a Mailbox server every 8 hours. This means that any additions, deletions, or changes made to mail-enabled recipients are only committed to the offline address book on these intervals, unless you modify the schedule to generate the offline address book more frequently, or manually update the OAB if needed.

Once Outlook clients download the OAB, any subsequent change to the address lists in Exchange are downloaded as a differential update instead of a full update; this optimizes bandwidth and performance of the client connections.

Outlook clients download the OAB every 24 hours if you do not modify the default on the client side. Users can also initiate an OAB download manually if needed.

If you plan to customize OABs, you should consider the following factors:

- Offline address book sizes can vary from a few megabytes to a few hundred megabytes. The OAB size depends on the following components:
  - The number of mail recipients and groups in your organization.
  - The usage of certificates. PKI certificates range from 1KB to 3KB.
  - The number of attributes used by each recipient object.
- The number of OAB downloads should be optimized to save bandwidth and server performance. For example, in a geographically distributed organization, clients in the United States should not download the OAB users in Europe.
- Overall changes to users and groups affect the size of the OAB downloads.

Before you install Exchange Server, you must assign a default offline address book (OAB) to all the Exchange mailboxes in your organization. Users with mailboxes that aren't assigned a default OAB when Exchange is installed will automatically download the new OAB generated by Exchange. This could potentially cause significant network traffic and additional server load, especially if your Exchange organization has hundreds or thousands of mailboxes.

OAB configuration tasks can be performed by Exchange Online PowerShell in Exchange Online and Exchanger Management Shell in Exchange Server. The most common OAB tasks include:

- **Create or remove an OAB.** For example, to create a custom OAB named "OAB_Contoso" that contains the "Marketing" address list, you would run the following command:

  ```
  New-OfflineAddressBook -Name "OAB_Marketing" -AddressLists "Marketing"
  ```

  To remove a custom OAB named "Contoso OAB", use the following command:

7 https://aka.ms/AA3r29l

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```
Remove-OfflineAddressBook -Identity "Contoso OAB"
```

- **Add or remove an address list to an OAB**. For example, if an OAB named "Finance" already contains the address list "Finance Managers" and you want to add an address list titled "Finance Employees," you would run the following command:

  ```
  Set-OfflineAddressBook -Identity "Finance" -Address Lists "Finance Manag-
  ers,"Finance Employees"
  ```

  For removing an address list from an OAB, repeat the previous command without the address list that needs to be removed. For example, if you need to remove the address list "Finance Managers" from the "Finance" OAB, you would run the following command:

  ```
  Set-OfflineAddressBook -Identity "Finance" -Address Lists "Finance Employees"
  ```

- **Change the default OAB**. In larger organizations, a default OAB can be very large; therefore, you may need to create a new OAB and configure it as a default. For example, to configure "Contoso OAB" as a default OAB in an organization, you would run the following command:

  ```
  Set-OfflineAddressBook -Identity "Contoso OAB" -IsDefault $true
  ```

- **Assigning OABs to users**. Once you customize your OABs, you must assign them to users according to your organization requirements. You can assign a custom OAB per mailbox or per address book policy. For example, to assign "Contoso OAB" to the mailbox "laura@contoso.com", you would run the following command:

  ```
  Set-Mailbox -Identity laura@contoso.com -OfflineAddressBook "Contoso OAB"
  ```

- **Assigning ABP to users.** To assign the "Contoso ABP" address book policy to the mailbox "laura@contoso.com", you would run the following command:

  ```
  Set-Mailbox -Identity laura@contoso.com -AddressBookPolicy "Contoso ABP"
  ```

Email address policies

In Exchange Server, email addresses for recipients are generated by email-address policies. Email address policies are not available in Exchange Online. By default, Exchange Server contains an email address policy that assigns one or more email addresses to every recipient. However, you can modify the default email address policy and create additional email address policies.

The recipient policy functionality is divided into two features:

- email address policies
- accepted domains

An accepted domain is an SMTP namespace that you configure in the Exchange organization so that Exchange servers accept messages sent to that SMTP namespace. By default, an Exchange Server contains an email address policy that assigns one or more email addresses to every mail-enabled user. This default policy specifies the recipients' alias as the local part of the email address and uses the default accepted domain.

The local part of an email address is the name that appears before the @ symbol. However, you can configure how your recipients' email addresses display. For example, you can specify that the addresses display as firstname.lastname@contoso.com. To specify additional email addresses for all recipients or just a subset of recipients, you can modify the default policy or create additional email address policies. For example, the user mailbox for Franklin Forester can receive email messages addressed to fforester@contoso.com and franklin.forester@contoso.com.

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Each time a recipient object is modified and saved, Exchange enforces the correct application of the email address criteria and settings. When an email address policy is modified and saved, all associated recipients are updated with the change. In addition, if a recipient object is modified, that recipient's email address policy membership is reevaluated and enforced.

Exchange Server always ensures that the email address that is assigned to the mailboxes are unique. For example, if two users have the same first name and last name, Exchange server appends a number to the local part of the second user's email address when the email address policy assigns an email address to the mailbox and the email address is based on the user's first name and last name.

Creating an Email Address Policy

The new Email Address Policy Wizard provides a default set of recipient scope filters, including:

- All recipient types
- Users with Exchange mailboxes
- Mail users with external email addresses
- Resource mailboxes
- Mail contacts with external email addresses
- Mail-enabled groups

You can also configure a rule that filters the recipients to which the email-address policy applies. Recipients can be filtered based on the following categories:

- Recipient container
- State or province
- Company
- Department
- Custom attributes

When creating an email-address policy, you can use the following email address types:

- **Default SMTP email address.** Default SMTP email addresses are commonly used email address types that Exchange Server provides for you.
- **Custom SMTP email address.** If you don't want to use one of the default SMTP email addresses, you can specify a custom SMTP email address. When creating a custom SMTP email address, you can use the variables in the following table to specify alternate values for the local part of the email address.

Variable	Value
%g	Given name (first name)
%i	Middle initial
%s	Surname (last name)
%d	Display name
%m	Exchange alias
%xs	Uses the x number of letters of the surname. For example, if x=2, the first two letters of the surname are used.

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Variable	Value
%xg	Uses the x number of letters of the given name. For example, if x=2, the first two letters of the given name are used.

Email address policies can be created using the Exchange Admin Center or the Exchange Management Shell.

Note: The email address policy feature is not available in Exchange Online.

Creating an Email Address Policy using the Exchange Admin Center

You should perform the following steps to create an email address policy using the EAC:

1. Navigate to **Mail flow** > **Email address policies**, and then click the **Add** icon.
2. In **Email Address Policy**, complete the following fields:
   - **Policy name**
   - **Email address format**
   - **Specify the types of recipients this email address will apply to**
3. Click **Add a rule** to further restrict the recipients that this policy will apply to. This creates a Boolean **And** statement.
4. Click **Preview recipients the policy applies to** to view the recipients the policy will apply to.
5. Click **Save** to save your changes and create the policy.
6. You'll receive a warning that the email address policy won't be applied until you update it. After it's created, select it, and then in the details pane, click **Apply**.

Creating an Email Address Policy using the Exchange Management Shell

The following example creates an email address policy that includes mailbox users in the "Southeast Offices" who will have email addresses that include their last name combined with the first two letters of their first name:

```
New-EmailAddressPolicy -Name "southeast offices" -IncludedRecipients Mailbox-
Users -ConditionalStateorProvince "Georgia","Alabama","Louisiana" -Ena-
bledEmailAddressTemplates "SMTP:%s%2g@southeast.contoso.com"
```

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Module Review

Module 8 Review

Multiple choice

Item 1. You are a messaging administrator for Contoso. You are using Outlook to create a mailbox for scheduling meetings in a certain conference room that can also accept automatic booking requests. What will you create?

- ☐ Shared mailboxes
- ☐ Room mailboxes
- ☐ Equipment mailboxes
- ☐ Microsoft 365 group
- ☐ Mail-enabled Public folders

Multiple choice

Item 2. You are a messaging administrator for Adatum Corporation. You must create mailboxes that can be accessed by multiple users as an additional mailbox in Outlook. What type of recipients do you need to create?

- ☐ Shared mailboxes
- ☐ Room mailboxes
- ☐ Equipment mailboxes
- ☐ Distribution groups
- ☐ Mail-enabled Public folders

Multiple choice

Item 3. You are a messaging administrator for Adatum Corporation. You need to create groups in Exchange Server that can be used to distribute messages and to provide access to resources. What type of groups will you create?

- ☐ Distribution groups
- ☐ Mail-enabled security groups
- ☐ Dynamic distribution groups
- ☐ Microsoft 365 groups

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Multiple choice

Item 4. You are a messaging administrator for Contoso. You want to use a custom SMTP domain in your Exchange Online organization. What should you do before you can add the email address to your users?

- ☐ Configure the custom domain as Internal Relay in Accepted Domains
- ☐ Verify the custom domain in your Microsoft 365 tenant
- ☐ Add the custom domain as a Remote Domain
- ☐ Configure the custom domain as Authoritative in Accepted Domains

Checkbox

Item 5. You are a messaging administrator for Adatum Corporation. You need to assign a permission to your users that gives them the ability to modify their display name and to create and modify distribution groups they own. What do you need to apply to each user account? (Select 2)

- ☐ End-user role
- ☐ Role group
- ☐ Role assignment policy
- ☐ Admin role

Multiple choice

Item 6. The following statement describes a recipient type: "The user can sign in to Active Directory, but the user's mailbox is located on an external email solution and not hosted by the Exchange organization." Which recipient type is described by this statement?

- ☐ Mail contacts
- ☐ Mail users
- ☐ Mail-enabled Public folders
- ☐ Site mailboxes

Multiple choice

Item 7. You are a messaging administrator for Contoso. Which type of group should you create where membership is defined by using recipient filters and conditions, and you will not need an Azure AD Premium license?

- ☐ Distribution groups
- ☐ Security groups
- ☐ Dynamic distribution groups
- ☐ Office 365 groups

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Checkbox

Item 8. Which of the following experiences can be provided by public folders? (Select 2)

- ☐ Support for every Outlook and OWA version
- ☐ Only admin users will see the full hierarchy in Outlook or OWA
- ☐ Once added to Favorites in Outlook, public folders can be made available offline
- ☐ View-Only Organization Management

Multiple choice

Item 9. You are a messaging administrator for Adatum Corporation. Adatum has more than 10,000 employees, and messaging administrators often receive complaints that it's difficult for them to locate a user in the address lists. You need to propose a solution that will provide users an easy way to find any recipient in the organization by using different criteria, such as department and geographical location. Which of the following solutions should you propose that meets these requirements?

- ☐ Create distribution lists based on department and location and inform users to send email to those distribution lists.
- ☐ Create an address lists for each department based on department and location and inform users that they can locate users in Outlook and Outlook on the Web based on department and geographical location.
- ☐ Crate shared mailboxes based on department and location and inform users to send email to those shared mailboxes.
- ☐ Instruct users to create their own contacts in Outlook for persons from different departments they frequently communicate with.

Multiple choice

Item 10. You are a messaging administrator for Adatum Corporation. You need to propose a solution that provides team collaboration functionality such as document storage, a centralized calendar, a OneNote notebook, working collaborative on project plans, and team email distribution in one space. What type of recipient should you create?

- ☐ Shared mailboxes
- ☐ Room mailboxes
- ☐ Equipment mailboxes
- ☐ Microsoft 365 Groups
- ☐ Mail-enabled Public folder

AUTHORIZED TRAINER USE ONLY. STUDENT USE PROHIBITED

Checkbox

Item 11. You are a messaging administrator for Adatum Corporation. Adatum recently deployed Microsoft 365, and it has now decided to convert distribution lists to Microsoft 365 Groups. What conditions should be met so that you will be able to convert the distribution lists to Microsoft 365 Groups? (Select 3)

- ☐ Only distribution groups can be converted; security groups or dynamic distribution groups are not able to be converted.
- ☐ Only distribution groups and security groups can be converted.
- ☐ Distribution group must only contain mailboxes, shared mailboxes, site mailboxes or mail users.
- ☐ Distribution group must be visible in global address list
- ☐ Distribution groups can be nested or moderated
- ☐ Distribution group can have a send on behalf setting

Multiple choice

Item 12. You are a messaging administrator for Contoso corporation. Contoso needs groups that will provide functionality to send email to multiple users. The group should not have any security permissions to any resource in the organization. Which type of groups are you going to create?

- ☐ Distribution groups
- ☐ Shared mailboxes
- ☐ Office 365 groups
- ☐ Security groups

Multiple choice

Item 13. You are a messaging administrator for Adatum Corporation, which is an Exchange Server 2019 organization. Adatum recently merged with Contoso, and Adatum will be hosting the messaging infrastructure for Contoso. You created user mailboxes for Contoso employees in the Adatum Exchange organization. However, they need to retain their email address SMTP domain of contoso.com. What should you do to meet these requirements?

- ☐ You must create a new accepted domain of contoso.com and create an email address policy that configures Contoso employees with contoso.com SMTP domain email addresses.
- ☐ You must create a new AD forest for contoso.com and install another Exchange Server organization with a contoso.com domain.
- ☐ You must edit each user's mailbox and add a contoso.com email address.
- ☐ You must create a forest trust with the Contoso messaging infrastructure.

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Multiple choice

Item 14. You are a messaging administrator for Adatum Corporation. Your organization has hundreds of meeting rooms worldwide and there is always a challenge when multiple teams want to use a meeting room that is already booked. You need to propose a solution that provides calendar information about room availability and the ability to automatically book a room if it's available. Which of the following solutions should you propose to meet these requirements?

- ☐ Create a contact for each room. Each time some users want to book a room, they should send a meeting request to the appropriate contact.
- ☐ Create a room mailbox for each room. Each time some users want to book a room, they should send a meeting request to the appropriate room mailbox.
- ☐ Create a shared mailbox for each room. Each time some users want to book a room, they should send a meeting request to the appropriate shared mailbox.
- ☐ Create a mail user for each room. Each time some users want to book a room, they should send a meeting request to the appropriate mail user.

Multiple choice

Item 15. Which of the following scenarios applies to linked mailboxes?

- ☐ Excellent for team or department collaboration where additional resources such as a OneNote or OneDrive is needed. Because it provides hooks into multiple Microsoft 365 services, it can be used as a tool to aid adoption and drive business value.
- ☐ Typically created for temporary employees or external vendors who need to access resources in the organization but use their own email address which is located outside your organization.
- ☐ When it's created, a disabled user account is created in the Exchange organization, and a user account from a trusted forest is given access to the mailbox.
- ☐ Team, department, or project mailboxes to centrally store, respond to, or manage messages. Organizations often use this to provide one specific mailbox for dedicated group of people such as sales, help desk, or general information requests.

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Answers

Multiple choice

Item 1. You are a messaging administrator for Contoso. You are using Outlook to create a mailbox for scheduling meetings in a certain conference room that can also accept automatic booking requests. What will you create?

- ☐ Shared mailboxes
- ■ Room mailboxes
- ☐ Equipment mailboxes
- ☐ Microsoft 365 group
- ☐ Mail-enabled Public folders

Multiple choice

Item 2. You are a messaging administrator for Adatum Corporation. You must create mailboxes that can be accessed by multiple users as an additional mailbox in Outlook. What type of recipients do you need to create?

- ■ Shared mailboxes
- ☐ Room mailboxes
- ☐ Equipment mailboxes
- ☐ Distribution groups
- ☐ Mail-enabled Public folders

Multiple choice

Item 3. You are a messaging administrator for Adatum Corporation. You need to create groups in Exchange Server that can be used to distribute messages and to provide access to resources. What type of groups will you create?

- ☐ Distribution groups
- ■ Mail-enabled security groups
- ☐ Dynamic distribution groups
- ☐ Microsoft 365 groups

AUTHORIZED TRAINER USE ONLY. STUDENT USE PROHIBITED

Multiple choice

Item 4. You are a messaging administrator for Contoso. You want to use a custom SMTP domain in your Exchange Online organization. What should you do before you can add the email address to your users?

- ☐ Configure the custom domain as Internal Relay in Accepted Domains
- ■ Verify the custom domain in your Microsoft 365 tenant
- ☐ Add the custom domain as a Remote Domain
- ☐ Configure the custom domain as Authoritative in Accepted Domains

Checkbox

Item 5. You are a messaging administrator for Adatum Corporation. You need to assign a permission to your users that gives them the ability to modify their display name and to create and modify distribution groups they own. What do you need to apply to each user account? (Select 2)

- ■ End-user role
- ☐ Role group
- ■ Role assignment policy
- ☐ Admin role

Multiple choice

Item 6. The following statement describes a recipient type: "The user can sign in to Active Directory, but the user's mailbox is located on an external email solution and not hosted by the Exchange organization." Which recipient type is described by this statement?

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- ☐ Mail-enabled Public folders
- ☐ Site mailboxes

Multiple choice

Item 7. You are a messaging administrator for Contoso. Which type of group should you create where membership is defined by using recipient filters and conditions, and you will not need an Azure AD Premium license?

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- ☐ Security groups
- ■ Dynamic distribution groups
- ☐ Office 365 groups

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Checkbox

Item 8. Which of the following experiences can be provided by public folders? (Select 2)

- ■ Support for every Outlook and OWA version
- ☐ Only admin users will see the full hierarchy in Outlook or OWA
- ■ Once added to Favorites in Outlook, public folders can be made available offline
- ☐ View-Only Organization Management

Multiple choice

Item 9. You are a messaging administrator for Adatum Corporation. Adatum has more than 10,000 employees, and messaging administrators often receive complaints that it's difficult for them to locate a user in the address lists. You need to propose a solution that will provide users an easy way to find any recipient in the organization by using different criteria, such as department and geographical location. Which of the following solutions should you propose that meets these requirements?

- ☐ Create distribution lists based on department and location and inform users to send email to those distribution lists.
- ■ Create an address lists for each department based on department and location and inform users that they can locate users in Outlook and Outlook on the Web based on department and geographical location.
- ☐ Crate shared mailboxes based on department and location and inform users to send email to those shared mailboxes.
- ☐ Instruct users to create their own contacts in Outlook for persons from different departments they frequently communicate with.

Multiple choice

Item 10. You are a messaging administrator for Adatum Corporation. You need to propose a solution that provides team collaboration functionality such as document storage, a centralized calendar, a OneNote notebook, working collaborative on project plans, and team email distribution in one space. What type of recipient should you create?

- ☐ Shared mailboxes
- ☐ Room mailboxes
- ☐ Equipment mailboxes
- ■ Microsoft 365 Groups
- ☐ Mail-enabled Public folder

AUTHORIZED TRAINER USE ONLY. STUDENT USE PROHIBITED

Checkbox

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- ☐ Only distribution groups and security groups can be converted.
- ■ Distribution group must only contain mailboxes, shared mailboxes, site mailboxes or mail users.
- ■ Distribution group must be visible in global address list
- ☐ Distribution groups can be nested or moderated
- ☐ Distribution group can have a send on behalf setting

Multiple choice

Item 12. You are a messaging administrator for Contoso corporation. Contoso needs groups that will provide functionality to send email to multiple users. The group should not have any security permissions to any resource in the organization. Which type of groups are you going to create?

- ■ Distribution groups
- ☐ Shared mailboxes
- ☐ Office 365 groups
- ☐ Security groups

Multiple choice

Item 13. You are a messaging administrator for Adatum Corporation, which is an Exchange Server 2019 organization. Adatum recently merged with Contoso, and Adatum will be hosting the messaging infrastructure for Contoso. You created user mailboxes for Contoso employees in the Adatum Exchange organization. However, they need to retain their email address SMTP domain of contoso.com. What should you do to meet these requirements?

- ■ You must create a new accepted domain of contoso.com and create an email address policy that configures Contoso employees with contoso.com SMTP domain email addresses.
- ☐ You must create a new AD forest for contoso.com and install another Exchange Server organization with a contoso.com domain.
- ☐ You must edit each user's mailbox and add a contoso.com email address.
- ☐ You must create a forest trust with the Contoso messaging infrastructure.

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Multiple choice

Item 14. You are a messaging administrator for Adatum Corporation. Your organization has hundreds of meeting rooms worldwide and there is always a challenge when multiple teams want to use a meeting room that is already booked. You need to propose a solution that provides calendar information about room availability and the ability to automatically book a room if it's available. Which of the following solutions should you propose to meet these requirements?

- ☐ Create a contact for each room. Each time some users want to book a room, they should send a meeting request to the appropriate contact.
- ■ Create a room mailbox for each room. Each time some users want to book a room, they should send a meeting request to the appropriate room mailbox.
- ☐ Create a shared mailbox for each room. Each time some users want to book a room, they should send a meeting request to the appropriate shared mailbox.
- ☐ Create a mail user for each room. Each time some users want to book a room, they should send a meeting request to the appropriate mail user.

Multiple choice

Item 15. Which of the following scenarios applies to linked mailboxes?

- ☐ Excellent for team or department collaboration where additional resources such as a OneNote or OneDrive is needed. Because it provides hooks into multiple Microsoft 365 services, it can be used as a tool to aid adoption and drive business value.
- ☐ Typically created for temporary employees or external vendors who need to access resources in the organization but use their own email address which is located outside your organization.
- ■ When it's created, a disabled user account is created in the Exchange organization, and a user account from a trusted forest is given access to the mailbox.
- ☐ Team, department, or project mailboxes to centrally store, respond to, or manage messages. Organizations often use this to provide one specific mailbox for dedicated group of people such as sales, help desk, or general information requests.

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Module 9 Managing Public Folders

Planning the Public Folder Hierarchy

Lesson Introduction

Exchange Server and Exchange Online fully support public folders. Public folders are designed for shared access and provide an easy and effective way to collect, organize, and share information with other people in your workgroup or organization. Public folders help organize content in a deep hierarchy that's easy to browse.

Public folders can also be used as an archiving method for distribution groups. When you mail-enable a public folder and add it as a member of the distribution group, email sent to the group is automatically added to the public folder for later reference.

In this lesson, you will learn about public folders in Exchange, review the planning considerations for deploying public folders, and discuss alternatives to public folders.

After this lesson, you will be able to:

- Describe public folders in Exchange
- Plan a public folder hierarchy
- Plan public folder mailboxes
- Explain public folder quotas
- Evaluate alternatives to public folders

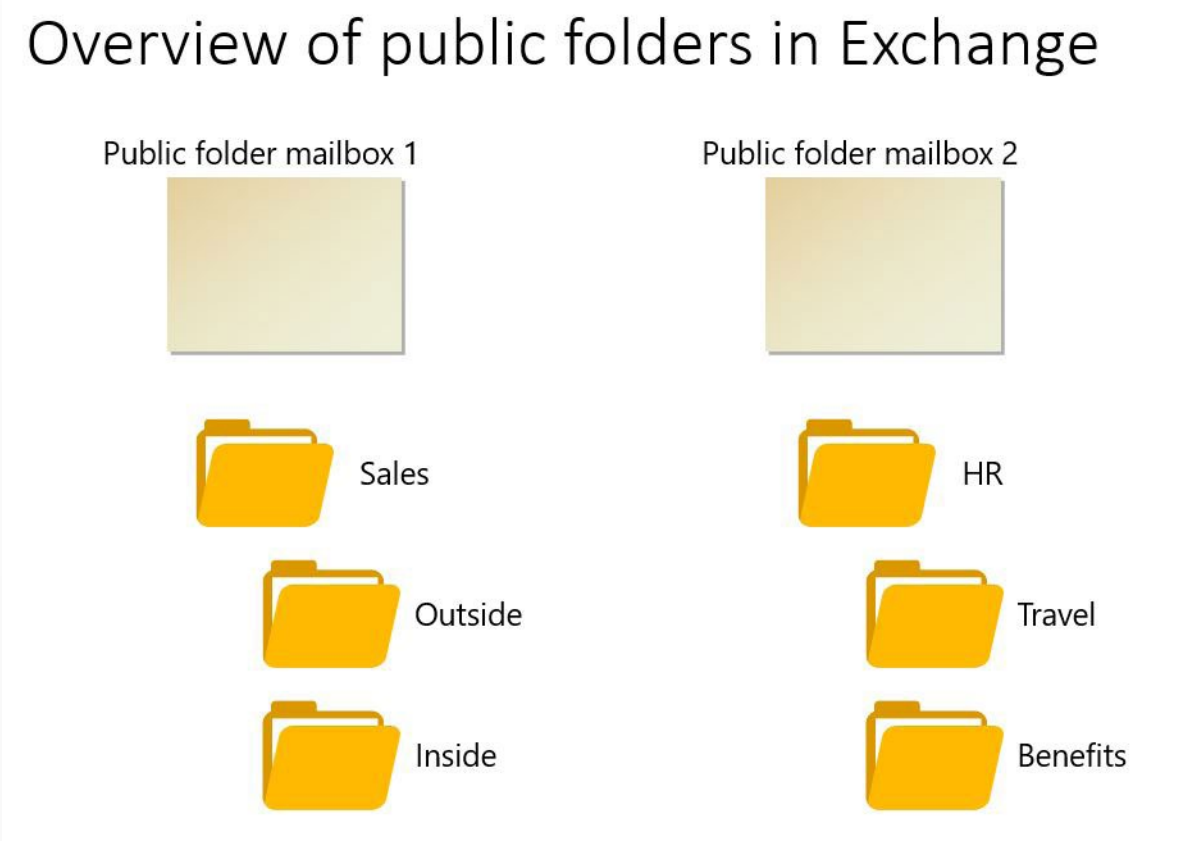
Overview of public folders in Exchange

Public folders are best suited for sharing access to data among a group of users. The following scenarios represent efficient uses of public folders:

- **Store email data for a project or distribution list.** Imagine that you have a Technical Support department, and that the department uses a distribution list for customers to send email messages for support. The distribution list can have a public folder as a member. All email messages sent to the

Technical Support department would be viewable and searchable in Outlook, which provides a good way for the team to access relevant support information.

- **Store publicly available information for all employees.** You can store information such as human resources (HR) information for all employees. The information is easily available, and users can access the information from Outlook on the web. If you're a small organization and do not have or use SharePoint Server, public folders often are a compelling option in lieu of SharePoint because of the ease in which public folders can be deployed and the positive user experience they generate.



As seen in the diagram, every public folder is associated to a Public folder mailbox. The diagram includes two Public Folder mailboxes, where the Sales department's public folder is in Public Folder mailbox 1, and Human Resources (HR) department's public folder is located in Public folder Mailbox 2. A public folder cannot span across two public folder mailboxes; thus, the size of all public folders in a public folder mailbox together is always limited to the total size of the public folder mailbox.

Some public folders can be locked down to a specific group of users or a department, such as the Sales department using Public Folder permissions. Other folders can be open to all employees, such as folders containing HR, travel, and benefits information.

Public Folder usage limitations

It's important to know the limitations of public folders so that you can opt for the right solution for collaboration. You should consider the following limitations associated with public folders:

- **Public folders are not designed for data archiving.** Archiving data in public folders impacts your mailbox database limits and settings, along with your backups. Instead, as you would with email data, you should archive data using in-place archiving in Exchange.

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- **Public folders do not support document versioning.** You cannot use public folders to collaborate on documents because there is a lack of document versioning. You cannot determine when somebody else is modifying data document, and you cannot provide any mechanism for workflow or rollback to an earlier version. Because of this limitation, you should consider SharePoint Server for document collaboration.

Public Folder characteristics

Public folders in Exchange Server have the following characteristics:

- Exchange Server 2013 or later no longer contains public folder databases. Instead, public folders use mailbox databases. Therefore, you can store the public folder mailboxes in mailbox databases that are part of a DAG to improve availability.
- Exchange Server stores public folders in special mailboxes that reside in mailbox databases. One mailbox, named the primary hierarchy mailbox, is the writable copy of the public folder hierarchy. The other type of mailbox is the secondary hierarchy mailbox, and it contains public folder content and a read-only copy of the public folder hierarchy.
- Public folders now support In-Place eDiscovery and Hold. You can search public folders in your organization, and you can place a hold on all public folders in a public folder mailbox. At the time of writing this course, you must place a hold on ALL public folders in a public folder mailbox; you cannot place a hold on a single public folder in a mailbox.
- Exchange Server 2013 or later does not support coexistence with legacy public folders located on Exchange 2007 or 2010 servers. Therefore, you must move and/or migrate all public folders stored on previous versions of Exchange.

Planning the public folder hierarchy

The public folder hierarchy is stored in a public folder mailbox. The public folder that holds the hierarchy information is known as the primary hierarchy mailbox, and it is the first public folder mailbox that you create. It is also the only copy that can be modified and written to. Other hierarchy mailboxes are secondary hierarchy mailboxes. While these mailboxes have public folder content, they only have a read-only copy of the hierarchy.

Microsoft has simplified the public folders implementation in Exchange Server. Although there are fewer considerations when planning, you must still consider some important areas when you plan your public folder hierarchy. You should plan your public folders based on the following information:

- **High availability.** If access to data in public folders is critical to your organization, you should plan to have the primary public folder mailbox stored on a database that is part of a DAG. You should also place secondary public folder mailboxes in a DAG. Using high availability helps prevent a situation where the public folder master hierarchy becomes unavailable, which prevents users from writing data to public folders.
- **Geographically dispersed public folders.** If you have multiple office locations, you need to examine how users at each location use public folder data, and then locate public folders closest to the users. For example, imagine that you have an office in Los Angeles and another in New York. You store all public folder mailboxes on mailbox servers in Los Angeles. Users in New York report poor public folder performance. By knowing in advance which users use which public folders, you can plan to locate the public folder mailboxes closest to the users. In this scenario, because the public folder users are in New York, you should locate public folders in New York.

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- **Public folder hierarchy location.** Update the **DefaultPublicFolderMailbox** property on users' mailboxes to improve performance. If you opt to deploy public folders in multiple geographic locations, you can use the **DefaultPublicFolderMailbox** property to ensure that users get the hierarchy from their closest public folder mailbox. If you do not use this property, users may get the hierarchy over a WAN link, which degrades efficiency and performance.
- **Heavily used public folders.** Improve performance by excluding heavily used public folders from providing the hierarchy. Each public folder can provide the public folder hierarchy to clients. However, heavily used public folders should be excluded from doing so to minimize the possibility of poor performance.

The following table identifies guidelines that you should consider when designing the public folder hierarchy.

Guideline	Reason
Create a hierarchical structure with logical and consistent groupings that are easy for users to explore and access.	A public folder hierarchy is typically organized according to a company's business model, so that each top-level folder represents one department within the company.
Use a consistent and logical naming scheme for public folders.	Users should be able to identify the contents of a public folder from the public folder name.
Create a public folder hierarchy that enables you to delegate administrative tasks.	By assigning the appropriate permissions at the top-level folders, you can allow users to perform tasks such as adding permissions or adding and removing folders within their department's top-level public folder.
Create a public folder hierarchy that can simplify administrative processes.	You can manage public folder settings such as permissions, folder size, and replication. Whenever possible, group public folders that require the same configuration under a top-level folder, so that you can apply the required settings to all the folders in the hierarchy simultaneously.

The following table identifies the limitations of public folders.

Configuration item	Supported limit inExchange Server	Supported limit inExchange Online
Number of public folder mailboxes	1,000	1,000
Total number of public folders	1,000,000	500,000
Max. size of single public folder	10 GB	10 GB
Max. size of single public folder mailbox	100 GB	100 GB (with E3/E5 plan)
Maximum number of public folders that can be migrated from Exchange 2010	500,000	500,000

Additional reading. For more information, especially on updated public folder limitations, see the following article on **Limits for public folders**[1].

1 https://aka.ms/AA3ramu

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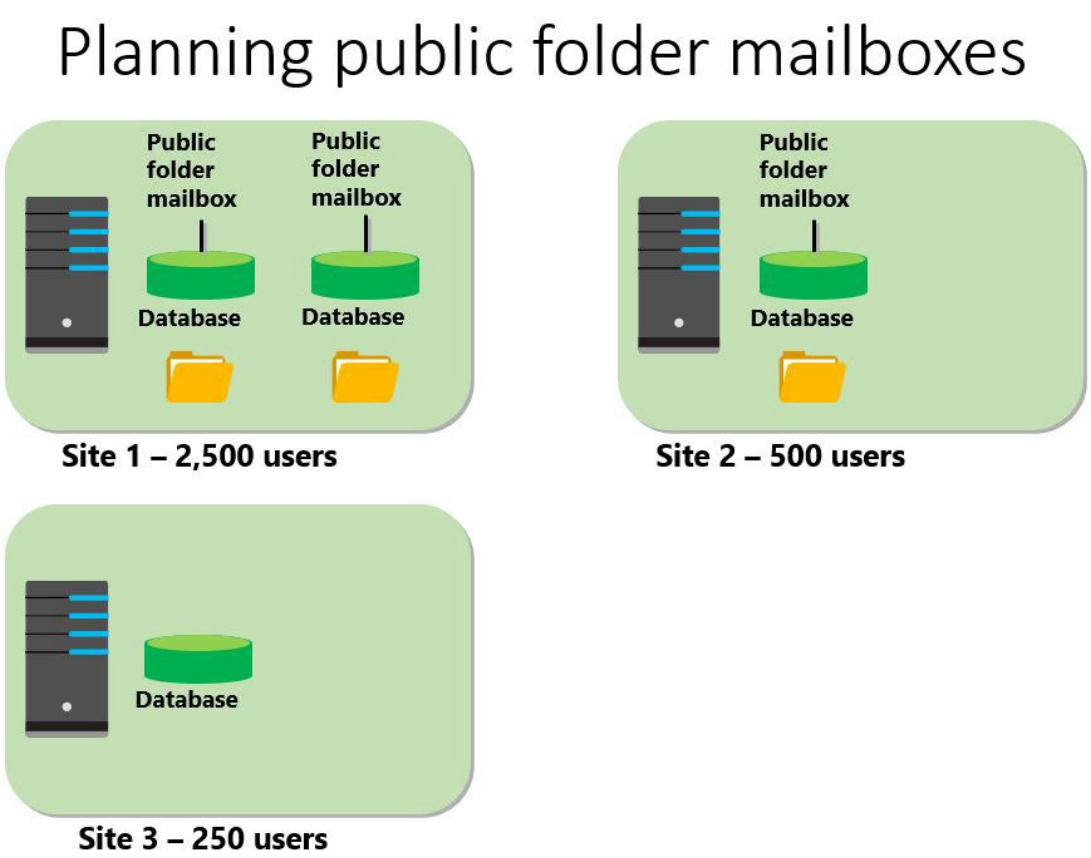
Planning public folder mailboxes

When planning for public-folder mailboxes, you must consider the following features:

- You need multiple public folder mailboxes when you have a large amount (47 GB or more) of public folder data.
- You need multiple public folder mailboxes when you have more than 2,000 concurrent user logons.
- You can place public folder mailboxes in a DAG for redundancy.
- The maximum number of supported public folder mailboxes is 1,000, but only 100 can serve as hierarchy mailboxes.
- You must monitor public folder mailbox sizes. It is a good practice to split public folder mailboxes when they reach or exceed 60 percent of the maximum mailbox size of 100 GB. This improves performance and minimizes the chance of users reporting poor performance.
- When you use multiple public folder mailboxes, you should separate the public folder mailboxes to use separate mailbox databases and separate storage. This maximizes performance and reduces impact during an outage.
- You should place your public folder mailboxes closest to the users of the public folders under that mailbox. For example, if your Accounting department is in Ireland, and those users use public folders for accounting-related data, you should consider having a public folder mailbox in Ireland to provide the best public folder performance for the Accounting department. In a decentralized model where you have Exchange servers and public folder mailboxes in multiple locations, your environment is more complex and requires more administrative overhead to manage. An alternative is to centralize your Exchange environment so that users go over the wide area network (WAN) to get to data. While this simplifies the administration, the overall user experience will be degraded based on the performance and latency of the WAN.

The following diagram depicts a public folder mailbox planning example.

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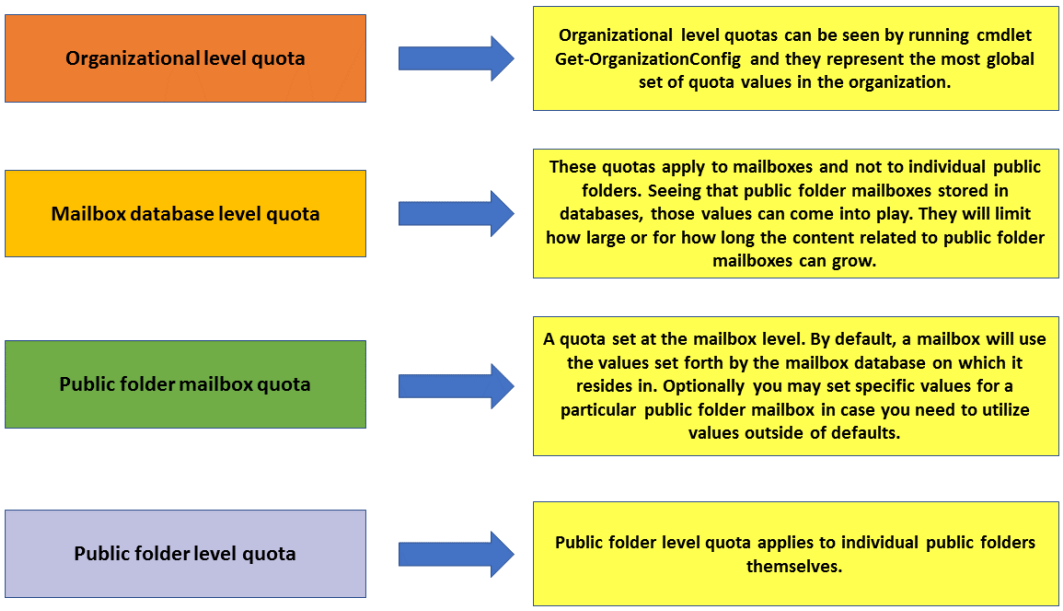
The diagram includes the following planning assumptions:

- **Site 1.** There are 2,500 users in Site 1. They use public folders regularly. When you have more than 2,000 concurrent logons, you need more than one public-folder mailbox. Therefore, you need two public-folder mailboxes. To increase performance, the public-folder mailboxes separate mailbox databases, each of which is on different storage.
- **Site 2.** There are 500 users in Site 2. They are heavy users of public folders. Site 2 could use the WAN to work with public folders, but that degrades the user experience. For sites with heavy public-folder users, you should have a public-folder mailbox and database at the site.
- **Site 3.** There are 250 users in Site 3. These users rarely use public folders. If they need to use public folders, they can go over the WAN to Site 1 or Site 2. While the user experience is degraded, it is adequate when public-folder usage is minimal.

Planning public folder quotas

Planning and setting quotas have always been important for Exchange administrators and is equally important when it comes to deployment of public folders. The following diagram provides an illustration of the types of quotas that can impact public folders that are available for Exchange Server and Exchange Online.

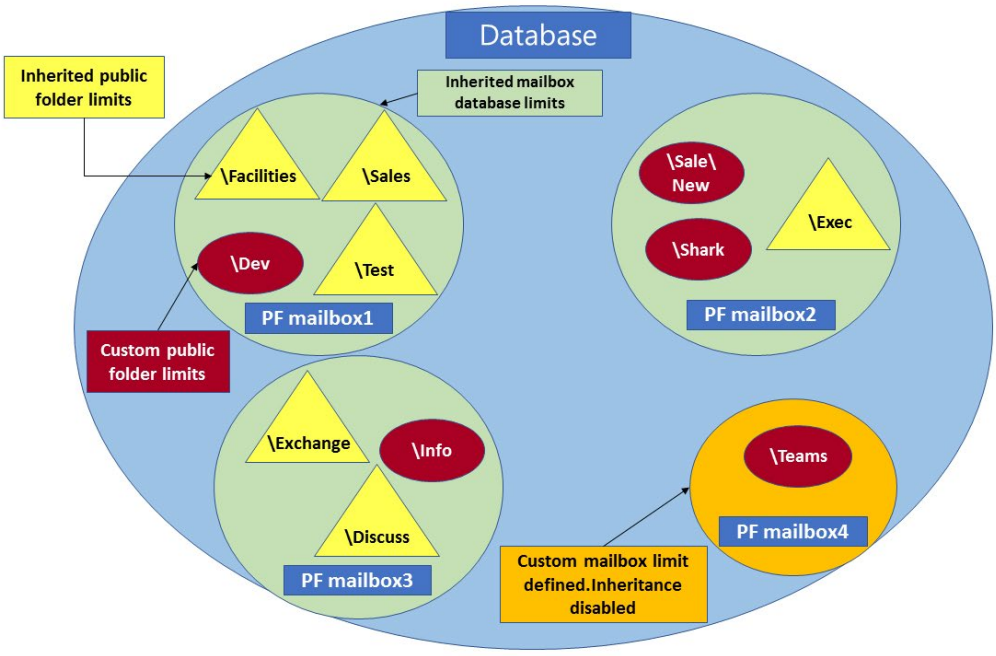
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A specific public folder limit (whether they are defined by the organization or explicitly on the public folder) can never exceed the limits being applied to a public folder mailbox containing it. For example, you should never set a public folder limit of 30 GB if the underlying public folder mailbox has a quota of 15 GB.

Important: Your goal should be that all public folders contained within a single public folder mailbox do not add up to a limit greater than the public folder mailbox they are contained in. To keep track of the size of the mailboxes you can use the method discussed earlier in this lesson.

To following diagram was created as an attempt to visualize how various quota settings relate:



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As shown in the diagram, the mailbox database named Database contains four of the organization's public folder mailboxes. Mailboxes 1 through 3 utilize the mailbox database size limits through inheritance, while Mailbox 4 uses a non-inherited, explicitly defined custom value. Each public folder mailbox has one or more public folders within them. Six public folders (the triangles) are using the organizational defined limits, and five public folders (the ovals) are all using different custom values.

Additional reading. For more information, see the following article on **Understanding modern public folder quotas**[2].

Evaluating alternatives to public folders

Microsoft continues to support public folders and will continue to do so as long as it supports Exchange Server. However, over the years Microsoft has developed new technology that has built on the foundation of public folders, and in several cases, offers significant advantages over public folders. Messaging administrator should explore alternatives to public folders before they consider creating additional ones.

The alternatives often mentioned to replace Public Folders include:

- Microsoft 365 Groups (only if you run a Hybrid or Exchange Online environment)
- SharePoint Server
- Shared mailboxes

Microsoft 365 Groups as a Public Folder replacement

Microsoft 365 Groups is the latest collaboration offering from Microsoft, which means there are many reasons why they would be a preferable solution over public folders, which is a much older technology. In Outlook, for example, Groups can replace mail-enabled public folders altogether. Compiling a list of every scenario in which Microsoft 365 Groups works better than public folders is impossible, but several scenarios stand out, including:

- **Collaboration over email**. Groups in Outlook has a dedicated **Conversations** space that stores all the emails and lets users collaborate over them. The group can even be set up to receive messages from people outside the group or from outside the organization. If you're currently using mail-enabled public folders to store project-related discussions, for example, or purchase orders that need to be viewed by a team of people, using groups would be an improvement. Groups are also better for situations when you simply want to broadcast information to a set of users.
- **Collaboration over documents**. In Outlook, Groups has a dedicated **Files** tab that displays all files from the group's SharePoint team site, as well as from mail attachments. You get one view of all the files, so you don't have to go searching for them like you would in public folders. Co-authoring also becomes easier. If you're using public folders for storing files meant to be consumed by multiple people, consider migrating to Groups.
- **Shared calendar**. Upon creation every group gets a shared calendar. Any member of the group can create events on that calendar. When you mark a group as a favorite, that group's calendar can be displayed alongside your personal calendar. You can also subscribe to a group's events, in which case events created in that group appear in your personal calendar. If you're using public folders to host calendars for your team, such as a schedule or a timetable, Groups would be an improved experience.
- **Simplified permissions**. When you assign users to a group, they immediately get the permissions they need, whereas with public folders you need to manually assign the proper permissions. Members can be added as "owners" or "members." Owners have full rights in the group, including the ability to

2 https://aka.ms/AA3rhtm

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perform group management tasks. Members can also create content and edit files like owners, but members cannot delete content that they have not created. If the public folders' permissions model is too overwhelming for you and you want something simple and quick, Microsoft 365 Groups is the way to go.

- **Mobile and Web presence**. Public folders can't be accessed through mobile devices, and they have a limited set of functionality on the Web. Microsoft 365 Groups, on the other hand, is accessible through Outlook mobile apps and has a richer set of features on the Web. If your team is on the move and requires mobile access, then you should be using Microsoft 365 Groups.
- **Access to a wide range of Microsoft 365 apps**. When you create a group, you unlock access to a wide range of apps from the Microsoft 365 suite. You get a SharePoint team site for storing files and a plan on Planner to track your tasks. Microsoft 365 Groups is the membership service that combines elements of the entire Microsoft 365 suite.

You should be aware of a few major differences that you'll notice after leaving the public folders experience. The two primary differences include:

- **Folder hierarchy.** While public folders are often used to organize content in deep rooted hierarchy, Office 365 Groups has a flat structure. All emails in the group reside in the Conversations space and all the documents go into the **Files** tab. Also, you can't create sub-folders in Microsoft 365 groups.
- **Granular permission roles.** While public folders have a variety of permission roles, Microsoft 365 Groups only provides two: owner and member.

If you decide to switch to Microsoft 365 Groups, you can use a process known as batch migration to move your email and calendar content from your existing public folders to Groups. The specific steps for running a batch migration depends on which version of Exchange currently hosts your public folder hierarchy.

Sharepoint as a Public Folder replacement

Another alternative to public folders is SharePoint Server. SharePoint Server has advanced features and a web-based platform for collaboration.

Some of the commonly used features in SharePoint Server include:

- **Document libraries.** Document libraries store documents that users can check in and check out, and the documents are tracked with version control.
- **Discussion groups.** Users can use the discussion-groups feature as a forum for communication, similar to using postings in a public folder.
- **Shared calendars.** Users can use the shared calendars feature instead of shared calendars in a public folder.
- **Contacts.** Users can link the Contacts feature with Office Outlook to provide a shared location for creating contacts.

You also can integrate SharePoint Server with Exchange Server to provide meeting workspaces. Meeting workspaces are created as a site to support a meeting and are created automatically as part of the meeting request. You can use the meeting workspace to store documents related to the meeting and to conduct project discussions.

Shared mailboxes as a Public Folder replacement

A shared mailbox is a mailbox that multiple designated users can access to read and send email messages and to share a common calendar. Shared mailboxes can provide a generic email address (such as

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info@contoso.com or sales@contoso.com) that customers can use to inquire about your company. It serves best as a common contact mailbox, such as a support team email or sales representatives general address.

Shared mailboxes are often found in company departments that have their own shared mailbox with a small group of users that have access to them.

When comparing shared mailboxes and public folders, both can store Outlook type items such as messages and calendar events, and both can receive emails (if public folders are mail enabled). However, public folders are considered a companywide collaboration tool which can be accessed by anyone in your organization; whereas a shared mailbox is designed to serve small- to medium-sized groups.

Additional reading. For more information, see the following article on **Collaboration in Exchange Online**[3].

3 https://aka.ms/AA3ravf

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Implementing and Managing Public Folders

Lesson Introduction

In this lesson, you will learn how to implement and manage public folder mailboxes and public folders. Implementing public folders also requires a solid understanding of public folder permissions as well as how to create and manage mail-enabled public folders.

After this lesson, you will be able to:

- Describe considerations for implementing Public Folders
- Deploy public folder mailboxes
- Know how to create additional public folders
- Manage public folder permissions
- Create and manage mail-enabled public folders

Considerations for Implementing Public Folders

Because of the new architecture for public folders in Exchange Server 2013 and later, your planning process for implementing public folders will differ from the process you used with previous versions of Exchange Server (2010 or before). You should consider the following differences between the different versions of public folders:

- In Exchange Server 2010, organizations with Exchange Servers in multiple locations often configured public folder replication to ensure that the public folder contents were available in each location. Starting in Exchange Server 2013, a public folder can only exist in a single public folder mailbox. If your organization has multiple locations, you will need to plan the location of the public folder contents to optimize user access.
- Planning the distribution of public folder contents may be complicated in organizations with a large amount of data in public folders. Exchange Server or Exchange Online (E3 or E5 plans) currently has a maximum mailbox size of 100 GB, so if your organization has more than 100 GB of data in public folders, you must create multiple public folder mailboxes and distribute the public folder contents across the mailboxes. Even if you have less than 100 GB of data in public folders, you should either decrease the mailbox size or distribute the public folder contents across geographic regions so that the contents are in the same location as the users who access the public folder contents.
- You can create up to 1,000 public folder mailboxes in Exchange Server or Exchange Online, but you are limited to a total of 50 TB public folder storage in Exchange Online.
- Generally, public folder access has not changed for users. Users will still use their Outlook clients to access public folders. If they have the required permissions, they will still be able to create new public folders and configure public folder permissions in their Outlook client. Technically, public folders in mailboxes are the same as public folders in older versions of Exchange Server. The storage of the public folders is different from an administration point of view, but that change is not visible to the users.
- It is recommended that you locate the primary hierarchy mailbox in a mailbox database with multiple mailbox copies in a DAG. If the primary hierarchy mailbox is not available, users can still read public folder contents, but they cannot make any changes to the public folders.

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Additional reading. For more information, see the following article on **Use favorite public folders in Outlook on the web**[4].

When deploying public folder mailboxes in your organization, you should take into account the following considerations to ensure they are protected and readily available to the clients:

- Public folder mailboxes, both hierarchy and content, should be protected by placing them in databases protected by multiple copies in a Database Availability Group. By doing this, mailboxes will remain protected in case of any outage and be available to end users.
- There should be no public folders hosted within the primary public folder mailbox. This way the primary public folder mailbox can focus on its job of replicating hierarchy changes to other public folder mailboxes.
- You should exclude the primary public folder mailbox from serving hierarchy to clients. This is done by setting the IsExcludedFromServingHierachy parameter to $True.
- The recommendation is to activate database copies hosting public folder mailboxes on mailbox servers which are geographically located close to the client location.
- The general recommendation is having one public folder hierarchy mailbox for every 2000 users accessing public folders. Additional hierarchy-only public folder mailboxes can always be created to divide the connection load among users to ensure that the 2000 connection limit is not reached.
- Plan and create more secondary hierarchy public folder mailboxes and content mailboxes to ensure there are fewer than 2000 active and concurrent connections to public folders and that they are close to the user's geographical location to ensure there are no latency issues and users have good experience.
- You can make use of up to 1,000 public folder mailboxes. Of the 1,000 public folder mailboxes, 100 of them can be used for hierarchy (or 99 once you exclude the primary PF) and the remaining 900 can be used for public folder data storage.

Additional reading. For more information, see the following article on **Modern public folder deployment best practices**[5].

Deploying public folder mailboxes

Public folders are available in all versions of Exchange Server. Many organizations use public folders as a means of sharing information between groups of users. With public folders, multiple users can access a public folder in Outlook.

Beginning with Exchange Server 2013, the underlying architecture for public folders changed entirely without significantly changing the user experience with public folders.

In Exchange Server:

- In Exchange Server 2010 and earlier, public folders are stored in a separate public folder database.
- In Exchange Server 2013 and later, public folders are stored in a special type of mailbox called a public folder mailbox. The public folder mailboxes are stored in regular mailbox databases. The public folder mailbox stores the public folder hierarchy as well as the public folder contents.

4 https://docs.microsoft.com/en-us/exchange/collaboration-exo/public-folders/use-favorite-public-folders
5 https://aka.ms/AA3r6r9

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- There are two types of public folder mailboxes: the primary hierarchy mailbox and secondary hierarchy mailboxes. Both types of mailboxes can contain public folder content but differ in the following way:
  - **Primary hierarchy mailbox**. The primary hierarchy mailbox is the one writable copy of the public folder hierarchy. The public folder hierarchy is copied to all other public folder mailboxes, but these will be read-only copies.
  - **Secondary hierarchy mailboxes**. Secondary hierarchy mailboxes contain public folder content as well and a read-only copy of the public folder hierarchy.
- Public folder mailboxes can be stored in mailbox databases that are part of a Database Availability Group (DAG). Before Exchange 2013, public folders used a public folder replication process to enable redundancy. By storing the public folder mailboxes in a mailbox database that is part of a DAG, you can provide high availability for the public folder deployment using the same mechanism as the one used for providing high availability for mailboxes.
- Public folders are spread across multiple public folder mailboxes. Before Exchange 2013, you could replicate public folder contents to public folder databases located in different locations to enhance client access to public folder contents. In Exchange Server 2013 or later, you can create public folders and store the public folders in different mailboxes, which can be located on Mailbox servers in different locations.

Note: In Exchange 2010, you can have multiple copies of the public folder contents, and public folder replication is a multi-master process. In Exchange Server 2013 and later, you can only store the public folder contents in one mailbox, and all clients must access that mailbox to see the public folder contents. If you put the public folder mailbox in a database that is part of a DAG, the mailbox is highly available, but all clients still only access the mailbox in the active copy of the database.

Public folders are accessed by clients only for Outlook 2010 or later. In Exchange Server or Exchange Online, you can only add public folders located on Exchange 2013 or later as Favorites in Outlook on the web to access them.

To implement public folders in Exchange, you first must create a primary public folder hierarchy mailbox. You can create a public folder mailbox by using EAC or Windows PowerShell.

The first (or primary) public folder mailbox contains the only writeable copy of the public folder hierarchy. After creating the primary public folder mailbox, you can create additional public folder mailboxes as secondary public folder mailboxes. These additional public folder mailboxes will contain a read-only versions of the public folder hierarchy, which is referred to as the "Secondary Hierarchy".

Additional reading. For more information, see the following article on **Create a public folder mailbox in Exchange Server**[6].

Creating public folders

After creating the first public folder mailbox, you can begin creating public folders. Public folders can be created by the Exchange administrator or by the users. As an Exchange administrator, you can create public folders either by using the Windows PowerShell cmdlet **New-PublicFolder**, or in the Exchange Admin Center (EAC) as shown in the following screenshot.

As a user, you need **Create subfolders permission** to create public folders in Outlook. A special requirement exists for top-level or root public folders - you require root permissions assigned by the Exchange administrator so that a user can create a root public folder.

6 https://aka.ms/AA3ri3r

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By default, all public folders are created in the primary public folder mailbox. If you create additional public folder mailboxes, you can create public folders in the additional public folder mailbox only if you create the public folder using the **New-PublicFolder** cmdlet with the **-Mailbox** parameter.

Additional reading. For more information, see the following article on **Create a public folder**[7].

Managing public folder permissions

Many organizations also configure public folder client permissions or access rights for users. These permissions are used to restrict the actions users can perform in the public folder. Client permissions have not changed compared to previous versions of Exchange Server. You can assign permissions to users by using roles such as Owner or Publishing Editor. You also can assign custom permissions by using a variety of the access rights.

In Exchange Server or Exchange Online, administrative permissions to manage public folders are enabled through Role-Based Access Control (RBAC). To grant users permission to manage public folders, you must add them to the Public Folder Management role group.

Public Folder client permissions

The following list describes public folder client permissions:

- **ReadItems.** The user can read items in the specified public folder.
- **CreateItems.** The user can create items in the specified public folder and send email messages to the public folder if it is mail-enabled.
- **EditOwnedItems.** The user can edit the items that the user owns in the specified public folder.
- **DeleteOwnedItems.** The user can delete items that the user owns in the specified public folder.
- **EditAllItems.** The user can edit all items in the specified public folder.
- **DeleteAllItems.** The user can delete all items in the specified public folder.
- **CreateSubfolders.** The user can create subfolders in the specified public folder.
- **FolderOwner.** The user is the owner of the specified public folder. The user can view and move the public folder, create subfolders, and set permissions for the folder. The user cannot read, edit, delete, or create items.
- **FolderContact.** The user is the contact for the specified public folder.
- **FolderVisible.** The user can view the specified public folder but cannot read or edit items in the folder.

Public Folder roles with permissions

The following table identifies the predefined public folder roles and the permissions that are included in each role.

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Role	Create Items	Read Items	Create Sub-folders	Folder Visible	Edit All-Items	Edit/ Delete Own-edItems	Delete All-Items	Folder Owner	Folder Contact
None				X					
Owner	X	X	X	X	X	X	X	X	X
Pub-lishingE-ditor	X	X	X	X	X	X	X		
Editor	X	X		X	X	X	X		
Pub-lishin-gAuthor	X	X	X	X		X	X		
Author	X	X		X		X			
Non-Ed-itingAu-thor	X	X		X					
Review-er		X		X					
Contrib-utor	X			X					

You can configure client permissions or public folder roles in the EAC by selecting the public folder and then clicking **Manage** under **Folder permissions**. You can also configure client permissions by accessing the public folder properties in Outlook, or by using the **Add-PublicFolderClientPermission** and **Remove-PublicFolderClientPermission** cmdlets.

When you create a public folder, it automatically inherits the same client permissions as the parent public folder. When you change the permissions on a parent folder, you have the option to enforce the permission change for all subfolders. The default permissions assigned to new root folders are **Author** for authenticated users and **None** for anonymous users.

Creating and managing mail-enabled public folders

Mail-enabling a public folder assigns an SMTP address to it and displays it (by default) in the GAL. Users can then post messages to the public folder by sending email messages to it. When a public folder is mail-enabled, you can configure additional settings on the public folder such as email addresses and mail quotas.

You can mail-enable a public folder using the **Enable-MailPublicFolder** cmdlet in Windows PowerShell. To mail-enable a public folder in the Exchange Admin Center, you should perform the following steps:

1. Navigate to **public folders** > **public folders**.
2. In the list view, select the public folder that you want to mail-enable.
3. In the details pane, under **Mail settings**, select **Enable**.
4. A warning box displays asking if you are sure you want to enable or disable email for the public folder. Select **Yes** to continue.

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5. You can then adjust the automatically created e-mail address for that public folder on the email address tab of the Public Folder properties.

Note: To ensure that users from the Internet can send e-mail messages to a mail-enabled public folder, that public folder needs to have at least the **CreateItems** public folder client permission granted to the **Anonymous** account.

Once you mail-enable a public folder, that public folder gets additional attributes such as custom attribute 1-12 or message delivery restrictions applied to it.

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Lab8-Ex1-2-Implement Public Folders

Lab8-Ex1-Create Public Folders

Holly Dickson wants to continue with her examination of messaging in Microsoft 365, and more specifically, with the use of public folders. In this lab, you will log into the Exchange admin center for Exchange Online and create a public folder mailbox and a public folder in Microsoft 365.

See your VM lab environment for the lab instructions.

Lab8-Ex2-Manage Public Folders

In the prior exercise, Holly Dickson logged into the Exchange admin center for Exchange Online on LON-CL1 and created a Microsoft 365 public folder mailbox and a public folder. In this exercise, she will manage settings and permissions for each of these objects using the Exchange Online EAC.

See your VM lab environment for the lab instructions.

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Troubleshooting Public Folders

Lesson Introduction

Once you enabled Public Folders for your Exchange organization, users will begin to create Public Folders on demand. As a messaging administrator, you must be aware of potential issues that typically arise with public folders, and you must be ready to handle them. This lesson examines how to monitor and troubleshoot Public Folder-related issues.

After this lesson, you will be able to:

- Monitor public folders
- Troubleshoot public folders
- Troubleshoot public folder access

Monitoring public folders

Monitoring your public folders is a task that must be performed in Windows PowerShell. The following table identifies the cmdlets provided by Exchange that can be used to monitor public folders.

Cmdlet	Description	What to monitor
Get-PublicFolderStatistics	Displays statistical information about all public folders, such as folder size and last logon time.	Identify the largest public folders and monitor their growth rate. Consider moving large public folders to their own public folder mailbox before storage quotas are exceeded.
Get-PublicFolderItemStatistics	Displays information about items within a specified public folder. The information includes the subject, last modification time, last access time, creation time, attachments, message size, and type of item.	Allows you to identify very large public folder items.
Get-PublicFolderMailboxDiagnostics	Displays event-level information about a public folder mailbox.	This information can be used to troubleshoot public folder hierarchy synchronization or item access issues such as whether an item in a public folder is not shown to every user.
Update-PublicFolderMailbox	Used to update the hierarchy for public folders.	This triggers a manual update of the public folder hierarchy.
Get-MailboxStatistics	Displays mailbox size information such as TotalItemSize or TotalDeletedItemSize (also available as Script).	You should monitor the size of public folder mailboxes and see which public folders might be getting more use than the others (as they might need to be moved to a different mailbox).

The following table identifies the additional areas related to public folders that you should monitor.

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Tool/Script	Monitoring area	Description
Windows Event Log	Public Folder mailbox size storage warnings	Warning events 10024 and 1077 will be logged on respective mailbox server (where the mailbox database hosting those public folder mailboxes is active) and the event ID will contain the GUID of the public folder mailbox.
Get-PublicFolderMailboxSize.ps1	Displays the size of a public folder mailbox	Can be used to identify storage quotas for public folder mailboxes.

Move a public folder to a different public folder mailbox

If the content of a public folder mailbox begins to exceed your mailbox quotas, you may need to move public folders to a different public folder mailbox.

Note: You can perform this task only on Exchange Server, as Exchange Online takes care of your public folder mailbox management automatically. For this reason, you cannot run any script or cmdlet to move public folders between Public Folder mailboxes in Exchange Online.

The following table identifies the cmdlets and PowerShell scripts (located in the Exchange Scripts folder by default located in **C:\Program Files\Microsoft\Exchange Server\V15\scripts**) that can be used to move Public Folders between Public Folder mailboxes or merge Public Folder mailboxes.

PowerShell Cmldet/Script	Description	When to use
New-PublicFolderMoveRequest	Moves one or more public folders to another public folder mailbox.	Use this cmdlet if you want to move one or more public folders to a different public folder mailbox.
Move-PublicFolderBranch.ps1	Moves a public folder including all subfolders to another public folder mailbox.	Use this script to move public folder branches to another public folder mailbox.
Split-PublicFolderMailbox.ps1	Moves content to other public folder mailboxes when 60% of quota reached.	You can schedule to run this script automatically so your public folder mailboxes never exceed their storage limit.
Merge-PublicFolderMailbox.ps1	Merges the content of public folder mailboxes with another.	Use this script when you want to reduce the number of public folder mailboxes in your environment.

Additional reading. For more information, see the following article on **Move a public folder to a different public folder mailbox**[8].

Troubleshooting public folders

Public Folders have been around for a long time; it's a very solid technology that's in its mature development phase. Even though issues with Public Folders are quite rare, messaging administrators must still

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know how to address potential issues. The following table identifies some of the key issues that may occur and how to troubleshoot them.

Public Folder Issues	Reason	How to troubleshoot
A Public Folder item is not available and cannot be seen in the public folder.	The item may have been deleted.	Try to use the **Recover Deleted Items option** in Outlook to recover the item to the Public Folder.
A Public Folder item is available in the folder, but it cannot be opened for all users.	The Public Folder item may be corrupt in the public Folder mailbox.	Try to run a repair on the public folder mailbox. This will trigger either a removal of the corrupt item or repair it and make it available.
A Public Folder is not available for one user (the folder is not shown in the Public Folder hierarchy).	The user may not have appropriate permissions to view the Public Folder. The Public Folder Hierarchy of the user may not be in-sync.	Check the permissions of the public folder to verify the user can see the folder. Identify the public folder mailbox the user is connected to and synchronize the public folder hierarchy. You can use the **Update-PublicFolderMailbox** cmdlet to start a manual synchronization.
A Public Folder is not available for all users.	The Public Folder may be corrupted or be deleted. The Public Folder permissions may be corrupted or removed.	Verify that the Public Folder exists in the EAC or with PowerShell. If it exists, run the **Get-PublicFolderItemStatistics** cmdlet to verify the items included in the folder. Also check the default permissions on that folder for any issues.
The Public Folders Hierarchy is not available for a user.	The Outlook client maynot be able to retrieve a Public Folder mailbox through Autodiscover. The user has a public folder mailbox assigned to his or her mailbox, and the mailbox has been deleted.	Troubleshoot Public Folder access on the client side as described in the next topic. Use the **Set-Mailbox** cmdlet to remove the DefaultPublicFolderMailbox setting.

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Public Folder Issues	Reason	How to troubleshoot
The Public Folders Hierarchy is not available for all users but can be viewed in the EAC or PowerShell.	The Public Folders may have been locked or incorrectly configured in the Exchange Organization Config.	Use the following command to identify any "false" statements: **Get-OrganizationConfig | fl *public*** For example, **PublicFolders-LockedForMigration** might be $true, or **PublicFoldersEnabled** might be Remote, even though the folders are stored locally on Exchange.
A Public Folder mailbox was deleted and is no longer available.	There may be various reasons why a public folder mailbox might disappear, the most common one is a failed mailbox move, or a soft-delete.	For more information on how to troubleshoot this issue, see the following article on **Restore public folders and public folder mailboxes from failed moves** *(https://aka.ms/AA3r71d)*.

Troubleshooting public folder access

The ability to estimate the number of connections being made to public folder mailboxes is very helpful as deployment guidance for public folders partially revolves around connection counts. The logging methods that are currently available will not reveal the names of the public folders that clients are connecting to, but they will contain information about public folder mailboxes being accessed by clients.

Depending on what information you're looking to gather, there are several types of logs you can consider.

- **Autodiscover logs.** These logs identify which public folder mailboxes Outlook clients get sent to during the Autodiscover process.
- **Outlook Web App logs.** These logs identify which default public folder mailboxes Outlook Web App clients get sent to during connection process. The default public folder mailboxes could either be the ones which are provided randomly to the requesting OWA client, or they could be a hard-coded default public folder mailbox assigned to a specific user's mailbox.
- **RPC Client Access logs & MAPI Client Access on Microsoft Exchange 2013 Mailbox Servers.** These logs identify which public folder mailboxes on a specific mailbox server the users are connecting to using RPC/HTTP and MAPI/HTTP protocols. These logs can be used with Microsoft Exchange 2013.
- **MAPI/HTTP logs in Microsoft Exchange 2016/19 Servers.** These logs identify which public folder mailboxes your MAPI/HTTP clients are connecting to. These logs should only be used with Microsoft Exchange 2016/19.

There are many different logs in Exchange Sever showing similar information. Depending on the protocol used by your organization, you may make decisions on the log type to parse. Autodiscover logs provide a combined view of what public folder mailboxes users are at least trying to access one time.

- If you have content-only public folder mailboxes in your environment that are excluded from the server hierarchy and not directly assigned to users as their default, you can determine if some are never accessed and may contain content worthy of purging.

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- If you need a more granular view of the world along with the ability to generate some sort of heat map, you may choose to go with more protocol-specific logs. These logs provide data about each time the client creates a new connection to a public folder mailbox, and they allow you to determine whether the client learned about it through Autodiscover, or if it's being used far more heavily by many users over time. The options are varied and up to you to choose based on your need.

Additional reading. For more information, see the following article on **Modern public folders logging and when to use it**[9].

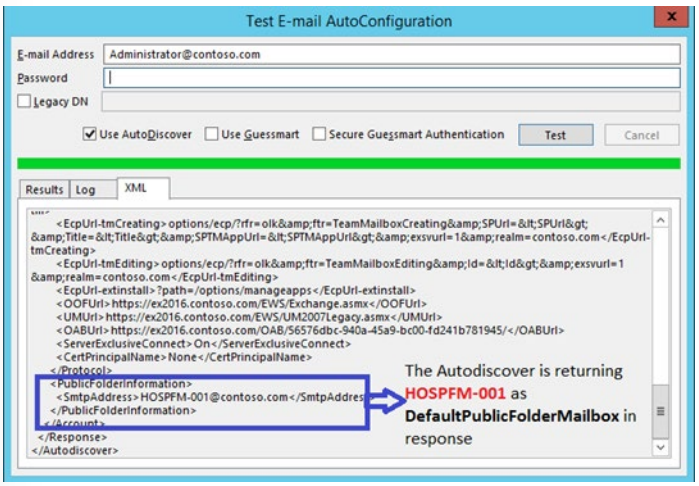
Identifying the Public Folder mailbox the client is connecting to

You can also use the **Test E-mail AutoConfiguration** tool from within the Outlook client to perform a single user test. This identifies which public folder mailbox is being returned to a single end-user by the Autodiscover service for hierarchy connections.

To start the **Test E-mail AutoConfiguration** tool, you should perform the following steps:

1. Start Outlook.
2. Hold down the Ctrl key, right-click the Outlook icon in the notification area, and then click **Test Email AutoConfiguration**.
3. Verify that the correct email address is in the **E-mail Address** box. You do not have provide a password if you're running a test for the currently logged in user. If you're testing a different user account than the one currently logged into the machine, then you must provide both the email address and password for that account.
4. In the **Test Email AutoConfiguration** window, click to clear the **Use Guessmart** check box and the **Secure Guessmart Authentication** check box.
5. Click to select the **Use Autodiscover** check box, and then click **Test**.

The following screenshot displays an excerpt from the XML file that was gathered from running the **Test E-mail AutoConfiguration** tool.



As you can see in this screenshot, the administrator@contoso.com user account used the public folder mailbox HOSPFM-001@contoso.com to make a hierarchy connection.

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Module Review

Module 9 Review

Checkbox

Item 1. You are a messaging administrator for Contoso Ltd., and you're currently working on a project to implement public folders throughout the company. Which of the following scenarios represents efficient uses for public folders for your organization? (Select 2)

- ☐ Store Word and Excel documents for simultaneous editing.
- ☐ Store email data for a project or distribution list.
- ☐ Store publicly available information for all employees.
- ☐ Store messaging data for users as personal archives.

Multiple choice

Item 2. You are working as a messaging administrator for Contoso Ltd., and you're planning to deploy public folders. What is the first step you must complete before you can deploy public folders for your users?

- ☐ Deploy a public folder database
- ☐ Deploy a public folder mailbox
- ☐ Deploy a user mailbox
- ☐ Assign permissions to users to use public folders

Checkbox

Item 3. Which of the following are characteristics of public folder mailboxes? (Select 3)

- ☐ You need multiple public folder mailboxes when you have more than 2,000 concurrent user logons.
- ☐ You cannot place public folder mailboxes in a DAG.
- ☐ You need public folder databases to create public folder mailboxes.
- ☐ You can place public folder mailboxes in a DAG for redundancy.
- ☐ The maximum number of supported public folder mailboxes is 1,000, but only 100 can serve as hierarchy mailboxes.
- ☐ Public folder mailboxes have a maximum size of 5 GB, which is the total size of all public folders inside of it.

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Multiple choice

Item 4. You work as a messaging administrator for Contoso Ltd. and you are responsible for public folders in your organization. Several users have reported poor performance when accessing certain public folders. You perform an analysis and discover that several heavily used public folders are all in the same public folder mailbox. What is the recommended action to fix your users performance issues?

- ☐ Move the heavily used public folders to separate mailboxes.
- ☐ Create writeable copies of the heavily used public folders in other public folder mailboxes.
- ☐ Move the heavily used public folders to user mailboxes that provide better performance.
- ☐ Move the affected public folder mailbox to a different database to increase performance.

Multiple choice

Item 5. You work as a messaging administrator for Contoso Ltd. and you're responsible for public folders. A user reports that saving new elements to a certain public folder is no longer possible, even after the public folder quota was recently raised. What is the most likely to reason why the public folder appears to be full, even if the quota for the folder was raised?

- ☐ The quota of the public folder mailbox is reached.
- ☐ Quotas for public folders cannot be raised and the user must move all content to a new public folder.
- ☐ The user tried to save unsupported file types to the public folder.
- ☐ The overall number of public folders in the public folder mailbox exceeded the maximum number allowed

Multiple choice

Item 6. You work as a messaging administrator for Contoso Ltd. You have been asked to evaluate different solutions for user collaboration in your organization. One of your requirements is that users should collaborate on editing documents in a centralized place. Which of the following solutions should you recommend to meet this requirement?

- ☐ Using public folders and public folder mailboxes per document type.
- ☐ Use SharePoint for collaboration on documents.
- ☐ Use Shared Mailboxes for collaboration on documents.
- ☐ Use Resource Mailboxes for collaboration.

Multiple choice

Item 7. Fill in the blank. Public folder mailboxes, including both hierarchy and content, should be protected by placing them in databases protected in a _____.

- ☐ Single server
- ☐ Backup set
- ☐ Hardware load-balancer
- ☐ Database Availability Group

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Multiple choice

Item 8. You are a messaging administrator for Contoso Ltd. You have been asked to plan the performance requirements for deploying public folders. You have a total of 10,000 users in your organization that will use different public folders, but none of the public folders will have more than 1,000 users accessing the public folders simultaneously. What is the maximum number of public folders that should be created per Public folder Mailbox?

- ☐ 10 heavily used public folders per public folder mailbox.
- ☐ 2 heavily used public folders per public folder mailbox.
- ☐ 40 heavily used public folders per public folder mailbox.
- ☐ As many as you like, there is no recommendation.

Multiple choice

Item 9. You are a messaging administrator for Contoso Ltd., and you're responsible for public folders in your organization. A user reports that everyone in his department can still see the public folders, but nobody can make any changes. Which of the following conditions would have resulted in users no longer being able to write changes to public folders?

- ☐ All public folder mailboxes are no longer available.
- ☐ The clients within the department have just received updates and need to restart their computers to get write access to their public folders again.
- ☐ The secondary public folder hierarchy mailbox is not available.
- ☐ The primary public folder hierarchy mailbox is not available.

Multiple choice

Item 10. Fill in the blank. Many organizations configure public folder _____ to restrict the actions users can perform in the public folder.

- ☐ Role-based Access Control
- ☐ Reviewers
- ☐ Anonymous users
- ☐ Client permissions

Multiple choice

Item 11. To ensure that users can send e-mail messages from the Internet to a mail-enabled public folder, that public folder needs to have which public folder client permission granted to the Anonymous account?

- ☐ Owner
- ☐ CreateItems
- ☐ FolderContact
- ☐ DeleteAllItems

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Multiple choice

Item 12. Fill in the blank. If the content of a public folder mailbox begins to exceed your mailbox quotas, you may need to move public folders to a different _____.

- ☐ Public folder database
- ☐ Public folder mailbox
- ☐ User mailbox
- ☐ Resource mailbox

Multiple choice

Item 13. You work as a messaging administrator for Contoso Ltd. Siegfried reports to you that he can no longer access the public folder hierarchy. Other users do not report any issues with accessing the hierarchy. Where do you start troubleshooting?

- ☐ Try to retrieve to retrieve the public folder mailbox with Autodiscover on the user's client.
- ☐ Move the public folder hierarchy mailbox to another database.
- ☐ Reinstall the Office suite and Outlook on the user's client.
- ☐ Check the remaining free disk space on the user's client.

Multiple choice

Item 14. Which tool can you use from within the Outlook client to perform a discover and connectivity single user test?

- ☐ File explorer
- ☐ Nslookup
- ☐ Get Help
- ☐ Test E-mail AutoConfiguration

Checkbox

Item 15. You are a messaging administrator for Contoso Ltd. After a recent upgrade, Contoso only employs Exchange 2019 servers. Which log files will you search on your servers to investigate connectivity issues with your organization's public folders? (Select 3)

- ☐ Autodiscover logs
- ☐ Event logs
- ☐ Outlook Web App logs
- ☐ MAPI/HTTP logs
- ☐ Administrator audit logs
- ☐ RPC Client Access logs
- ☐ Mailbox audit logs

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Answers

Checkbox

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- ☐ Public folder mailboxes have a maximum size of 5 GB, which is the total size of all public folders inside of it.

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■ Move the heavily used public folders to separate mailboxes.

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☐ Backup set

☐ Hardware load-balancer

■ Database Availability Group

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- ☐ Event logs
- ■ Outlook Web App logs
- ■ MAPI/HTTP logs
- ☐ Administrator audit logs
- ☐ RPC Client Access logs
- ☐ Mailbox audit logs

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Module 10 Planning a Hybrid Environment

Exchange Hybrid Deployment Requirements

Lesson Introduction

One of the main advantages of running an on-premises Exchange Server environment is that you're able to connect your environment in a seamless way to Exchange Online and Microsoft 365. This connection is called a hybrid deployment.

A hybrid deployment allows you to move mailboxes between your on-premises and cloud environments and share a common global address list (GAL). Your users have the same experience, no matter if their mailbox is located on a local Exchange server or in Exchange Online.

Even if you don't plan to migrate mailboxes from your on-premises environment to the cloud, a hybrid deployment still enables you to:

- provision cloud-based online archives for your Exchange users.
- benefit from Exchange Online Protection (EOP) and Advanced Threat Protection (ATP) functionality.
- allow users of services such as Microsoft Teams to benefit from rich, on-premises integration.
- provide secure, remote access using Microsoft 365 technologies through Modern Hybrid Authentication.

In this lesson you will examine the requirements necessary to implement a hybrid deployment, and you will learn about the features and components that are required when implementing a hybrid deployment.

After this lesson, you will be able to:

- Describe connection options that are available for connecting on-premises Exchange to Microsoft 365
- List and describe components of a hybrid deployment
- Describe Azure Active Directory Connect (Azure AD Connect)
- Identify Microsoft 365 identity options for Exchange hybrid
- Compare Exchange delegated federation vs. OAuth

Connection Options for Exchange to Microsoft 365

It's essential that you consider your coexistence requirements and functionality when choosing a migration strategy from on-premises Exchange to Microsoft 365. It's also important that you understand the consequences of your migration choice.

The following table identifies the different connection options and their coexistence impact.

Migration Option	Coexistence Impact
IMAP Migration	No coexistence available; this is basic mail data migration.
Cutover migration	Not a real coexistence; this is just a migration of the user account information and mailbox data. No mail routing coexistence is required; you just need to switch your DNS. Transition from on-premises Exchange Server to Exchange Online once the user accounts are migrated.
Staged Migration	Requires directory synchronization to be in place and mail routing between on-premises Exchange and Exchange Online.
Hybrid deployment	Requires Exchange 2010 or later to be installed in your on-premises Exchange environment. Requires directory synchronization using Azure AD Connect and mail routing between on-premises Exchange and Exchange Online. The SMTP email entry point can be on-premises or online. Minimal and Full Hybrid enable native mailbox migrations. Full Hybrid also enables rich coexistence, including free/busy data, secure SMTP, and Public Folder migrations.

Connection Options to Microsoft 365 Requirements

All connection options have requirements for which mail servers can be used to establish a connection. For a full hybrid feature set, you need the latest Exchange Server version to be running in your Exchange environment. If you run older Exchange Server versions such as Exchange Server 2000 or earlier or other messaging systems, the connection features are limited to e-mail-only migration support for IMAP migrations.

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	IMAP Migration	Cutover Migration	Staged Migration	Hybrid
Exchange 2007	X	X	X	X*
Exchange 2010 - 2019	X	X		X
Notes/Domino	X			
GroupWise	X			
Other	X			

* Exchange 2007 requires at least an Exchange 2010 Server for Hybrid

Components of a Hybrid Deployment

To plan for a hybrid deployment, you first need to understand the components of a hybrid deployment and the prerequisites for configuring a hybrid deployment.

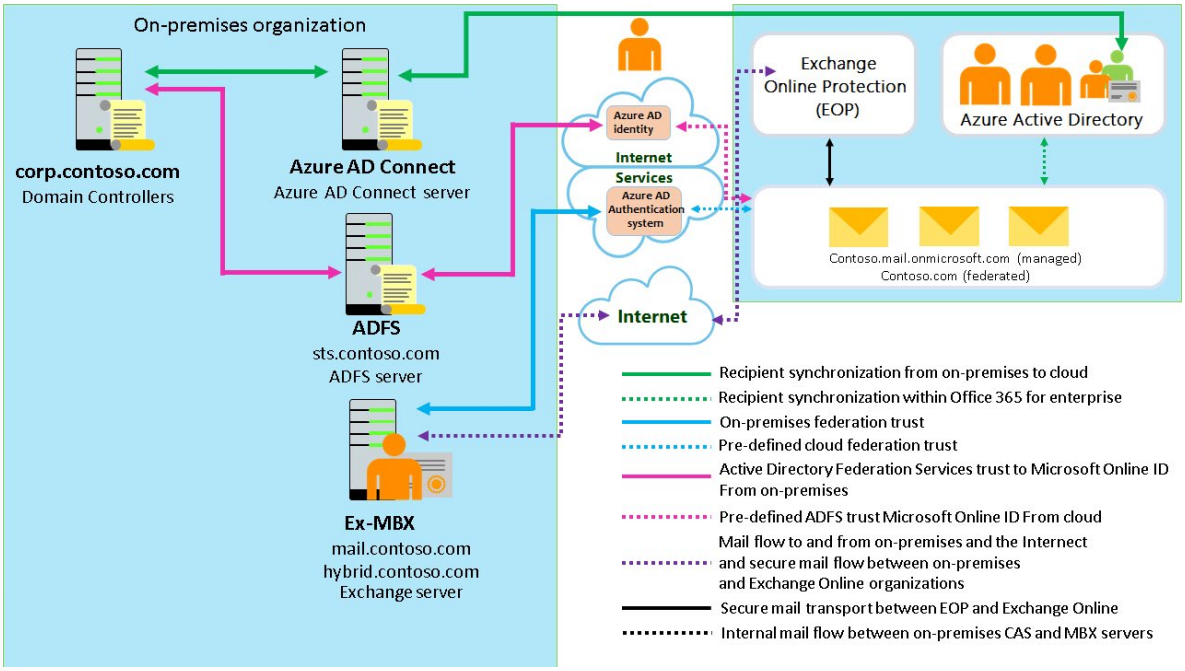
A hybrid deployment has the following components:

- **Exchange Server**. For the latest hybrid functionality such as attachment storage in OneDrive for Business, you need a version of Exchange Server on-premises within mainstream support, such as Exchange Server 2016 or 2019. Most organizations running Exchange 2010 or higher will however have all they need to enable Hybrid to support a migration.
- **Microsoft 365**. The Microsoft 365 service includes Exchange Online as a part of its subscription service. If you plan a hybrid deployment, you must create and configure a cloud-based Exchange Online organization.
- **Azure Active Directory (Azure AD).** The Azure AD service provides authentication and authorization for Microsoft 365 and other offerings within the Microsoft 365 suite.
- **Azure Active Directory Connect (Azure AD Connect)**. To support the unified GAL, synchronization of AD DS replicates information about mail-enabled objects from on-premises Active Directory to Microsoft 365. You must deploy Azure Active Directory Connect on an on-premises server before you can configure a hybrid deployment.
- **Exchange Online Protection**. By default, the Exchange Online Protection (EOP) service is included in all Microsoft 365 subscriptions for enterprise tenants. EOP works with on-premises Exchange servers to help secure message delivery between Exchange Server on-premises and Exchange Online. Depending on your configuration, EOP may also route incoming email from external recipients for Exchange Server on-premises and Exchange Online.
- **Microsoft 365 Hybrid Configuration Wizard**. You download the Microsoft 365 Hybrid Configuration Wizard (HCW) from the on-premises Exchange admin center and then run it on the on-premises Exchange Server. The HCW configures a hybrid deployment between the on-premises Exchange Server and Exchange Online.

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- **Azure AD authentication system**. The Azure Active Directory authentication system is a free, Azure-based service that acts as the trust broker between on-premises Exchange Server and Exchange Online for Federated Sharing (Free/Busy). If you are configuring a full hybrid deployment, you must have a federation trust with Azure AD authentication system. For minimal Hybrid deployments, this is not necessary.
- **Open authentication (OAuth)**. OAuth is an open standard for authentication and used in Exchange since version 2013 to authenticate servers especially between SharePoint and Skype for Business. In a Full Hybrid deployment OAuth is used to authenticate between your Exchange on-premises servers, and Exchange Online; for example, when you use In-place eDiscovery between premises.
- **Active Directory Federation Services (AD FS)**. Optionally, you can use AD FS to configure single sign-on (SSO) and centralize user management. However, consider that most organizations will find Azure AD Connect password hash sync, seamless sign-on and pass through authentication options (combined with Microsoft 365 functionality like Conditional Access) negate the need for AD FS.

The following diagram shows how the different components of a hybrid deployment interact with each other:



As you see in the diagram, the following data connections are required for a hybrid deployment:

- Azure AD Connect synchronizes your local Active Directory information to Azure Active Directory, which is used by Microsoft 365. This is a one-way synchronization of users, groups and contacts, with limited write-back of Exchange related attributes to support co-existence and Exchange Online protection.
- The AD FS server authenticates your Microsoft Online ID against your local domain controllers and provides access to your mailbox located on Exchange Online.
- Exchange federated delegation is used in a way where your on-premises Exchange Servers have a federation trust to the Azure AD authentication system; the same trust applies to Exchange Online. Using this trust, Exchange Online can access date such as free/busy information from your on-premises Exchange server, and vice-versa. Of course, the necessary configurations as mentioned earlier need to be in place.

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- Lastly, your on-premises Exchange Servers include a bi-directional connection to Exchange Online Protection which is the default anti-virus and anti-malware e-mail scanning functionality in Microsoft 365. This allows you to send e-mails back and forth.

Overview of Azure AD Connect

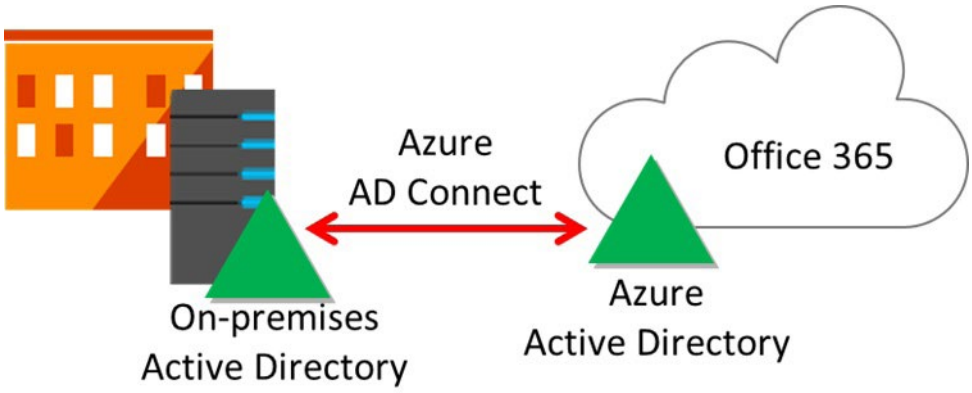
The Azure AD Connect tool, formerly known as DirSync, is the primary supported directory synchronization tool for Microsoft 365. Azure AD Connect is designed to operate as a software-based tool that you configure once, after which it automatically runs in the background without user interaction. For Microsoft 365, the purpose of the tool is to allow coexistence between your on-premises Active Directory Domain Services (AD DS) environment and Microsoft 365 in the cloud.

Azure Active Directory Connect is made up of three parts:

- Synchronization services
- The optional authentication piece that consists of either Pass-Through Authentication (PTA), Password hash synchronization (PHS), Seamless Single Sign On (SSO), or Active Directory Federation Services)
- The monitoring piece, which uses Azure AD Connect Health

Using Azure AD Connect for Directory Synchronization

Azure AD Connect is the primary supported tool to provide synchronization of objects between on-premises Active Directory and Azure Active Directory.



Azure AD Connect also supports the following scenarios:

- Synchronization of multiple Active Directory forest environments
- Synchronization for multiple Exchange organizations to one Office 365 tenant
- Synchronization of third-party on-premises LDAP directory services

When using Azure AD Connect for directory synchronization:

- New user, group, and contact objects in on-premises AD DS are added to Microsoft 365; however, Microsoft 365 licenses are not automatically assigned to these objects.
- Attributes of existing user, group, or contact objects that are modified in on-premises AD DS are modified in Microsoft 365; however, not all on-premises AD DS attributes are synchronized to Microsoft 365.

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- Existing user, group, and contact objects that are deleted from on-premises AD DS are deleted from Microsoft 365.
- Existing user objects that are disabled on-premises are disabled in Microsoft 365; however, licenses are not automatically unassigned.

In a cloud-only Microsoft 365 deployment, all Azure AD objects are created in the cloud, and they must be edited using cloud-based tools (either using the Office 365 portal or the Microsoft 365 admin center, or by using Windows PowerShell cmdlets). In this scenario, Azure AD is referred to as the source of authority for all Active Directory objects.

Azure AD requires a single source of authority for every object. Therefore, it's important to understand that when you deploy Azure AD Connect for Active Directory synchronization, you are mastering objects from within your on-premises AD DS by using tools such as Active Directory Users and Computers or Windows PowerShell. In this scenario, the source of authority is the on-premises AD DS. This is known as establishing Hybrid Identity.

After the first synchronization cycle has completed, the source of authority is transferred from the cloud to the on-premises AD DS. All subsequent changes to cloud objects (except for licensing) are mastered from the on-premises AD DS tools. The corresponding cloud objects are read-only, and Microsoft 365 administrators cannot edit cloud objects if the source of authority is on-premises.

Email address matching is used to identify the on-premises AD DS user object that relates to a Microsoft 365 user:

- If a user exists in your on-premises AD DS and no matching user exists in Microsoft 365, Azure AD Connect will create a new Microsoft 365 user with the same email address as the on-premises account.
- If a user already exists in both your on-premises AD DS and in Microsoft 365 and these objects have the same email address, then during the first synchronization these objects will become joined, or linked.

By synchronizing user, contact, and group objects, Azure AD Connect provides a unified GAL experience between Microsoft 365 and the on-premises AD DS or Exchange environment. Using the filtering features in Azure AD Connect, objects hidden from the GAL on-premises are also hidden from the GAL in Exchange Online.

Azure AD Connect supports the following mailbox migration scenarios:

- Where Microsoft 365 will replace on-premises Exchange Server.
- Where there are both on-premises and Exchange Online mailboxes in a hybrid deployment scenario.

In hybrid scenarios, Azure AD Connect allows mail routing between on-premises and Microsoft 365 with a shared SMTP domain namespace or to use archive mailboxes stored in Exchange Online for on-premises mailboxes. This scenario allows on-premises/cloud coexistence for both Exchange Server 2016 or later, Skype for Business Server 2015, or Lync Server 2013.

Additional reading. For more information, see the following article on **Hybrid identity documentation**[1].

1 https://aka.ms/AA3r8pd

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Microsoft 365 identity options for Exchange Hybrid

Azure Active Directory (Azure AD) is an online instance similar to Active Directory. Azure AD provides authentication and authorization for Microsoft 365 and Microsoft 365 services, such as Windows 10 devices registered with Azure AD, Intune, and Dynamics 365, and it's often used as a single sign on service for third party services.

Azure AD can authenticate cloud-only identities, or identities synchronized with an on-premises directory, with optional password synchronization. Alternatively, you can enable user authentication with on-premises user accounts through Active Directory Federation Services (AD FS) or other single sign-on (SSO) providers.

Authentication options for Microsoft 365 identities into one of three main categories:

- **Cloud-only.** Cloud-only identities are exactly as the name suggests; the user identity only exists in the cloud, so all password management and policy control are done through Azure AD.
- **Directory synchronization with optional Password hash synchronization (PHS).** With directory synchronization, you set up a directory synchronization server or appliance that provides one-way synchronization of users, groups, and attributes from on-premises Active Directory to Azure AD. In the case of Exchange hybrid environments, there is also synchronization of certain attributes from online to on-premises.

  However, it's important to remember that even with password hash synchronization, there are still two sets of security credentials (one on-premises, and one in Azure AD); it is just that directory synchronization and password synchronization keep them aligned. If you do not synchronize passwords, your users will have a separate password in Azure AD. They still authenticate to Azure AD to access Microsoft Exchange Online and other online services.
- **Directory synchronization with Pass-through authentication (PTA) or federated with AD FS.** Both PTA and federation delegate authentication control to your directory service. Therefore, users no longer authenticate against Azure AD but against your local Active Directory using a service such as Azure AD Connect pass-through authentication (PTA) or Active Directory Federation Services (AD FS).

  Consequently, with federation, when a user types user@adatum.com into the Microsoft 365 sign-in page, the user receives a message telling them that they have been redirected to their organization's sign-in page. They now enter their on-premises credentials and authenticate to the Microsoft 365 online services by using a delegated token that verifies to Office 365 that the user has been successfully authenticated by their on-premises directory service. For Pass-through authentication, the user is not redirected, but behind the scenes the Azure AD service validates the password entered against the organization's Active Directory.

Azure AD Connect pass-through authentication

It's very beneficial for users to use the same set of credentials to authenticate against both cloud and on-premises resources. Usually, this is achieved by using Azure AD Connect synchronization with a password synchronization. In scenarios where organizations want to perform all authentication on-premises, you should deploy an AD FS service and configure your Azure AD tenant in federated mode. In this scenario, each authentication request for resources on-premises or in a cloud is always directed to the AD FS server deployed locally. However, deployment and management of the locally deployed AD FS infrastructure might be too demanding and too complex for some organizations. A recent update for Azure AD Connect provided a new option to address this scenario; this new feature is called Azure AD pass-through authentication (PTA).

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Azure AD pass-through authentication helps ensure that password validation for services that rely on Azure AD is always performed against an on-premises Active Directory. If PTA fails, automatic failover to password sync is performed, if it's configured. Unlike the solution with AD FS, this solution is easy to implement and maintain.

Azure AD pass-through authentication is configured by using Azure AD Connect, and it works by using an on-premises agent that listens for external password validation requests. You can deploy this agent to one or more servers to provide high availability. There is no need to deploy this server to your network perimeter, as all communication is outbound only. A server that runs the agent for pass-through authentication should be joined to the Active Directory domain where users are located.

Achieving Single Sign On (SSO) without AD FS

For both password hash synchronization and pass-through authentication, organizations can take advantage of Seamless SSO. Seamless SSO allows domain-joined users to by-pass prompts for passwords and authenticate using traditional intranet methods.

Seamless SSO works by registering a computer account for Kerberos delegation within the local Active Directory. This allows a domain-joined user to utilize Windows Integrated Authentication when attempting to access Office 365 services.

However, Seamless SSO is not always necessary. For all scenarios, including cloud-only synchronization, if the Windows 10 device is Azure AD joined, users are signed in to Azure AD at PC sign-on and will benefit from Single Sign On to Microsoft 365 services. This scenario is typical for a full Microsoft 365 deployment that takes advantage of Intune for PC deployment and management.

User account provisioning

User objects in many directories such as the Active Directory include hundreds of parameters that define authorities, and the provisioning system can control these details in your environment. For example, your Human Resource directory can be the basis for any new user, and once synchronized, the user can be readily identified.

On-premises, cloud, and hybrid are the main deployment scenarios in which you must consider capabilities for user account management and provisioning.

- **On-Premises.** With Active Directory, you can create a scalable, secure, and manageable infrastructure for user and resource management, and provide support for directory-enabled applications such as Microsoft Exchange Server. Administrators can use access control to manage user access to shared resources for security purposes.

  In Active Directory, access control is administered at the object level by setting different levels of access, or permissions, to objects, such as Full Control, Write, Read, or No Access. Access control in Active Directory defines how different users can use Active Directory objects. By default, permissions on objects in Active Directory are set to the most secure setting.

- **Cloud.** You must create an account for every user who will access a Microsoft cloud service. You can also change user accounts or delete them when you no longer need them. By default, users do not have administrator permissions, but you can optionally assign them. Within Microsoft Azure Active Directory (Azure AD), one of the major features is the ability to manage access to resources. These resources can be part of the directory, as in the case of permissions to manage objects through roles in the directory, or resources that are external to the directory, such as software as a service (SaaS) applications, Azure services, and Microsoft SharePoint sites or on-premises resources.

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At the center of the Azure AD access management solution is the security group. The resource owner (or the administrator of the directory) can assign a group to provide certain access rights to the resources they own. The members of the group will be provided access, and the resource owner can delegate the rights to manage the group's members list to someone else—such as a department manager or a help-desk administrator.

- **Hybrid.** Extend Active Directory identities into the cloud through synchronization and optional Federation Service.

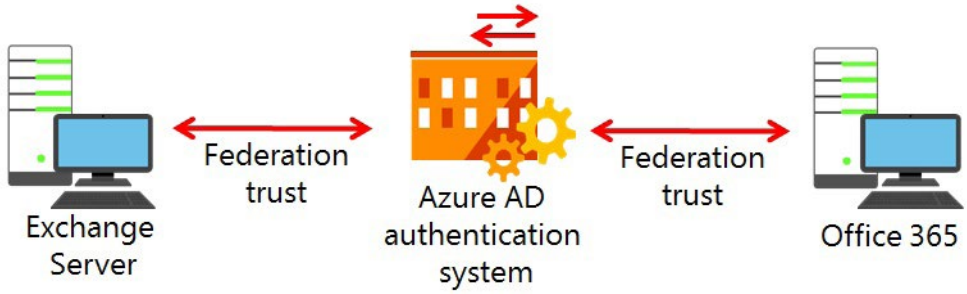
Additional reading. For more information, see the following article on **What is hybrid identity**[2].

Exchange Delegated Federation (DAuth) versus Open Authentication (OAuth)

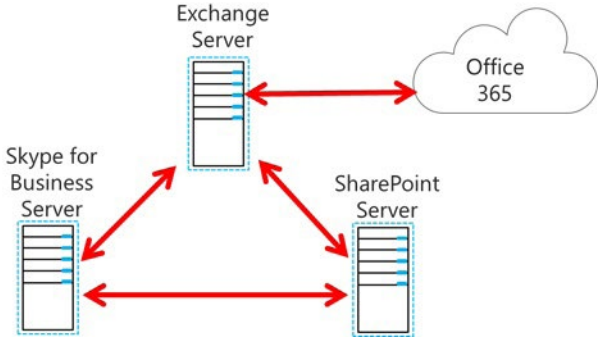
There are two federation methods in Exchange Server:

- Exchange Delegated Federation, otherwise known as Delegated Authentication (DAuth)
- Open Authentication (OAuth), which was introduced with Exchange Server 2013

DAuth uses the Azure AD authentication system for token generation. Organization relationships control which companies you share information with and allows for granular control of what features are available (free busy, mail tips, and so on).



Open Authentication (OAuth) uses Azure AD authentication server, which provides better resiliency and faster in-forest communications. It uses IntraOrgConnectors / Configuration to control what information you can share with other companies.



Both federation methods enable you to share information between two Exchange organizations, such as your local Exchange organization and Exchange Online. The biggest difference between DAuth and OAuth is the granularity of control. When using DAuth, you can configure which information to share,

2 https://aka.ms/AA3rcq2

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such as MailTips or Free/Busy. If you implement sharing when using OAuth, then you must share everything because OAuth does not provide granular control.

Delegated Federation

Exchange Federated Sharing, which is also known as Delegated Federation or Delegated Authentication (DAuth), uses standard federation technologies to allow Exchange organizations to establish trusted relationships with each other. To establish federation trust, organizations exchange certificates with public keys or with a trusted third-party, and then uses those certificates to authenticate and secure all communications between them. Exchange Federated Sharing has been available since Exchange Server 2010. Since Exchange Server 2010 SP1, you will typically use a self-signed certificate for the federation trust.

In Exchange Server, you use the Azure AD authentication system to establish the federation. The Azure AD authentication system is an identity service that runs over the Internet and works as a trust broker for Federated Sharing. To enable Federated Sharing, the organization must register with the Azure AD authentication system and then configure a Federated Sharing using an organizational relationship with another organization that also registers with the Azure AD authentication system.

Open Authentication

Open Authentication (OAuth) 2.0 is a framework that includes server-to-server authentication. OAuth is a standards-based framework that is widely used across the web services industry and within other Microsoft products such as Hotmail.

Open Authentication typically involves three components that must communicate with each other using the HTTPS protocol (Port 443):

- a trusted authorization server
- two services

The trusted authorization server, or token server, issues security tokens to the two services. These security tokens verify the authenticity of both services and ensure that user credentials and passwords do not pass between servers. Instead, these security tokens control authentication and authorization. For example, the trusted authorization server might issue security tokens that verify that users from a specific Skype for Business Server service are able to access a specific Exchange Server service and vice versa. In Skype for Business Server, the default SIP domain acts as the OAuth service.

As part of its support within the Microsoft Office family of server products (including Microsoft 365 and the on-premises versions of Exchange 2013 or later, Lync Server 2013 or later, and SharePoint 2013 or later), the OAuth framework supports on-premises and hybrid topologies. In an on-premises topology, there is no requirement to implement a trusted authorization server, as the use of partner applications establishes the trust. By creating the partner application, the server products directly swap security tokens and bypass the need for a third-party token server.

Functionality OAuth provides to Exchange Server 2016/2019

OAuth is required for some Exchange 2016/2019 related features, such as cross-premises discovery and automatic archive retention. Currently OAuth enables the following Exchange features:

- Message Rights Management (MRM) to move messages from an on-premises mailbox to an archive located in Office 365
- Exchange In-place eDiscovery, especially cross-premises searches

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- Exchange In-place Archiving
- On-premises mailboxes store attachments in OneDrive for Business

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Planning to Run the Hybrid Configuration Wizard

Lesson Introduction

Planning a hybrid configuration is crucial to preventing issues during deployment. One of the key components in a hybrid configuration is the Hybrid Configuration Wizard (HCW).

This lesson examines all planning aspects that are required before running the HCW. This includes the configuration options of the HCW, as well as the details on Organization Configuration Transfer (OCT) and the Hybrid Agent. The lesson concludes with a review of the mail flow options for a hybrid deployment.

After this lesson, you will be able to:

- Plan for Exchange Hybrid configuration
- Describe Organization Configuration Transfer
- Explain Exchange Modern Hybrid and Hybrid Agent
- Plan mail flow options for a hybrid deployment

Planning for Exchange hybrid configuration

An Exchange hybrid configuration is a connection from a single or multiple on-premises Exchange organizations to a single Exchange Online organization in a Microsoft 365 tenant. The term hybrid is used for cross-premises connectivity between local and cloud Exchange organizations.

Exchange hybrid is commonly used for mid-term to long-term coexistence, where mailboxes must reside in both on-premises and Exchange Online for compliance and security purposes. Exchange hybrid provides a smooth migration path to Microsoft 365 because you don't need to consider changing Outlook profiles when moving your users to the cloud; this is done automatically for you. A hybrid configuration offers a complete email experience, with the ability to extend your on-premises Exchange organization to Exchange Online. It provides a seamless look and feel, as if the users belong to a single Exchange organization.

The Hybrid Configuration Wizard (HCW) is the tool used for setting up a hybrid configuration between your on-premises Exchange organization and Microsoft 365. The HCW is an application and runs directly from a computer or server in your organization. You can use the HCW to configure the following two hybrid configuration options:

- Full hybrid configuration
- Minimal hybrid configuration

The following screenshot displays the HCW page that is used for selecting the hybrid configuration option you plan to deploy.

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Minimal Hybrid Configuration

Microsoft introduced the Minimal Hybrid Configuration for the following reasons:

- Because cutover and staged migration experiences are labor intensive on the client side.
- To provide a better user migration experience when a full hybrid configuration is not needed.

A minimal hybrid configuration provides a quick way to move to Exchange Online, as well as a reasonable option for both the administrator and users. Therefore, minimal hybrid configuration allows you to only perform migrations and administration in a hybrid deployment, but it does not include complex configurations such as Exchange Federation or secure email flow. The main benefit is that it allows you to have the same user migration experience as in a full hybrid configuration:

- User account credentials are synchronized to Azure AD and Exchange Online using Azure AD Connect.
- The mailbox move will be online; the client does not need to disconnect from their mail client during the mailbox move.
- The Outlook profile is automatically updated once the mailbox move is finished; therefore, it does not require any administrative intervention.
- No additional connectors will be created. The existing mail flow between your Exchange on-premises environment and Exchange Online is used.

The technical requirements for a minimal hybrid configuration are similar to a full hybrid configuration:

- Azure AD Connect for directory synchronization of your local Active Directory environment is required.
- All SMTP domains that you plan to use in Exchange Online must be registered and verified with the Microsoft 365 tenant.
- Unless you can use the Modern Hybrid (described later), you must configure your Mailbox Replication Services (MRS) Migration Endpoint using a trusted third-party SSL certificate. This is typically the URL you use to publish Exchange Web Service externally.
- You do not need any DNS TXT records for a Federated domain proof since no federation is configured in a minimal hybrid configuration.
- Your SMTP connectors do not need a certificate, nor do they need to be configured to allow TLS.

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Full Hybrid Configuration

A full hybrid configuration includes configuring the full feature set to configure the best Exchange hybrid experience for your users. You typically decide for a full hybrid configuration when you plan a longer coexistence between Exchange and Exchange Online, or if your Exchange organization is large and thus cannot migrate over a short period of time, such as a weekend.

A full hybrid configuration includes two hybrid connectivity options: Exchange Classic Hybrid and Exchange Modern Hybrid. The following table identifies the difference between the two approaches.

Hybrid Connectivity	Description	When to use
Exchange Classic Hybrid	This is the traditional hybrid approach, where Exchange Online can connect to your Exchange Servers over the Internet, and vice-versa. This requires your Exchange Servers be accessible over the Internet, and that you own an officially trusted 3rd party certificate.	Ideal if you already published your Exchange endpoints, such as OWA to the Internet.
Exchange Modern Hybrid	This installs the Hybrid Agent that takes over the communication between Exchange Online and your on-premises Exchange Servers.	You should consider this when you did not publish Exchange to the Internet, or you did not purchase official 3rd party certificates.

Note: Exchange Modern Hybrid is explained in detail in a later topic in this lesson.

A full hybrid configuration offers all functionality of a minimal hybrid configuration, as well as the following additional functionality once configured:

- Exchange Online and on-premises users can share free/busy calendar data.
- Administrators can use their familiar Exchange management tools, namely the Exchange admin center and Exchange Management Shell (or Windows PowerShell), to manage and move users to Exchange Online.
- When migrating mailboxes, Outlook profiles for users are automatically updated to the Exchange Online environment when the Exchange hybrid deployment and Autodiscover is configured properly. Administrators do not need to manually reconfigure Outlook profiles or resynchronize .OST files after moving users' mailboxes.
- Outlook on the web redirection allows users to use the well-known Outlook on the web URL to access their mailbox on the web. You must specify a target URL for your organization (for example, **https://outlook.com/owa/contoso.com**[3]).
- MailTips, internal out-of-office messages, and similar features are available to both cloud and on-premises mailboxes so the user cannot see a difference.
- Delivery reports and multi-mailbox search work with users who are on-premises and those working in Exchange Online.
- Authentication headers are preserved during cross-premises mail flow, so all mail looks and feels like it is internal to the company (for example, recipient names resolve in the global address list).

3 https://aka.ms/edx-cld222x-o365

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- If necessary, administrators can easily move mailboxes back to the on-premises Exchange environment.
- Support for archive mailboxes located in Exchange Online for on-premises users.
- Support for Public Folder coexistence with Legacy and Modern Public Folders located on Exchange on-premises.
- Cross-premises OAuth support since Exchange Server 2013 SP1.
- Cross-premises eDiscovery Search with SharePoint Online since Exchange Server 2013 SP1.
- Modern Attachments stored in OneDrive for Business Online since Exchange Server 2016
- Microsoft Teams integration for on-premises Mailboxes, including meeting and calendar integration when using OAuth and Exchange Server 2016 CU3 and higher.
- Support for Hybrid Modern Authentication with Exchange 2013 CU19 and higher, Exchange 2016 CU8 and higher and Exchange 2019, allowing Azure AD identities secured by Microsoft 365 technologies to be used for on-premises sign-in and use of the Outlook App.
- Limited Microsoft 365 groups support for on-premises mailboxes, as well as the following additional limitations:
  - On-premises users can't use links included in group message footers.
  - On-premises users can't become an administrator of a Microsoft 365 group.

As mentioned earlier, a full hybrid configuration can serve as an intermediate step to moving completely to an Exchange Online organization or to staying in long-term coexistence with your existing Exchange on-premises environment.

When should you use minimal and full hybrid configurations?

You should take into account the following considerations when determining whether to use a minimal or full hybrid configuration.

Hybrid configuration option	When to use
Minimal Hybrid Configuration	This is intended for small or medium sized customers that need a seamless migration experience for their users. Typically, you will use this in scenarios where hybrid is simply used as a migration option that does not require the enhanced features. One of the big advantages in using this migration option is that your users can keep their Outlook OST files and they don't need to resynchronize. You can also move mailboxes back to your on-premises Exchange. The minimal hybrid configuration is also an excellent solution for a quick onboarding during merger and acquisition scenarios.

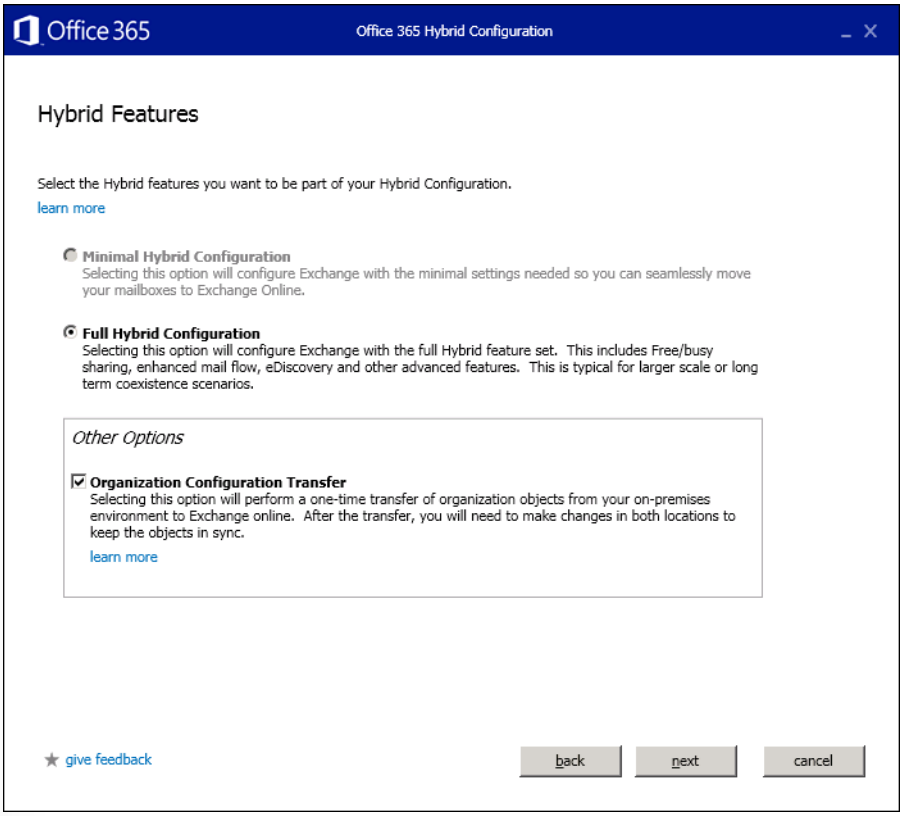
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Hybrid configuration option	When to use
Full Hybrid Configuration	This should be used if you plan a long-term coexistence between Exchange on-premises and Exchange Online that requires the most features available for your users. Organizations who require advanced coexistence functionality such as sharing of free / busy and calendar data between on-premises and Exchange Online, typically tend to implement a full hybrid configuration. In this scenario, users can collaborate as they're in the same organization. Also, and this is an important point, in a hybrid configuration you can move your mailboxes between your on-premises Exchange and Exchange Online in any direction.

Organization Configuration Transfer

In the past, after running the Hybrid Configuration Wizard (HCW) that configured a hybrid deployment for your Exchange environment, you needed to manually configure all Exchange organizational settings such as Active Sync Mailbox Policies in Exchange Online. If you did not configure these settings, the users probably experienced a difference situation once their mailbox was moved to Exchange Online.

For this reason, Microsoft added a feature to the HCW called Organization Configuration Transfer (OCT). OCT allows you to perform a one-time transfer of your organizational settings from your Exchange on-premises environment to Exchange Online.



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OCT allows organizations to transfer the following Exchange settings:

- Active Sync Mailbox Policy
- Mobile Device Mailbox Policy
- OWA Mailbox Policy
- Retention Policy
- Retention Policy Tag
- Active Sync Device Access Rule
- Active Sync Organization Settings
- Address List
- DLP Policy
- Malware Filter Policy
- Organization Config
- Policy Tip Config

OCT runs as part of HCW, and it reads your Exchange on-premises configuration as well as the settings in Exchange Online. You can decide whether or not you want to transfer each setting.



If an object with the same name already exists in both on-premises and online, you can choose to either overwrite the values of the objects in Exchange Online or keep them as is. And, just in case you overwrite the Exchange Online settings and didn't mean to, there's a rollback script to undo those changes.

For example, if you had an OWA mailbox policy on-premises that didn't exist in Exchange Online, HCW will perform a comparison between both organizations. The comparison is accomplished using the name attribute of the policy, which means if there is no OWA mailbox policy matching the name attribute, HCW will read all the configured properties on-premises, save it in the memory, and create a new one (using the New-* cmdlet) with the same properties, just like you had on-premises.

OCT supports Exchange Server 2019, 2016, 2013 and 2010 and requires the latest cumulative update or update roll-up available for the version of Exchange you have installed in the on-premises organization.

When using OCT, you must consider the following:

- OCT is a one-time transfer of your organizational settings.
- OCT cannot match all settings, so always make sure you review them.

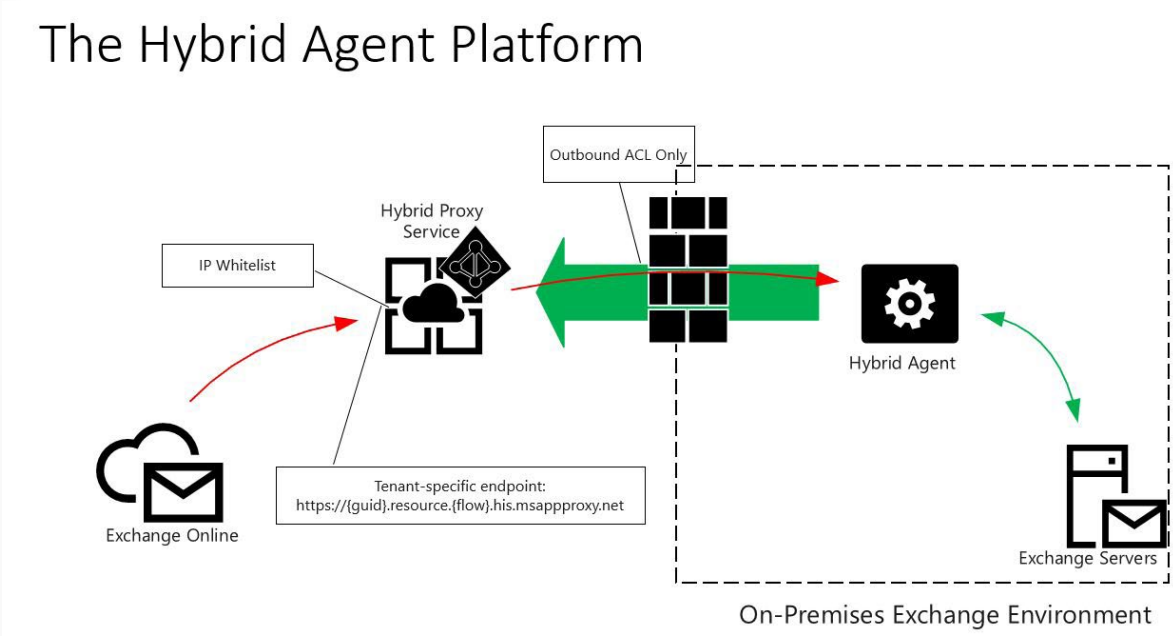
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- You can revert the OCT settings using the rollback script available in the logging folder of the HCW.

Exchange Modern Hybrid and Hybrid Agent

A core component of Exchange Modern Hybrid is the Hybrid Agent. The Hybrid Agent is based on Azure AD Application Proxy technology and will take over the communication between Exchange Online and your Exchange environment. The Agent is installed either on a stand-alone serve, or on your Exchange Servers. For failover reasons, it's recommended that you install it on multiple servers. For example, if you have four Exchange servers in your environment, it's recommended that you install the Agent on all four servers.

When the agent is started, it connects to both your Exchange Servers as well as the Hybrid Proxy Service located in your Microsoft 365 tenant. Any subsequent requests from Exchange Online that require a connection to your Exchange Server will then be communicated using the Hybrid Agent.



As the hybrid agent takes over the communication between Exchange Online and your Exchange environment, it provides the following advantages:

- **No customer DNS changes.** You do not need to change any DNS settings such as the DNS TXT record to allow Exchange Federated Delegation between Exchange Online and your environment.
- **No firewall/network changes.** You do not need to publish your Exchange Server Client Access endpoints to the Internet.
- **No certificate changes.** You do not need to purchase third-party certificates for your Exchange Servers.
- **Protect your Exchange servers.** With the hybrid agent you do not publish and allow any inbound communication from the Internet; therefore, you additionally protect your Exchange servers from attacks originating from the Internet.

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Hybrid Agent Functionality

There are two Exchange-related flows supported in private preview:

1. Free busy requests from cloud users to on prem
2. Mailbox migrations to/from cloud

Free busy requests from on-premises users to cloud users do not traverse the Hybrid Agent. These requests still require that your Exchange servers have outbound connectivity to Microsoft 365 end points.

MailTips, Message Tracking, and Multi-mailbox search do not traverse the Hybrid Agent. These Hybrid features would require the classic connectivity model where Exchange Web Services and Autodiscover are published on premises and externally available to Microsoft 365.

SMTP email traffic does not traverse the Hybrid Agent; therefore, it still requires a certificate for encrypted mail flow between Microsoft 365 and on premises.

Hybrid Agent Requirements

While Hybrid deployment using the Hybrid Agent, also known as "Modern Hybrid," has slightly different connectivity requirements, it does not mean that no outbound and inbound connectivity are required.

The Hybrid Agent is built on Azure AD Application Proxy technology, and the agent's core requirements are no different. These are summed up in the following article **Tutorial: Add an on-premises application for remote access through Application Proxy in Azure Active Directory**[4].

However, it's important to note that although the underlying technology is Azure AD Application Proxy, it is not simply an installation of the standard agent on the servers. Microsoft has customized the Hybrid Agent to meet requirements for Exchange, and to manage the configuration of the underlying Azure AD Application Proxy to ensure that connectivity is properly secured between Microsoft 365 and on-premises Exchange Servers.

Installation Location

The Hybrid Agent can be installed on either one or more machines specifically designed as the "proxy server" (preferred for large or complex organizations) or on an Exchange Server 2010 or later (recommended for smaller organization). The designated Hybrid Agent server is also supported in a perimeter network.

Installation Requirements

The following steps identify the hybrid agent's installation requirements:

1. Installation must be done as a local administrator account.
2. The machine hosting the Hybrid Agent install must be able to communicate:
   - To the Internet using HTTPS (port 443),
   - To Exchange Server for hybrid configuration using HTTPS and RPC
3. The machine hosting the Hybrid Agent should be Windows Server 2012 R2 or later, with .Net 4.6.2 installed.
4. The machine where the Hybrid Agent is installed must have either Edge or Internet Explorer installed.

4 https://aka.ms/AA3r13m

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5. The machine where the Hybrid Agent is installed must be able to communicate with a domain controller to authenticate your on-premises Exchange Org admin credentials. This means that you must be domain joined.
6. TLS 1.0 must be enabled on the machine where the Hybrid Agent is installed.

Internet Proxy Considerations

You should take into account the following considerations if your environment requires an Internet proxy server to connect to the Interne:

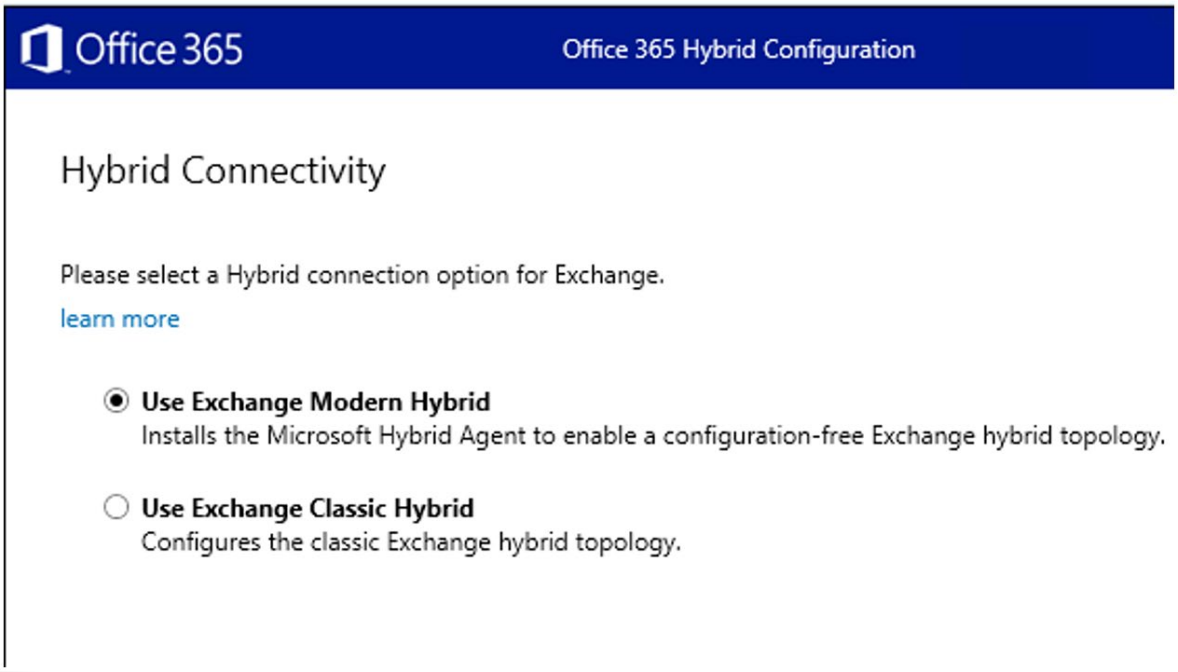
- **Hybrid Agent.** The agent does support using an Internet proxy but doing so requires modifications to the configuration file after installation. Also, a proxy which prevents registration will result in the connector failing to install. During installation, it is recommended that you allow the connectors to bypass the proxy until the HWC configuration changes can be made.
- **Client Access Server (CAS).** The CAS Server's proxy settings must be set correctly, or outbound free busy may fail. You should use the following cmdlet to retrieve the proxy settings:

```
Get-ExchangeServer | fl InternetWebProxy.
```

Hybrid Agent Setup

The Hybrid Configuration Wizard (HCW) is the application responsible for both installing and configuring the Hybrid Agent and setting the required configuration both on-premises and in Exchange Online to enable the traditional hybrid feature set (free busy, migrations, mail routing, and so on).

In the HCW, you must select the option to configure Full Hybrid Configuration because the Hybrid Agent is not available in a Minimal Hybrid Configuration. The Hybrid Agent will be installed once you select **Use Exchange Modern Hybrid** on the Hybrid Connectivity page as shown in the following screen:



You must run the HCW from the machine where you want the agent installed. After the Agent is installed and configured, the HCW locates a preferred server to connect to and runs the standard hybrid configu-

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ration steps. You do not have to run the HCW from the Exchange server directly; however, as stated above, the machine where the HCW is executed from must be able to HTTPS and RPS the selected CAS server.

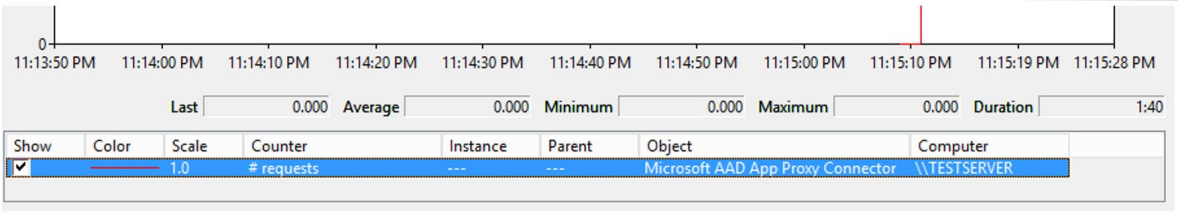
To set up the Hybrid Agent using the Hybrid Configuration Wizard, you must perform the following steps:

1. Download the hybrid agent installation package.
2. Install the agent on the local machine.
3. Register the agent to Azure, including creation of the URL used for proxying requests.
4. Test migration viability from your tenant to on-premises through the hybrid agent.

Testing & validation of the Hybrid Agent

After successful installation of the Hybrid Agent and HCW configurations, you should perform the following two simple tests to validate free busy and mailbox migration flow through the Agent.

- On the server where the Hybrid Agent is installed, open Perform Monitor. Add the object, Microsoft AD App Proxy Connector and the # requests counter to your view.



- Open an RPS session to your tenant and run the following test cmdlet:

```
Test-MigrationServerAvailability -ExchangeRemoteMove: $true -RemoteServer
'<your custom guid>.resource.mailboxmigration.his.msappproxy.net' -Creden-
tials (Get-Credential)
```

You should enter on-premises credentials in the pop up. After the test returns the success result, switch back to your Performance Monitor view and validate that the number of requests has increased.

In addition, you can optionally perform a test mailbox move from your on-premises deployment to the cloud.

Free Busy

The same validation can be performed by logging into a cloud mailbox and requesting free busy through a test meeting request for a mailbox located on-premises.

Mail flow options for a hybrid deployment

A hybrid deployment is usually configured to share a common SMTP email namespace. This requires routing messages in between your on-premises Exchange servers and Exchange Online. For this reason, you must consider how to organize the email flow to and from the Internet, and between Exchange Online and the Exchange Server on-premises organization.

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Inbound email flow from the Internet

Inbound email can be delivered to Exchange Server on-premises, or you can target the mail exchanger (MX) resource record to Exchange Online. You should choose a method depending on your organization's requirements. The following are some considerations for each scenario:

- **Inbound to Exchange Server on-premises**. Use this option if you want to keep full control of your email domains, such as message tracking and journaling of messages in your company. Configure the MX resource record to point to your organization's SMTP smart host. Then, the hybrid deployment automatically forwards all messages for mailboxes that are located on Exchange Online. If you currently use a third-party mail filter in front of your on-premises environment, then this scenario also applies.
- **Inbound to Exchange Online.** Use this option if you want Microsoft to handle your email domains, and you want to automatically take advantage of the antivirus and anti-spam scanning engines from Exchange Online Protection (EOP). To configure this, point the MX resource record to Exchange Online Protection. Exchange Online automatically delivers messages that are located on Exchange Server on-premises by using the Exchange Server that you define when you run the Hybrid Deployment Wizard.

Outbound email flow to the Internet

Just as with inbound email flow, you can also configure outbound email flow from your domain to the Internet by using one of the following options:

- **Deliver Internet-bound messages directly.** Use this option to send any outbound message that is targeted to the Internet directly from either Exchange Online or Exchange Server on-premises. If the mailbox is located on Exchange Online, the Internet messages are delivered directly to the target SMTP domain without passing through the Exchange Server on-premises environment. Messages sent from on-premises mailboxes are routed directly to Internet recipients without passing through Exchange Online. The benefit of this option is that the message traffic is optimized, but the drawback is that it is harder to track messages to the Internet because not every message flows through the on-premises Exchange servers.
- **Route all Internet-bound messages through your on-premises Exchange servers.** This option forces Exchange Online to send any message that is targeted to the Internet through the Exchange Server on-premises environment first. The on-premises Exchange servers then route the message to the Internet and deliver the message. The benefit of this option is that all messages pass through the Exchange servers, so you can use message tracking, journaling, and other compliance features.
- **Route all Internet-bound messages through Exchange Online.** This option forces the on-premises Exchange servers to send any message that is targeted to the Internet through the Exchange Online first.

Email flow between Exchange Online and the Exchange Server on-premises organization

Email flow between Exchange Online and an Exchange Server on-premises organization uses SMTP send and receive connectors that the HCW configures automatically. The connectors enforce the requirement that messages be encrypted by using the Transport Layer Security (TLS) protocol.

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Lab9-Ex1-Prepare Azure AD for a Hybrid Deployment

Lab9-Ex1-Prepare Azure AD for a Hybrid Deployment

In this lab you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. Adatum has decided to transition from their current Microsoft Exchange on-premises deployment to a hybrid deployment that utilizes Exchange Online within Microsoft 365. Adatum's CTO has tasked you with implementing this hybrid deployment. In this lab, you will perform the tasks necessary to prepare your messaging environment for your eventual hybrid deployment.

To complete this task, you must first prepare Azure Active Directory to support the hybrid synchronization between Exchange on-premises and Exchange Online. This will require that you:

- Configure your lab environment to support local mail transport
- Add an accepted domain to your Azure AD forest
- Configure the UPN Name for the new domain
- Configure Exchange to use the new domain
- Enable directory synchronization by installing and running the Microsoft Azure Active Directory Connect tool
- Migrate Adatum's on-premises user accounts to the new domain in Microsoft 365

While your trial tenant has already been set up by your lab hosting provider, you must ensure that your local, on-premises Active Directory is ready for hybrid synchronization before you create your hybrid deployment. You will do this by adding a custom, accepted domain to the Azure Active Directory forest and then configure Exchange to use the new accepted domain.

Once you finish configuring Azure AD for hybrid synchronization in this lab, you will then set up Exchange for a hybrid deployment and then test your new deployment.

See your VM lab environment for the lab instructions.

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Module Review

Module 10 Review

Multiple choice

Item 1. Contoso plans to migrate their mailboxes to Exchange Online. Currently they run Exchange Server 2013 in their environment. Which migration option provides them the flexibility to move mailboxes in batches to Exchange Online?

- ☐ IMAP Migration
- ☐ Cutover migration
- ☐ Staged Migration
- ☐ Hybrid deployment

Checkbox

Item 2. Your company currently runs Exchange Server 2016 and plans to migrate to Exchange Online. What migration options can you use? (Select 3)

- ☐ IMAP Migration
- ☐ Cutover migration
- ☐ Staged Migration
- ☐ Minimal Hybrid deployment

Checkbox

Item 3. What components do you require for a hybrid deployment when your company uses Exchange Server 2016? (Select 3)

- ☐ Azure Active Directory (Azure AD)
- ☐ Microsoft 365 Hybrid Configuration Wizard
- ☐ Azure AD authentication system
- ☐ Azure Multi-Factor Authentication (MFA)
- ☐ Exchange Advanced Threat Protection (ATP)

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Checkbox

Item 4. You are the messaging administrator for Contoso, which plans to implement a minimal hybrid deployment. What components should you consider for the implementation? (Select 3)

- ☐ Microsoft 365 Hybrid Configuration Wizard
- ☐ Open authentication (OAuth)
- ☐ Azure Active Directory Connect
- ☐ Azure AD authentication system
- ☐ Active Directory Federation Services (AD FS)
- ☐ Microsoft 365

Checkbox

Item 5. What statements about e-mail address matching are correct when using Azure AD Connect to synchronize your Active Directory objects with Azure AD? (Select 2)

- ☐ If a user exists in your on-premises AD and no matching user exists yet in Microsoft 365, Azure AD Connect will create a new **Exchange mailbox** with the same email address as the on-premises account.
- ☐ If a user exists in your on-premises AD and no matching user exists yet in Microsoft 365, Azure AD Connect will create a new **Microsoft 365 user** with the same email address as the on-premises account.
- ☐ If a user already exists in both your on-premises AD and in Microsoft 365 and these objects have the same email address, then during the first synchronization these objects will become **joined, or linked**.
- ☐ If a user already exists in both your on-premises AD and in Microsoft 365 and these objects have the same email address, then the object will be **ignored**.

Checkbox

Item 6. What objects are synchronized to Azure AD when using Azure AD Connect? (Select 3)

- ☐ Users
- ☐ Distribution Groups
- ☐ Contacts
- ☐ Dynamic Groups
- ☐ Microsoft 365 groups

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Multiple choice

Item 7. Which of the following statements about Open Authentication (OAuth) is correct?

- ☐ When using OAuth, the identity provider is responsible for verifying the identity of users and applications that exist in an organization's directory, as well as issuing security tokens upon successful authentication of those users and applications.
- ☐ OAuth uses standard federation technologies to allow Exchange organizations to establish trusted relationships with each other.
- ☐ OAuth is a framework that includes server-to-server authentication.
- ☐ OAuth is an architectural style for building distributed systems based on hypermedia.

Checkbox

Item 8. What Exchange federation methods are available when running Exchange Server 2016? (Select 2)

- ☐ Azure Active Directory Login
- ☐ Delegated Authentication (DAuth)
- ☐ Open Authentication (OAuth)
- ☐ Azure Multi-Factor Authentication (MFA)

Multiple choice

Item 9. Contoso plans to move mailboxes to Exchange Online. They do not require coexistence. What options in the Hybrid Configuration Wizard (HCW) should they use?

- ☐ Minimal Hybrid Configuration
- ☐ Full Hybrid Configuration
- ☐ Exchange Classic Hybrid
- ☐ Exchange Modern Hybrid

Multiple choice

Item 10. Contoso runs on Exchange 2016 without publishing their client access namespace to the Internet. The Exchange Servers do not have a public certificate but use a self-signed certificate. Contoso plans to move to Exchange Online over the period of one year. What option should Contoso implement in order to configure a hybrid deployment if it does not want to change its current Exchange configuration?

- ☐ Express Migration
- ☐ Minimal Hybrid Configuration
- ☐ Full Hybrid Configuration
- ☐ Exchange Classic Hybrid
- ☐ Exchange Modern Hybrid

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Checkbox

Item 11. What configuration settings from your Exchange Server 2016 organization can you transfer when using Organization Configuration Transfer (OCT)? (Select 4)

- ☐ Accepted Domains
- ☐ OWA Mailbox Policy
- ☐ Retention Policy
- ☐ Organization Config
- ☐ Malware Filter Policy
- ☐ E-Mail Address Policy
- ☐ Address Book Policy

Multiple choice

Item 12. You are the messaging administrator for Contoso, which recently implemented a hybrid deployment between its Exchange Server 2013 environment and Exchange Online. You recognize that some of the retention policies you configured in Exchange 2013 are not configured in Exchange Online. What functionality of the Hybrid Configuration Wizard would you use to configure the policies in Exchange Online?

- ☐ Minimal Hybrid Configuration
- ☐ Full Hybrid Configuration
- ☐ Organization Configuration Transfer
- ☐ Exchange Hybrid Configuration

Multiple choice

Item 13. Contoso plans to migrate their Exchange Server 2013 environment to Exchange Online. Their current security policy does not allow any inbound connections from the Internet. What hybrid connectivity feature do you need to implement in order to get full coexistence?

- ☐ Minimal Hybrid Configuration
- ☐ Exchange Classic Hybrid
- ☐ Exchange Modern Hybrid
- ☐ Express Migration
- ☐ Organization Configuration Transfer

Multiple choice

Item 14. You plan to deploy the Hybrid Agent in your Exchange environment. What option in the Hybrid Configuration Wizard do you need to select to install the Hybrid Agent?

- ☐ Minimal Hybrid Configuration
- ☐ Express Migration
- ☐ Full Hybrid Configuration
- ☐ Organization Configuration Transfer

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Multiple choice

Item 15. Contoso plans to implement a hybrid deployment in its Exchange Server 2016 organization. Contoso requires full control of its email domains, such as message tracking and journaling of messages. What Internet email flow option should Contoso use?

- ☐ Inbound to Exchange Online
- ☐ Route all Internet-bound messages through its on-premises Exchange servers
- ☐ Inbound to Exchange Server on-premises
- ☐ Inbound to a third-party smart host

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Answers

Multiple choice

Item 1. Contoso plans to migrate their mailboxes to Exchange Online. Currently they run Exchange Server 2013 in their environment. Which migration option provides them the flexibility to move mailboxes in batches to Exchange Online?

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- ☐ Cutover migration
- ☐ Staged Migration
- ■ Hybrid deployment

Checkbox

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Checkbox

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- ☐ Azure AD authentication system
- ☐ Active Directory Federation Services (AD FS)
- ■ Microsoft 365

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Checkbox

Item 5. What statements about e-mail address matching are correct when using Azure AD Connect to synchronize your Active Directory objects with Azure AD? (Select 2)

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- ■ If a user already exists in both your on-premises AD and in Microsoft 365 and these objects have the same email address, then during the first synchronization these objects will become **joined, or linked**.
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Checkbox

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- ■ Distribution Groups
- ■ Contacts
- ☐ Dynamic Groups
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- ☐ Express Migration
- ☐ Minimal Hybrid Configuration
- ☐ Full Hybrid Configuration
- ☐ Exchange Classic Hybrid
- ■ Exchange Modern Hybrid

Checkbox

Item 11. What configuration settings from your Exchange Server 2016 organization can you transfer when using Organization Configuration Transfer (OCT)? (Select 4)

- ☐ Accepted Domains
- ■ OWA Mailbox Policy
- ■ Retention Policy
- ■ Organization Config
- ■ Malware Filter Policy
- ☐ E-Mail Address Policy
- ☐ Address Book Policy

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Multiple choice

Item 12. You are the messaging administrator for Contoso, which recently implemented a hybrid deployment between its Exchange Server 2013 environment and Exchange Online. You recognize that some of the retention policies you configured in Exchange 2013 are not configured in Exchange Online. What functionality of the Hybrid Configuration Wizard would you use to configure the policies in Exchange Online?

- ☐ Minimal Hybrid Configuration
- ☐ Full Hybrid Configuration
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- ☐ Exchange Hybrid Configuration

Multiple choice

Item 13. Contoso plans to migrate their Exchange Server 2013 environment to Exchange Online. Their current security policy does not allow any inbound connections from the Internet. What hybrid connectivity feature do you need to implement in order to get full coexistence?

- ☐ Minimal Hybrid Configuration
- ☐ Exchange Classic Hybrid
- ■ Exchange Modern Hybrid
- ☐ Express Migration
- ☐ Organization Configuration Transfer

Multiple choice

Item 14. You plan to deploy the Hybrid Agent in your Exchange environment. What option in the Hybrid Configuration Wizard do you need to select to install the Hybrid Agent?

- ☐ Minimal Hybrid Configuration
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- ☐ Inbound to a third-party smart host

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Module 11 Performing Mailbox Migrations

Planning Mailbox Migrations

Lesson Introduction

Moving your existing mailbox data is crucial to the success of any migration to Exchange Online. It is important to consider the available migration options that you can use and to choose the best migration option for your users.

This lesson examines the options that are available for migrating email to Exchange Online, such as performing a migration or using FastTrack to move mailboxes from your existing mail servers to Exchange Online. This lesson summarizes the migration and co-existence options and recommends when to use which option.

After this lesson, you will be able to:

- Describe the migration and coexistence strategies with Exchange Online.
- Use FastTrack to move mailboxes.

Overview of migration options to Exchange Online

Before examining the migration options, it's important that you understand the difference between migration and coexistence.

- **Migration**. Migration is the process of moving user data from a source messaging system to Exchange Online. When migrating your email data, you want to move the mailboxes from your existing messaging system to Exchange Online as quickly as possible. You want to avoid maintaining mailboxes in both systems for a long period of time. There are various types of migrations, each of which has different requirements.
- **Coexistence.** Also known as a hybrid configuration, Coexistence is the process of maintaining mailboxes simultaneously in both your existing Exchange environment and Exchange Online. While this option is limited to Exchange environments, it typically deals with upgrades from on-premises deployments to hybrid deployments.

Coexistence does not require that you move your mailboxes all at once from your existing system to Exchange Online, as is the case with a migration. In fact, most organizations who implement a hybrid deployment do not plan on moving completely to the cloud in the short term. In a coexistence you can move mailboxes between your on-premises mail servers and Exchange Online as your business allows, and without the users noticing a difference. Therefore, coexistence provides the smoothest way to move mailboxes from the user perspective, but it also requires the greatest effort for mail administrators.

The following table summarizes when to use migration and coexistence. Please be aware that there may be other contributing factors, depending on your business needs.

Migration Approach	Source Email Server	Mechanism	When to use
Cutover migration	Exchange 2007 or later	Migrates all mailboxes at one time. Uses Outlook Anywhere connection to mailboxes. Requires manual Outlook profile updates.	If you can migrate all data within an acceptable time (for example, over a day or weekend). If your organization has fewer than 2000 users.
Staged migration	Exchange 2007	Requires Azure AD Connect for directory synchronization. Uses Outlook Anywhere connection to mailboxes. Requires manual Outlook profile updates.	If your organization has more than 2000 users. If you want to move your entire email organization to Exchange Online and manage user accounts in Microsoft 365, but the cutover option is not suitable for you.

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Migration Approach	Source Email Server	Mechanism	When to use
Minimal Hybrid Configuration / Express Migration	Exchange 2010 or later	Requires Azure AD Connect for directory synchronization. Configures hybrid but without federation related features like free/busy. Automatic Outlook profile updates.	If you use hybrid-only for migration. If you plan to move to Exchange online and do not need a full hybrid configuration. If you need to move users between your on-premises Exchange and Microsoft 365. If you need Exchange on-premises for recipient administration purposes.
Full Hybrid Configuration (co-existence)	Exchange 2010 or later	Requires Azure AD Connect for directory synchronization. Full featured Exchange federation between on-premises and Exchange Online organization. Automatic Outlook profile updates.	If you have the needs for a long-term coexistence between your on-premises Exchange and Microsoft 365. If you need to move users between your on-premises Exchange and Microsoft 365.
IMAP migration	Any IMAP-accessible email server	IMAP connection to user mailboxes. Needs manual Outlook profile updates.	If you only need to migrate your user's emails and its folder structure. If you are migrating from a third-party mail system providing IMAP access.

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Migration Approach	Source Email Server	Mechanism	When to use
PST migration	Any mail server that can store PST files	Migration tool to connect to PST files. PST files are transferred to Office 365 using either Network upload or Drive shipping.	If you have a large amount of mailbox data that needs to be migrated to Microsoft 365. If you run IMAP/POP3 servers and your clients can export PST files (e.g., use Microsoft Outlook). If you do not want to overutilize your Internet connectivity during migration.
Third-party migration (for example, IBM Domino, Novell Groupwise, and so on)	Any other email server	3rd party migration tool for migration.	If none of the above-mentioned options work for you and you want to migrate your user's mailboxes to Microsoft 365.

Determining which Migration Option to use

You should consider the following factors when determining which migration or coexistence option provides the best solution for your company:

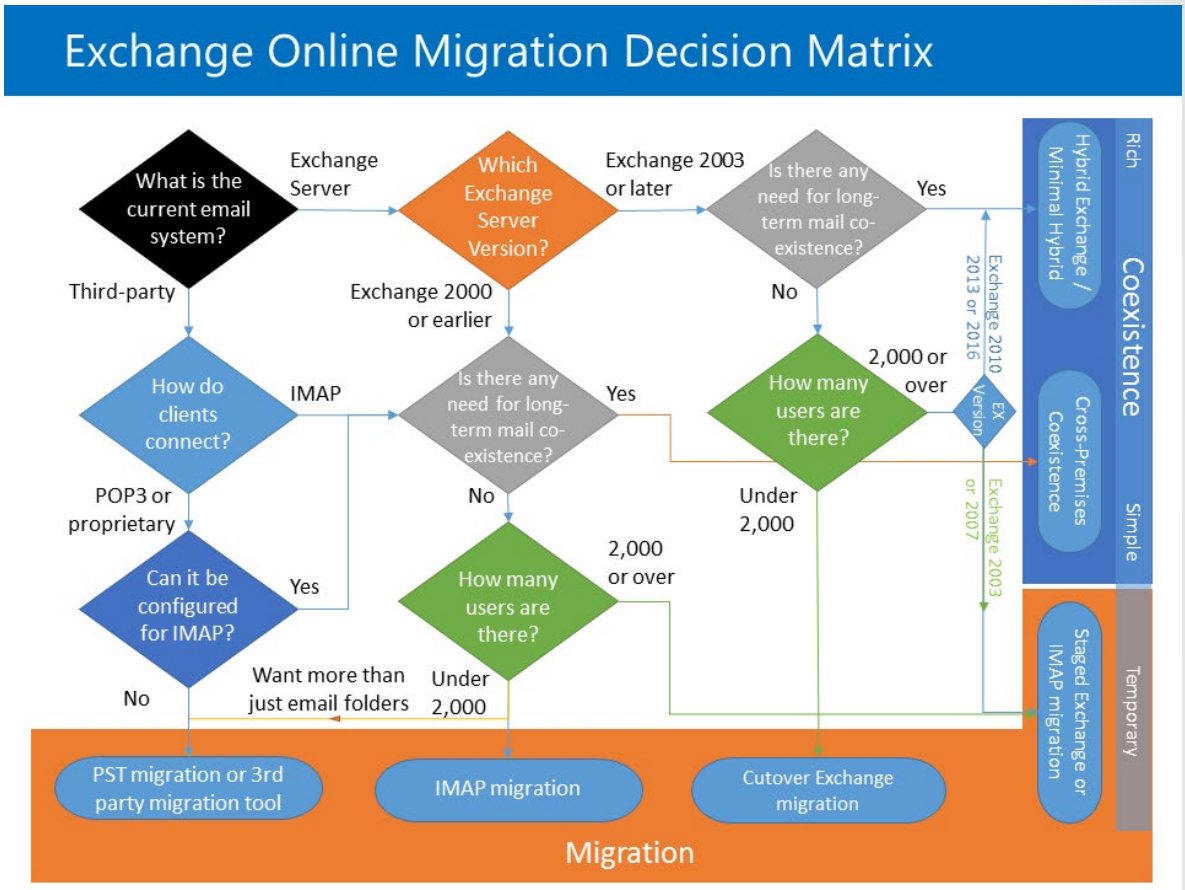
- **Current email system.** If your organization already has Exchange Server on-premises, and depending on the version of Exchange, you can consider the full range of migration options. If you currently employ a third-party mail system, you can only use IMAP or third-party tools as migration options.
- **Exchange Server version.** If you run Exchange Server on-premises, the version of Exchange provides guidance on the migration options available. If you run Exchange 2007 you can consider a cutover or staged migration, or a hybrid configuration (hybrid only, if you add at least Exchange 2010 to your organization).
- **Long-term coexistence with Exchange.** If the organization wants to keep its on-premises Exchange servers, the recommended approach is a hybrid configuration. This approach also works with Exchange Server 2007 SP3 RU10 or later. If you run Exchange 2007, you must have an Exchange Server 2010 or later for the hybrid configuration.
- **Number of users.** If the organization has Exchange but does not want long-term coexistence, the deciding factor is the number of users. If there are fewer than 2000 users, the recommended solution is an Exchange cutover migration. However, if there are 2000 or more mailboxes to migrate, then you need to review the version of Exchange Server again. If it's Exchange 2007, then staged migration is the answer. If the organization is using Exchange 2010 or later, you can consider implementing a minimal or hybrid configuration to migrate your mailboxes.
- **IMAP connections.** The most common types of messaging systems that are not based on Microsoft Exchange Server use the IMAP protocol to communicate with their messaging clients. For example,

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Yahoo Mail uses the IMAP protocol. For messaging systems using the IMAP protocol, an IMAP migration is possible. However, if IMAP is not available on non-Exchange Server systems, mail migration will need to be PST-based or use a third-party tool.

- **POP3 connections.** If the third-party email system only provides a POP3 protocol over which users connect to their mailboxes, the only migration option is to use a third-party migration tool.

The following diagram displays a decision matrix that can help Messaging Administrators determine which migration option is appropriate for his or her organization.



Changing the DNS MX records during a Microsoft 365 migration

The DNS MX record for a domain provides the main Internet SMTP message routing information and points to the target server for message delivery. For this reason, you must wisely decide when you want to change your domain's DNS MX record to point to Exchange Online in a Microsoft 365 migration.

You should consider the following factors when deciding which migration option to use:

- **Cutover Migration**. When performing a cutover migration, you must change the DNS MX record prior to your final synchronization run to ensure that no mail will be routed to your on-premises Exchange server after you have completed the mailbox migration. NOTE: The DNS MX record may have a high TTL of 24 hours. For migration purposes, it is recommended that you set the TTL of your DNS MX record to 60 minutes when starting the migration. This will help to propagate the change much faster.

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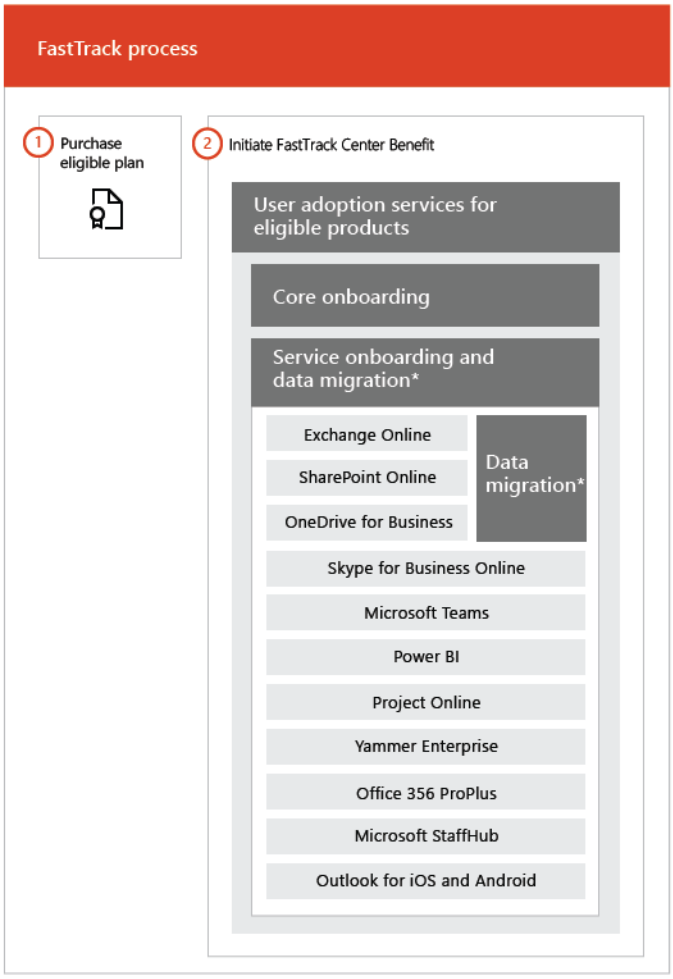
- **Staged Migration.** Because Azure AD Connect synchronizes your Active Directory to Microsoft 365, your DNS MX record needs to point to the on-premises environment for the entirety of the migration. You should NOT change the DNS MX record to point to Microsoft 365 until after all your migration batches are complete.
- **Full or Minimal Hybrid Configuration.** Because the Hybrid Configuration Wizard configures the mail flow to be seamless between Exchange on-premises and Exchange Online, you can decide when to switch the DNS MX record to Microsoft 365. While this commonly occurs once most of the mailboxes have been migrated, you may encounter reasons to switch earlier; for example, when you want to use EOP to scan messages on your on-premises servers. On the other hand, you may decide not to switch the DNS MX record, such as if you have a required third-party mail archiving system on-premises.
- **IMAP Migration.** With this method, you must pre-stage your mailboxes in Microsoft 365 and then synchronize mail between the two systems. Once your mail is synchronized and you want to make the switch, you should change the DNS MX records to point towards Microsoft 365. Once this change has been propagated, you can then stop the synchronization.

Using FastTrack Center to move mailboxes to Exchange Online

The FastTrack Center Benefit is included with eligible Microsoft 365 services and plans such as organizations that purchase at least 150 licenses for their Microsoft 365 tenant. If you want to use the FastTrack Center migration services, your organization must have 500 or more licenses. FastTrack helps organizations in onboarding to Microsoft 365, including core and service onboarding, as well as adoption guidance and data migration.

The following diagram displays a high-level overview of the FastTrack process.

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There are four steps in a typical FastTrack engagement:

1. **Start the onboarding process.** A FastTrack Center representative will contact you within 30 days of your eligible purchase of Microsoft 365, EMS, or Azure, or you can request assistance if you're ready to start sooner. Your FastTrack team and your in-house project team will meet to review your on-boarding needs.
2. **Assess your environment.** Your FastTrack engineer will work with your project team to assess your IT environment to identify any aspects of your current infrastructure that need to be changed to better ensure a great experience.
3. **Remediate any issues.** You'll receive a checklist and plan from your FastTrack engineer that you can use to help remediate any issues that were uncovered during the assessment process.
4. **Enable core capabilities.** Your FastTrack team will help you ensure that your cloud solution is ready to use, licenses are assigned appropriately, and eligible services are migrated.

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Onboarding consists of the following options:

- **Core onboarding**. Core onboarding helps you set up the basic service, which includes the configuration and synchronization of identities from your existing Active Directory with tools provided by Microsoft.
- **Service onboarding and migration**. Service onboarding includes the setup process for the specific Microsoft 365 workload you want to migrate. FastTrack helps you to understand the tasks needed to configure your source email environment, which can be either Exchange on-premises or selected third-party email systems, such as IBM Domino or Novell GroupWise.

User adoption services are a set of tasks that help you ensure your users are aware of the changes coming their way and how they can make use of the provided additional services. This benefit occurs in parallel to onboarding activities and provides you with guidance and information you need to distribute to end-users.

When to use Microsoft FastTrack

You can consider FastTrack when your existing messaging environment is one of the following source systems:

- A single or multiple Active Directory forests with a single or multiple Exchange organizations
- A single IBM Domino 7.0.3 or later environment
- A single IMAP-capable email environment
- A single Google Apps for Work environment (Gmail, Contacts, and Calendar only)
- A single Novell GroupWise 7.0.4 or later environment

Using FastTrack to move mailboxes

Migrating mailboxes to Exchange Online is probably one of the most time-consuming parts of a migration project. FastTrack Center migration services includes resources that helps organizations migrate their mailboxes as quickly as possible. They add migration batches on your defined schedule, and they monitor and complete the migration batches. This can help you save time and allow you to focus on helping your users explore the new services.

For more information on how to plan and use FastTrack to migrate to Exchange Online, see the training course titled Microsoft 365 Tenant and Service Management.

Additional reading. For more information, see the following article on **Microsoft FastTrack**[1].

1 https://aka.ms/AA3pva5

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Performing IMAP Migrations

Lesson Introduction

The most common types of messaging systems that are not based on Microsoft Exchange Server use the IMAP protocol to communicate with their messaging clients. Thus, the IMAP protocol is a very important factor when considering migrations to Exchange Online.

This lesson examines the requirements for running an IMAP migration, the migration options that are available, and the steps that are performed during a migration.

After this lesson, you will be able to:

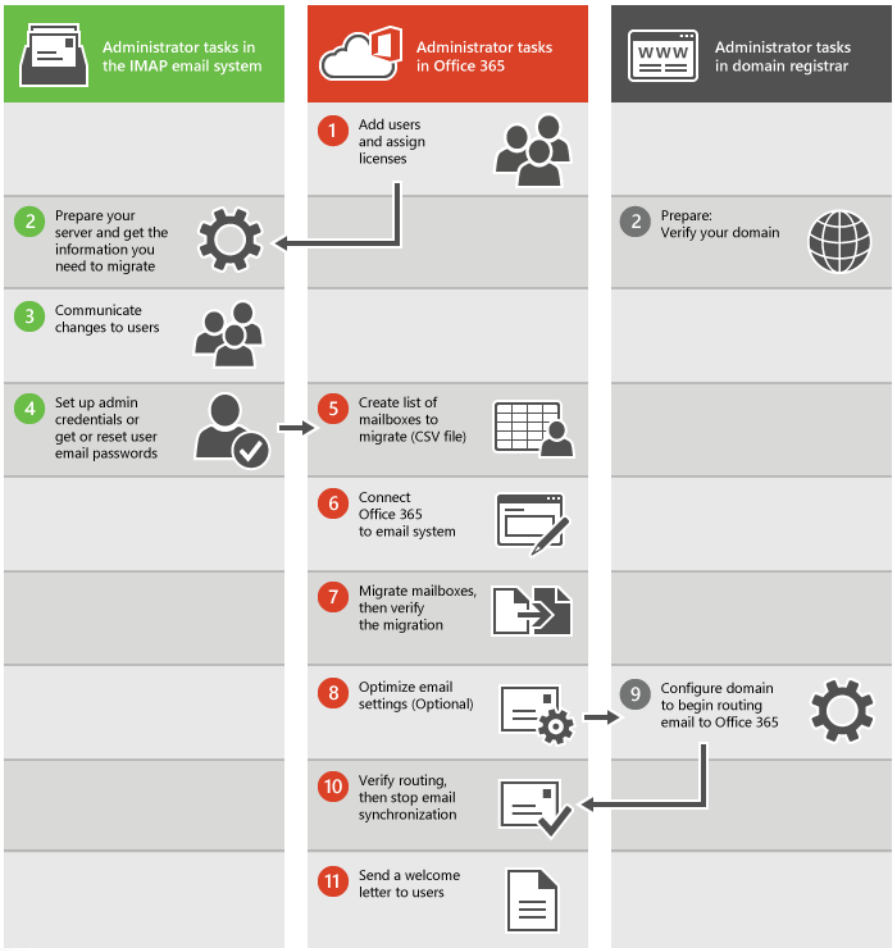
- Describe the requirements for an IMAP migration and how it's carried out.
- Move mailbox data using an IMAP migration.

Performing an IMAP migration

If your current messaging system only supports IMAP connections, the IMAP migration process may provide the proper solution to migrate to Exchange Online. It's important to note that an IMAP migration simply migrates emails and the folder structure, but it does not migrate any other objects, such as calendar items or contacts.

To successfully plan for an IMAP migration, it's essential that you understand the IMAP migration process. The following diagram identifies the steps required to perform an IMAP migration.

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Each step in this diagram is described in greater detail below:

1. **Add users and assign licenses.** Create your users in Microsoft 365 and assign Exchange licenses so that all mailboxes are created in Exchange Online. This is an important prerequisite for an IMAP migration because you cannot start the migration without having your target mailboxes available.
2. **Prepare your IMAP source system and Verify your domains in Microsoft 365**. You need to understand and prepare your IMAP source system in such a way that the Exchange Online migration wizard can log in to the mailboxes and read the messages. As you prepare for your migration, you must also verify all the existing SMTP domains in your Microsoft 365 tenant; otherwise, these SMTP addresses cannot be used in Exchange Online.
3. **Communicate changes to users.** You must let your users know about the migration and how it impacts them. You should provide them with instruction as to the tasks that must be completed before, during, and after the migration. As a best practice, you should ask your users to reduce their mailbox size as much as possible and empty their Deleted Items folder.
4. **Set up admin credentials or get or reset user email passwords.** To perform the migration, you either need an administrator account that can log in and access all mailboxes that need to be migrated, or you need the username and password to each mailbox. If you can't access user mailboxes, you'll

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have to reset your users' passwords. This lets you access the user mailboxes by using a new password that you know.

5. **Create a list of mailboxes to migrate.** You must create a CSV file that lists each mailbox that you want to migrate. For each source email address, you must include the corresponding Exchange Online mailbox and the user name and password.
6. **Connect Microsoft 365 to your source email system.** As a preparatory task, you can create a migration endpoint in Microsoft 365 that connects to your source email system. If you don't create a migration endpoint, this will be done for you by the migration wizard when it creates a migration batch.
7. **Migrate mailboxes and verify the migration.** To start migrating your mailboxes, you must use the IMAP migration wizard to create a migration batch. This will initiate a synchronization. Once the migration batch is finished, verify that all messages are successfully migrated. The migration batch will automatically do a delta-synchronization every 24 hours to migrate any newly arrived messages to their respective Exchange Online mailboxes.
8. **Optimize email settings (Optional).** There are some settings you can configure so that it doesn't take as long for email to begin showing up in your new Microsoft 365 mailboxes. For more information, see Tips for optimizing IMAP migrations.
9. **Change MX record to Microsoft 365.** You must change the DNS MX record of all your SMTP domains so that new email is delivered to Microsoft 365. As a best practice, you should wait a day or two to ensure the DNS settings are propagated to all the other SMTP servers on the Internet so that no messages are transferred to your old mail server.
10. **Verify routing.** After you verify that all email is being routed to Microsoft 365, you should perform a final delta-synchronization between your source email system and Microsoft 365. Once this is complete, you should delete the migration batch to stop the synchronization.
11. **Inform your users.** Once you have completed the migration, you should send a Welcome message to your users to let them know about Microsoft 365 and how to sign in to their new mailboxes.

Requirements for the IMAP migration

You need two things to migrate the contents of users' mailboxes from an IMAP messaging system to their Exchange Online mailboxes:

- A CSV file that lists all the mailboxes to be migrated.
- The IMAP Migration wizard that's located in the Exchange Admin Center.

Supported IMAP servers include Google Mail or other IMAP-based messaging systems.

Limitations of the IMAP migration process

There are several limitations that you must consider before running an IMAP migration, including:

- You can only migrate message items in a user's mailbox; you cannot migrate contacts, calendar items, or tasks.
- You can migrate a maximum of 500,000 mail items from a user's mailbox (emails are migrated from newest to oldest).
- The maximum size of a single message item that can be migrated is 35 MB.
- If you had previously limited the connections to your source email system, you should consider increasing them to improve migration performance. Common connection limits include client/server

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total connections, per-user connections, and IP address connections on either the server or the firewall.

Impact of an IMAP migration on users

To migrate email, you need access to the user mailboxes in your source email system. If you know the user passwords or can access their mailboxes by using administrative credentials, there won't be any impact to users until you shut down your source email system.

If you can't access user mailboxes, you'll have to reset your users' passwords. This lets you access the user mailboxes by using a new password that you know. Since your users won't know the new password, they won't be able to access their old mailboxes during the email migration. As a best practice, you should provide your users with their new Microsoft 365 credentials so they can log in to their mailbox in Exchange Online.

Moving mailboxes using IMAP migration

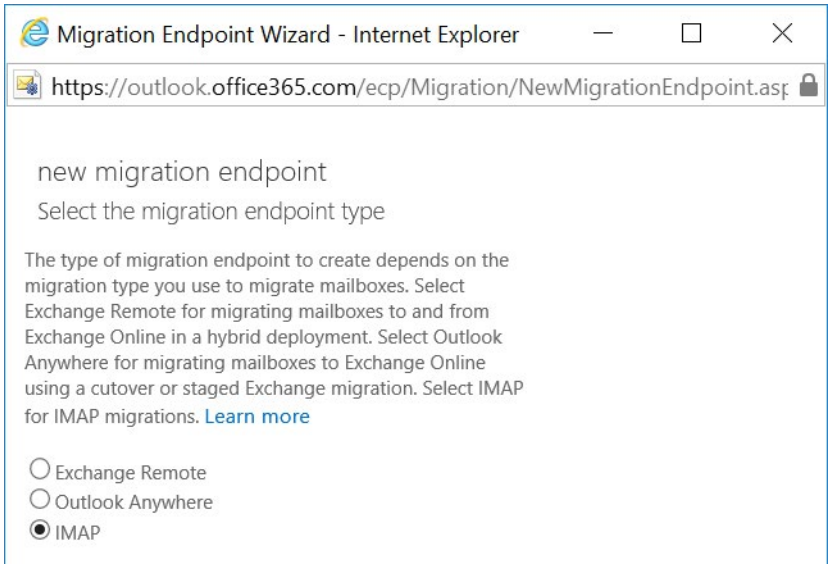
Once you've finished planning your IMAP migration, you're ready to move your user mailboxes. Performing an IMAP migration involves:

- Completing the migration prerequisites, such as registering any SMTP domains in Microsoft 365 and creating all your target mailboxes in Exchange Online.
- Creating an IMAP migration batch.
- Running the IMAP migration wizard.

You should perform the following steps to run an IMAP migration in the Exchange Admin Center (EAC):

1. Create a migration endpoint to connect Microsoft 365 to your email system.
    - In the EAC, navigate to **recipients** > **migration**.
    - On the menu bar, select the **More (ellipsis)** icon and then select **Migration endpoints**.
    - On the **migration endpoints** window, select the **+ (New)** icon to create a new migration endpoint as seen in the following screenshot.
    - On the **Select the migration endpoint type** page, select **IMAP** and then select **Next**.
    - On the **IMAP migration configuration** page, enter your source IMAP server name and configure the required access settings such as authentication, encryption, and port. Then select **Next**. The migration service uses the settings to test the connection to your email server.
    - On the **Enter general information** page, type a Migration endpoint name, such as **IMAP-endpoint**. Leave the other two boxes blank to use the default values. Select **New**.

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2. Create your migration batch and migrate your mailboxes.
   - In the EAC, navigate to **recipients > migration**.
   - On the menu bar, select the **plus (+) sign** icon and then select **Migrate to Exchange Online**.
   - On the **Select a migration type** page, select the **IMAP migration** option, and then select **Next**.
   - On the **Select the users** page, select **Browse** to specify the migration CSV file that you previously prepared. Once you select the file, Microsoft 365 validates it and displays the number of users listed in the file as the number of mailboxes to migrate. If the number displayed from the file doesn't match the number of users you planned to migrate, check your CSV file for any issues.
   - On the **IMAP migration configuration** page, select **Next** and select the migration endpoint you created.
   - On the **Move configuration** page, type the name of the migration batch. You can also optionally enter the names of the folders you want to exclude from migrating, such as Shared, Junk Email, and Deleted. Then select **Next**.
   - On the **Start the batch** page, select **automatically start the batch**. The migration starts as soon as you save the new migration batch. The batch status is first created and changes to syncing after the migration starts. As soon as the initial move is completed, the status will change to **synced**.
   - If you encounter any issues during the migration, you can use the **IMAP Migration Troubleshooter**[2] to investigate the problems.
3. Stop email synchronization. After your sync is completed, you must change the DNS MX record of all your domains to point to Microsoft 365. This might take up to a day to synchronize to all other DNS servers on the Internet. You can verify the change since no new messages should be sent to your old IMAP server.
4. To finish the migration, allow either 72 hours for the final delta-synchronization to take place or manually start the migration batch again. Once you are sure that all messages are migrated, you can remove the migration batch.

2 http://go.microsoft.com/fwlink/p/?LinkId=536482

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Additional reading. For more information, see the following article on **Migrate other types of IMAP mailboxes to Microsoft 365**[3].

3 https://go.microsoft.com/fwlink/?linkid=833320

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Performing Cutover and Staged Migrations

Lesson Introduction

Cutover and staged migrations are very useful options when you plan to migrate mailboxes from your on-premises Exchange Servers to Exchange Online.

This lesson examines how to plan and perform both a cutover and staged migration. We will compare each of these two migration approaches, and you will learn about the requirements, planning activities, and migration process for each option.

After this lesson, you will be able to:

- Describe the requirements for both cutover and staged migrations.
- Perform a migration and move mailboxes either with a cutover or staged migration.

Planning a cutover migration

If your organization has Exchange Server 2007 or later and fewer than 2000 users, you should implement a cutover Exchange migration. A cutover migration is an all-or-nothing approach where you migrate the entire Exchange organization in one step. In fact, the term "Cutover" migration comes from the fact that all on-premises mailboxes are migrated to Exchange Online in a single migration batch. A cutover migration does not require any sort of directory synchronization up front, but it does access the source address book to provision all mailboxes, distribution lists, and contacts.

For a cutover migration, the migration administrator account must either be a member of the Domain Admins group in Active Directory Domain Services (AD DS) in the on-premises organization, or it must be assigned one of the following two permissions:

- FullAccess permission for each on-premises mailbox
- Receive As permission on the on-premises mailbox database that stores the user mailboxes

When planning a cutover migration, you should consider the following:

- Your current Exchange environment must run Exchange 2007 or later.
- Directory synchronization, such as Azure AD Connect, must be disabled on your Microsoft 365 tenant.
- While you can migrate up to 2000 mailboxes, Microsoft recommends cutover migrations to be the ideal migration solution for up to 150 mailboxes.
- A cutover migration only moves mailboxes, mail users, mail contacts, and mail-enabled groups. Security Groups and system mailboxes are not migrated.
- Partial migrations, such as folder exclusions, time range, and so on are not possible with a cutover migration. You must either migrate all messages from all mailboxes, or you must remove the items that you don't want to migrate before you start the migration process.
- Outlook Cached-Mode OST files are not migrated; therefore, a full-resync must take place once the mailbox is migrated. If you want to preserve OST files, you should consider an Express migration with the Hybrid Configuration Wizard instead of using the cutover migration.
- Exchange ActiveSync devices must be manually reconfigured.
- You should estimate the time required to complete the mailbox migration based on the following metric: approximately 10 to 14 GB/hour, depending on your Internet connection speed.

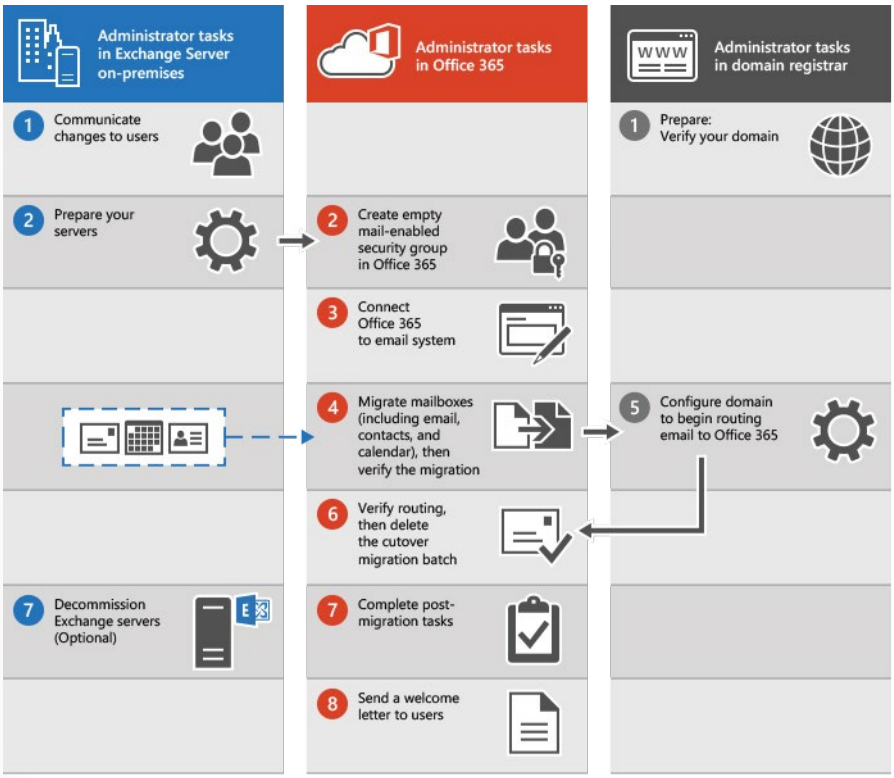
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- If the mailboxes you are migrating are enabled for Unified Messaging, you must disable UM on the mailboxes before you migrate them. You can enable UM on the mailboxes after the migration is complete.
- Cutover migrations will not migrate the following objects:
  - Any hidden objects in the address book, such as mailboxes, contacts, and so on.
  - Delegated permissions that were granted on a mailbox
  - Public folders
  - Dynamic distribution groups

During the cutover migration process, your on-premises accounts will be automatically created and matched with the corresponding objects in Exchange Online. If there is already an object with the same name, the migration process for this user or mailbox will fail.

Performing a cutover migration

The following diagram identifies the steps required to perform a cutover migration.



Each step in this diagram is described in greater detail below:

1. **Communicate changes to users and Verify your domains in Microsoft 365.** You must let your users know about the migration and how it impacts them. You should provide them with instruction as to the tasks that must be completed before, during, and after migration. As a best practice, you should ask your users to reduce their mailbox size as much as possible and empty their Deleted Items folder. You must also verify all the existing SMTP domains in your Microsoft 365 tenant; otherwise, these SMTP addresses cannot be used in Exchange Online.

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2. **Prepare your servers and optionally Create empty mail-enabled security groups**. Prepare your on-premises Exchange servers to allow Outlook Anywhere communication for Microsoft 365. Outlook Anywhere requires a third-party trusted SSL certificate with a principal name that matches the external host name of the published service. Self-signed certificates cannot be used. You can use the **Microsoft Remote Connectivity Analyzer**[4] to verify the connectivity.

   In preparing for the cutover migration, you can also optionally create empty mail-enabled security groups. Because the migration service can't detect whether on-premises Active Directory groups are security groups, it can't provision any migrated groups as security groups in Microsoft 365. If you want to have security groups in Microsoft 365, you must first provision an empty mail-enabled security group in Microsoft 365 before starting the cutover migration.

3. **Connect Office 365 to email system**. As a preparatory task, you can create a migration endpoint in Microsoft 365 that connects to your source email system. If you don't create a migration endpoint, this will be done for you by the migration wizard when it creates a migration batch.

4. **Migrate mailboxes**. To start migrating your mailboxes, you must use the Cutover Migration wizard to create a migration batch. This will initiate a synchronization. Once the migration batch is finished, verify that all messages were successfully migrated. The migration batch will automatically do a delta-synchronization for all mailboxes every 24 hours to migrate any newly arrived messages to their respective Exchange Online mailboxes.

5. **Change Autodiscover and MX record to Microsoft 365**. You must change the Autodiscover DNS record and DNS MX record of all your SMTP domains so that new email is delivered to Microsoft 365. As a best practice, you should wait a day or two to ensure the DNS settings are propagated to all the other SMTP servers in the Internet so that no messages are transferred to your old mail server.

6. **Verify routing**. After you verify that all email is being routed to Microsoft 365, you should perform a final delta-synchronization using your migration batch between your source email system and Microsoft 365. Once this is complete, you can delete the migration batch to stop the synchronization.

7. **Assign Exchange licenses and optionally Decommission Exchange servers**. In this first post-migration task, you must grant Exchange licenses to your migrated users in Microsoft 365; otherwise, they will be soft-deleted after 30 days. Once there are no messages being sent in your source Exchange environment, you can optionally decommission the servers since they are no longer needed.

8. **Send a welcome message to users**. Once you have completed the migration, you should send a Welcome message to your users to let them know about Microsoft 365 and how to sign in to their new mailboxes in Exchange Online.

Planning a staged migration

While a cutover migration migrates all your user mailboxes in one step, a staged migration migrates mailboxes in batches (up to 2000 mailboxes per batch), thereby moving some mailboxes to Exchange Online while maintaining the rest of the mailboxes in the on-premises mail environment. In this migration option, you must group the users that will be migrated together into "migration batches". In addition, you must implement directory synchronization using Azure AD Connect to ensure the Exchange Online address list is updated when a mailbox migration is finished.

A staged migration includes minimal coexistence because it requires directory synchronization using Azure AD Connect. Azure AD Connect synchronizes your local Active Directory to Microsoft 365 and though it requires you to manage all users, groups, and contacts in your Active Directory, it does provide you with a shared global address list (GAL).

4 http://aka.ms/RCA

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Staged migrations are intended for organizations that desire a short period of simple coexistence from their existing Exchange mail environment to Exchange Online. This enables you to move some mailboxes to Exchange Online while maintaining the rest of the mailboxes in your on-premises mail environment. However, it's important to note that the coexistence in a staged migration is a manual approach; therefore, it may require additional administrative work. Also, in a staged migration, you do not have any federation features such as free/busy available. If you require federation features, you should consider a full hybrid configuration.

A staged migration is performed in either the EAC or Windows PowerShell using a CSV file that includes the batches of users that should be migrated.

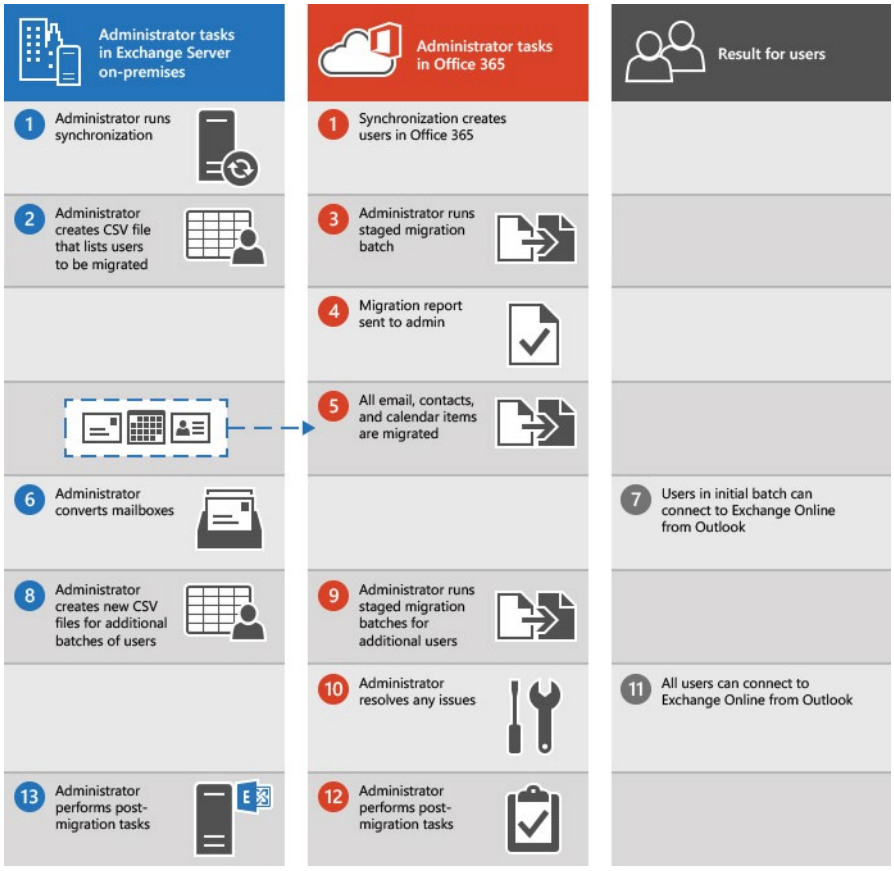
When planning a staged migration, you should consider the following:

- Your current Exchange environment must run Exchange 2007. Staged migrations are NOT supported in Exchange 2010 or later environments; in these instances, you must consider a minimal hybrid configuration instead.
- The account that you use to create the migration batch must have access to the user mailboxes in your Exchange source environment.
- A migration batch can migrate up to 2000 mailboxes. If you have more mailboxes, you need to split the batches up into smaller chunks.
- Because the Microsoft 365 migration service uses Outlook Anywhere to connect to the on-premises Exchange server, you must ensure that your source Exchange organization can be reached from Microsoft 365 using Outlook Anywhere.
- Users must create new Outlook profiles once their mailboxes are migrated.
- Outlook Cached-Mode OST files are not migrated; therefore, a full-resync must occur once the mailbox is migrated.
- Exchange ActiveSync devices must be manually reconfigured.
- You should estimate the time required to complete the mailbox migration based on the following metric: approximately 10 to 14 GB/hour, depending on your Internet connection speed.
- Verify that user mailboxes are not hidden in on-premises address lists. The migration of on-premises mailboxes that are hidden from address lists will fail.
- If the mailboxes you are migrating are enabled for Unified Messaging, you must disable UM on the mailboxes before you migrate them. You can enable UM on the mailboxes after the migration is complete.
- Staged migrations will not migrate the following objects:
  - Delegated permissions granted on a mailbox
  - Public folders
  - Dynamic distribution groups

Performing a staged migration

The following diagram identifies the steps required to perform a staged migration.

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Each step in this diagram is described in greater detail below:

1. **Administrator runs synchronization, which creates users in Microsoft 365**. You must set up directory synchronization between your on-premises Active Directory and Microsoft 365 using Azure AD Connect. This will synchronize all objects including users, groups and contact to Microsoft 365 and provide a common global address list (GAL). Azure AD Connect creates the mail-enabled users in Microsoft 365, which will automatically be converted to mailboxes during the mailbox migration.
2. **Create comma-separated value (CSV) file**. To start the migration batches, you need a CSV file that includes all users that will be migrated.
3. **Administrator runs staged migration batch**. To start migrating your mailboxes, you need to use the Staged Migration wizard to create a migration batch for the CSV file. This will create the Exchange Online mailbox and configures mail forwarding for each user in the CSV file.
4. **Migration report sent to admin**. Exchange Online sends a status email message to the administrator once the migration batch is complete. This status message lists the number of mailboxes that were successfully migrated and how many couldn't be migrated. The message also includes links to migration statistics and error reports that contain more detailed information.
5. **All mailboxes of migration batch are migrated**. Once the migration batch initiates a synchronization, copying all the items from the mailboxes that are in the migration batch will begin. Once the migration batch is finished, you must verify that all messages were successfully migrated. The migration batch will automatically perform a delta-synchronization for all mailboxes every 24 hours to migrate any newly arrived messages to their respective Exchange Online mailboxes.
6. **Administrator converts mailboxes**. After a migration batch is complete and you verified that all mailboxes in the batch are successfully migrated, you need to convert the on-premises mailboxes to

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mail-enabled users. Once you complete this conversion, you can manage these users using your on-premises Exchange admin tools. This is a manual process, but Microsoft provides the necessary scripts for Exchange 2007 in the site titled **Convert Exchange 2007 mailboxes to mail-enabled users**[5].

7. **Users in the initial batch can connect to Exchange Online.** The users that are included in the initial migration batch can use their mailboxes once they have been migrated. You must also remember to convert the mailboxes in the source environment to mail-enabled users so that you can manage them in your on-premises Exchange environment.
8. **Create comma-separated value (CSV) file for additional batches of users**. Once you finish the first migration batch, you can create new migration batches using a CSV file that includes all users that should be migrated.
9. **Run staged migration batches for additional users**. Start migrating your additional mailboxes using the Staged Migration wizard. Once these batches are finished, don't forget that you also need to convert the source mailboxes to mail-enabled users so that you can manage them in your on-premises Exchange environment (see step 6).
10. **Resolve any issue**. Resolve any issues that you see in the migration batches and continue to create new batches as needed.
11. **All users can connect to Exchange Online from Outlook**. At this step, all your users should be able to connect and use their Exchange Online mailboxes.
12. **Perform post-migration tasks in Microsoft 365**. To complete the transition to Exchange Online and Microsoft 365, you should perform the following post-configuration tasks:
    - Assign licenses to Microsoft 365 users (if not already done).
    - Configure the DNS MX record to point to your Microsoft 365 tenant so that email is delivered directly to the Exchange Online mailboxes. Allow up to 24 hours for DNS synchronization in the Internet.
    - Change the Autodiscover DNS record to point to your Microsoft 365 organization.
    - After you verify that all email is being routed to Microsoft 365, you should do a final delta-synchronization using your migration batches between your source email system and Microsoft 365. Once this is finished, you can delete the migration batches to stop the synchronization.
13. **Perform post-migration tasks in Exchange Server on-premises.** Once there are no messages being sent in your source Exchange environment, you can decommission all your servers but one, because you need one Exchange server to perform any necessary management tasks. It is recommended that you keep one Exchange server on-premises to use the Exchange Admin Center and Exchange Management Shell to manage your users, groups and contacts. This is necessary because you cannot manage all user attributes in Exchange Online after enabling directory synchronization.

Moving mailboxes using cutover and staged migrations

As is the case with any type of deployment or installation process, you should spend enough time planning your cutover or staged migration so that you're ready for any contingency that may occur. You must consider the cutover or staged migration planning factors, system requirements, and limitations that were covered previously, as well as your organization's business requirements. Once your plan is in

5 https://go.microsoft.com/fwlink/?linkid=833340

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place, you're ready to perform a cutover or staged migration using the Exchange Admin Center or Windows PowerShell.

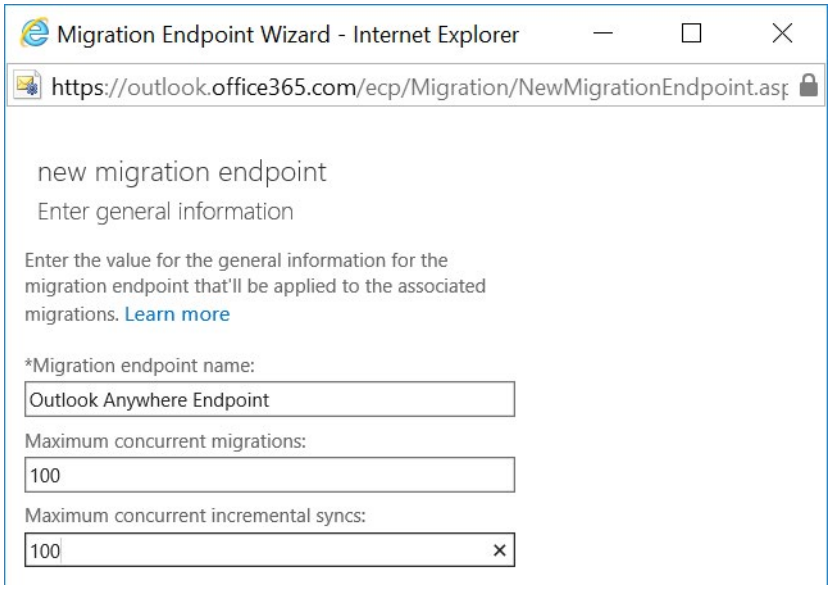
This topic examines how to create and manage migration batches in the Exchange Admin Center.

Creating a Migration Endpoint for Cutover or Staged Migration

As a prerequisite to creating a migration batch for a cutover or staged migration, you must create a migration endpoint to connect Microsoft 365 to your on-premises Exchange server, and then you must verify this connection. This can be done as a separate task in the EAC or during creation of the migration batch.

The following steps explain how you create a migration endpoint:

1. In the EAC, navigate to **recipients** > **migration**.
2. On the menu bar, select the **More (...)** icon and select **Migration endpoints**.
3. On the **Select the migration endpoint type** page, select **Outlook Anywhere** and then select **Next**.
4. On the **Enter on-premises accountcredentials** page, enter the e-mail address of an on-premises user, the login information and password of an on-premises Exchange administrator, and then select **Next**.
5. Microsoft 365 tests the connection to the source server and displays the connection settings. If Microsoft 365 was unable to detect your settings, you need to type an Exchange server name that you plan to use for migration, the RPC proxy server name, authentication, and your mailbox permission. Then select **Next**.
6. On the **Enter general information** page, type a Migration endpoint name, the maximum concurrent migrations (max. 300) and the maximum incremental syncs (max. 300). Then select **New** to create the migration endpoint.

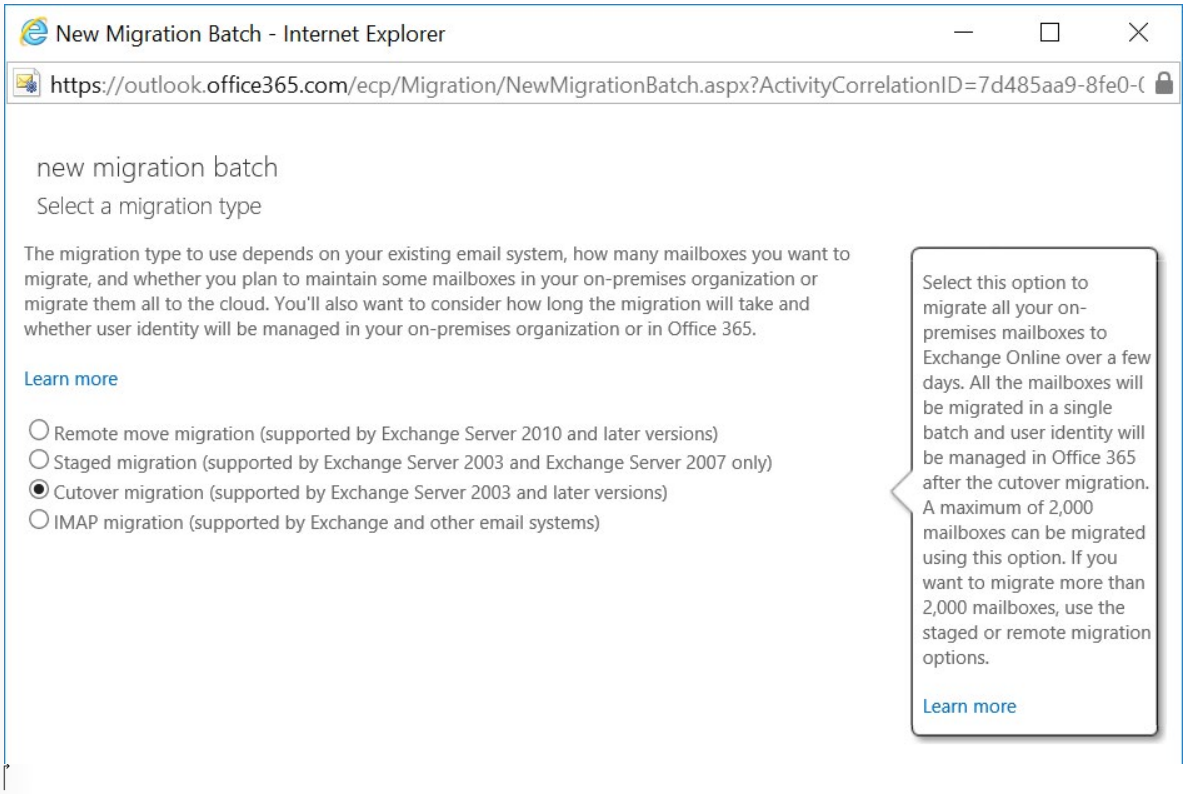


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Moving Mailboxes using a Cutover Migration

The following steps describe how to create a migration batch and migrate mailboxes using a cutover migration in the EAC:

1. In the EAC, navigate to **recipients** > **migration** .
2. On the menu bar, select the **plus (+) sign** icon and then select **Migrate to Exchange Online**.
3. On **new migration batch** page, select **Cutover migration**, and then select **Next**.



4. On the next page, you can either select a migration endpoint if you previously created one, or you can have the wizard create a new one for you. Then select **Next**.
5. On the **Confirm the migration endpoint** page, verify that the migration endpoint settings are correct, and then select **Next**.
6. On the **Move configuration** page, type the name of the migration batch, and then select **Next**. The batch name is displayed in the list of migration batches on the Migration page after you create the migration batch.
7. On the **Start the batch** page, select the recipient(s) that receive a report migration report once the migration is completed, and then choose one of the following options:
   - **Automatically start the batch**. The migration batch is started as soon as you save the new migration batch with a status of Syncing.
   - **Manually start the batch later**. The migration batch is created but is not started. The status of the batch is set to Created.
8. Select **New** to create the migration batch. The new migration batch is displayed on the migration dashboard.

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Once you have created a migration batch, you can perform the following tasks:

- **Verify the migration is working.** While the migration is in progress, you can follow the batch's synchronization status on the migration dashboard. If there are errors, you can view a log file that provides more information about each error. You can also verify whether the users were created in the Office 365 admin center as the migration proceeds. After the migration is done, the sync status is set to **Synced**.
- **Delete the migration batch.** After you change the MX record and verify that all email is being routed to Microsoft 365 mailboxes, notify the users that their mail is going to Microsoft 365. At this point, you can delete the cutover migration batch. When you delete a cutover migration batch, the migration service cleans up any records related to the migration batch and then deletes the migration batch. To delete a cutover migration batch from the list of migration batches on the migration dashboard, perform the following steps:
  1. In the EAC, navigate to **recipients** > **migration**.
  2. On the migration dashboard, select the batch and then select **Delete**.
  3. Verify that the migration batch is no longer listed on the migration dashboard.

Moving Mailboxes using a Staged Migration

Before you can create a migration batch and begin moving mailboxes using a staged migration, you must first create a CSV file that includes a user batch. Each record in the CSV file must include the following information (in this order), each of which is separated by a comma:

- **EmailAddress** (required). This is the email address for a user's on-premises mailbox.
- **Password** (optional). This is the password that will be set on the user's new Exchange Online mailbox. A password must have a minimum length of eight characters.
- **ForceChangePassword** (optional). When a user logs in to his or her new Exchange Online mailbox for the first time, this option specifies whether the user must change the password. Use either True or False for the value of this parameter. Other values are invalid.

The following is an example of a CSV file that would migrate two users from contoso.com to Exchange Online using staged migration (Note: the first record is the heading line that displays the sequence of fields in each record):

```
EmailAddress,Password,ForceChangePassword

SaraD@contoso.com,Pa$$w0rd,False

RobY@contoso.com,Pa$$w0rd,False
```

Additional reading. For more information, see the following article about **CSV files for staged Exchange migration**[6]s.

Once the CSV batch file is created, you can perform the following steps to create a migration batch and migrate mailboxes using a staged migration in the EAC:

1. In the EAC, navigate to **recipients** > **migration**.

6 https://go.microsoft.com/fwlink/?linkid=833341

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2. On the menu bar, select the **plus (+) sign** icon and then select **Migrate to Exchange Online**.
3. On the **new migration batch** page, select **Staged migration**, and then select **Next**.
4. On the **Select the users** page, select **Browse,** select the CSV file to use for this migration batch, and then select **Next**.
5. On the next page, you can either select a migration endpoint if you previously created one, or you can have the wizard create a new one for you. Then select **Next**.
6. On the **Confirm the migration endpoint** page, verify that the migration endpoint settings are correct, and then select **Next**.
7. On the **Move configuration** page, type the name of the migration batch and then select **Next**. The batch name is displayed in the list of migration batches on the Migration page after you create the migration batch.
8. On the **Start the batch** page, select the recipient(s) that should receive a migration report once the migration is completed, and then choose one of the following options:
   - **Automatically start the batch**. The migration batch is started as soon as you save the new migration batch with a status of Syncing.
   - **Manually start the batch later**. The migration batch is created but is not started. The status of the batch is set to Created.
9. Select **New** to create the migration batch. The new migration batch is displayed on the migration dashboard.

Once you have created a migration batch, you can perform the following tasks:

- **Verify the migration is working.** While the migration is in progress, you can follow the batch's synchronization status on the migration dashboard. If there are errors, you can view a log file that provides more information about each error. You can also verify whether the users were created in the Microsoft 365 admin center as the migration proceeds. After the migration is done, the sync status is set to Synced.
- **Delete the migration batch.** After the migration batch is synchronized and the users in the batch are connected to their Exchange Online mailboxes, you should wait at least another 72 hours and then delete the migration batch. To remove the batch from the list of migration batches on the migration dashboard, perform the following steps:
  1. In the EAC, navigate to **recipients** > **migration**.
  2. On the migration dashboard, select the batch and then select **Delete**.
  3. Verify that the migration batch is no longer listed on the migration dashboard.
- **Repeat the process until all your users are moved to Exchange Online.** If you have additional users whose mailboxes must be migrated to Exchange Online, create a new CSV file and start a new migration batch.

  NOTE: You can run batches simultaneously or one at a time. Remember, each migration batch has a limit of 2000 mailboxes.

Once all mailboxes have been moved to Exchange Online, you must change the DNS MX records of your SMTP domain to point to Microsoft 365 and then plan the decommissioning of your source Exchange servers.

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Performing Advanced Migrations

Lesson Introduction

In this module you have learned about the various migration options for mailboxes. However, there are some important migration tasks left. The focus of this lesson will be on migrating a PST file and the considerations for a Public Folder migration.

After this lesson, you will be able to:

- Import PST Files to Exchange Online mailboxes
- Migrate Public Folders to Exchange Online

Planning a PST migration

You can use the Import service in the Microsoft 365 Security & Compliance Center to bulk-import PST files to Exchange Online mailboxes in your Microsoft 365 organization. You can also use the Import service to bulk-import documents and other data from your on-premises SharePoint Server organization to SharePoint Online and OneDrive for Business sites.

To import the data to Microsoft 365, the Import service provides the following transfer options:

- **Network upload**. You can upload your PST files to the Import service using your existing Internet connection.
- **Drive shipping**. You can store your PST files on an encrypted hard drive and physically ship the drive to Microsoft. This option saves you from having to transfer the data over the Internet using the Network Upload option. This option is especially useful if you have a large number of PST files, or the files are just extremely large in size, either of which could cause a network upload to take many days or weeks to complete.

You should consider the following requirements when exporting your PST files and uploading them to Microsoft 365:

- To access the **Microsoft 365 import** page and view import jobs in the Microsoft 365 Security & Compliance Center, you must either be a member of the Global Administrators role in your Microsoft 365 tenant, or a member of the Mail Recipients role in Exchange Online.
- To create import jobs for a mailbox, your account must be assigned to the Exchange Online 'Mailbox Import Export' RBAC (Role Based Access Control) role.

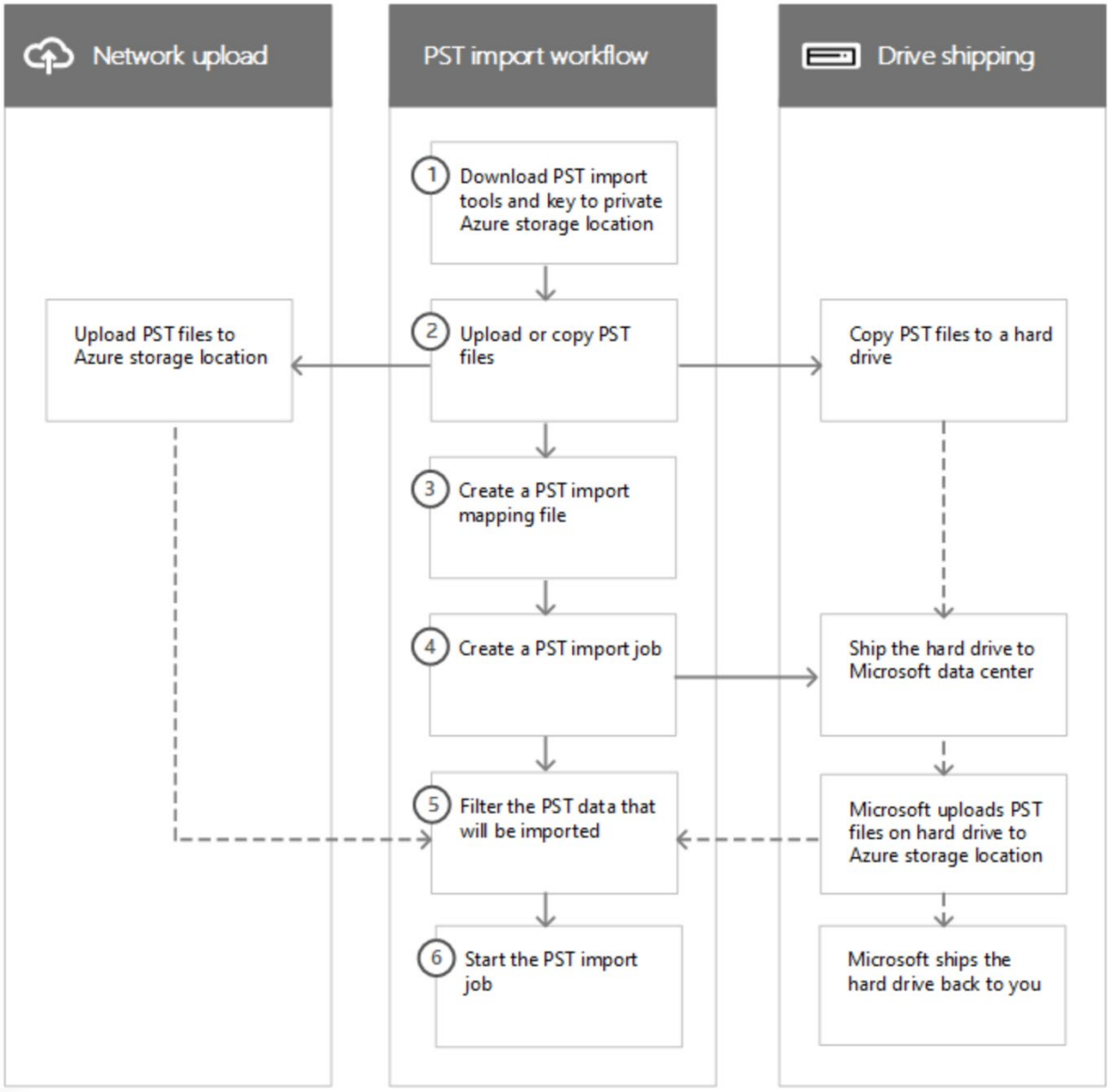
  NOTE: By default, this role is NOT assigned to any role group in Exchange Online.
- Your exported PST files must be located on a file server or shared folder within your on-premises environment.
- In Exchange hybrid deployments, you can outsource PST file data to Exchange Online archive mailboxes that have already been migrated or were newly created. Mailboxes which reside on-premises can access the imported data within the Exchange Online archive mailbox.
- You can also import PST files to inactive mailboxes in Exchange Online. The only requirement is that you must identify the GUID of the inactive mailbox to which the data will be imported.

Performing a PST migration

The steps involved in importing a PST file using the Import service in the Microsoft 365 Security and Compliance Center are illustrated in the following diagram. The Messaging Administrator should follow

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the steps defined in the PST import workflow (the center column in the diagram), although some portions of the process will vary depending on whether the Network upload or drive shipping option is selected.



The steps defined in the diagram are outlined in greater detail below:

1. **Download the PST import tools and key to a private Azure storage location.** The first step is to download the tool and access key used to upload the PST files or copy them to a hard drive. The access key is unique to your organization and provides you (or Microsoft data center personnel in the case of drive shipping) with the necessary permissions to upload PST files to a private and secure Azure storage location.
2. **Upload or copy the PST files.** You must then use the appropriate tool to upload or copy the PST files:
   - **Network upload.** The **AzCopy.exe** tool is used to upload and store your PST files in a private Azure storage location in the Microsoft cloud.
   - **Drive shipping.** The **WAImportExport.exe** tool is used to copy your PST files to the hard drive. This tool encrypts the hard drive with BitLocker and then copies the PST files to the hard drive.

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3. **Create a PST import mapping file.** You must then create a comma separated value (CSV) file that specifies which user mailboxes the PST files will be imported to.
4. **Create a PST import job.** You're now ready to create a PST import job on the **Import** page in the Security & Compliance Center and submit the PST import mapping file.
    - **Network upload**. Because the PST files have been uploaded to Azure, Microsoft 365 analyzes the data in the PST files and then gives you an opportunity to set filters that control what data gets imported to the mailboxes specified in the PST import mapping file.
    - **Drive shipping**. When shipping a physical hard drive to Microsoft, the following steps are involved:
        - You physically ship the hard drive to a Microsoft data center.
        - When Microsoft receives the hard drive, data center personnel will upload the PSTs files on the hard drive to the Azure storage location for your organization. The PST files on the hard drive are uploaded to Azure within 7 to 10 business days after Microsoft receives the hard drive.
        - Microsoft ships the hard drive back to you.
5. **Filter the PST data that will be imported to mailboxes.** After the import job is created, Microsoft 365 analyzes the data in the PST files (safely and securely) by identifying the age of the items and the different message types included in the PST files. You can then decide if you want to import all data or just the messages that fall within a specific date range.
6. **Start the PST import job.** After the import job begins, Microsoft 365 uses the information in the PST import mapping file to import the PSTs files from the Azure storage location to the user mailboxes.

Additional reading. For more information, see the following article on **Overview of importing your organization PST files to Office 365**[7]

Performing a public folder migration

If you currently use Microsoft Exchange Server in your organization, your users may also use Public Folders. In a migration scenario, once you migrate the mailboxes to Exchange Online, you can configure Public Folder coexistence if you are running in a hybrid configuration, whether your users are in Exchange Online or on-premises. Alternatively, if you want to remove the on-premises Exchange servers, you must migrate the Public Folders to Exchange Online.

When planning your Public Folder migration to Exchange Online, you should keep in mind the following limitations:

- Maximum number of public folders that you can migrate: 250,000
- Maximum number of public folders you can create in Exchange Online: 500,000
- Maximum number of subfolders per public folder: 10,000
- Maximum size of a public folder mailbox: 50 GB; however, for E3 or E5 plans you can have 100 GB
- Maximum number of public folder mailboxes: 1000
- Maximum total size of all public folder mailboxes: 50 TB.

There are two types of Public Folders – legacy folders and modern folders.

- Legacy Public Folders are Public Folders that are in a Public Folder database on Exchange Server 2010 or earlier.

7 https://aka.ms/AA3r8oo

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- Modern Public Folders are in a Public Folder mailbox on Exchange Server 2013 or later.

This distinction is important because the migration process is different for each Public Folder type. The following options are available for migrating Public Folders:

- Migrating Legacy Public Folders to Exchange Online
- Migrating Modern Public Folders to Exchange Online
- Migrating Public Folders to Office 365 Groups

Migrating Legacy Public Folders to Exchange Online

Legacy Public Folders are Public Folders stored on Exchange 2010 or earlier. To migrate legacy Public Folders to Exchange Online, your on-premises Exchange servers must meet the following requirements:

- Your on-premises Exchange Servers must run at least:
  - Exchange 2007 SP3 RU15 or later
  - Exchange 2010 SP3 RU8 or later
- It is recommended that you move all your mailboxes to Exchange Online before migrating legacy Public Folders.
- You do not need to configure a hybrid deployment to migrate legacy Public Folders.

You should perform the following steps to migrate legacy Public Folders to Exchange Online:

1. **Download the migration scripts**. You need to download the Public Folders Migration Scripts package that consists of the following three PowerShell scripts:
   - Export-PublicFolderStatistics.ps1
   - PublicFolderToMailboxMapGenerator.ps1
   - Create-PublicFolderMailboxesForMigration.ps1
2. **Prepare for the migration**. You need to prepare your existing Exchange servers as well as Microsoft 365 for the migration. For example, you need to make sure that your existing Public Folder names do not include special characters like the "\" (backslash).
3. **Generate the CSV files**. You use the Export-PublicFolderStatistics.ps1 and PublicFolderToMailboxMapGenerator.ps1 scripts to create the mapping CSV file that includes the required Public Folder mailboxes and their name.
4. **Create the public folder mailboxes in Exchange Online**. Use the Create-PublicFolderMailboxesForMigration.ps1 script together with your CSV mapping file to create all required public folder mailboxes in Exchange Online.
5. **Create and start the migration request**. Create a new Public Migration Batch using your CSV mapping file. Then start the migration batch to begin synchronizing the Public Folder content to Exchange Online.
6. **Lock down the public folders on the Exchange server for final migration**. Once the initial synchronization is finished, you must lock-down your legacy Public Folder environment for the final synchronization. During this lock-down, your users cannot access the Public Folders until you unlock them. You should take this into consideration when planning your migration schedule.
7. **Finalize the public folder migration**. This includes the final delta-synchronization and completes the Public Folder migration batch. At this point, on-premises Public Folders and modern Public Folders in Exchange Online are identical.

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8. **Test and unlock the public folder migration**. After you finalize the migration, you must run tests to ensure the migration was successful and then unlock the Public Folders so all mailboxes in Exchange Online can access those folders.

Additional reading. The Legacy Public Folder migration steps were simplified for this topic; however, for detailed information, see the following article titled **Use batch migration to migrate legacy public folders to Office 365 and Exchange Online**[8].

Migrating Modern Public Folders to Exchange Online

Modern Public Folders are Public Folders stored on Exchange 2013, 2016 or 2019. To migrate modern Public Folders to Exchange Online, your on-premises Exchange servers must meet the following requirements:

- If you intend to migrate users to Microsoft 365, you should complete your user migration prior to migrating your public folders.
- Your need to prepare your Active Directory with Exchange Server 2016 CU4 or later.
- Your Exchange Servers must run at least:
  - Exchange Server 2013 CU15 (only in a mixed environment with Exchange Server 2016 or 2019)
  - Exchange Server 2016 CU4 or later
  - Exchange Server 2019
- You do not need to configure a hybrid deployment to migrate modern Public Folders.

The migration process modern Public Folders follows the same 8 steps as described in the Migrating Legacy Public Folders to Exchange Online section.

Additional reading. For more information, see the following article titled **Use batch migration to migrate Exchange Server public folders to Exchange Online**[9].

Migrating public folders to Microsoft 365 Groups

Once you migrated your local Public Folders to Exchange Online, or if you plan to remove your on-premises Public Folders, you can optionally move your Public Folder data to Microsoft 365 Groups. Microsoft 365 Groups is the latest collaboration offering from Microsoft, which means there are many reasons why they would be a preferable solution over public folders, which is a much older technology. In Outlook, for example, Groups can replace mail-enabled public folders altogether.

You can migrate your Public Folders to Microsoft 365 Groups from the following environments:

- Exchange Online
- Exchange on-premises (Note: If your public folders are on-premises, you must have an Exchange Hybrid environment set up)
  - Exchange 2010 SP3 RU8 or later
  - Exchange 2013 CU15 or later
  - Exchange 2016 CU4 or later
  - Exchange 2019

8 https://go.microsoft.com/fwlink/?linkid=833342
9 https://aka.ms/AA3r11l

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Migrating your Public Folder data to Microsoft 365 Groups is a fairly simple process consisting of the following steps:

1. **Select source.** Choose the public folders that you want to migrate. You can choose any folder containing mail or calendar content.
2. **Create target.** Create corresponding groups for your folders. Groups can be based on the desired configuration, such as members, privacy settings, and data classification.
3. **Copy data.** Use the migration batch cmdlets to copy data from public folders to Groups.
4. **Lock source.** Lock the public folders once you have verified the data in Groups.
5. **Cutover.** Copy any new data that has been created between steps 3 and 4.

Note that your public folders and their corresponding groups will remain online for your users during steps 1 through 3 above. After step 3, you can evaluate whether to proceed with the rest of the migration, based on the Groups experience and whether it suits your users and your organization.

- If you're not satisfied with the Groups experience, you can roll back your migration and resume using public folders at that point.
- If you're satisfied with the Groups experience, then proceed with the migration. It is recommended that you back up your original public folders following step 5. Should you decide that you want to go back to public folders following the migration, you can still roll back to public folders, provided you saved your backup files from the migration process and you have not deleted your original public folders. Conversely, if you decide to move forward with Groups, you can delete the original public folders after completing step 5. However, please note that once the original public folders are removed, you cannot roll back to them even if you saved your backup files.

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Exercise - Planning a Mail Migration to Microsoft 365

Exercise - Planning a Mail Migration to Microsoft 365

You have recently been hired as the Messaging Administrator for Fabrikam, Inc. Fabrikam is moving to Microsoft 365 as its cloud solution.

Fabrikam's Enterprise Administrator has requested that you review the company's footprint and then develop a plan for migrating a group of on-premises mailboxes to Microsoft 365 as part of its mail migration pilot project. As part of that analysis, he has asked that you address some specific issues that concern him.

Download the Paper-Based Lab Document

For this exercise, you must first **download the Paper-Based Lab document**[10] that contains the list of Adatum's company assets, as well as the assessment questions that you must complete.

Steps to complete the exercise

Once you have the document downloaded, perform the following steps to complete this exercise:

Step 1. Each student should review the company asset information in the document and answer the questions in the space provided.

Step 2. After completing the assessment, each student should exchange his or her solution with another student and review each other's assessment. Use the space provided in the document to identify the key differences between your two proposals.

Step 3. Once you are finished collaborating with your fellow student, discuss the assessment as a class and come to a final solution for Fabrikam.

10 https://github.com/MicrosoftLearning/MS-203T00-Microsoft-365-Messaging/blob/master/Allfiles/MS-203T00-ENU-M11-L05-PaperBasedLab.docx

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Module Review

Module 11 Review

Multiple choice

Item 1. What Exchange Online Migration option migrates all Exchange on-premises mailboxes in one run?

- ☐ Cutover Exchange Migration
- ☐ Staged Exchange Migration
- ☐ Full Hybrid Configuration
- ☐ PST migration

Multiple choice

Item 2. What Exchange Online Migration option requires Azure AD Connect for directory synchronization and provides full coexistence between Exchange on-premises and an Exchange Online organization?

- ☐ Minimal Hybrid Configuration
- ☐ Cutover Exchange Migration
- ☐ Staged Exchange Migration
- ☐ Full Hybrid Configuration

Multiple choice

Item 3. What Exchange Online Migration tool connects to personal folder data files?

- ☐ Cutover Exchange Migration
- ☐ Staged Exchange Migration
- ☐ PST migration
- ☐ IMAP migration

Multiple choice

Item 4. Which of the following migration options should be used to migrate mailboxes that are currently hosted on-premises on Exchange 2013 to Exchange Online using the Outlook Anywhere protocol?

- ☐ Minimal Hybrid Configuration
- ☐ Cutover Exchange Migration
- ☐ Staged Exchange Migration
- ☐ Full Hybrid Configuration
- ☐ IMAP migrationn

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Multiple choice

Item 5. You are the messaging administrator for Contoso. You plan to migrate your Exchange 2016 mailboxes to Exchange Online. During the migration, you want the Outlook profiles to update automatically, and you do not need Exchange coexistence. Which of the following migration options would accomplish this migration task with the least amount of effort?

- ☐ Minimal Hybrid Configuration
- ☐ Cutover Exchange Migration
- ☐ Staged Exchange Migration
- ☐ Full Hybrid Configuration
- ☐ IMAP migrationn

Checkbox

Item 6. Which of the following Exchange Online Migration options require a manual update for each user's Outlook profile? (Select 3)

- ☐ Cutover Exchange Migration
- ☐ Minimal Hybrid Configuration
- ☐ Staged Exchange Migration
- ☐ IMAP migration
- ☐ Full Hybrid Configuration

Checkbox

Item 7. Which of the following are valid FastTrack options? (Select 2)

- ☐ Core offboarding
- ☐ Core onboarding
- ☐ Core migration
- ☐ Service onboarding and migration
- ☐ Service offboarding and migration

Multiple choice

Item 8. Which of the following statements about the supported source messaging systems in Microsoft FastTrack is correct?

- ☐ Only a single or multiple Active Directory forests with a single or multiple Exchange organizations or Novell GroupWise 7.0.4 or later
- ☐ A single or multiple Active Directory forests with a single or multiple Exchange organizations, IBM Domino 7.0.3 or later, Novell GroupWise 7.0.4 or later
- ☐ Only single or multiple Active Directory forests with a single or multiple Exchange organizations and IBM Domino 7.0.3 or later
- ☐ Only a single or multiple Active Directory forests with a single Exchange organization, IBM Domino 7.0.3 or later and Novell GroupWise 7.0.4 or later

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Checkbox

Item 9. Which of the following are limitations when doing an IMAP migration? (Select 3)

- ☐ Reply to messages that were migrated is not possible
- ☐ You can migrate a maximum of 100,000 mail items from a user's mailbox
- ☐ You can only migrate message items (no contacts or calendar items)
- ☐ You can migrate a maximum of 500,000 mail items from a user's mailbox
- ☐ The maximum size of a single message item is 35 MB
- ☐ No delta-synchronization is available for the migration batch

Checkbox

Item 10. You are the messaging administrator for Contoso. You plan a cutover migration to Exchange Online. Which of the following statements do you need to consider for your planning? (Select 4)

- ☐ Your current Exchange environment must run Exchange 2003 or later.
- ☐ If directory synchronization is enabled, it must be disabled in your Office 365 tenant.
- ☐ Exchange ActiveSync devices need to be manually reconfigured.
- ☐ The migration option supports migrations larger than 2,000 mailboxes.
- ☐ Security Groups cannot be migrated.
- ☐ Your current Exchange environment must run Exchange 2003 or Exchange 2007.

Checkbox

Item 11. You are the messaging administrator for Contoso. You plan to migrate to Exchange Online using the staged migration option. Which of the following issues do you need to consider to have a successful migration? (Select 3)

- ☐ If directory synchronization is enabled, it must be disabled in your Microsoft 365 tenant.
- ☐ Exchange ActiveSync devices need to be manually reconfigured.
- ☐ The migration option supports migrations larger than 2,000 mailboxes.
- ☐ Security Groups cannot be migrated.
- ☐ Your current Exchange environment must run Exchange 2003 or Exchange 2007.

Checkbox

Item 12. You are the messaging administrator for Contoso. Your users have a lot of local data stored in their PST files in Outlook. You are evaluating the options that will allow you to import the PST data to Microsoft 365. Which of the following options are valid transfer options? (Select 2)

- ☐ Ship data using a physical floppy disk
- ☐ Ship data on a physical hard drive
- ☐ Upload data over the network
- ☐ Upload data using OneDrive

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Checkbox

Item 13. When using the Microsoft 365 Import service to upload PSTs, which mailbox types are valid targets for the data import? (Select 3)

- ☐ Active mailboxes
- ☐ Archive mailboxes
- ☐ Site mailboxes
- ☐ Inactive mailboxes
- ☐ Microsoft 365 Groups

Checkbox

Item 14. Which of the following are valid limits for Public Folders in Exchange? (Select 2)

- ☐ You can have up to 10,000 Public Folders in your Public Folder hierarchy.
- ☐ A public folder mailbox can have a maximum size of 100 GB.
- ☐ You can have up to 1,000 public folder mailboxes.
- ☐ The maximum size of a single Public Folder is 30 MB.

Checkbox

Item 15. You are the messaging administrator for Contoso. You have been evaluating the options for Public Folder migrations to Exchange Online that are available using Microsoft tools. Which of the following are valid options that you can use? (Select 3)

- ☐ Migrating Legacy Public Folders to Exchange Online Mailboxes
- ☐ Migrating Legacy Public Folders to Exchange Online Public Folders
- ☐ Migrating Modern Public Folders to Exchange Online Public Folders
- ☐ Migrating Public Folders to SharePoint Online
- ☐ Migrating Public Folders to Microsoft 365 Groups

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Answers

Multiple choice

Item 1. What Exchange Online Migration option migrates all Exchange on-premises mailboxes in one run?

■ Cutover Exchange Migration

☐ Staged Exchange Migration

☐ Full Hybrid Configuration

☐ PST migration

Multiple choice

Item 2. What Exchange Online Migration option requires Azure AD Connect for directory synchronization and provides full coexistence between Exchange on-premises and an Exchange Online organization?

☐ Minimal Hybrid Configuration

☐ Cutover Exchange Migration

☐ Staged Exchange Migration

■ Full Hybrid Configuration

Multiple choice

Item 3. What Exchange Online Migration tool connects to personal folder data files?

☐ Cutover Exchange Migration

☐ Staged Exchange Migration

■ PST migration

☐ IMAP migration

Multiple choice

Item 4. Which of the following migration options should be used to migrate mailboxes that are currently hosted on-premises on Exchange 2013 to Exchange Online using the Outlook Anywhere protocol?

☐ Minimal Hybrid Configuration

■ Cutover Exchange Migration

☐ Staged Exchange Migration

☐ Full Hybrid Configuration

☐ IMAP migrationn

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Multiple choice

Item 5. You are the messaging administrator for Contoso. You plan to migrate your Exchange 2016 mailboxes to Exchange Online. During the migration, you want the Outlook profiles to update automatically, and you do not need Exchange coexistence. Which of the following migration options would accomplish this migration task with the least amount of effort?

- ■ Minimal Hybrid Configuration
- ☐ Cutover Exchange Migration
- ☐ Staged Exchange Migration
- ☐ Full Hybrid Configuration
- ☐ IMAP migrationn

Checkbox

Item 6. Which of the following Exchange Online Migration options require a manual update for each user's Outlook profile? (Select 3)

- ■ Cutover Exchange Migration
- ☐ Minimal Hybrid Configuration
- ■ Staged Exchange Migration
- ■ IMAP migration
- ☐ Full Hybrid Configuration

Checkbox

Item 7. Which of the following are valid FastTrack options? (Select 2)

- ☐ Core offboarding
- ■ Core onboarding
- ☐ Core migration
- ■ Service onboarding and migration
- ☐ Service offboarding and migration

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Multiple choice

Item 8. Which of the following statements about the supported source messaging systems in Microsoft FastTrack is correct?

- ☐ Only a single or multiple Active Directory forests with a single or multiple Exchange organizations or Novell GroupWise 7.0.4 or later
- ■ A single or multiple Active Directory forests with a single or multiple Exchange organizations, IBM Domino 7.0.3 or later, Novell GroupWise 7.0.4 or later
- ☐ Only single or multiple Active Directory forests with a single or multiple Exchange organizations and IBM Domino 7.0.3 or later
- ☐ Only a single or multiple Active Directory forests with a single Exchange organization, IBM Domino 7.0.3 or later and Novell GroupWise 7.0.4 or later

Checkbox

Item 9. Which of the following are limitations when doing an IMAP migration? (Select 3)

- ☐ Reply to messages that were migrated is not possible
- ☐ You can migrate a maximum of 100,000 mail items from a user's mailbox
- ■ You can only migrate message items (no contacts or calendar items)
- ■ You can migrate a maximum of 500,000 mail items from a user's mailbox
- ■ The maximum size of a single message item is 35 MB
- ☐ No delta-synchronization is available for the migration batch

Checkbox

Item 10. You are the messaging administrator for Contoso. You plan a cutover migration to Exchange Online. Which of the following statements do you need to consider for your planning? (Select 4)

- ■ Your current Exchange environment must run Exchange 2003 or later.
- ■ If directory synchronization is enabled, it must be disabled in your Office 365 tenant.
- ■ Exchange ActiveSync devices need to be manually reconfigured.
- ☐ The migration option supports migrations larger than 2,000 mailboxes.
- ■ Security Groups cannot be migrated.
- ☐ Your current Exchange environment must run Exchange 2003 or Exchange 2007.

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Checkbox

Item 11. You are the messaging administrator for Contoso. You plan to migrate to Exchange Online using the staged migration option. Which of the following issues do you need to consider to have a successful migration? (Select 3)

- ☐ If directory synchronization is enabled, it must be disabled in your Microsoft 365 tenant.
- ■ Exchange ActiveSync devices need to be manually reconfigured.
- ■ The migration option supports migrations larger than 2,000 mailboxes.
- ☐ Security Groups cannot be migrated.
- ■ Your current Exchange environment must run Exchange 2003 or Exchange 2007.

Checkbox

Item 12. You are the messaging administrator for Contoso. Your users have a lot of local data stored in their PST files in Outlook. You are evaluating the options that will allow you to import the PST data to Microsoft 365. Which of the following options are valid transfer options? (Select 2)

- ☐ Ship data using a physical floppy disk
- ■ Ship data on a physical hard drive
- ■ Upload data over the network
- ☐ Upload data using OneDrive

Checkbox

Item 13. When using the Microsoft 365 Import service to upload PSTs, which mailbox types are valid targets for the data import? (Select 3)

- ■ Active mailboxes
- ■ Archive mailboxes
- ☐ Site mailboxes
- ■ Inactive mailboxes
- ☐ Microsoft 365 Groups

Checkbox

Item 14. Which of the following are valid limits for Public Folders in Exchange? (Select 2)

- ☐ You can have up to 10,000 Public Folders in your Public Folder hierarchy.
- ■ A public folder mailbox can have a maximum size of 100 GB.
- ■ You can have up to 1,000 public folder mailboxes.
- ☐ The maximum size of a single Public Folder is 30 MB.

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Checkbox

Item 15. You are the messaging administrator for Contoso. You have been evaluating the options for Public Folder migrations to Exchange Online that are available using Microsoft tools. Which of the following are valid options that you can use? (Select 3)

- ☐ Migrating Legacy Public Folders to Exchange Online Mailboxes
- ■ Migrating Legacy Public Folders to Exchange Online Public Folders
- ■ Migrating Modern Public Folders to Exchange Online Public Folders
- ☐ Migrating Public Folders to SharePoint Online
- ■ Migrating Public Folders to Microsoft 365 Groups

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Module 12 Deploying and Troubleshooting a Hybrid Environment

Deploying and Managing an Edge Transport Server

Lesson Introduction

Edge Transport servers handle all inbound and outbound Internet mail flow by providing mail relay and smart host services for your Exchange organization. Agents running on the Edge Transport server provide additional layers of message protection and security. These agents provide protection against spam and apply transport rules to control mail flow. All these features work together to help minimize the exposure of your internal Exchange environment to threats from the Internet.

In this lesson you will learn the key areas to plan for Edge Transport servers. This includes the infrastructure requirements as well as EdgeSync.

After this lesson, you will be able to:

- Describe the purpose and functionality of Edge Transport servers
- Explain the infrastructure requirements for Edge Transport servers
- Describe EdgeSync
- Plan for message flow with an Edge Transport server

Overview of the Edge Transport server

The Edge Transport server provides a secure SMTP gateway for all incoming and outgoing Internet email in an organization. As an SMTP gateway, the primary role of the Edge Transport server is to maintain a secure border between external SMTP mail and internal Exchange Servers and perform anti-spam protection. You also can use the Edge Transport server to apply messaging policies to messages that are sent through the Internet.

The Edge Transport server was developed primarily for the medium-to-large business market to replace an existing SMTP gateway or smart host. You should evaluate the Edge Transport features against your

existing solution, and then decide if your organization would benefit from using Edge Transport servers. Using Edge Transport servers is not a mandatory requirement for Exchange Server organizations.

The Edge Transport server in Exchange Server 2019 does not provide a graphical management interface. You can only configure and manage it using the Exchange Management Shell.

Note: If your organization's security requirements dictate that all SMTP emails must be relayed to Exchange Online using an SMTP gateway in your organization's perimeter network, you must consider the Edge Transport server as the only Microsoft fully-supported option to connect your Exchange organization to Microsoft 365.

Edge Transport server functionality

The Edge Transport server provides the functionalities listed in the following table.

Feature	Description
Internet message delivery	The Edge Transport server accepts all email coming into the Exchange organization from the Internet as well as from servers in external organizations. The Edge Transport server routes all accepted inbound messages to Exchange servers inside the organization. It also routes all outbound messages to the Internet.
Antispam protection	The Edge Transport role in Exchange Server helps prevent spam messages from reaching your organization's users by using a collection of agents that provide different layers of spam filtering and virus protection. It uses these agents to filter email messages based on the source or destination recipients, source SMTP server, attachments, and message contents. Exchange Server 2019's Edge role does not include anti-malware agents. If you do not wish to rely upon Exchange 2019 Mailbox servers to perform anti-malware protection, you must use either Microsoft products such as Exchange Online Protection, or 3rd party software that integrates with Exchange Server 2019 to provide anti-malware protection. The perimeter network is the ideal location for antivirus protection because it's the entry point from the Internet to the company network.

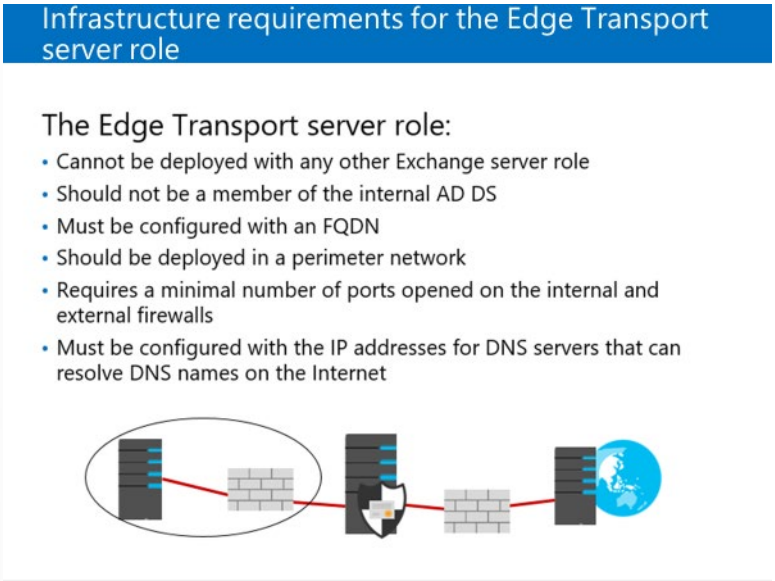
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Feature	Description
Edge Transport rules	Edge Transport rules control the flow of messages that are sent to, or received from, the Internet. These rules function much like Transport rules. Edge Transport rules apply actions to messages that meet specified conditions. The Edge Transport rule conditions are based on data such as specific words or text patterns in the message subject, body, header, or from address, the spam confidence level (SCL), or attachment type. Actions determine how the message is processed when a specified condition is true. Possible actions include quarantining a message, dropping or rejecting a message, appending additional recipients, or logging an event.
Address rewriting	Address rewriting enables SMTP address modification for any of your organization's message senders or recipients. Address rewriting can be useful in scenarios where an organization wants to hide internal domains, to enable multiple organizations to appear as a single organization, or to integrate services that another company provides to an organization.

Infrastructure requirements for the Edge Transport server

The Edge Transport server role differs from the Exchange Mailbox role because you can install the Edge Transport Server role on Windows Server 2019 computers that are not members of the internal Active Directory. This configuration makes it easier and more secure to deploy Edge Transport servers in a perimeter network.

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Edge Transport server deployment considerations

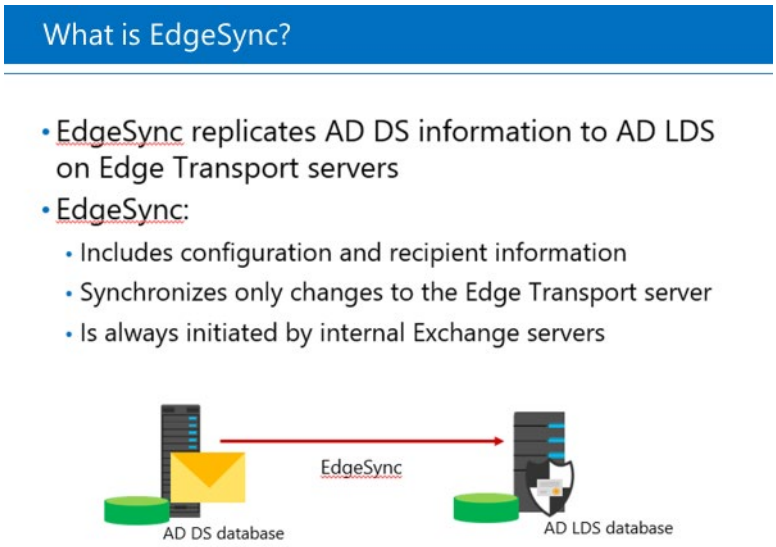
When you plan to deploy Edge Transport servers, you must consider the following factors:

- You cannot combine the Edge Transport server with any other Exchange Server role. To provide increased security, you must install the Edge Transport server on a separate computer, which can be virtual or physical.
- You should not install the Edge Transport server on a computer that is a member of the internal Active Directory, but you can install it in a perimeter network forest or workgroup.
- The computer running the Edge Transport server must have an FQDN.
- You should not allow Active Directory communications through the firewall that protects the internal network from the perimeter network. This can cause security issues such as allowing an unauthorized user to retrieve your email addresses directly from Active Directory and then use them for spam. Instead, the Edge Transport server uses Active Directory Lightweight Directory Services (AD LDS) to store configuration and recipient information. AD LDS does not contain all the information from Active Directory but synchronizes only the required information such as email addresses.
- You should deploy the Edge Transport server on a perimeter network to ensure network isolation from both the internal network and the internal Exchange servers. This configuration provides a higher level of security for the rest of the Exchange Server organization, which should be located behind a firewall on a separate internal network.
- You must configure the external firewall on the perimeter network to allow inbound and outbound SMTP traffic to and from the Edge Transport server. The internal firewall must allow SMTP traffic between the Edge Transport server and one or more internal Exchange 2016 servers (Port 25/TCP). The firewall must also allow outbound traffic towards the perimeter network for synchronization from Active Directory to AD LDS (Port 50636/TCP).
- The firewall configuration required for Edge Transport servers is greatly simplified because the server does not need to be an internal domain member.
- If the Edge Transport server routes email directly to the Internet, you must configure the server with the IP addresses for DNS servers that can resolve DNS names on the Internet.

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Overview of EdgeSync

EdgeSync (as part of an Edge Subscription) is a process that replicates information from Active Directory to AD LDS on Edge Transport servers. Edge Transport servers are not joined to the internal Active Directory domain. Therefore, they cannot directly access the Exchange Server organization configuration or recipient information that is stored in Active Directory. The EdgeSync feature enables the shared information to replicate from Active Directory to AD LDS.



You can deploy Edge Transport servers without using EdgeSync; however, using EdgeSync can decrease the effort needed to administer the Edge Transport servers. Active Directory contains much of the configuration information required by the Edge Transport server. For example, if you configure accepted domains in your Exchange admin center or Exchange Management Shell, these accepted domains can replicate automatically to the Edge Transport servers. To enable any filtering or transport rules that are based on recipients, you must implement EdgeSync to replicate the recipient information to AD LDS.

Important: When you deploy Edge Transport servers, it's recommended that you also deploy EdgeSync.

How to configure EdgeSync

You need to manually configure EdgeSync between Edge Transport servers and Mailbox servers. Additionally, you can only deploy Edge Transport servers to one Active Directory site at a time.

Configuring EdgeSync on each Edge Transport server requires the following steps:

1. Create an Edge subscription file on an Edge Transport server.
2. Import an Edge subscription file on a Mailbox server.
3. Start and verify the EdgeSync process.

Note: To verify that EdgeSync is configured correctly, you should run the following command:

```
Test-EdgeSynchronization -FullCompareMode
```

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Information replicated by EdgeSync

After you enable EdgeSync, the EdgeSync process establishes connections between an Exchange server and the Edge Transport server, and then synchronizes configuration and recipient information between Active Directory and AD LDS. Once the initial synchronization completes, only the changes are synchronized to the Edge Transport server.

Important: The internal Exchange servers - and not the Edge Transport servers - always initiate EdgeSync replication. EdgeSync replication traffic always encrypts using secure LDAP.

During synchronization, EdgeSync replicates the following data from Active Directory to AD LDS:

- Accepted domains
- Remote domains
- Recipients (hashed)
- Safe Senders Lists (hashed)
- Blocked Senders Lists
- Send connector configuration
- List of send and receive domains used in domain secure communications with partners
- List of SMTP servers listed as internal in your organization's transport configuration
- List of Mailbox servers in the subscribed Active Directory site (for dynamic connector generation)

Note: The recipient and the safe senders are hashed by using a one-way hash, which prevents an attacker from retrieving recipient information from the Edge Transport server.

Additional reading. For more information, see the following article on **Edge Subscriptions**[1].

Message flow with an Edge Transport server

The primary function of the Edge Transport server is to secure both inbound and outbound Internet email. After you configure an Edge subscription between your organization's Exchange servers and the Edge Transport servers in the perimeter network, both inbound and outbound Internet email is enabled.

Default SMTP connectors

By default, five SMTP Receive connectors are created per server when you install an Exchange Server in an Exchange organization. When you install an Edge Transport server, just one SMTP Receive connector is created. When you enable an Edge subscription, two additional SMTP Send connectors are created as listed in the following table.

1 https://aka.ms/AA3qvok

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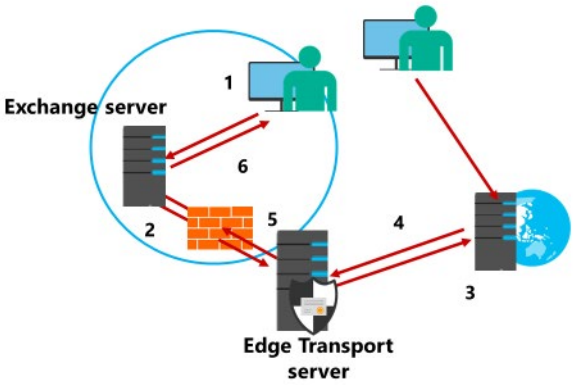
Connector name	Connector type	Description
EdgeSync - Inbound to sitename	SMTP Send connector	Created on the Edge Transport server by Edge subscription. Created in AD DS, and then replicated to the Edge Transport server by EdgeSync. Settings such as smart hosts and address space are defined by the Edge subscription. The connector is configured to use an address space that includes all internal domains.
EdgeSync – sitename to Internet	SMTP Send connector	Created on the site that is defined by the Edge subscription. Created in AD DS, and then replicated to the Edge Transport server by EdgeSync. Source server is the Edge Transport server on which the Edge subscription is enabled. Address space of SMTP:*Uses DNS to locate SMTP servers on the Internet.

Default message transfer

After you enable EdgeSync, email flows through the Exchange Server organization in a predefined order, as indicated in the following diagram.

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Message flow with an Edge Transport server



The following steps explain a default message transfer between Exchange Server and the Internet using Edge Transport Servers:

1. A user submits a message to the Mailbox server. If the message recipient is outside the organization, the Exchange server retrieves the message, and then categorizes it for delivery. The Exchange server determines that to send email to the Internet it must use the EdgeSync sitename to the Internet Send connector. It locates the Edge Transport server that is configured as the bridgehead server for the connector.

   Note: If the Exchange server that retrieves the message is not in the AD DS site that is included in the Edge subscription, or it has not been added to the subscription, the Exchange server forwards the message to an Exchange server that is in the subscribed site.

2. The Exchange server forwards the message to the Edge Transport server, which sends the email message to the Internet by using the EdgeSync sitename to Internet Send connector.

3. For inbound messages, the sending SMTP connector connects to the Edge Transport server. The Edge Transport server accepts this connection by using the Default Internal Receive connector SERVER-NAME, which is configured to accept anonymous connections on port 25 from all IP addresses. The Edge Transport server applies all antivirus and spam-filtering rules.

4. If the message is accepted, the Edge Transport server uses the EdgeSync Inbound to sitename connector to forward the message to a Hub Transport server that is configured to accept Internet messages.

5. The Exchange server uses the Default SERVERNAME connector to receive the message.

6. The Exchange server then delivers the message to the appropriate Mailbox database.

Note: You can modify the default message flow by creating additional SMTP connectors. For example, you may need to create a new SMTP Send connector to send email to a specific destination domain. You can do this by creating a new Send connector, and then configuring the destination domain name as the address space for the connector. You must then configure the connector to support the unique message-routing requirements for messages sent to the domain.

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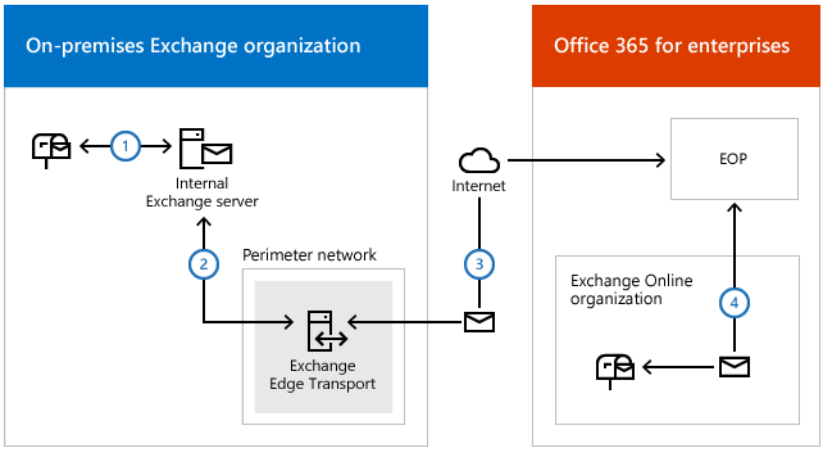
Mail flow for hybrid deployments

When you deploy an Edge Transport server in a hybrid deployment, Exchange Online, through the Exchange Online Protection service, connects to your Edge Transport server to deliver messages. The Edge Transport server (or more typically, servers, in a highly available deployment) then delivers messages to the on-premises Exchange Mailbox servers, which in turn delivers the message to the recipient's mailbox.

The following process describes the path messages take between an on-premises organization and Exchange Online when there is an Edge Transport server deployed. Messages from the on-premises organization to recipients in the Exchange Online organization are sent from the internal Exchange server:

1. Messages from the on-premises organization to recipients in the Exchange Online organization are sent from a mailbox on an internal Exchange server.
2. The Exchange server sends the message to an Edge Transport server running a supported version and release of Exchange.
3. The Edge Transport server sends the message to EOP.
4. EOP delivers the message to the Exchange Online organization.

Messages sent from the Exchange Online organization to recipients in the on-premises organization follow the reverse route.



Additional reading. For more information, see the following article on **Edge Transport servers with hybrid deployments**[2].

2 https://aka.ms/AA3qvpd

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Configuring a Hybrid Deployment using the HCW

Lesson Introduction

Configuring a hybrid deployment is the first step for your Exchange organization, regardless of whether you want to connect your Exchange on-premises and Exchange Online organizations for long-term coexistence or as part of a cloud migration strategy.

In this first lesson you will learn about the requirements and best practices to configure a hybrid deployment.

After this lesson, you will be able to:

- Describe the prerequisites to run the Microsoft 365 Hybrid Configuration Wizard
- Explain best practices for implementing a hybrid deployment

Preparing to run the HCW

If you're configuring an Exchange hybrid deployment, you should plan to spend at least a couple of days on this process, depending on the complexity of your organization's deployment due to factors such as DNS replication and certificate verifications.

Hybrid deployment requirements

You should perform the following high-level steps to configure a hybrid deployment:

1. **Sign up for Microsoft 365.** Register your Microsoft 365 enterprise tenant if you don't have one yet.
2. **Register your custom domains with Microsoft 365.** Register with Microsoft 365 the SMTP domains that you want to use for Exchange Online. Each domain needs to be verified with a DNS service (TXT) resource record, so this step might take a while.
3. **Remediate identity within Active Directory.** Use the **IDFix tool**[3] to identify and remediate issues with your local Active Directory, such as invalid User Principal Names or duplicate SMTP addresses.
4. **Deploy Azure AD Connect.** Activate directory synchronization in Microsoft 365, and then deploy the Azure AD Connect tool for Active Directory synchronization. Activating directory synchronization may take a while to replicate the information throughout Microsoft 365; therefore, you should first install the Azure AD Connect tool once Microsoft 365 is activated correctly.
5. **Ensure Exchange is properly deployed.** Verify Exchange is working properly and then run the **Exchange Deployment Assistant**[4].

   If you plan to implement Exchange Classic Hybrid for an Exchange Modern Hybrid deployment (where you install the Hybrid Agent), you should perform the following additional steps:

6. **Install Trusted Digital Certificates for HTTPS services.** You must install and assign Exchange services to a valid digital certificate that you purchase from a trusted public certification authority (CA). The easiest way to verify that Exchange Online trusts your certificate is to run the Microsoft Remote Connectivity Analyzer against the Exchange Server on-premises environment.

3 https://aka.ms/AA3rk8z
4 http://aka.ms/ExDeploy

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7. **Publish your Exchange Server**. Ensure that the correct certificates are installed, that the Exchange Server is correctly published in the firewall, and that AutoDiscover is working. The easiest way to verify the AutoDiscover and Mailbox server configuration from the Internet is to use the **Microsoft Remote Connectivity Analyzer**[5].

Once you followed all these steps, you are ready to run the Hybrid Configuration Wizard in your Exchange organization.

Hybrid deployment prerequisites

Before you can configure a hybrid deployment with Exchange Server using the HCW, you must meet the following prerequisites:

- **Exchange Server on-premises.** You can configure a hybrid deployment for an on-premises organization that is based on Exchange Server 2010 or later and you don't need to implement the latest version of Exchange Server to enable Hybrid. For Exchange organizations that include Exchange Server 2007, you must install at least one Exchange Server 2010 with the Client Access, Hub Transport and Mailbox roles or Exchange Server 2013 with the Client Access and Mailbox server roles in the on-premises organization to be able to run the HCW.
- **Microsoft 365 for Enterprise or Business.** You must have a Microsoft 365 for Enterprise or Business tenant administrator account and user licenses available on the tenant service.
- **Register custom domains.** You must register any custom domains that you want to use in the hybrid deployment with Microsoft 365. You can register the domains by using the Microsoft 365 admin center portal.
- **Active Directory synchronization.** You must deploy the Azure AD Connection in the on-premises organization to synchronize Active Directory to Microsoft 365.
- **AutoDiscover DNS records.** If you aren't deploying a Modern Hybrid deployment, you must configure the AutoDiscover DNS records for your existing SMTP domains on the Internet to point to an on-premises Exchange Server 2010 server or later. As a general rule, always point AutoDiscover DNS records to the latest version of Exchange Server you run in your environment.
- **Trusted Digital Certificate.** You must install and assign Exchange services to a valid digital certificate that you purchase from a trusted public certification authority (CA) if you plan to use Exchange Classic Hybrid. If you use self-signed certificates for Exchange Servers, you must use Exchange Modern Hybrid because Exchange Classic Hybrid does not support self-signed certificates.
- **Ensure Mail Flow is configured correctly to support Hybrid Mail flow**. If you are planning a full Hybrid deployment, you must enable SMTP connectivity between either Exchange Mailbox Servers or Edge Servers and Exchange Online Protection IP addresses, allowing TCP Port 25 connectivity inbound and outbound, secured by a trusted digital certificate.
- **Edge Synchronization (for Edge Transport).** If the on-premises organization has Edge Transport servers and you want to configure the Edge Transport servers for hybrid secure mail transport, you must configure Edge Synchronization before you configure the hybrid environment. With Edge Synchronization, you can automatically configure the Edge Transport servers from the Exchange Admin Center.
- **Unified Messaging-enabled (UM) mailboxes.** If you have UM-enabled mailboxes and you want to move them to Microsoft 365, you need Lync Server 2010, Lync Server 2013, or Skype for Business Server 2015 or later integrated with your on-premises telephony system or Skype for Business Online

5 http://aka.ms/RCA

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integrated with your on-premises telephony system or a traditional on-premises PBX or IP-PBX solution.

Additional reading. For more information, see the following article on **Hybrid deployment prerequisites**[6].

Best Practices for implementing a hybrid deployment

Messaging administrators should consider the following best practices when implementing a hybrid deployment:

- **Use the Hybrid Configuration Wizard (HCW) app**. You configure a hybrid deployment in two ways: either manually or by using the HCW app. It is highly recommended that you use the HCW app because the wizard automatically handles all necessary configuration settings for you.
- **Understand the requirements of Minimal Hybrid, Exchange Classic, and Exchange Modern Hybrid**. To test the Exchange Server on-premises environment from the Internet, use the **Microsoft Remote Connectivity Analyzer**[7].
- **Understand why your organization wants to implement a hybrid deployment**. Do not try to use "everything" if your organization only wants to move archive mailboxes to Exchange Online.
- **Start slowly, and then speed up when everything works**. At the beginning, move mailboxes only for people who can acceptably work with a short outage. After you gain confidence that the hybrid deployment works reliably, move the other mailboxes. Always move test mailboxes first, and then consider moving production mailboxes.
- **Do not change the MX resource record at first.** Change it only after you know that the hybrid deployment works.
- **Use multiple MRS Proxy endpoints where possible.** The Mailbox Replication service (MRS) has a proxy endpoint that is required for cross-forest mailbox moves and remote move migrations between your on-premises Exchange organization and Microsoft 365. Depending on your organization's Internet connection, it is wise to have multiple MRS proxy endpoints if you also have multiple Internet inbound connections and certificates.

Additional reading. For more information, see the following article on **Office 365 Mail Flow best practices**[8].

6 https://aka.ms/AA3r3c5
7 http://aka.ms/RCA
8 https://aka.ms/O365MailFlowBP

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Implementing Advanced Hybrid Functionality

Lesson Introduction

Once you implement a hybrid deployment to connect your on-premises Exchange environment to Exchange Online, you must then manage the deployment. However, you can implement additional advanced functionality that would not be possible without a hybrid deployment.

In this lesson, you will learn how to manage a hybrid deployment and implement advanced hybrid functionality. We will cover the features that require a successful hybrid deployment such as Public Folder coexistence or OneDrive for Business attachment storage for on-premises mailboxes.

After completing this lesson, you will be able to:

- Manage a hybrid deployment
- Describe when you must configure Public Folder coexistence with Microsoft 365
- Explain how to configure Oauth for a mixed Exchange environment
- Describe how to configure OneDrive for Business attachments for on-premises mailboxes

Managing a Hybrid Deployment

A hybrid deployment includes special management and monitoring requirements. This is due to the complex configuration of hybrid deployments that automatically exchange information between the Exchange Server on-premises organization and Exchange Online. The complexity of this configuration is not visible to end users, but you need to monitor specific areas of the deployment to ensure that it functions properly.

Monitoring a Hybrid Deployment

You should perform the following tasks to monitor a hybrid deployment:

- **Ensure that Azure AD Connect is running reliably**. Azure AD (AAD) Connect synchronizes the underlying Active Directory that supports the Exchange Server on-premises environment with Exchange Online. For example, if you configure a personal archive for a mailbox that is stored in an Exchange database on-premises, AAD Connect synchronizes the properties of the mailbox so that Exchange Online recognizes the archive. If AAD Connect is not running, Exchange Online does not recognize the change and the user cannot use their archive. Microsoft 365 automatically monitors your Directory Synchronization activity, and it sends a message to the technical account if Directory Synchronization does not occur for a day.
- **Manage synchronization between Exchange Server on-premises and Exchange Online.** Use the Exchange Admin Center (EAC) of the on-premises Exchange Server environment to manage Exchange Server on-premises and Exchange Online recipients so that Directory Synchronization synchronizes them correctly. If you use the Exchange Admin Center to synchronize users, distribution lists, and contacts, keep in mind that synchronization occurs in one direction only—from the Exchange Server on-premises organization to Exchange Online. For example, if you create an on-premises user mailbox, Directory Synchronization creates the user mailbox in Exchange Online. But if you create a user mailbox in Exchange Online, Directory Synchronization does not synchronize or create the user mailbox in AD DS.
- **Monitor message routing between on-premises and Exchange Online.** Message routing between Exchange Server on-premises and Exchange Online is one of the most important factors that makes a

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hybrid deployment successful. Make sure that the messages flow successfully and do not queue somewhere. For this reason, it's recommended that you monitor the queues in the Exchange Server on-premises environment so that you can react quickly if messages queue for too long.

- **Use monitoring software to monitor the federated delegation.** Federated delegation is the basis for the information exchange between Exchange Server on-premises and Exchange Online. If federated delegation does not work correctly, users cannot retrieve any free/busy information, MailTips, or other information between the on-premises and cloud deployments. Consider testing federated delegation with the monitoring software so that you're notified immediately if federated delegation does not work. Also consider using the following test cmdlets:

```
Test-FederationTrust
Test-FederationTrustCertificate
Test-OrganizationRelationship
```

- **Verify the configuration by running the Microsoft Remote Connectivity Analyzer.** The Microsoft Remote Connectivity Analyzer (RCA) is a Microsoft tool that can verify your configuration, such as the Exchange Web Services or Exchange ActiveSync settings, and ensure that all settings are configured properly. This tool helps prevent issues that can easily be overlooked. Because a hybrid deployment uses those services to communicate between Exchange Online and on-premises, it's recommended that you occasionally run the RCA to verify that the configuration has not changed in any way.
- **Monitor the middle-tier components.** A hybrid deployment involves not only Exchange servers but other components as well, such as the new Hybrid Agent and firewalls. Therefore, you must ensure that these components function correctly by monitoring any middle-tier components that are involved in the deployment. These components include Active Directory Federation Services, third-party firewalls, and other products.

Updating a Hybrid Deployment

There are various situations when you must update your hybrid deployment by running the Hybrid Configuration wizard again, including:

- When you add an accepted domain to your on-premises Exchange environment.
- When you add a domain to your Microsoft 365 tenant.
- When you update your Exchange certificate.
- When you change your Exchange organizational settings (consider running OCT again, to match settings).

Configuring Public Folder Coexistence with Microsoft 365

A hybrid Exchange deployment also supports coexistence with on-premises Public Folders. This enables you to configure your cloud mailboxes to access Public Folders located on your local Exchange servers. This is required during migration if your organization uses Public Folders. There are two types of public folders:

- **Legacy Public Folders.** These are Public Folders on Exchange Server 2007 or 2010 that are located in a Public Folder database.
- **Modern Public Folders.** These are Public Folders on Exchange Server 2013 or later that are located in a Public Folder mailbox.

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Note: If not specifically mentioned, the configuration settings are for both Public Folder configurations.

How Public Folder coexistence works

For Public Folder coexistence with Microsoft 365, you must configure Exchange Online to access your on-premises Public Folders located on your Exchange servers. Outlook clients with mailboxes located on Exchange Online can access the on-premises Public Folders using the your local Autodiscovery process and the protocol Outlook Anywhere. The Public Folder proxy mailbox SMTP addresses will be returned by Exchange Online to the Outlook client. The client then uses the Autodiscover process to find and establish a direct connection to Public Folders hosted on the on-premises Exchange environment.

Public Folder coexistence requirements

You must satisfy the following requirements to configure Public Folder coexistence with Microsoft 365:

- AAD Connect must provide directory synchronization between your on-premises environment and Microsoft 365.
- A hybrid Exchange deployment is required to enable email forwarding between on-premises and Exchange Online as well as all other functionality.
- Legacy Public Folders (Exchange 2007/2010) or modern Public Folder (Exchange 2013 or later). Public Folder coexistence is supported for both legacy Public Folders and modern Public Folders.

Planning legacy Public Folder coexistence

When planning for legacy Public Folder coexistence for Exchange Server 2010, you must take into account the following considerations:

- Each server that hosts a Public Folder database requires the CAS role to be installed. You must verify if additional resources, such as memory or CPUs, are needed.
- You must create one mailbox database per Exchange server that holds Public Folder databases (that is one proxy Public Folder mailbox per Public Folder database server).

If your on-premises Exchange environment uses modern Public Folders, you do not need to plan the additional steps.

Public Folder coexistence prerequisites

You must meet the following prerequisites before configuring Public Folder coexistence in Exchange Online:

- The DNS records used for Autodiscover (for example, autodiscover.adatum.com) reference an on-premises end-point.
- Outlook Anywhere is enabled and functional on the on-premises Exchange servers.
- Implementing Public Folder coexistence may require you to fix conflicts during the import procedure. Conflicts can occur for a variety of reasons, such as non-routable email addresses assigned to mail-enabled Public Folders, conflicts with other users and groups in Microsoft 365, and other attributes.
- In Exchange 2010, you must be a member of the Organization Management or Server Management RBAC role groups to be able to create and configure mailbox databases and mailboxes.
- In Exchange Online, you must be a member of the Organization Management role group.

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- If your Public Folders are on Exchange 2010, then you must install the Client Access Server role on all mailbox servers that have a Public Folder database. This allows the Microsoft Exchange RpcClientAccess service to be running, which allows for all clients to access Public Folders.

Configuration steps for Public Folder coexistence

Once you have satisfied the prerequisites, you can configure Public Folder coexistence by performing the following steps:

NOTE: If you use Public Folders on Exchange 2013 or later, skip to Step 5. If you host Public Folders on Exchange 2010, start with Step 1.

1. Create an empty mailbox database on each Public Folder server by running the following command:

   ```
   New-MailboxDatabase -Server <pfservername> -Name <newdbforpf> -IsExcluded-
   FromProvisioning $true
   ```

   This command excludes the mailbox database from the mailbox provisioning load balancer, and it prevents new mailboxes from automatically being added to this database.

2. Enable AutoDiscover to return the proxy Public Folder mailboxes by running the following command:

   ```
   Set-MailboxDatabase <newdbforpfs> -RPCClientAccessServer <PFServer-Name_
   with_CASRole>
   ```

3. Create a proxy mailbox within the new mailbox database and hide the mailbox from the address book by running the following commands:

   ```
   New-Mailbox -Name <pfmailbox1> -Database <newdbforpf>
   Set-Mailbox -Identity <pfmailbox1> -HiddenFromAddressListsEnabled $true
   ```

   The SMTP of this mailbox is returned by AutoDiscover as the DefaultPublicFolderMailbox SMTP, so that when you resolve, the SMTP client can reach the legacy exchange server for Public Folder access.

4. Repeat the preceding steps for every Public Folder server in your organization.

5. The final step in this procedure is to configure the Exchange online organization and to allow access to the legacy on-premises Public Folders. It is important that the previously created proxy Public Folder mailboxes have been synchronized to Exchange Online by directory synchronization. To enable the exchange online organization to access the on-premises Public Folders, you point to all of the proxy Public Folder mailboxes that you created for Exchange Server 2010, or to your Public Folder mailboxes located on Exchange 2013 or later.

   You should run the following cmdlet in Exchange Online to configure this:

   ```
   Set-OrganizationConfig -PublicFoldersEnabled Remote -RemotePublicFolderMail-
   boxes PFMailbox1, PFMailbox2 ,…
   ```

Synchronize mail-enabled Public Folders

Because AADSync does not synchronize mail-enabled Public Folders to Exchange Online, all email enabled Public Folders must be synchronized manually using the **Sync-MailPublicFolders.ps1script**[9].

9 http://go.microsoft.com/fwlink/?LinkId=789470

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Configuring OAuth for a Mixed Exchange Environment

OAuth is required for some Exchange-related features, such as cross-premises discovery and automatic archive retention. The following situations are examples of when you must manually configure Oauth:

- Your Exchange organization includes Exchange 2007 or 2010 servers. The Hybrid Configuration Wizard (HCW) does not configure OAuth in this situation.
- The HCW app failed to configure OAuth successfully.

If you are not sure if Oauth is already configured in your environment, you can use the **Test-OAuthConnectivity** cmdlet to verify it. For example, run the following cmdlet in your Exchange Management Console connected to your Exchange Server to verify Oauth is correctly configured and working:

```
Test-OAuthConnectivity -Service EWS -TargetUri https://outlook.office365.com/
ews/exchange.asmx -Mailbox <on-prem>
```

Manually configuring OAuth for your Exchange organization

You should perform the following steps to manually configure OAuth:

1. Create an authorization server object for your Exchange Online organization.
2. Enable the partner application for your Exchange Online organization.
3. Export the on-premises authorization certificate.
4. Upload the on-premises authorization certificate to Azure Active Directory ACS.
5. Register all hostname authorities with Azure AD.
6. Create an IntraOrganizationConnector from your on-premises organization to Microsoft 365.
7. Create an IntraOrganizationConnector from your Microsoft 365 tenant to your on-premises Exchange organization.
8. Configure an AvailabilityAddressSpace for any pre-Exchange 2013 SP1 servers.
9. Test your OAuth implementation using the Test-OAuthConnectivity cmdlet.

Additional reading. For more information, see the following article on **Configure OAuth authentication between Exchange and Exchange Online organizations**[10].

Configuring OneDrive for Business Access for On-Premises Mailboxes

One of features of Exchange Server 2016 or later is the ability to configure on-premises mailboxes that can store attachments in OneDrive for Business in Microsoft 365. This functionality preserves space on your on-premises Exchange servers because user attachments are not stored in the mailbox but in OneDrive with a link in the e-mail to the original attachment, should the user choose to upload them.

This functionality requires the following prerequisites:

- The user's mailbox must be located on an Exchange 2016/19 server.

10 http://go.microsoft.com/fwlink/?LinkId=789471

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- Your on-premises Exchange environment must be configured as a hybrid deployment.
- OAuth must be configured between your on-premises Exchange environment and Exchange Online. You can test this by running the following command:

```
Test-OAuthConnectivity -Service EWS -TargetUri https://outlook.office365.com/
ews/exchange.asmx -Mailbox <on-prem>
```

Once you have verified the prerequisites, you must perform the following steps to enable the ability to store attachments from an on-premises mailbox on OneDrive for Business:

1. Decide which OWA Mailbox Policy you want to use for this configuration. The following example uses the Default policy.
2. Run the following command:

```
Set-OwaMailboxPolicy Default -InternalSPMySiteHostURL https://<tenant>-my.
sharepoint.com -ExternalSPMySiteHostURL
```

3. Run the following command:

```
Set-CASMailbox <user mailbox> -OwaMailboxPolicy Default
```

Optionally, you can run **Restart-WebAppPool MSExchangeOWAAppPool** on your local Exchange 2016/19 server so that the changes are effective immediately. **Note:** This command will disconnect all your users from OWA.

Note: If you are running a multi-geo Microsoft 365 tenant, you may need to create OWA Mailbox Policies for users aligned to the preferred location in which their data is stored and provide the specific geo-related URL when configuring this feature.

Additional reading. For more information, see the following article on **Configure document collaboration with OneDrive for Business and Exchange on-premises**[11].

11 https://aka.ms/AA3qkx3

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Lab10-Ex1-Configure Your Hybrid Deployment

Lab10-Ex1-Configure Your Hybrid Deployment

In this lab you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. Adatum has decided to transition from their current on-premises Exchange Server 2019 deployment to a hybrid deployment that also utilizes Microsoft 365 Exchange Online.

In the prior lab, you prepared Azure Active Directory to support the hybrid synchronization between Adatum's on-premises Exchange Server 2019 environment and their cloud-based Exchange Online deployment. In this lab, you will set up Exchange for a hybrid deployment. This will require that you:

- Run the Hybrid Configuration Wizard (HCW) to create your hybrid deployment
- Configure Exchange admin center (EAC) settings to accommodate cloud and on-premises users within the same domain
- Configure the Outbound Connector from Microsoft 365 to your Exchange Server

To set up Exchange for a hybrid deployment, you must first run the Hybrid Configuration Wizard on your Exchange Server (LON-EX1). Once your hybrid deployment is installed, you will then test your hybrid deployment in the next exercise to verify that it's functioning properly.

See your VM lab environment for the lab instructions.

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Troubleshooting Hybrid Deployments

Lesson Introduction

Troubleshooting a hybrid deployment can be a complex task as it involves a solid understanding of all the components involved.

In this lesson you will be introduced to troubleshooting techniques for a hybrid deployment. You will learn how to troubleshoot directory synchronization issues including pass-through authentication (PTA) and single sign-on, Exchange transport, and client access troubleshooting as well as mailbox replication service troubleshooting.

After this lesson, you will be able to troubleshoot:

- Directory synchronization
- Pass-Through Authentication and Single Sign-On
- Transport with Exchange Online
- Client Access in Coexistence
- Mailbox Replication Service

Troubleshooting Directory Synchronization

Once you have established directory synchronization, you may occasionally see issues with the synchronization or with individual objects syncing. You must know how to troubleshoot and resolve common sync issues to maintain a healthy sync environment. In this topic, you'll examine several common sync issues and how to troubleshoot and resolve them.

Sync fails completely

When your directory synchronizations are not working, it's often the result of a backend configuration or communication problem, and less likely a problem with user or group objects. The following list identifies some common issues that cause synchronization to fail:

- **The sync service is stopped.** Synchronization relies on the Microsoft Azure AD Sync (ADSync) service. If the service isn't running, then the sync won't run. Use your standard troubleshooting steps to validate the status of the service; that is, check the account that it uses to be sure the password hasn't expired, check to see if the service account is still valid, enabled, and not locked out, and try to start the service. The service depends on the Windows Management Instrumentation service, so you should check that service too. Then, check the System event log for errors related to the service, especially when you try to start it and it fails.
- **Connector credentials are invalid.** You can use the Synchronization Services Manager to look at the Connectors tab. You should have at least one connector for your on-premises Active Directory forest and one connector for Azure Active Directory (represented by the *.onmicrosoft.com domain). Each of the connectors has credentials specified. If one of the accounts gets locked out, disabled, or has a similar problem, then the entire sync might fail to run.
- **Configuration is set incorrectly.** For configuration issues related to syncs, the most common issues include:
  - A server is enabled for Staging mode. If this occurs, it doesn't sync
  - The scheduler is suspended. In this case, the sync won't run on a schedule.

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- The sync cycle interval is set to a large number of hours. The default is to sync every 30 minutes, but if the configuration is set to sync every 90 hours, it might appear to not be syncing at all. You can review your existing settings by running the Get-ADSyncScheduler command on the sync server. The following table displays sample output after running this command.

Sync configuration parameter	Configuration parameter value	Notes
AllowedSyncCycleInterval	00:30:00	Sync every 30 minutes
CurrentlyEffectiveSyncCycleInterval	00:30:00	Sync every 30 minutes
CustomizedSyncCycleInterval		Null by default
NextSyncCyclePolicyType	Delta	Next sync is delta sync
NextSyncCycleStartTimeinUTC	5/30/2017 3:35:20 PM	Next sync date/time in UTC
PurgeRunHistoryInterval	7.00:00:00	Keep run history for 7 days
SyncCycleEnabled	True	Sync is enabled
MaintenanceEnabled	True	Built-in process to update certificates & keys and purge logs
StagingModeEnabled	False	Server is in active mode
SchedulerSuspended	False	Scheduler is active
SyncCycleInProgress	False	Sync is not in progress

Single user object does not sync

When you have a functional sync but you have a single object that doesn't sync, it typically means that there's a problem with that object. The following are some common issues that result a single object not syncing:

- **There is a duplicate object in Azure Active Directory.** A duplicate object is any object that has one or more attribute values that match the attribute values of the user object that isn't syncing. For example, imagine that you have two users. One is named Ana Cantrell. The other is named Anastasia Rojas. Both go by the first name "Ana". Ana Cantrell has a user account in Azure Active Directory. Anastasia doesn't. Ana's user account in Azure Active Directory has a user principal name (UPN) of Ana@Adatum.com. When you try to sync Anastasia's account, it attempts to use her on-premises UPN of Ana@Adatum.com. However, that UPN already exists, so the sync of the object fails.

  Or, if you are running a new version of Azure Active Directory, the object will sync but it will require you to fix the duplicate data before using it. You can fix these scenarios by using unique values for the core attributes such as proxyAddresses and UserPrincipalName.

- **The object is filtered out.** When you sync objects, you can either sync all objects, or you can sync some objects. There are a few ways to sync a partial set of objects. You can use attribute-based filtering, group-based filtering (although only supported in non-production environments), and OU-level filtering. If an object isn't syncing and you don't see errors, you should look at the filtering configuration to find out if the object is in scope. With attribute-based filtering, you can view the cloudFiltered attribute. If it's set to True, then the object won't sync. If the attribute is empty (null), then it will sync.

- **Invalid characters are present on the object.** In this scenario, invalid characters are present on the user object. These invalid characters are often represented in a phone number. You can use Microsoft IdFix tool to scan and fix these attribute issues prior to the sync.

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Troubleshooting Pass-Through Authentication and Single Sign-On

When Azure AD Connect pass-through authentication (PTA) or single sign-in aren't working or are in a degraded state, users will be quick to report it. And it's important to be able to quickly troubleshoot and resolve the issues to maintain a good user experience. In this topic, we'll look at pass-through authentication troubleshooting and single sign-on troubleshooting.

Pass-through authentication troubleshooting

The pass-through authentication (PTA) feature enables your on-premises domain controllers to authenticate users when they access Azure cloud services such as Exchange Online. When pass-through authentication isn't working, users are still prompted to input their credentials, but often, there will be a long pause during validation, and it will fail with an error message. Let's walk through a couple of common pass-through authentication failure scenarios:

- **All users are unable to sign in.** In this scenario, the problem is typically related to the backend components or connectivity. Pass-through authentication relies on a service named Microsoft AAD Application Proxy Connector (WAPCSvc). If it isn't running, then pass-through authentication will be unavailable. If port 443 is not enabled for outbound from the PTA server to Microsoft Azure, then pass-through authentication will be unavailable.
- **One user is unable to sign in.** In this scenario, the problem is typically related to the user account. In many cases, the user won't be able to sign in because his or her user account hasn't synced to Azure Active Directory. In such a case, you should look at the sync logs and troubleshoot that portion. Otherwise, you can use standard account sign in troubleshooting. Verify that the user account is enabled and not expired and ensure that the password is not expired.

Pass-through authentication relies on TCP port 443. From the PTA server, you can validate connectivity to the Microsoft cloud by running the **Invoke-WebRequest -Uri**[12] command. If successful, you should get a status code of 200. If not successful, then you should look at your network connectivity, firewall, and proxy configuration.

By default, pass-through authentication components are automatically updated to the latest versions. This is helpful to keep the components up-to-date which ensures stability and compatibility with the latest improvements. Automatic updating uses TCP port 80 and a service named Microsoft AAD Application Proxy Connector Updater. Be sure that you have TCP port 80 open for automatic updates and that the service is running.

Single sign-on troubleshooting

Single sign-on enables users to authenticate without having to specify their password. Instead, their device sign-in is used and the user is signed in seamlessly. When SSO isn't working, users must manually specify their password, and this degrades the user experience. The following list identifies some common reasons why SSO fails:

- **Service URL is not configured in the Intranet zone.** A prerequisite to using SSO is having https://autologon.microsoftazuread-sso.com/ defined in the Intranet zone (Note: This is a service URL that cannot be accessed directly; you will receive an HTTP 404 Not Found error if you try clicking on it). Adding this URL to the Intranet zone is commonly configured by using Group Policy. However, it can also be done manually, especially in troubleshooting scenarios. This service URL needs to be added to

12 https://aka.ms/AA3rkb1

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your users' Intranet zone settings because by default, browsers will not send your Kerberos tickets (obtained during device sign-in) to the internet (instead, only to intranet locations), unless you explicitly add this service URL to the browser's Intranet zone.

- **Client devices are not joined and connected to the on-premises domain.** SSO only works when the client devices are joined to the domain configured for SSO. And the client devices must be able to contact a domain controller for the domain configured for SSO. Finally, the user must be signed into the device using their domain account, not a local account.
- **SSO prerequisites are not met.** There are a few prerequisites to establish SSO for the first time and to maintain functionality thereafter. The Azure Active Directory Connect server must be able to communicate with *.msappproxy.net (any subdomain of the msappproxy.net domain) for ongoing authentication. Initially, the server must be able to communicate to Azure Active Directory on TCP port 443 and to make connections to the Azure data center IP address ranges (for enabling SSO). See the article titled **Azure Active Directory Seamless Single Sign-On**[13] for more details on the prerequisites and setup process.
- **The SSO optional feature is not configured.** When you deploy Azure AD Connect, you specify which options you want to enable. If you want SSO, then you must use a custom installation and manually choose to enable single sign-on. In a troubleshooting scenario, you should check to be sure SSO is enabled.

The **Remote Connectivity Analyzer's**[14] Microsoft 365 Single Sign-On Test is a tool to help you troubleshoot SSO issues. Note that you should create a test account for the Remote Connectivity Analyzer test, or optionally change the password of an existing account to a temporary password before running the test. Once finished, change the password back.

Troubleshooting Transport with Exchange Online

Having a thorough understanding of how transport works in a hybrid environment is helpful for troubleshooting. First, you must understand which type of transport is currently in use. Is centralized mail flow being used, or is decentralized mail flow being used? Let's review the key areas that impact transport in a hybrid environment:

- Mail flow configuration (decentralized or centralized)
- Connectivity (network and firewalls)
- DNS
- Third-party hardware and software

Mail flow configuration

If you are using a centralized mail flow, then all email sent or received from the internet is routed to your on-premises environment first. When troubleshooting mail flow, you can start your troubleshooting at the edge (often Edge Transport servers or similar). But, with this configuration comes a dependency on the on-premises environment. A server or network problem at your company could result in a problem for mail flow from the internet to Exchange Online. This isn't something that administrators typically think through, especially when their hybrid environment is new.

If you are using a decentralized mail flow, then all email sent to the internet is routed directly to the recipient from your Exchange Online environment. The mail protections in Exchange Online might differ

13 https://aka.ms/AA3qw0g
14 https://aka.ms/RCA

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from what you used on-premises. The mail flow limits (for example, attachment size limits) might be different in Exchange Online too. If you are using a third-party email protection product to scan all inbound emails, you might need to make changes to facilitate the flow of email to Exchange Online (instead of to your on-premises environment).

Connectivity

When you set up your hybrid environment, you must ensure that you have connectivity to Exchange Online, and the firewall must allow the email communication to go between on-premises and Exchange Online. The two key ports are TCP port 25 for mail flow and TCP port 443 for free/busy information and mailbox migrations.

If you are in a high security environment, your firewalls might be configured to allow communication to and from specific IP addresses. With Exchange Online, that can be challenging because of the large number of IP addresses required and the potential for changes to the IP address ranges. To simplify things, you can configure your firewalls to use hostnames for the rules (when supported by the firewalls).

DNS

Mail flow is controlled by using MX records and can be restricted with SPF and DKIM records. When you implement a hybrid environment, you must make changes to your MX records and potentially your SPF and DKIM records. If your mail routing sends all inbound email to your on-premises environment, then you won't need to make a change to your MX records immediately. If Exchange Online is used for outbound email to the internet, then your SPF and DKIM records, if you use them, must be updated to ensure that your configuration enables Exchange Online to send email for your domain.

Third-party hardware and software

Some organizations have third-party hardware and software in their Exchange environments. These third-party technologies are often used for anti-spam, anti-virus, anti-malware, and security or compliance related tasks (such as auditing). Sometimes, these solutions are configured to be in-line with the mail routing. For example, email from the internet might first pass through an anti-malware solution before it gets delivered to your Exchange servers. These solutions are sometimes used for all outbound email too, often as smart hosts.

When you move to Exchange Online, whether in hybrid or completely, you must decide if you will continue to use the third-party solutions or migrate to an Exchange Online solution. In a troubleshooting scenario, you must factor in any third-party solutions that you have in place. You should always ensure that the solutions are supported for Exchange Online and for hybrid environments (if applicable). Sometimes, these solutions must be regularly updated to stay in support for Exchange Online.

Troubleshooting Client Access in Coexistence

When your on-premises environment runs Hybrid coexistence, troubleshooting client connectivity becomes more complex. Not only are you managing your existing Exchange on-premises servers, but you are managing Exchange Online mailboxes. Clients like Outlook will connect both to the on-premises servers, using your existing Autodiscover records to find their mailbox location and potentially to the Exchange Servers for services like Public Folder co-existence, or accessing Shared Mailboxes and Read/Write access to Calendars.

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To prepare for coexistence, you need to consider the following points:

- **Firewalls.** If you aren't using Modern Hybrid and the Hybrid Agent, you must connect public Exchange Servers to the internet. This might be through a supported Reverse Proxy or Load Balancer device. Connectivity from Exchange Online IP address ranges or URLs is required for classic hybrid. With Modern Hybrid, firewall rules to support inbound Exchange Online connectivity are not necessary. However, in all scenarios, outbound HTTPS connectivity from Exchange Servers is required.
- **Certificates and DNS.** If you are running Modern Hybrid, then your clients will access the on-premises environment using the same methods they previously used, and you won't need to update URLs or Digital Certificates. However, if you are using classic Hybrid, then you will have ensured you use trusted, third party Digital Certificates for HTTPS services. As part of this process, you will typically have ensured you use valid Internal and External URLs and configured Autodiscover correctly. If you haven't, then you may experience client certificate prompts or a failure for Hybrid functionality like Free/Busy or Mailbox moves to work. Ensure your Digital Certificates have Subject Alternative Names that match the configured Internal and External URLs, and DNS names. You can use the Remote Connectivity Analyzer to validate this.

Often when you troubleshoot client access in coexistence, the problem is limited to a single protocol. For example, Outlook on the Web might not work but Outlook will work. When troubleshooting, test client access across all the protocols to narrow down the protocol(s) that are not working. This will help you focus your troubleshooting to speed up the time for resolution.

The following troubleshooting steps often help find the problem for client access in co-existence scenarios:

- **Run the Outlook Test Email AutoConfiguration tool.** Run the tool to look for Autodiscover issues and namespace or certificate issues.
- **Test a new profile setup in Outlook.** This test indirectly validates Autodiscover and initial connectivity to the mailbox server. It also validates credential and permissions to the mailbox.
- **Run the Remote Connectivity Analyzer tool.** The **RCA tool**[15], which is useful in many client access scenarios, can help pinpoint configuration issues by automating a series of connectivity and discovery requests.

Additional reading. For more information, see the following article on **Troubleshoot a hybrid deployment**[16].

Troubleshooting Mailbox Replication Service

When you move mailboxes or public folders, the backend process that handles the data move is the Mailbox Replication Service (MRS). MRS also handles importing and exporting of .PST files. You will typically troubleshoot MRS when you have issues moving mailboxes or public folders, or you may troubleshoot MRS when move performance is poor.

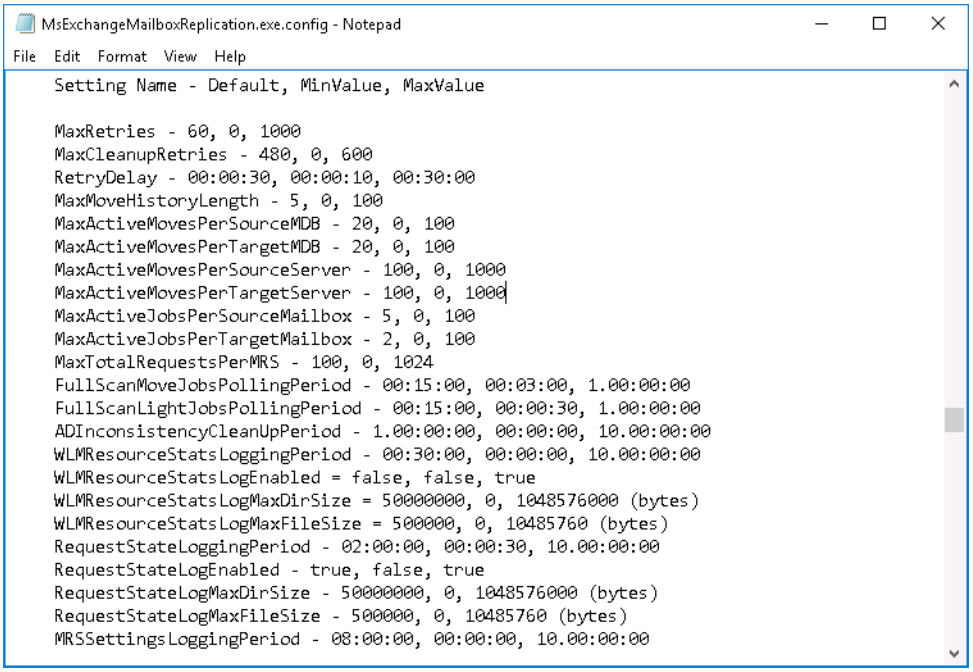
The default MRS settings are usually sufficient for most organizations. The default settings enable Exchange Server to lower the priority for mailbox and public folder moves, mostly when the server is under heavy load from other activities. In some cases, MRS can stop all current mailbox and public folder moves until the server has adequate resources to handle the moves.

Exchange Server uses dynamic throttling. However, you can disable dynamic throttling and instead, customize throttling by adjusting several settings.

15 https://aka.ms/rca
16 https://aka.ms/AA3qs3a

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The following screenshot displays the file which controls the settings, which is named MsExchangeMailboxReplication.exe.config (stored in the %ExchangeInstallPath%\Bin folder).



Note at the top of the file where it shows that the first number is the default value, the second number is the minimum value, and the third number is the maximum value. Remember, these values are not used with dynamic throttling. The values in the config file are only used if you disable dynamic throttling. After you make changes to the config file, restart the Microsoft Exchange Mailbox Replication service (MSExchangeMailboxReplication service).

Some indirect troubleshooting of MRS is often beneficial as well, such as collecting statistics to find information about move requests.

For example, if you want to get information on Beth's move, you would run the following command:

```
Get-MoveRequest Beth | Get-MoveRequestStatistics | FL
```

If you have a migration batch with a bunch of users, named Batch1, you can run the following command to get information about the batch:

```
Get-MigrationBatch -Identity Batch1 -IncludeReport | Select Report | FL
```

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Lab10-Ex2-Test Your Hybrid Deployment

Lab10-Ex2-Test Your Hybrid Deployment

In the prior lab, you set up Exchange for a hybrid deployment. In this lab, you will test your new deployment. This will require that you:

- Test the hybrid configuration by sending emails between on-premises and cloud users
- Migrate a user mailbox from Exchange on-premises to Exchange Online to test your connectors
- Test the newly migrated mailbox

IMPORTANT: Migrating an on-premises mailbox to Microsoft 365 can take up to an hour to complete. Once you have started the migration at the end of Task 2, proceed to the final assessment lab in Exercise 3. You cannot test the newly migrated mailbox in Task 3 until the migration is complete; therefore, once you have finished the final assessment, then return to this lab and perform Task 3 to test the newly migrated mailbox.

See your VM lab environment for the lab instructions.

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Lab10-Ex3-Final Assessment

Lab10-Ex3-Troubleshoot Your Hybrid Deployment

In this final lab exercise, you will continue in your role as Holly Dickson, Adatum's Messaging Administrator. You have been tasked with resolving a messaging issue in Adatum's newly configured hybrid deployment.

Problem

In your role as Adatum's Messaging Administrator, you have completed Adatum's hybrid Exchange deployment. However, several Adatum users have just informed you that emails are being delivered to their Junk Email folder rather than their Inbox. In fact, your CTO (Alex Wilber) is one of those users; he just stopped by your office to let you know that he did not receive an expected email from Allan Yoo, only to find out later that the email had been delivered to his Junk Email folder. This has added additional urgency to the problem.

The CTO wants you to figure out why emails are not being delivered properly.

Resolving the Problem

In this final lab exercise, you should use all the skills you have learned during this course to identify the issue and mitigate the problem that Adatum is currently experiencing. While a step-by-step lab solution is available, you should try and determine what is causing this problem and then solve it on your own.

If you find yourself struggling with this problem and unable to determine how to resolve it, you can download the **step-by-step solution guide for this Final Assessment lab**[17]. The solution is in this pdf document rather than in your VM environment.

However, in the spirit of learning, please try to resolve the issue on your own using the knowledge you have acquired in this course before turning to the solution guide, which should only be used as a last resort.

Hint: Review the information in this course relating to message delivery. Compare that to what you did in the previous labs when you configured Adatum's hybrid deployment to determine what needs to be fixed to solve this problem.

Reminder - Complete the mailbox migration lab after finishing this lab

Prior to starting this Final Assessment lab, you initiated a mailbox migration that is migrating Allan Yoo's on-premises mailbox to Microsoft 365. That migration process normally takes about an hour to complete. You should work on this Final Assessment lab while the mailbox migration is in progress.

Once you finish this Final Assessment lab, you should return to the final task in the prior lab exercise to test whether Allan's on-premises mailbox migrated successfully to Microsoft 365.

17 https://github.com/MicrosoftLearning/MS-203T00-Microsoft-365-Messaging/blob/master/Allfiles/MS-203T00-Lab10-Ex3-Final%20Assessment%20Solution%20Guide.pdf

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Module Review

Module 12 Review

Multiple choice

Item 1. Contoso currently runs Exchange Server 2010 and plans to deploy a hybrid environment. The company wants to ensure that all email messages that are sent to and from Exchange Online are processed in its perimeter network. What server should Contoso deploy in its perimeter network to accomplish this?

- ☐ Exchange Mailbox server role
- ☐ Exchange Client Access server role
- ☐ Exchange Edge Transport server role
- ☐ Exchange Hub Transport server role

Checkbox

Item 2. What features are included in the Exchange Edge Transport server role? (Select 3)

- ☐ Database Availability Group
- ☐ Internet message delivery
- ☐ Transport rules
- ☐ Availability service
- ☐ Address rewriting

Multiple choice

Item 3. You are the messaging administrator for Contoso, which was recently purchased by a competitor named Fabrikam. The Marketing department has since informed you that you must ensure that all emails sent to the Internet use the @fabrikam.com domain; however, internally you do not want to change the existing @contoso.com addresses. What feature do you need to implement to satisfy this requirement?

- ☐ Use Exchange Transport rules to configure address rewriting.
- ☐ Deploy an Exchange Edge Transport server to your perimeter network and configure address rewriting.
- ☐ Deploy an Exchange Mailbox server to your perimeter network and configure address rewriting.
- ☐ Implement an E-Mail Address Policy that includes @fabrikam.com.

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Checkbox

Item 4. You are planning to deploy Edge Transport servers in your perimeter network. Which of the following are valid requirement considerations? (Select 3)

- ☐ You cannot combine the Edge Transport server with any other Exchange Server role.
- ☐ You can combine the Edge Transport server role with the Exchange Mailbox role.
- ☐ The computer running the Edge Transport server must have an FQDN.
- ☐ The Edge Transport server must be member of an Active Directory Domain.
- ☐ The Edge Transport server must be able to resolve DNS names on the Internet.
- ☐ The Edge Transport server must be able to communicate directly with your Domain Controllers.

Checkbox

Item 5. You are planning to set up Exchange Edge Transport servers and Edge Synchronization for your company. What firewall ports do you need to open between your Exchange Mailbox and your Edge servers to allow mail flow and directory synchronization? (Select 2)

- ☐ Port 25/TCP
- ☐ Port 443/TCP
- ☐ Port 50636/TCP
- ☐ Port 50737/TCP

Checkbox

Item 6. How do you describe the functionality of EdgeSync? (Select 2)

- ☐ Synchronizes information from Active Directory to AD LDS on Edge Transport servers
- ☐ Synchronizes information from AD LDS to Active Directory on Edge Transport servers
- ☐ EdgeSync is always initiated by internal Exchange servers
- ☐ EdgeSync is always initiated by Edge Transport servers

Checkbox

Item 7. What are the steps required to configure EdgeSync? (Select 3)

- ☐ Create an Edge subscription file on a Mailbox server.
- ☐ Import an Edge subscription file on an Edge Transport server.
- ☐ Create an Edge subscription file on an Edge Transport server.
- ☐ Import an Edge subscription file on a Mailbox server.
- ☐ Start and verify the EdgeSync process on the Edge server.
- ☐ Start and verify the EdgeSync process on the Mailbox server.

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Checkbox

Item 8. What additional connectors are created in your Exchange organization when you configure EdgeSync? (Select 2)

- ☐ EdgeSync - Outbound to sitename
- ☐ EdgeSync - Inbound to sitename
- ☐ EdgeSync – sitename to Internet
- ☐ EdgeSync – Internet to sitename

Checkbox

Item 9. Contoso is planning to configure a hybrid deployment. Which of the following are hybrid deployment requirements that it needs to consider? (Select 3)

- ☐ Add your custom domains to Accepted domains in the Exchange Admin Center
- ☐ Register your custom domains with Microsoft 365
- ☐ Configure an Inbound connector in the Exchange Admin Center
- ☐ Deploy Azure AD Connect
- ☐ Ensure Exchange is properly deployed
- ☐ Configure an Outbound connector in the Exchange Admin Center

Checkbox

Item 10. What additional requirements do you have if you plan to implement Exchange Classic Hybrid compared to an Exchange Modern Hybrid deployment? (Select 2)

- ☐ Register your custom domains with Microsoft 365
- ☐ Trusted digital certificates for HTTPS services
- ☐ Self-signed digital certificates for HTTPS services
- ☐ Deploy Azure AD Connect
- ☐ Publish your client access namespace to the Internet

Checkbox

Item 11. You are the messaging administrator for Contoso, which runs a hybrid deployment with Exchange Server 2016. This hybrid deployment was configured using the hybrid configuration wizard (HCW). After which of the following actions do you need to run the HCW again? (Select 3)

- ☐ When you update your e-mail address policy.
- ☐ When you add an accepted domain to your on-premises Exchange environment.
- ☐ When you add a domain to your Microsoft 365 tenant.
- ☐ When you update your Exchange certificate.
- ☐ When you add a mailbox database.
- ☐ When you add a new send connector.

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Checkbox

Item 12. Your company runs a hybrid deployment with Exchange Server 2016. Which of the following management tasks should you perform to ensure your hybrid deployment is healthy? (Select 3)

- ☐ Verify that Azure AD Connect is running reliably.
- ☐ Monitor message routing between on-premises and Exchange Online.
- ☐ Use monitoring software to monitor the federated delegation.
- ☐ Verify that your users are created in the Microsoft 365 admin center.
- ☐ Replace expired trusted certificates with self-signed certificates on Exchange mailbox servers.
- ☐ Enable journaling to verify mail flow.

Multiple choice

Item 13. Contoso runs a hybrid deployment with mailboxes on both Exchange Server and Exchange Online. Users of mailboxes located in Exchange Server complain that they cannot retrieve any free/busy information from users located in Exchange Online. They can retrieve free/busy information from users located in Exchange Server. What should you do to troubleshoot the problem?

- ☐ Verify Oauth
- ☐ Verify mail flow
- ☐ Verify federated delegation
- ☐ Run Mailbox Repair Request

Checkbox

Item 14. In order to preserve hard disk space on your Exchange 2013 and 2019 servers, you plan to give your users the ability to store attachments in OneDrive for Business in Microsoft 365. What requirements do you need to fulfill before you can configure this feature? (Select 3)

- ☐ The user's mailbox must be located on an Exchange 2013 or later server.
- ☐ The user's mailbox must be located on an Exchange 2016 or 2019 server.
- ☐ You must configure a hybrid deployment.
- ☐ OAuth must be configured.
- ☐ Federated delegation must be configured.
- ☐ You must deploy SharePoint Server 2016 on-premises.
- ☐ You must deploy Office Online Server on-premises.

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Checkbox

Item 15. You recognize that Azure AD Connect is not correctly synchronizing a user named Vladimir to Azure AD. New users are synchronized correctly. You want to troubleshoot directory synchronization. Which of the following conditions should you check for to try and find the cause of the issue? (Select 3)

- ☐ Whether the Azure AD Connect console is started.
- ☐ Whether the sync service is stopped.
- ☐ Whether Connector credentials are invalid.
- ☐ Whether there is a duplicate object in Azure Active Directory.
- ☐ Whether the object is filtered out.
- ☐ Whether invalid characters are present on the object.

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Answers

Multiple choice

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